

The Thiele Machine

A Computational Model with Explicit Structural Cost

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The point

Here's the whole thing before any of the careful parts. I'm going to start with one small picture, take it apart, and keep asking what each piece means, as far down as I can get before I hit something that holds or something I can't get under. I'll tell you which. Then I'll put it back together at the end and you can check whether it still stands.

One warning about how this went, because it explains the shape of everything after it. Most of the time I had the hunch first and found the names later. Over and over, somebody had been here before me. Sometimes that made my day. Sometimes it pissed me off. You'll see both.

Two computers that end up the same

Put two computers on a table. Hand each of them the same list of a thousand numbers.

The first one sorts it. It compares, swaps, walks the list again and again until nothing is out of order. Then it does one more thing: it walks the whole list front to back and checks that every number is no bigger than the one after it. That check could have failed. It passes. Then it tidies up after itself, wipes its scratch work, and leaves the sorted list in memory.

The second one never looks at your list. Somebody typed the sorted version into it earlier, same thousand numbers, same order, and it puts them in memory.

Now wait until both are done and look inside. Same bits in memory. Same bits on the screen. Set it up right, and you can, and it's the same everything, down to the last bit. One of them found something out about your list and stood behind it. The other one is holding identical bits and found out nothing. Wheel them out on a cart, swap them, wheel them back, and ask me which is which. I can't tell you. You can't tell me. And the part that bugs me most: built the ordinary way, neither machine can tell you either.

Sure, while they ran they were different. One did thousands of comparisons, the other did a copy. But the runs are over and the runs are gone, and nobody was being sloppy. Ordinary computers are built to forget how their bits got there, on purpose. Later in this chapter there's the sentence where a physicist said so in 1961, and why he said it costs heat.

Who cares? Anybody who's been handed a sheet that says "all tests passed." Anybody who's looked at an inspection sticker on a windshield: the sticker looks the same whether somebody checked the brakes or somebody sold the sticker. An audit, a proof, a ledger, an AI that sounds very sure of itself: every one of them is a pile of bits that could have come out of either computer, and the question that matters is the one the bits can't answer. Did something establish this, or did it just print it? I'll come back to every one of those at the end of this chapter and say which of them this book

actually helps with. Fair warning: fewer than you'd hope.

I'm going to call the difference *established*. The first machine established that your list is sorted. The second has the same answer and established nothing. The rest of this chapter is me taking that one word apart.

Look harder

The obvious answer is: just look harder. Don't look at the end, look at everything. So make the computers as lookable as computers get. Scale them up to the size of a warehouse, every wire a lamp that's lit or dark, every switch something you can walk up to and touch. Nothing hidden in a chip too small to see. Put both of them side by side and walk the floor.

While they run, sure, you'd see the difference. One has a whole wing lighting up, comparing and swapping, and the other just copies. But they finish. The comparing wing goes dark. Now walk the floor and look at every single lamp. The sorted list is spelled out in lit and dark lamps in both, the same pattern, and everything else is dark in both, because both cleaned up.

There is no lamp anywhere that says "a check ran here and passed." Not hidden. Not encrypted. Just not there. You can see everything there is, and the thing I'm after isn't one of the things there is. Looking harder doesn't help, because the problem isn't that you can't see enough. It's that nothing was kept.

That's the shape I want you to carry through the rest: it's not a visibility problem, it's a keeping problem.

So keep the trace

Then the fix is obvious, isn't it? Don't let the run go. Keep the trace: every comparison, every swap, the check and its answer, written down as it happens. Wheel the computers out on the cart and back, read the traces, and you'll know which one sorted. Problem solved?

It's the right instinct, and it gets most of the way to this book. But take it apart, because "just keep the trace" has three things wrong with it, and each one is a piece of what comes later.

First, a trace is bits, and bits can be typed in. Whoever typed the sorted list into the second machine can type a trace in too: "compared, swapped, checked, passed," a few thousand lines of it, every one of them plausible. A trace written by a program is something the program produced, and the machine running the program doesn't care whether it's true. So keeping the trace doesn't answer the question. It moves it. Did something establish this trace, or did it just print it?

Second, the way out of that is to take the pen away from the program. Don't let the program write its own trace. Have something underneath it write the trace, one

step at a time, as the steps actually happen, so nothing a program does can add a line that didn't happen or erase one that did. Airplanes do a version of this. Flight recorders, which go back to a prototype David Warren built in Melbourne in the 1950s, are fed by the aircraft's own instruments rather than by anybody's account of events, and they're built to survive the crash, so the investigators don't have to depend on anyone's memory of what happened. That's the right shape, and it isn't mine. Later in this chapter it gets its name from computer security, a reference monitor, and I'll say outright that a machine built that way would be doing what this book describes. What the book adds is narrower than a rulebook for everything underneath. It's about the records that go up and never come back down: what that something has to keep for good, and what keeping it costs.

Third, you can't keep all of it. A full trace grows with every step and never stops growing, and a machine with finite memory runs out. Even the flight recorder doesn't keep everything. The cockpit voice recorder is a loop: for decades it held the last thirty minutes, now it holds at least two hours, and everything older gets recorded over. Every real machine that keeps a trace keeps a window of it, or a fingerprint of it, and forgets the content.

The fingerprint is the sharpest reply, so take it seriously. Fold every new line of the trace into a fixed-size scramble of everything before it, and you've got a hash chain. Haber and Stornetta's timestamps, later in this chapter, work that way; so do the audit logs Bruce Schneier and John Kelsey designed in 1999; and so do the measurement registers in the security chip a lot of computers carry, the TPM, which a running program can only extend, never set, until the next reboot. That really does keep something about the whole trace in a few dozen bytes. But look at what it keeps. It keeps a binding, not the content: to read the history you still need the trace stored somewhere, and the fingerprint only tells you whether a copy somebody hands you is the one that went in. The binding holds only because nobody knows how to find two traces with the same fingerprint, which is a bet on how hard a search is, not a law. That part isn't only my opinion. Moni Naor and Guy Rothblum proved a sharp version of it in 2005, and they even call the thing a fingerprint: keep a small private fingerprint of a big file somebody else stores, check the file by reading only a little of it, and against an opponent with unlimited computing the size of the fingerprint times the amount you read has to be on the order of the whole file. Do even slightly better than that against an opponent with limited computing, and you've shown that one-way functions exist, which is the hardness bet in its barest form. Their setting is a stored file, not a growing trace, but it's the same bet with its terms written down. And squeezing traces that keep growing into a fixed number of bits has to land many traces on one fingerprint. That's a merge, the same merge as the chairs further down, hidden behind how hard it is to find which traces share a seat. Fingerprints are great. They're not a way out of forgetting. They're a very clever way of forgetting.

So the real question was never "save everything." It's this: what's the smallest thing

worth keeping for good, and what does keeping it cost? The answer this book ends up with is not the trace but the commitment (in the database's sense, a question closed, not the cryptographer's sealed envelope; both come up later). Not every comparison the sort made, just the one fact that a check of this very claim passed and the question got closed. And here's the part that matters most: on a finite machine, keeping even that one fact for good forces the machine to forget something somewhere else. That's the chairs, further down. And keeping things for good fills up, which Part II turns into a count.

So the premise was never that computers can't keep records. They keep records all day. It's that nothing in the rules of an ordinary computer makes the record paid for, permanent, or earned, and those three come in at different places, so here's where. Paid for is the model's one rule: in the Thiele Machine, the step that raises the record has to cost something. Permanent is a premise, the one the chairs need, and one of the conditions Part III puts on a record. Earned, meaning a check of this very claim passed first, isn't in the model's core at all; it's what the step of the small machine in Part IV enforces, proved for that machine and every program on it. A machine that charges one mark for every step it takes meets the first rule and none of the others, and I'll keep pointing that out.

Computed, stored, established

Three words that get used like they're one.

Computed is about the steps. The first machine computed the sorted list. The second didn't.

Stored is about the bits. Both machines stored it.

Established is neither, and that's the whole problem. It isn't the steps, because they're gone. It isn't the bits, because they're identical. It's something like this: the first machine has a right to lean on "this list is sorted," because it ran a test that could have failed, the test passed, the test was about this very list, and then the question got closed. The second machine has the same sentence and no right to it. That's a true fact about the two of them, and it's written down nowhere, in either one. (Whose right, strictly? A machine doesn't lean on anything, any more than a filing cabinet does. It's the right of whoever reads the first machine to lean on what it says, and the machine is where the grounds for that right would have to be kept. "The machine has a right" is shorter, so I'll keep saying it, and now you know what it's short for.)

Four words I'll hold still from here on. A *check* is the test, the thing that could say no. A *commit* is the closing, the question shut for good, about that edition of that thing. The *record* is whatever stays behind to say so. And *established* is all three, about one thing. That's the working sense. The book cashes it at three depths, a paid closing in the bare model, an object with a checked relation and a paid event in Part III, and the whole chain on the small machine in Part IV, and each place says which one it means.

Which raises a fair question. What if the second machine, handed the typed-in list, runs the check on it? Then it has established that the list is sorted. Full stop. It didn't do the sorting and it doesn't need to. The line I'm drawing is between checked and unchecked, not between computed and copied, and "found out nothing" up there was about the copier that never looked. So whatever this book ends up counting, it isn't how hard anybody worked.

People say "a right to lean on it" about a lot of things that turn out to be vibes, so I had to find out whether this one was anything. So I went looking. Here's what I ran into.

The clerk who forgets everything

The picture everybody passes around is a person with a pencil, a strip of paper and a short list of rules, and that's what a computer is. It's worth going back to the source, Alan Turing's 1936 paper, and seeing how far he actually takes it.

He takes it a long way. Back then a "computer" was a job, a human doing sums by hand, and Turing modelled that person. He's careful with the person's mind: whatever the clerk knows, beyond a handful of states, has to go down on the paper. Then he goes all the way. He imagines the clerk getting up, walking off, forgetting everything, and coming back later. To pick up where he left off, the clerk leaves himself "a note of instructions," and Turing says that note *is* the clerk's state of mind, as far as the computation is concerned. Marks on paper and a note. That's the whole state.

My two computers are sitting right inside that. Suppose the clerk didn't write the note. Somebody slipped it under the door, in the right form, with the right instructions. He comes back, reads it, carries on, and every mark he makes from then on is exactly what he'd have made if he'd written it himself. Turing's model can't tell the clerk who did the work from the clerk who was handed a note saying he did. It was never asked to. Turing wanted something you could check without crawling into anybody's head, and a note on a desk is about as checkable as it gets. That's why the paper still works ninety years later. It's also why there's no place in it for the thing I'm after.

So the founding picture of computing is a man who leaves his desk, forgets everything he was doing, and trusts a sticky note. I relate to him deeply.

And now the question nobody needs answered to follow the argument, which is exactly the kind I can't leave alone: who were those clerks? Real people. Before machines took the word, a "computer" was a job title. In the late 1800s the Harvard Observatory hired rooms full of women to do the arithmetic of the sky, cataloguing stars by hand for a fraction of a man's wage, and some of them found things the astronomers who hired them had missed. Rooms like that ran artillery tables in both world wars. So when Turing modelled "the computer," he was modelling somebody's actual afternoon. Does that change the argument? Not a bit. But it changes the picture, for me anyway: the sticky note was always a person trusting another person's work,

and the question of whether the work behind it was really done is older than any machine.

Information doesn't have it either

My next guess was information: one machine got some, the other didn't. That one died fast. Claude Shannon, who turned information into a number in 1948, says on his first page that the meaning of a message is "irrelevant to the engineering problem." His number measures how many messages you *could* have sent, not what the one you sent did, which is exactly right for building telephone lines and exactly no help to me. Same message out of the same set, same number, for both computers. He'd give them the same number if one of them got hit by lightning and came out right by luck.

Fine, look at the one message instead of the set. In the 1960s some people (Ray Solomonoff, Andrei Kolmogorov, Gregory Chaitin, separately) worked out how to measure the information in one string: the length of the shortest program that prints it. Same two computers, same string, same number. And then there's the thing in that theory I keep going back to and standing on, to see if it creaks. Running a program can't add this kind of information. What comes out can't hold more than what went in plus the program, give or take a little. Leonid Levin, who proved a general version of this, has a one-liner for it that's too good not to repeat: you can torture an uninformed witness all you want, you won't get information about the crime.

Which means, as far as information goes, sorting your list told you nothing. The sorted list holds nothing that "your list, plus the word *sort*" didn't already hold.

Logic says the same thing in a different coat. The standard logic of knowing, the one Jaakko Hintikka worked out in 1962, has a property people call logical omniscience: if you know some facts, you automatically know everything that follows from them. So by that logic, once you knew every number on your list, you already knew whether it was sorted. Hintikka flagged the problem himself, in the same book, and people have been trying to fix it since. The fix that comes closest to me is Sergei Artemov's, from 1995: you only get to know a thing if you've built an actual justification for it, and the justification rides along with the claim. That's a reason traveling with the claim, which is almost a record. But it's a way of writing things down and reasoning about them. People have taught it to count, too: Ho Ngoc Duc, in 1997, made knowing take reasoning steps, and Artemov and Roman Kuznets, in 2014, asked how big a justification can get before knowing by it stops being feasible. It has a computational side, too: Francisco Bavera and Eduardo Bonelli (2018) turned justification logic into a lambda calculus whose terms carry, and can inspect, the history of the computation that made them, and Samuel Bucheli, Meghdad Ghari and Thomas Studer (2017) put justifications on a time line, where they persist and change. What none of it has is the closing of a question as a step with a price.

So the two most careful accounts I could find, one about bits and one about knowing, say the same thing: the sort added nothing.

And yet you know something after the sort that you didn't before. You can read the smallest number off the first slot. You can stop worrying. Something happened. That's the clue, and it's the first solid thing I've got: whatever *established* is, it isn't new bits. It's the same bits, *settled*.

The guy who got closest

Somebody had that clue before me, and he's the closest neighbour I've found, so he gets the most room and the hardest push.

Charles Bennett, a physicist at IBM, wrote a paper in 1988 that starts where I just ended up: information isn't the same as value. His example is an almanac of where the moon and planets will be every day for a hundred years. It has no more information than the equations it was calculated from, but it saves you doing the calculating. So the value is the work. He called the amount of work a thing carries its *depth*.

Was this my idea, already done, by a guy at IBM? The only way to find out was to find where it parts from mine, because if I couldn't find that place I'd just be renaming his idea. And it's one word: *plausibly*. Depth is the amount of work a thing was *plausibly* made with. Bennett says straight out that you can't know how a message actually came to be; for all you know it came from flipping coins. Depth is read off the finished thing, the way a fossil's age is read off the fossil.

Bring the two computers back. The sorted list in the first one has some depth. The identical list in the second has exactly the same depth, because depth looks at the list and nothing else. A photocopy of a deep thing is just as deep. Depth tells you *some* work was done, somewhere, by someone. It's not a record that *this* machine did it, and it's not a receipt for what doing it cost.

So Bennett's standing right next to the line, on the other side of it, and as far as I can tell he got his own question right. He asked how much work a thing embodies. I'm asking who has the right to lean on it and what they paid to get there. His answer lives in the object. Mine, if I've got one, has to live in the machine.

Statues that walk off

If it isn't in the information, maybe it's in the knowing, so I went and read the philosophers. And there it was again: people had been arguing about the difference between having the right answer and knowing it for a couple thousand years.

Plato's version is a joke about statues. There was a story that the sculptor Daedalus made statues so lifelike they'd walk off if you didn't tie them down. Plato says true opinions are like that: they wander off, and they're not worth much until you tie them down with the reason why. Once they're tied down, they're knowledge, and they stay put. Knowledge tied to a reason, and knowledge that stays. Of everybody I'd have bet on to hand me that shape, Plato wasn't on the list.

Then in 1963 a philosopher named Edmund Gettier broke the tidy version in three

pages. Smith and Jones are up for the same job. Smith has good reason to think Jones will get it, and he's counted the coins in Jones's pocket: ten. So Smith figures, fairly, that the man who'll get the job has ten coins in his pocket. Then Smith gets the job, and, unknown to him, Smith also has ten coins in his pocket. True, believed, backed by a reason. Not knowledge. He was right by luck, because his reason was about the wrong pocket.

That's the push that matters for a machine. Checking one pocket and committing to a claim about another is right by luck. So whatever a machine commits to, the check underneath has to be about *this very claim*, and I'd better say exactly what that means, because it carries a lot of weight later.

A check is about this very claim when four things hold. It tests the same property the commit names, not a cousin of it. It tests that property on the same thing the commit names, not the thing next to it. Nothing has written to that thing between the check and the commit, so it's still the same edition the check looked at. And the check is honest: it says yes when the property is true and no when it isn't, on every value it could ever be handed, not just this one. Break the first three and you're Smith with his ten coins. Break the fourth and the check is just a rubber stamp. Later in the book I build a machine whose step won't let a commit through unless the first three hold, and whose checks come with a proof of the fourth.

Two notes on those four, because they matter later. The fourth asks for more than it needs. To rule out luck, a check only has to never say yes when the property is false; saying yes every time it's true is a bonus. And the third is a very old bug in computing wearing Gettier's coat. Security people call it time-of-check-to-time-of-use: check a file, use it, and something swaps the file in between (Matt Bishop and Michael Dilger wrote up how to hunt for it in 1996). Databases guard against the same thing by making sure, just before a commit, that nothing they read has been written since; H. T. Kung and John Robinson called that optimistic concurrency control in 1981. A philosopher's pocket and a sysadmin's race condition, same bug. One limit, said now: a machine can only promise "nothing has written to that thing" about its own memory; for brake pads or a voter roll the third condition becomes a premise about whoever measured (Section 12).

And here's where I went further than this chapter needs, because the obvious next question is: does anything fix Gettier? People tried for years. The first patch is "no false steps in the reasoning": Smith's mistake was believing Jones would get the job, so rule that out. Then a philosopher named Alvin Goldman, crediting Carl Ginet, came back with barns. You're driving through a county that, unknown to you, is full of fake barns, just painted fronts propped up to look like the real thing. You look at the one real barn in the county and think "that's a barn." True. Believed. Good reason, your own eyes, and no false step anywhere. And still you were lucky: if you'd glanced at the next one over you'd have believed the same thing and been wrong.

Every patch gets a new barn, and as far as I can tell the arguing hasn't stopped. Which tells me something I'll keep: whether a check was good isn't only about this one case. It's about what the check would have done on the cases next door. Hold on to that, because it's the next section.

What "could have failed" means

I keep saying the first machine's test *could have failed*. Take that apart, because it's the piece everything else hangs off.

The machine is deterministic. Given your list, the check was always going to pass. So in what sense could it have failed? In this sense: if your list had been out of order, the check would have said no. That's a claim about a different list, not about chance. Run the second machine on a scrambled list and it prints the same sorted list anyway, because it never looked. Run the first on a scrambled list and its check catches it. The first machine's answer follows the truth about your list; the second machine's would be the same either way.

Here's the shape. A check is a question asked of every possible input, and it splits them into two piles: the ones it says yes to and the ones it says no to. "Could have failed" just means the no pile isn't empty. A check that says yes to everything has an empty no pile, and it establishes nothing, however long it runs and however official it looks. A philosopher, Robert Nozick, made a version of this a condition for knowing, in 1981: you know a thing only if, had it been false, you wouldn't believe it.

Push on it once more, because this is where it really gives. "Could have failed" is a fact about the *procedure*, about what the check would have done on other lists. It isn't a fact about the end state at all. Look at the first machine's memory when it's done: a sorted list. Nothing in those bits says "and whatever made me would have caught a mistake." That's the warehouse again, every lamp visible and none of them saying it. The one thing that makes the first machine's answer worth more is exactly the kind of thing an end state can't hold. The only way a machine can own it is to write something down at the moment the check passes, and to have that something mean what it says. More than anything else in this chapter, that's why I ended up needing a record and not just a better answer. Even then the record only carries half of it. It can say this check ran on this edition and passed. That the check would have said no on a bad list is a fact about the checker, and that half rides on a proof about the checker, which somebody outside the machine has to read and accept. Where that kind of chain of reasons stops is Popper's swamp, a little further down.

One more thing that tripped me up and then didn't. Rig a check to pass only on your one list. Is that cheating? No. It's a perfectly honest check of a different claim, "this is that exact list," and it has a big no pile: every other list. It establishes exactly that and nothing more. What a check establishes is whatever it's honest about.

Put that next to the empty no pile and the two fit together: a check establishes what

it's honest about, and a check whose no pile is empty is honest about nothing, so it establishes nothing. One thing I'll say plainly now, so it doesn't ambush you in Part IV: the small machine I build there doesn't police this. Its step makes sure a commit was checked first, about the same thing, by a checker proved honest. It doesn't stop you asking a question that can't come out no, like "is this number at least zero." Ask one and the whole chain runs, three marks paid, and nothing is established but the chain. The certificate I show off there asks one that can fail.

Settled, whether it's true or not

Then I ran into Charles Sanders Peirce, writing in 1877, and he said the thing I'd been running together without noticing. The point of inquiry, he says, is to get out of doubt and settle on a belief. And when someone objects that surely what people want is a *true* belief, he answers like a guy who's watched a lot of arguments end: "as soon as a firm belief is reached we are entirely satisfied, whether the belief be true or false."

That's funny, and then it isn't, because he's right. Settling and being right are two different events. A question can be closed on a wrong answer, and once it's closed it acts closed. Peirce cares a lot about which ways of settling hold up (science wins, for him, because it has to answer to something outside any one person), but the settling itself, the question snapping shut, is there every time, and it never comes with a guarantee.

Karl Popper said it from the other side. Every test, he says, has to stop somewhere, at some statement somebody decides to accept, and nothing in the evidence forces you to stop there instead of one step further. You stop when you're satisfied. His picture is a jury: the verdict is a decision about what happened, and the jury deciding it doesn't make it true. Science, for him, is a building on piles driven into a swamp, and the piles stop where somebody's satisfied they'll hold, not where the ground gets firm.

A real jury has a problem Popper doesn't mention, because a real jury is twelve people. Lewis Kornhauser and Lawrence Sager (1986) noticed that a court that votes by majority on each premise of a case and also on the verdict can end up contradicting itself: every judge is consistent, and the court isn't. Christian List and Philip Pettit (2002) turned that into a theorem. No way of combining people's yes-or-no judgments on connected questions meets a short list of reasonable-sounding conditions and always hands back a consistent, complete group judgment. A group's settling is a rule's settling, and which rule you pick decides which of the honest settlings sticks. I haven't tried to put a group on the machine, and that's where I'd expect the first wall to be.

Popper's swamp sent me further down, because it's the oldest problem in the book and it doesn't have a bottom. Ask why you believe something and you get a reason. Ask why you believe the reason and you get another one. Sextus Empiricus wrote down the ancient version, credited to someone called Agrippa, and in the 1960s a follower of Popper's, Hans Albert, gave it a name with a joke in it: the Münchhausen

trilemma, after the baron who claimed he pulled himself out of a swamp by his own hair. Every chain of reasons does one of three things. It goes on forever. It loops back on itself. Or it stops somewhere, because somebody decided it stops. People have argued for a very long time about which of the three to live with. So “when do I actually know, and when am I done checking?” doesn’t get a clean answer from anybody. What you can do is be honest about where you stopped and why. That’s a lot of what the rest of this book is: every place it stops is written down, with its name on it, so you can decide whether you’d have stopped there too.

Lay those two on top of each other and here’s what shows through. The check is one act. The stopping is another. The check can run as long as you like and it never *makes* you stop; at some point something decides, and the deciding is its own event. That’s the second machine’s whole trick: it skipped the check and kept the answer.

And a logician said the part I didn’t expect a logician to say. Per Martin-Löf, whose ideas sit underneath the kind of proof checker that checks this book, says a proof isn’t an object, it’s an act, and once it’s done it’s done, *irrevocably*. You could say that’s true of everything that ever happened. Fine. But it’s the two computers again: the first machine’s check happened, irrevocably, and the machine right now carries no trace of it. The past can’t be undone. The *record* of it can be lost, unless something keeps one. That gap is where this book lives.

Then the intuitionists got there first, which by now I should’ve expected. Brouwer, who held that a mathematical truth is something a mind makes evident, argued about an idealised mathematician working in stages, the creating subject. Georg Kreisel (1967) wrote it down as a formula: the subject has evidence for A at stage n . Anne Troelstra (1969) added that the subject never forgets what it has made evident. Mark van Atten (2018), going through the whole history, reads the formula as “by stage n the Creating Subject has made A evident.” Look at what that carries. It’s tied to a stage; at any stage you can ask whether it’s happened yet; it never goes away; and it’s never wrong. Van Atten even says the subject can’t just accept a proof handed to it from outside: it has to go through the steps and rebuild the evidence itself. That’s the copier who runs the check. And Göran Sundholm, who has long kept a proof as an act apart from a proof as an object (1998), worked through how that theory sits beside the thesis that every effective procedure is a program (2014), which is the side this book’s machines live on. So it’s the closest thing to my *established* that I’ve found, and it’s around sixty years old. Where they differ, the difference is honest: their subject is an idealised mind with unlimited time and memory, and mine is a machine with a finite table that establishes only the claims its program checks.

Not a switch

So far I’ve been talking about *established* as if it were a light switch: on or off, the first computer on and the second one off. That bothered me the whole time I was writing it, and for a while I couldn’t say why. It felt too small, the way a word can be technically

right and still not cover what you're pointing at. I don't trust that feeling on its own, because I've had it plenty of times about things that turned out to be nothing. So I went looking for something outside my own head that would either give the feeling something to stand on or show me I was imagining it.

I didn't know what I was looking for, at first, beyond a rough shape: two things that look exactly the same and differ in how far something about them has been settled. So I started collecting anything that seemed to have that shape, and most of what I collected fell apart as soon as I looked at it closely. How each one fell apart taught me more than the thing itself did, so I'll go through them in the order it happened.

The first batch was all hidden things. A pea under one of three cups. A tree, whose age you can't see from the outside. A forged painting that looks right until somebody tests the paint. A glass of gas, where the thermometer shows one temperature and underneath there are uncountably many arrangements of molecules that would all show the same reading. Plato's prisoners, who see shadows on a wall and not the things casting them. Every one of these looked promising for about a minute. Then I noticed they all fail the same way: look harder and the hidden thing shows up. Lift the cup. Cut the trunk and count the rings. Run the paint test. Look at the molecules, at least in principle. Turn around in the cave. In each case the hidden part is a fact about the object that better eyes could find. That isn't what the two computers have. You can see every lamp in the warehouse and not find the check.

That failure gave me the first real tool I had, and it's a crude one. Give an observer perfect vision of the visible thing, as much detail as they want, and ask whether they could read the difference off it. If they could, it's something hidden, and that's not what I'm after. If they couldn't, it might be. I want to be honest about how much weight that test can hold. It's a judgment, and I'm the one making it, with whatever blind spots I have. Somebody careful can look straight at the evidence and still read it wrong, because of what they expected to see. So I treat the test as a way of sorting my own examples, not as a verdict, and if you find a case where it sorts wrongly, that tells me something about the test and about me.

The second batch got past that test and failed a different way. A clock face reads three o'clock and doesn't show how many days have gone by; no amount of staring at the face tells you. A computer can keep a complete history of what it did, and Charles Bennett showed back in the 1970s, as I understand it, that a machine can keep such a history while it works and then erase it again without losing anything. Those are real extra directions, not hidden facts. But they're free, and they can be run back down, and neither of those fits what happened to the first computer, which did something that can't be undone and paid for it in work. If the standing I'm after could be added for free and taken back for free, it would just be bookkeeping. So that batch taught me what else I needed: something that only moves one way, and costs something to move.

The third batch came closer and still didn't quite fit. A thermometer that records the highest temperature it has seen only moves one way, until somebody resets it, and resetting is easy and free. The labels on vaccine vials that change colour if the vaccine has been too warm only go one way, but they're chemistry, not a check anybody ran. A measurement only counts as traceable if there's an unbroken chain of calibrations behind it, which is a lot like what I want, except the chain can lapse, and even the official reference kilogram turned out to drift over the years. The rules of chess say (as best I can read them) that a king that can still castle and one that can't are in different positions even on an identical board, which passes the perfect-vision test, except that in chess the hidden part changes which moves are legal, and in the machine I'm building the record never touches the ordinary computing at all. Each of these got something right and missed something else.

The one that fit best, and I want to say clearly that it only fit best among the ones I found, was the standing of a mathematical statement. Here's the version I could piece together from accounts, so check me on the details. In 1611 Kepler, in a short New Year's pamphlet about snowflakes, guessed that you can't pack equal balls any tighter than the way grocers stack oranges. He couldn't prove it. Around 1998 Thomas Hales announced a proof that leaned on a great deal of computer work. As I understand it, the referees spent years on it and came back saying they were about ninety-nine percent sure and couldn't check the computer part. Hales went on to have the whole proof checked by a proof assistant, the same kind of tool that checks this book, and that work finished around 2014. The sentence never changed. Whether it's true never changed, as far as anyone can tell. What changed was how far it had been established, in stages: a guess, a claim, a referee's ninety-nine percent, a publication, a machine check. A perfect reader of the sentence can't see which stage it's at. It only moved one way. And each stage cost real work, by people running checks that could have failed.

Then I tried to break that one too, because by this point I'd watched a lot of good-looking examples fall over. In the four-colour story, as I've seen it told, Alfred Kempe published a proof in 1879 that was accepted, and about eleven years later Percy Heawood found a hole in it. So standing can fall. I went back and forth on this for a while. Where I've landed for now is that the check that got Kempe's proof accepted couldn't see that particular hole. In the sense of the previous few pages, it was never a check that could have said no to that mistake, so what fell was never really earned in the first place. I'm aware that's a convenient reading for me, and I'm wary of convenient readings, especially my own. Take it as where I ended up rather than as a settled answer. It is, though, the reason the machine later in this book only lets a record go up on a check that can fail.

Laying all of that out together is what changed my picture. If there are stages, then at each one there's a line, and the line has two sides: not this far yet, and at least this far. Each line amounts to its own yes-or-no question, and the answer to each can only ever change from no to yes. Maybe the switch wasn't wrong so much as incomplete. Maybe

it was just the first of many lines, and the real shape is a direction with lines drawn across it. I've gone around many times on whether the lines create the direction or the direction comes first and the lines are just places where you cut it, and on whether yes and no are even different from each other until something draws the line between them. I don't have an answer to that. The one part I could actually settle is that a single switch isn't enough once there are three or more stages, and that is proved (Section 10 and the sections after it). The rest of this paragraph is me thinking out loud.

Flatland, where this hunch started (Section 1), turned out to belong in the first batch. When the sphere passes through the flat world, the narrator sees a line whose ends fade wrong and whose length keeps changing, and when he feels it he finds a circle that grows and shrinks. The hidden direction leaks into his view, faintly, and a sharp enough flat observer could in principle work out where the sphere is. In this machine, as far as I can make out, you could move a state all the way along the established direction without changing a single bit of the ordinary machine. That may be exactly why the ordinary machine can't see it. It may also be why, if the direction is worth having, someone has to build it in on purpose and keep it on purpose. Whether keeping it costs anything is what the rest of this chapter tries to find out.

None of these people were writing about what I'm writing about, and none of them is the reason any of this is true or false. I went through them because I needed something to test the feeling against, and most of what I tested failed. The failures told me what the thing isn't: not hidden, not free, not reversible, not steering. What's left after all that is mostly a list of what it isn't, and I'm still not certain I've described what it is.

Computing already had a word for it

Here's the slightly embarrassing part. Computing already has a word for this and uses it every day. Anyone who's used a database has seen it: COMMIT.

A database does a pile of tentative work, and then there's one moment, the commit, written to a log, after which the change counts. Jim Gray, who more or less built the theory behind that, opens his 1981 paper on it by saying the idea comes from contract law, and his example is a wedding. People negotiate, somebody asks, two people say "I do," and from then on it's binding. If it was a mistake you don't erase it; you do another transaction to fix it, the way a bad contract gets a second contract and not a time machine.

The wedding was a philosopher's example first. J. L. Austin, lecturing in 1955, said "I do" doesn't *describe* a marriage, it *makes* one, and sayings like that aren't true or false. They can misfire (the wrong person says it, a step gets skipped) but they can't be false. That's a commit exactly. Writing the commit record doesn't report that a question got closed. Writing it *is* the closing. And a commit can't be false, only misfire: committed without the check, committed about the wrong pocket. Gettier and Peirce again, from a third direction. Then somebody else reads the record, and for them it stops being a performance and turns into a report: "this was checked and closed." A report can be

false. On the small machine in Part IV it can't be, about the machine's own history: if its flag is up from a clean start, some claim was checked and closed, and that claim held of that edition when it was checked. That's the one place in the book where a commit and the truth get tied together, and it took a proof to do it.

Robin Milner, building a proof checker in the 1970s, ran out of memory, so he decided to keep only the finished theorems and throw the proofs away, like a teacher rubbing out the blackboard to make room. The obvious worry is forgery: if the proofs are gone, what stops someone just writing a theorem down? His answer was beautiful. Theorems got their own type in the programming language, and the only way to make one was through the rules of logic. You couldn't forge one because the language wouldn't let you build it any other way. He named that language ML, for Meta Language. I wanted to call my machine the meta machine. I'm choosing to treat that as a coincidence and not a sign, which is more restraint than I usually manage.

Bitcoin is the loud modern version. Its records can't be changed without redoing all the work that came after them, and a computer that drops off the network and comes back trusts the longest chain as the story of what happened while it was gone. That's Turing's clerk again, coming back to a note. The difference is that the clerk's note costs nothing to forge and Bitcoin's costs a mountain of work. It's the clearest case I know of somebody making a settled fact expensive to fake, and they got there by paying for a search, not by putting a price on the settling.

That's a different job from the one the toll does, and it's worth saying before you read the toll as a deterrent. One mark deters nobody. Bitcoin's work is tied to something scarce, electricity and hardware; a mark in this book is tied to nothing. Eric Budish (2025) worked out what that tie has to cost: for Bitcoin's kind of trust to hold, the honest work has to stay expensive at every moment compared with what an attacker could win, and the bill grows in step with the value being protected. A flat mark can't do that job, and it isn't trying to. What makes a raised record mean something, on the machine where it means anything, is an order rule: check, then commit, then certify. That rule would work just as well at a price of zero. The price is there because, on the premises of the chairs a little further on, a finite machine that prices its merges can't avoid it, not because it scares anybody off. The machines I actually build don't meet those premises, so on them the price is a choice made to match (Section 14 says why).

And Bitcoin wasn't first. Pull on its footnotes: three of the eight references in the paper that started it point to Stuart Haber and Scott Stornetta, who in 1991 worked out how to stamp a document with a time nobody could fake later, by chaining each stamp to the ones before it. A company built on their idea has been printing a short fingerprint of its chain in the classified ads of the New York Times, once a week, since 1995. Think about that for a second. To make a record nobody could quietly rewrite, they put it somewhere a million people would keep a copy of without even meaning to. That's the same move Zurek describes the world making with a measurement, and I'll get to

him at the end.

There are more fences like these: logs that can't quietly change, certificates you can check. Later in the book I take five of them apart. I lean on all of them. They work. I want that said plainly, because the next thing isn't "they're broken."

The next thing is where they sit. The database's commit is bits in a log, kept by a database program, on a machine whose own step will overwrite those bits the moment some instruction says to. Milner's guarantee is a type checker; it holds inside programs written in his language and nowhere else. Bitcoin is a protocol on ordinary computers that will do whatever they're told. Every one of them is a program or a rule, on top of a machine that can't tell my two computers apart. The machine underneath keeps nothing and charges nothing for what got settled.

The obvious comeback: so put the rule one level lower, in the operating system, or what security people call a reference monitor, the small always-on piece everything has to pass through. Yes. Exactly that. A monitor that wouldn't let any program raise its record without paying would be doing the thing I'm talking about. In this book's words it would be *implementing* the model, and I'd count that as agreement. What I care about isn't who builds the fence. It's what the fence has to obey, wherever it's built, and whether that's forced or just a good idea.

That comeback has a history, and it's older than me by a lot. In 1972 James P. Anderson's planning study for the US Air Force described a reference monitor: a piece every access has to pass through, that can't be tampered with, and small enough to check. In 2000 Fred Schneider worked out exactly which rules a monitor that only watches a run as it goes can enforce: the ones whose breaking shows up at some finite point, the kind computer scientists call safety rules. "No certificate without an earlier passing check and a commit" is one of those. And in 2015 Arthur Azevedo de Amorim and six coauthors built monitors that check a rule at every single instruction, and proved them correct in Coq, the same proof checker this book uses. So putting a rule under every program, so nothing can get around it, is their move, not mine. What I'm putting under that move is narrower: a record that has to be paid for, on the small machine one that has to be earned, and a reason, from counting, why a finite machine that prices its merges can't make that price zero.

What settling looks like

The chapter on what structure is (Section 2) ends with a pile of things sorted into buckets. Settling a question is that, plus one more thing.

Picture every state the machine could be in as a field of dots. A yes-or-no question about the state splits the field into two rooms: yes and no. That's a sorting, the smallest one there is. And remember the no pile: if one of the rooms is empty, there's no sorting at all, just a field. Now the extra thing that makes it a settlement: once the machine is in the yes room, *no step it can take leads back out*. The door opens one way. A settled

question is a sorting plus a door that only swings in.

At its best, what goes through the door is a relation: these numbers are in this order, these two things belong together. There's a chimpanzee named Sultan, later in the book, who fitted two short sticks into one long one and reached the food. Nothing new got added to the cage. Two sticks became one tool. That's what I'd want the door to be for.

But the door doesn't care, and this is the honest edge of the whole picture. Hook the door to a check that says yes to everything and the machine walks through, settled, and wrong. Settled whether true or false, like Peirce said. A door that only swings one way tells you a question was closed. It doesn't tell you it was closed right. Price and correctness are two different books, and I keep both, separately.

One more edge, the one Gray's wedding already hinted at. What stays shut for good is that the question got closed, about that edition of that thing. Whether the claim is still true is another matter. Change the list after the commit and the record still says, truthfully, that the old edition was checked and closed. And if the claim turns out bad, the fix is another act on top, a second record that says "withdrawn," never the first one quietly coming back down. So "entitled to lean on it right now" is three things at once: closed, not withdrawn since, and the thing unchanged.

Why it has to cost

This is the part I'm proudest of, because once you've picked a record that stays, the choosing stops and the counting takes over. And the counting is chairs.

Take any machine with a finite number of states. Its step is a function: give it a state and a move, it gives you back one state. Pick a yes-or-no reading of the state and make it permanent: no step ever turns a yes back into a no. Now suppose some step takes a state that reads no and lands it on one that reads yes. Then that step has to land two different states on one. It *merges*.

Do it with chairs. Draw a box on the floor, three chairs inside, one chair outside by the door. Four people, one per chair: the four states the machine could be in. The three inside already read yes. One step moves everybody at once, each to the chair the rule sends them to. Nobody inside may leave, because yes is permanent. The person by the door has to come in, because this is the step that flips them to yes. Now finish the move, one person per chair. You can't. Four people, three inside chairs: somebody shares. Maybe the newcomer ends up on a lap, maybe two insiders squeeze onto one chair; either way, two people end on one chair, and from where everyone's sitting you can't tell which two. That's the pigeonhole principle, which is old and not mine, and the checked version, with exactly those three conditions (finite, permanent, one flip), is in the chapter "Why the toll isn't arbitrary." No physics in it anywhere. Counting.

(How old is it? Older than you'd guess. A French Jesuit, Jean Leurechon, used it in

1622 to argue that two people somewhere must have exactly the same number of hairs on their heads, which is the kind of thing you can be completely sure of and never check. Its first proper name came two centuries later, from Dirichlet, who called it the drawer principle; the pigeons came later still. Is 1622 the bottom? I can't tell you. History is like that: it's where you start digging, not where you stop.)

Turn it around and it gets better. A *reversible* computer, one where every step can be run backwards, never merges anything, because a step you can undo never puts two people on one chair. Bennett, the same Bennett, showed in 1973 that any computation can be done reversibly if you give it room to keep its history while it works. But give a reversible computer a finite memory and, however much it computes, it can never raise a flag that stays up. It can always back up to before the flag. Settling needs exactly the kind of step a reversible computer refuses to take. (Where does the merged state go, physically? Out of the machine and into the room, as heat; the machine and the room together never merge. Section 4 comes back to that.)

Now the physics, as an assumption with its name on it. Rolf Landauer, at IBM, 1961, said that a step that can't be run backwards, that merges, "requires a minimal heat generation." That's the rule people call Landauer's principle. On a finite machine, settling something for good takes a merge, and if Landauer's right, a merge costs heat whenever the machine doesn't already know which of the merged states it was in. So on that assumption, a device that has to settle things correctly without knowing its own state ahead of time can't do it for free.

And here's the rest of his sentence, the one I promised at the start. That heat, he wrote, "serves the purpose of standardizing signals and making them independent of their exact logical history." That's my two computers, in 1961. Ordinary machines spend their merges, and their heat, on *forgetting how the bits got there*, on purpose, so a 1 is a 1 no matter what path made it. That's why the sorter and the copier end up identical, and why the comparing wing goes dark. And the very same kind of step, a merge, is what it takes to close a door for good.

So here's the deepest thing I can say about settling. Not every forgetting settles anything: wiping a scratch pad closes no question. But on a finite machine, every settling forgets. The step that closes a question for good lands two states on one, somewhere, maybe not where you happen to be standing, and afterward nothing can tell which one it was. A verdict keeps the decision and lets go of the arguing. An ordinary computer pays for the forgetting and keeps nothing to show for it. What I want to know is what a machine looks like that pays for the same forgetting and keeps the receipt.

Every assumption in that is a way out, and you should know where they are. Give the machine unlimited memory and it never has to merge; it can write its history down forever, and the chapter on the toll builds exactly that machine. Let the flag come back down and the chairs lose their grip. And the heat isn't owed by every run. Landauer's

principle, in the general form Part II adopts, charges for what you didn't know, so one run from a known start, with its controls timed from outside, owes no heat at all; the floor bites for a device that has to cope with states it can't tell apart, which is how devices get built (Theorem 4.12). The toll still charges its mark on that run, because the toll is a rule about the move, and the move doesn't know who's watching. And Landauer's principle is physics, not math. A group in Lyon measured the heat of erasing one bit in a tiny bead held by a laser, in 2012, and it has serious critics too, the philosopher John Norton among them. So the counting is proved and the price is an assumption, and I never let those two swap places.

Somebody got there first, goddamn it

Bennett also says looking can be free. His famous move with Maxwell's demon (the whole story's later in the book) is that you can write down which side of a box a molecule is on without paying anything, as long as you write on a blank page. What costs is wiping the page to make room for the next one.

I'd been sitting on this idea that *committing* is what costs, not learning, and I thought: this one's mine, I finally had an original thought. Then I found Carlo Rovelli's 2024 paper, "The Thermodynamic Cost of Choosing." Nope. Somebody got there first, goddamn it. A real physicist, with a real paper, saying that making a choice that sticks costs at least the same heat Landauer charges for erasing a bit. He even quotes a German saying, *Wahl ist Leiden*, to choose is to suffer, and at that point what are you going to do but laugh.

And then, taking him apart properly, it got good, because he seems to contradict Bennett. Bennett says measuring can be free; Rovelli says any measurement costs. So which is it? Look at what Rovelli calls a choice. A coin in the air hasn't chosen anything; it has to hit the ground and stop. And then the sentence I'd put on my wall: "stabilization must be irreversible, otherwise it is not stabilization." Bennett's free measurement is one that could still be undone; that's *why* it's free. Rovelli's choice is one that's been made to stay; that's why it costs. The way I read them, they're standing on opposite sides of the same door: learning on Bennett's side, committing on Rovelli's. (Bennett's demon still pays in the end, when it wipes the page.) Neither of them puts it that way, so push on it. But if it holds, it's the best sign I've got that the line I keep drawing between knowing and settling isn't something I made up.

Push on it, then, with his paper open, and it gives in four places: his choice is unpredictable, it's a branching where mine is a merge, it's stable only for a while, and he stands against Bennett's free measurement, not beside it (Section 5 goes through all four). So the two-sides-of-one-door picture is my reading, not his, and the piece of it that's really his is the one sentence about stabilizing. That sentence I'll still put on my wall.

So he's my closest neighbour, and here's the difference, exactly. His is physics: a price in heat, at a temperature, for a choice that sticks. Mine is a rule in a model of

computation: a charge in the machine's own count, on the step that flips a permanent reading on, with no temperature anywhere in it. Same instinct, different place. I'll take it. It's annoying to be second. It's worse to be alone and wrong. And he's the closest only on choosing. On copying there's an older neighbour, in the physics of molecules, and Section 5 gets to it.

What the Thiele Machine is

So here's what I built, and which parts I was pushed into and which parts I chose, because you should know which is which.

The Thiele Machine is a model of computation, the way Turing's machine is: math you can reason about, not a box you can buy. It has a few parts. A state. A step that moves the state. A record: a yes-or-no reading of the state that says whether something's been established. A charge on each kind of move, added up on a counter that only climbs. One rule, which I call the toll: the step that flips the record from no to yes has to cost at least one mark on that counter. And a window, what an outside observer gets to see, which is there because the two computers already showed you the problem: the usual window, the bits, can't see the record. The toll's first name, the one I gave it before I understood it, is No Free Insight.

One what, and why one? One mark, in the model's own count. One because it's the least a whole-number count can charge and still not be free, and the toll only says *not free*. The model doesn't say a mark is worth a joule or a dollar; whether it's worth anything physical is Landauer's assumption, named, and nowhere else. And one is just the floor: Part II shows that if you charge one mark per halving, a step that squeezes a lot together pays at least the logarithm of the squeeze, roughly how many times it halves the number of states it can tell apart. Don't read that backwards, though. One mark isn't one halving. A single state flipping into a big room of states that already read yes squeezes things together very little, much less than a halving, and the whole-number count rounds that up to one.

What was I pushed into? That on a finite machine, a record that never comes down can only be raised by a merge: counting, proved. That merges cost: physics, assumed. What did I choose? To put the count *inside the step*, as part of what a step is, instead of in a log off to the side. Nothing forces that. It's my claim about where the account belongs, and the rest of the book is the case for it.

About the name. I named it after myself, on purpose. Half of that is a joke and half of it is a stake in the ground. Turing got a machine, so here's mine. The longer story, meta machines and all, is in "How I got here" (Section 1).

The names work the way Turing's do, and I'm going to be a little annoying about them. *The Thiele Machine*, capital letters, is the model and nothing else. *A Thiele machine*, lower case, is any particular system proved to fit the model, the way one rule table is a Turing machine; a two-box toy on a sheet of paper counts. A Thiele machine is *Thiele-complete*

when the machine under the record can compute anything a Turing machine can, for free and without ever touching the record, and the record goes up only after a check that could have failed, then a commitment to the same claim, then a certificate, with exactly one mark charged for each of those three acts. And a *universal* Thiele machine is one fixed program that runs other machines as guests, with each guest's record earned again by the host's own steps and each guest's toll paid on the host's own ledger, not left to the guest's good manners.

The first bar, being a Thiele machine at all, is low, and I'd rather you hear it from me. A machine that charges one mark for every step it takes, whatever the step does, obeys the toll for any record you like; a clock is enough to make something a Thiele machine. If Thiele-complete only meant a Thiele machine that can compute anything, a clock bolted to a computer would pass that too, and so would a machine whose record never goes up. That's why Thiele-complete asks for more, and both of those fail it, by theorem: a clock charges for computing, and Thiele-complete says computing is free (Section 14). The small machine of Part IV meets it, and so does the same machine with any property whose checker is proved to mean what it says, sorted lists included. And every machine that meets it has the same blind spot built in, also by theorem: the part a Turing machine would call the whole machine, the line it's on and the numbers it holds, can't tell you its record or its ledger.

The two computers, put back together

Here's the two-computer story again, rebuilt as Thiele machines, with the record and the counter written out, so you can watch it instead of taking my word. The prices are the ones on the smallest machine I build later: one mark each to check, to commit, and to certify (raise the record), and nothing for ordinary work like moving numbers around.

step	sorter does	record	counter	copier does	record	counter
start	nothing yet	no	0	nothing yet	no	0
1	sorts (lots of ordinary steps)	no	0	copies in the typed list	no	0
2	checks "sorted" on this list, this edition: passes	no	1	(no check)	no	0

step	sorter does	record	counter	copier does	record	counter
3	commits to “sorted,” about this list, this edition	no	2	tries to commit: nothing checked, refused	no	1
4	certifies	yes	3	tries to certify: nothing commit- ted, refused	no	2

At the end, the bits on both machines are still the same sorted list. Everything else differs. The sorter’s record is up and its counter reads 3. The copier’s record is down, and it’s paid 2 for two refusals, because on that machine a refused attempt still costs what it costs. (Read those as two separate attempts. Written as one stored program, the copier would stop at its first refusal and pay 1.) Nothing the copier can do raises its record without first running a check of that very claim that passes. And no outside window that only shows the list can tell you which is which; the difference is all in the part the window doesn’t show. Walk that warehouse now and there’s finally a lamp that’s lit in one and dark in the other.

Now how literally to take that table. It’s a picture of the chain, and the chain is what’s proved. The small machine’s own checks are three questions about one counter: is it zero, is it even, is it at least some number. What its proofs need from a question is only a checker proved to say yes exactly when the answer is yes. That’s enough for “sorted”: a version of the machine that takes any questions with checkers like that proves the same chain once for all of them, and with a list packed into one counter it certifies a sorted list for three marks and refuses an unsorted one forever (Section 14). Without a proved checker for “sorted,” the chain would be honest about a counter and only by convention about your list, which is Gettier’s pocket again, one level down.

And a lamp for whom? The record is bits too. It’s a lamp on the machine, for anybody walking that warehouse. Hand somebody a printout of it across a desk and you’re back at the sticker on the windshield: they have to trust whoever hands it over, or have some way to check the box that says it ran. That crossing is a different problem with its own tools, signatures, attestation and proofs a stranger can check, and this book doesn’t solve it. Section 7 says what those tools rest on, and the small machine’s record, like any record kept in a machine’s state, carries nobody’s signature. What the step changes is the question, from “did it run?” to “do I trust the box that says it ran?”

That's a smaller question. It isn't no question.

Now the honest half. Take away the small machine's rules and keep only the bare toll, and the copier has an easier move: pay one mark, flip the record to yes, done. The toll is satisfied. The receipt is real. It's a receipt for nothing. The bare toll prices the closing of the door, not the check behind it.

What putting it in the step buys

So here's the hardest objection, and I'd rather make it myself. You defined a flag and a charge, then proved the charge gets paid when the flag goes up. That's your definitions agreeing with each other.

On the bare model, that's fair, and the copier's receipt for nothing is the proof. What makes the record mean *established* is the small machine's step, which won't let a commit through unless a check of that very claim passed, in the four-part sense from Gettier's pocket. And what it proves isn't about one program. It's this: on *every* list of instructions anybody can write, starting clean, if the record is up, then somewhere earlier a check of exactly that claim, about exactly that edition, passed; then a commit of the same claim; then the certificate; with nothing touching the thing between the check and the commit.

That "every" is the whole answer to what the step buys. A log kept by a program gives you a fact about that program: if the program's right, the log's right. The database's guarantee holds for work that goes through the database. A rule in the step gives you a fact about every program the machine can run, the buggy ones and the hostile ones too, because there's no way to run anything on it that isn't a string of its steps. That's what moving the rule down a level changes. Not what can be computed. *Which* programs the guarantee covers: this one becomes every one. Most of the rest of the book is how far that "every" goes and where it stops.

Two limits on that "every," said now rather than found later. First, it's every program, not every everything. The guarantee covers any list of instructions, run from a clean start, one step at a time, on a step that does what its definition says. Hostile programs, yes. A hostile start (a table of facts nobody checked, a reset), hostile hardware (a soldering iron on the flag's bit, a second writer the step can't see) or a hostile reader, no; Section 20 lists the whole fence. That's the same shape as the database's guarantee, true if the layer underneath is faithful, with a much bigger class of programs inside it.

Second, the flag and the counter on their own are what people who verify programs call ghost state: erase them and every program computes exactly what it did before (Section 6 has where that idea comes from). What isn't ghost on the small machine is the table of checked facts, which the step reads and refuses on, at run time, for every program. And the move of putting a rule under every program isn't mine either; it's the reference monitor's, and its history is a few sections back.

What it isn't

It isn't more computing power. Anything a Thiele machine computes, an ordinary computer computes too. The book proves that over any machine at all, an honest record is a switch the machine's own events flip once and nothing flips back; it watches the computing, it doesn't add to it. It isn't faster. It isn't quantum. It says nothing about which problems are hard.

And an ordinary computer can keep a record like this. Give a Turing machine a spot for the record and a track for the counter, write its rules so it never sets the spot without bumping the counter, prove that, and it *is* a Thiele machine, not a counterexample. What an ordinary programmable computer, as built, doesn't do is make its own step enforce a record *for the programs it runs*. When it runs another program, that program's record is just data in memory the program can write, and a buggy program can flip it for free, because the host's step has no opinion about it. Could it be built otherwise? Sure. A universal machine with one fixed table, the kind Turing built, could have that table written so it mirrors each guest's record and charges for the raise, and it would qualify. Nothing about ordinary computers stops that. They just aren't built that way. That gap, between the machine doing the stepping and the program it runs, is where the databases and the proof checkers and Bitcoin built their fences, and it's the gap the universal Thiele machine later in the book closes, first for the programs of the small machine it's built on, and then for every Thiele machine whose workings a program can compute.

Be clear about what that universal machine is, because there's a cheap version. A host with a move that means "run one guest step" runs any guest because I wrote it that way, and the book has one of those, as the simple picture. The universal Thiele machine doesn't get that move. It's one fixed program, written in the small machine's own six kinds of instruction on a host with more counters, and it fetches, decodes and carries out each guest instruction itself, the way Turing's universal machine does. The theorems are about that program: it halts exactly when its guest halts, with the guest's answer in its counters; its flag goes up exactly when the guest's would, by its own check, its own commit and its own certificate; its ledger matches the guest's mark for mark; and the machine it runs on is Thiele-complete (Section 14.13). Its guests are the small machine's programs, which compute everything. Add one move that pays a mark and does nothing else, and the same kind of program runs every Thiele machine whose step, prices and reading are computed by programs: it halts when the guest's run stops, its flag goes up exactly when the guest's reading has said yes, on a chain whose check is the guest's own reading, and its ledger is the guest's prices plus at most two marks at the raise, a surcharge no program on that host can avoid when the raising move costs less than three (Section 14.14). A Thiele machine whose reading no program decides isn't covered, and I say exactly where the edges are.

What's proved, what's argued, and what I just believe

Every claim in this book has a status and I'll tell you which, every time. Here's the whole list at once, so you know what I'm on the hook for.

Checked by a proof assistant, with every assumption printed, for any system with the named parts: anything that obeys the toll pays at least one mark to get from not-established to established (short enough that you'll prove it yourself on paper in Part II). On a finite machine whose record never comes down, the step that raises it merges, which is the chairs; with merging priced, the toll follows, and the price is an assumption with a name. Finding out can be free: you can learn something from a step that merged nothing and cost nothing, so the toll lands on committing, not learning. A window that shows a paid state and an unpaid one as the same can't price the step between them exactly, however clever it is. And over any machine, an honest record is a one-way switch on an event, with two choices left free: the price list and which event.

On the smallest machine I could build, six instructions on two counters: the chain in the table above (check, commit, certify, same claim, same edition), for every program, not just that one, with claims about any property that comes with a checker proved to mean it, "sorted" included (Section 14). A raised record always went through a passing check, then a commit, then a certificate, of exactly the same claim about exactly the same edition. Nothing gets into its list of checked facts without a check writing it. Every certificate costs at least three marks, and three is enough: check "this counter is zero," commit to it, certify, and when the counter is zero that pays exactly three; when it isn't, the check traps with the flag down and nothing after that ever raises it. The machine doesn't stop you asking a question that can't come out no, and I say so where it matters. It's Thiele-complete in the strong sense from a few pages back, and that sense has teeth: every clock-style machine, one that charges for computing, fails it, and every machine that meets it hides its record and its ledger from its own window, both by theorem. Its computing is Marvin Minsky's two-counter machines, which Minsky proved can compute anything a Turing machine can. And there's a universal Thiele machine on top of it: one fixed program, with no guest step built in, that runs every program of the small machine, halts when the guest halts, raises its own flag exactly when the guest's would, on its own earned chain, and keeps a ledger equal to the guest's. With one paid move added, one fixed program runs every computably presented Thiele machine, a machine from anywhere whose step, prices and decided reading are programs: halting matched, its record carried as a latch and raised on the host's own earned chain about the guest's own reading, its prices matched to within a surcharge of at most two, which can't be avoided in general.

Argued, not proved: that an account of computation *should* carry what was established, and what it cost, in how it steps. The proofs show that it hangs together, what it follows from, and what it costs (six instructions and three marks, on the small machine); whether it ought to be adopted is argued, and it's a claim about what matters, which

no proof checker can check.

Guessed at, with the ways to kill it written down: that the event worth recording is certification, because its records are the ones that get copied and relied on all over the place (the pointer conjecture, Part III).

Not proved, and I'll say so every time it's tempting: any physical price for one mark, anything about a physical device running any of this, or anything about which problems are hard.

Which of the proved part is new? Not one big thing, and I'd be suspicious of me if I said otherwise. By my own sorting at the back (Appendix C) it's four smaller ones, each with its neighbours standing right there. The toll derived from finiteness, permanence and a price on merging: the pigeonhole principle, Landauer and Bennett on merging, and the finite heart of Zermelo's 1896 recurrence argument are all next to it. A window that shows a paid state and an unpaid one alike can't price the step between them exactly: an old fact about functions, Moore's 1956 experiments and Kalman's observability, with a price put in. An honest permanent record, over any machine, is a latch on one event: the 1918 flip-flop and write-once memory are its neighbours. And the small machine, a definition of Thiele-complete that a clock can't meet, and a universal Thiele machine written as one fixed program: a record that's priced, and earned, under the reference monitor's every-program move (Anderson in 1972, Schneider in 2000, Coq-checked monitors from Azevedo de Amorim and coauthors in 2015), carried to every program of the small machine and, with one paid move added, to every computably presented Thiele machine, built and checked rather than new in idea. The first three are small exact statements I didn't find stated before, where I looked. The two-player game check later in the book is careful engineering on known mathematics, not a fifth.

At the bottom: what I suspect and can't prove

This is underneath the proofs, so read it with the lights on. None of it is a theorem, and I'm not going to dress it up as one. It's why I care, and I'd rather you know that going in than catch me sneaking it in later.

Physics has a word for the moment a question closes: *measurement*. And the people who thought hardest about it keep describing it in this chapter's shape. Before a measurement, it could go either way. After, it's settled, and the result gets copied out into the world, the light and the air, so many times that it can't be taken back and everybody who looks agrees. Wojciech Zurek has spent decades on how that copying works. Rovelli's coin has to hit the ground.

Niels Bohr said seeing anything at that scale depends on an apparatus that works irreversibly, like permanent marks on a photographic plate: a door that swings one way. John Wheeler went further and said every physical thing gets its existence from yes-or-no questions answered and registered, and boiled it down to "it from bit." And

Peirce, way before any of them, said the kind of settling worth trusting is the kind that has to answer to everybody.

Lay that next to everything above and the parts line up: a check that could have come out either way, a result that's now settled, a price paid in forgetting, a record that stays. I suspect that isn't a coincidence. I suspect the event this book prices on a machine is the same kind of event physics calls a measurement, showing up in a different place. One piece already sticks out, and I'll leave it sticking out: in Zurek's picture the world settles a result by copying it, and copying merges nothing (Section 13 pushes on that). Which picture a settling on a machine matches is part of the suspicion, not settled by it. That's as far as I'll take it here, and it's a suspicion, not a finding.

Now let me point the biggest gun I can find at it, and that's John Bell. In 1990 Bell wrote a short, furious essay against the word "measurement," with a list of words he said have no place in a physics that wants to be precise. On the list: system, apparatus, environment, irreversible, information, and measurement, which he called the worst of all. I used half of them in the last few paragraphs without even trying. Bell would have taken my suspicion apart in an afternoon, and he'd have been right to, because every piece of it leans on words physics hasn't pinned down. Nothing in this book proves any of it. The proofs are about models and machines, and whenever physics shows up in them it shows up as an assumption with a name on it. The suspicion is why I went looking. It isn't what I found.

Where this goes

So here's the picture, put back together. Two machines can hold the same bits and differ in what they've established, and you can't see it even walking the floor between them, because it was never kept. The difference isn't information; information theory says the sort added nothing new. It isn't depth; depth belongs to the bits, and a copy has just as much. It's an event: a check that could have said no, about this very claim, then a question closed for good, with no promise the answer's true. The could-have-said-no part lives in the procedure, not in the end state, so only a record can carry it. On a finite machine, a record that never comes down can only be raised by a merge, a kind of forgetting, the same kind ordinary computers pay heat for so their bits forget where they came from. If Landauer's right, that costs, for any device that has to settle things without already knowing which state it's in. The Thiele Machine puts the record and the charge in the step itself, and what that buys isn't more computing. It's a guarantee about every program instead of one.

Every time I thought I'd found something new on the way down, somebody had a name for part of it already, and that last sentence is no exception: a rule under every program is the reference monitor's move, from 1972. Fine. What I think is put together here is smaller, and more than one thing: the four I listed under what's proved, several small things, each with its neighbours named, not one big one.

And the stuff from the start, the test sheet, the sticker, the confident AI? Here's the

straight answer on which of those this helps. The sticker is a question of whether a piece of paper is genuine, and nothing here touches it. The “all tests passed” sheet fits only if the test is a checker proved to mean what it says, in the machine’s own language; a test suite that can miss a bug certifies “this procedure said yes,” which is true and smaller. The AI that sounds sure of itself isn’t addressed at all. What the book gives you is a record on the machine itself, with a proof of what a raised one means. Getting that record to somebody across the room who has to believe it is a separate job, and Section 20, under “Outward consequences,” says how little of the outside world I’ve reached. If you came for a tool, that’s a short list. If you came for the question, this is where it starts.

The rest of the book is me trying to break all of it. First where I came from, then the word I couldn’t define when I started, structure. Then the smallest thing that could carry the bill, on a sheet of paper, with one rule I’ll hand you without a reason, and then the chapter where I find out whether I had to pick it. If it breaks there, the book ends there. I’d rather know.

Before the book: the short version

The book in one paragraph. The Thiele Machine is an abstract model of computation: a state, a step rule, a counter that only climbs, a rule that the step turning a certification reading from no to yes costs at least one unit on that counter (I call that rule the toll, and its first name, the one I gave it before I understood it, is No Free Insight; Part II builds it), and a window that says what an observer sees. Here's what's proved, in Coq (a proof assistant: software that checks every step of a proof), with every premise printed. Any system that follows the toll pays something to get from uncertified to certified. On a finite machine whose certificate is never taken back, the certifying step merges states, so the toll follows from pricing merges, with Landauer's principle (the physicist's rule that forgetting costs heat) as a named premise. A measurement that forgets nothing can be priced at zero. A window that shows a certifying step and a quiet one as the same can't price certification exactly. Over any base machine, an honest permanent record is a latch on an event (a switch the event flips once and nothing flips back). And no procedure a substrate can represent decides which programs admit a shortcut, given a recursion theorem (the fact that a program can act as though it held its own code). A substrate is the model's bare shape, a kind of state and a rule for stepping, with whatever extra pieces a given theorem asks for; here the extra pieces are programs written as states and that recursion theorem. All of that is about the model: any system with the named pieces. One machine carries the model in this book, and it's a Thiele machine in the sense a few pages on pins down: six instructions on two counters, where commitments are earned and universality is proved. It's Thiele-complete under a definition no clock meets (computing free, the record earned, each of the three acts priced at exactly one), every machine that meets that definition hides its record and its ledger from its own window, and one fixed program in its own instruction kinds, with no step of any guest built in, runs every program of it with halting, flag and ledger matched exactly: a universal Thiele machine. Add one move that pays a mark and does nothing else, and one fixed program runs every Thiele machine whose step, prices and reading are computed by programs, the reading decided on every state: halting matched, the guest's reading carried as a latch on the host's own earned flag, and the guest's prices matched to within a surcharge of at most two marks, which can't be avoided in general. The lambda calculus L gets a recursion theorem by proof, and so does the machine that universal program runs on, with Rice's theorem beside it (Rice's also holds on the two-counter machine itself, for what a program does from the clean start). An integer check on eight counts is earned on the small machine: a raised flag can stand on a passing check of the counts, and the check passes exactly when every question pair was sampled and a fixed moment matrix is positive semidefinite, which forces Tsirelson's bound, the ceiling quantum mechanics puts on the score of a two-player game (Part IV builds the game, the matrix and the check from zero; it's there as a check with a striking result, to show the toll charges the same however deep the check is, not as a physics tool). The count of what narrowing a list of rivals costs holds for the record moves of every Thiele-complete machine; the eight-state

finite machine runs on the small machine and pays exactly its own charges; and on every Thiele-complete machine no checker that sees only the clean start and the final window is both sound and complete about the record. Not proved: a physical price for one mark, any separation of complexity classes, that certification is the event worth recording (that's a conjecture), or anything about a physical device running any of it. Of all that, what I'd call new is several small things, not one: the toll derived from finiteness, permanence and a price on merging; a window that can't price it exactly; the latch shape of a permanent record; and, built and checked rather than new in idea, a priced record, earned on the small machine, under the reference monitor's every-program move, carried by one fixed universal program to every program of it and to every computably presented Thiele machine. Appendix C puts each beside the work it leans on.

I didn't invent a machine. I found one. That's how I see this work, and I'll be straight about the difference between that conviction and a theorem before I do anything else.

That sentence walks straight into one of the oldest fights in mathematics, so here's a short trip into it first. G. H. Hardy, in *A Mathematician's Apology* in 1940, came down on the found side about as hard as anyone ever has: mathematical reality, he wrote, lies outside the people who study it, and the theorems they like to call their creations are really just notes on what they saw there. Leopold Kronecker supposedly split the difference: God made the whole numbers, and everything else is the work of people. Supposedly, because push on that line and it gets thin fast. It comes down through Heinrich Weber, in a memorial piece written after Kronecker died, pointing back to a lecture from 1886, and that's the only source anybody cites. A great line with one witness. (The first chapter of Part II stops early on to ask how many witnesses a claim ought to need. The answer is more than one.) So does any of this settle whether I found the Thiele Machine or made it up? No. A proof checker doesn't settle it either; Coq checks a proof the same whether you think the theorem was out there waiting or you think you built it. That's why it's labelled a conviction. What it changes is how I hold the thing: like something that was already there, which means I can be wrong about its shape and the shape won't care. And to be exact about what "found" covers: the parts the counting forces on you, the latch shape and the merge. The event to record, the price list, and putting the count inside the step are choices, mine, and "The point" said so where it sorted what I was pushed into from what I chose.

What the Thiele Machine is. An abstract model of computation. A machine the way Turing's machine is a machine: mathematics you can reason about, not a device. It isn't a CPU or an instruction set; those are things you build to carry it, and Part IV builds one, as small as I could make it. The model is small. A state. A step rule. A counter that only climbs. A rule for which steps have to pay. And a window: what an observer is allowed to see. Like a Turing machine, it's a precise way to say what a computation is. Unlike a Turing machine, it computes nothing new. It adds what a computation *establishes*, and what establishing it cost. The event it accounts for happens on any

computer, whether anybody keeps track of it or not. What Turing's model doesn't have is a place for the record of that event, or anything that enforces one. A Turing machine can be programmed to keep the record, and then it obeys the toll and it's a Thiele machine in its own right. But when it runs some other program's record as data on its tape, nothing in its own step enforces it (Section 14.16). The event's physical cost comes in only under Landauer's premise, with its name on it. The record exists only if something keeps it, and keeping it, priced, is what the model adds.

What this book is. An argument, made with that model, about what an account of computation should preserve. The usual account describes a computation by what it reads and writes. I think that account is a shadow, and that there's another axis, holding what the computation established and what that cost. Here's the argument in five steps, each with where it stands.

Holding a sorting costs something; standing behind one might. That's Part I: one word, structure, pushed on with you through as many of the people who wrestled it before me as I could reach. It finds three different acts that people run together. Building and storing a sorting you can read back costs work and space. Forgetting where things came from costs heat, if Landauer is right. And standing behind a sorting, vouching for it, is a third act that nobody I read had put a price on. Whether it has one is the question the rest of the book takes up.

Establishing is not free. If the step that turns uncertified into certified has to pay, every run from uncertified to certified has paid. Proved, for any system that follows that rule, whatever its states and steps. And the rule isn't arbitrary: on any system with finitely many states, a certificate nothing takes back can only be switched on by a step that merges two states. That part is pure counting, proved. If merging is charged for, the way Landauer's principle charges for it in its worst case, the rule follows. That part is proved with the charge as a named premise, its critics named, and every premise refusable with a counterexample built. Learning isn't what costs: a measurement that forgets nothing can be priced at zero by an honest price list, and an observer still learns from it. Proved. What can't be free, on those premises, is committing to what you learned in a record the machine can't take back.

Two more things belong to that step. First, which steps a price on merging has to charge: exactly the ones that merge. Every step that writes a record nothing takes back is one of them, and on a finite machine a step that switches a certificate on escapes the price only when the same step switches one off somewhere else. The tidier guess, that the priced steps are exactly the permanent records, is false, and three states show it. No theorem picks out a wider class of events every account must price. With one mark per halving, the step that writes a permanent record costs at least the base-two logarithm of how much it squeezes together (how many times it halves the number of states it can tell apart), and the same bound holds in bits of Shannon entropy, the standard measure of information. Proved, with the price as a premise. Second, where

the machine I built stands, which is a Part IV fact and not a Part II one. The small machine of Part IV isn't such a machine: its counters have no top, so it isn't finite, and some of its free steps merge states too. On it the toll is a design choice the chairs don't force. What the chairs do reach is a finite piece: an eight-state machine, four slots and a flag, on which every premise is a theorem and the toll follows outright (Section 4). And the small machine runs that piece: seen through a window that shows only the line number and the flag, a program of the small machine does what the eight-state machine does, one move at a time, and pays exactly what the eight-state machine charges (Section 4, with the construction in Section 14.17).

A window can hide what was paid. Two runs can look identical through the usual window and differ in what they established and paid, and nothing reading only the window gets the difference back, so no price read off the window charges the certifying step exactly. Proved, for any window with such a pair; and the machine I built has such a pair for the window a Turing machine would point at it, the line and the counters, and so does every machine that meets the definition of Thiele-complete.

Leaning on a fact takes a paid history. A computation is entitled to lean on a structural fact when it holds the object, a checked relation, and a paid event on its record. Defined, for the model, with a counting fact beside it: narrowing a list of rivals down takes at least the yes-or-no questions that name the survivors. The cost floor that ties those questions to steps a run actually paid for holds for any system with a counter, given lists, a tree and a covering that somebody supplies: if a run paid at least as many charged steps as the covering tree is deep, those payments cover the index bits. No run is shown to have to pay them. On every Thiele-complete machine, the small machine included, the paid steps are the record moves, each exactly one mark, and a run from a clean start that ends with the record up, having narrowed a list that way, carries an earned chain and pays at least the index bits in record moves (Section 14.17); and when the questions are counted as the run's own checks, the floor needs no tree to be supplied, with a limit stated where it's proved. Entitlement is a license to lean on a claim, not a proof that the claim is true.

So the axis belongs in the step. An account of computation should carry what was established and what it cost as part of how it steps, not as something bolted on after. Proposed. The proofs around it show it hangs together and what it costs; whether it should be adopted is argued, not tested. What's proved around it: over any base machine, an honest record is a latch on an event, with two free choices, the price list and the event; no procedure a substrate can itself represent decides which programs admit a shortcut, given a recursion theorem, by the same self-referring trick, the diagonal, that put a hole in Gottlob Frege's foundations for arithmetic (Part I tells that story); and which event deserves the record is a conjecture, with its refutations listed.

Certification is one point I could pin down. It gave me a law I could state, a machine I could build and run, and proofs I could put in front of someone who didn't share

my intuition. It's one witness in a larger argument about structure and observation. It isn't the argument.

How the book is laid out. Part I is the picture: how I got here, and what structure is. Part II is the axiom: the smallest thing that could carry a bill, and why its one rule isn't arbitrary. Part III is the logic: the counter, the windows, entitlement, insight, the latch, the diagonal, five real systems, and a conjecture. Parts I to III never use my machine; every claim in them is about any system with the pieces it names, and the pictures are toys. Part IV builds one machine, the smallest honest machine I could make: two counters, six instructions, a table of checked facts, a ledger and a flag. On it a raised flag provably went through a passing check, a commitment and a certification of the same claim, in that order, and it computes everything Minsky's two-counter machines can, so whether it halts is undecidable. There Thiele-complete gets its definition, the one a clock fails, and the universal Thiele machine gets built as one fixed program that runs every program of the small machine and earns its own flag, and then, with one paid move added, every computably presented Thiele machine. Then a check worth earning, the two-player game and its integer test, and what the counter has to do with physics. Part V is how to check all of it and how to break it. The vocabulary I had to learn is at the back.

If you want the argument and not the whole tour, read "The point" and this short version, then [Section 3](#), [Section 4](#), [Section 10](#), [Section 14](#) and [Section 21](#). If physics is your thing, add [Section 5](#) and [Section 16](#). If you mean to check me, add [Section 18](#) and [Section 19](#). The appendices are reference: everything in them is there to be looked up when a claim points at it. Nothing gets cut for that. It's a map, not a diet.

One more thing before the book starts, because people get it backwards. I used language models to write the code and the proofs in this project, and I trust language models about as far as I can throw them, which is exactly why every formal declaration is checked by a proof assistant that doesn't care who or what wrote it. Coq rejects a wrong proof the same whether a tenured professor or a model hands it over, and it accepted these, with none of my own axioms anywhere in them, re-checkable by the procedure in Part V. Being self-taught and using modern tools didn't make this looser. It's the reason I made a machine check the formal corpus, and then built an auditor to read over the checker's shoulder for every way I might have cheated, on purpose or by accident (Part V says what it looks for). It doesn't trust me either. Good.

I think there's something foundational here. The results in this book have statements, premises, and proof terms. The larger claim is a conjecture, not a theorem. I built the machine so that the work could be done in public, with something firmer than my confidence to argue over. If you're the skeptical sort, and I hope you are, the two places to aim are the appendix that sorts what's mine from what isn't ([Appendix C](#)) and the chapter of ways to break it ([Section 18](#)).

Let me be straight with you

What this project is. An argument about what an account of computation should preserve, made with an abstract model built to carry it. Its central proposal is that what a computation established, and what establishing it cost, belong in how it steps. The working form of that proposal is structural entitlement: a computation is entitled to lean on a structural fact when it holds the object, a checked relation, and a history that earned it. The model, its laws, and the instructions of any one build are different things, and I keep them apart. The generic flag-raising step pays to set a flag; it checks nothing. A claim that something is actually *true* needs explicit witnesses and a checker proved sound, and some results here are of that kind; the generic accounting law isn't one of them. It's a law about marks.

The proved core, in three steps. Anything that follows the toll pays something to get from uncertified to certified. A window that shows two states as the same, when they disagree on the counter or on some other question, can't answer that question on every state. Those two are about any system with the named pieces, and so is the third, argued in three lines in Part III: a checker that only sees a window where two states collide can't be both sound and complete about the claim that separates them (Section 7). All three are statements about named laws and named windows, and the third is checked too, over abstract transcripts and then for every Thiele-complete machine (Section 7). An ordinary machine can keep a richer encoding, or enforce the same invariant through its own transition table. What I'm proposing is about why the extra distinctions matter. It isn't a theorem that every Turing-equivalent system lacks them. And if you want the list of what I think is new, it's the short one at the end of "What's proved, what's argued, and what I just believe" in "The point," the same four things the short version names, and nothing bigger.

Substrate versus scaffolding. Two layers run through the book, and confusing them makes the theorems quietly stop meaning what they say. The *substrate* is the shape: a kind of state and a rule for stepping. Different theorems ask that shape for different extra pieces, and the pieces aren't the same. The universal floor (the "pays something" result above) asks for a certification reading and the toll on it. The counter's uniqueness (only one count fits a chosen start and price list) asks for a starting value and a schedule of increments; Part III gives the argument, a short induction that works for any system with those two pieces, and on the small machine of Part IV the ledger is the sum of the charges by theorem (Section 14). The diagonal asks for a way to write programs as states and a recursion theorem for the transformations the substrate can express, and needs no toll at all. "Substrate-level" doesn't mean one fixed plug that every theorem shares. A theorem only gets to use the parts it asks for. Ask it what it needs and it'll tell you, and whatever else is lying around the workshop isn't the theorem's. So each result is stated over exactly the extension it uses. The substrate with the extension a theorem names is what I mean by *the model*, and the name Thiele Machine belongs to the model. The *scaffolding* is everything that holds the shape up

so you can touch it, and when I say *my machine*, *the small machine*, *the machine I build* or *the build*, this is what I mean: the machine of Part IV, with its two counters and their versions, its table of checked facts, its channel, trap, ledger and flag, its six instructions and their prices, the host with a register for every whole number, the universal programs that run on that host, and the Coq files that define all of it. The build fills in several extensions at once and borrows, for any one theorem, only the extension that theorem names. Let that line blur and a theorem stated over one extension starts reading as if it carried premises it never uses. “Substrate” never means the build, and it never means the side of a simulation doing the stepping; for that I say “the machine doing the stepping,” as against “the program it runs.”

So here’s how the book keeps the two apart. Parts I to III are about any system with the pieces each result names, and their pictures are toys you can hold (two boxes on a sheet of paper, chairs in a room, a row of dots), not my machine. My machine first appears in Part IV. Every Part III claim that has a machine instance gets that instance in Part IV, with its own checked name, and Part III says so in one sentence at most. When the origin story, a few pages on, says the shape “was the substrate the entire time,” it means this layer, the model’s shape, and not any piece of the build. Same story with two more of its words. When it says “the machine” about the thing it named the Thiele Machine, or calls that thing a meta machine, it means the model; “my machine” is the build, and the build doesn’t show up until Part IV. And when it lists what the model asks for (a state, a step, a ledger, a window), read the ledger and the window as two of the extra pieces: a result asks for them only if it uses them.

Part IV has one piece of scaffolding, and I’d rather you know up front how small it is. Section 14 builds the smallest honest machine I could: two counters, six instructions, a table of checked facts, a ledger and a flag. Earned commitments are proved there (a raised flag was checked, then committed, then certified, on the same edition, with nothing in between), and so is universality, through Minsky’s two-counter machines and a halting result borrowed from the Coq undecidability library. Then one fixed program, on a host made of the same kinds of instruction with more counters, runs every program of it, and with one paid move added, every computably presented Thiele machine. None of that is the model. It shows the model fits in six instructions.

The names again, this time with every part pinned down, because I’ll hold to them for the rest of the book. *The Thiele Machine*, with capitals, is the model and nothing else. *A Thiele machine*, lower case, is any particular system proved to be an instance of the model’s core definition, a certification system with its five parts and its toll (Definition 3.1), the way one rule table is a Turing machine. The toys on paper in Part II count, the same way trivial Turing machines do. The small machine counts too, with its toll proved, and Part IV names the theorem. A Thiele machine is *weakly* Thiele-complete when its base is Turing-universal, and that bar is nearly empty: a machine that charges one for every step is a Thiele machine for any record, so a clock bolted to a computer clears it, and so does a machine whose record never rises. So

Thiele-complete asks for four things (Definition 14.7). A universal base whose moves are free and can't touch the record. A record that rises only after a passing check of a claim, then a commitment to that same claim with nothing it's about changed in between, then a certificate, with the checker proved to mean what it says. An exact toll: one mark for each of those three acts and nothing for anything else. And a claim whose check could have failed. Every clock-style machine fails that, whatever its record, and every machine that meets it hides its record and its ledger from its own window, both by theorem. The small machine meets it, and so does the same machine over any property with a proved checker, sorted lists included. A *universal* Thiele machine is one fixed program that runs any machine of a stated class as a guest, with the guest's record earned again and its toll paid by the host's own steps. Part IV builds one, U : a fixed program in the small machine's own six kinds of instruction, on a host with more counters, with no step of any guest built in. Its guests are the small machine's programs, so every computation runs on it, and it matches each guest's halting, answer, flag and ledger exactly, raising its own flag by its own check, commit and certificate. Then U_P , the same program with one more handler for a paid move that does nothing else, takes as its guests every computably presented Thiele machine, one whose step, prices and reading are computed by programs, with the reading decided on every state. It matches the guest's halting, raises its own flag on its own earned chain exactly when the guest's reading has said yes, and keeps the guest's prices to within a surcharge of at most two, which no program on that host can avoid in general. And a computer that runs the small machine, your laptop say, isn't a Thiele machine either. It implements one, inside bounds, the way a laptop implements a Turing machine. Push on that comparison, because it's loose, and the looseness is the point. Does a laptop implement a Turing machine? Not exactly. Turing's tape never runs out, and a laptop's memory does; strictly, a laptop is a very large machine with finitely many states, and it behaves like a Turing machine only on computations that fit. Nobody minds, because the bound is known and most computations people care about fit inside it. That's the same deal any device running the small machine gets: its counters stop somewhere, and the theorems speak only for runs that fit. The bounds aren't fine print. They're the whole difference between implementing a model and being one.

How much can a difference in width matter? There's a famous answer, and it's worth the detour. On 4 June 1996 the first Ariane 5 rocket lifted off and broke apart about forty seconds after its engines lit. The inquiry board, chaired by the mathematician Jacques-Louis Lions, traced it to one conversion in the inertial reference software: a sixty-four-bit floating-point number, an alignment value called BH that's tied to horizontal speed, squeezed into a sixteen-bit signed integer. On Ariane 4, where the software came from, that value could never get big enough to overflow, because of how that rocket flew in its first forty seconds. Ariane 5 flew differently. The value got too big, the backup unit failed on it, the main unit failed the same way about seventy milliseconds later, and the function doing the converting didn't even serve a

purpose on Ariane 5; it was kept for commonality with the old rocket. The board's report says the assumption that made the conversion safe had been agreed on and never exposed to outside review. A width premise that holds on one machine is a fact about how that machine gets used, not about the code, and it travels with the code without anybody noticing. The only defence I know is the dull one: write the premise into every statement that needs it, where the next person will trip over it.

What "new model of computation" means here. It adds something to how a machine runs: structure it keeps, and an account of what that structure cost. It doesn't add computable functions. It's an abstract model the way the Turing machine is, but it doesn't hand you a new primitive to compute with; whatever base machine sits under it still does all the computing. When I say "shadow" I mean the particular window I name, not every way a conventional machine could store things. Enforcing the law, storing it, and letting an observer see it are three separate questions, and the book keeps them separate.

Which leaves a fair question hanging: if it computes nothing new, does it even get to call itself a model of computation? I think so, and not only on my say-so. Computer science has long kept models that compute exactly what Turing's does and differ only in what they count, how long a step takes, what reaching into memory costs, because the counting is the reason to have them. Part III lays the toll over a register machine of that kind, and over a Turing machine and a lambda calculus, and none of them computes one function more afterward (Section 10). What each of them gains is a record and a bill.

What this project is not. Not a solution to P versus NP, and not a complexity-class separation of any other kind. Not a theory of everything. Not a claim that computation is physics, or that physics is computation. Not a physics result: the game result (the integer check and Tsirelson's bound from the short version) is pure mathematics, and Landauer's principle enters only as a premise with its name on it, never as something derived. Not a quantum computer or a claim about quantum behaviour. Not a verified chip or a verified device of any kind; a checked model and a checked machine, on paper and in Coq. Not a claim to priority on any idea in it. Not closed on every question: the ones I couldn't settle are listed at the end.

What I do and don't know about prior work. I came to this with no formal training in the fields it turned out to need, and I've read what I could reach, with every tool I had. The prior-work map below is orientation, not scholarship: for each piece of this book, who did the thing it leans on, and what, if anything, this development adds. I'd welcome correction on any row.

There's a law about rows like these, and it's a joke that keeps turning out true. In 1980 the statistician Stephen Stigler proposed that no scientific discovery is named after the person who first found it, and then credited the idea to the sociologist Robert K. Merton, who had been writing about fights over scientific priority since 1957. So

Stigler’s law is an example of itself, which is either the best joke in the history of science or a very tidy accident. Is it a law? No. It’s a pattern, not a theorem. But this book has already tripped over it once, in the opening chapter: the pigeonhole principle wears Dirichlet’s name, not Leurechon’s. And I put my own name on a machine. By Stigler’s law, that makes it somebody else’s. The table below is my best attempt at finding out whose.

What I compare each time is what a result is *about*. Landauer’s bound is about heat, a physical cost under a thermodynamic model. No Free Lunch (an optimisation result, in the list below the table) is about averages over a whole spread of problems. The toll is a floor on one kind of step, the step that turns a certification reading on. Exact pricing adds a second rule on top: never charge more. So the comparisons are about the shape of each result. They aren’t claims about who got somewhere first, and they don’t show that certification is the only event worth pricing.

The right-hand column uses words the book only builds later (the latch, the fibre, the universal program, the integer check, the matrix test), so read it as a list of places to go, not as something you’re meant to follow yet.

Prior work	What this book leans on	What this development adds
Landauer 1961; Bennett 1973, 1982, 2003; Zermelo 1896; Bocchieri and Loinger 1957; Pin 1992; Brodsky and Pippenger 2002; Ouldrige and Wolpert 2023; Murashita, Funo and Ueda 2014	erasure and merging cost; measurement can be free; recurrence (a step that loses nothing, on finitely many states, comes back around); machines whose moves all reshuffle states recognise only group languages; a language whose smallest automaton has to merge states costs entropy to recognise; a step with no way back has a logarithmic entropy floor	the ledger form of a counting core that’s classical (recurrence’s finite cousin, folklore for permutation automata) and already priced in entropy for automata: the price as a named rule on a ledger, a certification toll derived from finiteness, permanence and that rule, with merge-or-revoke, the squeeze and entropy forms, and each premise’s counterexample built
Maroney 2009; Sagawa and Ueda 2009; Sagawa 2014	thermodynamic bounds that depend on the actual spread over states; a known state can be reset for free under control timed from outside; time-symmetric control charges every nonreciprocal move	the certificate-specific entropy bound with its support and permanence premises out in the open; heat still needs the Landauer premise, and no new physical law is proved
Shannon 1948; Hartley 1928	entropy and log-of-count bounds	the same squeeze bound in counted states and in bits, for a permanent flip
Tarjan 1985; Lyapunov 1892; Nipkow; Charguéraud and Pottier	the potential method, already mechanized	the toll as that method with the flag as potential

Prior work	What this book leans on	What this development adds
Katsumata 2014; Orchard, Liepelt and Eades 2019; Danner and Licata	cost carried as a grade that composes with the computation	a run's cost as a writer cost and as a grade fixed before the run; the counter rides in the state, put there by the step rule, and it is their writer exactly, while their type systems and before-the-run checking of grades are not embedded, by design
Lamport 1978; Floyd 1967; Turing 1949	one-way counters give order and termination	the receipt's three properties (it only climbs, it equals the sum of the charges, it's the only such total) stated and proved for one schedule
Moore 1956; Kalman 1960; Cousot 1977; Codd 1970; Sampath and coauthors 1995; Saboori and Hadjicostis 2007 Smith 2009 (quantitative information flow)	what a window loses; when an observer can work out a hidden event, and when a secret state stays hidden how much an observer learns about a secret from what it sees	the fibre fact with a price in place of a decoder; the small machine's named collision, and the window every Thiele-complete machine is blind through none for leakage: the collisions rule out exact decoding through named windows, which isn't zero leakage, and no leakage measure is supplied
Gettier 1963; Necula 1997; Cook 1971	entitlement is not truth; proof-carrying checks	entitlement as object, relation, paid event, with a counted bound
Dwork and Naor 1992; Back 1997 (Hashcash)	make establishing something cost, in a way anyone can check cheaply	a comparison, not an identification: proof of work checks a found value against a difficulty target, while the toll charges one mark at the step that certifies and verifies nothing about effort
Eccles and Jordan 1918; Plotkin 1989; Shapiro and coauthors; Abadi and Lamport 1991; Ahman and coauthors 2018 Cantor, Gödel, Turing, Kleene, Rice; Lawvere 1969	the latch; write-once and grow-only state; history variables; monotonic state whose witnessed facts can be recalled later the diagonal	the latch form over any base, with its two counterexamples and converse the shortcut wall on any substrate; L's recursion theorem by construction

Prior work	What this book leans on	What this development adds
Bell, CHSH, Tsirelson, Wehner, NPA, Landau, Masanes, Fine; Echenim and Mhalla 2023 and Lean's mathlib (Tsirelson's bound already machine-checked)	the game, its walls, and the matrix test	an integer check on counts that implies the pinned matrix is PSD, which implies the bound
Buterin and Griffith; Wood; TCG; RFC 9162; Nacula	five record disciplines	five fragments read through one question: is the write priced
Zurek 2009; Ollivier, Poulin and Zurek 2004; Blume-Kohout and Zurek 2006	pointer states and redundancy	a conjecture borrowing the shape, with the gap to the physics stated
Leroy 2009; Klein et al. 2009	the discipline of saying where the proof stops	the list of what's trusted, in one place
Minsky 1961, 1967; Forster, Larchey-Wendling and coauthors 2019, 2020	two-counter machines are universal, and their halting is undecidable, checked in Coq	a six-instruction machine on two counters where earned commitments, no forging and an exact minimum price of three are theorems, with halting reduced to the library's
Forster and Smolka 2017; the Coq Library of Undecidability; Norrish 2011	L as a model of computation in Coq; counter-machine constructions; a compiler from computable functions to counter machines	L's recursion theorem, Rice and halting, checked; the library's compiler used to put every computably presented Thiele machine on the universal host
Turing 1936; Post 1947; Popek and Goldberg 1974; Hamlen, Morrisett and Schneider 2006	the universal machine as one fixed table; hosts that keep guests apart; which policies a monitor can enforce	a definition of Thiele-complete that every clock fails; one fixed program in the small machine's own instruction kinds that runs every program of it with halting, flag and ledger exact, earning its own flag; and one fixed program, with a paid move added, that runs every computably presented Thiele machine, its ledger exact to within an unavoidable two

Everything in the right column is stated with its premises in the body of the book, and everything in the middle column is cited where it's used. The table is the short form. The long form is Appendix C, which takes each row, says what was already known with sources, and sorts what's added into new result, new proof, mechanization,

implementation, synthesis, or conjecture. Most of the right column is mechanization, implementation, and synthesis.

A few more labels I went and looked up don't get a row, because nothing in the book leans on them. They're neighbours, and I list them so you can see why I don't think the toll is one of them in a hat. Each is a claim inviting correction.

- Linear logic (Girard 1987) treats an assumption as a resource that gets used up. It constrains what a program, or a proof, may consume. The toll constrains the model's step function instead. Same instinct, different layer.
- Foundational proof-carrying code (Appel 2001) checks safety proofs against the smallest trusted base it can: a logic plus the machine's semantics. Here no proof rides along with the program. The step rule writes the receipts as the run goes.
- Session types (Honda 1993) constrain how programs talk to each other so the conversation can't go off script. The toll is about what one computation has paid to establish. Different problem.
- No Free Lunch (Wolpert and Macready 1997): averaged over every problem, whatever an optimiser gains on some problems it gives back on others. That's an average over a whole problem space, and the edge is paid for elsewhere. The toll charges one step in one run, with no refund anywhere.
- Kolmogorov complexity (Kolmogorov 1965): the structure in an object is the length of the shortest program that prints it. That's a property of a finished string, with no steps and no moment of paying. The toll is a law about a step.
- Cook and Levin (Cook 1971; Levin 1973): checking can be cheap while finding looks expensive, and whether finding is really harder is the open P-versus-NP question, a gap in how difficulty grows with size. The toll puts a fixed floor under the one step that commits, however easy or hard the finding was.
- The thermodynamic cost of choosing (Rovelli 2024): settling a binary choice at temperature T gives off at least $kT \ln 2$. That's the closest thing to the toll itself I've found, because it charges a commitment rather than an object or an average. But it's physics, indexed to a temperature, and its choice is unpredictable, a branching, and stable only for a while, where the toll's commit is a deterministic merge that stays. The toll charges the commitment in a model of computation, with no temperature in it. Closest neighbour, not the same chair.
- Persistent copies (Ouldrige and ten Wolde 2017; Ouldrige, Govern and ten Wolde 2017): making a copy that stays correlated with its original after the two come apart has a thermodynamic bill. The nearest physics I've found to "learning free, committing priced." It's heat for a correlation, in chemistry; the toll is a count in a model of computation.
- Thermodynamics of finite automata (Ouldrige and Wolpert 2023): if the smallest automaton for a language has a letter that sends two states to one, recognising

that language on a device that repeats one local step produces entropy, and if the smallest automaton only reshuffles, recognition can be free. That's the chairs with a physical price on them, and one of their worked examples is the latch's language turned inside out. It's the closest neighbour of all to the road the toll is derived on; Section 4 says exactly what's left over for me.

- CerCo, the Certified Complexity project of Amadio and coauthors, builds a verified C compiler that carries the object code's execution cost back up to annotations on the source. There the cost decorates the program and survives compilation. Here it lives in the model's step rule.
- Reference monitors and execution monitors (Anderson 1972; Schneider 2000; Azevedo de Amorim and coauthors 2015). A small piece every action passes through, enforcing a rule on every program. Schneider showed that a monitor watching a run enforces exactly the rules whose breaking shows up at a finite point, and "no certificate without an earlier passing check and commit" is one of those; the micro-policies check a tag rule at every instruction, proved in Coq. That's the move the small machine's step makes, so this one isn't a neighbour, it's the house. What's added is the edition-bound check-then-commit with a paid, permanent record on top, and the derivation of why the payment can't be zero. Kevin Hamlen, Greg Morrisett and Fred Schneider (2006) went on to sort which rules monitors, program rewriters and static checkers can each enforce, and they differ, which is one reason this book doesn't claim that every enforcement mechanism is a Thiele machine underneath.
- Trusted components in replicated services (Chun and coauthors 2007; Levin and coauthors 2009; Veronese and coauthors 2013; Decouchant and coauthors 2022; Clement, Junqueira, Kate and Rodrigues 2012): a small record inside each replica that a replica taken over still can't make say two things, with a proof of exactly what that buys. The record is theirs. The toll isn't in their picture.
- Ghost state (Owicki and Gries 1976; Filliâtre, Gondelman and Paskevich 2014): variables added for a proof that can be erased without changing what the program computes. The flag and the counter are ghost in that sense. The small machine's table of checked facts isn't, because the step reads it and refuses on it, for every program.
- Verifiable computation and proof-carrying data (Parno, Howell, Gentry and Raykova 2013; Chiesa and Tromer 2010): a short proof, checkable by a stranger, that a computation's output is right. That establishes the claim for the stranger, and anyone with the inputs can make one, the copier included. The record here is about one machine's own history. A proof like that is one way to carry the local record out of the box.
- Commitments and entitlements (Brandom 1994): talk as score-keeping, tracking who is committed to what and who is entitled to it. Brandom's entitlements can be defeated and can be passed on by testimony. Here entitlement is priced, step

by step, and a copy carries it only as far as the copy is genuine. Same words, different machinery.

- Correspondence assertions (Woo and Lam 1993; Gordon and Jeffrey 2003). In protocol verification, “this end event must be preceded by a begin event for the same thing” is a correspondence assertion, and a type system can prove a protocol keeps it against any opponent. “No certificate without an earlier commit, and no commit without an earlier check of the same claim” has that shape. Theirs is proved about programs by typing. Mine is enforced by the step, which is the LCF comparison again, at the level of single instructions.
- Monotonic parallel composition (Åman Pohjola and Parrow 2016) and persistence in the pi-calculus (Palamidessi, Saraswat, Valencia and Victor 2006). Åman Pohjola and Parrow separate models where a neighbouring process can never stop a step from models that can solve consensus, and test-and-set, an instruction that writes a cell and reports what was there, is on the consensus side. Palamidessi and coauthors compare messages that are used up with messages that stay, and the end of their study is a reduction from Minsky machines. A store that can refuse permanently, which is what the version rule does to a stale commit, looks like the non-monotonic side. I haven’t written that encoding, so that’s a pointer and not a claim.
- Constructor theory (Deutsch and Marletto 2015; Marletto 2016; Marletto, Vedral and coauthors 2022; Violaris and Marletto 2022): physics as which tasks a device that runs in a cycle can do, with irreversible meaning possible while the inverse isn’t. It’s the nearest physics I know to “committing costs” stated for a cycling device. It’s quantum and approximate, and the chairs are classical, finite and exact (Section 4).
- Accountability and fault detection (Haeberlen and Kuznetsov 2009; Küsters, Truderung and Vogt 2010; Treitz and Künnemann 2026): when an outside party can tell who misbehaved. Their windows are this book’s windows, and the trust they leave over, an honest log or a trusted step, is left over here too (Section 7).
- Transferability (Easwaran 2009): a deductive proof lets you learn from someone you don’t trust because you can check every step, and a probabilistic proof doesn’t. That’s the argument for carrying the object and re-running the check (Section 7).

And then the silly question that list raises, which I’m going to chase anyway: why is an optimisation theorem named after lunch? Wolpert and Macready borrowed an old saying, “there ain’t no such thing as a free lunch,” and the saying is older than almost everyone who gets credit for it. American saloons in the 1800s put out a free lunch for anyone who bought a drink, and the drink paid for the lunch. A Washington newspaper was already saying in 1909 that somebody has to pay for a free lunch; a newspaper fable by Walter Morrow had the whole line in 1938; Robert Heinlein’s

1966 novel *The Moon Is a Harsh Mistress* squeezed it into TANSTAAFL; and Milton Friedman put it on the cover of a book in 1975. People usually credit Heinlein or Friedman. Neither coined it. Stigler's law again, at lunch. Here's what it changes. No Free Insight sounds like a cousin of the saying, and the sound misleads in one way. The saloon's lunch was paid for somewhere else, in the price of the beer, and that's Wolpert and Macready's shape: a gain here, handed back over there. The toll isn't that. Nothing gets paid somewhere else. The step that raises the flag pays, on the spot.

Part I

The picture

1 How I got here

What I am is curious, past the point of good sense.

I pick a thing up and cannot set it down until I have turned it over enough times to see how it holds together, and this time it did not let go.

I am not a programmer and I was never formally taught any of this.

I was not trying to build a formally verified CPU.

I would not have known what one was if it bit me.

I was trying to build a 3D renderer.

There was no plan behind it.

Just a pull to make something real with my own hands, and the kind of stubbornness that will not let a thing go until it stands up on its own.

I have worked on it in off hours every day since.

Once I started digging in, two things came clear fast. I knew nothing about computers, and the field was already saturated, every obvious road paved and crowded. So I needed an angle nobody else was standing on, and the angle I had was not a strategy, it was just the only way my head works. I did not know what compositional meant. I had never heard of category theory. I called the thing I was building digital DNA, because that is what it felt like in my hands, and I thought about it in the only language I had: gravity pipes and directions and vectors, things that were linear and somehow not linear at the same time, structure that flowed and joined and came apart without my owning one proper word for any of it. I built a custom rendering pipeline out of those intuitions, by feel, throwing out pieces that did not sit right even when I could not have told you the rule they were breaking. And then, the way I always do, I went

looking for what I was actually doing, and found out it already had a name. An old one. A deep one. What I had been reaching for in the dark was composition. It was category theory, a whole field I had never once heard of, sitting there describing the exact shapes I had been shoving around. The moment that landed, the rendering pipeline stopped being a rendering pipeline and started turning into a categorical engine, and I started swallowing the real theory as fast as I could find the words for things I had already built.

The method underneath was crude and it held. I wrote a long structured spec in JSON (a plain-text format for describing data) as a context document, fed it to Copilot, and had it build one Python file per concept, one at a time, each with its own tests, and I would not move to the next until I could watch the current one actually run. Category. Object. Morphism. Functor. Monoidal product. It came together in pieces and never in chapters, the way a picture resolves out of static instead of getting drawn in left to right. The Yoneda lemma (a result buried deep in category theory) was the piece that made the rest snap into place, the night I quit borrowing other people's runtimes and started writing my own around it, with Z3 (a constraint solver out of Microsoft Research) wired into the middle.

Then within weeks I found out the engine I had built to draw pictures was general enough to run a physics simulation on the exact same primitives, without my rewriting any of it. I wrote Newton's laws as a program in it, half expecting nothing, and it ran. I want to be careful about what that was and what it was not, because this is exactly the kind of thing that gets misread. It was not me deriving physics, and I am not claiming the machine explains gravity. It was the compositional structure being general enough to express a physics simulation the same way it had expressed a renderer, and that generality is what told me the structure was the point and the pixels never were. Somewhere in those weeks I crossed from debugging why one pixel would not show up to understanding that I had quietly built a categorical computing engine, and that the rendering had only ever been the first thing I aimed it at. The pixel did eventually show up, for what it is worth.

I kept going, because by then I could not not keep going. I built a small custom language for describing scenes (the name for that is a domain-specific language, a DSL), then a program that read the language and caught my mistakes in it (a linter), then a way to type commands at the whole thing from a terminal (a command-line interface, a CLI). The night the engine first ran a program declaratively, meaning I told it what should happen and let it work out the how instead of my spelling out every step, I told my wife, Julie, I thought I was onto something. She said okay.

Then I hit a wall, and it is the wall this whole document is about. Z3 could settle the small questions and fell over on the exact ones I needed answered, which left me staring down an infinite hallway of guess-and-test with no proof waiting at the end of it, ever. One night, not even working, I was watching a video about Flatland, the

old story about a creature living in two dimensions who cannot so much as picture a third. And it landed on me sideways, all at once, the way the real ones always do. I said it out loud: I get it now. There is another axis. Classical computation only ever sees what it can read and write off its own surface, and there is a whole direction it is blind to, not because it is broken but because it was never built with an eye that points that way. The two computers on the first pages of this book are my attempt to show you that blindness with something you can hold. I opened a new repo that same night and called the thing in it the Thiele Machine. The name began as a working title and became the name of the model. It carries the larger claim plainly: the machine treats structural state as part of computation.

What survived the whole trip from there to here is not the code. Not one line of it. By the time the Coq proofs finally landed, every line of that first renderer had been thrown out and rewritten at least twice, probably more, and good riddance to all of it. What survived was a single architectural commitment I made early and never once let anyone talk me out of: structure is the state itself, the actual thing the machine runs on, not a metaphor laid over some realer state underneath. Category theory was the first language I found for that structure. It is not what the model is. The model asks for a state, a step, a ledger, and a window, and category theory appears nowhere in that list. It lived on for a while in a bigger machine I built on the way here, a fifty-one-instruction virtual machine with its own chip design, whose partition graph (groups, and arrows between them) was, in different words and different syntax, the same object as the category structure that came together in that first week and first ran morphisms correctly at two in the morning. That machine was useful for building and confusing for everything else, so it's set aside now, kept but out of this book. What building it showed me is that six instructions were enough for everything this book needs, and that the big one was mostly in the way. I did not write that code, not then and not now. I directed the models and refused everything they handed me until the morphisms composed the way the shape said they had to. The morphisms in that early code were objects that remembered where they came from; the records in the small machine I ended up with remember where they came from too, and you can walk up to one and challenge it. The receipt grew teeth. But the shape itself never moved. The shape was the substrate the entire time, and everything else, every renderer and language and proof, was me slowly learning to see what I had already built. By the shape I mean structure held as state. Not category theory. Category theory was just the first language I had for it.

Formal verification was a graveyard of dead ends further on. The hardware came after that. Every step pulled back to show another layer underneath that I had not known was there, and I mean that as plainly as it sounds: I did not know what a Turing machine was when I started, I did not know what Coq was, I did not know most of the words in this document existed. I learned each one the same way, by walking face-first into a wall that needed it and not being let past until I had it. The needing

always came first. The learning came after, dragged along behind the needing because it had no say in the matter.

Large language models were my hands the whole way through, and I want the division of labor exact, because it is the thing people most want to get wrong about work like this. The hands had nothing like this to draw on: no reference implementation, no prior project to point at. The ideas, the architecture, the direction, the demand that the thing be true: those are mine, all of them, no asterisk. The symbol-pushing, the actual Coq and the actual math, the models produced and the proof checker judged. The proofs compile or they do not. I refused every one that did not. Coq is an obedient idiot. It checks the proof matches the statement. It does not check that the statement means what you think it means. A theorem can claim “every cat that is also a prime number has whiskers” and the proof can be three characters long and Coq will nod and stamp it. So I read every statement. The proofs you can’t fake. The statements you can fake by accident, which is the more dangerous failure mode. I refused both kinds. However the symbols got onto the page, the argument on it is checkable by reading, and it would be exactly as right or as wrong if a million monkeys had typed it. The monkeys could not have refused.

A word on what I actually do, because it runs under everything after it. I do not write code or math by hand. I do not pick constants. I do not grind through derivations. I have never written a line of code on paper or in a notepad in my life. What I do is see, and direct, and refuse. When I take hold of an idea it does not stay a point. It opens up. It becomes a shape I can blow up and turn over and take apart, lay across another shape to find where the two of them touch, run forward and then back to where it started, zoom into until it is the only thing in the world and pull back until it is one dot in a field of other dots, and the whole time it stays wired to everything I have ever connected it to, the right connections and the wrong ones both, all of them live at once. I am not reading that field neutrally, the way a search would. The thing doing the looking is me, the whole of me, my gut and my mood and my stubbornness picking what lights up and which way I move through it, and that is the part you cannot lift out without turning me back into a search engine. So I see where a thing has to fit, the shape it is obligated to have, usually well before I could write it down in a single symbol. Then the model gives me the symbols, the proof checker takes them or throws them back, and I refuse anything that will not compile or does not match the shape I am holding. That refusal is the one thing in this entire document that is mine and only mine: the demand that it be true, and the throwing-out of everything that wasn’t.

I am telling you all of this because it is not background color, it is the engine. I did not come up through anywhere that hands you a foundation and says trust us, this part is settled. No professors, no colleagues, no field full of authority to lean back on when a claim got heavy. From the outside that is a deficit. From the inside it is the whole reason any of this is solid: when I meet a claim I either find the proof or I do

not believe it, and that goes double for my own, because my own are the ones I most want to wave through. It is how I am built. It is the only way I know to hold a thing and actually trust that I am holding it.

So everything in here gets built from the ground up, and not as a courtesy to you. I build it because I do not believe I understand a thing until I have built it with my own hands, watched it stand up on its own, and tried to knock it over. I built this document the exact way I built the machine, brick by brick, refusing each brick until it held weight. That includes the parts you already knew cold before you opened this. I did not skip them for you, and I could not skip them for me.

The vocabulary I had to go look up to do any of this sits at the back of the book, so the argument comes first. If a word stops you, it is there. If you already know this material, you never have to open it. But if you catch something in the basics that is wrong, I want to know, and I mean that as a real ask and not a polite one. Nothing in here is correct just because I wrote it. Especially the parts I wrote.

On the name. I wanted to call it the meta machine. Then Facebook bought the word “meta”: a company renaming itself after a term for self-reference and draining it of any in the same motion. So I went one level up, where this machine already lives. The most meta move available, for a machine that runs on meta-level reasoning, was to put my own name on the cover and take the word back. That’s the whole of it, a meta move on a meta machine. It happens to be my dad’s name too, and my wife’s now. So I asked them before I did it. They said go ahead.

2 What structure is

One word. That’s this whole section, and I’m aware of how that sounds, so here’s the word and you can decide whether to stay. Structure.

I used that word for most of my life without once looking at it, which I suspect is true of most people and most words, and it’s fine, right up until you do something stupid. The stupid thing I did was build a machine that keeps a running bill, and the bill has one rule I couldn’t talk it out of. You can arrange things however you like for whatever price you name, including free. But the moment you ask the machine to *vouch* for an arrangement, to put its own name on the claim that these things belong together, there’s a charge, and it isn’t zero. How much, charged in what, why I picked that and not something else, and whether I had any business picking it at all: that’s all later, and some of it I’m still not sure of. What I was sure of, standing there with the thing built, was that I’d made something that charges for standing behind structure and I could not have told you what structure was. It didn’t care. It can’t nod along the way a person does. It needed the word to hold still, and the word had never held still for anyone, least of all me. (The machine itself stays off the page until Part IV, on purpose. Until then, the one rule is all of it that matters, and I’d rather you met the rule in a toy you can hold than in a machine you’d have to take on trust.)

So I went looking at the people who'd wrestled it before me. Some of them have been dead two thousand years and are still, I'd bet money, better at this than I am. I used everything I had: language models, more than one, books, libraries, my own gut, anything in reach. I don't know if I found enough. I never do. There's always another shelf behind the one I was reading. And being dead and famous is not the same as being right, so I pushed on each of them, and what follows is what I could reach, pushed on as hard as I know how. The one thing I'm asking is that you push on it with me, out loud, at your own kitchen table, because the first time through I was the only one checking, and that's not enough people.

The order I ran into them was a mess. Category theory came first, because I'd built one by feel before I knew it had a name, and everything else came in whatever order the next question dragged it in. I'm going to put them in the order they happened instead, because that's the order where each one leans on the last. And none of it is an oracle. History is where I start, not where I stop. It gets retold, smoothed over, and now and then made up outright (there's a drowning coming up that never happened), so every piece of it is a lead to check, and I check the parts I lean on.

2.1 A string, a square, and a drowning that didn't happen

Start with the oldest one, and start by doing it. Take a string. A guitar string, a rubber band over an open box, anything that twangs. Pluck it. Now press it down exactly at the middle and pluck half. That's the same note, an octave up, the one that makes a round sound like a round. Press at a third of the way along and pluck the long side: that's the fifth, the second note of every bugle call you've ever heard. Half. Two thirds. Small whole numbers, and your ear says yes before your brain does. About twenty-five hundred years ago, in the south of Italy, the followers of a man named Pythagoras are credited with finding exactly that, and it's as true in your kitchen tonight as it was on their hillside.

Here's what they did with it, as best anyone can tell. Nothing survives that's surely from Pythagoras's own hand. The earliest fragments are from a later member, Philolaus, and scholars argue over those. Mostly what's left is Aristotle, one to two centuries later, writing in his *Metaphysics* about them the way you'd write about the neighbours' religion, and he says they took the elements of numbers to be the elements of all things: everything is number. Not that everything can be counted. That the parts of numbers are the parts of all things.

Stop and ask what I asked. What does that even mean? Is a chair a number? Does a chair have a number tucked inside it somewhere? Aristotle doesn't say, and he's reporting a belief he doesn't hold. So don't take it on his word, or theirs, or mine. But before you push on it, know what it was holding up, because you can't test a load-bearing wall without knowing what sits on it. These weren't people doing math for fun on weekends. They were a brotherhood, with rules about what you could eat and when you had to keep quiet, and that plucked string was their scripture. Halve

the string, hear the octave: to them that was proof that under the mess of the world there were whole numbers in clean ratios, that the cosmos was a chord and they'd heard it. "Everything is number" wasn't a slogan. It was what they'd organized their lives around. Which is exactly the kind of sentence you should be suspicious of, mine included.

Now push, and push where they loved most: the square. Draw one, with a side one unit long, any unit you like, and draw the line from one corner to the opposite corner. Call that the diagonal. If everything is number, then the diagonal is some whole number of parts long when the side is some whole number of parts, so the diagonal divided by the side is a fraction, a whole number over a whole number. Cancel anything the top and bottom share, the way four-sixths becomes two-thirds, until they have no common factor left. I'll call the top p and the bottom q , and the letters are purely so I don't have to keep writing "the top" and "the bottom."

You need one fact about squares, and it's the one Pythagoras is famous for. The square built on the diagonal has exactly twice the area of the square built on the side. If you want to see it rather than believe it, cut a square along both diagonals into four triangles and arrange them around; it's a five-minute job with scissors. So p times p is twice q times q . People write "times itself" as a small raised two, so that's $p^2 = 2q^2$, and that is the only piece of notation in this argument.

Twice anything is even. So p times p is even. An odd number times itself is always odd (three times three, five times five, try a couple), so p itself must be even: it's two times something, and I'll call that something r . Put it back in. Two r times two r is four times r times r , and that equals twice q times q . Halve both sides: two times r times r equals q times q . So q times q is even, so q is even too, by the same step as before. But you already cancelled everything the top and bottom had in common, so they can't both be even. The argument has eaten itself, and when an argument eats itself the thing you supposed at the start is false. There is no such fraction. The diagonal of a square can't be named by any ratio of whole numbers, ever. Aristotle reaches for this very argument in his *Prior Analytics* as his example of proof by contradiction (suppose the diagonal can be measured, and odd numbers come out equal to even ones), and you just did it with a pencil.

There's a story that's been told for two thousand years: that a member of the brotherhood named Hippasus let this secret out, and his brothers took him to sea and drowned him for it. I went and checked, because it's a good story and good stories are the ones to check. The Stanford Encyclopedia of Philosophy's entry on Pythagoreanism finds no ancient writer who connects Hippasus to this discovery at all. It's a legend. But notice that the legend exists. Somebody, long after, felt that a man *should* have been drowned for this. That's how much was riding on it. To them, the diagonal wasn't a math problem. It was the end of the world.

So the claim is dead, and you killed it, not me. Now the question that matters: what's

left? Look at your square again. The diagonal is still there. It still stands in a perfectly definite relation to the side; you drew it, and nobody drowned. That relation was real before anyone tried to put a number on it, and it stayed real after the number failed. The relation came first. The number, when there is one, is a name for the relation, and sometimes there isn't a name. I'm going to keep that, and I want you to keep it too, because you're going to watch the same move get made again and again as this book goes on: somebody says "everything is X," and it's true enough to do real work with, and it sounds right when you pluck it, and then somebody draws the diagonal. I don't know yet whether the thing I built survives its own diagonal. That's a real question, and I'll get to it.

2.2 Shapes poured into nothing

About a century and a half after the brotherhood, Plato, in a dialogue called the *Timaeus*, tried to build the whole physical world out of geometry. Fire, air, water and earth were four regular solids (the pyramid, the eight-faced one, the twenty-faced one, and the cube), and the faces of those solids were built from exactly two kinds of right triangle. You can make one of each in a minute: cut a square from corner to corner for the first, and cut an equilateral triangle from a corner to the middle of the opposite side for the second. Everything else, in Plato's story, was a receptacle, a featureless medium that takes on shape the way wax takes a seal. Structure, for Plato, is shape laid into something that has no shape of its own.

Why did he care? He wanted a world made on purpose, by a craftsman working from a plan, and triangles were the plan. Is it true? No, but not for the cheap reason. You can't refute a claim about what things are made of at the smallest scale by picking one up; the smallest scale is exactly where your hands don't reach, and Plato knew that. What sank it is that it made no prediction anyone could check and the people who eventually did reach that scale, two thousand years on, found atoms, and atoms aren't triangles. The idea was a guess with no way to lose, until it lost. But push on it and two things survive. One: structure is something *laid in*, and the stuff it's laid into doesn't carry it by itself. That's a real idea, and I'd ask you to keep it loosely for now, because later I'll have to ask whether the state a machine holds is the wax or the seal. Two: Plato broke his own claim, in a later dialogue called the *Parmenides*, with an objection that Aristotle later nicknamed the third man, and it's the most useful mistake in this section. Take all the large things. Plato says they share one thing, Largeness. But Largeness is itself large, so now there's a further thing that all the large things *and* Largeness share, and that one is large too, and it never stops. Try it with any property you like; it runs away every time. The lesson, and I think it's right though Plato never quite accepted it himself: the structure of a collection is not one more member of the collection. The plan isn't a brick. Hold on to that one. It'll come back in a place you won't expect.

2.3 Prove it from what you and I both accept

Around 300 BC, in Alexandria, Euclid wrote the *Elements*, and the thing to take from it is not the geometry. It's the method, and the method is the whole reason I trust anything in this book, so give me a paragraph on it. He starts with definitions ("a point is that which has no part"), five postulates (things he asks you to grant, like "a straight line can be drawn from any point to any point"), and five common notions (things nobody argues with, like "things equal to the same thing are equal to each other"). Then he derives everything else, four hundred and sixty-five propositions, from only those, in order, each meant to use nothing but what came before (he sometimes leans on things he never stated, and later geometers filled those in). Not "trust me." Not "it's well known." Here are the rules you and I both accept; now watch. If you've ever been in an argument that ended with "well, that's just your opinion," Euclid is the man who found a way out of that room, and I'm not embarrassed to say I learned the way out from someone dead for two thousand three hundred years. Four hundred and sixty-five times he does the same trick, and the trick is everything.

And then there's the one rule Euclid seems almost embarrassed by, and it's a hole worth falling into. His fifth postulate, the one about parallel lines, is long and awkward next to the others, and for two thousand years people tried to prove it from the rest so it wouldn't have to be assumed. Proclus tried. Mathematicians in Persia and the Arab world tried. In 1733 an Italian priest named Girolamo Saccheri spent a whole book assuming it was false so he could watch the contradiction arrive, and he got pages deep into what is now a perfectly good geometry before declaring that it was repugnant and stopping. A century later three people, separately, took the step he wouldn't: drop the postulate, and you don't get nonsense, you get other geometries that hold up just as well. Farkas Bolyai begged his son János to stay away from the problem, warning that it had swallowed years of his own life; the son went in anyway and came out with one of those geometries. Here's the part I keep: they found something real by refusing to believe one of the rules everyone accepted, and checking what happened without it. That's the push, two thousand years long.

What is structure, to Euclid? Whatever the starting rules force. And when are two shapes the same? Here's the one you can do: cut out two paper triangles and lay one on the other. If they coincide, they're equal. That's his fourth common notion, and it's how he proves his fourth proposition, by picking up one triangle and setting it down on another. Sameness is coinciding under a move. Keep that; it's about to get very general, and I didn't see that coming either.

2.4 The statue and the bronze

Aristotle, Plato's student, pulled the forms down out of heaven and put them inside things, which was a bit rude to his teacher and also, I think, correct. His example, from the *Metaphysics*: a bronze statue is bronze plus a shape, and being made of bronze is no part of what it is to be a statue. The same shape in marble makes a different statue

with the same form. Matter and form, and form is what the thing *is*.

Push on it. Is that true? Take anything on your desk. Could the same thing be made of something else? A coffee cup, sure. A sandwich, no; a sandwich made of wood is a sculpture of a sandwich, and if you've ever tried to eat one you know the difference matters. So the claim doesn't hold everywhere. It holds exactly where the material was never doing the work, and that's a sharper thing than Aristotle said: structure is what survives a change of material, in the places where the material didn't matter. Why do I care? Because this whole book bets that the record is a form and not a material: the same rule whether it sits on a tape, on a table of counters or on a lambda term, and I don't get to just say it is. Section 10 is where that bet gets checked, over any base at all, and Part IV builds it once, on the smallest base I could find (Section 14). Aristotle is the one who told me what the bet means.

2.5 Where things are, without numbers

In 1679 Gottfried Leibniz, who co-invented calculus and had opinions about everything, wrote to the physicist Christiaan Huygens asking for something that didn't exist: a way to reason about *where* things are directly, the way algebra reasons about *how much*, without ever laying down coordinates. He called it the analysis of situation. Try it yourself. Describe where the furniture is in the room you're sitting in without using a single number. "The couch is against the wall. The table is between the couch and the window. The lamp is next to the chair." You just did it, and what you used was relations between places and nothing else. Huygens wrote back that November and was not impressed. It took roughly two hundred years for Leibniz's idea to become a field, and the field is called topology, and Huygens is remembered for other things.

What survives the push is the same thing that survived the square, coming at you from the other side. Position is a relation among places. The numbers you lay on top are a convenience, and they're a convenience you can take away and still have the room. I'll be honest about why this one landed for me: the thing I built keeps track of which things belong together, and I spent a long time thinking that was a list of numbers before I understood it was a relation that the numbers were merely naming. Leibniz had that in 1679. Huygens told him no. I'd like to think I'd have sided with Leibniz. I probably would have sided with the famous guy.

2.6 Naming a thing by saying when two are the same

In 1884, in a short book called *The Foundations of Arithmetic*, Gottlob Frege noticed something you can check in your head right now. "Line *a* is parallel to line *b*" and "the direction of *a* is the direction of *b*" say exactly the same thing. There is no fact in one that isn't in the other. But the second sentence has a new object in it, a *direction*, that the first one doesn't. Where did it come from? From the sameness. Once you can say when two things count as the same, you've earned the right to name the thing they share. Frege used exactly this move to say what a number is: the number of things in a pile is whatever every pile you can match up with it, one to one, has in common.

That's a residue with teeth: a sameness relation lets you name a new object.

Now push, because Frege pushed too hard, and it cost him. In his big system he also assumed that any condition you can state picks out a collection, the collection of everything that satisfies it. In June of 1902, a young Bertrand Russell wrote him a short letter. Consider the collection of all collections that are not members of themselves. Is it a member of itself? If it is, then by its own rule it isn't. If it isn't, then it qualifies, so it is. Do it in your head; it flips every time you look at it, like one of those drawings that's a vase and then two faces and then a vase. Frege's floor had a hole in it, and he knew it the day he opened the letter, and his second volume was already at the printer. The repair came from Ernst Zermelo in 1908 and it's the one mathematics still stands on: you may only cut a collection out of a collection you were *already given*. You don't get to sort "everything." You sort a pile. I'll tell you now that this rule shows up again, later, in a place where I had to obey it or fall through the same hole, and I'll point at it when the book gets there.

Frege's reply is worth hearing, because it's the best example I know of how to lose. He wrote back within days. He didn't hide it or argue around it. He said, more or less, that arithmetic was tottering, and in an appendix to the volume that was about to come out he wrote that hardly anything worse can happen to a scientific writer than having a foundation give way right after the work is finished. Then he tried to fix it. If I'm wrong about something in this book, and I'd bet I am somewhere, that's how I'd like to take it.

2.7 Choose the changes, keep what survives

By 1872 there were too many geometries. Euclid's, and ones where parallel lines meet, and ones with no such thing as distance, and nobody could say what made them all geometry rather than a pile of unrelated hobbies. Felix Klein, twenty-three years old and taking up a professorship at Erlangen, published an answer in a program paper that people still call the Erlangen program. Pick a set of allowed changes: rotations, stretches, whatever you decide. Then a geometry is the study of the properties that *stay the same* under those changes. Pick a different set of changes and you get a different geometry on the very same page. Euclid's is the one where you may slide and turn a shape but not stretch it. And here's the part I'd underline if this were my book, which it is: in Klein's scheme any group of changes can define a geometry. The choice is free. Nobody tells you which changes to allow.

You can do this one at your desk. Take any drawing and ask what's still true after you turn it upside down (lengths, angles), after you stretch it (not lengths anymore, but which lines cross which), after you crumple it (almost nothing but what touches what). Three answers, three structures, one drawing. The residue: structure is what survives a chosen set of changes, and "chosen" is doing real work in that sentence. There's no structure of a thing from nowhere. You have to say what you're allowed to do to it first. I didn't know that when I started, and I'd been saying "the structure of" things

for years.

2.8 An invariance comes with a bill

In 1918 Emmy Noether, who wasn't allowed to hold a professorship because she was a woman and for a while lectured under a male colleague's name, proved something that physicists still treat as close to holy. Take a physical system whose laws can be written the way most of physics is written, as "of all the paths the system could take, it takes the one that makes a certain total smallest" (the total is called the action, and that's all I'll say about it here). For such a system, every continuous symmetry of the laws comes with a quantity that is conserved. If the laws don't care *where* you are, momentum is conserved. If they don't care *when* it is, energy is. A symmetry isn't decoration. It has a consequence downstream that you can measure with instruments. I can't hand you this one to do at the table; the proof needs calculus I'd have to build first and I'd lose you and probably me, and the "written as an action" condition is doing real work that I'm not going to unpack here. What I can hand you is the shape. Structure, once it's really there and the setting is the right kind, has consequences you can measure. I'll admit that's the sentence that made me think a machine could carry a bill for structure in the first place, and I'll also admit I don't yet know whether that hope was reasonable. Noether makes it reasonable to ask. She doesn't make it true, and nothing I build in this book is a system with an action.

2.9 All of mathematics as structures

Starting in the 1930s, a group of young French mathematicians began publishing a vast treatise under one made-up name, Nicolas Bourbaki, as if he were a single person. He wasn't. He also applied for membership in the American Mathematical Society and was turned down, which I find comforting. Their goal was to rebuild all of mathematics on one idea: a structure is a set of elements "whose nature has not been specified," together with one or more relations on them that satisfy stated axioms. They believed three kinds of structure were the mothers of all the rest: algebraic, order, and topological.

Push on it, and here's what the historian Leo Corry found when he went through the entire treatise looking for the place where this formal definition of structure actually does any work: it "plays no actual role in the presentation of theories within the treatise." Outside the chapter that defines it, almost nowhere. And I found no argument in their writing for why the mother structures number three. And the chapter on category theory, which they announced, never appeared. Why? The thing their definition left out was the *maps*, the ways of carrying a structure from one set to another, and the maps were where the weight was. So I keep two things from Bourbaki. The right thing: relations on elements whose nature doesn't matter, which is Aristotle's bronze in different words. And the warning: a structure stated without its maps won't carry the work. I learned that one the slow way, and the origin story you just read tells it.

2.10 A set with some named relations

What mathematical logic eventually settled on, with Alfred Tarski giving the field its name, “the theory of models,” in 1954, is almost boring, and the boring ones are the ones that last. A structure is a set, called the domain, plus whatever relations and functions a chosen vocabulary names on it. Pick the vocabulary (Klein again: you choose), and a structure is anything that gives those names a meaning. Now I want to say “the smallest one,” and the word “smallest” has no meaning until I say what I’m measuring, so here is the ruler: count the vocabulary. One name, taking one thing at a time, with two possible answers. That’s a domain with one yes-or-no property named on it. One question, put to each thing. Two honest limits, and I’d rather give them to you than have you find them. A vocabulary with one named constant (“this one, here”) is exactly as small by my ruler and it doesn’t sort anything, so “smallest that sorts” is the claim, not “smallest.” And one question only splits the domain into two groups when some things answer yes and some answer no; if they all agree there’s one group and nothing was sorted. With those limits said, and with the ruler said: by the count of names, the smallest structure that sorts anything is one question. Someone with a different ruler (fewest symbols to write down, say, or least information) might land somewhere else, and I’d want to see where. Keep it. It’s about to turn into the whole section.

2.11 The structure is in the arrows

This is the one I met first, backwards. I’d built it by feel, in a rendering engine, as “digital DNA,” and only went looking for its name after the thing already ran. In 1945 Samuel Eilenberg and Saunders Mac Lane published a paper that founded a field now called category theory, and they said in plain words that it was a continuation of Klein, which I’d have loved to know at the time. The move: stop asking what a thing is, and ask how it’s carried. A category is some objects and some arrows between them, where arrows can be followed one after another and every object has a do-nothing arrow to itself, plus two rules that make the following dependable: following three arrows in a row gives the same result whether you pair up the first two or the last two, and the do-nothing arrow really does nothing when you tack it onto either end. That’s the whole definition, and I know it sounds like nothing. The two rules are the part people skip, so slow down on them for a second, because each one is there for a reason you can see.

Picture three dots in a row, A, B and C, an arrow f hopping from the first to the second and an arrow g from the second to the third. Composition says there’s an arrow taking the whole stride from A to C in one, g after f . Without it, every multi-hop path has to be named on its own, and the structure falls apart into a pile of unconnected pairs, a phone book without a search function. Associativity says that with a third arrow h on the end, pairing up the first two and then adding h gives the same arrow as adding h to the last two first. Both spell the same three-hop path, and if they came out different, two people who built the same chain in different orders would hold two different

arrows, which is a bug: the chain is the thing, and how you assembled it isn't. The identity laws say that tacking the do-nothing loop onto either end of an arrow gives back the arrow. If the loop changed things, it wouldn't be the loop. Six words, objects, arrows, composition, identities, associativity, identity laws, and that's all of Eilenberg and Mac Lane's definition.

Composing relations is older than the definition, too. Who put a name on that move first? People older than you'd guess. Augustus De Morgan, writing on the logic of relations around 1860, took the "is" out of the old syllogism, put a general relation in its place, and composed two relations to get the conclusion. The example that gets quoted from that tradition is " x is the lover of a servant of y ," which is relational composition with better gossip. When exactly he read that paper to the Cambridge Philosophical Society, and on which pages it was printed, the sources disagree, which is a small reminder that history is where you start digging. Then in 1870 Charles Sanders Peirce, the same Peirce from the opening chapter, carried George Boole's algebra of logic (the arithmetic of true and false that Part IV comes back to) over to relations, with composition as one of its basic operations, and he's usually counted as the real founder of the theory. So following one arrow after another is older than computers, older than categories, older than Turing's clerk: a nineteenth-century logician's product of relations.

And where did they get the word? Not from mathematics. They borrowed *category* from the philosophers, from Aristotle and Kant, and *functor* (a map from one whole category to another) from the philosopher Rudolf Carnap, and then gave both words exact meanings none of those philosophers had in mind. So Aristotle walks back into this chapter without being asked: the man who put the form inside the bronze lent his word to the field that keeps track of forms by their arrows. Does that change the laws? No. It's a reminder that a word can start in philosophy and arrive in mathematics with a precise meaning bolted on, which is the thing I've been trying to do with *established* since the first page.

What a thing *is* is how it maps. For now the residue is the one Bourbaki missed: structure lives in the maps. Nothing later in this book needs more category theory than that sentence, and the latch of Section 10, where a record is pinned down by the maps between a machine and the base it extends, is where the sentence does its work.

2.12 Structure is the part that compresses

Play this one with me. I write "th" and you guess the next letter. You said "e," and you were right. Now "the q": you said "u." Congratulations, you just measured the structure of English. In 1948 Claude Shannon measured it properly, in a paper that founded information theory, and found that about half of written English is "determined by the structure of the language." It's the reason a zip file of a long letter is so much smaller than the letter. Be careful what that says. It does not say that every other letter is fixed by the ones before it; you just guessed "e" after "th" and you

could have been wrong (think “than”). It says that, on average and over long stretches, about half of what’s written could be rebuilt from the rest: the text is roughly twice as long as the choices in it. And he said why that matters in one line that I’d frame if I framed things: the structure “allows one to make a saving in time.” The more you can predict, the less you have to read.

Thirty years later Jorma Rissanen turned that saving into a price. His idea is called the minimum description length principle, and it says: the best model of some data is the one that makes the total description shortest, where the total is the length of the model *plus* the length of the data once the model has explained what it can. A model is paid for in bits. It’s worth stating only if it saves more than it costs. Peter Grünwald, in a tutorial you can find online, adds the footnote that keeps this honest: the price of a model depends on the code you chose to write it in, so the price itself needs a principle behind it, and nobody hands you that principle for free.

That’s a lot of “length of” to swallow in one paragraph, so do one with me, with numbers small enough to check on the back of an envelope. Every number in it comes from a code I’m about to make up on the spot, and I’ll say every rule of the code out loud, because Grünwald’s footnote is the whole point: change the rules and every number moves.

Here’s the data. A row of thirty-two digits, each a 0 or a 1, and you and I agree in advance that the row is thirty-two long, so nobody has to spend anything saying so:

0101010101010101 0101010101010101

(The space is only there to rest your eyes.) Now the code. Every description starts with one digit that says which kind of description follows: 0 means “here’s the row, written out,” 1 means “here’s a pattern, then the exceptions to it.” Each digit is one yes-or-no answer, one bit, which is what Shannon’s unit is. Written out, the row costs that first digit plus the thirty-two digits themselves: thirty-three bits, and there’s no model in it at all. A pattern description goes like this. First five digits giving the length of a little block, anything from 1 to 32, because five yes-or-no answers can pick out one of thirty-two lengths. Then the block itself. The rule is “repeat the block until you have thirty-two digits.” Then five digits for how many places the row disagrees with the pattern (up to thirty-one, which is plenty here), and five more digits for each of those places, saying which of the thirty-two it is.

Do it for the row above. The block is 01, two digits long. So the model costs 1 (which kind) plus 5 (the block’s length) plus 2 (the block): eight bits. Then what’s left to say about the data once the model has said its piece: the row disagrees with “repeat 01” nowhere, so that’s five digits saying “zero exceptions” and nothing else. Five bits. Total, thirteen, against thirty-three for writing it out. The pattern pays for itself, with twenty bits to spare. That’s Rissanen’s sentence happening on paper: a model is worth

stating when it saves more than it costs.

Now spoil the row. Flip two of its digits anywhere you like. The model doesn't change, eight bits, but now the leftover says "two exceptions, here and here": five plus five plus five, fifteen. Total twenty-three. Still cheaper than thirty-three; the pattern still pays, it just pays less, because the data has more to say that the model can't. Flip four, and the total is $8 + 5 + 4 \times 5 = 33$, a dead tie. Flip five and it's thirty-eight, and now writing the row out is cheaper than explaining it, and by Rissanen's rule you shouldn't state the pattern at all, even though it's still sort of there if you squint. So "is there structure here?" has turned into a question with a number on each side, and a break-even point you can find yourself: somewhere between four and five spoiled digits, in this code.

And now change the code, because I promised you'd see the numbers move. Say I'd used six digits for each exception's place instead of five (wasteful, but nobody stopped me). Then four spoiled digits cost $8 + 5 + 4 \times 6 = 37$, and the tie is gone; the pattern stopped paying one spoiled digit sooner. Same row, same pattern, different verdict, and the only thing that changed was a convention I chose. That's Grünwald's footnote, and it's why I won't let you take any of those numbers anywhere else. They're the price of a description in one made-up code, a length of a written-down message, and nothing more. They are not the charge I opened this book with. That charge, when the book finally states it, is on one step a machine takes, not on how long anybody's description is, and if the two ever turn out to be related, that has to be shown, not assumed from both of them being counted in whole numbers.

Now push, because this is where "structure means information" gets born, and it's wrong, and it's an easy thing to believe. Flip a fair coin two hundred times and write down the string. Almost always, there's no shorter way to say that string than to say the whole thing. You can watch it happen in the little code above, cut to thirty-two digits: a coin-flip row disagrees with "repeat 01" in about sixteen places, which would cost eighty-odd bits to list, and even the one block guaranteed to fit with no exceptions, the whole row used as its own block, costs $1 + 5 + 32 + 5 = 43$. The written-out row at thirty-three wins, and no pattern earns its keep. Nikolai Vereshchagin and Paul Vitányi, in a 2004 paper on exactly this question, put it without mercy: every bit of a string like that is "meaningless information, random noise." Maximum information. Zero structure. So in this sense structure is the part that compresses, the part where the "th" tells you about the "e," and most information and most structure are different things. Hold that one tightly, because if you built something that charged for information, you'd be charging for the wrong thing, and for a while I wasn't sure which one I'd built.

And now notice that the word just changed meaning on you, and on me, and I'd rather point at it than let it slide. Shannon and Rissanen and Vereshchagin mean *regularity*: a pattern that lets you say something shorter. Tarski, a few pages back, and Halmos,

coming up, mean a set with named relations on it, and those two are not the same thing. Here's the test, and you can run it: number a thousand objects, flip a coin for each, heads goes in group one, tails in group two. That's a perfectly good sorting in the set theorist's sense, every object in exactly one group. And the list of who went where is a thousand coin flips, which is to say it compresses about as well as a thousand coin flips, which is to say not at all. Feed it to a code built like the little one, with the fields widened to fit a thousand, and you get the coin-flip row again, only longer: no block earns its keep, the written-out list wins, and by Rissanen's rule there's no pattern worth stating. By the set theorist's rule, the sorting *is* the structure, and it's every bit as much a sorting as "evens in one group, odds in the other," which compresses to almost nothing. Same kind of object, wildly different description lengths. So the two meanings don't even rank things the same way: one asks "is there a sorting?" and the answer is yes for both; the other asks "does a short rule say who went where?" and the answer is yes for evens and odds and no for the coins. Structure in the set theorist's sense, none in Rissanen's. I'll keep both meanings, and I'll try to say which one I'm using, and when I slip you'll know what to look for.

2.13 The four parts of every computer

In 1936 Alan Turing described the machine that every computer you've ever touched is a version of, and he did it before any of them existed. It's the clerk with the pencil from the opening chapter, written down as a machine. Write down its parts; there are four. A tape, divided into squares. A head that reads one square at a time and can write on it and move left or right. A finite list of states the head can be in. A table of rules: in this state, reading this symbol, do this. That's all, and two different things are true about it that get run together. Every other formal recipe for "a procedure" that anyone has written down since computes exactly what this one does; those are theorems, proved one pairing at a time, and Turing proved the first of them himself. That it also captures everything a person could ever carry out by rote, everything "computable" in the plain-English sense, is a claim, called the Church-Turing thesis, and nobody has proved it, because the plain-English side isn't a mathematical object. Turing himself kept those two apart in the 1936 paper, and so will I. There's a third claim that gets stacked on top, and it's a different kind again: the physical Church-Turing thesis, that every device you could build out of real stuff computes no more than this machine does. That one is a claim about physics, not mathematics. Nobody has built a device that breaks it. Your laptop, your phone, every server in every data center compute nothing a Turing machine can't; what differs between them is speed, space, energy, and cost. Hold on to that last word. Cost is where this book lives.

Now do the thing I did, and do it honestly. Look through those four parts for a place that says "these squares belong together." I couldn't find one. Any grouping has to live in the table, as a rule, or on the tape, as marks, which is to say as bytes like any other bytes. Maybe you can find a slot I missed. I'd want to know. I looked for a long time and I kept not finding it, and not finding something is weaker than proving it

isn't there, so I'll be careful about what I claim from that.

In 1972 David Parnas, a software engineer, wrote a short paper that every programmer should be made to read and most haven't, on how to cut a program into modules. He took one small program and cut it two ways. Both worked. But one was easy to change later and the other wasn't, and the easy one could be "much less efficient" if talking between modules cost anything. His point: a module is "a responsibility assignment," a decision about who answers for what, and how you cut decides what every later change costs. If you've ever maintained someone else's code, you've paid Parnas's bill, probably at two in the morning. What survives the push: a division is a commitment, not a caption. It changes what the work costs. It is not a label stuck on afterwards. I'd like to say the thing I built understood that from the beginning. It didn't. I did, eventually.

2.14 Laying them all down

Now lay every one of those on top of the others and see what shows through. Here are the seven things I think I see. Be clear about what this is: it's my overlay, the places where I see the sources agreeing, and not a thing the sources proved together. They didn't; some of them never met, and some of them disagree (Bourbaki left out the maps that Eilenberg and Mac Lane built everything on; Shannon's structure and Tarski's are two different words, as I just showed you). You've pushed on each source yourself by now, so you're in a position to tell me if I'm seeing things.

A list of seven slogans would be easy to nod at and impossible to check, so I'm going to carry one thing through all seven and let you watch each one add to the one before it. The thing is the square you drew for the Pythagoreans, side, diagonal and all. Draw a few more on the same sheet: one small, one big, the small one again but tilted, and one stretched out into a long rectangle. You'll want a pile.

Structure is about relations among things, not the things. The diagonal, the bronze, Leibniz's room, Bourbaki's unspecified elements, Tarski's named relations, the arrows. On your sheet, what makes a square a square was never the ink. It's that the four sides match and the corners are square, and that the diagonal stands to the side the way it does whether or not any number names it. Every one of those is a relation among parts. But on its own this first thing is too loose to use, because a sheet of paper is drowning in relations: this square is left of that one, this one is nearer the coffee ring. Which relations count?

Structure is what stays the same under a chosen set of changes. Euclid's triangles laid together, Klein's allowed moves, Noether's symmetry, Frege's direction that every parallel line shares. That's the answer the first thing was missing. The relations that count are the ones that survive the moves you've decided to allow. Slide the small square across the sheet, turn it, and the diagonal still stands to the side exactly as before; "left of the coffee ring" doesn't make it across. The second thing is a filter on the first.

The same structure can be carried by different stuff. The statue in marble. Two sets that answer the same vocabulary the same way. This is the second thing again with one more change allowed: swap the material. Cut the square out of cardboard, or lay it out in string on the table, and if the relations survive, it's the same square in every way you agreed to care about. Aristotle's bronze is one more thing you agreed not to look at, and the wooden sandwich is what happens when the material was never on that list.

The first act of structure is a sameness that sorts a pile you were already given. Euclid's first common notion, Frege's direction, Tarski's one question, and Russell's letter saying you must start from a pile. This is where the overlay stopped being a list for me and turned into a shape, so slow down here with me. Look at your pile of drawings and say: two of them count as the same when an allowed move carries one exactly onto the other. Watch what happens to the pile. It falls into groups. The small square and its tilted twin go together; the big one sits on its own; the rectangle sits on its own. And it falls cleanly, nothing half in one group and half in another, for a reason you can check. Klein's allowed moves always come with three guarantees: doing nothing counts as a move, every move can be undone, and two moves in a row is again a move. Those three are, word for word in different clothes, "everything is the same as itself," "if this matches that, then that matches this," and Euclid's first common notion, "things equal to the same thing are equal to each other." So the second thing, sameness under chosen moves, *is* a sorting of the pile the moment you ask which drawings it calls the same. Frege did it with lines and parallels. Tarski's one question does it in the crudest way there is, by cutting the pile in two. And the pile has to be there first, because a sameness has to be a sameness among something; that's Russell's letter, and Zermelo's repair.

Adding structure rules possibilities out. Every one of Euclid's postulates is a door closed. Half of English is already decided. A conserved quantity limits where a system can go. On your sheet: once the small square and its tilted twin are in one group, you've given up the right to answer any question the moves respect differently for the two of them. Tell me the side of one and the diagonal of the other isn't an open question any more. A sorting is a bundle of doors closed at once.

Held structure gets used without re-deriving it. Shannon's saving in time. Rissanen's compression. Parnas's one-module-at-a-time. Leibniz, who hoped his algebra of position would spare people a great many fruitless tries. This is the fifth thing paying out. The doors you closed are doors you never have to check again: know which group a drawing is in and you know everything the group shares without getting the ruler out, the same way the row of 01 never needed writing out. With one catch that I'll need a whole section for: it only pays if you kept the sorting somewhere you can read it back.

And structure is always relative to a chosen frame. Klein's group. The vocabulary. The

code you compress against. Parnas's criteria. Nobody gets to say "the" structure of a thing without saying from where, and I do it constantly, so watch me for it. On your sheet: allow stretching as one of the moves and the rectangle walks over and joins the squares; allow only slides and turns and it doesn't. Same pile, same ink, different groups, because I chose different moves. The groups were never in the paper. They were in the paper and the choice, together.

So here's how I think they hang together. Relations are what structure is made of. The chosen moves pick which relations count, and a change of material is just one more move. Asking which things the moves call the same sorts the pile. The sorting closes doors, and closed doors are what you get to skip later. And because the moves were chosen, all of it is relative. Read that way, the fourth thing sits in the middle: it's where the first three turn into something you can write down, and where the last three get their start.

That's my reason for starting the next subsection with a sorting, and I want to be exact about what kind of reason it is, because it's mine and not theirs. None of these people said "start with a partition." Klein would tell you to start with the moves, Eilenberg and Mac Lane with the arrows, Shannon with the probabilities. What I can actually point at is narrower. In the framings about when two things are the same shape or the same kind (Euclid laying one triangle on another, Klein's moves, Frege's directions, Tarski's question) there's a moment where somebody decides what counts as the same, and that decision, written down, is a sorting of a pile. In Shannon, Rissanen and Noether I can't find that moment as cleanly, and I'm not going to bend them until I do. So the overlay doesn't prove that a sorting comes first. It shows me where the framings I trust most keep putting their first foot down, and I'm putting mine there too.

I looked for a framing that denies any of the seven and didn't find one, which could mean there isn't one or could mean I didn't look where it lives. If you know of one, I'd like to see it.

2.15 The smallest piece

Now the sentence I wanted to say at the top and had no right to until here, and even here I have to split it in two, because half of it is a textbook fact and the other half is me. The piece of structure I'm going to start from is a partition.

The textbook half: a partition and an equivalence relation are the same object. That's Paul Halmos, *Naive Set Theory*, 1960, section seven, and you could read that section before dinner. Here it is in words. Take any pile of things. An equivalence relation on it is a way of saying "these two count as the same" that behaves the way sameness should: everything is the same as itself; if a is the same as b then b is the same as a ; and if a is the same as b and b the same as c , then a is the same as c . Every such relation sorts the pile into groups, the things that count as the same go together, and every sorting into groups hands you back a relation, "same group." Do one and then

the other and you land where you started, in both directions. So a partition and an equivalence relation are one object seen from two sides, and that's the fourth thing on the overlay, the sorting of a given pile, now with a name and a page number.

Here it is in the exact form, since this is the first place the book uses one. A *set* is a pile, $x \in X$ means “ x is in the pile X ,” and a *relation* on X is any rule $\sim$ that, for each pair of things, says whether $x \sim y$ holds. A few more marks show up below, and each is a word in disguise. $S_1, \dots, S_k$ are the buckets, numbered one to k . A cap, $S_j \cap S_l$, is “what the two buckets share,” a cup, $S_1 \cup \dots \cup S_k$, is “everything in any bucket,” and $\emptyset$ is the empty pile. And curly braces with a colon, $\{y \in X : y \sim x\}$, mean “the pile of every y in X for which $y \sim x$ holds”; the theorem writes that pile $[x]$, the things that count as the same as x .

Definition 2.1 (Equivalence relation, partition). A relation $\sim$ on a set X is an *equivalence relation* when it is reflexive ($x \sim x$), symmetric ($x \sim y$ implies $y \sim x$), and transitive ($x \sim y$ and $y \sim z$ imply $x \sim z$). A *partition* of X is a collection of non-empty subsets of X , the blocks, such that every member of X lies in exactly one block. Spelled out for blocks $S_1, \dots, S_k$: each S_j is non-empty, no two share a member ($S_j \cap S_l = \emptyset$ when $j \neq l$), and together they cover everything ($S_1 \cup \dots \cup S_k = X$).

Theorem 2.1 (Sameness and sorting are one object). *Every equivalence relation on X determines a partition, whose blocks are the classes $[x] = \{y \in X : y \sim x\}$; every partition determines an equivalence relation, “ x and y lie in the same block”; and the two constructions undo each other.*

Proof. The classes are non-empty ($x \in [x]$ by reflexivity) and cover X . If z lies in $[x]$ and in $[y]$, then $z \sim x$ and $z \sim y$, so $x \sim y$ by symmetry and transitivity, and then $[x] = [y]$ by transitivity again; so the blocks don't overlap. “Same block” is plainly reflexive, symmetric, and transitive. Going relation to partition to relation returns $x \sim y$ exactly when x and y share a class, which is the original relation; going partition to relation to partition returns the original blocks. $\square$

This is textbook mathematics (Halmos, section 7), and the book uses it as a foundation, not as a contribution.

And the smallest structure that sorts, by my ruler in Tarski's setup, the one question, is the same object a third time. Put a yes-or-no question to every thing in the pile. The ones that say yes are one group and the ones that say no are the other. Going the other way, a two-group partition becomes a question the moment you point at one group and call it “yes.” The only thing a question has that the partition doesn't is which group got the label. So the set theorist's first step and the logician's first step are the same step, and nobody has to pick a side, which is a relief, because it's an argument you can lose with yourself for a long time.

That's the textbook half. Now mine, and hold it looser, because Halmos says nothing about "smallest" or "first" and I'd be lying if I let his page number carry my sentence. In what I'm going to build, the order goes: a pile, then groups in the pile, then whatever connects one group to another, and the groups come first because a question needs something to be about and a connection needs somewhere to land. That's an order for *my* construction. It is not a law that every account of structure has to start by sorting. You could start with an ordering instead ("this one comes before that one"), or with arrows between the things themselves and no groups at all, and people do, and nothing I've shown you rules that out. So the honest sentence is: a partition is where I chose to start, it is the smallest start by the ruler I gave you, and I chose it because it's the first thing that can be right or wrong about a pile and the last thing I could take away without everything I build on it coming down. If you have a smaller start by a ruler you can state, I want to see it. It would change this book.

2.16 Buckets

So here's the smallest piece, done once, for real. Take everything in front of you and sort it into buckets: nothing in two buckets, nothing left on the floor. That's the whole of the definition. It's the thing you do with a kindergartner and a pile of blocks, and I am aware that I've just marched you past Pythagoras, Plato, Euclid, Frege and Noether to arrive at a pile of blocks. Keep the blocks anyway. They're about to cost you something.

Ten blocks. Red ones in one bucket, square ones in another, everything else in a third. Now a red square turns up, as they do, and you've got your first crisis: it belongs in two buckets and the rule says one. So this isn't a partition yet, and you have to decide which bucket wins. That decision is the whole thing. A partition is a pile of small rulings like that one, and once you've made them all, the pile is holding a fact it wasn't holding a minute ago. The formal version says the same thing with fewer words and more squiggles, and you've already seen it: it's the definition of a partition a few pages back (Definition 2.1). Non-empty buckets, nothing in two, nothing on the floor.

Here's the part that isn't kindergarten. Why would that fact cost anything? Watch what it does to one question. Before the buckets, "are these two in the same group?" means working it out: go find the first one, go find the second one, apply the rule to each, compare. After the buckets, it's a glance: read the first one's bucket, read the second one's bucket, done. And the glance costs the same when the pile goes from ten blocks to a billion. That's Shannon's saving in time and Parnas's one-module-at-a-time, the sixth thing on the overlay, showing up in a kindergarten.

Now push on it, because that paragraph sells more than it should, and I want you to see where. The partition didn't give you the glance. A *table* gave you the glance: a written-down line for each block saying which bucket it went in. The partition is the fact; the table is one way of holding the fact so it can be read back fast, and you had to

build the table, and you have to store it, and it has to be the kind of thing you can read in one look (a bucket number, say, and not a list of every block in the bucket that you'd have to search through again). Anybody who's used a database has met this table; it's what an index is, and an index is a thing you build, store, and keep up to date, not a thing you get for free. Take the table away and keep the partition and you're back to working it out. And sometimes there was never a search to save: if the rule is "evens here, odds there," you can answer "same group?" by looking at the last digit of each, no buckets needed. So here's what survives: holding a sorting in a form you can read back is worth something, and building that form cost something, and those are already two different events. Building the table, storing it, checking that it's right, and deciding to *rely* on it are four different things. The question this book is about lives in the gap between the third and the fourth, and nothing in the kindergarten has shown you yet that there's a charge in that gap. It's shown you there might be a place for one.

So who, in everything I read, says what the bill is? I found three answers that come close, and I'll be exact about how close, because this is the edge of what I know. Rolf Landauer, in 1961, and Charles Bennett after him, say that any step which takes two possible situations and makes them one, erasing a bit or merging two paths a computation could have taken, has to push some disorder out into the world, which in a real machine means heat. That's not a metaphor; it's the one place where a fact about information turns into a fact about your electricity bill, and it's been measured. Is sending blocks to buckets a step of that kind? Push, because it depends on what you keep. If you keep each block and write its bucket next to it, you've lost nothing; you can read the block right back off the pair, and Bennett's whole point in his 1973 paper was that a computation which keeps its inputs can be run without any such step at all. If instead you throw the block away and keep only the bucket label, then two different blocks can leave the same label behind, and that is a many-to-one step, the kind Landauer charges for. Both are things people call "sorting into buckets." Only the second loses a distinction. So the bucket by itself doesn't hand you a Landauer bill; the *forgetting* does, when there is forgetting, and whether a given machine forgets is a question about that machine, which has to be answered on the page and not waved at here. Rissanen says a model is paid for in the bits it takes to state, and Grünwald says the price depends on the code. Parnas says a bad cut is paid for later, at two in the morning. And that's where it stops. I couldn't find anyone who says that the moment you *vouch* for a grouping, the moment you stand behind "these belong together" instead of just arranging them that way, has a price of its own. That could mean nobody says it. It could mean I didn't find the person who does. I honestly can't tell you which.

So that's where I am at the end of one word. Holding a sorting in readable form is worth something and cost something, forgetting where things came from costs heat, and standing behind a sorting is a different act from either, and nobody I found has priced it. You can walk back through all of that and tell me where it breaks. Maybe

there's no such bill. Maybe I'm about to spend a book chasing a charge that nobody owes, which would be an extremely me thing to do. The next section is me trying to find out whether that charge is real on any machine at all, not mine, any machine, and if the answer comes back no, you should stop reading there. I would have stopped writing there too.

Part II

The axiom

3 A record and a toll

I left you with a question I couldn't answer: something pays for the glance, and I couldn't find anyone who says the moment of vouching for a grouping has a price of its own. Before I go looking for that price in anything complicated, I want to try the opposite. Build the smallest thing that could possibly carry such a bill, on paper, with a pencil, write the bill into its rules, and see what follows from it. If nothing follows in something with five parts, nothing is going to follow in something with five thousand, and I'd rather find that out on page one than on page four hundred.

One thing about where the buckets went, because you'll notice they vanish for a while. Part I ended on vouching for a grouping, and what I'm about to build prices vouching for anything at all: a flag goes up, and the rule doesn't ask what the flag is standing behind. A sorting is the first kind of claim I care about, not the only kind, and the book comes back to what a flag stands behind in Section 8, and on a machine in Section 14, where a flag stands on a checked claim. Until then nothing leans on the buckets. The floor in this section, the reason for it in the next, and the latch in Part III all work for any yes-or-no reading whatsoever, partition or no partition.

But first the word, because I've got a new one and I don't trust it either. Record. Receipt. Certificate. Vouch. People have been making marks that other people are supposed to rely on for a very long time, and I wanted to know what they thought they were doing, and what it cost them, and whether any of them ever said the thing I need said. So, same deal as before: here's what I could reach, pushed on with you, and I'll tell you where the search ran out.

3.1 Clay, seals, and seven witnesses

Start about five thousand years ago, in the Near East, with people who counted sheep with little clay tokens. To send a count across a distance without the courier changing it, they put the tokens inside a hollow clay ball, pressed the shapes of the tokens onto the outside while the clay was soft, and sealed it. The archaeologist Denise Schmandt-Besserat has spent a career on these; the ball is called a bulla. Here's the push, and

you can do it with a cookie tin: the thing that makes it a record isn't the tokens, it's that the outside shows what's inside and you have to break it to get in. Opening it destroys it. Tampering shows. That's a real idea and it will show up again in a form you won't expect: a record is worth something only if it can't be silently undone.

Now law, which is where people get serious about who's allowed to say what happened. The book of Deuteronomy, chapter nineteen: nobody is convicted on the word of one witness; it takes "two or three." The Roman jurist Gaius, writing in the second century, says a will the praetor will honour is one sealed by seven witnesses. Seven. Why did they care that much? Because one person can lie, and a will moves a fortune, and the dead man can't object. So the record isn't the paper. It's the number of independent people who have put their own seal, their own name, their own risk, next to the claim. Try it: write "paid five dollars, signed A" on a napkin and ask a friend to forge it. Takes them ten seconds. Now require two other people to initial it at the moment of payment and each keep a copy. Your forger needs two accomplices or has to match two copies held in two other pockets. The claim didn't change. What stands behind it did.

Which brings me to seals, and the medieval historian Michael Clanchy, whose book is called *From Memory to Written Record*. England between the Norman Conquest and the early 1300s was learning to trust writing instead of trusting people who remembered, and the lesson was painful: seals were forged, charters were forged, and historians today have a whole craft of telling real charters from fake ones because the fakes are so many. A seal is only as good as whoever holds the stamp. Keep that one too, because in 2011 a certificate authority called DigiNotar found out, at the cost of its existence, that a mathematically perfect signature is worth exactly what the custody of the key is worth, and intruders had taken control of the machines that signed. Nine hundred years, same failure. I find that oddly comforting.

And then the obvious next question: if seals could be forged, what caught the forgeries? Not the seals. The most famous forged record in European history is the Donation of Constantine, a document in which the emperor Constantine supposedly handed rule over Rome and the western empire to the Pope. It was most likely written in the eighth century, some four hundred years after Constantine died, and it was leaned on for centuries. Around 1440 a scholar named Lorenzo Valla took it apart by its Latin: the language belonged to a much later age, and it used words, "satraps" among them, that no Roman of Constantine's day would have used for his officials. (Valla wasn't even first; Nicholas of Cusa had already called it a forgery. Valla's is the version people remember, which is a pattern that keeps turning up.) Look at what kind of check that was. Not "is the stamp real" but "could these words have been written when they say they were." A record can pass the first and fail the second, which is a hint that what I keep calling one thing, a record, is doing more than one job.

3.2 Two columns and a stamp

Here's the one I'd never have guessed mattered. Bookkeeping. Around 1300, the ledger of a Florentine company called Farolfi, kept at its branch in Provence, shows the earliest complete example anyone has found of a trick merchants were working out: write every transaction twice, once where the money came from and once where it went, and the two columns have to balance. Genoa's public accounts of 1340 do it. And in 1494 a friar named Luca Pacioli printed the method in a mathematics book, and it spread. That's double-entry bookkeeping, and it's easy to take it for just a tidy habit.

Do it for one day. Keep two columns of every dollar that moves, and total them at night. Now have a friend secretly change one entry in one column. The totals disagree; you catch it. Now have them change the matching entry in both columns. The totals agree. You've caught nothing. Push on that and you've found what double entry actually is: a trap for slips, not for careful liars. Pacioli knew. The scholar Forrester reads him as reporting that in many places the books were taken to the local consul to be authenticated before a single entry went in. The stamp is a witness again, from outside. The two columns are the discipline; the stamp is the claim that someone other than you will stand behind the fact that these books, and not some others, are the books. Keep both halves separate in your head. One is a method. The other is a commitment.

And then I found the place where this got turned into a motto. The Royal Society, founded in London in 1660 and still running, picked as its motto *Nullius in verba*: take nobody's word for it. (Where did they get it? From somebody's words, naturally. It's half a line of the Roman poet Horace, from a verse letter to his patron, saying he isn't bound to swear by the words of any master. So the society that takes nobody's word for it took its motto from a poet and cut the line to fit. That's not a contradiction, and it's a nice picture of the method: borrow what's useful, say whose it was, and don't swear by it.) The historian Steven Shapin wrote a paper about what that meant in practice, and the answer is more interesting than the slogan. Robert Boyle, pumping the air out of a glass globe to see what happens to a candle or a bird, had three problems: the machine, the report, and the witnesses. He solved the third by having gentlemen present who'd sign their names, and the second by writing the report so richly that a reader who wasn't there could picture the scene and count as a witness anyway. Shapin calls that virtual witnessing, and he notes, dryly, that the air pump was very expensive, seventeenth-century big science. So there *was* a cost. It just wasn't a cost of declaring the fact. It was the cost of the instrument and of getting enough people into the room. Hold on to that distinction. It's going to turn out to be the whole question.

3.3 A demon, and the step that pays

Now the strangest entry, and the one that gets closest. In 1867 the physicist James Clerk Maxwell, in a letter to a friend, imagined a “very observant and neat-fingered being” sitting at a tiny door between two halves of a box of gas, letting fast molecules through one way and slow ones the other, so that one side heats up and the other cools with no work done, which is supposed to be impossible. The being got called Maxwell’s demon, and physicists have been arguing about it for a hundred and fifty years, which should tell you something.

Szilard’s one-molecule engine, in 1929, made information part of the accounting. The tempting move is to put the whole bill on finding out which side the molecule is on. Bennett pulled that apart from restoring the memory for another round: measuring onto a blank record can, in principle, be thermodynamically reversible, and resetting an unknown record is a different operation. That does not make a real measuring device free to build or run. It says that learning and resetting do not carry the same unavoidable entropy cost, and I’m about to pick a step to charge, so I need that distinction before I do.

Bennett’s version of the rule, in a 2003 paper you can read, is that what must cost is “the erasure of a bit or the merging of two computation paths.” Erasing, or taking two possible situations and making them one. And in the same paper he lists copying something onto a blank place to write as an operation that can be thermodynamically reversible. So, in the one place in all of this where a record is priced by physics, the unavoidable erasure charge belongs to *forgetting* the unknown record, rather than to every act of making a record. Read that twice. It’s not what I expected either.

I’ll put the rest of the lineage quickly, because it all points the same way. Regulators who make brokers keep records on media that can’t be rewritten (a rule the SEC had for decades) are pricing the hardware, not the act. Haber and Stornetta in 1991, the timestamping that Bitcoin’s footnotes lead back to in the opening chapter, chained time-stamped records together with hashes so that changing an old entry breaks every entry after it (and Ralph Merkle’s hash tree of 1979 lets one short value vouch for a great many records); try it with five lines and a made-up sum, change line two, and watch every number below it move. That makes a change *visible*. It doesn’t make it cost. Proof-of-work, which Dwork and Naor proposed in 1992 to make junk mail expensive and which Bitcoin later chained, is the one framing where a record genuinely costs, and what it charges for is a search: how many tries it took to find a lucky number. An empty block costs the same work as a full one. So the work prices the search, not the moment anything becomes certified. Floyd and Hoare, in 1967 and 1969, attached claims to points in a program (“whenever you get here, this is true”), and the claim is backed by a proof that’s paid for off the machine. And the certificates of complexity theory, which go back to Stephen Cook’s 1971 work on NP, are strings designed to be cheap to check: hand me the certificate and I can confirm the claim quickly. Whether they’re also expensive to *find* is the famous open question of the

field, P versus NP, and nobody knows; what's settled is only the cheap half. That's the opposite of a price on vouching: the whole design goal is that checking a claim costs nearly nothing. And from statistics, a ledger that runs the other way: Dean Foster and Robert Stine's *alpha-investing* (2008) runs a long string of tests against a budget of allowed false alarms, where every test spends some of the budget and every discovery, a rejection that stands, pays some back. Records that don't come down, a ledger, and the price on the test, with the commitment earning.

3.4 What they all agree on, and who they rule out

Lay them down, and this time what shows through is not one thing. It's five different jobs that the word "record" has been doing, and I'd been letting one word carry all five. Each has a corpse next to it, and the corpses are different too.

Durability: the record can't be silently undone. The bulla, the write-once disk, the hash chain, Bennett's card that has to be wiped. The corpse is the editable log, and the company that kept its debts off the books: Enron hid its debt in side entities and its auditor shredded the paper, and those books recorded nothing. But notice this job doesn't need a witness at all, only a copy: a hash chain nobody signed is still hard to change quietly, so long as somebody else holds the latest link, because a chain with no copy anywhere can be rewritten from end to end and recomputed, and nobody would know.

Authenticity: the record came from who it says it came from. Seals, the signature only a key-holder can make, the consul's stamp. The corpse is the forged charter and DigiNotar. This job is about custody of a stamp, and it's a different job from durability; a perfectly authentic entry can still be erased, and a perfectly durable one can still be a forgery.

Evidence: enough independent people stand behind the claim that lying is expensive. Two witnesses, seven seals, Boyle's gentlemen. It rules out anonymous assertion, and it rules out "the universe keeps the receipt," which I mention because people say it: information spread into the environment is not a record anyone can read or rely on. This one is about numbers of people and their risk, not about paper.

Accounting: the record checks itself for slips. The two columns. And here's a thing it's easy to get wrong: the second column is *not* an outside witness. It's you, checking you. It catches a dropped digit; it doesn't catch a careful liar, as you saw. It belongs on this list as its own job, not as a cousin of the seven seals.

Cheap checking: a record whose correctness can be confirmed for much less than it cost to make. Cook's certificates, the signature, the hash. This one isn't universal either, and I thought it was. Whether checking needs to be cheaper than making depends on who you're worried about and how much a wrong record costs you; plenty of records are checked by redoing the whole thing, and that's fine when the thing is small.

Now the thing they don't agree on, because none of them says it. Not one of these jobs is pricing the act of certifying. Bennett prices merging and erasing, which a certifying step might or might not be. Proof-of-work prices a search the designer chose. Bookkeeping prices a discipline. Witnesses and seals price showing up and guarding the stamp. I looked, with everything I had, and that's where it ran out. If the bill exists, nobody in this lineage wrote it down, and that's either because it isn't there or because the one person who did write it down is on a shelf I never found. So I'm going to have to try it myself, on the smallest thing I can build, and let you watch. And I'll have to be honest about which of the five jobs the small thing even attempts, because it's not going to be all of them.

3.5 The smallest thing that could carry the bill

Take a sheet of paper. Draw two boxes and label them NO and YES. That's the reading, a single yes-or-no question asked of the state, and I'm going to be careful about what the question is, because after five thousand years of seals and witnesses the word "vouch" is carrying a lot and this toy carries almost none of it. The question is only: "has the designated flag been set." Not "is some claim about a grouping true." Not "did anyone check." There is no claim in this toy, no evidence, no checker, nobody standing behind anything. There is a flag. Whether a flag can ever be made to *mean* that something was checked and can be relied on is a real question, and it's not this section's; I'll build toward it later, and until then please read YES as "the flag is up" and nothing more. Where the state is: which box you're in. That's the whole state, one bit. Then draw a line under the boxes for tally marks. The marks are not part of the state; they're the running total you keep beside it as you play, the way you'd keep score beside a board game.

Now give yourself three moves. STAY: you don't move, no mark. WORK: you don't move either (it's a stand-in for every ordinary step a machine takes), no mark. VOUCH: from NO you move to YES; from YES you stay in YES; either way you make one tally mark, because in this toy the price belongs to the move's name and not to where you were standing. Play. Start in NO, make any sequence of moves you like, and stop in YES. Count the marks. Now try to reach YES with zero marks. You can't, and you can see exactly why: the only move that crosses from NO to YES is VOUCH, and VOUCH makes a mark. Now erase the mark from VOUCH's rule and play again. YES for free, every time. That's the entire content of what I'm about to call a rule, and I want you to see how little it is before I dress it up: I've decided that the move which crosses the line makes a mark, and nothing else about the machine forces that. It's a choice. The next section is about whether I had to make it.

Why tally marks, by the way? Because they're about the oldest receipt there is. The real ones, the English treasury's notched sticks split down the middle, get their own story in Part III (Section 6), fire and all. The tally on your paper is nothing like those. Nobody holds the other half. You keep it, beside the board, and you could rub a mark out and nobody would ever know. Hold on to that, because by the end of this section

it's the reason the toy does none of the five jobs a record does.

Here is the same toy, stated once, carefully, because this is the one place in the book where I state it and everything later points back here. The thing on your paper has five parts.

A set of states. (Here: NO and YES, and nothing else.)

A set of move names. (STAY, WORK, VOUCH.)

A step: for each state and each move name, where you land.

A cost for each move name: a whole number of marks, zero or more. (Zero, zero, one.)

A reading: for each state, yes or no.

That's it. Add the one rule below and I'll call it a certification system; in the checked record the rule is a sixth field, so nothing counts as one without it. And I want to be exact about what's not in it. There's no counter stored inside the state that's forbidden from going down; the marks are the side tally from the board game, and the tally only ever grows because the costs are whole numbers, zero or more. (A machine could keep the total inside its state too; that's an extra part, not one of the five, and Section 6 builds it.) There's no window, no observer who sees only part of the state; that's a later question and it gets its own section. And the cost depends only on the move's name, not on where you are when you make it, which is a real simplification and I'll say so when it matters.

One rule, which I'll call the toll. For every state and every move: if the reading is NO before the move and YES after it, the move's cost is at least one mark. That's all. It says nothing about any other move. It says nothing about how big a mark is in anything you could measure. One mark is one mark; what a mark is worth, and why the smallest charge is one and not two or a half, and whether I had any right to decide that, come next, and I'm not sure of all of it.

And one theorem, which is the only thing the toy actually proves, and you've already proved it by playing. Take any run of moves that starts in a NO state and ends in a YES state. Then the marks it made add up to at least one. Here's the argument in words, and it's short enough that you can hold the whole thing. Walk the run move by move, reading the box at each step. It read NO at the start and YES at the end, and every move either leaves the reading alone or flips it, so at some move it went from NO to YES. The toll says that move cost at least one. Every other move cost zero or more. Add them up: at least one. Notice what the argument did not need. It didn't need the reading to *stay* YES afterward; if a later move flipped it back to NO and another flipped it to YES again, that's two tolls, not a problem. It didn't need the machine to be finite. It didn't need to know what any move does besides what the reading says before and after. It needed exactly one thing: the toll.

That's the whole theorem. Here it is once more in the exact form, because from here on the book carries two layers side by side: what I mean, in words, and what's stated and checked, in symbols you can hold me to. The symbols are few. $\mathbb{N}$ is the whole numbers 0, 1, 2, and so on. S and I are just names for "the set of states" and "the set of move names." Writing $\text{step} : S \times I \rightarrow S$ means "hand it a state and a move name and it hands back a state." A *run* is a list of move names, $t = [i_1, \dots, i_n]$, played from a start s_0 , one after another. The double arrow $\implies$ reads "then": the thing on its left forces the thing on its right. And "by induction on the run," in the proof, is the formal name for the way you'd check it by hand: settle the empty run, then show that one more move at the front can't break what you've settled.

Definition 3.1 (Certification system). A *certification system* consists of a set S of states, a set I of move names, a step function $\text{step} : S \times I \rightarrow S$, a cost $\text{cost} : I \rightarrow \mathbb{N}$, and a reading $\text{cert} : S \rightarrow \{\text{no}, \text{yes}\}$, subject to one rule, the *toll*: for every state s and move i ,

$$\text{cert}(s) = \text{no} \text{ and } \text{cert}(\text{step}(s, i)) = \text{yes} \implies \text{cost}(i) \geq 1.$$

The run of $t = [i_1, \dots, i_n]$ from s_0 ends at $\text{run}(t, s_0)$, the result of applying the steps in order, and its total cost is $\text{cost}(t) = \text{cost}(i_1) + \dots + \text{cost}(i_n)$, with the empty run costing 0.

Defined as the record `CertificationSystem` (`UniversalCertificationCost.v`).

Theorem 3.1 (The universal floor). *In any certification system, every run that starts at a state reading no and ends at a state reading yes has total cost at least 1: if $\text{cert}(s_0) = \text{no}$ and $\text{cert}(\text{run}(t, s_0)) = \text{yes}$, then $\text{cost}(t) \geq 1$.*

Proof. By induction on the run. If t is empty, the run ends where it starts, at a state reading no, so the hypothesis "ends at yes" can't hold and there's nothing to prove. Otherwise $t = i :: t'$ (the first move, then the rest); let $s_1 = \text{step}(s_0, i)$. If $\text{cert}(s_1) = \text{yes}$, the first move is a crossing, the toll gives $\text{cost}(i) \geq 1$, and the rest cost at least 0, so $\text{cost}(t) \geq 1$. If $\text{cert}(s_1) = \text{no}$, the rest of the run starts at no and ends at yes, so by the induction hypothesis $\text{cost}(t') \geq 1$, and $\text{cost}(t) = \text{cost}(i) + \text{cost}(t') \geq 1$. Nothing else is used: not finiteness, not that the reading stays yes, not what any move does beyond the reading before and after. $\square$

Where to check it: the record's six fields are `cs_state`, `cs_instr`, `cs_step`, `cs_cost`, `cs_cert`, and the toll `cs_cert_costs`; the theorem is `universal_nfi_any_substrate`, in the same file; and the checked proof is the induction above, `step` for `step`. These names identify the companion formalization. The argument is the induction on this page. One honest edge: the theorem does not assume that a YES state exists. If no state ever reads YES, the theorem is true for no runs at all, which is a

mathematician's way of saying it's true and useless. The toy on your paper has a YES box, so it isn't useless there. It would be natural to assume the system has to have at least one NO state and one YES state; the checked statement has no such clause, and that's on purpose, because a theorem that holds for no runs is still a theorem and the toy is where you supply the runs.

The theorem also runs backwards, and that direction is one line. Suppose every run from a state reading no to a state reading yes costs at least one, whichever state you start from. A single move is a run of length one, so a move that takes no to yes costs at least one, which is the toll. The floor from every start and the toll are the same rule, and nobody has to choose between them. From one fixed start the floor asks for the toll only at the crossings you can reach without paying anything; a crossing you can only reach by paying has already cost at least one by the time you get there.

And one more edge, because it's the first thing to bite when you ask a machine to stand behind more than one thing. The definition has *one* reading in it. One question, one flag, one toll on its crossing. Once the flag is up, a second claim has nowhere to go: raising a flag that's already up doesn't move anything. So a second claim needs a reading of its own, and a third claim a third: one reading per claim, each with its own toll. Part III builds that properly, as a family of flags, one per threshold, each charged when it goes up (Theorem 10.10), and the same theorem sends the bill for finiteness: a record held in k bits that only ever turns bits on can take at most $k + 1$ different values, so it can go up at most k times. A finite machine can keep only so many things up for good in its whole life. That's the opening chapter's "you can't keep all of it," turned into a count.

3.6 What this does and doesn't buy

I want to be very plain about how small this is, because the temptation is to look at the word "theorem" and feel something has been established about the world. Nothing has. What's been established is that *if* you adopt the toll, *then* the flag can't be raised for free, in any system with those five parts that obeys it, no matter what the states and moves are. The "if" is doing all the work. I chose the rule. A reasonable person could ask why, and the only honest answer right now is "because that's the thing I wanted to be true," which is not a reason.

Notice also what it doesn't ask about how strong the thing is, in either direction. A system is *Turing-complete* when it can imitate a Turing machine, and no stronger than one when a Turing machine can imitate it. The theorem asks neither. Stronger than a Turing machine, weaker, dead even: none of it reaches the floor. Bring the five parts, charge the toll on the crossing, and you're in scope. The floor doesn't check what you can compute, because it was never about what you can compute. It's about what a move is allowed to do for free.

Which brings up the objection a theorist makes in about four seconds, so I'll make it first. The toll is a floor, and a floor is easy to meet. Charge one mark for every move,

whatever it does, and the toll holds for any reading you like, since every crossing is a move and every move costs one. So any computer with a step counter beside it, and any yes-or-no question at all about its state, is a certification system. A clock passes. That isn't a loophole; it's what the definition says, and it means the definition, on its own, doesn't tie the marks to certifying anything. It only says a crossing can't be free. (It's also why "a base that can compute anything plus a flag that obeys the toll" would be a cheap thing to call Thiele-complete, and why the definition that gets that name, in Part IV, asks for free computing and an earned record, which no clock meets; Section 14.) Two things later in the book do the tying. One is a ceiling: demand that a run's total never exceed the number of times the flag went up, and together with the floor exactly one bill survives, one mark per raise and nothing on any other step (Section 7); a clock fails that the first time it charges a step that raised nothing. The other is order, on the smallest machine I build: there the flag can't go up unless a passing check of a claim came first, then a commitment to that same claim, and a clock can't fake that, because it's about which moves happened, not how many marks they made (Theorem 14.9).

And the other side of that: nothing is in scope just for being a computation. A Turing machine's run isn't an instance of this theorem because it's a run. It becomes one when somebody supplies all five parts for it, says which states read yes, and proves the toll for that cost. Until then the theorem says nothing about it, and I won't pretend it does.

That's also where the other axis from the Flatland night ended up, once I checked it. It isn't a direction an ordinary computer can't reach. Hand a Turing machine a reading, charge the toll on its crossing, and it's in scope like anything else. What the usual rules leave out is the price, not the possibility.

The same goes for a system with no reading at all. If nothing in it answers "is the flag up," then nothing in it can say, from the inside, that anything is entitled to be relied on. Whatever entitlement it ends up with gets carried in from outside, by somebody who signs their name to it, and nothing in the system priced it. Smuggled entitlement isn't entitlement, no matter whose name is on it, not for the system. Be careful which way that cuts, though. Whoever reads the signature can be perfectly entitled on their own account; that's what an auditor has, and it's honest as long as it's called the auditor's and not the system's. On the smallest machine I build later, the reason to believe its checker says yes only when the claim is true is exactly that kind: a proof, checked outside any record the machine keeps, and I'll say so when I get there (Section 14). What outside entitlement never is, is something the system itself earned.

And here's a question I can only half answer: bolt two certification systems together, and do you get one? Run them side by side, each move being a move of one or the other, and add up the marks. It depends on the reading you give the pair. Take a reading that can only go up when one of the parts' readings goes up, like "either flag is up" or "both are": then any crossing of the pair is a crossing of a part, that part's

toll charged it, and the pair obeys the toll. Take “the first flag is up and the second is down,” and it doesn’t: that reading turns on when the second flag falls, and nothing charges a fall. That’s an argument on this page, not a checked theorem. One product of this kind is checked later, a host machine running a guest, read through “the host’s own flag or its copy of the guest’s” (Theorem 14.35). Whether the class is closed under composition or simulation in general, I don’t know. It’s open.

So why bother with the toy at all? Not because the lineage was looking for it. The bulla was after durability, the seven seals after evidence, the consul’s stamp after authenticity, the two columns after catching slips; none of them was trying to put a price on the moment a flag goes up, and I shouldn’t pretend they were. The reason is narrower. Everywhere in that lineage where somebody *did* locate a cost, it sat on a step next to the flag and not on the flag: Szilard on the measurement, Bennett on the wiping, proof-of-work on the search, Boyle on the instrument and the room, Foster and Stine on the test. The toy is the first thing I could build where the cost sits on the crossing itself, and it proves that if you price that step, the floor follows everywhere, for any machine with the five parts. That’s not an answer. It’s the question made exact enough to be wrong. And of the five jobs a record does, the toy attempts none of them. Not even durability: the flag can be lowered by any move you care to write, and the marks that don’t go away are kept by *you*, the scorekeeper, beside the board, not by the machine. What the toy has is a scorekeeper’s receipt for the crossing. Whether the machine can be made to keep that receipt itself, where a step could reach in and change it, is Section 6’s question. Authenticity, evidence, accounting, cheap checking: none of them is in here, and I’ll say so again when it’s tempting to forget.

And here’s what I’ll owe you by the end. Whether the toll is a choice or a necessity: the next section, and I’ll tell you now that the answer surprised me. What a mark is worth in anything physical: later, and only partly. What an observer who sees only a window can and cannot recover, including whether they can even tell which step paid: its own section. And whether a full-size machine, the kind you could actually run a program on, can be made to obey the toll without cheating, and what that machine looks like, and what it costs to vouch for structure in it: that waits for Part IV, on purpose, and takes most of the rest of the book, and I’m still not sure I pulled it off.

For now you have a sheet of paper with two boxes, three moves, one rule you were handed without a reason, and one theorem you proved yourself in about a minute. Don’t let go of it. It’s the smallest version of everything that follows, and if it’s wrong here, it’s wrong everywhere.

4 Why the toll isn’t arbitrary

I handed you a rule without a reason. The move that raises the flag makes a mark, because I said so. If that’s all it is, this book is a long way of describing a preference, and you should treat it the way you’d treat someone telling you their favourite colour is a law of nature. So this section is me trying to find out whether I had to pick it.

I'll tell you the shape of the answer up front, because I don't want you reading forty paragraphs to find out whether there's a point: on a machine with a finite memory, where the flag once raised stays up, the step that raises it has to forget something, and forgetting is the one step Landauer's principle puts a price on, when what's forgotten wasn't already known (that last clause takes a lot back, and I'll show you exactly how much). Three "ifs" in that sentence. Every one of them is a door you can walk out of, and I'm going to show you each door, because the doors are where the actual content is.

4.1 A closed room with no spare chair

Do this one with real objects, because the whole section rests on it and it's about ninety seconds of work. Draw a box on the table and call it the certified room. Put three chairs inside the box and one chair outside, by the door. Four people, one per chair; the people are the states your machine could be in, and the three inside are the ones whose flag is already up. One move of the machine moves everybody at once: each person stands up and goes to the chair the rule sends them to. Two rules. First: nobody inside the room may end up outside, because the flag, once up, stays up. Second: the person by the door must end up inside, because this move is the one that raises their flag.

Now try to finish the move with one person in each chair.

You can't. Three people start inside and have to stay inside, so they take the three inside chairs between them, with nobody left out, because nobody may leave. The person from the door has to come in. There is no fourth chair inside. Somebody shares. Two of the machine's possible states land on one state, and after the move you can no longer tell, from where everyone is sitting, which of the two it was. The move forgot something.

That's the whole argument, and I want to name the piece of mathematics that did the work, because it's old and it's not mine. If you put more things into fewer boxes than there are things, some box gets two. People call it the pigeonhole principle; it's the one from the opening chapter, where Jean Leurechon used it in 1622 to argue that two people somewhere have the same number of hairs, and the name comes from Dirichlet calling it the drawer principle in 1834. It's the kind of fact a child accepts and a mathematician spends a career finding new uses for. Here the use is: a step that forgets nothing is a one-to-one shuffle, a one-to-one shuffle of a finite set of chairs hits every chair, so the inside chairs are all taken by insiders, so there's no room for the newcomer. Finite. Permanent. Flip. Those three and nothing else, and the conclusion is "two states became one."

The checked version of the chairs is a theorem a few lines down, and it needs one word first. "Injective" is the mathematician's word for "forgets nothing": a step is injective when two different starts always give two different ends. The theorem says exactly what the chairs said and takes exactly the three premises: a duplicate-free list

that names every state (finite), a reading that no step ever turns off (permanent), and one state that reads NO before the step and YES after it (the flip). There is no price in it anywhere. It is pure counting. I'll come back to what counts as physics, but notice that nothing so far has been.

Here are the words, exactly. For this whole section the setting is the five parts from Section 3 without the toll: states S , move names I , a step, a reading, and (later) a cost.

Definition 4.1 (Finite, permanent, injective). The machine is *finite* when there is a list of states with no repeats that contains every state. The reading is *permanent* when no move ever turns it from yes back to no: for all s and i , $\text{cert}(s) = \text{yes}$ implies $\text{cert}(\text{step}(s, i)) = \text{yes}$. A move i is *injective* (forgets nothing) when different starts give different ends: $\text{step}(a, i) = \text{step}(b, i)$ implies $a = b$. A move that is not injective *merges*.

Defined as `permanent` and `step_injective` (PermanentCertification.v).

Theorem 4.1 (A permanent flip merges). *On a finite machine with a permanent reading, if $\text{cert}(s) = \text{no}$ and $\text{cert}(\text{step}(s, i)) = \text{yes}$, then the move i merges.*

Proof. Suppose i were injective. Let C be the list of states that read yes; it is finite and has no repeats. By permanence, $\text{step}(\cdot, i)$ sends every member of C into C . Being injective, it sends the members of C to members of C with no two landing on the same one, so its images are as many as C itself, and a list of that many distinct members of C is all of C . So the yes-state $\text{step}(s, i)$ equals $\text{step}(c, i)$ for some c in C , and injectivity gives $s = c$, so s reads yes. It reads no. That's the contradiction, and it's the chairs. $\square$

Checked as `permanent_flip_is_not_injective` (PermanentCertification.v). The proof there uses the standard-library fact that a duplicate-free list of n members inside a list of n is the whole list, which is the pigeonhole in the form Leurechon would recognise.

Is that new? No. It has been sitting in two places for a long time, one in physics and one in the theory of automata, and a physicist will recognise it on sight, so start there. In 1876 Josef Loschmidt pointed out, against Ludwig Boltzmann's account of how a gas settles down, that the laws of mechanics run just as well backwards: reverse every molecule's velocity and the gas retraces its steps, so mechanics alone can't make entropy only ever climb. Twenty years on, in 1896, a young assistant of Max Planck's in Berlin pushed harder. He took Henri Poincaré's recurrence theorem of 1890, which says a bounded mechanical system comes back as close as you like to wherever it started, again and again, and used it to argue that nothing in such a system can climb forever: give it time and it comes back down. The assistant was Ernst Zermelo, the same Zermelo who'd patch Frege's floor twelve years later. Boltzmann answered

the same year, not very politely, that the times you'd have to wait for a return are so enormous that nobody would ever see one.

Now look at the chairs again. A step that forgets nothing, on a finite set, is a reshuffle: it moves everybody around in loops, and every loop comes back to where it started. A flag that goes up and stays up is exactly a quantity that climbs and never comes back down. So a reshuffle can't carry one, which is Zermelo's objection with the infinities taken out. On a finite set you don't need Poincaré's theorem, only the loops, and you don't wait an enormous time, only as many steps as there are chairs. Zermelo was aiming it at the second law, not at a flag. The quantum version came in 1957, when P. Bocchieri and A. Loinger proved the same return for a closed quantum system whose energies come in separate levels, so going quantum isn't a way out of the room either, not for a closed system that never sheds anything.

The second place stings more, because it's in computer science. Somebody got here first, again. Picture a finite machine that reads letters, one move per letter. If every letter just reshuffles the states one for one, automata theorists call it a permutation automaton, and the sets of inputs such machines can pick out are called group languages. Jean-Éric Pin's 1992 paper on reversible automata, machines whose letters act as one-to-one maps, sorts out exactly which languages those machines recognise, and in that world the chairs are folklore. Here's the short version. Start at a state that reads no, and collect every input that ends at a yes. Because yes is permanent, adding letters to the end of a collected input keeps it collected. A group language with that property is either empty or everything, and everything would include the empty input, which ends where it started, at no. So it's empty: no run of reshuffles raises the reading. That's Theorem 4.1 with the moves as letters. Pin's paper is where I'd send you to check the automata side. It isn't about flags or prices, and I'm not claiming it says anything about them.

And if you like your facts in the most general clothes going, category theorists have this one too. Treat the moves as a monoid acting on the states, and a permanent reading as a set of states the moves never leave. In the topos of that monoid's actions, the truth value of "reads yes" at a state is the set of move sequences that carry it into the yes-states, and that topos is Boolean, so every such truth value is all or nothing, exactly when every move can be undone, which is when the monoid is a group. Morgan Rogers lays out that fact in his 2019 and 2021 papers on toposes of monoid actions. On a finite set, a move that merges nothing can be undone, so it's the chairs again, in a third language. Does it change anything here? No. It's the same fact wearing a bigger coat, and I mention it so nobody thinks the chairs are mine.

Going quantum doesn't get around the automata version either. Alex Brodsky and Nicholas Pippenger (2002) showed that a quantum finite automaton that measures only once, at the very end, accepts with bounded error exactly the group languages, the reshuffle kind. One that's allowed to measure after every letter, and to stop on

what it sees, does better: it accepts every piecewise testable language, and “has seen a b somewhere” is one of those. A measurement that can stop the machine is a record that stays, and it’s the extra thing that lets the bigger class in.

The third place is the closest, and it isn’t old at all. In 2023 Thomas Ouldridge and David Wolpert worked out the thermodynamics of a finite automaton built the way real computers are: the same physical step run again and again, each time touching only the part of the machine that changes. They sorted languages into two kinds by the smallest automaton that recognises each one. If that automaton has a letter that sends two states to one, every automaton for the language has such a letter too, and recognising the language on a device like that produces entropy, waste no cleverer design gets back, unless the inputs are so finely arranged that only one of the merged states ever actually turns up. If the smallest automaton only reshuffles, the language can be recognised with no entropy produced at all. One of their worked examples is the strings with no b in them, which is my flag turned inside out: the runs that raise it are exactly the strings that have seen a b . So the chairs, a real price on them, and the escape when you already know which chair you’re in are all in their paper, three years before this book, with the price derived from the physics of the device rather than assumed.

There’s a fourth neighbour, more a cousin than a relative, because it speaks another language. Constructor theory is David Deutsch’s and Chiara Marletto’s programme for stating physics as which transformations a device that runs in a cycle can and can’t bring about, and in it a task is irreversible when it’s possible and its inverse isn’t. Marletto’s version of thermodynamics (posted in 2016) leans on exactly that: tasks that are possible while their transposes aren’t. Marletto, Vedral and coauthors (2022) made an irreversible task out of the quantum homogeniser, a machine of reservoir qubits that interact by partial swaps, and ran it with photons. Maria Violaris and Marletto (2022) then pointed out that the same machine, run in a cycle, can turn a pure qubit into a mixed one but can’t turn a mixed qubit into a pure one, even though every interaction in it is reversible, and argued that this suggests erasure may cost something beyond Landauer’s bound. The chairs are a classical, finite, exact cousin of that sentence: a set of states a step can’t leave, a task that’s possible, and no way back. Theirs is quantum and approximate and aimed at physics’ price; mine is counted, and assumes its price. Deutsch and Marletto’s paper on information (2015) has a neighbour of the chairs argument in its section 8.6: measuring one observable disturbs another irreducibly, and nothing can undo the disturbance afterwards while the outcome stays in an information variable. That’s a second, theory-independent way of saying that a record that stays makes the step irreversible.

So what I can claim here is narrower again, and I’d rather say exactly how. Is even the pricing from theirs? In physics, yes: a merge the language can’t avoid costs entropy, and they derived that. What I didn’t find in their paper is the price as a named rule about a move, one a machine charges on a ledger whether or not anybody’s physics is

right (put a price on merging and the flip has to pay it, which is the next subsection), and what I get out of that rule: the toll, as a theorem about the ledger, the merge-or-revoke split for a single move, the exact match between “forced to be priced” and “merges,” the squeeze bound counted in whole marks and its entropy form, a built machine walking out of each door, and all of it checked. The core move is older than the computer, it had two homes before it got to mine, and its physical price had a third.

4.2 Now charge for forgetting, and the toll falls out

Here’s the step I’d been circling since the kindergarten buckets. In the last section I told you Bennett’s rule as he states it: what has to cost is erasing a bit or merging two paths a computation could have taken. I didn’t ask you to believe it. I’m not going to ask you now either. I’m going to write it down as a rule you can accept or refuse, in the smallest form that does any work: *any move that forgets something costs at least one mark*. Any move that is not injective has cost at least one. That’s it. Note what it does not say. It says nothing about certifying. It says nothing about flags. It’s a price on forgetting and only on forgetting.

Put the two together. The room says: on a finite machine with a permanent flag, the move that raises the flag forgets something. The price says: a move that forgets something costs at least one. So the move that raises the flag costs at least one. That is the toll, the rule I handed you without a reason, and now it has one. The proof is two lines: the first theorem says the flipping move merges, the premise says merges are priced, done. I’ll keep saying “premise,” because that’s what it is, and because the next section of this book is about what happened to some very smart people who put a price where a premise goes and then treated it as a law of nature, and about half of this one is me trying not to do that.

Premise 4.2 (Merge pricing). Every move that merges costs at least one: for all i , if i is not injective then $\text{cost}(i) \geq 1$. (Checked name: `merging_steps_priced`.)

Theorem 4.2 (The toll from finiteness, permanence, and merge pricing). *On a finite machine with a permanent reading whose cost satisfies merge pricing, every move that turns the reading from no to yes costs at least one. That is, the machine with that cost is a certification system in the sense of Definition 3.1, and the universal floor (Theorem 3.1) applies to it.*

Proof. A move that flips the reading merges, by Theorem 4.1; merge pricing charges it at least one. □

Checked as `a2_from_merging_price_and_permanence`; the packaging of such a machine as a certification system is `certification_system_from_merging_price`, and the run-level floor for it is `permanent_certification_trace_floor` (`PermanentCertification.v`).

One more way to see the same thing, because it answers a rival I'd otherwise owe you. The rival says: don't price certification, price *erasure*, and keep an honest flag that reports when something was erased. Fine; but a system can set its certificate at cost zero with its erasure flag reporting nothing, exactly because the certifying move merged two states and the flag didn't say so (`commit_without_erasure_system_is_not_honest`). Make the erasure flag honest, so that it reports every move that forgets, and on a finite machine with a flag that stays up the erasure law sends the same bill the toll does (`honest_erasure_accounting_implies_a2`). The toll isn't a rival to honest erasure accounting. It's what honest erasure accounting says about the certifying move.

So, is the toll arbitrary? Here's the honest version. The toll is a consequence of three things: a finite memory, a flag that stays up, and a charge on forgetting. If you accept all three, you don't get to refuse the toll; it's forced. If you refuse any one of them, the toll goes back to being my preference. So the question "did I have to pick it" has turned into three smaller questions, which is what progress usually looks like: not an answer, but a better set of things to argue about. Let me take the three doors in order, and for each one, build the thing that walks out through it, because a premise you can't see the counterexample to is a premise you're taking on faith.

4.3 Door one: keep everything, and nothing ever has to merge

Take a strip of paper that never runs out of boxes. The machine's whole state is the strip. Each move writes a 1 in the next empty box, and the flag is up as soon as any box has a 1 in it. Check the three premises. Permanent: once there's a 1, there's always a 1. Flip: the empty strip reads NO, and after one move it reads YES. Forgets nothing: every move can be undone by erasing the last 1, so two different strips never become the same strip. So here is a machine that raises a flag, keeps it up, and never merges anything. The room argument has no grip on it, and the toll doesn't follow. The checked version is `unbounded_history_escapes`, and its state is a list of bits of any length, which is the formal way of saying the strip has no end.

If the endless strip feels like a cheat, it's a famous one. David Hilbert, in a lecture in January 1924, told a story about a hotel with infinitely many rooms, every one of them taken, where a new guest still gets a room: everybody moves from room n to room $n + 1$, and room 1 comes free. That's the chairs argument losing its grip, told as a joke about a front desk. And here's the part worth checking: Hilbert never published it. The hotel became famous because George Gamow put it in a popular book in 1947, and for a long time it wasn't clear Hilbert had told it at all, until the historian Helge Kragh traced it, in 2014, back to that 1924 lecture. A story everybody repeated, and it turned out to be true, which is rarer than it should be. Bring it back to the table. In the room, the person from the door had nowhere to sit because the chairs ran out. In Hilbert's hotel they never do, so everybody shifts one over and nobody has to share. The strip machine is that hotel. It doesn't change the argument; it shows you the argument was about running out of chairs all along.

What walked out the door is the finiteness premise. Now cut the strip to eight boxes and run it for real: eight boxes, nine moves. On the ninth move there's nowhere to write. To keep going you have to reuse a box, and the moment you reuse a box two different histories end on the same strip, and you're back in the room with a shared chair. This is exactly the move Bennett made in 1973, when he showed that any computation can be done without forgetting if you let it keep a copy of everything it did; Landauer, back in 1961, had already written that for a program that doesn't stop, the space to hold every intermediate result "cannot be there." The history escape is real. It costs you a memory that never fills. Every machine you can buy has a memory that fills.

One thing I have to say plainly, because the file is more careful than I'd like to be. The theorem shows the strip machine has the other two properties and forgets nothing. It does not contain a sentence saying "and this machine is infinite"; that's something you and I can see about a list of unbounded length, but no Coq statement says it. And it contains no cost at all, so "certifies for free" is an inference I'm drawing (a move that forgets nothing is allowed to cost zero under the price premise), not something written in the theorem. I'd rather mark where the words outrun the file than let them run.

4.4 Door two: let the flag come back down

Take one coin. The rule is: turn it over. Heads is YES, tails is NO. Every move is undoable (turn it over again), so nothing is forgotten. Tails goes to heads: that's a flip, NO to YES. And heads goes to tails: the same move takes the flag back down somewhere else. Two states, finite as it gets, and the move that raises the flag forgets nothing, so a schedule that charges only for forgetting is allowed to charge it nothing. The file states the one move's bijection and its on-then-off behaviour (`revocable_certificate_escapes`); that the coin is finite and that the price may be zero are read off the picture, not written in the theorem. The thing that walked out the door is permanence.

Now here's a fact that I think is sharper than the premise it replaces, and it was the thing in this section that surprised me. Drop permanence entirely. Keep only finiteness. Take any move that raises the flag anywhere. Then one of two things is true: the move forgets something, or the same move lowers the flag somewhere else. There's no third option, and the proof below is a search: go through the finitely many YES states looking for one this move turns to NO. Find one, and that's the revocation; find none, and the move is permanent at least for itself, which is all the room argument needs. So a flag that goes up for free is a flag the same move can take back. A flag that nothing takes back is paid for. You can't have a free raise that stays raised, not on any finite machine, not with any price schedule that charges for forgetting. That's a narrower sentence than "every certificate costs," and I'll take narrow and true over broad and mine.

Theorem 4.3 (Merge or revoke). *On a finite machine (no permanence assumed), if $\text{cert}(s) = \text{no}$ and $\text{cert}(\text{step}(s, i)) = \text{yes}$, then either i merges, or there is a state t with $\text{cert}(t) = \text{yes}$ and $\text{cert}(\text{step}(t, i)) = \text{no}$.*

Proof. Check every yes-state on the finite list for one that i sends to no. If there is one, that is the second alternative. If there is none, then i keeps every yes-state at yes, which is permanence for the single move i ; the argument of Theorem 4.1 uses permanence only for the move it is about, so it applies, and i merges. $\square$

Checked as `flip_merges_or_revokes`, with the single-move form of the room argument as `permanent_at_flip_is_not_injective` (`PermanentRecordPricing.v`).

I'll give you the three-card version too, because it shows the converse fails and I don't want you to think the picture is tidier than it is. Three cards, T0, T1, T2, with a star on T1. The rule: T0 goes to T1, T1 goes to T2, T2 goes to T1. T0 gains the star (a flip). T1 loses it (a revocation). T0 and T2 both land on T1 (a merge). So this move is priced, by the forgetting rule, and yet it writes nothing permanent; it takes a star away in the same breath. "The priced moves are exactly the permanent records" is false, and the file proves it false with these three cards. What's true is one direction: every permanent raise is a priced move. The other direction has the three cards standing in it. And "priced" here has an exact meaning worth one sentence: a move is *forced to be priced* when every schedule that charges for forgetting charges it, and that holds exactly when the move forgets something, with no finiteness needed at all, only a way to tell two move names apart.

Definition 4.3 (Forced to be priced). A move i is *forced to be priced* when every cost function satisfying merge pricing (Premise 4.2) charges i at least one.

Defined as `forced_priced` (`PermanentRecordPricing.v`).

Theorem 4.4 (Forced price is merging). *If any two move names can be decided equal or different, then a move is forced to be priced exactly when it merges. No finiteness is needed.*

Proof. If i merges, merge pricing charges it at least one, whatever the cost function. If i does not merge, define the cost that is 0 on i and 1 on every other move (this is where telling move names apart is used). It satisfies merge pricing, since every merging move is some move other than i and costs one, and it charges i nothing, so i is not forced. $\square$

Checked as `forced_priced_iff_merges` (`PermanentRecordPricing.v`).

Corollary 4.5 (Permanent records are forced). *On a finite machine, a move that never turns the reading off and turns it on somewhere is forced to be priced. (By Theorem 4.3 it merges; by Theorem 4.4 it is forced.) The converse fails: the three cards are forced, because they merge, and they revoke.*

Checked as `forced_priced_iff_merges, permanent_record_write_is_forced_priced`, and the three-card counterexample `forced_price_without_permanent_record` (`PermanentRecordPricing.v`).

4.5 Door three: forget for free

The third door is the easiest to describe and the hardest to argue about, so I'll describe it and then spend the rest of the section on the argument. Take two sheets of paper, one blank and one stamped. One move: stamp whatever you're handed. Finite, two states. Permanent: stamped stays stamped. Flip: blank goes to stamped. Forgets: both sheets end up as the same stamped sheet. Everything the room argument needs is here, and the move merges. Now set the price of that move to zero. The toll fails, and you can watch it fail: the flag went up for nothing. That's `free_merge_escapes`, and what walked out the door is the charge on forgetting. (The same file shows that a price of one on that move satisfies all three premises, `priced_reset_satisfies_premises`, so the premises can hold together; door three is about whether they must.) Nothing in counting stops you from setting that price to zero. Only a claim about the world does. The nearest such claim is Landauer's, and I have to be exact about the gap between the two: the premise as written charges *every* forgetting move, whether or not you knew which state you were in, and Landauer's own heat, as the coin under the cup below will show, is owed only for what you didn't know. So the premise is Landauer's claim taken in its worst case over what the machine might be holding, and that's the version the counting needs.

Here is what Landauer said, in 1961, as exactly as I can put it after reading the paper rather than the summaries. A logical operation with no single-valued inverse (a step that forgets, in this book's words) must produce heat, "typically of the order of kT " per operation, where k is Boltzmann's constant and T is the temperature; and for the specific case of resetting one bit whose value you don't know, the minimum is kT times the natural log of 2, about 0.69 kT . I want you to hear two things in that. First, "of the order of": he was not handing out a number to three decimals. Second, and this surprised me, he did not measure anything. The paper is argument, with models. He writes that he can "show, or at least very strongly suggest" the conclusion, and that his reasoning gives "no guarantee that this minimum is in fact achievable." That's the founding paper of the whole subject, and it's a careful man saying "I think this is right and here's why." I like him more for it.

Was it measured, eventually? Yes, and only recently. In 2012 a group at the École Normale Supérieure in Lyon, Antoine Bérut and Sergio Ciliberto's lab working with

Eric Lutz, trapped a single glass bead, two millionths of a metre across, in a double well made of laser light, and used the two wells as the two values of a bit. Then they erased the bit, by lowering the barrier, tilting the trap, and raising it again, many times over, slower and slower, and measured the heat each time. The average heat came down toward Landauer's $kT \ln 2$ and sat there as the erasure got slower. On average. Single runs went under it; the bound is about the average, not any one erasure, and if you only succeed in erasing half the time you owe nothing, which the paper's formula shows and which I think is the most honest line in it. Others have done it since with different devices. So: predicted in 1961 by argument, seen in a laboratory fifty-one years later.

And is it settled? No, and I'd be lying by omission if I left that out, because the people who doubt it are not cranks. John Norton, a philosopher of physics, has argued since 2005 that the usual proofs of the principle quietly assume what they're trying to prove, and that the kinds of process they assume, combined, could be made to break the very law the principle is supposed to guard. Charles Bennett, who did more than anyone to build the subject, rated Earman and Norton's 1999 form of this objection the one of "greatest merit," and then disagreed with it. Others replied to Norton; Norton replied back. The Stanford Encyclopedia entry on the topic, written by a philosopher of physics who works on exactly this, ends without a verdict. I've read what I could of both sides and I'm not qualified to referee them, and I'm not going to pretend the laboratory settles a question about what the principle means in general, because measuring one bead doesn't do that.

So here's what I did with it, and I think it's the only honest thing to do. The charge on forgetting is written into the checked work as a *named premise*. It's a definition the theorems take as an assumption, and nowhere is it proved as a general fact, because it isn't provable from counting; for particular machines with particular price lists it's checked (the stamp at cost one, above), which is a different thing. If Norton turns out to be right, every theorem in this section still holds; they just have one premise fewer that the world agrees to. If Landauer's side turns out to be right, the premise is physics. Either way, nothing in this section says a device does anything until someone measures the device. The counting is mine to prove. The physics is a door I've labelled and left open.

Labelled, though, isn't the same as unplaced, and a hedge isn't a position, so here's mine. Earman and Norton's objection is a dilemma. Either an information-theoretic exorcism of the demon assumes the second law, which they call the *sound* horn, and then it adds nothing the second law didn't already give; or it rests on a new principle tying information to entropy, the *profound* horn, and nobody has given that principle independent grounds. A named premise is the sound horn, on purpose. That's where Bennett ended up in 2003, and Norton's "Too Good to Be True" (written in 2025, in print in 2026) says the careful literature almost always accepts that horn, and argues that whatever succeeds there is ordinary thermodynamics relabelled. I'll take

that. Ordinary thermodynamics, relabelled onto a ledger, is all this section asks of the physics. And on the counting I'm on Norton's side outright. His 2018 paper "Maxwell's Demon Does Not Compute" exorcises the demon with Liouville's theorem and nothing else, no information and no computation: dynamics that keeps every patch of states the same size can't crowd a big set of states into a small one. The counting core of this section, that a step losing nothing can't carry a no-state into a set that stays up, is that point in finite, discrete form. And if you want the best case for the premise itself, the one I found is James Ladyman, Stuart Presnell, Anthony Short and Berry Groisman (2007): from the Kelvin statement of the second law they derive that a logically irreversible operation can't be carried out thermodynamically reversibly by any of their L-machines. That's the merge-pricing premise with an argument under it, and if you want to argue with the premise, that's the paper to argue with.

One question a physicist asks before any of that, and it's a good one: the laws underneath never merge anything. Classical mechanics carries each exact state forward to exactly one next state and could carry it back again, and Liouville's theorem adds that a patch of states keeps its size as it moves; quantum mechanics is unitary, which is one-to-one too. So when this section says a step merged two states, where did the difference go? Out of the machine. The merge is in the machine's own description, the states its reading is about. The machine together with everything around it still runs one-to-one, and the distinction the machine forgot doesn't vanish; it gets pushed out into the surroundings, mostly as heat. The surroundings are door one's endless strip. That door is always open for the universe as a whole and shut for the machine, because the machine isn't the universe and its memory fills. Read that way, Landauer's premise is the claim that pushing a distinction out of a finite machine into its surroundings costs heat, and that's all it claims. Which also means the bill is heat *handed over*, not heat *wasted*. Takahiro Sagawa showed in 2014 that a step that forgets can, in principle, be run slowly enough that no entropy is produced overall: the heat leaves the machine, the room's entropy goes up by what the machine's went down, and nothing is lost that didn't have to be. So on the premise the forgetting step can't be free, but it doesn't have to be wasteful. When I call it a cost, I mean a transfer you can't avoid, not a loss you could have engineered away.

4.6 How much: the price grows with the squeeze

The toll says "at least one mark." That's a floor for raising one flag. The room argument says more if you let it. Say there are m people already inside and k more come in through the door in one move. (That's a different k from the one in Landauer's kT ; when the two meet, below, Boltzmann's constant gets a little B hung under it, k_B , so you can tell it from a head count.) All m plus k of them end up in m chairs. That's a squeeze, by a factor of $(m + k)$ over m , and if you accept a slightly stronger version of the forgetting price, that each mark pays for at most one *halving* of the number of distinguishable states, then the move costs at least the number of halvings in that squeeze, which is the base-two logarithm of $(m + k)/m$.

Back to door three's stamp, with four sheets this time: three blank, one stamped, and the stamp-everything move. Four states go to one. Before the move, to say which sheet you had took two yes-or-no questions (is it one of the first two? then which of those two?); after the move, it takes none. Two bits gone. Two halvings. So that move costs at least two marks, and two marks is enough: a cost of exactly two satisfies the halving rule (both facts are checked for these four sheets, `quad_stamp_needs_two` and `quad_stamp_cost_two_is_priced`). The general statement is $m + k$ at most 2 to the cost times m , and next to it in the file sits a logarithm form that rounds both logarithms up and is weaker for it (three states and one newcomer round to the same two bits on both sides, and the rounded form says nothing). The halving premise, like the one before it, is a definition used as an assumption. The units are halvings, not joules. To turn a halving into heat you multiply by $k_B T \ln 2$, and that multiplication is the physics, not the counting.

That little k_B deserves a short trip of its own, because it sits exactly where counting turns into physics, and its name is a small history lesson. Ludwig Boltzmann's grave in Vienna has $S = k \log W$ carved on it (that k is the same k_B , before anyone hung a B under it): entropy is a constant times the logarithm of the number of ways the parts could be arranged. The idea is his. The constant, and the equation written that way, came from Max Planck, in his work on radiation around 1900. Planck named k after Boltzmann, and in his Nobel lecture in 1920 he said that, as far as he knew, Boltzmann had never introduced it himself, because Boltzmann never thought anyone would try to measure it exactly. So the most famous constant in this corner of physics wears the name of a man who didn't write it down, as a gift from the man who did. Does that change the bill? No. But look at the job the constant does in that equation. Take it away and what's left is the logarithm of a count, which is a number of halvings, which is what the squeeze bound hands you. The constant is the exchange rate, and whether the world honours that rate for a merge is the premise, not a theorem.

Exactly, with one new piece of notation: for a list D of states, write $|D|$ for how many it has, and $\text{img}_i(D)$ for the number of *different* states the move i sends them to.

Premise 4.4 (Halving price). Each mark pays for at most one halving: for every move i and every list D of states with no repeats, $|D| \leq 2^{\text{cost}(i)} \cdot \text{img}_i(D)$. (Checked name: `compression_priced`. It implies merge pricing, since a merge sends two different states to one and so needs $\text{cost}(i) \geq 1$.)

There's a plainer way to read that premise, and it's worth having because it says what the premise is *about*. Look at one landing spot of move i and count how many states land there; call the biggest such count the move's worst pile-up. The premise holds exactly when every pile-up has at most $2^{\text{cost}(i)}$ states in it, which is to say the cost is at least the base-two logarithm of the worst pile-up. One direction: take D to be one whole pile, which lands on a single spot, and the premise says the pile has at most

$2^{\text{cost}(i)}$ members. The other: any list D splits by where its members land, each part fits inside one pile, so $|D|$ is at most the number of landing spots times the worst pile-up, which is the premise. So the halving price asks a move to pay for the worst it could ever do, whichever state it's actually handed. That's worst case on purpose, and it has a home in physics. In the branch of thermodynamics that prices one run that has to succeed every time, rather than an average over many, Philippe Faist, Frédéric Dupuis, Jonathan Oppenheim and Renato Renner worked out in 2015 the least work any logical process needs, and when every input might occur and no error is allowed, their bound comes to $k_B T \ln 2$ times exactly that logarithm. Take an AND gate as the example: three inputs land on 0, so it needs at least $\log_2 3$, about 1.6, halvings' worth of work (two marks, in whole numbers). The premise asks for that bound, counted in marks instead of joules, of a device that has to work whatever it's handed.

Theorem 4.6 (The squeeze bound). *On a finite machine with a permanent reading and a cost satisfying the halving price, let m be the number of states that read yes, and let a move i turn k different no-states to yes. Then*

$$m + k \leq 2^{\text{cost}(i)} \cdot m, \quad \text{equivalently} \quad \text{cost}(i) \geq \log_2 \frac{m + k}{m}.$$

Proof. Put the m yes-states and the k newcomers in one list; it has $m + k$ different members. By permanence every yes-state lands on a yes-state, and by hypothesis every newcomer does too, so all $m + k$ land among the m yes-states: $\text{img}_i \leq m$. The halving price gives $m + k \leq 2^{\text{cost}(i)} \cdot \text{img}_i \leq 2^{\text{cost}(i)} \cdot m$. $\square$

The four sheets are the case $m = 1, k = 3$, where the bound is reached. With $k = 1$ the bound says $\text{cost}(i) \geq 1$, which is the toll again (a2_from_compression_price_and_permanence). Checked as permanent_flips_compression_bound, with the rounded-logarithm form permanent_flips_log_bound and the counting-only cousin permanent_flips_collapse_at_least (at least k states collapse, with no price premise at all).

Four more facts about the halving price, all checked, and the last one is a limit. First, the plainer reading above is a theorem on a finite machine: the halving price holds exactly when no move has a landing spot with more than $2^{\text{cost}(i)}$ states landing on it (ent2_compression_priced_iff_fibres). Second, for the eight-state machine described later in this section that's a theorem about its own price list: no move has a larger pile-up than 2^{cost} , so the halving price holds, and the toll follows from it (ent2_frag_compression_priced, ent2_frag_toll_by_compression); and the price list is the least that meets the halving price, since any price list that does charges every move at least what this one charges (ent2_frag_cost_minimal). Third, the squeeze bound is met with equality there: the stamp turns four blank sheets on beside four already on, and $4 + 4 = 2^1 \cdot 4$ (ent2_frag_bound_attained). Fourth, and this is the limit, the small machine of Part IV as a whole does not meet

the halving price, whatever procedure you use to decide whether two of its states are equal, because a free DEC lands two different full states on one state (`ent2_small_not_compression_priced`; the pair is in Theorem 14.38). So on that machine the toll is the rule I chose, as I said it was, and now that's a theorem and not a remark (`CompressionSmall2.v`).

4.7 The same bill, written in bits, and when it's zero

There's one more way to say all this, and I include it because it's the way physicists say it and because it hands you the one case where the bill is exactly zero, which I find more instructive than any case where it isn't.

Claude Shannon, in 1948, gave a number to "how much you don't know": if something could be in any of n states and you have no idea which, your uncertainty is log base two of n bits, and that's the most it can be for n states. If you push your uncertainty through a step that forgets nothing, it stays the same; through a step that merges two states you might have been in, it drops. Now run the room. Before the move, you might be in any of the $m + k$ states in play. After it, you're in one of m . So your uncertainty afterward is at most $\log_2$ of m , whatever you believed before. If you started with no idea at all, every one of the $m + k$ states equally likely, your uncertainty was exactly $\log_2$ of $(m + k)$, and the move removed at least $\log_2$ of $(m + k)/m$ bits: the same number as the halving count, now in Shannon's units. Landauer's principle, in the form physicists write it, then says a step that removes that many bits from the state gives off at least $k_B T \ln 2$ of heat per bit. Put the two together and the flip gives off heat, and the exact statements below say how much and under which side conditions, one of which (the temperature has to be taken as zero or above) is easy to skip and shouldn't be.

Why does Shannon's number share a name with the physicists' entropy at all? There's a story about it. Shannon wanted to call it uncertainty, and John von Neumann told him to call it entropy instead, for two reasons: the same formula already lived in statistical mechanics under that name, and nobody really knows what entropy is, so in an argument he'd always have the upper hand. It's a great story. It comes from Myron Tribus, who said Shannon told it to him in 1961, and who printed it in *Scientific American* in 1971. Then in 1982 an interviewer, Robert Price, asked Shannon about it, and Shannon said he didn't think von Neumann had told him any such thing. So it's a legend with one witness, and the person the witness quoted doesn't remember saying it. What doesn't hang on the story: the two formulas really do have the same shape, and it still takes Landauer's premise, a named assumption, to turn a count of bits into heat.

Exactly, with the notation built first. A *spread* (probabilists say distribution) is a way of giving each state a chance $p(x) \geq 0$, with the chances adding to 1. Its Shannon

entropy, in bits, is

$$H(p) = - \sum_{x: p(x) > 0} p(x) \log_2 p(x),$$

where the big sigma means “add up, over every state x with a chance above zero,” and each term is a state’s chance times the bits it would take to single that state out. That comes to $\log_2 n$ when n states share the chance evenly and less when the chance is lopsided. Pushing a spread through a move i gives the spread of where you land: $q(y) = \sum_{x: \text{step}(x,i)=y} p(x)$. The bits the move *removes* are $H(p) - H(q)$.

Start with the half of the claim that costs nothing, Bennett’s half. A move that forgets nothing sends each state somewhere different, so the chances just get carried to new addresses, one for one. Nothing piles up and nothing spreads out. The uncertainty is the same number before and after.

Theorem 4.7 (A move that forgets nothing removes no bits). *Let the states be finite, listed once each, and let f send states to states with $f(a) = f(b)$ only when $a = b$. Then for every assignment of chances p , the pushed spread q has $H(q) = H(p)$.*

Proof. An injective map on a finite list of distinct states is a reshuffle of that list: every state is hit exactly once. So $q(f(x)) = p(x)$ for every x , and the sum defining $H(q)$ is the sum defining $H(p)$ with its terms in a different order. $\square$

Checked as `step_entropy_invariant_if_injective` (PermanentCertificationEntropy.v). That’s why a price on removed bits can leave every reversible move free, which is the shape the toll needs: the bill lands only where something was forgotten.

Theorem 4.8 (Entropy after a permanent flip). *On a finite machine with a permanent reading, let a move i turn $k \geq 1$ different no-states yes, beside m yes-states. For any spread p whose chance lies on those $m + k$ states, the spread after the move has $H(q) \leq \log_2 m$. For the even spread on the $m + k$ states, the move removes at least $\log_2 \frac{m+k}{m}$ bits.*

Proof. After the move, all the chance sits on the m yes-states (permanence for the old ones, the hypothesis for the new). There is at least one yes-state: the image of a state that flips. Thus $m \geq 1$. Write $q_1, \dots, q_r$ for the positive output probabilities, where $r \leq m$ and their sum is 1.

For $y > 0$, $\ln y \leq y - 1$: the derivative of $y - 1 - \ln y$ is $1 - 1/y$, so its minimum is 0 at $y = 1$. Apply this with $y = 1/(mq_j)$, multiply by q_j , and add:

$$-\ln m - \sum_{j=1}^r q_j \ln q_j \leq \sum_{j=1}^r (1/m - q_j) = r/m - 1 \leq 0.$$

Dividing by $\ln 2 > 0$ gives $H(q) \leq \log_2 m$. The even input spread has $m + k$ probabilities each $1/(m + k)$, hence $H(p) = \log_2(m + k)$. Subtracting the output bound gives

the claimed loss $\log_2((m+k)/m)$. □

Checked as `permanent_step_entropy_ceiling` and, for the even spread, `permanent_flip_uniform_entropy_drop` (`PermanentCertificationEntropy.v`).

Premise 4.5 (Landauer’s principle, as a premise). A step that lowers the entropy of the machine’s state by ΔH bits, at temperature T , gives off heat $Q \geq k_B T \ln 2 \cdot \Delta H$, where k_B is Boltzmann’s constant. (Checked name: `landauer_heat`, with $k_B T$ as one variable.)

Corollary 4.9 (Heat floor of a permanent flip). *Under Premise 4.5, with $k_B T \geq 0$, the flip of Theorem 4.8 from the even spread gives off at least $k_B T \ln \frac{m+k}{m}$; with $k_B T > 0$ that is strictly positive.*

Proof. Theorem 4.8 gives an entropy loss of at least $\log_2((m+k)/m)$ bits. Multiply by the nonnegative factor $k_B T \ln 2$ and use Premise 4.5. The identity $\ln 2 \log_2 x = \ln x$ gives the displayed floor. Here $m > 0$ and $k > 0$, so $(m+k)/m > 1$ and its logarithm is positive; if $k_B T > 0$, the heat floor is strictly positive too. □

Checked as `permanent_step_entropy_ceiling`, `permanent_flip_uniform_entropy_drop`, `entropy_le_log_support`, `permanent_flip_heat_floor`, and `permanent_flip_heat_positive` (`PermanentCertificationEntropy.v`).

How big is that? Take the case where the squeeze is two, one state flipping in beside one that was already yes, so $m = k = 1$ and the floor is $k_B T \ln 2$. At room temperature, about 300 kelvin, that’s roughly three zeptojoules: three thousandths of a billionth of a billionth of a joule. And that’s the *biggest* bill a single flip gets from this corollary, not the smallest. With $k = 1$ the floor is $k_B T \ln \frac{m+1}{m}$, and it shrinks as more states already read yes: with a thousand of them it’s about a thousandth of $k_B T$, some seven hundred times less than the case of two. So one mark is not one halving. It’s one halving exactly when the squeeze is two; for any thinner squeeze the physics asks for less than a halving, and the toll still charges a whole mark, because marks come in whole numbers (that’s the rounding the entropy road to the toll leans on, a little further down). Whatever the size, it’s a bill only if you grant the premise; nothing on this page measured anything. And even granted, it’s a bill nobody will ever see on a meter: real logic gates spend many thousands of times more than this on every switch, even the frugal ones. The point of the number isn’t that certifying is expensive. It’s that, on the premise, it isn’t zero.

Physicists have a name for a step with no way back, and that logarithm has turned up in their work before. Yûto Murashita, Ken Funo and Masahito Ueda (2014) studied processes where some histories of the reversed process have no forward history at all. They call that *absolute irreversibility*, and they showed that the entropy such a process

produces averages at least $-\ln(1 - \lambda)$, where λ is the chance of those orphaned reverse histories. Their first example is a gas let out of a box that held a fraction l of the room, where the bound comes to $\ln(1/l)$. Now picture the even spread of my flip played backwards: the chance sitting on the m yes-states spreading back out over all $m + k$, a gas let out of a box holding $m/(m + k)$ of the room. Their bound for that is $\ln \frac{m+k}{m}$, the same logarithm as the heat floor, in units of $k_B T$. So is my squeeze a special case of their bound? No, and the reason is worth a minute, because the match is a good trap. The even spread gives a real chance to every state the flip starts from, so no forward history is missing: the commit itself isn't absolutely irreversible at all, and their λ for it is zero. The heat floor is ordinary Landauer heat, the bill for squeezing a spread over $m + k$ states into m of them, the same kind of bill as any reset into a smaller set. The two logarithms match because a squeeze's Landauer heat equals the entropy produced by the free expansion that undoes it, and that free expansion, the flip played backwards, is the process their bound is about. A mirror, not a family tree. Their idea does reach the commit, just somewhere else: on a device whose rates never change, committing from a state it knows, which turns up at the end of this section. Gonzalo Manzano, Gülce Kardeş, Édgar Roldán and David Wolpert (2024) carried absolute irreversibility over to computations, with one-way transitions and machines that stop at random times, worked on finite automata, and the same $-\ln(1 - \lambda)$ term sits in their second-law bounds.

The even spread is the easy case, and you might worry the heat depends on picking it. It doesn't. Any spread that gives a real chance to every yes-state and to the state that flips will do.

Corollary 4.10 (Positive heat for any full spread). *On a finite machine with a permanent reading, let move i turn a no-state s yes. Take any spread p that gives positive chance to s and to every yes-state. Under Premise 4.5 with $k_B T > 0$, the heat the move gives off is strictly positive.*

Proof. After the move, s and the yes-states all land among the yes-states, and there is one more of them than there are yes-states, so two distinct states with positive chance land on the same state. That merge removes a positive number of bits (Theorem 4.13 below), and a positive temperature times a positive number of bits is positive heat. $\square$

Checked as `permanent_flip_full_support_entropy_drop_positive` and `permanent_flip_full_support_heat_positive` (PermanentCertificationEntropy.v).

There's a third road to the toll here too, and it's the same shape as the merge road, with bits in place of marks for forgetting.

Definition 4.6 (Entropy-priced). On a finite machine, a cost is *entropy-priced* when, for

every move i and every spread p , the cost of i is at least the bits the move removes: $\text{cost}(i) \geq H(p) - H(q)$, where q is p pushed through i .

Theorem 4.11 (The toll from entropy pricing). *On a finite machine with a permanent reading and an entropy-priced cost, every move that turns the reading from no to yes costs at least one. So the machine with that cost is a certification system (Definition 3.1), and every run from a no-state to a yes-state costs at least one.*

Proof. Let move i turn s yes, and let m be the number of yes-states; $m \geq 1$, since $\text{step}(s, i)$ is one. Take the even spread on the m yes-states and s . By Theorem 4.8 with $k = 1$, the move removes at least $\log_2 \frac{m+1}{m} > 0$ bits, so the cost of i is a whole number above a positive number, which makes it at least one. The run-level floor is the universal floor (Theorem 3.1) applied to that system. $\square$

Checked as `a2_from_entropy_price_and_permanence`, with the packaging `certification_system_from_entropy_price` and the run-level floor `entropy_priced_trace_floor` (`PermanentCertificationEntropy.v`).

Now the zero. Put a coin under a cup without looking. Rule: set it heads up. You might have had tails; now you have heads; one bit of what you didn't know is gone, and that's the bit the principle charges for. Now lift the cup, see heads, put the cup back, and apply the rule again. Nothing you didn't know changed. No bit was lost. And the principle, applied to a state you already knew, demands nothing: the premise holds exactly when the heat is zero or more, which is to say it sets no floor at all. The principle charges for *uncertainty removed*, not for the shape of the move. This is the generalized form of Landauer's principle, worked out by Maroney in 2009 and Sagawa in 2014, and Landauer himself had the seed of it in 1961: a row of bits all known to be one can be reset with no entropy change and no heat.

Theorem 4.12 (A known state owes nothing). *On a finite machine, if the spread puts all its chance on one state s , then any move removes zero bits, and Premise 4.5 for that move holds exactly when $Q \geq 0$.*

Proof. A spread on one state has entropy 0, and so does its image, which is again on one state. So $\Delta H = 0$ and the premise reads $Q \geq 0$. $\square$

Checked as `known_state_step_removes_no_entropy` and `known_state_flip_forces_no_heat` (`PermanentCertificationEntropy.v`).

Be careful with that one, because it's easy to overread. "A known state owes nothing" does not mean a stamping machine run on a known-blank sheet gives off no heat. A fixed device that merges states dissipates on known input too; Bennett calls that wasteful, because entropy goes out into the room with nothing taken from the data.

What the theorem says is only that the *premise* sets no floor there. The floor comes from what you didn't know. That's a sentence about the bill, not about the device. And notice that the toll still charges the known-state stamp one mark, because the toll is a rule about the *move*, and the move doesn't know what the observer knows. That's the worst-case gap from door three showing up again, and it's the price of having a rule a machine can follow without asking who's watching.

There's a second piece of fine print on that theorem, and it's about who works the controls. The generalized principle can charge a known state nothing because it lets the step be driven by a protocol that changes in time: lower this barrier, tilt that well, raise the barrier again, each at the right moment. Something outside the step keeps that time, an outside clock or an outside hand, and the premise doesn't bill it. Take the hand away and the bill changes. Paul Riechers, Alexander Boyd, Gregory Wimsatt and James Crutchfield (2020) worked out what happens when the control is *time-symmetric*, the same played forwards or backwards, which covers a clocked pulse and any device whose rates never change. Then every move that doesn't come back to where it started when you apply it twice, which they call *nonreciprocal*, costs at least $k_B T \ln(1/\epsilon)$ on top of the change in free energy, to run with error at most ϵ . A flip that stays up is always nonreciprocal: apply it twice from the no-state and you're still at yes. So under that kind of control the known-state stamp isn't free after all, and the more reliable you want it, the more it costs. Their class is bigger than mine, though. Erasure is nonreciprocal, and so is a counter of three or more values that steps forward and wraps around, which merges nothing, so their bound isn't about records; it's about what the controls are allowed to do. So read Theorem 4.12 with its control written in: a known state owes nothing on Landauer's premise when the step can be driven by a protocol timed from outside it. The same goes for Theorem 4.4. The moves every merge-pricing schedule must charge are exactly the merging moves, and that's true against merge pricing; measured against time-symmetric control, the forced moves would be the nonreciprocal ones, the wrapping counter included. Which moves the physics forces a price on depends on what the controls can do.

Now follow that all the way down, because it reaches further than one stamp. A machine that runs a fixed rulebook from a start it knows is in a known state at every step: no coin under any cup, ever. So on the premise as Maroney and Sagawa state it, with the controls timed from outside, the step where such a run raises its flag owes no heat at all. Not "might owe nothing." Owes nothing. And spreading the start out over many inputs doesn't rescue much. The merge the chairs guarantee is between the state that flips and some state already reading yes. Run the step only when it's known that the flag is down, which is the ordinary first raise, and the spread puts no chance on any state already reading yes, so that particular merge removes nothing: Theorem 4.13's strict term needs *both* states to carry chance, and the heat for a full spread (Corollary 4.10) needs chance on every yes-state, which is exactly what knowing the flag is down rules out. Whether the step also merges states that do carry

chance depends on the step, and nothing here forces it to.

So here's what this section actually earns. A device that has to raise a flag and keep it up correctly *from every state it might be in*, including states where it can't tell whether the flag is already up, can't do it for free, on Landauer's premise; that's the design envelope, the worst case. A particular run from a start it knows can, on the same premise, as long as something outside the step does the timing. What stays forced in every case is the toll as a rule about the move, because the rule can't ask what the run knows. Wherever this book says that settling something for good can't be free if Landauer's right, it means the device sentence, not the run sentence.

Here's the general case, and it's the one that shows you where the bits go. Split the bill state by state. Each state x you might be in carries its chance $p(x)$ along to where it lands, $\text{step}(x, i)$, and finds whatever other chance landed there too. If nobody else landed there, it owes nothing. If others did, it owes the difference between how crowded its landing spot is and how much of the crowd it brought.

Theorem 4.13 (The bits removed, one term per state). *On a finite machine, for any move i and any chances $p(x) \geq 0$, with q the pushed spread,*

$$H(p) - H(q) = \sum_{x: p(x) > 0} p(x) (\log_2 q(\text{step}(x, i)) - \log_2 p(x)).$$

Every term is zero or more, so a move never adds bits. If two different states $x \neq x'$, both with positive chance, land on the same state, the sum is strictly positive.

Proof. Group the sum for $H(q)$ by where things land: the chance $q(y)$ at a landing spot is the sum of the $p(x)$ that land there, so $H(q) = -\sum_{x: p(x) > 0} p(x) \log_2 q(\text{step}(x, i))$. Subtract from $H(p)$ term by term. Each term is non-negative because the landing spot holds at least what x brought, $q(\text{step}(x, i)) \geq p(x)$, and the logarithm keeps order. If x and x' share a landing spot, that spot holds at least $p(x) + p(x') > p(x)$, so the term for x is strictly positive. $\square$

Checked as `entropy_drop_as_point_sum`, `entropy_drop_nonneg`, and `entropy_drop_pos_of_support_merge` (PermanentCertificationEntropy.v).

4.8 What you're entitled to, and what you're not

Let me total it: what have you earned the right to conclude, and what is still a door?

You've earned this: on any machine with finitely many states, a move that raises a flag either forgets something or takes a flag down somewhere else. Pure counting, no physics, no premise beyond finiteness, and the half of it where nothing comes down is the four chairs. You've earned this: on a finite machine, if a flag, once up, stays up, then raising it forgets something, and a move that raises k states collapses at least k of them. Counting again. And you've earned this: if forgetting is charged for, the toll

follows, and if forgetting is charged per halving, the size of the toll follows too; each of those derivations hands back a system with the five parts and the toll, so Section 3's floor applies to it as it stands. That's the rule from Section 3, derived rather than declared, with every premise named, for any finite machine with a flag that stays up.

Here's what you haven't earned, and I'd rather list it than have you assume it. You haven't earned that forgetting costs anything in the world; that's Landauer, it's measured for beads and argued over for the general case, and in the checked work it's an assumption with a name. And even granting it, Landauer charges a run that knows where it is nothing at all (the lifted cup), so what the physics forces is a floor on a device that has to work whatever it's holding, not on any one run. You haven't earned that any machine you'd actually want to build is of this kind, either, and there are two easy ways to miss. A machine that keeps its running total inside its own state, as a number with no ceiling, has infinitely many states, so the finite premise fails for it as a whole. And you can't patch that by giving the total a ceiling. A total that only climbs, held in finite memory, eventually either stops at its top, and then two different histories leave the same top value, which is a merge, or rolls back over to zero, and then it doesn't only climb. So any running total a finite machine keeps for itself has a horizon: it's honest up to its width and not past it. And a price list that charges the step that raises the flag, but lets some other step merge two states for nothing, raises the flag honestly and still doesn't charge for forgetting in general, so the merge-pricing premise fails for it. Either one puts a machine outside these theorems as stated, though a finite piece of it can still sit inside. The machine I build later misses in both ways. It (Section 14) keeps counters with no ceiling, so it's infinite outright, and some of its ordinary steps, counting a counter down among them, can land two different states on one for free. So for it the toll is a rule I chose; the chairs force it only on finite machines that price every merge, and it isn't one.

A finite piece can still sit inside, and here's the smallest one I could build where every premise is a theorem rather than a hypothesis. Four slots times a flag, eight states. Three moves: stamp (set the flag, go to the next slot, cost one), jump to slot a (cost two), next slot (cost zero). It's finite (`fin_finite`) and permanent (`fin_permanent`); the next-slot move forgets nothing (`fnext_injective`); the stamp and the jump merge (`fcertify_merges`, `fjump_merges`); merges are priced (`fin_merging_priced`), and priced per halving (`fin_compression_priced`). So this section's two conclusions hold on it outright (`fin_a2_from_merging_price`, `fin_a2_from_compression_price`, `FiniteCertMachine.v`). The counting lemma underneath, that a duplicate-free list with at most K preimages per image has at most K times as many members as images, is proved generically (`fiber_bound_compression`). The jump's price of two is a choice, not something the chairs force; any price of at least one would do. One more fact about it, also checked: each move's price is exactly the number of halvings in its largest squeeze. For a move that costs c , no state is landed on by more than 2^c states, and some state is landed on by exactly that many

(stamp squeezes at most two states into one, jump to a slot squeezes four, and next-slot squeezes nothing).

The machine Part IV builds runs this eight-state machine. Seen through a window that shows only a line number, read modulo four as the slot, and a flag, a short program of that machine does what the eight-state machine does, one move at a time, and pays exactly what the eight-state machine charges (Section 14.17 has the construction and the exact statements). The machine as a whole is still not merge-priced: some of its free steps merge states, so the finite piece is a piece, and the machine around it keeps the choice the chairs don't force.

And you haven't earned anything about *what* the flag means. Four chairs don't know what they're certifying. The bill is for raising a flag that stays up. Whether raising that flag ever amounts to standing behind a structural fact, the "vouching" I keep promising and keep not delivering, is still the same open question it was in Section 3, and it has moved exactly this far: if that flag stays up, it was paid for.

One last thing, because it should nag at you. The toll is forced by the premises, but the premises were chosen, and "permanent" in particular was chosen by me. Why should a certificate stay up? Why not let it come down, pay nothing, and be honest about it? That's a fair question, and the three-card machine is the start of an answer: a flag that comes down for free is a flag the same move takes back, and a record that can be silently taken back is the thing the bulla and the hash chain and the write-once disk were all built against. "Permanent" here is exactly that: the flag can't be lowered. The toy in Section 3 didn't have it; the lineage in Section 3 was built around it; I chose it. Whether that was the right job to pick out of the five is something you can argue with me about, and I'd want you to.

And "permanent" is an idealization, which I should say before a physicist does. No physical record lasts forever. A bit held behind an energy barrier leaks over it at some rate, however slow, and a closed finite system comes back around, which is Zermelo's point from the chairs. The theorems in this section need permanence on every state, including states the machine never visits, forever. Weaken that to "the flag stays up on the states that actually turn up, for as long as anyone cares," and merge-or-revoke tells you what goes wrong: the move could take the flag down somewhere nobody ever looks, and then it isn't forced to merge at all. A version of the toll stated in chances and time spans would be a different theorem, and I don't have it.

One piece of it I can hand you, argued here on the page and not in the checked files. Take one move and repeat it from one start, forever. The states that run passes through are finitely many, so it falls into a loop. If the move forgot nothing on those states, it would be a reshuffle of them, and a reshuffle brings every one of them back around, the start included, where the flag was down. So if the flag goes up along that run and stays up from some point on, the move merges two states *the run itself visits*. The shared chair is one the machine actually sits in. And it doesn't touch the known-state

point above: a merge among states the run visits still costs Landauer nothing if the run knew where it was and something outside timed the step.

Runs that mix different moves split two ways, and the answer is short enough to give here, argued on the page like the last one. If the moves are picked from outside, in any order at all, the answer is no. Take two states, one reading no and one reading yes. Move a swaps them; move b does nothing. Run a once and then b forever: the flag goes up at the first step and stays up for good, and neither move merges anything anywhere. The swap would take the flag down from the yes-state if it ever ran there; on this run it never does, and that's exactly the weakened permanence this paragraph is about. But if the machine is closed, picking its next move from its own state, the run is the orbit of one map, the step with whatever move the state picks, and the single-move argument applies to that map as it stands. And if the moves come in a pattern that eventually repeats, one word w of moves over and over, and the flag, as before, goes up and stays up while having been down somewhere after the repeating starts, run the same argument on the composite of w 's moves, at the states where a repetition begins. It merges two of those states, both visited; follow the two through the letters of w , and the first letter where their paths meet merges two states the run visits at that letter's place in the pattern. So: moves picked from outside, no; a closed machine or a repeating program, yes.

And one limited piece of the chances-and-time-spans version, also argued here and not checked, because it shows exactly where the known-state escape stops. Take a device with no outside hand at all: a little random process hopping among finitely many states at rates that never change, with the usual physical conditions (every hop has a reverse hop, and the rates obey local detailed balance), able to get from any state to any other. Left to run, it settles into a long-run spread π and spends stretches with the flag up and stretches with it down; call their mean lengths T_{up} and T_{down} . Start it in a no-state you know. Then the entropy it produces from then on is at least $\ln(1 + T_{\text{up}}/T_{\text{down}})$, in units of k_B . The argument is three lines once the pieces have names. With fixed rates, the entropy produced is at least the drop in relative entropy between the device's spread and π ; that's half of Massimiliano Esposito and Christian Van den Broeck's split of entropy production (2010). A start known to be in state s has relative entropy $-\ln \pi(s)$, which is at least $-\ln \pi(\text{no})$, where $\pi(\text{no})$ is the long-run chance of the flag being down, and the relative entropy runs down to zero as the device settles. And the long-run chances of up and down stand in the ratio of the mean stretch lengths, because the crossings up and the crossings down balance, so $-\ln \pi(\text{no}) = \ln(1 + T_{\text{up}}/T_{\text{down}})$. Read it this way: each doubling of how long the record stays up, against how long it sits down, costs about one more $k_B \ln 2$, and a record that stays up forever on a fixed-rate device costs without bound. That's Murashita, Funo and Ueda's absolute irreversibility landing on the commit at last, with $1 - \lambda$ the long-run chance of the starting state. It isn't new physics: nothing in it needs a record, since an autonomous reset to a fixed state obeys the same bound,

and the time-symmetric-control result above covers the same ground with an error rate in place of the time ratio. And it isn't the theorem I said I didn't have. It's a first honest step toward it, and it pins down where "a known state owes nothing" needs its outside clock.

The next section is about a different door entirely: not whether forgetting costs, but whether *finding out* does, because the demon's whole trick was that it knew which way the molecule was going, and I owe you a straight answer on whether knowing is free.

5 Learning is free, committing is not

I ended the last section owing you an answer about knowing. The bill so far sits on forgetting: a flag that stays up was paid for because raising it merged two states. But there's a thing that looks like the opposite of forgetting and feels like it ought to cost more, not less. Finding out. Every time a machine rules out a possibility, it knows more than it did. If forgetting a bit costs, does learning one cost too? There's a strong feeling about this, and the feeling is wrong, and the way it's wrong is the content of this section, so here's the feeling first: learning is the expensive part. It's a common feeling, and most physicists had it too. It's why the demon took a hundred and fifteen years to exorcise. Or didn't. "Exorcise" is the word the people who think it worked like to use. John Earman and John Norton, the doubters from the last section, called their two-part paper on it "Exorcist XIV: The Wrath of Maxwell's Demon," and I read that Roman numeral as a joke about how many times the demon has been declared gone. So hold the word loosely. I'm going to tell you the version where the exorcism worked, because it's the version that says where the bill goes, and the door from the last section stays open behind it.

5.1 A small being at a small door

You met Maxwell's demon in Section 3, in one paragraph. It gets a whole section here, because the step it seemed to pay for and the step it actually pays for are what this section is about.

In December 1867 James Clerk Maxwell, who had published his equations for electricity and magnetism two years before and had as good a claim as anyone to being the most important physicist alive, wrote a letter to his friend Peter Tait describing a problem he couldn't get rid of. Take a box of gas at one temperature, with a wall down the middle and a tiny door in the wall. Now imagine, he said, a "very observant and neat-fingered being" sitting at the door, who can see individual molecules coming. When a fast one approaches from the left, it opens the door and lets it through. When a slow one approaches from the right, same. Otherwise it keeps the door shut. It never pushes anything; it only opens and closes a frictionless door at the right moments. After a while the right side is hot and the left side is cold, and you've got a temperature difference you could run an engine off, made out of nothing but *looking*.

That's supposed to be impossible. It's the thing the second law of thermodynamics says can't happen, and the second law is the law physicists are least willing to give up; Eddington said that a theory found to contradict it has nothing for it but to collapse in deepest humiliation, and nobody has had to test that yet.

Maxwell put it in his textbook in 1871, and it got named Maxwell's demon, though not by Maxwell. William Thomson, later Lord Kelvin, called it a demon in *Nature* in 1874, and Maxwell, in an undated note to Tait headed "Concerning demons," tried to take the name back: "Call him no more a demon but a valve." Nobody listened, which is probably for the best, because "Maxwell's valve" would not have lasted a century. But keep his correction, because it's exactly right and it's the whole problem: the being does no work. It opens and shuts a door. So physicists spent the next century trying to find where the being was cheating. I want you to feel the pull of the obvious answer, because I felt it and so did most of them: the being has to *see* the molecules. Seeing costs. Case closed. In 1929 Leo Szilard built a one-molecule version of the box and did the arithmetic, and said the second law survives if finding out which side the molecule is on costs at least as much entropy as the being gains. In 1951 Léon Brillouin made it concrete: to see a molecule you have to bounce light off it, and bouncing a photon costs at least $kT \ln 2$, Landauer's number before Landauer, and there was your toll on looking. For thirty years that was the answer. Information costs because getting it costs.

That's a price on one way of looking, and it doesn't show that every possible way of making a record has to pay it. A device can burn energy without that being a law.

Szilard is worth one more minute, for a reason that has nothing to do with the demon. Four years after the one-molecule engine, in September 1933, by his own later telling, he was waiting at a traffic light on Southampton Row in London, having just read a newspaper report in which Rutherford called the hope of getting power out of atoms moonshine, and as the light changed it occurred to him that an element which gave off two neutrons when it swallowed one could keep a reaction going by itself. That's the nuclear chain reaction, thought up crossing a street. It's his memory of himself, and Richard Rhodes's history of the bomb is where most people meet it, so hold it as the man's own story and not as a timestamp. Does it change anything about the demon? Nothing at all. It changes the picture: the same mind looked at one molecule in 1929 and one neutron in 1933, and both times the question was what a single small step forces downstream.

5.2 Bennett moves the bill

In 1982 Charles Bennett, who'd already shown in 1973 that computation can be done without forgetting, took the demon apart one step at a time. The being looks at a molecule and writes down which side it's on. Is writing that down a merge? Picture the being's notepad, which starts blank. After looking, the notepad says "left" or "right." Two possible worlds before (molecule left, pad blank; molecule right, pad

blank). Two possible worlds after (left, “left”; right, “right”). Nothing merged. Two in, two out, and you can read the before off the after. Copying what you see onto a blank page forgets nothing, and a step that forgets nothing is a step Landauer doesn’t charge for. Brillouin had priced one particular way of looking (bouncing light) and then said all looking costs. Bennett showed a way of looking that doesn’t.

So where is the bill? The being has one notepad and a lot of molecules. After the first molecule the page is full. To do the next one it has to wipe the page. And *wiping* takes two possible worlds (pad says “left”; pad says “right”) to one (pad blank). That’s the merge. That’s the step with no way back. The demon pays, every cycle, not for seeing but for making room to see again, and the amount it pays is at least what it gained, and the second law sleeps soundly. That’s the rule from his 2003 notes I quoted in Section 3: erasing and merging cost, and copying onto a blank place to write doesn’t.

Section 3 separated making a record from resetting it. Now you can see exactly which step and why it was tempting, because looking *feels* like the active part. It’s the thing the being does. Wiping feels like housekeeping. The housekeeping is where the money goes. If there’s a sentence in this section I’d want you to carry out of it, that’s the one, and I didn’t write it, Bennett did.

5.3 Two different things called “narrowing”

Here is what I had to get straight in my own head before any of this made sense, and I’ll give it to you as a picture because that’s how I eventually got it.

Draw the machine’s possible states as a row of dots. Now draw a second row above it: the dots that *you*, standing outside, still think it might be in. Those are two different rows. The first is a fact about the machine. The second is a fact about you, about what you’ve seen and what you’ve ruled out. Both rows can shrink. They shrink for different reasons, and only one of them costs.

The machine’s row shrinks when two of its states land on the same state. That’s a merge. That’s what the price charges. Under the halving rule from Section 4, a run can’t shrink the machine’s own row by more halvings than it paid for. That’s the theorem just below, and it’s the Section 4 squeeze bound run forward along a whole program instead of one step. No new idea; the same chairs.

Exactly. Write $\lceil \log_2 n \rceil$ for “log base two of n , rounded up,” the number of yes-or-no questions it takes to pick one of n things. For a run t and a list D of starting states, let $\text{img}_t(D)$ be the set of states the run can end in, starting from somewhere in D , and write $|\text{img}_t(D)|$ for how many there are (in Section 4 the symbol for one move already meant the count; here I need the set and its size both). Let $\text{cost}(t)$ be the run’s total cost.

Theorem 5.1 (The machine’s own narrowing is priced). *If the cost satisfies the halving*

price (Premise 4.4), then for every run t and every list D of starting states with no repeats,

$$\lceil \log_2 |D| \rceil \leq \text{cost}(t) + \lceil \log_2 |\text{img}_t(D)| \rceil.$$

Proof. By induction on the run. A single move of cost c shrinks a list with no repeats by at most a factor 2^c (that's the halving price), which costs at most c in rounded logarithms; the states reachable after the rest of the run from the shrunk set are exactly those reachable after the whole run from the original. Add the costs. $\square$

Checked as `run_narrowing_priced_log (KnowledgeNarrowing.v)`. The run itself is the witness; nothing else is handed in. That will matter in Section 8, and I'll say why at the end of this section.

Your row shrinks when you see something that some of your candidates couldn't have shown you. You cross those candidates off. Nothing happened to the machine. It didn't merge anything; it didn't know you were watching; it would have done the same thing with nobody in the room. The physicist Edwin Jaynes had a name for mixing these up, and I think it's one of the most useful names in all of science: the mind projection fallacy, treating what you know about a thing as if it were a property of the thing. Your list of candidates is in your head. The machine's row is in the machine. The halving price is about the machine's row, and crossing a candidate off yours doesn't move it.

And here I have to be fair to Jaynes, because he'd argue with where I just took his phrase. In a 1965 paper on Gibbs's and Boltzmann's entropies he wrote that entropy is "an anthropomorphic concept" (a phrase he credited to Eugene Wigner): a property not of the physical system but of the particular experiments you or I choose to do on it. So for Jaynes the second law is about what's known too, and the line I'd have liked to draw, that the second law has opinions about the machine and none about you, is exactly the line he'd refuse. Who's right is one of the oldest fights in statistical physics, Jaynes's reading against Boltzmann's, and I can't settle it. I don't have to. The spreads in Section 4's entropy theorems are spreads over the states the device might be in when the step runs, and every theorem there holds whether you read those chances as ignorance or as physics; "the floor comes from what you didn't know" is the same sentence in either reading, with "you" being whoever built the spread. What I need from the fallacy is only this: the two rows are different rows, and only one of them is the machine's.

5.4 The demon on a sheet of paper

Build it, because the whole claim is four dots and two arrows and you should see it merge or not merge with your own eyes.

The machine has a hidden bit (think of it as the molecule's side) and a display (the notepad). Four states: write them as pairs, hidden then display. 00, 01, 10, 11. Two

moves.

MEASURE copies the hidden bit onto the display, by adding it in the way that a light switch adds: if the hidden bit is 1, flip the display; if it's 0, leave it. Draw the arrows. 00 goes to 00. 01 goes to 01. 10 goes to 11. 11 goes to 10. Look at the arrowheads: every state has exactly one arrow landing on it. Nothing merged. MEASURE forgets nothing.

WIPE sets the display to 0 whatever it held. 00 goes to 00; 01 goes to 00 as well. 10 goes to 10; 11 goes to 10 as well. Two arrows land on 00 and two on 10. That's a merge, twice over. Under any price that charges per halving, WIPE costs at least one. MEASURE is allowed to cost zero.

Now be the observer. You can see the display and not the hidden bit. What you're shown, the display here, is what I'll call a *window*: the part of the state an outside observer gets to see. Section 7 is about windows in general; here the window is just the display. The display starts blank, so your list of candidates is {00, 10}: two possibilities, one bit you don't know. The machine runs MEASURE. You look at the display. If it shows 1, the only candidate that could have produced it is 10. If it shows 0, only 00. Your list went from two to one. You learned the hidden bit. The machine's price for that move: zero. Its own row of possible states went from {00, 10} to {00, 11}, still two; nothing it could have been got merged with anything else. You narrowed. It didn't. That's the measuring half of Bennett's argument on four dots, and WIPE's charge is the wiping half. So the demon just broke something: one bit gained, nothing charged. Here's the thing it broke, written out so you can see exactly where the crack runs: for every list of starting candidates, every run, and every actual start in the list, the index bits the observer gains ($\log_2$ of the candidates before, rounded up, minus $\log_2$ of the candidates after) are at most what the run was charged. Four dots, and that sentence is false.

Definition 5.1 (What an observer knows). Fix a window w from states to what can be seen. Given a list Ω of candidate starting states, a run t , and the actual start $s_0 \in \Omega$, the observer's *knowledge* $K(\Omega, t, s_0)$ is the list of candidates $s \in \Omega$ whose run shows the same thing through the window at every step as the actual run does.

Defined as `knowledge` (KnowledgeNarrowing.v).

If that definition looks familiar, it should. It's the standard way logicians of knowledge set it up: what you know is whatever holds of every case you can't tell apart from the actual one, given everything you've seen (Fagin, Halpern, Moses and Vardi's *Reasoning About Knowledge*, 1995, builds a whole subject on it, runs and observations included). And it's the logically omniscient kind from the opening chapter, the kind where knowing some facts means knowing everything that follows from them, for free. I'm using it on purpose, flaw and all. This section is about learning, and learning

in that sense is exactly the thing that's allowed to be free. Settling is the separate act with a price, and nothing in this definition settles anything.

Theorem 5.2 (An observer can learn for free). *There is a four-state machine (the demon), a cost satisfying the halving price, a run of cost 0, and candidates with $|\Omega| = 2$, for which $|K(\Omega, t, s_0)| = 1$. So the claim “for all Ω, t, s_0 : $\lceil \log_2 |\Omega| \rceil - \lceil \log_2 |K(\Omega, t, s_0)| \rceil \leq \text{cost}(t)$ ” is false for that machine and that cost. In the same machine, every cost satisfying the halving price charges WIPE at least one.*

Proof. The machine and run are the four dots above: MEASURE is a bijection, so cost 0 satisfies the halving price; the observer's two candidates show different displays after MEASURE; WIPE sends two pairs of states to one each, so the halving price needs its cost ≥ 1 . $\square$

Checked as `observer_narrowing_can_be_free`, `demon_observer_learns`, `demon_machine_spread_kept`, `measure_forgets_nothing`, `wipe_merges`, and `wipe_costs_at_least_one` (`KnowledgeNarrowing.v`).

I have to be exact about one thing the file is exact about. The halving rule is a floor. It says merges cost at least this much; it does not say anything costs zero. So what the theorem exhibits is *one* price schedule that meets the halving rule and charges MEASURE nothing. A different schedule could meet the rule and still charge MEASURE; nothing in counting forbids overcharging. The claim is “learning is not *forced* to cost,” not “learning is free on every machine.” That's weaker than the section title, and the section title is what the strong feeling says before you check, so the title stays and this paragraph stays next to it.

The physicists had this freedom on the books long before me. Paul Fahn (1996), doing the entropy accounting of Szilard's one-molecule engine step by step, found a symmetry between the measuring step and the erasing step: which of the two you hand the bill to is partly a choice of bookkeeping, and the total is what's fixed. Sagawa and Ueda (2009) gave that total its bound, with the split depending on the memory. So “a schedule could charge MEASURE instead” isn't a loophole I found; it's a known freedom in how the books are kept. And the other reading is very much alive. Ruth Kastner (2025) argues that Brillouin was essentially right after all: once the physics of a real measurement is put back in, it's measurement and not erasure that pays and foils the demon, and Bennett's free copy fails as physics. The theorem above is about price lists, so it stands either way. What hangs on her is whether a real demon can look for nothing, which is the same door the next paragraph's doubter pushes on.

And the doubter from the last section cuts this side too, so I owe it to you, since I leaned on him there. In 2013 John Norton argued for a no-go result: at the scale of molecules, thermal jiggling keeps any process that's supposed to be thermodynamically reversible from ever actually finishing, and finishing it takes more entropy than

Landauer's amount. If he's right, Bennett's free measurement isn't free in practice either, because it's one of those processes. The theorem above survives that, because it's about price lists: a schedule that meets the halving rule isn't forced to charge the copy, whatever a real device spends. What wouldn't survive is the physics I've been hanging on it, that a real demon can look for nothing. If Norton is right, the line between learning and committing keeps its counting and loses its physics. I can't referee that one either.

5.5 Who holds the pen changes the bill

Now the part that I think is the most beautiful fact in here, and the way to find it isn't by thinking; it's by playing with the four dots. Run MEASURE *twice*. The first copies the hidden bit onto the display. The second adds the hidden bit in again, and adding a bit twice with a light-switch add gets you back where you were: the display is blank again. Check the arrows: 10 to 11 to 10. The display went full and then blank, and nothing merged, and the price was zero both times. So a display that was blank before the first MEASURE *can* be put back to blank for free, by a second MEASURE. Just not by WIPE.

What's the difference? MEASURE *consults the hidden bit* when it clears the display. It knows what's written there, because it has access to the thing it was copied from, and so it can undo the copy instead of erasing it. WIPE doesn't look. It sets the display to 0 regardless, and "regardless" is exactly the many-to-one. Bennett has the general form of this in the 2003 notes: erasing one of two copies that you *know* match is reversible, because you're not destroying information, you're removing a duplicate. And it goes further than Bennett in work that came after him: del Rio and colleagues in 2011 showed that the cost of erasing a system depends on how much the eraser already knows about it, down to zero and in the quantum case past zero. The bill for clearing a record is sent to whoever clears it, and it's smaller the more they know. Try it with a card under a cup and a sticky note: write the colour on the note, then erase the note *without lifting the cup* (two possible notes become one blank: a merge) versus lift the cup, check the note matches, and then erase (nothing unknown was destroyed). Same blank note. Different bill.

5.6 The smallest machine that teaches for free, and the one that can't

Two more small facts, and then the point. The demon measures what you learn by comparing the start to the end, but an observer's very first look already rules things out before any step runs, so the fairer question is: after that first look, can a *step* teach you more, for free? Yes, and the demon still does it. And the smallest machine that can is three states in a ring: U goes to V, V goes to W, W goes to U, and the window shows "true" only at W. Start with candidates {U, V}; both show false. Take one step. From U you now see false; from V you now see true. One step told them apart, and a ring forgets nothing, so a zero price meets the halving rule, and the file takes it.

Two states can't do it, ever, with any arrows, and this is the one that needs no price

premise at all. If a machine has only two states and you can't tell them apart at the first look, then the window shows the same thing on both, and those two are *all the states there are*, so the window shows the same thing everywhere, forever, and you never learn anything. Try to draw a counterexample. You'll draw the same two dots a few times and give up, and the giving up is the proof. Three is the smallest number of states on which two states that look alike at the first look can come apart at a later look, and that's what teaching is.

Theorem 5.3 (Three states are the minimum for free teaching). *Call a machine a free teacher when it has a cost satisfying the halving price, a list of candidates with no repeats containing the actual start, and a run of cost 0 after which the observer knows strictly more than after the first look, $|K(\Omega, t, s_0)| < |K(\Omega, [], s_0)|$. There is a free teacher with exactly three states, and there is none with two or fewer.*

Proof. Three: the ring $U \rightarrow V \rightarrow W \rightarrow U$ with the window true only at W , candidates $\{U, V\}$, one step, cost 0 (a ring is a bijection, so cost 0 satisfies the halving price). Two or fewer: two candidates that agree at the first look are, on a two-state machine, all the states there are, so the window takes one value everywhere and no later look can separate them. This half never uses the price at all. $\square$

Checked as `free_incremental_narrowing_with_three` and `no_free_incremental_narrowing_below_three` (KnowledgeNarrowingMinimal.v), and the demon-against-the-first-look form as `demon_refutes_incremental`.

5.7 What the toll prices, said at last

Now I can say, more precisely than I could two sections ago, what the mark in Section 3 is a mark for. Not for learning. What you know can be narrowed at no cost, with nothing merged, on some machines under some honest price lists, and the demon is one. Not for looking, not for copying, not for finding out. The mark is for the one thing in Section 4 that couldn't be done without merging, on a machine with finitely many states: raising a flag that stays up. Learning is the demon writing on a blank pad, which merges nothing. What the demon pays for is the wipe, a step with a shared chair. The mark in Section 3 is paid for the same reason, because raising a flag that stays up also has a shared chair, but it's a different step: the demon machine has no flag, and its file prices merges, not commitments. Same chair. Different room.

This is also where I settle up with Rovelli, my closest neighbour from the opening chapter, because I put him and Bennett on opposite sides of one door, and I owe you how much of that is mine. All of the reconciliation is mine. Rovelli's paper doesn't make it, and in four places his own framing points somewhere else. He reserves "choice" for an outcome that is stable and *not predictable*, and he rules out a thermostat because it runs on a predictable mechanism; the commit in this book is a step a rule fixes in advance, so by his own definition his result doesn't cover it. His choice

is a branching, one possibility splitting into two, and he calls choosing in a sense the inverse of erasing, because branching is the inverse of merging; the toll comes from a merge, the opposite shape. His stability is for a while: he says stabilization means something only on times that aren't too long next to the time it takes things to thermalize, and past that nothing stays chosen; my permanence is forever (and the last section owned up that forever is an idealization). And he doesn't stand beside Bennett at all: he holds that any measurement is dissipative, credits the point to Reichenbach, and doesn't cite Bennett. So what Rovelli and I share is a sentence, that a choice that sticks costs something like Landauer's bill, and not a mechanism. Push on it and his coin hitting the ground, a fact gained from something nobody could predict, sits closer to what this section calls learning than to committing. I still think it's the same instinct. That's my reading, not his, and you now have everything you need to disagree with it.

And on one point Rovelli isn't the nearest physics. The nearest I've found is in biology. Thomas Ouldridge and Pieter Rein ten Wolde (2017) worked out the fundamental costs of making a *persistent* copy of a polymer, one whose sequence stays correlated with its template after the two have physically come apart, and of destroying one. A copy like that is a state out of equilibrium, and making it has a thermodynamic bill they measure efficiency against. A companion paper the same year, with Christopher Govern, does the same kind of accounting for the molecules a cell uses to read out the state of its receptors, which is a measurement in all but name. The way I read them, the copy that has to stay is the one with the bill, and that's the closest thing in physics I've found to the line this section draws between finding out and committing. The differences, so it doesn't get borrowed for more than it is: theirs is heat, in chemistry, for a correlation between two molecules, and their title has the destruction of copies in it too, so their copies don't last forever; mine is a count in a model of computation, for a reading that never turns off, with no temperature anywhere. Same instinct again, from a different direction.

The chemistry has the sharp form of it, too, and it came two years later. Jenny Poulton, Pieter Rein ten Wolde and Thomas Ouldridge (2019) studied minimal models of polymer copying and showed that when the copy grows at equilibrium, with the weakest chemical driving, a persistent copy's accuracy and the efficiency of making it both vanish; only driving the growth out of equilibrium, paying, buys a copy that still carries information about its template after the two let go. That's finding out against committing, in chemistry, with the bill attached, and I'd rather say plainly that biology had the distinction before I did. What stays different is the kind of thing the record is: theirs is a correlation in a chemical ensemble, mine is a reading enforced by a step in a model of computation.

Two more neighbours have "learning" and "commit" in their titles, so the slogan shouldn't be read against them by accident. Sebastian Goldt and Udo Seifert (2017) did the stochastic thermodynamics of a neural network learning a rule, and found that

learning has a thermodynamic cost: the entropy the network produces bounds how much it can learn. That doesn't contradict this section, and here's why. Their learner isn't copying onto a blank page. Its weights are a driven thermal system, jostled by noise and dragged toward what the data push them to, and what they price is that physical process. This section's free learning is narrower: one way of finding out, a copy onto a blank, is *allowed* to cost nothing. Nothing here says every way of learning is cheap, and theirs isn't. From statistics, Aurélien Garivier, Emilie Kaufmann and Tor Lattimore (2016) proved that in a two-armed bandit with Gaussian rewards, a strategy that explores first and then commits for good ends up, as the horizon grows, with at least twice the regret of the best strategy that keeps deciding as it goes. That's the closest published sentence I've found to "committing isn't free" with a number on it. Their price is reward lost, not a mark on a ledger, so it's a cousin, not a twin.

So here's exactly what "No Free Insight," the name I gave the toll before I understood it, covers, because I don't want the name doing work it hasn't earned. Insight in the sense of *finding out* can be free. The insight the toll prices is the kind you've *committed to*, written into a record the machine can't silently revise. Whether that narrower thing deserves the name is a fair thing to argue about; I kept the name because it's the question I started with, and because the narrowing is the finding, not a retreat.

It also tells you how to read a bound that's coming. Section 8 counts what it takes to narrow a list of candidates down to a shorter one: at least as many yes-or-no questions as it takes to point at the survivors among their rivals, a counting fact that holds for anything with a list in it. Turning that count into a *bill*, saying a run actually paid that many marks, is a different claim, and when the book makes it, for any system with a counter in Section 8, the payment goes in as a hypothesis: a decision tree (a branching chart of yes-or-no questions, which Section 8 builds) no deeper than the number of paid steps the run took is handed in, along with the lists themselves. This section is the reason it has to go in that way. The narrowing alone doesn't force anyone to pay; the demon just showed you a narrowing that cost nothing. So wherever narrowing gets a price in this book, the payment is a premise, and I'll keep calling it one.

And one more boundary. This whole section is about a machine that's finite and deterministic and a watcher who crosses candidates off a list. There's no quantum anything in it. People sometimes reach for "observation collapses the state" when they hear "watching narrows things down," and I want to head that off: the narrowing here is bookkeeping in the watcher's list, the machine never notices, and the result would be the same if the watcher were a filing cabinet. In 2010 a group in Tokyo (Toyabe, Sagawa, Ueda, Muneyuki and Sano) did run a real demon, a tiny particle walked up a staircase of potential by feedback on what a camera saw, and the energy it gained was paid for by the information the camera took, with no violation of anything. The demon is real, it works, and it pays. Where it pays is the part that took a century.

What I still owe: everything so far says "at least one mark" and I've been keeping

the marks on the side of the page, a running total you tally as you go. What happens when the machine keeps that total itself, inside its own state, where a step could in principle reach in and change it? That's the counter, and it's the next thing I have to build.

Part III

The logic

6 The counter

Everything so far has kept the marks on the side of the page. You played the toy, you made a tally as you went, and the tally lived in your hand and not in the machine. That's fine for a toy. For anything that's going to run on its own, with nobody tallying, the total has to live somewhere the machine can reach, which means the machine can, in principle, reach it. So this section is about what a counter has to be before you're allowed to call it a receipt. Not the counter of some particular machine: the counter of *any* machine whose state carries a running total that every step bumps by whatever a price list says that step costs. Section 3's five parts don't need one (there the total was a sum you did on the side), so this is one piece added to the five, and the facts below are about anything that has it. I'll give you the answer first, because it's short, and then spend the section on why each word is there: a receipt is a total that lives in the state, that only climbs, that equals the sum of what was charged, and that nothing else could have been.

6.1 A stick, split down the middle

The oldest receipt I could find that you can still hold is a piece of wood. From about 1100, under Henry the First, until 1826, the English treasury recorded what it was owed on tally sticks: a strip of hazel, notched across with cuts of set widths (a thumb's width for a hundred pounds, a swollen barleycorn for a pound, a bare cut for a penny), then split lengthwise so that each half carried the same notches. The payer kept one half, called the stock, which is one story for where the word stockholder comes from. The treasury kept the other. When the time came to settle, you laid the two halves together, and only the two original halves fit, because wood splits along a grain nobody can copy. Add a notch to your half and the other half shows a gap where the partner should be. Try to remove a notch and you find that wood can't be uncut. When they finally stopped using them, the clerks burned centuries of old tallies in the furnaces under the House of Lords, overheated the flue, and burned down the Palace of Westminster, which I mention because it's the best fact I picked up for this book and because it tells you how many of them there were.

Do it: notch a strip of card once for every coin you hand a friend, split it, keep half

each. Try to cheat, either direction. The three things you'll find are the three things I want. The total is the notches, added up, and nothing else (I'll call that conservation). You can add a notch but you can't take one away (it only climbs). And two halves that were notched together must agree; a half that disagrees is a forgery, and the wood says which one (there is only one honest count). The treasury didn't have the words. They had the stick.

Economists read the same stick another way, and they're not wrong either. Alfred Mitchell-Innes (1913) made the exchequer tally his star witness that money is credit: the crown paid the people it owed by "raising a tally" on some branch of its revenue, a wooden acknowledgment of debt. A receipt to me, money to him, and it's the same piece of wood. Narayana Kocherlakota (1998) later proved a version of his point: in a wide class of economies where nobody can be held to a promise, anything money lets people do, a memory of everyone's past dealings lets them do too. His title says it in three words: money is memory.

6.2 Why it has to climb

Start with the second property, because it's the one that does the most work and the one people fight hardest. Why can't the counter ever go down?

Because of what the toll is for. The floor it buys, back in Section 3, says a run from NO to YES costs at least one. That sentence is a *floor*: it says the run paid at least this much. If the counter could come back down, a run could pay the toll, get its flag, and then take the payment back, and the floor would be a floor on nothing; "this run cost at least one" would be true of a run that ended having cost nothing. So a counter that can fall is an accountant you can lean on to revise the books, and a floor stated in terms of it isn't a floor. The counter only climbing is not a flourish. It's what makes "at least" mean something. (In Section 3's abstract theorem the total is a sum of whole numbers and climbing comes for free; this paragraph is about why a total *stored in the state*, where a step could in principle write it, has to be designed so that no step ever lowers it.)

The same rule turns up in two places that have nothing to do with each other, pointing in opposite directions, and that's a big part of why I trust it. In 1978 Leslie Lamport asked how a set of computers that pass each other messages, with no shared clock, could ever agree on what happened before what. His answer is a counter per machine with two rules: add one before you do anything, and when a message arrives carrying the sender's count, jump your own counter above it (never below your own). Those counters, he showed, respect cause and effect: if this caused that, this has the smaller number. And they do it "with no actual timing mechanism," his words; they're not clocks at all. Try it with three people passing notes and you'll see it work in two minutes, and then let one of them reset their counter to zero after reading a note, and watch a reply come out numbered *before* the note it replies to. A counter that goes down can't be leaned on for order.

Where did Lamport get it? In the notes he keeps on his own papers, he says the timestamps came from an earlier note by Paul Johnson and Bob Thomas on keeping copies of a database in step, whose algorithm wasn't quite right: it let things happen in an order that broke cause and effect. What let him see the fix, he says, was special relativity, which teaches that there's no one order of events everybody agrees on, only the order of what could have caused what. And he passes on a line from Jim Gray, the same Gray as the commit at the start of the book: Gray had heard two opinions of the paper, that it was trivial and that it was brilliant. Lamport's answer is that he can't argue with the first and isn't inclined to argue with the second. And what's under the relativity? His account stops there, so the chase does too. What it changes is only the picture, and the picture is a good one: a number that's allowed to climb and never to fall is enough to write "this came after that" with no clock anywhere.

Now the other direction. In 1949 Alan Turing, and in 1967 Robert Floyd, each asked how you prove a program *stops*. The answer, both times: find a quantity that goes *down* on every step and can't go down forever (a whole number that can't go below zero, say). If every step lowers it and it can't fall forever, the steps can't go on forever. Up-only gives you order. Down-only gives you stopping. They're not the same rule (one needs the jump-above-the-message step, the other needs a floor it can't fall below), but they're the same shape seen from the two ends: a one-way quantity is something you can build on, and a two-way quantity is a number that happens to be written down somewhere.

And the real-world case where a counter was allowed to go down, and got gamed exactly there: Ethereum, one of the largest blockchains, charges "gas" for every step of a program, and for years it *refunded* gas when a program freed up storage. Someone noticed you could fill storage when gas was cheap, hold it, and cash the refunds when gas was dear, and a token was built to do exactly that, and the real work in a block could reach nearly twice the block's stated limit. In 2021 a change called EIP-3529 cut the refunds sharply (none for self-destruct, a smaller one for clearing storage, and a cap of one fifth of the gas used). The written reasons are the gas token and those overfull blocks. I read both as the same lesson: a counter that goes down was being leaned on at the place where it went down. Ethereum comes back in Section 12 as one of five small models of systems people actually run, and it's the only one of the five that prices a write at all, and it's instructive that the part they had to fix was the part that could fall.

6.3 Conservation: the total is the charges and nothing else

The first property sounds too obvious to state. The counter after a step is the counter before plus what that step was charged. Add up a whole run and the middle values cancel out, the way a sum of differences collapses to last minus first, and the total is the sum of the per-step charges. Why state it?

Because it's the property that fails quietly. Say a machine charges one for raising a

flag, and also, somewhere in its error handling, forgets to add the charge when the step fails halfway through. Or charges two in one code path and one in another for the same move. Or has one step that writes the counter directly. Then the counter is *mostly* the sum of the charges, and “mostly” is the whole gap between a receipt and a note that says “trust me, about this much.” Conservation is the sentence that there is no “mostly”: every step, every outcome, every error, adds exactly its charge, and a step that fails is charged the same as one that succeeds, because the charge is on the attempt. For a machine built the way I’m describing, that sentence is the rule the counter is updated by, so it can’t fail by definition. What can fail is the claim that a real machine *is* built that way, and checking that means going through every move it has, the failure branches included, one at a time, which is dull and is the whole job. The run-long form, counter at the end equals counter at the start plus the sum of the charges, then follows from the one-step rule by the cancelling sum; it’s below. It’s the kind of theorem that looks boring and is the one you’d most want if you were going to rely on the number. One thing it doesn’t say: that every mark bought something. The ledger is exact about what was spent, not about what the spending was worth; whether a mark bought a real narrowing is Section 8’s question, and it needs a witness on top of the ledger entry.

6.4 Only one honest count

Here’s the property that surprised me by being provable at all. Suppose you fixed the schedule: this move costs one, that move costs zero, and so on, for every move. And suppose someone else keeps their own books: their own function from states to totals, which starts at zero and, on every step, goes up by exactly what the schedule says. Could their books disagree with the machine’s counter, somewhere, on some state you can actually get to?

No. And the proof is the tally stick. Both totals start at zero. On every step both go up by the same amount. So on every state you can reach by steps from the start, they’re equal. That’s it; it’s an induction, the one-step-at-a-time argument from Section 3’s floor. The mathematician’s word for this is *initiality*: the counter is the one bookkeeping that all the others, if they’re honest about the schedule, turn out to be. Given the schedule, the count is forced. Two things about that sentence I had to be careful with, and they’re worth saying before the symbols. It doesn’t assume the other books climb: climbing follows from the schedule, since the schedule never charges a negative amount, so “only one honest count” and “it only climbs” aren’t two separate demands on the other bookkeeper. And it holds on *reachable* states only. A state you can’t get to from the start is one where it says nothing about whether the other books agree, which is the right thing for a receipt to say.

Here are the properties exactly. First the piece being added to Section 3’s five, a counter and a price list, and then the two facts about it. The counter gets the Greek letter μ (mu), which is what the checked work calls it.

Definition 6.1 (A counter kept by a price list). A *counter machine* has a set of states, a set of moves, a step function apply that takes a state and a move, a start state s_{init} , a schedule c giving each move a cost in $\mathbb{N}$, and a counter μ from states to $\mathbb{N}$, such that $\mu(s_{\text{init}}) = 0$ and $\mu(\text{apply}(s, i)) = \mu(s) + c(i)$ for every state s and move i , with no exceptions. A *run* of a list t of moves from s applies them in order, and $c(t)$ is the sum of their costs. A state is *reachable* when it is the start or is $\text{apply}(s, i)$ for a reachable s and some i .

Theorem 6.1 (Conservation). *In a counter machine, for every state s and every list t , μ at the end of the run of t from s equals $\mu(s) + c(t)$. In particular μ never decreases along a run.*

Proof. Induction on t . For the empty list both sides are $\mu(s)$. For a list i followed by t' , the one-step rule gives $\mu(\text{apply}(s, i)) = \mu(s) + c(i)$, and the induction hypothesis from $\text{apply}(s, i)$ adds $c(t')$. Since every $c(i) \geq 0$, the counter never falls. $\square$

Theorem 6.2 (Only one honest count). *In a counter machine, let B (for “books”) be any function from states to $\mathbb{N}$ with $B(s_{\text{init}}) = 0$ and $B(\text{apply}(s, i)) = B(s) + c(i)$ for all s and i . Then $B(s) = \mu(s)$ for every reachable state s .*

Proof. By induction on why s is reachable. At the start both are 0. If $s = \text{apply}(s', i)$ with $B(s') = \mu(s')$, then $B(s) = B(s') + c(i) = \mu(s') + c(i) = \mu(s)$, the last step by the counter’s rule. $\square$

Those two proofs are the whole of it, and they’re about any counter machine. There’s no Coq statement of them over arbitrary states, so here they’re argued, on the page, where you can check every line. The machine I build later is one of these, and on it the run form is a checked theorem: the ledger after any list of instructions is the start plus the sum of their costs (Theorem 14.5). The uniqueness half stays argued, here.

I want to flag what initiality does *not* do, because the word sounds grander than the fact. It doesn’t pick the prices. The schedule is an input: somebody decided that raising the flag costs one and adding two numbers costs zero, and nothing here says that decision was right (Section 4 is where I argued the floor isn’t arbitrary; the rest of a schedule has to be defended by whoever writes it). It doesn’t turn marks into anything physical. And it isn’t the category theorist’s “initial object,” though the word is borrowed for a related reason.

There *is* a universal property nearby, and I’ll say exactly how far it reaches so the borrowed word doesn’t reach further. Take any other machine that understands the same moves (call it the *target*), and pick a state of it to start from. Walking a list of moves through it, one at a time, is the one and only way to send lists to its states that starts where you said and keeps in step as the list grows by one. Programmers call that walk down a list, one item at a time, a *fold*, and I’ll borrow the word. That’s an

induction on the list from its end, the same shape as the proof just above, and it never looks at the other machine's reading or its prices, which tells you how much the toll is doing here: nothing. It's a fact about *lists of moves*, not about states.

Coming down from lists to the states of the machine running them is where it gets interesting, and where the borrowed word stops. Two different lists can leave a machine in the same state. If they leave the target in two different states, there's no single place to send that state. That's the whole obstruction, and it's exact: the map from states exists on the reachable ones precisely when equal states always come from lists that agree on the other side (and the readings agree). Call that the *descent* condition, because it's what lets the fold come down from lists to states. A toll plus a reading that agrees doesn't force it to happen. A machine that keeps its whole history as its state obeys any toll you like and still never comes down. So the fold is unique on lists, and that's what the word "initial" earns. It doesn't make anybody's states initial among everything that obeys the toll, and I won't say it does. The fold and the descent condition are argued here, over any machine, and not stated in Coq; the history-keeping machine is the example that shows where descent fails.

And the borrowed word has a home, which I'd rather point at than pretend I built. That lists of moves are the starting object, and that a machine's behaviour factors through its reachable states exactly when a condition like descent holds, is the theory of minimal realization. Joseph Goguen put it as a universal property in 1973, in a paper whose title says it outright, "Realization is universal," and Thomas Colcombet and Daniela Petrişan redid automata minimization in the same functorial terms in 2020. Their machines can have any output at all, which is the honest point: the fold and the descent condition never use the record, so the toll is doing nothing here, just as I said.

6.5 The toll is an old trick with a new name

Now a thing that sounds like a let-down and isn't. It would be nice to have a new kind of cost argument. This isn't one. The toll is a case of a technique that algorithm people have used since 1985 and that physicists had, in a different costume, in 1892, and I'd rather tell you it's old than let you think I invented it.

The trick is called the potential method, and here's the whole of it. Give every state a number, call it the potential, that measures how much "owing" is stored up in that state. Then look at each step and ask: does the cost of the step, plus the change in potential it causes, ever come out negative? If it never does, you've got a bound: any run costs at least the amount the potential *dropped* from start to finish, because the drop had to be paid for by the steps along the way. Robert Tarjan and Daniel Sleator used it in 1985, going the other way (potential as prepaid credit, to bound a run from above), to analyse data structures that are occasionally expensive and usually cheap; and the decreasing-function idea underneath it is the one Aleksandr Lyapunov used in 1892 to prove that a system settles down. Do the coin version: a stack of coins, PUSH

adds one and costs one, POP-ALL removes them all and costs as many as it removed, potential equals the number of coins. PUSH costs one plus one potential; POP-ALL of k coins costs k and drops the potential by k , net zero. So any sequence from an empty stack costs at most twice the number of pushes, however many expensive pops are in it. That's the method. Three lines.

Push on that 1892 for a second, because a date like that hides a lot. Lyapunov wrote the idea up as his doctoral thesis, in Russian. A French translation came out in 1907. The English one came out in 1992, a hundred years after the original, and it was translated from the French: a translation of a translation. So for most of the twentieth century a reader with only English met the idea secondhand or not at all. Did the algorithm people of 1985 get it from him? I can't tell you, and nothing here needs them to have. Whether anybody walked from one to the other on purpose is a question I'd like answered. What it changes is the picture: the same idea, a number that only goes one way along a run, turned up in a thesis about whether a moving system settles down and in a paper about data structures, close to a century apart.

Now point it at the flag. Potential: one if the flag is down, zero if it's up. A step from down to up drops the potential by one, so for the step's cost plus potential change to be non-negative, the step has to cost at least one. That's the toll, exactly, and the two are the same statement: the toll holds for a machine if and only if every step has non-negative amortized cost under the flag potential ("amortized cost" is just that sum, the step's cost plus the change in potential). Then the universal floor from Section 3, "any run from NO to YES paid at least one," is just the potential method's standard conclusion, the drop in potential from one to zero, and every system with the five parts that obeys the toll is an instance (the checked record carries the toll as a field, so that's every certification system). Two more facts fall out of the same picture. Under the toll, a run raises the flag at most as many times as its total cost; each raise uses up a unit and a unit isn't used twice. And if you demand the upper-bound version too, that the potential before a step covers its cost plus the potential after, then the two together hold exactly when raising the flag costs exactly one, everything else costs zero, and the flag never comes down; that's the sharpest version of the schedule, and notice that Section 3's toy doesn't meet it: the toy's VOUCH move, played from YES, makes a mark although it raises nothing, which the floor allows and the exact version forbids. (These two theorems let the price depend on the state as well as the move's name, which is wider than Section 3's per-name cost; a per-name cost is a special case.) Two older tools sit beside these, named for completeness: running a list of moves in a "writer" that accumulates a number returns the final state and the run's cost (`run_writer_is_run_and_cost`), and a per-move schedule is a "grade" fixed before the run (`run_graded_is_run`); neither embeds those frameworks' full type systems, and the files say so. The writer is the move Danner and Licata make in their cost semantics, a translation that carries a cost alongside a program's result (Danner, Licata and Ramyaa, ICFP 2015; Danner and Licata, *Journal of Functional Programming*,

2022). The grade is Katsumata's: a computation carries its resource as a grade, and the grades compose as the computations do ("Parametric effect monads and semantics of effect systems," POPL 2014). Orchard, Liepelt and Eades put grades into the types of a working language ("Quantitative program reasoning with graded modal types," ICFP 2019). Read against them, the counter is a grade carried in the state by the step rule, not in a type. And what does that difference even mean? A cost that lives in a type gets checked before the program runs, by whoever checks the types; a cost that lives in the state gets written during the run, by the step itself. Same number, different moment of truth.

So is mine an embedding of their frameworks, or something weaker? Half of it is exactly theirs, and the other half is left out on purpose. The counter *is* their writer: a step that moves the rest of the state without looking at the counter, hands back its cost, and has the cost added on is a computation in the writer over whole numbers with addition, and running a list of moves in that writer gives back the final state and the total cost, nothing more and nothing less (`run_writer_is_run_and_cost`). Every system in this book whose step never reads its own counter has that shape. What doesn't come along is the part of their frameworks that works before anything runs: the type systems that carry grades, Danner and Licata's bounding relation and the recurrences they extract for a program's cost. That isn't a hole waiting for a patch. It's the moment-of-truth difference I just named. Putting the count in the step is choosing the other moment, so the grade here is written by the step at run time and never checked ahead of it.

How close do the two moments come? Close enough to say exactly, in a limited form, argued here and not checked. Take a small structured language over a machine with this kind of counter: sequencing, branches whose two arms carry the same grade, and paid acts each graded with their charge, and grade every program the way Katsumata's discipline does, adding grades along the program. On loop-free programs the two agree on the nose: on every run that doesn't trap, the counter rises by exactly the grade. The argument is conservation set beside the grade rule; the counter's rise is the sum of the charges along the run, and the grade rule adds the same charges in the same order. Loops break the agreement, and in one direction only. A loop that runs a paid check that always passes once for every unit in some register costs that register's value on entry, and no fixed whole number is an exact grade for it; a grade that's a function of the inputs, the kind cost-aware type theories like calf (Niu, Sterling, Grodin and Harper, 2022) allow, gets exactness back. So the counter carries their semantics exactly, agrees with fixed grades wherever a fixed grade can be exact, and where a run's cost depends on its data it holds the exact number that a fixed grade can only bound. What has no counterpart in the step is the checking ahead of time: a type checker turning a program away before it runs.

There's one more neighbour, closer than any of those, and I'd rather name it than wait for somebody to name it at me. A number the step keeps up to date and nothing

else ever reads is what program-proof people call an *auxiliary variable*, or *ghost state*. Susan Owicki and David Gries added auxiliary variables to parallel programs in 1976, because some true facts about such a program can't be proved without one, under the rule that an auxiliary variable never steers the program. Jean-Christophe Filliâtre, Léon Gondelman and Andrei Paskevich, in 2014, worked out when ghost code can be erased without changing what the program computes. By that test this section's counter is ghost whenever no step consults it, and Section 10's honest extensions are built exactly that way: the base moves the same whatever the ledger and the record say. So what does putting it in the step buy, if the number is ghost anyway? Two things, as far as I can see. The number is there while the machine runs, where a window can show it (Section 7), and not only inside a proof somebody wrote afterwards. And the rule that keeps it is one rule for every run of every program, not an annotation on the one program somebody happened to prove. Whether a record stays ghost depends on what else the step does with it; the machine in Part IV has a part that isn't ghost, its table of checked facts, and that's where I'll say why. One last relative, from the graph side: Section 3's floor is a cheapest-path bound in a system with weights on its moves. Every path from the no-states to the yes-states crosses the line between them, and every edge across that line weighs at least one. Weighted automata go back to Marcel-Paul Schützenberger in 1961, and priced timed automata, which add clocks, were set out in 2001. The floor is about the smallest case of what they compute.

Exactly, for any certification system with states S , moves I , step, per-name cost, and reading:

Definition 6.2 (Potential, amortized cost). A *potential* is a function $\Phi : S \rightarrow \mathbb{N}$. A step has *non-negative amortized cost* under Φ when $\text{cost}(i) + \Phi(\text{step}(s, i)) \geq \Phi(s)$. The *flag potential* is $\Phi_{\text{cert}}(s) = 1$ if $\text{cert}(s) = \text{no}$ and 0 if it reads yes.

Defined as `amortized_nonneg` and `cert_potential` (`CostSemanticsComparison.v`).

Theorem 6.3 (The toll is the potential method). (1) *The toll holds exactly when every step has non-negative amortized cost under Φ_{cert} .* (2) *For any Φ with non-negative amortized cost and any run t from s , $\text{cost}(t) + \Phi(\text{run}(t, s)) \geq \Phi(s)$.* (3) *Hence, under the toll, a run from a no-state to a yes-state costs at least $1 - 0 = 1$.*

Proof. (1) Check the four combinations of the reading before and after: only no-to-yes has Φ_{cert} dropping, by one, so the inequality there is exactly $\text{cost}(i) \geq 1$; in the other three it holds for any cost. (2) Induction on the run, adding the per-step inequalities so that the middle potentials cancel. (3) Apply (2) with Φ_{cert} , which is 1 at the start and 0 at the end. $\square$

Checked as `a2_iff_nonnegative_amortized_cost`, `potential_telescoping`, `nfi_by_potential`, and `certification_system_is_potential_method` (`CostSemanticsComparison.v`), with the state-dependent-cost results `flips_le_cost` and `a2_and_aara_iff_exact` (`CostFrameworks.v`).

So what’s new, if the method isn’t? Only where it’s pointed. The potential method is a way of proving things about costs. What this book adds is a claim about *which* potential, the flag, and about what happens to that claim when you can only see a window of the state. The method is Tarjan’s. The reading is mine, and I’m about to find out what it can and can’t survive.

The flag potential has relatives in physics-flavoured mathematics too, which I found after the fact. Resource theories study what you can turn into what when some operations are free: Bob Coecke, Tobias Fritz and Robert Spekkens gave the general definition in 2016, Fritz worked out its algebra in 2017, and Tomáš Gonda and Spekkens catalogued the numbers that never go up under free operations, which they call monotones, in 2023. Read with “certified” as the resource and one mark as the currency, the flag potential is the kind of cost monotone they build, measured in that currency, and the floor is its simplest case. Does that make the toll a resource theory? Only in the trivial way: the conversion costs exactly one mark and that’s the whole theory. It’s a neighbour to know about, not a result.

6.6 What the counter is, said once

A receipt is a total that lives in the state, only climbs, equals the sum of the charges on every path including the failures, and is the only such total for its schedule. Each of those four, counting “lives in the state” as the one the others are about, had a home before this book: the stick, Lamport’s clocks and Floyd’s ranking functions, the telescoping sum, the induction. None of them is physics. None of them says what the charges should be. What they say is that once you’ve chosen the charges, the counter isn’t a second opinion; it’s the arithmetic. And the toll on top of it is the potential method with the flag as potential, which is to say a trick from 1985 wearing the question from Section 2.

Here’s one thing the counter can’t do just by existing: if you can only see *part* of the state, can you tell what the counter is? Can you even tell, from outside, which step paid? The stick works because you can lay the two halves together. What if you’re only ever shown one half, through a window, and the window doesn’t show the notches?

One piece of that has a short answer, and I’ll give it now so the next section can start from it. A window, for now, is just the part of the state an outside observer is shown (Section 5’s display was one, and Section 7 is about windows in general). Call a window *counter-complete* when some rule, applied to what the window shows, returns the counter on every state; *flag-complete* likewise for the flag. Say a window *forgets the counter* when changing the counter and nothing else never changes what the window shows, and likewise for the flag.

The short answer: a window that forgets the counter is never counter-complete, whatever else it shows, and a window that forgets the flag is never flag-complete. Take any state and make two copies of it, one with the counter set to 1 and one with it set to 2, everything else held still. The window shows the same thing for both, so any rule gives the same answer for both, and one of the two answers is wrong. The flag is the same argument with yes and no. On the stick: a window that shows you the wood's colour and its length but not the notches can't tell you what's owed, and staring harder at the colour won't help. (The argument needs a state space where the counter can be changed on its own, the rest held still; that's what "forgets" quietly assumes. The machine I build later is such a space, and it has a checked pair of its own: two runs with the same window, one paid and one not (Theorem 14.14).)

Be careful what that doesn't say. It says neither part can be dropped. It doesn't say a window showing both is the smallest window that works, because nothing here orders windows, so there's nothing for it to be smallest in. And it's about reading the counter back, not about which step paid. That second question is harder, and it's the next section's.

7 Windows and shadows

I left you with the stick split in half and asked what happens if you can only ever see one half through a window that doesn't show the notches, and with a harder question under that one: can you tell, from outside, which step paid? This section is about windows. It's a section where the mathematics is so small that I'm going to state it in full and then spend most of the pages on what it's *not*, because the smallness is the trap: the fact is a one-liner and people keep trying to climb over it with cleverness, and you can't, and seeing why you can't is the content.

7.1 A cave, and the people who got good at it

This one is Plato's, and the part I need is near the end. In the seventh book of the *Republic*, Plato has Socrates describe prisoners chained in a cave since childhood, facing a wall, with a fire behind them and people carrying objects along a walkway between the fire and their backs. All the prisoners have ever seen is the shadows on the wall. To them, Socrates says, the truth would be nothing but those shadows. Plato did believe the real world is hidden from us, and the cave says that too, but it's not what I need from the story.

What I need is one line further on. The prisoners, Socrates says, hand out honours to whoever is quickest at spotting the shadows as they pass and best at guessing which comes next. Read that again. The prisoners *get good at it*. They build a whole science of shadows, with experts and prizes, and it works, on shadows. What they can't do is say anything about a carried object that leaves no mark on the wall. Not because they're stupid. Because the information isn't on the wall. Skill inside a window doesn't reach past the window, no matter how much skill. That's the fact. Plato dressed it as a

lesson about education, and it is one, but strip the dressing and it's a sentence about functions, and it's going to be the whole section.

7.2 The shadow of a cube

Do this one tonight with a lamp. Hold a cube face-on to the light: its shadow is a square. Now hold up a square of card the same size: same shadow. A person who sees only the wall can't tell you whether what's casting the shadow has any thickness at all. Ask them, and they'll tell you, truthfully, that they can't know, and if they're honest they'll also tell you that no amount of staring at that square will change it.

Now turn both a quarter turn. The card's shadow collapses to a line. The cube's stays a square. Now the wall-watcher knows. What changed? Not their cleverness. The *window*. A second look from a different angle is a bigger window than one look, and it was bought with a turn and a second look. Edwin Abbott wrote a novel about a window like this in 1884, *Flatland*. A Flatlander never sees a shape. Abbott's narrator says it in his first section: nothing was visible to them, or could be, except straight lines. Put a coin flat on a table and lower your eye to the table's edge, and the coin turns into a line; that's the whole view. What gives a Flatlander anything more is fog. A line that's near is bright and a line that runs off into the fog goes dim, so a trained eye can read a triangle or a pentagon off how fast the brightness fades toward the ends. The training takes years, and in the story only people with money get it. That's a window that got bigger, and it was paid for in time.

Then a sphere comes through the narrator's plane. What his eye gets is a line whose ends fade too fast for any flat figure he knows, changing size in a way no flat figure could. He only settles that it's a circle by feeling it, and the circle grows from a point to its widest and shrinks back to nothing as the sphere rises through. From one slice he can't put a sphere together. From the slices one after another, seen and felt and argued over, he starts to. The escape from a window is never "think harder." It's always "a different window," and a different window always costs something: a turn, a second look, years of training, time.

Who was Edwin Abbott? Not a mathematician by trade. He was headmaster of the City of London School for twenty-four years, a clergyman, and the author of more than forty books, a Shakespeare grammar among them. He put *Flatland* out under the name "A Square." His full name was Edwin Abbott Abbott, two A's, and the pen name may be a joke on that, A squared; nice if true, but take it as a lead. Here's the part I like best, from a sketch of his life on the geometer Thomas Banchoff's pages: *Flatland*, by far the most read thing he wrote, isn't mentioned at all in his obituary in the *London Times* or in his entry in the *Dictionary of National Biography*. The official record of the man was a window too: it showed the headmaster and the grammarian and left out the one book people still read. That changes nothing in the argument. It's the argument, acted out by an obituary. (*Flatland* already turned up once in this book, in Section 1.)

7.3 The whole fact, in one paragraph

Here's the mathematics, and I promise it's all of it. You have a set of states, the real things. You have a window, which is a function that takes a state and gives you what you can see of it: the shadow, the slice, the half-stick without notches. And you have a question about the state, which is also a function: it takes a state and gives an answer. The thing you want is a *decoder*: a rule that takes what you can see and gives the right answer to the question, every time. When does one exist?

Think about two states that cast the same shadow. The decoder sees the same shadow for both, so it gives the same answer for both. So if the question has *different* answers on those two states, the decoder is wrong on at least one of them, and there's nothing it can do about it, because it never saw the difference. That's the whole fact. If a decoder exists, then the question must already give the same answer on every pair of states that look the same through the window. Mathematicians call the set of states that cast one particular shadow the *fibre* over that shadow (think of the shadow as a point on the wall and the fibre as the bundle of all the things that could have cast it), and the sentence is: a decoder exists only if the question is constant on every fibre. That's the theorem just below, and the "only if" is the direction that bites. The other direction, "if the question is constant on fibres then a decoder exists," is true too, but it needs you to be handed, for every possible shadow, one actual state that casts it, so you can look up the answer there, and the handing is not free either, since someone has to supply a representative for every shadow. (You can always fill a gap with a default answer. The default doesn't recover what was really there.)

Exactly. Let S be any set of states, V any set of views, $w : S \rightarrow V$ a window, and $q : S \rightarrow A$ a question with answers in any set A . A *decoder* is a function $d : V \rightarrow A$ with $d(w(s)) = q(s)$ for every state s . The question is *constant on fibres* when $w(s) = w(s')$ implies $q(s) = q(s')$.

Theorem 7.1 (Decoders need constant fibres). *If a decoder exists, the question is constant on fibres. Conversely, if the question is constant on fibres and you are handed, for every view v , a state $r(v)$ with $w(r(v)) = v$, then $d(v) = q(r(v))$ is a decoder.*

Proof. If $w(s) = w(s')$ then $q(s) = d(w(s)) = d(w(s')) = q(s')$. Conversely, $d(w(s)) = q(r(w(s)))$, and $r(w(s))$ has the same view as s , so by constancy its answer is $q(s)$. $\square$

Checked as `decoding_requires_fiber_constancy` and `selected_representatives_give_decoder` (`ObservationPolicy.v`), for arbitrary types.

Try the smallest case, by hand, because it'll keep coming back. A state is a pair: a position, 0 or 1, and a hidden flag, up or down. The window shows the position only. Two moves from (0, down): one sets the position to 1 and raises the flag; the other sets the position to 1 and leaves the flag alone. Both land on position 1. Now write down every possible decoder from "what I see" to "is the flag up": there are exactly four

functions from two positions to two answers. Check each against both results. None of the four gets both right, because both results show position 1 and one has the flag up and one doesn't. That exhaustive check is the theorem on this example; the general statement is the one-line argument above, and that's what's checked. There's no fifth function hiding somewhere.

7.4 People who found this on their own

The fact is so plain that it got discovered separately in every field that has to deal with seeing part of something, and I want to lay them beside each other, because that's the kind of evidence I trust: not that one smart person said it, but that people who'd never met kept walking into the same wall.

In 1956 Edward Moore, working on the small switching machines that became computer hardware, asked when an experimenter who can only feed a machine inputs and watch its outputs can tell two of its internal states apart. His answer gave a precise meaning to "indistinguishable from outside," and a theorem with a sting in it: even when every pair of states can be told apart by *some* experiment, there are machines where no single experiment tells you which state you started in. What you can always do is run an experiment after which you know which state you're in *now*. In 1960 Rudolf Kalman, whose filter is in the navigation of aeroplanes and phones, asked the same question for continuous systems, called it *observability*, and gave a test (a matrix rank, which you can compute) for whether watching the output over time pins down the internal state. His unobservable states are exactly a fibre: states whose difference never reaches the output, and, this is the part people miss, never will, however long you watch. More samples of the same window only repeat the same fibre. Try it on a car: if your only instrument is the speedometer, two cars driving side by side at the same speed read the same forever, and nothing on that dial will ever tell you which mile marker either one started at. Swap the speedometer for an odometer and you can recover both. Same car, different window.

Where did Kalman's filter go first, before aeroplanes and phones? Toward the Moon. In the autumn of 1960 Kalman called an acquaintance, Stanley Schmidt, at NASA's Ames Research Center, and came to visit, not knowing what Schmidt's group was working on. The group, a branch with eight people doing the analysis, was stuck on how a spacecraft going around the Moon and back could work out where it was from what its crew could measure through an optical sight: a few angles to the Earth and the Moon. That's a window problem with a capsule in it. The state is position and velocity, the view is a handful of angles, and the job is to keep a running best guess at the state from a sequence of views. Kalman presented his new filtering paper to them, and with real work (their problem wasn't linear and his theory was) it fit, and it went into the feasibility studies for navigating Apollo. Schmidt and Leonard McGee, telling the story in 1985, call it a fortunate sequence of events and say the solution might have gone undiscovered but for that meeting. What it changes: it shows the limit from the useful side. The filter didn't make the angles show more. It made a

sequence of views, over time, pin down the state, and it can only do that when the state is observable from those views, which is Kalman's own condition. The odometer, not a cleverer speedometer.

In 1977 Patrick and Radhia Cousot built a whole discipline of program analysis, called abstract interpretation, on throwing information away *on purpose*: run the program over coarse descriptions of its values instead of the values, and accept that the answer you get has to be true of *every* real value the coarse description could stand for. That last clause is fibre constancy as a design rule. And in 1970 Ted Codd, inventing the relational database, made programs independent of how the data is stored, and the relational systems that followed gave users *views*, deliberately partial windows onto the stored data. A decade after Codd, Bancilhon and Spyratos set out the view-update problem: if a user changes what they see through a view, which of the many underlying states that cast that view should the database move to? Several are legitimate. The fibre, from the writing side.

And Parnas, from Section 2, hid things on purpose: a module is characterised by the design decision it *hides* from its callers. Hiding is a window. The caller can't later read back what was hidden, including the things it later wishes it could check, and that's the price side of the coin whose benefit side Parnas was selling. None of these people were doing philosophy. They were trying to build filters, databases, compilers, and modules, and the wall is the same wall in each.

One more, from information theory. Oliver Kosut and Jörg Kliewer (2018) asked when a receiver on a channel with a possible attacker, and no shared key, can always either get the message right or notice the attacker. The answer is all or nothing: the rate is zero, or it's everything the channel would carry with no attacker at all, and which one depends on a condition they call *overwritability*, that the attacker can make what the receiver sees look exactly like an honest transmission of some other message. That's a printer, a way to put the same thing in the window without the event behind it. The fibre fact is their condition in its plainest form, with the probabilities taken out.

And one from machine learning, where the two computers turn up with a lawyer attached. Anvith Thudi, Hengrui Jia, Ilia Shumailov and Nicolas Papernot (2022) asked how a company could prove a trained model had forgotten someone's data, and found that the same model can come out of different training sets, so the finished model can't show whether a point was used or removed. The only claim they found you can audit is about the procedure that was run. Same wall: the finished state is a window, and what happened is in the run.

Philosophers who study testimony, which is the trade of being told things, have the same wall from the other side. Kenny Easwaran (2009) asked why mathematicians will accept a proof too long for anyone to check by hand and refuse a probabilistic one, even one far more certain. His answer is a property he calls *transferability*. Given a deductive proof you can learn from the prover without trusting the prover, because

every step is there for you to check; a probabilistic proof hands you only the prover's word about what the dice did. That's the clerk's problem below in his vocabulary, and it's the book's answer too: don't take the transcript, carry the thing and run the check yourself. (That reading is mine. He never mentions machines.) John Locke complained about the alternative in 1690, in the first book of the *Essay*: believing something because someone with authority said it is not knowing it. The opposite default has a champion as well. Tyler Burge (1993) argued that you're entitled to accept what you're told, if it's presented as true and it makes sense to you, unless you have a reason not to. That's the default this book declines. A claim is on the record because its object and its paid event are there, or it isn't on the record. Jennifer Lackey (2008) gave a teacher who doesn't believe the theory she's required to teach and still passes knowledge of it to her students, so what the hearer gets can't just be what the speaker believes. Sanford Goldberg (2010) argued that whether your word justifies me can depend on how reliable the processes behind your word were, which sit outside my head. Christoph Jäger and Nicholas Shackel (2025) showed, in a Bayesian model, that the speaker knowing is neither necessary nor sufficient for the hearer to know. Wesley Holliday (2009) put testimony into a logic of belief change, with an authority graph recording whose word each agent takes and complete rules for what hearing a speaker does to a belief. Security people had a leash of that kind two decades earlier: in the logic of authentication of Michael Burrows, Martin Abadi and Roger Needham (1990), if I believe an authority has jurisdiction over a statement and I believe the authority holds it, I believe it. That's Burge's default with the authority named in advance. None of them is about machines. Each is about what survives the trip from one head to another, which is the copier's problem with philosophers' names on it.

Distributed-systems people hit the wall too, and one pair of them measured how far it goes. Andreas Haeberlen and Petr Kuznetsov (2009) asked which faults an outside observer can detect at all. The problem is solvable for exactly two classes, omission faults, where a node fails to send something it should have, and commission faults, where it sends something it shouldn't. A fault whose effects never reach anything observable can't be detected by any algorithm, and neither can one that looks from outside the same as something innocent. That's the fibre fact with an auditor attached. Their group's PeerReview system (Haeberlen, Kouznetsov and Druschel, 2007) is the matching construction: every node keeps a secure record of the messages it sent and received, and a checker replays the record against a reference implementation, which exposes the faults a correct node got to see. That's close to the whole-state escape from the three escapes below, built: ship the log, and the checker re-runs it. Ralf Küsters, Tomasz Truderung and Andreas Vogt (2010) gave a general definition of accountability, a verdict naming who misbehaved, and showed that verifiability, the property that an outsider can check a result, is a weak form of it. Radha Jagadeesan, Alan Jeffrey, Corin Pitcher and James Riely (2009) built an operational model of systems that enforce a policy after the fact by audit, and studied the trade between the power of the auditor, the cost of the audit and what's asked of the agents. Robert Künnemann, Ilkan Esiyok

and Michael Backes (2019) made the idea a mechanized check, with Certificate Transparency among their examples. In September 2026 Timo Treitz and Künnemann gave CT the full treatment (Section 12 has my fragment of it) and showed that plain CT gives accountability only if the log is assumed honest, an assumption the SCT auditing extension can drop and the gossiping extension can't. An honest log is the commitment bit anyone can set, from the three escapes below, with a name on it: somebody's contract, trusted at the step.

7.5 The clerk with two photographs

Now point it at the counter, because that's why this section is here. Suppose you want to bill for raising the flag, exactly: at least one mark for every step that raises it, and nothing for a step that doesn't. And suppose the person doing the billing is a clerk who sees only a *window* on the state, a photograph before the step and a photograph after. The clerk's bill is a function of the two photographs. Can the clerk bill exactly?

Not if the window has the wrong kind of blind spot. Here's why, and it's the cube again. Suppose there are two steps that give the clerk the same before-photo and the same after-photo, and one of them raised the flag and the other didn't. (The two-move toy above is exactly such a pair; so are "a stamp goes onto a document inside a closed drawer" and "someone opens and closes the drawer.") The clerk has one bill for that pair of photographs. If the bill is at least one, it overcharges the step that didn't raise the flag. If it's zero, it undercharges the step that did. Any bill that never misses a raise bills something that wasn't a raise, and no bill from those photographs is exact. The fix is the plain one: put the flag in the photograph. If the window shows the reading, then "one when the reading goes from down to up, zero otherwise" is exact, and such a window has no pair like the one above. That's enough. It isn't the only way: a window can hide the flag and still have no colliding pair, in which case a bill from the photographs can be exact; what the file proves the other way is only that a colliding pair rules exact pricing out. And "exact" here has a definition: at least one on every step that raises the flag, nothing on any step that doesn't. That's all the clerk's problem needs, and the bill "one for a raise, zero otherwise" meets it (`step_price_is_exact`). So it's exact about *which* steps get charged. It isn't exact about *how much*, and it doesn't pick out one bill: "two for a raise, zero otherwise" meets it too. Pinning the bill down takes one more demand, a ceiling: over any run, the total never exceeds the number of raises. With the floor and that ceiling there's exactly one bill left, one per raise and zero otherwise; that's an if and only if, proved for any system whose bill is decided step by step (`exact_commitment_pricing_characterization`, `CommitmentPredicateAdequacy.v`). These are stated for any states, any steps, any window; they're the cube theorem with a price in place of a decoder.

Same clerk, now in symbols, so you can check every word I just used. Take the five parts of Section 3 with a window $w : S \rightarrow V$, and say a step (s, i) raises the flag when $\text{cert}(s) = \text{no}$ and $\text{cert}(\text{step}(s, i)) = \text{yes}$. A *price* is any function of a state and a move. It *meets the floor* when every raise costs at least one, and *never overcharges*

when every non-raise costs zero; together, it is *exact* (about which steps are charged; the amount is pinned only by the whole-run ceiling above). A *window price* is one computed from the two photographs only: $\text{price}(s, i) = b(w(s), w(\text{step}(s, i)))$ for some bill $b : V \times V \rightarrow \mathbb{N}$. A *collision* is two steps (s_1, i_1) and (s_2, i_2) with $w(s_1) = w(s_2)$ and $w(\text{step}(s_1, i_1)) = w(\text{step}(s_2, i_2))$, the first a raise and the second not.

Theorem 7.2 (No exact window price across a collision). *If the window has a collision, every window price that meets the floor charges at least one on some step that is not a raise; so no window price is exact. If instead the reading is a function of the view, $\text{cert}(s) = r(w(s))$ for some r , then the window price “one when r goes from no to yes, zero otherwise” is exact, and the window has no collision.*

Proof. Across a collision both steps get the same bill $b(v, v')$. The floor makes it at least one because the first step is a raise; the second step, not a raise, pays it. If the reading is r of the view, the window price reads the raise directly, and two steps with the same photographs either both raise or both don’t, so there’s no collision. $\square$

Checked as `shadow_floor_overcharges`, `shadow_cannot_price_exactly`, `window_showing_reading_prices_exactly`, and `window_showing_reading_has_no_collision` (`ShadowPricing.v`), for arbitrary types.

So the exact price of raising the flag lives in the step itself, or in a window that shows the flag. A window that contains a pair like that one can’t carry it. Whether a particular machine has such a pair, for a particular window, has to be checked machine by machine and window by window; the theorem only tells you what a pair does once you’ve found one.

There’s a tidy way to say what the best window prices are, and it isn’t mine. Give each pair of before-and-after photographs the highest charge of any step that casts that pair, and you have the cheapest window price that never undercharges; give it the lowest, and you have the dearest one that never overcharges. Those two are the “there is” and “for all” of logic taken along the window, which F. William Lawvere showed in 1969 (“Adjointness in foundations”) are the two adjoints of pulling a statement back along a map. Cockett, Geng, Scandolo and Srinivasan (2023) used the same move, as Kan extensions, to extend resource measures along a map. In those terms the theorem says: a window price is exact exactly when the two coincide, and a collision pulls them apart.

Control engineers have been hunting pairs like that for a long time, under other names. Discrete-event systems is the field of machines that jump from state to state on events, some of which an outside observer sees and some of which it doesn’t. In 1995 Sampath, Sengupta, Lafortune, Sinnamohideen and Teneketzis asked when such an observer can always work out, from what it sees, that a hidden fault event has happened, within a bounded number of further steps. They called that *diagnosability* and gave a test

for it on finite machines. Saboori and Hadjicostis (2007) asked the question the other way round, for secrets: a system is *opaque* when an observer can never be sure it's in a secret state, because some run outside the secret always looks the same. Read the raise as the hidden event and the window as what the observer sees. Then a collision is the raise failing to be diagnosable with no delay allowed, and a window that can't tell the flag is the flag's state being opaque to it (Theorem 14.15, later, is that, for the small machine). That's my reading, not their words, and their versions are richer than mine: they allow delays, look at whole runs, and come with algorithms. What this section adds is only the price: what a bill computed through the window does across one such pair.

Why can't the clerk just watch the step itself? A clerk who does, who sees the whole state before and the move, bills exactly with no trouble at all: "one for a raise" is a function of exactly those two things, and it's the step price above. So the collision bites an observer who doesn't see the step, or who's handed an account of the step and can't trust the account. That second kind is the opening chapter's trace problem come back. If the names of the moves arrive in a log the program wrote, the program can write whatever names it likes, and a clerk reading that log is reading one more window, one the program painted. Seeing the move means being underneath the program, where the step is.

7.6 The three escapes, and what each one costs

Suppose you're the person outside, and you want to be convinced that a run paid. The window you've got doesn't show the counter. Here's how bad that is, in the smallest setting I can build. Take a counter machine from Section 6, give its states a position and a flag as well, and give it a window that shows the position and nothing else. Now take two runs from the same start, with the position at 0, the flag down and the counter at 0. Run *A* makes one move that raises the flag, costs one, and lands at position 1; its counter reads 1. Run *B* makes one move that does nothing but step to position 1, at cost zero; its counter reads 0. Call the pair of photographs, position 0 then position 1, the *transcript*. Both runs leave exactly that transcript. A *verifier* is a rule that takes a transcript and says accept or reject. It is *sound* when every transcript it accepts comes only from states where the claim holds, and *complete* when every transcript from a state where the claim holds is accepted. The claim is "the counter is 1."

No verifier on these transcripts is both, and the proof is three lines. Suppose one were both. Run *A* ends with the counter at 1 and leaves the transcript, so completeness says the verifier accepts it. Now soundness, on that same accepted transcript: every state that leaves it satisfies the claim, and run *B*'s does. So run *B*'s counter is 1. It is 0. That's the cube again, with a verdict in place of an answer: a verifier sees only what the window kept, and two states that the window can't tell apart get one verdict between them.

Now dress the transcript up. Take any richer kind of transcript, with a way to squash one down to its bare photographs, and say a verifier on the rich ones *only looks at the bare part* when two rich transcripts that squash to the same photographs always get the same verdict. If some rich transcript left by run *A* and some rich transcript left by run *B* squash to the same photographs, then no verifier that only looks at the bare part is sound and complete either: it gives those two the same verdict, completeness makes that verdict accept, and soundness then lies about run *B*. Notice what that needs. The pair has to survive into the richer kind of transcript; a design whose extra part tells *A* from *B* doesn't meet the premise, and the argument says nothing about it. So it doesn't classify every transcript design. What it says is that a verifier that only reads what the bare window keeps fails on that pair.

Economists got to the same wall by another road. Jerry Green and Jean-Jacques Laffont (1986) asked what a mechanism can do when people can't claim just anything about themselves, only what their situation lets them back up, and Jesse Bull and Joel Watson (2007) built that into mechanism design as *hard evidence*. In their terms, an audit can tell apart only the cases the evidence on offer tells apart. That's the cube with a ledger in it.

What can you do? Here are three things that work, and nothing says they're the only ones; as far as I can tell, cryptography uses all three. Each rests on something you have to trust, and I'll say what, because "a price" in the counter's sense is not what any of them costs.

You can be shown the *whole state*. Then there's nothing to decode; you read the counter off the state. The price is trust: you have to believe the state you were shown is the state there was.

You can be shown a *commitment*: the ordinary window plus one extra bit that is set when, and only when, the claim holds (here, that the counter reads exactly one). Reading that bit gives a verifier that is sound and complete for that claim, *if* the bit is binding (set only when the claim holds) and honest (set whenever it holds). The trust is in those two "ifs," which are a contract about who can set the bit. The contract can be met, and a bit anyone can set breaks it, which is the one-line version of why an unsigned log is just another window. Nothing in that picture has a hardness assumption in it; whether a real commitment scheme meets the contract is a question for cryptography, not for this book. (A clash of words I should own: "commitment" here is the cryptographers' word, something bound to a claim, which in their schemes also hides the claim until it's opened, and this bit doesn't bother hiding anything. The opening chapter's commit is a different act, the closing of a question. Same word, two jobs. The sealed envelope has an exact theory of its own: Andreas Winter, Anderson Nascimento and Hideki Imai (2003) found the best rate at which a noisy channel can carry commitments, and Claude Crépeau, Rafael Dowsley and Nascimento (2020) worked out how that rate drops as the committer gets some control over the noise,

which is the opening chapter's "take the pen away from the program" with a formula on it.) In 1999 Bruce Schneier and John Kelsey worked out how to make a log that an intruder who takes over the machine can't quietly rewrite, and the honest scope in their paper is the honest scope here: it protects what was written *before* the break-in, through keys, and nothing after. Mihir Bellare and Bennet Yee had named that property *forward integrity* in a 1997 report and proved their scheme has it. Point it at an AI agent that keeps its own log of what it did, and the line moves to the very start: an agent that writes the wrong thing from its first step is on the far side of the break-in for the whole log. A 2026 preprint on tamper-evident audit trails for such agents, built from hashes anchored on a public blockchain, is even called Agent Flight Recorder (Bindschaedler, Botha and Siebenbrunner), which is the opening chapter's word, and the opening chapter's fingerprint paragraph is the part of this book that applies to it.

Or you can *ask*. Interaction: the outside party gets to put a question and gets an answer that pins the thing down. The price is the honesty of the answerer; a liar's answer is another shadow. Moore's experimenter chose inputs; the zero-knowledge proofs of Goldwasser, Micali and Rackoff in 1985 are the refined form, where a verifier is convinced by a conversation designed to cut exactly the line between true and false and nothing else, with a mathematical guarantee against a cheating prover. I mention zero-knowledge to say what I'm *not* claiming: the asking in this section is one round, with one fixed question ("tell me your counter"), and nothing in it models a cheating prover; the honesty is assumed, not enforced. That's a toy of the real thing, and I'm not going to let the real thing's reputation stand behind the toy. (Joseph Halpern, Yoram Moses and Mark Tuttle gave zero knowledge a knowledge-based analysis in 1988, which is where I'd start for the logic of it.)

The real thing even has a cave in it, which after Plato I can't resist. In a paper for the 1989 CRYPTO conference called "How to Explain Zero-Knowledge Protocols to Your Children," Jean-Jacques Quisquater and Louis Guillou, with their families on the byline and Tom Berson, told it as Ali Baba's cave: a ring-shaped tunnel with a magic door at the back that opens only for a secret word. Peggy goes in down one side while Victor waits outside; then Victor walks to the fork and shouts which side she should come out of. If she knows the word, she always can. If she doesn't, she gets away with it only when Victor happens to call the side she went in. Play it at the table with a coin for Victor's shout: one round, a cheat gets through half the time; twenty rounds, about one time in a million. And here's the part that belongs in this section. In the story, Victor ends up convinced, but a video of the whole thing convinces nobody else, because a staged run, where the two agreed beforehand which sides he'd shout, makes exactly the same recording. The recording is a window, and the real run and the fake one cast the same shadow. That's the cube in the cryptographers' favourite bedtime story, and it's why the real thing has to ask live, many times, against a prover who might be lying: everything my one round leaves out.

Each escape is a window with something added: the whole state, a bound bit, an

honest answer. None of them is “a smarter verifier on the old window.” Cleverness lives on the wrong side of the projection. What’s true of any machine is the cube.

Here are the three constructions exactly, for the pair above, with the relations written down. The relation matters: it says which state a transcript is allowed to describe. If it quietly assumes a truthful answer, the verifier has not earned the right to trust a liar. Write A and B for the states runs A and B end at, μ for the counter, and τ for their shared transcript.

Definition 7.1 (Three richer transcripts and their explanations). Keep the states A, B and their shared bare transcript τ , with $\mu(A) = 1$ and $\mu(B) = 0$. Write $E_0(s, t)$ for $t = \tau \wedge (s = A \vee s = B)$. The claim is $\mu(s) = 1$.

A *full-state transcript* is a finite list t of states. Define $E_S(s, t)$ to mean that t is nonempty and its last entry is s . The verifier rejects the empty list; otherwise it accepts exactly when the last entry’s counter is 1.

A *commitment transcript* is a pair (t, b) of a bare transcript and a Boolean. For any chosen explanation relation E_C , the *commitment contract* is the pair of implications, for all s, t, b :

$$E_C(s, (t, b)) \wedge b = \text{true} \implies \mu(s) = 1,$$

$$E_C(s, (t, b)) \wedge \mu(s) = 1 \implies b = \text{true}.$$

These are binding and honesty, respectively. The verifier accepts exactly when b is true. One relation meeting the contract is $E_H(s, (t, b)) \iff E_0(s, t) \wedge (b = \text{true} \iff \mu(s) = 1)$.

A *response transcript* is a pair (t, r) of a bare transcript and a natural number. Define $E_I(s, (t, r)) \iff \mu(s) = r$. The verifier accepts exactly when $r = 1$. This relation imposes no condition on the bare component t ; the response alone carries its entire truth condition.

Each construction assigns verification the constant abstract cost 1.

Theorem 7.3 (Three conditional escapes). *The full-state verifier is sound and complete relative to E_S . For every E_C satisfying the commitment contract, the commitment verifier is sound and complete relative to E_C . The response verifier is sound and complete relative to E_I . Each has cost 1 under the stipulated cost assignment. The honest relation E_H satisfies the commitment contract; the unchecked relation $E_U(s, (t, b)) \iff E_0(s, t)$ does not.*

Proof. For the full-state verifier, an explaining state is precisely the last entry. Reading that entry’s counter therefore accepts exactly when the explaining state’s counter is 1. An empty transcript is rejected and has no explaining state.

For the commitment verifier, acceptance says b is true. Binding then gives the claim

for every explaining state, which is soundness. Conversely, honesty says that any explaining state satisfying the claim has a true bit, which the verifier accepts. The equivalence in E_H gives both implications directly. Under E_U , however, B explains (τ, true) even though $\mu(B) = 0$. That violates binding.

For the response verifier, explanation gives $\mu(s) = r$. Acceptance adds $r = 1$, so $\mu(s) = 1$. Conversely, $\mu(s) = 1$ and explanation give $r = 1$, hence acceptance. The three cost claims follow from the assigned constant. $\square$

Checked, for any pair like A and B , over arbitrary types of states, claims, transcripts and explanation relations; nothing in it uses more about the two states than their claim values and their shared transcript. The collision argument is `ver_collision_blocks`, the richer-transcript one is `ver_no_factor` (a verifier sound and complete for the claim under an explanation relation cannot give the same verdict to two transcripts with the same bare part when one is explained by a state with the claim and the other by a state without it), and the three escapes are `ver_full_state_escape`, `ver_commitment_escape` (under the contract, with `ver_honest_meets_contract` and `ver_unchecked_breaks_contract` for the two relations), and `ver_response_escape` (`VerifierSmall.v`). The cost 1 in the definition above is not in any of those statements; it is a number I wrote down, and no proof reads it.

On a real machine the collision is a theorem, and it doesn't depend on the toy. Give a run from a loaded start $(1, a, b)$ the *bare transcript* made of a, b and the base window the run ends with, and say a state *explains* a bare transcript when some run from that start ends at it showing that window.

Theorem 7.4 (No bare verifier on a Thiele-complete machine). *Let a machine be Thiele-complete through an interface (Definition 14.7). (1) No function from bare transcripts to Booleans is both sound and complete, under that explanation relation, for the claim "the record is up." (2) There is a clean loaded state such that no function from bare transcripts to Booleans is both sound and complete for the claim "the ledger is at least 3 above what it was at that start." (3) For the small machine's own pair, a run CHECK, COMMIT, CERTIFY from $(0, 0)$, which ends with the ledger at 3, and three DEC's of a zero counter, which end with the ledger at 0, end with the same window, and no function on bare transcripts is sound and complete for "the ledger reads 3."*

Proof. Theorem 14.22(2) gives a loaded state and two runs from it that end with the same window, one with the record up and one with it down, and, from one loaded state, two runs whose ledgers are at least 3 apart. Both end states explain the same bare transcript. A verifier complete for the claim accepts that transcript, since a state where the claim holds explains it; a verifier sound for the claim then makes the claim true of every state that explains it, including the one where it's false. That's (1), and (2) is the same with the ledger. For (3) the two runs are written out in the statement: three

moves land on line 4 in both runs with both counters at 0, and the same two-state argument applies. $\square$

Checked as `ver_complete_no_record_verifier`, `ver_complete_no_ledger_verifier` and `ver_small_no_ledger_verifier` (`VerifierSmall.v`), the first two from `complete_hides_record` and `complete_two_runs` of `ThieleCompleteWindow.v`. Put with the three escapes, the whole picture is one statement for every Thiele-complete machine, `ver_corollary`, and for the small machine `ver_small_corollary`: the bare-transcript verifier doesn't exist; the collision survives into any richer transcript that keeps it, so no verifier that only reads its bare part works either; a verifier on full states exists; one on the bare transcript with a committed bit exists, sound and complete when the bit is honest, and the unchecked bit breaks the contract; and one on a truthful response exists. The collision is a premise of the richer-transcript half, and the contract is assumed; no fourth way is claimed to be impossible, and no hardness is proved.

The collision is gone, and no trick did it: the extra information pulls *A* and *B* apart. The full-state transcript shows the counter outright. The honest commitment for *A* carries a true bit and the one for *B* a false bit. The honest responses are 1 and 0. In none of the three is the verifier asked to tell identical inputs apart, which was the impossible job.

Now the limit, and I'd rather say it before you find it. A list of states has no authentication built into it, so nothing in it stops a liar from handing over whatever the verifier wants to see. The commitment contract is assumed, and the unchecked bit above is what happens when the assumption is dropped. The response relation only allows a truthful counter value, so asking a question here does not buy resistance to a cheating prover. And the cost 1 is an abstract price I wrote down, not a measurement of what it takes to authenticate a state, check a cryptographic proof or send a response. What these three show is which extra distinctions a verifier can use. They are not an adversary model, a security parameter or a hardness reduction.

So what would it take to make that commitment bit binding for real, against somebody who wants to cheat? Cryptographers had to face that one in the open, and the place I'd point is a short paper Manuel Blum presented at the CRYPTO meeting in 1981, "Coin Flipping by Telephone." His setup: Alice and Bob have just divorced, they live in different cities, and they want to flip a coin over the phone for who gets the car, with nobody they both trust to do the flipping. If Alice flips and announces, Bob has to take her word. If Bob calls it first, Alice can lie about how it landed. Blum's way out is for one side to commit first: send something that fixes the choice without showing it, and open it only after the other side has called. And what keeps that commitment binding isn't anybody's promise. It's a bet that a certain kind of big number is hard to factor. Push on that and you're standing on the honest edge of this subsection. In Blum's protocol, binding is earned, on the condition that factoring is hard. Here it's

written into the contract and assumed outright, and the unchecked bit above shows what's left without it. That's why I won't say these escapes beat a cheater. Blum's paper is the kind of thing it would take, and nothing like it is in this chapter.

Now point the three escapes back at the record this book is about, because that's where they bite hardest. A record kept in a machine's state is bits too. Hand someone outside a copy of those bits with a note saying "the flag is up," and they're reading a window: the run that really raised the flag and a box built to print that it did cast the same shadow. So putting the record in the step answers "was this earned?" for whoever can see the state and trusts the step that wrote it. For anybody else it moves the question, from "did it run?" to "do I trust the box that says it ran?" Crossing that line takes one of the escapes: the whole state, from a box you already trust; a bound bit, which in practice means a signature over the record or a quote from measured hardware (and Section 12 shows a quote vouches only for what was measured); or asking, against a box that might lie, which is the cryptographers' business. One more relative deserves its name here, *verifiable computation*: the box hands back its answer with a short proof that the answer is right, which a checker can verify without redoing the work (Bryan Parno, Jon Howell, Craig Gentry and Mariana Raykova's Pinocchio, 2013), and proof-carrying data passes proofs like that along a whole chain of parties who don't trust each other (Alessandro Chiesa and Eran Tromer, 2010). That's the strongest export there is, and notice what it exports: the claim, not the history. The proof shows the answer is right for that input. It doesn't show which machine did the work, or when, and any box with the input can make one. Nothing in this book supplies any of these escapes. The model's record is local to the machine that keeps it, and the small machine of Part IV keeps its record in its state with no signature on it at all.

7.7 What this isn't

Three things people reach for, and I'd rather take them off the table now.

It isn't mysticism. Nothing here says the window is false or the state is unknowable. The wall is a function, the objects are its inputs, and the only claim is that the function isn't one-to-one on the thing you asked about. A richer window recovers it. Plato's prisoners could have turned around.

Somebody always says Rice's theorem at this point, so let me say it first. It isn't Rice's theorem, though it rhymes with it. The logician Henry Rice, in 1953, proved that no program can decide any interesting property of what another program *does* from its code, and a cousin of the fact here is that a run which hasn't halted yet looks, so far, exactly like one that never will. But Rice is about what's *computable*. The cube is about what's *there*. The cube theorem holds for a toy with two states and nothing to compute; the obstruction is missing information, not missing cleverness, and I'd be overclaiming if I let the famous theorem lend its weight to the small one.

And it isn't "with enough data you can recover anything." Kalman's unobservable

states stay unobservable for all time. The question is never how much of the window you've got. It's which window.

So: the counter lives in the state, and whether anyone outside can read it depends entirely on what they're shown, and a window with a pair like that in it can't price the flag exactly, not with any rule, ever. Which raises the question I've been pushing in front of me since the buckets. Fine, the flag went up and was paid for and I can see it in the right window. What does that *entitle* me to? A paid flag is a paid flag. What's the claim behind it, and when am I allowed to lean on it?

8 Structural entitlement

I keep writing the word "vouch" and then taking it back. Let me stop doing that and say what it would take. A reading went from NO to YES, the toll charged the move that did it (Section 3), maybe the reading stays up (Section 4 says what it costs to make it stay), and the right window shows it (Section 7). Now: what is a later step *allowed to do* because of that? That's a different question from "is the thing true," and the gap between those two is the subject of this section, and I'd argue it's the subject of most of the arguments people have about evidence, so give me the long way round.

8.1 The coin that moved

Here's an experiment you can run on a friend, and I'd ask you to actually run it, because the feeling of it is the point. Put a coin under one of three cups while they watch. Say, cup one. Send them out of the room. Move the coin to cup three. Then, and this is the trick, put a *second* coin under cup one. Bring them back and ask: "Is there a coin under cup one?"

They say yes. They're right; there is. And they have a reason: they watched you put it there. True, and believed, and justified by what they saw. And it's not knowledge, and you can feel that it isn't, because their reason points at a coin that left. The coin that makes them right is not the coin they're thinking of.

That's a three-page paper from 1963 by Edmund Gettier, "Is Justified True Belief Knowledge?", which broke a definition people had leaned on since Plato, in about the time it takes to read it. The definition, which people read into Plato's *Theaetetus*, was that you know something when it's true, you believe it, and you've got good reasons. Gettier's two cases, and the cups, show a belief that passes all three tests and still isn't knowledge, because the reasons and the fact aren't connected; they just happen to agree. Bertrand Russell had a related case in 1948: somebody looks at a stopped clock at the one moment it happens to be right, and they're right about the time, and they know nothing.

Is 1963 the bottom? Not even close, and 1948 isn't either. The Stanford Encyclopedia's entry on the analysis of knowledge, which is where I'd send you to check me, gives a case from the eighth-century Buddhist philosopher Dharmottara: you're looking for

water on a hot day, you see water ahead, it's a mirage, and when you get there you happen to find water under a rock. And one from Peter of Mantua, in the fourteenth century: Plato is running right next to you, you take him for Socrates, so you believe Socrates is running, and Socrates is in fact running, in Rome, where you can't see him. Older still? Maybe. This is where the trail goes cold, which isn't the same as where it ends. What it changes is how much I trust the shape. People in different centuries, languages and traditions kept building the same trap by hand: a reason that points at one thing, and a truth that happens to be sitting on another.

What's missing is a wire from the reason to the fact. Ask your friend what they'd need, to be *entitled* to "there's a coin under cup one," and they'll tell you: lift the cup. Not "remember putting it there." Lift it, now, and look. That lifting is an event, it happens on the record, and after it the reason and the fact are the same object.

8.2 Plato nearly had it

Who was Theaetetus, though? The dialogue is named for a real person, and that side of it usually gets left out. He was a mathematician. In the dialogue he's a teenager, and he tells Socrates that his teacher Theodorus had been showing them, one case at a time, that the side of a square of area three, or five, and so on up to seventeen, can't be named by any fraction: the Pythagoreans' diagonal from Section 2, done again case by case. The young man's idea was to stop going one case at a time and find one word for all of them. Euclid's tenth book, on lengths that share no common measure, is thought to be largely his work, and he's credited with being the first to study the eight-faced and twenty-faced solids, two of the four Plato built air and water from. The dialogue opens with him being carried home to Athens from the army at Corinth, badly wounded and sick with the dysentery going round the camp, barely alive; the usual date is 369 BC. And the conversation itself has come down as a record. Euclid of Megara (not the geometer) says he took notes as soon as he got home, filled them out later, and had Socrates correct them whenever he came to Athens, and he has the result read aloud. So the dialogue about what turns a true belief into knowledge survives as a checked record of what was said. That changes nothing in the argument. I just like that the book about knowledge is named for the man who took the diagonal and made it general.

I want to be fair to the *Theaetetus* here, because it's easy to be quick with the ancients and they deserve better. Socrates, in that dialogue, doesn't just say "true belief plus reasons." He tries three different senses of what the "account" behind a true belief could be: a statement you can say aloud; a list of the thing's parts; and, third, the mark that sets this thing apart from everything else it could be confused with. He tries each, and each fails, and the dialogue ends with nobody knowing what knowledge is, which is one of the more honest endings in philosophy. But the third sense is the one I'd underline. An account as *the thing that distinguishes*: the test that gives the real case a value no rival gives. He couldn't make it work because he had no way to say what the account had to be attached *to*. Two and a half thousand years later, here's

this book's guess: the thing it has to be attached to is a state, and the account is an object sitting in it. I'm not claiming that settles Plato. I'm claiming he was pointing at the right thing, and that's my reading of the dialogue, which you can check against it in an afternoon.

Push on that once more, because Socrates gives his own reason for dropping the third account, and it isn't the one I just gave. If the account means *knowing* the mark that sets the thing apart, then knowledge has been defined as true belief plus knowledge of something, and the word you were trying to define is back inside its own definition. A circle. Does a check get out of it? It gets the word out: a test that ran and found the mark isn't defined in terms of knowing. Whether the test was any good is still a real question, and it's the question the rest of this section puts a price on. So I'd say the circle moves rather than vanishes. It moves somewhere you can look at it, which is the most I'd claim.

8.3 The shape, said once

So here's what I mean by entitlement, and it has three parts, and I'll try to earn each.

A run is entitled to a claim when three things are present. First, an *object*, carried in the state: a grouping, a relation between groups, a counter, a package of constraints. Not a belief; a thing you could print. Second, a *checked relation* between that object and the claim: a test that was actually run, with its result on hand, not "a test exists." Third, an *event on the record* that says this object was put there and paid for, so that anyone auditing the run can point back at the moment the claim was marked.

Said as a definition, so the rest of the section has something to point at:

Definition 8.1 (Entitlement). A run is *entitled* to a claim when three things are on hand: an *object* carried in the run's states; a *checked relation* between that object and the claim, meaning a test whose result is on the record, not a test that merely exists; and a *paid event*, a move of the run at which the object was put in place and which the price list charged at least one. In a certification system (Definition 3.1) a move that takes the reading from no to yes is such a move, because the toll charges it. This is the book's word, not a Coq definition.

Without the third part, the object may be sitting right there and the claim may be true, and the run still hasn't *earned* it. That's the cups: the coin is under cup one; the reason is disconnected from it; nobody lifted the cup. Put it on Section 3's sheet of paper, the one with the NO box, the YES box and the tally line, and grow the toy by one thing: the state is now the box you're in plus whatever is written on the sheet beside it. Next to the boxes, write a note: "coin under cup one, checked." Writing the note is a WORK move, and WORK costs nothing. The note is an object, it's on the paper, and it might even be true. What it doesn't have is a crossing on the record: no move took the reading from NO to YES when the note went down, and there's no mark under

the line for it. The note is raw data, forged in one write, and any system with the five parts lets you write one, because nothing in the five parts says what an ordinary move may put in the state. A note put down by the move that crosses has a receipt, and the receipt is the event. One limit on that word “receipt”: the crossing records that something was paid for at that moment. What the crossing move actually looked at before it crossed is up to whatever the move does, and the five parts don’t say.

And here’s the part to hold on to. Nothing in the five parts makes a later move *read* the reading. The step function is handed the state and a move name, and it’s allowed to ignore the YES box completely; plenty of systems will. So when I say a run is *entitled*, I mean what someone auditing the run may conclude from the reading and the receipt. The system doesn’t have to enforce it, and unless somebody builds it to, it won’t.

This is a good place to line up the three things the word *established* has had to mean so far, because they blur easily, and a blur here would make the book look like it proves more than it does. The yardstick is the opening chapter’s four conditions for a check being about this very claim: the same property, the same thing, the same edition, and an honest check.

What’s asked	Bare toll	Entitlement	Earned certification
A check ran	not required	a test result on the record	a passing check before the commit, on every run from a clean start
Same property	not asked	not asked	enforced by the step
Same thing	not asked	not asked	enforced by the step
Same edition, nothing written between	not asked	not asked	enforced by the step
Honest check	not asked	not asked	proved of the checker, once, outside the record
Could have failed	not asked	not asked	not policed
Paid for	at least one mark	at least one mark	at least three marks for the chain

The first column is Section 3’s bare toll: a paid crossing, and nothing says a check happened at all (Section 9 has a remark saying exactly that). The second is this section’s entitlement. The third is the small machine of Part IV, whose step won’t let a commit through without a passing check of the same property, on the same thing, at the same edition, for every program from a clean start (Theorem 14.9), whose checker comes with a proof that it says what it means (Lemma 14.1, Theorem 14.11), and whose certified runs cost at least three (Theorem 14.16). Even there two things are missing, and I want them in the table and not in a footnote. The step doesn’t police whether a check could have failed: a property every value has is a perfectly legal check, sound

and useless. And in all three columns the status belongs to whoever reads the record. The machine doesn't know anything. What its flag means is assigned from outside, by whoever says what each property means.

Why does the honest check carry so much? Linda Zagzebski argued in 1994 that any account of knowledge whose justification leaves even a small gap between having good reasons and being right lets a Gettier case through: take a case where bad luck makes the reasons point the wrong way, then add a stroke of good luck that makes the belief true anyway. The first three conditions plus an honest check close that gap by construction, and only one half of honesty is needed for it, the half that says no when the property is false. Saying yes whenever it's true is extra. A check that can be wrong, a test suite that sometimes passes broken code, puts you straight back in her gap, and then the record certifies something smaller and still true: this procedure said yes.

And who certifies that the check is honest? Not the record. Honesty is a fact about what the check does on every input, and the opening chapter already said a finished state can't hold that kind of fact. It has to be carried by a proof about the checker, done once, outside every run, and nothing on the record pays for it. So an established claim is two things stuck together: a run property the record carries and a program property a proof carries, and the proof is entitlement for the auditor, not for the machine. Section 14 shows where that chain of reasons stops on a real machine.

The engineering version of this was already built, in 1997. George Necula's *proof-carrying code*: when you ship a program to a host that doesn't trust you, you ship a *proof* with it that the program obeys the host's safety rules, and the host *checks the proof that arrived* before running anything. Not your reputation. Not "a proof exists somewhere." The one in the box. A quarter century earlier Stephen Cook's certificates had the same bones, from Section 3: a claim comes with a witness that's cheap to check, and the checker checks *that* witness. And David Hume, in 1748, arguing about miracles, laid out the thing the receipt replaces: he weighed testimony against evidence, and said a wise person believes in proportion to the evidence, and the whole of his argument is a method for how much to trust a witness when you have no receipt. The move in this book is to stop weighing. A claim on the record is either backed by its object and its paid event, or it's testimony from outside, and a run that leans on testimony hasn't earned it itself. It can inherit it, but only as far as the copy it leans on is genuine, which is Section 7's problem of a record shown from outside.

The philosopher who built the most on these two words is Robert Brandom, and his *Making It Explicit* (1994) is where I'd send anyone who suspects I invented the pair. In his picture, people talking keep score on each other: what each speaker is *committed* to, what they've put themselves on the hook for, and what each is *entitled* to, what they're licensed to claim. You can be committed without being entitled (a reckless assertion), many claims count as entitled by default until someone challenges them, and entitlement can be borrowed: I can claim a thing on your say-so and defer to

you when I'm challenged. Lay mine beside his and the differences are the content. His score is kept by the other people in the room; here it's a reading in the state. His entitlement often comes by default; here it's an event that has to happen and is charged. And his deferral is my testimony, which I've just let back in on a leash: borrowed entitlement is only as good as the copy is genuine. Same two words, and his came first.

There's a whole field that would want a word here too, the logic of belief revision, and its nearest neighbours are worth naming. Krister Segerberg (1998) built a logic of irrevocable revision: accepting something so that it can't be given up. Isaac Levi (1991) split adding a belief into a deliberate kind and a routine kind, where you take on whatever a procedure you committed to in advance hands you, the way you'd accept a reading off a thermometer you've already decided to trust. A passing CHECK is routine in exactly that sense, with a checker proved honest once, outside every run. Neil Tennant (2008) showed that the best-known impossibility in that field, Peter Gärdenfors's theorem that a natural rule for conditionals can't live with rational revision, goes through only by treating some additions as revisions. Gösta Grahne (1998) drew the line this book leans on hardest: revising is changing what you believe about a world that stayed put, updating is changing it because the world moved, and Gärdenfors's theorem bites only the first. The version rule is an update. A write moved the counter, so every fact about the old edition stops counting for the new one. And Alexandru Baltag, Bryan Renne and Sonja Smets (2014) gave a logic that keeps apart three things Gettier's cases blur: justified belief, explicit knowledge and conclusive evidence. Put them together and they say something exact about this book's record. A record that's never wrong and never corrected sounds like a dogmatist's dream, and it would be one if its sentences were about the counter. They aren't. Each says a property held of a thing at one edition, and that sentence stays true after the counter moves on. The record can be infallible and never corrected only because its sentences are edition-stamped. Strip the stamps, and the first write would turn it into a liar that can't take anything back.

And the logicians the opening chapter named in passing, the ones who taught the logic of knowing to count. Here they are properly. Ho Ngoc Duc, in 1997, gave the logic of knowing an operator for "true after some train of thought": an agent doesn't know every consequence of what it knows now, but it can come to know any of them by taking steps of reasoning. Steps. And Sergei Artemov and Roman Kuznets, in 2014, turned logical omniscience into a question of cost: a logic escapes it when every valid claim that something is known comes with a proof whose size is bounded by a polynomial in the size of the claim. The usual modal logics of knowledge can't pass that test unless NP equals coNP, which most people would bet against, and justification logic passes it for knowledge stated together with its justification, because the size of the justification pays for the search. A price, of a sort. So steps and a price are both there, in their own form. What I still don't find in them is the one I care about:

a single act that closes a question for good and is charged as that act. Their steps are steps of reasoning, and their price is on finding a proof, not on settling.

Three things entitlement is *not*, which people reach for. It isn't permission: an access control system says who may write to what, and that's granted from outside and says nothing about truth; a process with permission to write the note still has no entitlement. It isn't "true, therefore usable": that's the cups. And it isn't "sound by luck": a shortcut that gives the right answer because the inputs happened to cooperate is the stopped clock. The count below has one condition aimed at it, and I'll tell you what the condition says and how much it actually catches.

8.4 What narrowing costs: count the rivals

Now the number, because a book about a bill has to say how much. Here's the idea in a form you can do with three playing cards. Shuffle cards marked 1, 2, 3 face down. Sort them using only questions of the form "is this one bigger than that one?", and count the questions. There are six orders the cards could be in. Each question has two answers, so after one question you've split the six into at most two piles, after two questions at most four, and you need to get down to one. Four is less than six, so two questions can't always do it, and three can, and you can draw the tree of questions on paper and see it: six leaves, three deep. The number of questions you need is at least the base-two logarithm of the number of possibilities you started with, rounded up: that's $\log_2$ of 6, which is a bit under 3, rounded to 3. Donald Knuth, in the volume on sorting of *The Art of Computer Programming*, gives this as the lower bound for all comparison sorting, and the shape of the argument is just: a tree f questions deep has at most 2 to the f leaves, so it can't tell apart more than 2 to the f cases.

And then the obvious next question: is the floor always the price? Try five cards. There are 120 orders, $\log_2$ of 120 is a bit under 7, so you need at least seven questions, and seven is enough, though not by the obvious method. Finding the way is a good evening with a deck of cards (Knuth has it, under the name merge insertion, from Lester Ford and Selmer Johnson in 1959). Now twelve cards. They can be in 479,001,600 orders, $\log_2$ of that is about 28.8, so the count says at least 29. Ford and Johnson's method takes 30, and in 1965 Mark Wells showed that nothing does better: twelve cards need 30. The floor was 29. The price is 30. So the count is a floor and only a floor; it never promises the floor can be reached, and the bound later in this section has to be read exactly the same way.

Ralph Hartley in 1928, before Shannon, measured information exactly this way: the logarithm of the number of possibilities, no probabilities, just a count. That's the measure here, and I'll say plainly it's Hartley's and not Shannon's: there's no distribution over the list, only the list's length.

Here's the same count with something you can hold. Put three coins under three cups. Each coin is heads or tails, so there are eight possible rows under the cups. Someone hands you two answers: heads under the first cup, tails under the second. Lift the

first cup and compare. A matching answer leaves four possible rows. Lift the second and compare. A match leaves two. The last coin can still be either way. The list really shrank: eight, four, two. Nothing in that count needed a claim about which way the last coin faces.

This is a parlour game, and an old one. In *A Christmas Carol*, in 1843, Scrooge's nephew Fred plays a game called Yes and No at his Christmas party: he thinks of something, and everybody else asks questions he may only answer yes or no. He's thinking of an animal, a disagreeable, savage one that growls and grunts and lives in London, and the answer is Uncle Scrooge. Most people know the game as twenty questions, and twenty is a good number for it: each yes-or-no answer can at best cut the field in half, and twenty halvings take a million or so possibilities, $2^{20} = 1,048,576$, down to one. Push on that, though. The halving only happens because Fred answers truthfully about what's actually in his head. Lie once and the count still shrinks; it shrinks around the wrong animal. That's the cups again: the four and the two are real because you lifted the cup and compared, not because somebody told you what was under it.

Now put it on a run. Say a run comes with a list of states that are "still on the table," the candidates, and a shorter list, with some ruled out. Call the first the prior and the second the posterior, and I want you to hear those as *a list you could print*, not "the space of possible worlds." Picture the ruling-out as observations, each observation that ruled something out being one branch point in a tree like the card tree. And here's the claim: if that tree is no deeper than the number of paid steps the run took, and the tree has enough leaves to cover the narrowing in a way I'll spell out in a moment, then the counter rose by at least $\log_2$ of the prior's length minus $\log_2$ of the posterior's length. At least as many marks as *index bits* the list lost, where the index bits of a list are the yes/no questions it takes to point at one entry of it, the card count again. That's the bound.

"Paid step" needs its own sentence. It means a step the price list charges at least one for. In a certification system the toll guarantees that every crossing of the reading is one; a price list is free to put a floor under more steps than that (a look, a send, a check), and the count takes every paid step there is. The tree is a bare shape, branches and leaves with nothing written on them, and "no deeper than the number of paid steps" is a count along the run that happened. It isn't a walk down the tree. A paid step doesn't have to have asked any question at all.

Notice, too, what the bound does and doesn't say about the run. It takes the two lists, the tree and the covering as given. It doesn't say the run produced the posterior from the prior; nothing in it reads the lists off the run. What it says is that, given lists, a tree and a covering that someone supplies, the paid steps of the run cover the rounded drop in list length. That's Section 5's warning come due: watching can narrow for free, so the payment can't be dug out of the narrowing. It goes in as a premise, the tree's depth against the paid steps, and I'll keep calling it one.

8.5 What “narrower” buys: a stronger test

Before the count, one link that’s pure logic, and Plato’s third account is the whole of it. Take a list of candidates and an observation, a function that says what each candidate looks like from outside. The list’s *membership test* accepts an observation when some candidate on the list shows it. Shrink the list, and the test can only get pickier: anything the short list’s test accepts, the long list’s accepts too. The word for that, and it runs backward from English, is *stronger*: the stronger test is the one that lets *fewer* things through.

Pickier, though, isn’t the same as strictly pickier, and the cards show the gap. Take all six orders as the prior, and as the posterior the three where card 1 sits left of card 2: 1-2-3, 1-3-2 and 3-1-2. First observation: “is 1 left of 2?” Every survivor says yes, and the removed 2-1-3 says no, so the prior’s test accepts yes and no, the posterior’s only yes. Strictly stronger. Second observation, same two lists: “is the first card a 3?” Now 3-1-2 says yes and the other two survivors say no, and among the removed orders 3-2-1 says yes and 2-1-3 says no. Both tests accept yes and no. Same strength, even though three orders are gone. What made the difference is a removed candidate whose observation *no survivor shares*. That’s Plato’s mark, the thing that sets this case apart from everything it could be confused with, turned into a function. Narrow the list and keep one distinguishing removed candidate, and the test is strictly stronger; narrow it without one, and the test may not have moved at all.

8.6 The count, exactly

Here are the words and the count, for any system with Section 6’s counter. A *list* is a finite list of candidates, repeats allowed, and $|\Omega|$ is the length of the list Ω , repeats counted. $\lceil \log_2 x \rceil$ is the base-two logarithm of x rounded up: the smallest $k \in \mathbb{N}$ with $2^k \geq x$. At $x = 0$ this index-bit count is defined to be 0 as well; the notation $\lceil \log_2 0 \rceil = 0$ here is a convention for empty lists, not the real logarithm of zero. And $a \dot{-} b$ is subtraction on whole numbers that stops at zero, so $3 \dot{-} 5 = 0$.

Definition 8.2 (Decision tree). A *decision tree* is a leaf, or a branch with two subtrees. Its depth is 0 for a leaf and one more than the deeper subtree for a branch; its leaf count $L(T)$ is 1 for a leaf and the sum of the two counts for a branch. Nothing is written on a tree: no question, no answer, no candidate.

Lemma 8.1 (A tree is no wider than two to its depth). *For every decision tree T , $L(T) \leq 2^{\text{depth}(T)}$, and so $\lceil \log_2 L(T) \rceil \leq \text{depth}(T)$.*

Proof. Induction on the tree. A leaf has one leaf and depth 0, and $1 \leq 2^0$. A branch with subtrees of depths d_1, d_2 has at most $2^{d_1} + 2^{d_2} \leq 2 \cdot 2^{\max(d_1, d_2)}$ leaves, which is 2 to its own depth. The second half: $L(T) \leq 2^{\text{depth}(T)}$ means $\text{depth}(T)$ is one of the k with $2^k \geq L(T)$, and the rounded-up logarithm is the smallest of them. $\square$

Checked as `decision_tree_leaves_le_pow2_depth` and `decision_tree_log2_up_leaf_bound` (DecisionTreeBound.v). Both are stated over the tree type alone, a leaf or a branch, with nothing about any machine in them.

Theorem 8.2 (Paid narrowing, counted). *Take a system with a counter μ and a price list c that obey conservation (Theorem 6.1): each step raises μ by exactly its charge. Let Ω and Ω' be lists with $|\Omega'| > 0$, let T be a decision tree with $|\Omega| \leq L(T) \cdot |\Omega'|$ (the tree covers the narrowing), and let t be a run from s ending at s' in which at least $\text{depth}(T)$ steps are charged at least one each. Then*

$$\lceil \log_2 |\Omega| \rceil - \lceil \log_2 |\Omega'| \rceil \leq \mu(s') - \mu(s).$$

Proof. The rounded-up logarithm respects order, so the covering gives $\lceil \log_2 |\Omega| \rceil \leq \lceil \log_2 (L(T) \cdot |\Omega'|) \rceil$. It is also subadditive on products of positive numbers, $\lceil \log_2 ab \rceil \leq \lceil \log_2 a \rceil + \lceil \log_2 b \rceil$, because $2^{\lceil \log_2 a \rceil} \cdot 2^{\lceil \log_2 b \rceil} \geq ab$. So

$$\lceil \log_2 |\Omega| \rceil - \lceil \log_2 |\Omega'| \rceil \leq \lceil \log_2 L(T) \rceil \leq \text{depth}(T),$$

the last step by Lemma 8.1. By hypothesis at least $\text{depth}(T)$ steps of the run were charged at least one, and no step is charged less than zero, so the sum of the charges is at least $\text{depth}(T)$. By conservation that sum is $\mu(s') - \mu(s)$. $\square$

Argued here for any system with a counter that obeys conservation, and checked, with the sum of the charges in place of the counter's rise, as part of Theorem 8.3 below. The tree half is the checked lemma above; the rest is arithmetic on three whole numbers, the two lengths and the leaf count, and on the counter. Notice what's missing from the hypotheses: anything tying the lists to the run. That's the honest shape of the bound.

Three honest small prints, all of them visible in the statement. Both logarithms round up separately, and the subtraction stops at zero, so when the prior and the posterior round up to the same number of index bits (four candidates narrowed to three, say) the bound says nothing. The lists are plain lists, not sets, so "index bits" is a difference of rounded list lengths with repeats counted; a list can get shorter by dropping a duplicate without losing a single candidate, and "a candidate was ruled out" and "the list got shorter" are two different claims. And the difference isn't the number of questions anyone asked.

The covering condition, $|\Omega| \leq L(T) \cdot |\Omega'|$, is the nearest thing the count has to an anti-stopped-clock clause, and I have to be exact about how near. Picture each survivor standing for a little crowd of the candidates that were dropped, a crowd no bigger than the tree has leaves. Then the condition says every candidate in the prior fits in some survivor's crowd. That limits how much can be deleted per leaf you paid for. It doesn't say why anything was deleted. So the condition rules out deleting too much for the tree you've paid for. It doesn't rule out deleting for no reason. A run can satisfy

every hypothesis of the count and have ruled nothing out on purpose; the count prices the size of the drop, not the honesty of it. And the covering can't just be dropped. On the small machine one yes-or-no question with a big constant in it narrows n bits down to one candidate for a ledger of three, so the bound is false without it (Section 14.17).

Here's a side trip worth taking, because it looks like it breaks the count and doesn't. Rudolf Ahlswede and Gunter Dueck (1989) noticed that asking "is it this one?" is a different job from asking "which one is it?" If each receiver only wants to know whether the message is theirs, a noisy channel used n times can serve a number of receivers that grows like two raised to two to the nC , with C the channel's capacity, where naming the message only gets you two to the nC . That's identification, and on its face it says checking can be exponentially cheaper than naming. But their tests are allowed to say yes wrongly once in a while. Make the yes sound, a test that never says yes to the wrong message, which is the only kind of check this book lets raise a record, and the double exponential collapses. The argument is two lines of pigeonhole, argued here and not checked. Each receiver's test says yes on some output that its own message can produce and that no other message can produce, or the yes wouldn't be sound; so those outputs are all different, one per receiver, and there can't be more receivers than outputs, $|Y|^n$ for an output alphabet Y . Single exponential, the same order as naming. So for a check that never lies, the card count is the right scale after all, and the cheap trick needs a check that can be fooled.

8.7 What you've got

A paid flag is not a true claim; Section 7 left it at "a paid flag is a paid flag," and Section 9 will say it flatly. What a paid flag with an object and a checked relation behind it gives you is *entitlement*: anyone auditing the run may treat the claim as established, and can point at the event where that became legitimate. (Nothing in the five parts makes the system itself read the flag; I said that above and I'm saying it again because it's the sentence most likely to get lost.) Entitlement costs at least the index bits of narrowing it commits to, given lists, a tree and a covering somebody supplies, and given a run that paid for the tree's depth. The cups say why entitlement and truth are different. Plato says what the checked relation has to do: distinguish. Necula says how to build it. Hartley says what to count.

Put the three pieces on one hypothesis list (narrow the list with a test that separates, find the paid event, count the bill against a tree the run paid for) and you get a single statement with a long name, structural entitlement represented, once over any certification system and once for every Thiele-complete machine, the small machine included (Section 14.17). The first one reads as follows.

Theorem 8.3 (Structural entitlement, over any certification system). *Take a certification system (Definition 3.1), a state s that reads no, and a run t from s that ends in a state that reads yes. Take a type of candidates, a type of observations with an exact equality test, an*

observation obs of each candidate, and a decoder d that turns a run into an observation (what its record shows). Take lists Ω and Ω' of candidates and a decision tree T such that: (H1) Ω' strictly narrows Ω : every member of Ω' is in Ω , and some member of Ω isn't in Ω' ; (H2) some dropped candidate w , in Ω and not in Ω' , shows an observation different from every survivor's; (H3) the decoded run $d(t)$ is shown by some survivor; (H4) $|\Omega'| > 0$ and $|\Omega| \leq L(T) \cdot |\Omega'|$; (H5) at least $\text{depth}(T)$ steps of the run are charged at least one each. Then (1) the membership test of Ω' is strictly stronger than the membership test of Ω ; (2) some step of the run took the reading from no to yes, and that step was charged at least one; and (3) the sum of the charges of the run's steps is at least $\lceil \log_2 |\Omega| \rceil - \lceil \log_2 |\Omega'| \rceil$.

Proof. (1) is the cards' argument: the dropped w shows something no survivor shows, so the posterior test says no where the prior test says yes, and every survivor was in the prior, so nothing the posterior accepts is refused by the prior. (2) The run starts at no and ends at yes, so at some step the reading goes from no to yes, and the toll charges that step at least one. (3) is the proof of Theorem 8.2 with the sum of the charges in place of the counter's rise: the covering and Lemma 8.1 bound the index bits by $\text{depth}(T)$, and the sum of the charges is at least the number of steps charged at least one. On a system that obeys conservation the sum of the charges is the counter's rise. $\square$

Checked as `ent_narrowing_strengthens`, `ent_cs_count` and `ent_cs_entitlement` (EntitlementSmall.v); the tree lemma underneath is also there, as `ent_leaves_le_pow2` and `ent_log2_leaves_le_depth`. Theorem 8.2 is the same count with the counter's rise, so for a system that obeys conservation it's the special case. Two things in the hypotheses feed nothing, and I'd rather say so than leave you to find it: (H3) isn't used by any of the three conclusions, and from (H1) only the inclusion is used, not the gap.

A statement like that is true by packaging: it takes the hypotheses in and hands the conclusions back, and the right word for it is a checklist. A checklist for leaning on a claim, not a discovery about leaning.

And a checklist isn't nothing. On 30 October 1935, at Wright Field, northeast of Dayton, Ohio, the Army was evaluating Boeing's new four-engine bomber, the Model 299. It took off, pitched up, stalled and crashed, and two men died of their injuries, the Army's pilot, Major Ployer Hill, and Boeing's chief test pilot, Leslie Tower. The airplane was fine. The crew had left the gust locks on, the clamps that hold the control surfaces still while the plane sits on the ground. A newspaper, as the story is usually told, called it too much airplane for one man to fly. The answer wasn't more skill, it was a card: a list of things to check, in order, every time, and the story usually told is that this is where the pilot's checklist comes from. The airplane went on to be the B-17. Nothing on that card was a discovery. Every item on it was something the pilots already knew. What the card did was make sure none of it got left out when it mattered. That's what putting entitlement's hypotheses on one list does: it finds

nothing new about leaning on a claim, and it makes the hypotheses impossible to drop quietly.

Before any machine, I owe you the word the toll is named after, the “insight” in “No Free Insight” (Section 5), because I’ve been using it as if it meant one thing.

9 What insight is

I named the rule in this book “No Free Insight,” and the name ran ahead of any careful account of what insight is. The book has been narrowing it ever since: not learning (Section 5), not looking, not copying. It’s time to say what’s left, and to say it in a way that someone who studies actual insight, in actual heads, wouldn’t laugh at. So who are those people, and what did they find? Here’s what I ran into, pushed on as I go, because the founding stories of a field get retold like any other stories, and a retold story is a lead to check.

9.1 Sultan and the sticks

In 1913 the psychologist Wolfgang Köhler went to Tenerife to run a research station with a colony of chimpanzees, and the First World War then kept him there until 1920 with nothing to do but watch them. The famous one is Sultan. Food is outside the cage, out of reach; two sticks are inside, each too short, but one fits into the end of the other. Sultan tries each stick, fails, sulks, plays with the sticks, and then, by Köhler’s account, at a moment you can see, joins them and reaches the food. Köhler’s word was *einsichtig*, which came into English as “insight,” and the whole school of psychology he belonged to, the Gestalt school, said that’s what thinking is: not stacking up tries until one works, but a *restructuring*, the moment two sticks stop being two sticks and become one tool. Karl Duncker gave the human version, in German in 1935 and in English in 1945, five years after he died at thirty-seven; he’d spent his last years working with Köhler at Swarthmore. You can set it up yourself: a candle, a box of tacks, matches, and the task of fixing the candle to the wall. Almost nobody sees that the *box* is a shelf, until they do. And then he found that if you hand people the box *empty*, with the tacks poured out beside it, they solve it far more readily. Same objects. What changed was which relation among them was on the table.

I like both of these for the same reason: the insight didn’t add an object. Sultan had both sticks the whole time. The box was there the whole time. What arrived was a *relation*, which is to say a piece of structure in Section 2’s sense, and the arrival was sudden. Hold that, because it’s going to turn out that the thing the toll prices also arrives in one step, and at its best it’s also a relation.

9.2 It isn’t a feeling

But here’s the finding I’d hand anyone who thinks insight is a feeling, and it comes from someone who measured it. In the 1980s the psychologist Janet Metcalfe asked people to solve problems while rating, every fifteen seconds, how “warm” they felt, how close to the answer. On ordinary problems, algebra for instance, the warmth

climbs steadily and peaks at the answer, which is what you'd expect. On *insight* problems, riddles of the candle-box kind, the warmth stays flat, flat, flat, and then the answer arrives, and if anything the people who felt warmest beforehand were *more* likely to be wrong. The feeling of being close is not a gauge of being close. Try it yourself: write a number from one to seven every fifteen seconds while doing a long sum, then while doing a riddle, and compare the curves.

And the most famous insight story there is fits Metcalfe uncomfortably well. Archimedes, asked whether a goldsmith had cheated the king on a crown, gets into a bath, sees the water rise, and runs home naked shouting "eureka," I've found it. That comes from Vitruvius, writing about two centuries later. Is it true? Galileo, at twenty-two, in 1586, thought the method as Vitruvius tells it was too crude to catch a clever goldsmith, and wrote a little tract, *La Bilancetta*, proposing a balance weighed in air and in water instead, which he thought was probably what Archimedes actually did, since it rests on Archimedes's own results. So the eureka is a legend, wrapped around a real man and probably a real problem. Notice what the legend kept: the bath, the shout, the running home naked. The feeling. What it got shaky is the part that would count as the check, the test that could have caught the goldsmith. The bit everybody remembers is the bit that tells you least.

So any account of insight that keys on the sensation, the "aha," is keyed to something that the data says is unreliable. Whatever insight is, it isn't the feeling of having it. That's a relief, because a system of states and moves doesn't have feelings, and a word that only meant a feeling would be cheating if I hung it on one. The version this book can price has to be keyed to an *event*, and the question is which.

9.3 What happens when a computation learns

Here's the split, and I'll say what the toll does with each part, and then I'll say who decided the split, because it wasn't me. Keep Section 3's sheet of paper in front of you: the NO box, the YES box, the tally line, and the three moves STAY, WORK and VOUCH.

The first thing: *raw observation*. Something comes in and gets written down. A counter ticks. Nothing is claimed about what it means, nothing is marked as resting on it, and the reading doesn't move because of it. Later moves can read what it wrote, but that reading is a move of its own. This is the demon's pad in Section 5, and it's what the statistician Dennis Lindley measured in 1956 as "the information an experiment provides," the expected drop in your uncertainty: a real quantity, and a quantity about what was *seen*. It can be free. The toll says nothing at all about a move that doesn't cross from NO to YES, so a price list may charge it zero: on the paper, STAY and WORK cost nothing, and Section 5's watcher narrowed a whole row of candidates without paying a mark. "Can be" is the honest word. A price list is allowed to charge for looking anyway, and whether looking costs anything in the physical world is a different question that this sheet of paper doesn't touch.

Lindley's measure has a back story, and it runs through Bletchley Park. I. J. Good, who was Alan Turing's main statistical assistant at Bletchley Park in 1941, wrote in 1979 that Turing had worked out the expected weight of evidence an experiment would give, to judge one of the codebreaking procedures before running it, and so partly anticipated Lindley by about fifteen years, in work that stayed secret. Turing measured evidence in *bans*, logarithms to base ten, with a tenth of one called a deciban, and the name came from the town of Banbury, where tens of thousands of the sheets used in that procedure were printed. It's the same base-ten unit other people call the hartley, after the Hartley from the last section. And Good adds that Turing didn't know the logarithm he was using had already appeared in 1878, in a paper by Charles Sanders Peirce, the same Peirce from the opening chapter, who'd called it the weight of evidence. Peirce is as far back as Good takes it. What it changes: nothing in the argument, but it shows how well trodden this first kind of thing is. For a century and a half careful people have measured what an observation tells you, and every one of those measures is about what was seen. That's the side of the line the toll leaves alone.

The second thing: *certified insight*. Some move turns the reading from NO to YES. Here is where the toll lives. In any system with the five parts that obeys the toll, that move costs at least one mark, and every run that starts at NO and ends at YES pays at least one somewhere along the way, which is Section 3's universal floor. What's being charged for, at its best, is exactly the Gestalt thing: a relation, committed to. Not two sticks; one tool, claimed. "At its best," because Section 3 was careful about what YES means: the flag is up, and nothing more. A crossing that writes down which relation it's committing to carries Sultan's tool with it. A bare crossing, VOUCH on the paper, commits to no relation at all, and it pays exactly the same.

The third thing isn't a kind of event. It's an observation about the price: it's *flat*, whatever is certified. Say the YES state also shows something striking, a deep relation, a result nobody could have guessed, Sultan's two sticks become one tool. The toll charges the crossing at least one. It doesn't charge the striking crossing more, because the floor's proof never looks at anything but the reading before and after. On the paper: VOUCH from NO makes one mark whether you were thinking of Duncker's box or of lunch. Meaning is not cost. The depth of what's certified adds meaning. It doesn't add a mark.

The candle box has a footnote that says the same thing from the other side. In 1962 Sam Glucksberg ran Duncker's problem with money on it: some people were told they could win five or twenty dollars depending on how fast they solved it. With the tacks in the box, the people chasing the money did worse than the people who weren't. With the box empty, the money made no clear difference. Paying more didn't buy the insight, and on the hard version it got in the way. That's a fact about heads, not about records, and I won't push it further than that. But it's a nice mirror for the paper. On the paper a deeper crossing isn't charged more, and in Glucksberg's room a bigger prize didn't bring the crossing sooner. In neither place does the price measure

the depth.

Here they are, exactly, for any certification system (Definition 3.1). Runs are written as lists, the way Section 3's proof wrote them: $[]$ is the empty run, and $i :: t$ is the run that starts with move i and carries on with the run t .

Definition 9.1 (Certified insight event, raw observation). In a certification system, a step from state s by move i is a *certified insight event* when it turns the reading on: $\text{cert}(s) = \text{no}$ and $\text{cert}(\text{step}(s, i)) = \text{yes}$. A move i is *raw observation* when no step by it, from any state, is a certified insight event.

Remark 9.1 (What the toll prices, and what it doesn't). Read off Definition 3.1 and Theorem 3.1; there is no further Coq theorem at this generality. (1) The toll puts no bound on the cost of a raw-observation move; on Section 3's board STAY and WORK are raw observation and cost 0. (2) Every certified insight event is charged at least 1, by the toll, and every run from a state reading no to one reading yes has total cost at least 1, by the universal floor. (3) The price is flat: the floor's proof uses nothing about the end state but its reading, so adding any hypothesis about what else the end state shows leaves the bound at 1 and cannot raise it. A system that wants to charge more for a deeper certification has to write that into its price list, move by move; the toll doesn't do it for anyone.

So the account comes down to two facts and a warning. Looking can be free. Turning the reading on can't. And putting something remarkable behind the reading doesn't make it cost any more.

Who decided the split? Not the costs alone. The toll prices exactly one thing, the crossing. A price list is free to put a floor under more moves than that (a look, a send, a check) and free to leave other moves at zero, so the sorting into raw observation and certified insight is by whether the reading moves, and that sorting is this book's. The part that's *forced* is the floor under the crossing, and Section 4 is the argument for it; anything a price list charges beyond that is a choice, and any concrete machine owes you its whole table so you can see which choices it made.

9.4 Test, then commit, and nothing forces the second to follow the first

The philosopher Karl Popper said a theory earns its standing by being *exposed*: you make a conjecture, you try to kill it, and what survives a real attempt counts for something, where "real" means the test could have failed. His word for that earned credit is *corroboration*. I've leaned on that idea through this whole book, and here's where a certification system has a version of it, and here's where the version is weaker than Popper and I have to say so.

Add a fourth move to the paper: CHECK. It looks at something (a sum, a coin under a

cup, a formula against a row of values), and it can come out pass or fail, and it doesn't move the box. Charge it whatever you like; charging it on the attempt, pass or fail, fits Popper, because what made his test count was that it was really run and could have failed. That's the test. It's raw observation in the sense above, because it never turns the reading on. VOUCH is the commit. So the honest protocol is two moves: test, then commit, and the costs are separate, one for the checking work and one for the crossing.

But nothing in the five parts makes the second move wait for the first. The step is handed the current state and a move name; the toll looks only at whether the reading crossed. Play VOUCH from NO with no CHECK anywhere before it, and you land in YES and make one mark, the same box and the same mark as the player who checked first. The two tally lines differ by exactly what the CHECK was charged, and if you charged it nothing they don't differ at all. The mark for the crossing is the same mark either way. The receipt says a commitment was paid for. It does not say it was earned. Popper's corroboration needs the test; the toll doesn't.

You can build a system that makes the commit wait. Let CHECK leave a mark in the state when it passes, and write the step so VOUCH only crosses when that mark is there. That's a design, written into the step function, and it's a perfectly good one. Nothing in the five parts or the toll asks for it. That's a real gap between the word and the bare system, and the only honest thing is to leave it visible: the toll prices the commit, and whether the commit was preceded by a test is a fact about the run, or about a design somebody chose, not a guarantee of the five parts.

Remark 9.2 (Nothing makes the crossing wait). Read off Definition 3.1; there is no Coq theorem. The toll constrains $\text{cost}(i)$ for moves that cross, and nothing else; the step function is any function of the current state and the move. So a certification system may have a move that crosses from a state no check has touched, and the toll still holds. On Section 3's board extended with CHECK, the run [VOUCH] from NO crosses with one mark, and so does [CHECK, VOUCH]. Making a crossing depend on an earlier check is a condition on the step function, which a particular system can impose and the definition does not.

9.5 What the word means

So: insight, in the sense this book prices, is a relation committed to, in a move that flips a reading from off to on, and that move costs at least one mark, in any system with the toll, for any relation, whatever its depth. (And one edge from the paper: a bare crossing that commits to nothing pays the same mark, so "a relation committed to" is what the crossing is at its best, not what the toll guarantees.) Three boundaries, each from a different place. From Metcalfe: it isn't a feeling, and a system's version has to be an event. From the proof: the price is flat; a shallow commit and a deep one both cost at least one, and cost is not depth. A system can pay heavily to check

a claim that turns out trivial, and Duncker's box is one small commit. And from the two-move protocol: a paid flag proves nothing on its own. Price and correctness are separate books.

Which leaves a suspicion I'd have in your place. Everything so far has been one reading on one sheet of paper, with a rule I chose and a count I bolted on. Is the shape of that record something I picked, or is it what any honest record has to look like, on any machine at all? That's the next section: take *any* machine that steps, bolt a record onto it, and see what shape it's forced into. How a concrete machine hangs actual claims on its parts, a checked property of one counter at one edition, and where those sit in the state, comes in Part IV.

10 Records over any base: the latch

For the whole book so far I've been building up a thing with a flag and a counter, and a fair reader is entitled to suspect that the whole apparatus is special pleading for one machine, the one I keep promising to build. So here's the section that answers the suspicion with no particular machine in it at all, and sees what's left. Take *any* machine that steps deterministically: a Turing machine, a register machine (a few numbered boxes that each hold a whole number, and a program that adds, subtracts and jumps), a lambda calculus (Alonzo Church's way of computing by nothing but plugging expressions into expressions; Section 11 builds one), anything. Call it the base. Now bolt a record onto it, under rules I'll state, and ask: what can the record be?

The answer is one shape, and I'll show you the shape first because it's a thing you can build from a cup and a die.

10.1 A cup and a die

Put a cup upside down on the table. The rule: the first time a coin lands in front of it, turn it right side up. There is no rule for turning it back over. Now roll a die, and put a coin down on every six. Roll twenty times. The cup turns over on the first six and stays turned over. Write down which roll did it. That number, the roll that turned the cup, is where the record happened, and the cup afterward is the record.

That's a *latch*. Electrical engineers have had the circuit since 1918, when William Eccles and Frank Jordan applied for a patent on a two-valve arrangement that holds whichever state you last put it in; it became the flip-flop, and the registers and caches in the device you're reading this on descend from it. The simplest version has one input, "set," and the rule *new state = old state or set*: once set, it stays set, and there's no reset wire. The cup is that circuit: a flip-flop with its reset wire cut off.

(Eccles and Jordan didn't call it a flip-flop. Their 1919 paper calls it a *trigger relay*. Where the newer, sillier name came from I can't source, so I'll leave it at that.)

Is a flip-flop with its reset wire cut off only a picture, or did anybody ever build one on purpose? Somebody did, and for a reason that's very close to this book's. In the mid-

1950s Wen Tsing Chow, an engineer at the Arma division of the American Bosch Arma Corporation in Garden City, New York, was asked by the United States Air Force for a more flexible and secure way to hold the fixed numbers, the targeting constants, inside the guidance computer of the Atlas missile. What he and William Henrich patented (filed in 1957, granted in 1962, under the plain title “Storage matrix”) was a grid with a connection at every crossing, where you set the numbers by taking out the connections you didn’t want, physically or by burning them out with too much current. The word *burn* is right there in the patent, and people still say they burn a chip. The descendants are called programmable read-only memories, PROMs, and a fuse blown in one stays blown: there’s no current you can send to grow it back. A whole memory built out of cups with no reset wire, because somebody wanted numbers nobody could quietly change later. Does that prove anything here? No. The theorem below doesn’t care what a cup is made of. But it’s good to know the shape wasn’t invented for a sheet of paper. Somebody needed it, for stakes a good deal higher than a die.

Now the claim, stated loosely and then exactly: *every honest record over any base is a cup with no reset wire, turned by some event the base passes through.*

10.2 What “honest” has to mean

Here’s where the content is, and I’ll give you the five rules, because the theorem is only as strong as they are.

A machine *extends* a base when there’s a map from the bigger machine’s states down to the base’s, and that map starts in the base’s start states, reaches every one of them, moves in step with the base (one step up top is one step down below, through the map), and agrees about halting. The bigger machine may carry as much hidden state as it likes beyond the base. That’s the “any base” part: the base doesn’t get to see the extension, and the extension doesn’t get to change what the base does.

The extension carries a *record*, a yes/no reading, and a *ledger*, a counter. It’s an honest extension when five things hold. One: the record’s next value depends only on the base’s current state and the record’s current value. (I’ll call this *driven by the computation*. It says the record answers to what the machine is doing, not to a clock on the wall or someone’s hand.) Two: the ledger never goes down (Section 6). Three: the step that turns the record on costs at least one; that’s the toll (Section 3). Four: once on, the record stays on; that’s permanence (Section 4). Five: some run from a start state actually reaches a step that turns the record on; otherwise the record is decoration.

Here are the words exactly. On/off is true/false, $\vee$ is “or” (on when either side is on), and next^k is k steps.

Definition 10.1 (Base, record-carrying machine, cover). A *base* B is a set of states with a step function next_B , a set of starting states, and a halting predicate. A *record-carrying machine* M is a set of states with a step function next_M , a set of starting states, a *record*

rec from states to Booleans, a *ledger* μ from states to whole numbers, and a halting predicate. The cost of a step is $\text{cost}(m) = \mu(\text{next}_M(m)) - \mu(m)$. A *cover* of B by M is a map π from states of M to states of B that sends starting states to starting states, has every starting state of B as the image of some starting state of M , commutes with the steps, $\pi(\text{next}_M(m)) = \text{next}_B(\pi(m))$, and agrees about halting: m halts exactly when $\pi(m)$ does.

Definition 10.2 (Honest extension, latch). M with cover π is an *honest extension* of B when, for every state m : (R1, driven by the computation) there is one function f with $\text{rec}(\text{next}_M(m)) = f(\pi(m), \text{rec}(m))$; (R2, ledger) $\mu(m) \leq \mu(\text{next}_M(m))$; (R3, toll) if $\text{rec}(m)$ is off and $\text{rec}(\text{next}_M(m))$ is on, then $\text{cost}(m) \geq 1$; (R4, permanence) if $\text{rec}(m)$ is on, so is $\text{rec}(\text{next}_M(m))$; and (R5, a reachable write) some starting state m_0 and some k have $\text{rec}(\text{next}_M^k(m_0))$ off and $\text{rec}(\text{next}_M^{k+1}(m_0))$ on. An *event* is a function h from base states to Booleans, and its *latch* is the step $(b, r) \mapsto (\text{next}_B(b), r \vee h(b))$. M *factors as the latch of h* when, for every state m (reachable or not),

$$(\pi(\text{next}_M(m)), \text{rec}(\text{next}_M(m))) = (\text{next}_B(\pi(m)), \text{rec}(m) \vee h(\pi(m))).$$

Now the theorem. Under those rules, the base state and the record move *exactly* as a latch: the record switches on at the first base state where some event holds, and never off, and the event is read off the extension's own rule by asking it what it does when the record is off (stated exactly below). Everything else the extension carries, all the hidden state, shows up only in its ledger; it can't leak into the record's behaviour.

I promised the content was in the definition, and here's what that means: the proof is a short case split on whether the record is already on, and it uses only *two* of the five rules, "driven by the computation" and "permanent." The ledger, the toll, and the reachable-write clauses play no part in it. A latch is what you get from "answers only to the base" plus "never comes back down," full stop. The other three rules matter for the *converse*: for any base and any event the base can reach, the latch of that event, priced at one unit per write, *is* an honest extension. Put the two together and you've got: honest records are exactly latches. One shape, no others, over any base.

Theorem 10.1 (The record is a latch). *Let B be any base and M an honest extension of it through π , with f the function of (R1). Then M factors as the latch of the event $h(b) = f(b, \text{off})$.*

Proof. The base half of the equation is the cover's step clause. For the record, split on $\text{rec}(m)$. If it is on, permanence (R4) makes the next value on, and $\text{on} \vee h(\pi(m))$ is on. If it is off, (R1) makes the next value $f(\pi(m), \text{off}) = h(\pi(m)) = \text{off} \vee h(\pi(m))$. Only (R1), (R4) and the cover are used; (R2), (R3) and (R5) are discarded. $\square$

Checked as `record_axis_is_latch_holds` (StructuralRecordAxis.v), with the

definitions `BaseMachine`, `BaseCover`, `computation_driven`, `HonestBaseExtension` and `latch_factorization` (`StructuralCoreAnyBase.v`), and the clauses (R2) to (R5) as `ledger_carried`, `rc_a2` (`StructuralCore.v`), `record_permanent` and `reachable_record_write` (`StructuralCoreCover.v`).

One thing about that proof deserves a second look, because it cuts both ways. It never asks what the base is. The cover decides that: whatever the map sends down is base, and the record's rule is allowed to read all of it; whatever stays up top is extension, and (R1) keeps the record's rule from reading it. Draw the line low, so the base is only the computation's own working state, and (R1) is a real demand. Draw it high, put every table and every refusal bit into the base, and (R1) is easy to meet, and the latch shape comes out exactly the same. So the theorem holds wherever you draw the line, and that's exactly why it can't tell you where a record's meaning lives. The meaning lives in what the event h reads, and that's in the base by construction. The pieces above the line, the record and the ledger, are the ones that never steer anything; the pieces that make a flag mean something sit below it, where this theorem doesn't look. So when the machine in Part IV arrives, ask it first which parts it puts above the line.

Theorem 10.2 (Every reachable latch is an honest record). *Let B be any base and h any event. Let Latch_h have states (b, r, x) , starting states $(b, \text{off}, 0)$ with b starting in B , step $(b, r, x) \mapsto (\text{next}_B(b), r \vee h(b), x + \delta)$ with $\delta = 1$ when r is off and $h(b)$ is on and $\delta = 0$ otherwise, record r , ledger x , halting as b halts, and cover $(b, r, x) \mapsto b$. Suppose some starting b_0 and some k bring Latch_h from $(b_0, \text{off}, 0)$, in k steps, to a state whose record is still off and whose base state satisfies h . Then Latch_h is an honest extension of B .*

Proof. (R1) holds with $f(b, r) = r \vee h(b)$. (R2): each step adds 0 or 1. (R3): a step that turns the record on has r off and $h(b)$ on, so it adds 1. (R4): $\text{on} \vee h(b)$ is on. (R5) is the supposition. The supposition asks for the first time h holds on some run; any run from a start that reaches an h -state has such a first time, though the file states only the first-time form. $\square$

Checked as `latch_core_honest` (`RecordAxisDiscrimination.v`).

A reversible base needs one more thing to carry a record, and the files say exactly what. If the base never merges two states, keep every earlier record value in a growing history beside the latch, and the whole step merges nothing either; but on a finite state space a permanent record can't be written by a step that merges nothing at all. A step that merges nothing is called *injective*: no two states go to the same next state. And $r :: H$ is the history H with r put on the front, as in Section 9.

Theorem 10.3 (Records on reversible machines). *(1) Let B be a base whose step is injective. The machine with states (b, r, H, x) , step $(b, r, H, x) \mapsto (\text{next}_B(b), r \vee h(b), r :: H, x + \delta)$ (δ as in Theorem 10.2), starting states with r off, H empty and $x = 0$, has an injective step,*

and under the supposition of Theorem 10.2 it is an honest extension of B . (2) On a finite state space, a single injective step with a permanent reading never turns that reading from off to on.

Proof. (1) Equal images have equal base parts, so equal base states by injectivity; equal histories $r :: H$, so equal records and histories; and equal ledgers once the equal δ is subtracted. Honesty is the proof of Theorem 10.2 with the history carried along. (2) is Theorem 4.1 of Section 4 with a single move. $\square$

Checked as `history_latch_injective`, `history_latch_honest` and `finite_reversible_cannot_write` (`RecordAxisDiscrimination.v`), the last from `permanent_flip_is_not_injective` (`PermanentCertification.v`).

So a base that never merges can still carry a record, if it's allowed to grow a history that never ends. A finite one can't. On a finite machine the record's first write is a merge, and Section 4 already put the price on that.

Process-calculus people ran into the same thing from the other side. Vincent Danos and Jean Krivine (2005) took a reversible version of CCS, where every step can be rolled back, and modelled a transaction as a process that wanders freely through everything that can be undone until it performs an action declared irreversible, which is the commit. In their calculus, as in the theorem, a permanent effect needs a step with no way back, and the calculus has to say which steps those are.

There's an arrow of time in that theorem, and I should say where it came from, because I put it there by hand. Every starting state has the record off. That's the only reason a raised record means the event happened: it was down at the start, and only h lifts it. So the clean start is an assumed ready state, a past hypothesis in miniature. The past hypothesis is the physicists' name for the low-entropy beginning of the universe that every trusted record leans on, and nothing in this book derives one; every theorem here about records assumes its small version and starts from there. David Wolpert sorted memory systems in 1992, and one of his kinds works only because it begins in a prepared state set up by a process that runs one way, which is how it leans on the second law. Wolpert and Jens Kipper (2024) put that sorting on a formal footing, and they say plainly that they have no formal proof this kind can only remember the past. Aram Eftekar and Marcus Hutter (2024) built a reversible toy world, imposed a past hypothesis on it, coarse-grained it, and got records of the past and none of the future; Thomas Blanchard (2024) got the same lopsidedness of knowledge out of the causal Markov condition. And Carlo Rovelli, two years before the paper on choosing, studied traces in "Memory and entropy" (2022) and concluded that a memory turns low entropy into available information. Read against the latch, that's the blank the record uses up: the flag can say yes once because it started at no. So does the latch explain why records point backwards? No, and I'd rather say so than let it look like it does. The step runs forwards and the runs start clean, and a theorem built on those

two can only hand the arrow back. Under a step that forgets nothing, on a finite set, there are no permanent records at all, past or future, which is Zermelo again.

Constructor theory, which Section 4 brings in as a neighbour of the chairs, has a paper on time (Deutsch and Marletto, 2025) saying that laws stated as which transformations a device can and can't bring about can't mention time at all, so duration has to come from relations between physical systems, clocks, that have their own dynamics. That's the position this book is in by construction. The step count is the only time in it, and it comes from the machine's own steps.

Physicists have built this latch on paper too, in quantum mechanics, and it behaves the way the last theorem says. Klaus Hepp (1972) studied a particle that flips spins in a long chain as it passes, now called the Coleman-Hepp model, and John Bell (1975) took it apart: the record is exactly definite only for an infinitely long chain, which is the reversible escape above, a history that never ends, in quantum clothes. Jonathan Halliwell (1999) used a two-state detector that flips once when a particle crosses a point and is kept from flipping back by a large environment, which is a latch with the environment holding the history. And Murray Gell-Mann and James Hartle (1995) gave the property a name inside their framework of decoherent histories: the permanence of the past, past alternatives that stay settled as the story is extended into the future.

10.3 Both rules do work: the toggle and the clock

A theorem with two hypotheses should have two counterexamples, one for each hypothesis dropped, and it does, and they're the two ways a record goes wrong in the world.

Drop permanence. Take a base that just counts, and a record that flips at every step: on, off, on, off. It's driven by the computation; it's not permanent; and it's not a latch of any event. This is a *toggle*, a piece of working state. A record you can revoke is working state wearing a record's clothes, and if a revocation has to stick, that's a *second latch* ("revoked"), which is how certificate revocation lists actually work: you add an entry; you never delete the certificate. Alin Tomescu and coauthors (2019) built append-only authenticated sets and dictionaries for transparency logs, and revocation fits that shape exactly: a revocation goes in once and stays. The pair theorem (stated below) covers two latches that can each read the other; do it with two cups where the second only turns over on an even roll *if the first is already up*, and you'll see the second can never turn before the first.

Dialogue theorists built a version of that two-latch rule for arguments. C. L. Hamblin's *Fallacies* (1970) has a chapter on formal dialectic where each speaker has a *commitment store*, a list of what they're on the hook for, and the rules of the dialogue update it, not the speaker. Henry Prakken (2005) built dialogue games for argumentation on stores like that and made commitments sticky: a claim can be retracted only when it has been challenged, and claims and backtracking carry a price. He also floats a

convention that once a player has retracted a proposition, that player can't take it on again, which is a revocation that stays revoked, the way an entry on a revocation list does. The part of that I'd keep is the first one, that the store belongs to the rules and not to the speaker. Take the pen away from the program. Erik Krabbe (2001) names the awkward part of that: retraction is frowned on, and yet resolving a dispute takes somebody retracting a doubt or a point of view, so the rules have to say when taking a commitment back is allowed. This book answers with a price.

The same field has a notion that looks a lot like earning. In Phan Minh Dung's (1995) abstract argumentation, arguments attack one another, and the *grounded* semantics starts from the arguments nobody attacks and keeps adding the ones they defend. Baroni and Giacomin (2007) introduced, and Martin Caminada (2014) revisited, *strong admissibility*: a set is strongly admissible when every argument in it is defended by others in the set that don't lean back on it, so the support bottoms out. That's the argumentation form of a claim that's earned and not defaulted, and Caminada shows it plays a vital part in the dialogue games that decide grounded acceptance. Notice the base case, though. An argument nobody attacks gets in on a check that can't fail, and this book says a check that can't fail establishes nothing. Whether a machine's earned record certifies exactly the strongly admissible sets is the kind of statement I'd have to prove before I said it, and I haven't.

A record that's allowed to change its mind isn't lawless, though. It's somebody else's subject. Hilary Putnam and E. Mark Gold, in back-to-back papers in 1965, looked at questions a machine answers by guessing and revising, where the answer is whatever the guesses finally settle on. Yuri Ershov (1968) sorted those by how many revisions they're allowed. Allow one, from no to yes, and you get exactly the sets a program can list, which is what a permanent record raised by a search catches. Allow two, three, and so on, and every extra change buys new sets: the rungs of the ladder are all different. A record that can come back down lives on that ladder. This book stays on the bottom rung, and I'm pointing at the ladder, not climbing it.

The ladder doesn't stop at finite time, and the people who followed it past infinity have names for the two kinds of record I just described. A definition that only ever adds, stage after stage, and keeps everything it has at each limit, is an *inductive definition* (Richter and Aczel's 1974 chapter is the long treatment). Let it take things back, and keep what stays put from some stage on, and you have what John Burgess (1986) called *quasi-inductive* definitions, which he met in the revision theories of truth that grew up beside Kripke's; his paper measures how complicated the truths of those theories get. A permanent record is the first kind. A toggle that's allowed to settle is the second. Joel Hamkins and Andy Lewis (2000) let a Turing machine keep running through the ordinals, with a cell at a limit reading 1 if it has switched to 1 unboundedly often, and the trick they use again and again is the toggle: flash a flag on and off every time a guess changes, and at the limit the flag is on exactly when the guess changed infinitely often. A flag that can come back down is how that machine learns what

no finite run could tell it. A permanent one can't do that trick, because at a limit a cell that only ever goes up just stays up. One finite-time fact belongs beside it, so nobody concludes too much from permanence. Hao Wang (1957) described a machine whose only write is to mark a square and which can never erase one, and it computes everything a Turing machine does. Permanence of a machine's own memory costs it no computing power. What the book prices is a permanent reading raised by a step that has to work whether or not the reading is already up, which is a different thing.

Drop "driven by the computation." Take a base that counts, and a record that switches on at the fifth tick of a *hidden clock*, whatever the base is doing. It's permanent; it's not driven by the computation. This is a *timestamp*. A timestamp is not evidence; it says *when*, not *what happened*, and a record switched by an outside party or a clock records the party or the clock, not the computation. Try it with the die: play twenty rolls with the rule "turn the cup after roll seven no matter what," and notice you've recorded nothing about the dice.

The file proves exactly the facts I named for each: the toggle is driven, not permanent and not a latch; the clock is permanent and not driven. That each also keeps the other rules (the ledger is the step count, the first switch-on costs one, a run reaches it) is true by inspection, and isn't a separate theorem.

So: permanent but undriven is a clock, driven but impermanent is a toggle, and a record is the thing that's both. That's the whole discrimination, and I think it's the most useful three-sentence thing in this book for looking at systems other than the one I build, which is why it's here.

The pair theorem and the two counterexamples, exactly. The base for both counterexamples is the counter: states are whole numbers, the step adds one, every state is a starting state, and nothing halts. Two more bits of shorthand: $\neg r$ is r flipped, and a statement in square brackets, like $[k = 5]$, reads on when it's true and off when it isn't.

Lemma 10.4 (Two records are two latches). *Let c_1, c_2 be two records on M , with cover π onto a base B , such that: there is one function F with $(c_1(\text{next}_M(m)), c_2(\text{next}_M(m))) = F(\pi(m), c_1(m), c_2(m))$; the ledger never goes down; a step that turns either record on costs at least one; and each record is permanent. Then there are h_1, h_2 with $c_1(\text{next}_M(m)) = c_1(m) \vee h_1(\pi(m), c_2(m))$ and $c_2(\text{next}_M(m)) = c_2(m) \vee h_2(\pi(m), c_1(m))$ for every m .*

Proof. Take $h_1(b, r_2)$ to be the first part of $F(b, \text{off}, r_2)$ and $h_2(b, r_1)$ the second part of $F(b, r_1, \text{off})$, then split on each record as in Theorem 10.1. $\square$

Checked as `record_pair_is_two_latches_holds (StructuralRecordAxis.v)`.

Lemma 10.5 (The toggle). *Let the toggle have states (n, r) , step $(n, r) \mapsto (n + 1, \neg r)$, every state starting, record r , ledger n , cover $(n, r) \mapsto n$ onto the counter. It is driven by the*

computation, with $f(b, r) = \neg r$; it is not permanent; and it factors as the latch of no event.

Proof. At $(0, \text{on})$ the next record is off, which breaks permanence; and any latch gives $\text{on} \vee h(0) = \text{on}$ there. $\square$

Checked as `toggle_not_permanent` and `toggle_not_latch` (`StructuralRecordAxis.v`).

Lemma 10.6 (The clock). *Let the clock have states (b, k, r) , step $(b, k, r) \mapsto (b + 1, k + 1, r \vee [k = 5])$, every state starting, record r , ledger b , cover $(b, k, r) \mapsto b$ onto the counter. It is permanent and not driven by the computation.*

Proof. Permanence: $\text{on} \vee x$ is on. Not driven: the states $(0, 5, \text{off})$ and $(0, 0, \text{off})$ have the same base state and the same record, and their next records are on and off, so no f of the base state and the record can give both. $\square$

Checked as `toggle_computation_driven`, `toggle_not_permanent`, `toggle_not_latch`, `clock_record_permanent` and `clock_record_not_driven` (`StructuralRecordAxis.v`). “Driven” is a condition on every state, reachable or not, and that is what the clock’s proof uses.

10.4 Is there only one such machine? No. One structure, two free choices.

Here’s a sentence that’s tempting and wrong: “there’s only one honest way to carry a record.” Three different things get squeezed into that breath, and I’ll pull them apart.

First claim: given the toll and a fixed price list, is there one way to keep the books? Yes; that’s Section 6’s initiality, and it assumes the price list.

Second claim: is every honest extension of a base *the same machine*, charges and all? No, and the counterexample is embarrassing in its simplicity. Take any honest extension and add one more coin to the bill on every step, whatever the step is. Now walk the five rules. The record still answers only to the base and its own value (R1); the bill never touches it. The ledger still never goes down (R2); it climbs faster, if anything. The step that turns the record on still costs at least one (R3), because the toll is a *floor* and a surcharge only raises the price. Once on, the record stays on (R4), and the same run still reaches the write (R5). So the billed machine is honest, and it isn’t the machine you started with: wherever the original sat still for free, the billed one pays a coin a tick for sitting still. Do it with the cup and the die. Charge a coin for every roll, six or not. The cup turns over on the same roll as before. You’ve changed the bill and nothing about the record. A machine that charges for CPU time is honest and isn’t the one that doesn’t.

But the billed machine isn’t a different *structure*. It’s the same machine with a different price list. So the claim worth keeping is sameness *up to the price schedule*: compare two

machines by their record, their halting and their steps, through their maps down to the base, and don't compare charges. Fix which event the record is, and the bill stops mattering; a flat surcharge, or any surcharge that depends on the current state, sits on the same structure. That's the first free choice. The walk above is the whole argument, and it's argued here, not checked over an arbitrary base.

And then the second free choice, and this one still surprises me. *Which event* is the record? Nothing in the five rules picks it. The converse above already says so, if you read it twice: for any event the base can reach, the latch of that event, priced at one unit per write, is an honest record. Do it with the die. One cup turns over on the first six. Another turns over on the first one. Both are cups with no reset wire; both are honest. Roll a one first, and until a six turns up the two cups disagree about whether anything has been recorded. So *which permanent reading counts as the record* is a second thing nobody forces on you. The price list is one free choice. The event is the other. That second choice is the question Section 13 is about: what, if anything, makes certification the event worth recording, rather than "the ledger passed one" or anything else.

Corollary 10.7 (The event is a free choice). *Let B be any base and h_1, h_2 two events, each meeting the supposition of Theorem 10.2. Then Latch_{h_1} and Latch_{h_2} are both honest extensions of B . If some starting base state b has $h_1(b)$ on and $h_2(b)$ off, then one step from $(b, \text{off}, 0)$ leaves the first record on and the second off.*

Proof. Honesty is Theorem 10.2 applied twice. The step is the latch's own rule: $\text{off} \vee h_1(b)$ is on and $\text{off} \vee h_2(b)$ is off. $\square$

Argued from `latch_core_honest` (RecordAxisDiscrimination.v) applied to each event; the corollary itself is not a separate Coq statement.

Now let the cup wobble. Say whatever turns it over sometimes does and sometimes doesn't, with odds you could write on a card. Does fixing the price list fix the odds? No, and the counterexample is two cups with the same price tag and differently loaded coins. Here's the setup, said carefully. Let a record be a single bit that moves at random: from each value, a list of weighted next values, every weight positive, never empty, never leaving on once on, and with the write from off to on charged at least one. Call two such records *the same schedule* when the same moves are possible and every move is charged the same, and *the same in probability* when the weighted lists are equal.

Theorem 10.8 (The schedule doesn't fix the probabilities). *(1) There are two honest probabilistic records with the same schedule that are not the same in probability. (2) There is an honest probabilistic record that no deterministic latch on its current value describes.*

Proof. (1) From off, one record moves to off or on with weights 1 and 1, the other with

weights 1 and 2; from on, both stay on with weight 1; both charge 1 for the write. Same possible moves, same charges, different weights. (2) The first record moves from off both to off and to on, while a latch $r \vee h(r)$ gives one value. $\square$

Checked as `schedule_determines_probabilities_refuted` and `deterministic_latch_handles_branching_refuted` (`ProbabilisticRecord.v`), with the definitions in `ProbabilisticRecordCore.v`. These are finite-weight models on one bit with no base underneath; they say nothing about any particular machine.

Third claim: is there one state space, up to renaming, that every honest record-carrier has to have? I don't claim it. Nothing needs it.

10.5 Bases from outside, and a relation I have to be honest about

The theorem is stated for any base, and the checked work instantiates it on three models of computation that other people built: a small Turing machine (any program); a register machine of the kind Cook and Reckhow described in 1973, with unit-cost indirect addressing (any program); and a lambda calculus (call-by-value, meaning an argument is worked out before it's plugged in, and made deterministic by letting a term with no step left to take step to itself). On each, the latch form holds, by the one generic theorem applied to that base (the checked names are listed with the formal statement below). So a register machine with an honest record is a register machine plus a cup. So is a lambda calculus. (There's one more instance, the machine I build later, where the flag is shown to be a latch directly, in Section 14.)

Here are the three bases explicitly. A base needs a state, a next-state function, a set of starting states and a halting predicate; the latch theorem asks for nothing else about its instruction vocabulary.

Definition 10.3 (Three concrete bases). The **tape base** has a finite Boolean list t , a head index h , a program counter q and a natural-number cost field. Reading beyond the list returns false. Writing at h extends the list with false cells if needed, then changes that cell. Moving right increments h and extends the list through the new head position. Moving left decrements a positive h ; at $h = 0$, it prepends a false cell and keeps the head at zero. A fixed finite program contains these instructions:

Instruction	Next state
Write a bit	Perform the write and advance q by one.
Move left or right	Perform the move and advance q by one.
Branch to j	Set $q := j$ if the current cell is true; otherwise advance q by one.
Halt	Leave the entire state unchanged.
ClaimTapeIsZero with parameter δ	Advance q by one and do nothing else.

An out-of-range program fetch is Halt. Every state is allowed as a starting state, and

halting means the fetched instruction is Halt. The cost field is preserved by every instruction in this base. In particular, its ClaimTapeIsZero instruction neither erases the tape nor charges the cost field; the name supplies no extra semantics.

The **random-access base** has a register function $X : \mathbb{N} \rightarrow \mathbb{N}$ and a program counter q . For a fixed finite program its instructions are:

Instruction	Next state
Constant (i, c)	Set $X(i) := c$.
Add (i, j, k)	Set $X(i) := X(j) + X(k)$.
Subtract (i, j, k)	Set $X(i) := \max(0, X(j) - X(k))$.
Indirect load (i, j)	Set $X(i) := X(X(j))$.
Indirect store (i, j)	Set $X(X(i)) := X(j)$.
Positive branch (j, a)	Set $q := a$ if $X(j) > 0$; otherwise advance q by one.
Jump to a	Set $q := a$.
Halt	Leave the entire state unchanged.

Each register-writing instruction advances q by one; its operands are evaluated in the old state, and all other registers are preserved. A fetch outside the program also leaves the state unchanged. Starting states have $q = 0$ and arbitrary register contents. A state is halted when it fetches Halt or is outside the program. Unit cost here concerns instruction counting, not the certification ledger.

The **lambda base** uses the terms and weak call-by-value rule defined in Definition 11.3. To compute its next state, first try to reduce the function part of an application. If that part cannot step and is an abstraction, either substitute an argument that is already an abstraction or take a step in the argument. If neither applies, the term cannot step. Variables and abstractions cannot step by themselves, and there is no reduction under an abstraction. The base leaves a term unchanged when it cannot step and calls precisely those terms halted. Every term is allowed as a starting state, including open terms.

These definitions make the examples instances of the base record. Applying the latch theorem to them does not assert universality, an execution-time bound, or a simulation between the three bases.

There's also a relation defined between bases, *weak base equivalence*: two bases are equivalent when some correspondence covers their start states both ways, related states agree on an observation and on halting, and one step on one side is matched by some number of steps, possibly zero, on the other. That "some number of steps" is the *stuttering* idea Leslie Lamport (the same Lamport as Section 6's message counters) put forward in 1983: a fine-grained machine and a coarse one shouldn't count as different because one takes three ticks to do what the other does in one. The relation carries no

cost; a machine that multiplies huge numbers in one step and a Turing machine differ in price, not in records. It's proved to be an equivalence relation, and the latch form is preserved across it.

There is a limit to that statement. The “preserved across it” proof throws away its equivalence hypothesis; the latch holds on every base regardless, so the preservation is true for a boring reason. And nowhere in the checked work are *two* of the bases actually related by this relation. The relation is defined, it's an equivalence, and it is not used to connect the Turing machine to the register machine to the lambda calculus, or any of them to the machine in Part IV. It would be easy to read that list as if it did. What's proved is that each base, separately, carries a latch. That's the honest sentence.

Definition 10.4 (Weak base equivalence). Bases B_1 and B_2 , with observations o_1 and o_2 into one common set, are *weakly equivalent* when some relation R relates every starting state of each to some starting state of the other, and any related x, y have $o_1(x) = o_2(y)$, the same halting, some $n \geq 0$ with $R(\text{next}_{B_1}(x), \text{next}_{B_2}^n(y))$, and some $n \geq 0$ with $R(\text{next}_{B_1}^n(x), \text{next}_{B_2}(y))$.

Theorem 10.9 (Outside bases, and the relation). (1) *Weak base equivalence is reflexive, symmetric and transitive.* (2) *If two bases are weakly equivalent, Theorem 10.1 holds over one exactly when it holds over the other.* (3) *Theorem 10.1 holds over each of the three bases in Definition 10.3, for every fixed program where the base has one.*

Proof. (1) Reflexive with equality as R and one step against one. Symmetric by reversing R . Transitive by composing the two relations; a step on the first base is matched by n steps on the second, and those n steps by some number on the third, by induction on n . (2) and (3) apply Theorem 10.1 directly. In (2) the equivalence hypothesis is discarded. $\square$

Checked as `weak_base_equiv_refl_holds`, `weak_base_equiv_sym_holds`, `weak_base_equiv_trans_holds` (through `weak_match_left_runs` and `weak_match_right_runs`), `weak_equiv_preserves_record_latch_holds` and `record_axis_is_latch_on_tm_holds` (`CrossBaseGranularity.v`), `record_axis_is_latch_on_ram_holds` (`CrossBaseGranularityRAM.v`) and `record_axis_is_latch_on_l_holds` (`CrossBaseGranularityL.v`), with the definitions in `CrossBaseGranularityCore.v`.

10.6 Records with more than two values

A cup has two positions. Some records have more: a counter of how many checks passed, a level that only rises. The same reasoning goes through one threshold at a time. For each value a , “the record has reached a ” is a yes/no reading, and each of those readings is a latch. What doesn't go through is squeezing the whole record into one cup. “Rises” needs one word: the values come with an order $\leq$, “is at or below,”

which needn't compare every pair (that's what makes it a *partial* order), and $[a \leq x]$ is the yes/no reading "x has reached a."

Definition 10.5 (A growing record). Let the record take values in a set A with a partial order $\leq$ decided by a Boolean test. A record ρ on M , with cover π onto a base B , is an *honest growing record* when: there is one function f with $\rho(\text{next}_M(m)) = f(\pi(m), \rho(m))$; it grows, $\rho(m) \leq \rho(\text{next}_M(m))$; the ledger never goes down and every step that changes ρ costs at least one; and some run from a starting state reaches a step that changes it.

Theorem 10.10 (Growing records are families of latches). (1) For every honest growing record there is an h with

$$[a \leq \rho(\text{next}_M(m))] = [a \leq \rho(m)] \vee h(a, \pi(m), \rho(m))$$

for every value a and state m : each threshold is a latch, whose event may read the current value. (2) The thresholds determine the value: if $[a \leq x] = [a \leq y]$ for every a , then $x = y$. (3) For a growing record, charging every change is the same as charging every threshold that switches on. (4) One latch is not enough: a three-valued record that climbs $0 \rightarrow 1 \rightarrow 2$ over a one-state base, each climb charged one, is honest, and no single Boolean latch with a decoding from the base state and the latch reproduces it. (5) A chain of distinct k -bit words, each word's set bits contained in the next one's, has at most $k + 1$ members.

Proof. (1) Take $h(a, b, x) = [a \not\leq x] \wedge [a \leq f(b, x)]$. If $a \leq \rho(m)$ already, growth and transitivity give $a \leq \rho(\text{next}_M(m))$; otherwise the equation is the definition of h . (2) Antisymmetry, with $a = x$ and $a = y$. (3) Growth plus antisymmetry turn "the value changed" into "some threshold switched on," and back. (4) Over one base state, a latch and a decoding can produce at most two values, and the record takes three. (5) Each strict step up the chain adds at least one set bit, and there are only k bits. $\square$

Checked as `growing_record_decomposes_holds, thresholds_determine_record_holds, record_price_iff_threshold_price_holds, one_latch_refuted` and `chain_needs_bits_holds` (GrowingRecord.v), with the definitions in GrowingRecordCore.v. The order need not be finite; (5) is the finite-memory bound.

10.7 What the base being free buys you

The reason this section exists is to answer the suspicion I started with. The record isn't special pleading for one machine. Over any base, under rules you can read, a record that answers to the computation and never comes down is a latch on an event, and the proof doesn't care what the base is. What a particular machine adds is not the latch; it's a *choice of event* and a *choice of price list*, and both are choices. The surcharged cup shows the first: the bill can change and the record not notice. The two cups on one die show the second: the rules are happy with either event. Any machine I build

will have to make both choices, and whether its choice of event is the right one is an open question, and I'll get to it. Before that, the question every machine with a record has to face, and that no amount of good design makes go away. A record is worth keeping because something later leans on it instead of working the thing out again: that's Section 2's glance, reading a block's bucket off the table instead of applying the rule from scratch. Call any move like that a *shortcut*, and call it *sound* when it gives the answer the long way would have given. (That gloss is about cost, leaning instead of redoing, and the wall I'm about to show you doesn't look at cost at all; the next section says exactly what it does look at.) So: can a machine *decide*, from inside, whether its own shortcuts are sound? That's a wall, and it's the same wall Gödel and Turing hit, and I have to show you that it's there before I show you anything that pretends to be on the other side of it.

11 The diagonal

In Section 2 I said you'd watch the same move get made again and again as the book went on: somebody says "everything is X," it's true enough to do real work with, and then somebody draws the diagonal. I owe you the diagonal itself now, because Section 10 just handed a record to *any* system that steps, and the next thing anyone wants from a system with a record is for it to tell sound shortcuts from unsound ones by itself. Before I let anything claim that, I have to show you the wall every such claim runs into, and then show you exactly where a system with a record stands relative to the wall. Not on the far side. Nobody's on the far side.

11.1 Five names and a table

Do this first, with a pencil, because the whole family of arguments is this one trick and you should own it. Write five friends' names down the left of a page. Next to each name, write a five-digit string of zeros and ones, anything you like. Now build a new string: its first digit is the *opposite* of the first digit of the first row; its second digit is the opposite of the second digit of the second row; and so on down the diagonal. Look at what you've made. It differs from row one in position one, from row two in position two, from every row at that row's own position. So it's nobody's string. Your table of five missed something.

With five rows and five digits that's a cute fact. Georg Cantor's point, in 1891, was that it works for *any* list, however long, even infinite: hand me a complete list of all infinite strings of two symbols, I'll walk the diagonal flipping as I go, and I'll hand you back a string that's on no row. So there is no complete list. There are more such strings than there are places on any list, and that's where "some infinities are bigger than others" comes from. (He wrote the two symbols as m and w, not 0 and 1, and that paper wasn't even his first proof of the fact; the 1874 one was harder. The diagonal was him finding the easy way.)

Now look at what the argument *did*, because it's going to do the same thing again

and again, in new clothes, and I want you to recognise it each time. (Not every name below is a flipped diagonal in the table sense. Kleene's theorem is the tool that makes the diagonal buildable, Rice's is the general form, and Thompson's is an attack that uses the same trick.) There was a table: rows are the things being listed, columns are questions, each cell says "does row i do column j ." There was the diagonal: the cells where a thing is asked about *itself*. There was a flip: something with no fixed point, like turning yes into no, applied along the diagonal. And the thing that got refuted was *the assumed object*, the complete list. Nothing outside the assumption was touched. Table, diagonal, flip, and the assumption eats itself. Hold those four.

11.2 The same move, again and again

Russell, 1902, from Section 2: the collection of all collections that don't contain themselves. That's Cantor's flipped diagonal with "contains" as the table, and Russell said as much; he got there by running Cantor's argument on a supposed collection of everything. What got refuted was the assumed object, the collection of everything, and Zermelo's fix was to stop assuming it.

Jules Richard, 1905: some English phrases define real numbers ("the ratio of a circle's circumference to its diameter"); there are only countably many phrases, so list the numbers they define in order; run Cantor's diagonal down their decimal expansions and you get a real number no phrase defines, except that you just defined it, in the phrase that describes the diagonal. The lesson that came out of that one is the lesson this whole section turns on: keep apart the language a system *talks in* and the language you use to talk *about* it. Mix them and the diagonal bites you.

Richard's paradox has a cousin small enough to do at the table, and it comes with a footnote I can't resist. Russell, the first to put it in print, in 1906, credited it to G. G. Berry, a junior librarian at the Bodleian Library in Oxford. A paradox that has outlived empires, and it came from the library desk. In the version in *Principia Mathematica* it's about syllables. Take the phrase "the least integer not nameable in fewer than nineteen syllables." There are only so many phrases shorter than nineteen syllables, so they name only so many numbers, so some least number is left over, and the phrase picks it out. Now count the phrase: the, least, in-te-ger, not, name-a-ble, in, few-er, than, nine-teen, syl-la-bles. Eighteen. In eighteen syllables it has named the number that can't be named in fewer than nineteen. Same lesson as Richard's, in a smaller coat: "nameable" is a word about the language, used inside the language, and that's where the bite is.

Kurt Gödel, 1931, with an improvement by Barkley Rosser in 1936: the big one, and I'll say exactly what it says because it's misquoted more than any theorem alive. Take any consistent formal system with a mechanically checkable list of axioms that can do a bit of arithmetic. Then there are sentences it can neither prove nor refute. The engine is a lemma that lets you build, for any property you can write down, a sentence that the system proves *equivalent* to that property holding of the sentence's own code number.

Apply it to “not provable” and you get a sentence equivalent, inside the system, to its own unprovability. Table: sentences by code. Diagonal: a sentence about its own code. Flip: provable to not-provable. Two cautions the Stanford Encyclopedia stresses and I’ll repeat: the sentence doesn’t literally “say” anything about itself, it’s provably *equivalent* to a claim about its code, which is less mystical and more exact; and the theorem doesn’t say the sentence is unknowable, it says *this one system* can’t settle it.

And how did the world take it? Mostly, at first, it didn’t. Gödel, twenty-four years old, announced the first theorem during a round-table discussion on the last day of a conference in Königsberg in September 1930, and by the account of Solomon Feferman, one of the editors of Gödel’s collected works, the room met it with general incomprehension. Except for one man. John von Neumann pulled Gödel aside to talk about it, kept thinking about it, and a couple of months later saw that the argument gave something more: a consistent system of this kind can’t prove its own consistency. He wrote to Gödel with it in a letter dated 20 November 1930. Gödel had already got there. His paper had reached the journal three days earlier, with that second theorem announced in it, and von Neumann left the credit with Gödel. It’s the “somebody got there first” shape from the opening chapter, with one difference: here the man who got there second was von Neumann.

Alan Turing, 1936: and here the historical record is not the story you usually hear. The usual story is that Turing proved the halting problem undecidable. Joel David Hamkins and Theodor Nenu went back to the 1936 paper and found that the word “halting,” the modern statement, and the self-referential proof everyone teaches are all, strictly, absent from it. What Turing proved undecidable were two other questions about his machines (whether a machine prints infinitely many digits; whether it ever prints a given symbol). The second is *equivalent* to halting and the first is strictly harder, and the name “halting problem” came later, probably from Martin Davis. The textbook version is still right as mathematics, and here it is, on two index cards. Card A is the Checker: hand it any program card and it announces “halts” or “runs forever.” Card B is the Contrarian: “Hand me to the Checker. If it says I halt, run forever. If it says I run forever, stop.” Hand B to A. Either answer is wrong. There’s no Checker. Table: programs by programs. Diagonal: a program run on itself. Flip: halt to loop.

Stephen Kleene, 1938, and this is the one that turns the paradox into a tool, so slow down. Everything above needs the self-referring object to *exist*: Gödel’s sentence has to be buildable, the Contrarian has to be able to “hand me to the Checker.” Kleene’s second recursion theorem says: for any transformation of programs you can compute, there’s a program that behaves exactly as if it had been handed its own source code. Any program may be written as though it already had a copy of itself in hand. He stated it almost in passing, in a paper on something else, and proved it in two lines, and Yiannis Moschovakis calls it a little gem, and it’s the reason “programs that print their own source” exist in every programming language (they’re called quines, after the philosopher; try “Print the following twice, the second time in quotes: ‘Print the

following twice, the second time in quotes:''' and obey it). Self-reference in computation isn't a paradox. It's a construction. The paradoxes happened only when someone assumed an *unrestricted* collection or language; the theorem happens when you build the self-referrer honestly.

And where does the word quine come from? Not from the philosopher himself. Douglas Hofstadter coined it in 1979, in *Gödel, Escher, Bach*, in honour of Willard Van Orman Quine, who had made a long study of self-reference that works indirectly. Quine left a sentence you can take apart at the table, from a 1962 essay on paradox. Here it is, all of it: "yields falsehood when preceded by its quotation" yields falsehood when preceded by its quotation. Do what it says. Take the part in quotation marks, put a quoted copy of it in front of itself, and look at what you've built: it's the whole sentence, word for word. So the sentence says that it itself is false. That's the liar paradox, but look at how it got there: no "this sentence" anywhere, no pointing at itself, just an instruction to quote and copy, the same move as the print-twice instruction above. That's Kleene's trick in plain English, and it's why the paradox and the tool look so alike. The paradox is what you get when the copy asserts something about its own truth. The tool is what you get when the copy just does something.

Henry Rice, 1953, whom you met briefly in Section 7: the general form. Any property that depends only on what a program *does* (not how it's written), and that's true of some programs and false of others, has no algorithm that decides it for all programs. "Does this program halt" is one such property. "Does this program ever output 7" is another. "Does this program contain an if-statement" isn't, because that's about the text. One standard proof goes straight through Kleene.

And the one that's actually an attack: Ken Thompson, in his Turing Award lecture, published in 1984. Build a compiler that, when it compiles the login program, plants a back door, and when it compiles *a compiler*, plants both habits into that one too. Then delete the trick from the source. The compiled compiler still carries it, and recompiling a clean source with the poisoned binary re-plants it, forever. Reading the source never reveals it. A locksmith teaches an apprentice to cut a spare for every town-hall key and to teach *that* to every apprentice they train, and never to write it in the manual; years later the manual is spotless and every locksmith in town cuts the spare. That's the recursion theorem with a knife in it, and his moral was that you can't trust code you didn't write entirely yourself, which is a sentence I'll come back to in Section 17 when I tell you what the trusted base of this book is.

And Thompson wasn't first either, and he said so. Pull on the lecture's acknowledgment the way the opening chapter pulled on Bitcoin's footnotes: he writes that he first read of the possibility in an Air Force critique of the security of an early implementation of Multics, a time-sharing system from the 1960s, and that he couldn't find a more specific reference to it, and he asks anyone who can to send him one. The critique is Paul Karger and Roger Schell's 1974 security evaluation of Multics, which already

described a trap door planted in a compiler that, because the compiler is written in its own language, keeps itself alive. So the knife was in the drawer ten years before the lecture. Then the obvious next question: is there any way out, or is “trust no binary you didn’t build” the bottom? There’s a partial answer, and it took a while. David A. Wheeler, in a 2005 paper and a 2009 dissertation at George Mason University, worked out *diverse double-compiling*: compile the suspect compiler’s source with a second, independently made compiler, use the result to compile that source again, and compare the output bit for bit with the suspect binary. If they match, the binary really is what its source says, unless both compilers were poisoned in the same way. The trust doesn’t vanish. It moves, from one compiler to the claim that two unrelated ones weren’t rigged together. That’s the shape of every stopping point in this book: you don’t reach the bottom, you reach a place where you can say exactly what you’re trusting.

Bill Lawvere, 1969, showed all of these are one theorem in disguise: if a set can be mapped onto all the functions from itself to some target, then every function on the target has a fixed point; so if the target has a function with no fixed point (a flip), no such covering exists. Cantor, Russell, Gödel, halting, as instances of one shape. Noson Yanofsky retold it in 2003 without the category theory. I mention it because “it’s the same wall” is a claim I make below, and I want you to know it’s fair at the level of *proof shape*, and that every instance has its own boundary of *who counts as a decider*, and that boundary is exactly where most overclaims live.

11.3 What it doesn’t say

Before I point this at a system with a record, the four wrong turns, because they’re the ones people take.

“Gödel shows minds aren’t machines.” John Lucas argued it in 1961 and Roger Penrose revived it twice. The standard replies, as far as I can tell, have held up: the Gödel sentence is true *only if the system is consistent*, so a human who “sees” its truth has to know the system is consistent, which is the hard part; and nobody can build the Gödel sentence of a system they can’t specify, so humans might be machines too complicated to diagonalise. The theorem has a conditional in it and the argument drops the conditional.

And here’s the part that’s easy to miss: Gödel himself didn’t go as far as Lucas. In a lecture on the day after Christmas in 1951, at a meeting of the American Mathematical Society at Brown University, he drew from his own theorems only an either-or. Either the human mind, even inside pure mathematics, infinitely surpasses the powers of any finite machine, or there are problems about whole numbers, of a certain simple kind, that no proof the human mind can conceive will ever settle. Feferman, writing about that lecture, says Gödel privately believed the first half and still stated only the disjunction, because he didn’t have a knock-down argument against the machine side. Gödel wrote soon afterward that he meant to publish the lecture, and never did; the

text sat among his papers until after his death. So the man whose theorem gets quoted for “minds aren’t machines” stated an either-or and left the deciding to whoever came next. That’s the conditional again, kept, by the person with the most right to drop it.

“Undecidable means unknowable.” No. Two senses share the word: a problem with no single algorithm correct on every instance, and a sentence independent of one particular system. Neither means nobody can know a given answer. Any single instance is decidable, trivially: for any one program, one of “say yes” and “say no” is right. What’s ruled out is one *inside* procedure that’s right on *every* case.

“Everything interesting is undecidable.” Rice is about behavioural properties over *all* programs. It says nothing against checking the text, nothing against deciding for specific programs, nothing against analyses allowed to say “I don’t know,” nothing against finite-state model checking or type checking. Real compilers are limited by cost and by choosing to be sound-but-incomplete, not by the diagonal.

“Self-reference is a paradox.” It’s a theorem with a construction. Kleene, quines, Thompson.

11.4 Now point it at the record

Here’s the question Section 10 left open, with nothing in it but what Parts I to III have built. A system steps; a record rides along and latches on an event (Section 10); when the event is a certification, Section 8 said the reading, with its object and its paid event behind it, is what lets an auditor treat a claim as established, and for a claim that a shortcut is sound, that’s what makes the shortcut *admissible*. So: can a system like that have, as one of its own programs, a procedure that correctly decides, for every program, “does this program admit a sound shortcut”? That would be autopilot. Hand it any program and it says yes or no, and it’s always right.

That’s the question in plain words, and the checked statement is narrower, so here it is before the proof. In the theorem, “admits a shortcut” is any property of programs with two features: two programs that behave alike agree on it, and there’s one program that has it and one that doesn’t. That’s all the argument gets to look at. The diagonal doesn’t see soundness. Each system that wants the wall picks its own property, and whether that property lines up with being entitled to soundness in Section 8’s sense is a separate fact about that system, checked one system at a time if it’s checked at all. A procedure that returns soundness records for some programs, or only for restricted inputs, or with outside evidence, is not what the wall rules out.

So in the theorem “shortcut” is a label. The plain-words gloss at the end of Section 10, leaning on a record instead of working the thing out again, is about *how* a program gets its answer, and how is exactly what extensionality throws away: the theorem only gets to see what programs do, and it treats any two that do the same as one. A question like “is there a cheaper program that gives the same answers?” isn’t the theorem’s question, and a wall for it would need a different theorem. The nearest thing I know

of is Manuel Blum’s speedup theorem, from 1967, and it goes somewhere stranger: there are computable functions with no fastest program at all, where every program for the function has another that’s faster on all but finitely many inputs. That’s a fact about cost, not a wall about deciding, and nothing in this book proves anything in that territory.

The answer is no, under premises. The proof is the two index cards, and I’ll give you the premises because they’re the whole content. Suppose there were such a decider, one that says yes exactly when a program admits a shortcut. Build the flip: “if the decider says yes about me, behave like a program that admits no shortcut; if it says no, behave like one that does.” If the system has a recursion theorem (a program can be handed its own code), and if that flip is one of the transformations the system can carry out, and if “admits a shortcut” depends only on behaviour (two programs that behave the same agree on it), and if there’s at least one program each way, then there’s a program that behaves exactly like its own flip, and whatever the decider says about it is wrong. Both cases close in a line each. That’s `structural_shortcut_undecidable`, stated for *any* substrate with those fields. (A substrate here is the bare shape with exactly the pieces this theorem asks for: programs, a run that may fail to give back a state, a counter, an encoding of programs as data, a notion of “representable” transformation, and the recursion theorem for those. Nothing about what the states are made of. The smallest checked instance is the nat family below.) It’s Rice’s trick one floor over, and I should be careful with that. Rice’s theorem is stated for the functions Turing machines compute. This wall is the same diagonal with different premises: it carries its own recursion theorem and its own list of representable transformations, and it picks the property. So it isn’t a corollary of Rice as Rice stated it, and it doesn’t separate anything from classical halting. And here’s something about its proof that I want you to have: it never touches the counter, the toll, or the step relation. It uses the recursion theorem, representability of the flip, the two witnesses, and extensionality, and nothing else. (The file’s header calls the substrates it covers ones that respect the toll, which reads as if the toll were needed. The proof uses neither the toll nor the counter’s climbing. It’s the diagonal, clean.)

Here’s the setting and the wall, with every word pinned down so you can poke at it.

Definition 11.1 (Substrate). A *substrate* has a type of states, a type `Prog` of programs, and a run map that takes a program and a state and either gives back a state or gives nothing. It has a counter μ on states and a step relation along which μ never decreases; an encoding `enc` of programs as states with a decoder `dec` such that $\text{dec}(\text{enc}(p)) = p$; and a predicate `Rep` on maps $f : \text{Prog} \rightarrow \text{Prog}$, the transformations the substrate counts as its own. Two programs are *equivalent*, $p \simeq q$, when $\text{run}(p, s) = \text{run}(q, s)$ for every state s . The last field is the *recursion theorem*: for every f with `Rep(f)` there is a program p with $p \simeq f(p)$.

Definition 11.2 (Shortcut predicate). A *shortcut predicate* on a substrate is a property A of programs with two programs y and n such that $A(y)$ holds, $A(n)$ fails, and A is extensional: $p \simeq q$ implies $A(p) \iff A(q)$. For a Boolean function d on programs, its *flip* is $f_d(p) = n$ if $d(p)$ is true and $f_d(p) = y$ if $d(p)$ is false.

Theorem 11.1 (No inside decider). *Let a substrate carry a shortcut predicate A with witnesses y and n . Then there is no Boolean function d on programs such that $\text{Rep}(f_d)$ holds and, for every program p , $d(p) = \text{true} \iff A(p)$.*

Proof. Suppose d is such a function. Since $\text{Rep}(f_d)$, the recursion theorem gives a program p with $p \simeq f_d(p)$. Split on $d(p)$. If $d(p)$ is true, then $A(p)$ holds by correctness, and $f_d(p) = n$, so $p \simeq n$, and extensionality turns $A(p)$ into $A(n)$, which fails. If $d(p)$ is false, then $A(p)$ fails by correctness, and $f_d(p) = y$, so $p \simeq y$, and extensionality turns $A(y)$ into $A(p)$. Both cases are contradictions. The argument uses the recursion theorem once, the two witnesses, extensionality, and nothing about μ , the step relation, or the encoding. $\square$

Checked as `structural_shortcut_undecidable` (`StructuralUndecidability.v`; the correctness it rules out is $d(p)$ true exactly when $A(p)$), stated over the class `Substrate` (`Substrate.v`) and its extension `WithShortcutPredicate`. Every field named above is a premise; the theorem is about every substrate that supplies them.

So the wall is there. Now the honest part: *who counts as a decider*. The theorem rules out a decider whose flip the substrate can itself represent. It says nothing about a decider standing *outside*, and an outside decider can exist for the very same property, correct on every program. That's not a loophole; it's the boundary of the sentence, and the boundary is Richard's lesson: the language you talk about a system in is not the language it talks in. You'll see it in miniature in a minute, on the `nat` family, with a pencil.

Two forms of the wall live in this chapter, and they're honest in different ways.

The *nat family* (`nat_self_undecidable`): a tiny substrate built so that program two is the Contrarian for a given candidate decider d , and for every d , d is wrong about program two. It's a *family*, one instance per d , and its recursion theorem holds by construction. It's the index cards made formal and nothing more. Don't let it sound like more.

Theorem 11.2 (The `nat` family). *Fix any Boolean function d on $\mathbb{N}$. Let the substrate N_d have programs and states $\mathbb{N}$; program 0 returns 1, program 1 returns 0, program 2 returns 0 if $d(2)$ is true and 1 if it is false, and every other program gives nothing; μ is constantly 0; $\text{Rep}(f)$ holds when $f(p) = 1$ for every p with $d(p)$ true and $f(p) = 0$ for every other p ; and $A(p)$ holds when program p returns 1 on input 0. Then it is not the case that $d(p) = \text{true} \iff A(p)$ for every p .*

Proof. The recursion theorem of N_d holds with the fixed witness 2: for a representable f , $f(2)$ is 1 or 0 according to $d(2)$, and program 2 returns exactly what that program returns. The witnesses for A are 0 and 1. The flip of d is representable by the definition of Rep, so Theorem 11.1 applies to d itself. Directly: $A(2)$ holds exactly when $d(2)$ is false. $\square$

Checked as `nat_self_undecidable (NatSubstrateInstance.v)`, with the built-in recursion theorem `nat_recursion_theorem`; the general form, no decider whose flip is representable in N_d , is `nat_structural_shortcut_undecidable` in the same file.

Now push on it once more, because the nat family shows the outside decider in three lines, and you can do them yourself. Stand outside N_d , knowing d . Write down a new decider e : say yes on program 0, say “not $d(2)$ ” on program 2, and say no on everything else. Check it against A . Program 0 returns 1, so $A(0)$ holds and e says yes. Program 1 returns 0, so $A(1)$ fails and e says no. Program 2 returns 1 exactly when $d(2)$ is false, and that’s what e says. Every other program gives nothing, so A fails and e says no. So e is right about every program. Does that break the wall? No, and here’s the bit that matters: in N_d a flip is representable only if it says 1 exactly where d says yes, so the flip of e is representable only if e and d agree everywhere, and they disagree at program 2. The correct decider exists; its flip just isn’t one of the substrate’s own moves. (That one isn’t a separate theorem in the Coq. It’s three lines and a pencil, and you just did it.) Undecidable from inside and decided cleanly from outside, at the same time, about the same property. That’s not a paradox. It’s two different people asking.

And the one where the recursion theorem is *proved*, not assumed, which is why it exists: L , the weak call-by-value lambda calculus that Yannick Forster and Gert Smolka presented in 2017 as a model of computation in Coq, and which a group at Saarland has since used as the seed of a whole library of undecidable problems. The development here builds its own copy of L from the reduction rules, so the L results below assume nothing from that library. There, `L_recursion_theorem` is a theorem with a proof, Rice’s theorem follows (`L_rice`), halting follows (`L_halting_undecidable`), and the diagonal is reproved in the shape of the substrate theorem (`L_structural_shortcut_undecidable`), for any property of programs that depends only on what they compute and that holds of one program and fails of another. The statement mentions no shortcut. That’s the diagonal for a fixed model with nothing assumed. It isn’t an instance of the generic class (L ’s runs aren’t total in the way the class wants), so it’s the argument *reproved*, not the generic theorem *applied*. Both are true; they’re different sentences.

Here’s what that string of words means, because L is the strangest thing in this book that computes, and the smallest. The lambda calculus is Alonzo Church’s, from the 1930s, and it’s a programming language with one idea in it: functions. A term is one of

three things. A variable. A function, written λ followed by its body, the body saying what to do with whatever you hand it. Or one term applied to another, written side by side. Running a program is one move, over and over: when a function meets an argument, put the argument into the body where the variable was. No numbers, no memory, no instructions, and Turing showed in 1937 that it computes exactly what his machines compute. “Weak call-by-value” is two house rules about the order of moves: an argument is run down to a finished function (a *value*) before it’s put in, and nothing runs inside a function that hasn’t been handed anything yet. Variables are written as numbers counting how many λ s back they were bound (Nicolaas de Bruijn’s trick, so two variables can never get mixed up by sharing a name), and a term is *closed* when every variable in it is bound inside it, so it leans on nothing outside. Last piece: to hand a program to a program, L needs a way to write any term as data, the way a program’s source is a string you can pass around. The one used here is the *Scott encoding*, named for Dana Scott, and all you need about it is that every term t gets one *code*, written $\ulcorner t \urcorner$, and the code is itself a closed value. Two tools get built on top of that in the proof: a *quoting* term Q that takes the code of a term to the code of that code (a program that prints its own source needs exactly this move), and a *fixed-point combinator*, a term that lets a function call itself, which is how Q works its way through a code of any size.

Definition 11.3 (L, as built here). A term of L is a variable (a de Bruijn number), an application st , or an abstraction λs ; the values are the abstractions. One step is weak call-by-value, left to right: $(\lambda b)v$ steps to b with v substituted when v is a value, a step inside the left part of an application is a step of the application, and so is a step inside the right part once the left part is a value. A term s *evaluates to* v when s reaches v in some number of steps and v is a value; $p \equiv q$ when p and q evaluate to the same values. Every term t has a code $\ulcorner t \urcorner$, its Scott encoding, which is a closed value. A map f on terms is *L-computable* when some closed value s has $s \ulcorner p \urcorner \equiv f(p)$ for every p . A term D *decides* a property P of terms when, for every p , $D \ulcorner p \urcorner$ reaches $\lambda\lambda.1$ (true) if $P(p)$ holds and $\lambda\lambda.0$ (false) if it fails. P is *extensional* when $p \equiv q$ and $P(p)$ give $P(q)$.

Theorem 11.3 (Recursion, Rice and halting for L). (1) For every closed value s there is a closed term t that reaches $s \ulcorner t \urcorner$; hence every L-computable f has a term p with $p \equiv f(p)$. (2) Let P be extensional, with $P(y)$ and not $P(n)$ for closed terms y and n . Then no closed term decides P . (3) No closed term decides halting, the property of evaluating to some value.

Proof. (1) A closed quoting term Q takes $\ulcorner t \urcorner$ to the code of $\ulcorner t \urcorner$; its constructors, recursive program and terminating induction are given in Appendix G. A closed term mkapp takes two codes to the code of their application. Let $A = \lambda x. s (\text{mkapp } x (Q x))$ and $t = A \ulcorner A \urcorner$. One step gives $s (\text{mkapp } \ulcorner A \urcorner (Q \ulcorner A \urcorner))$, and the argument reduces to $\ulcorner A \ulcorner A \urcorner \urcorner = \ulcorner t \urcorner$. For the second half, take s computing f ; then $t \equiv s \ulcorner t \urcorner \equiv f(t)$. (2) Given a closed decider D , the flip $\lambda x. D x (\lambda_. n) (\lambda_. y) (\lambda z. z)$ is a closed value; on the

code of p it reaches n when D answers true and y when D answers false. Take its t from (1). If $P(t)$, then $t \equiv n$ and $P(n)$; if not, $t \equiv y$ and $P(t)$. Both are contradictions. (3) Halting is extensional; apply (2) with $y = \lambda z.z$ and $n = (\lambda x.x x)(\lambda x.x x)$, which steps only to itself and so never reaches a value. $\square$

Checked as `second_recursion`, `L_recursion_theorem`, `L_rice`, and `L_halting_undecidable` (`LRecursion.v`), with the quoting term's specification `Q_spec` and the substrate-shaped form `L_structural_shortcut_undecidable` (no decider whose flip L can compute says true exactly when the property holds) in the same file. The file uses only Coq's standard library.

That's the wall on paper, on a family built to fit it, and on one outside model with nothing assumed. The machine built in Part IV fills in this same shape with nothing assumed either: on its host machine, the one with a counter for every number that the universal program runs on, programs are numbers, Kleene's recursion theorem is a theorem, and so is Rice's, and the fixed point can be taken to agree on the ledger and the flag, not just on what the program computes (Section 14.18).

Here's the picture all of that fills in, lined up against Turing's, one column per axis.

Time axis	Structural axis
counts: steps	counts: μ
asks: does it halt?	asks: does it admit the shortcut?
the wall: no Turing machine decides halting	the wall: no decider whose flip is representable, under a recursion theorem

Read the table as two uses of one diagonal, each under its own premises, and not as more. The structural wall never uses μ ; the μ entry only says what that axis counts. The nat result is a family indexed by the candidate decider, and L 's results are about one fixed model and properties that mention no shortcut. Neither hands you a second fixed undecidable problem, and neither separates Turing degrees: a property can sit behind the wall from inside and be decided cleanly from outside, as the pencil argument on N_d showed. And "admits the shortcut" in the right-hand column is the theorem's sense, any behavioural property with a yes and a no, not "has a cheaper way to the same answer"; the column has no wall for that second question.

11.5 How to kill it

Three ways, and I mean them as instructions.

One: produce, for some substrate with a shortcut predicate, a decider whose flip the substrate can represent and that's correct on every program. That breaks the generic proof at a named step, and I've named the steps.

Two: for the nat family, produce a d that correctly decides the predicate of the substrate built from d .

Three: for L , produce a closed term that decides some extensional property holding of one closed term and failing of another. That breaks L_{rice} , which rests on the reduction rules and Coq’s standard library and nothing else.

An outside decider alone doesn’t count against any of the three; the pencil argument on N_d is why.

11.6 What this does to the book

It does two things. It keeps me honest: nothing in what follows claims that a system with a record can check its own shortcuts on autopilot, because that’s a wall, the same wall, and I showed you the bricks. And it tells you what a record *can* do, which is the thing Part III has been about since the counter: it can carry a shortcut *with its receipt*, so that the question “was this earned” is answered by an event you can point at rather than a decider that can’t exist. The diagonal says you can’t automate the judgement. The latch says you can record the commitment. Those are different jobs, and a record does the second.

The diagonal has a famous cousin that seems like it ought to bite a record too, so let me walk right up to it. Gödel’s second theorem says a consistent system strong enough for arithmetic can’t prove its own consistency, and Martin Löb sharpened that in 1955: if such a system proves “if this is provable, it’s true,” it already proves the thing itself. So can a record vouch for itself, and does Löb stop it? Here’s where I landed, and it’s a relief more than a discovery. Löb’s theorem is about *provability*, the bare claim that some proof exists. A record of the kind this book keeps is a different animal: a dated event, this check, of this claim, passed. Sergei Artemov drew exactly that line in 2001, between implicit provability (“there is a proof”) and explicit evidence (“this is a proof”), and showed that for explicit evidence the reflection step “if this is a proof of F , then F ” is harmless, while for the bare “there is a proof” it’s what Gödel and Löb forbid. A record sits on the explicit side. Where its trust really rests, on the checker meaning what it says, that’s proved once from outside, and Gödel’s second theorem says that’s where it has to rest for anything of arithmetic strength. Try to move that proof inside the system, so it can certify its own checker or a successor’s, and you hit exactly the wall Eliezer Yudkowsky and Marcello Herreshoff named the Löbian obstacle in 2013, writing about agents that want to trust their own rewrites.

One more thing falls out of the same picture, and it’s a short argument rather than a checked theorem, so take it at that weight. Can a record rest on itself, the way the sentence “this sentence is certified” would like to? Not if it’s earned and permanent. The check that a certificate stands on happened while the record was still down, because the record never comes back down and it went up only afterwards. So no claim whose truth needs the record already up can be the claim a certificate stands on. Saul Kripke’s 1975 theory of truth calls a sentence like that, one that only ever leans on itself, ungrounded, and earned records are grounded in his sense: every one bottoms out in a check that didn’t lean on it. Caspar Oosterheld’s 2019 work on programs that

cooperate by simulating each other needed a grounding step of the same kind to stop an infinite regress. I haven't checked this in Coq, and I'm not claiming more than the paragraph says.

One last distinction, because the two are easy to conflate and conflating them is the easiest way to misread this book. There's a *second* wall, nothing to do with the diagonal. It's Section 7's kind of wall, a window with a colliding pair in it. Two states that differ in their certification reading can look identical through a window that doesn't show that reading, the stamp inside the closed drawer and the drawer opened and shut, and then no function of the window recovers the reading (Theorem 7.1: a decoder has to give one answer on everything the window lumps together). That's missing information, not an undecidable question; it holds for finite things with nothing to compute; and it sits *beside* Rice, not under it. The property the diagonal uses (any behavioural property of programs with a yes and a no) is a different kind of thing from the reading the window hides, and nothing here says the two are the same. Two walls, two axes. The diagonal is about what no inside procedure can decide. The window is about what no outside observer can see. I'll need both, and I'll try not to call one by the other's name again. The machine I build later has a colliding pair of exactly that kind (Theorem 14.14).

12 Five small models

A toll, a counter, two kinds of wall and a latch, and every one of them shown on paper, on cups and cards and dice I made up. I've been building a toy and claiming it's the smallest version of something people actually do. Time to go and look at what people actually do. I picked five systems that keep records other people rely on, with real money or real security behind them, and asked each the one question this book is about: *is there a price on the moment the record is written?* Not on the work around it. On the write. I'll tell you the answer first, because it's a single sentence and it surprised me: of the five specifications, read through the transcriptions I made of them, only one prices the write, and that one has a leak at exactly the place Section 6 said a counter leaks, where it goes down.

Three things are easy to blur in a chapter like this, so I'll name them now and keep them apart all the way through. There's the *real specification*: the document real money and real security run on. There's my *transcription*: one mechanism copied out of that document into Coq, small enough to check. And for some of the five there's a *wrapper*: a tiny abstract model that takes one premise as given and says what follows from it. A claim proved at one level isn't a claim at the level above it. Near the end I'll walk one system, gas, through all three, so you can see exactly where each claim stops.

What I built for each is a *fragment*: a small Coq model of the one mechanism I'm asking about, transcribed from the system's own specification where there was one, and nothing more. Not the system. I'll say for each one what the fragment is and what it isn't, because "Ethereum has been formalised" is the kind of sentence that gets a

book thrown across a room, and rightly.

12.1 Proof-of-stake finality: the price is on going back

Ethereum and chains like it decide which blocks are final by having validators, people who've locked up money (their *stake*), vote on checkpoints: blocks picked out at regular numbered rounds called *epochs*. A vote isn't for one checkpoint. It's a *link* from a checkpoint the validator already trusts, the source, to a later one, the target, and every chain starts from one first block, its *genesis*, which counts as trusted from the start. In the design Vitalik Buterin and Virgil Griffith published in 2017, called Casper, a checkpoint is *justified* when validators holding two thirds of the stake vote a link to it from a justified source, and *finalized* when a justified checkpoint's direct child is also justified. The record is "this checkpoint is final."

What's priced? Two rules: don't vote for two different targets in the same epoch, and don't cast a vote whose span, source to target, surrounds the span of another of your votes. Break either and your deposit can be destroyed, which is called *slashing*. The theorem they prove, called accountable safety, says that if two *conflicting* checkpoints are ever both finalized, at least a third of the stake broke a rule and can be slashed. So the price is on *reversal*, on making the record contradict itself. The write itself, finalizing, is free: honest validators finalize all day and nobody loses anything.

Do it with three validators. Let A and B vote one branch, B and C vote a competing branch. Each pair is two of three. Both branches finalize. Now read the rule: B cast two different votes with the same target epoch. B is the third that gets slashed; A and C are clean. That's what a fork costs and who pays, and the checked work has exactly this instance (`casper_fork_exists`, `casper_fork_slashable`), so the theorem's premise isn't vacuous: forks of that kind can be built, and the theorem fires on them.

The fragment: a transcription of a published Coq proof of Casper's accountable safety (the casper-proofs development from Runtime Verification and the University of Texas at Austin, itself based on Yoichi Hirai's proof in Isabelle, another proof checker; proofs redone on the standard library), with validator sets as predicates and quorum intersection as a premise. A *quorum* is a set of validators big enough to act, here two thirds of the stake, and *quorum intersection* is the arithmetic of the three-validator example: two thirds plus two thirds is four thirds, so any two quorums share at least a third. The fragment doesn't do that arithmetic; it takes the overlap as given. No deposits, no rewards, no fork choice (the rule for picking a branch to build on), no connection to anything else in this book. `accountable_safety`: a finalization fork implies a slashed quorum. And `finalization_without_slashing`: a one-validator chain finalizes and nobody's slashed, which is the "write is free" sentence in a degenerate setting; I'll say so rather than let it sound like a result about every setting.

Here's the fragment, small enough to read in one sitting. Every name below is from the transcription.

Definition 12.1 (Casper setting and votes). A *setting* has a type of validators, a type of block hashes with a parent relation and a genesis hash, and two classes of validator sets, the “two-thirds” sets Q_1 and the “one-third” sets Q_2 . It carries two premises: *quorum intersection*, for any two sets in Q_1 some set in Q_2 lies inside both; and *at most one parent* per hash. A *state* says, for each validator, target hash h , target epoch t and source epoch u , whether that validator voted the link (h, t, u) . A link from p at epoch u to h at epoch t is *justified by q* when $q \in Q_1$, every member of q voted (h, t, u) , h is the $(t - u)$ -th descendant of p , and $t > u$. Genesis is justified at epoch 0, and a block reached from a justified block by a justified link is justified. Block h is *finalized* at epoch v with child c when c is a child of h , h is justified at v , and the link from h at v to c at $v + 1$ is justified. A *finalization fork* is two finalized blocks $h_1 \neq h_2$, neither an ancestor of the other. A validator is *slashed* when it voted two different targets with the same target epoch (a double vote), or voted links (h_1, t_1, u_1) and (h_2, t_2, u_2) with $t_1 > t_2$ and $u_2 > u_1$ (a surround vote). The state is *quorum-slashed* when some set in Q_2 consists of slashed validators.

Theorem 12.1 (Accountable safety). *In every setting (so under quorum intersection and at most one parent), every state with a finalization fork is quorum-slashed.*

Proof. Let the fork be h_1 finalized at v_1 and h_2 at v_2 . If $v_1 = v_2$, the two links into epoch $v_1 + 1$ are each voted by a set in Q_1 ; quorum intersection gives a set in Q_2 that voted both; their targets differ, since equal children would give h_1 and h_2 the same child and so, by at most one parent, $h_1 = h_2$. That set double-voted. If $v_1 \neq v_2$, say $v_1 < v_2$, walk down the chain of justified links that justifies h_2 , by strong induction on the epoch gap. Every block on that chain is an ancestor of h_2 , so none descends from h_1 . Some link in the chain starts at or below v_1 and lands above v_1 . If it lands at $v_1 + 1$, it and the finalization link of h_1 have distinct targets at that epoch, and the intersection set double-voted. If it starts at v_1 , its source is a justified block at v_1 other than h_1 , and the links that justified those two blocks give a double vote the same way. Otherwise it starts below v_1 and lands above $v_1 + 1$, so it surrounds the finalization link of h_1 , and the intersection set made a surround vote. $\square$

Checked as `accountable_safety` (CasperFFG.v), a transcription of the casper-proofs Core/AccountableSafety file at a pinned commit with proofs redone on the standard library; validator sets are predicates, and both premises are fields of the setting record, so they are hypotheses of every theorem. The record reading is `conflicting_records_are_priced` (CasperRecordReading.v).

Theorem 12.2 (The premise can hold, and the write is free). (1) *In the three-validator setting where each validator has stake 1, Q_1 is the sets of stake at least two thirds and Q_2 the sets of at least one third, the state where A and B vote one branch and B and C the other has a finalization fork; B is slashed, A and C are not, and $\{B\}$ is in Q_2 .* (2) *In the one-validator*

chain setting, one vote finalizes genesis and no validator is slashed.

Proof. Both are computations on fixed finite data. In (1) B voted two different targets at epoch 1, which is the double vote; A's and C's votes are one link at epoch 1 from source 0 and one at epoch 2 from source 1, which neither double-vote nor surround. $\square$

Checked as `casper_fork_exists` and `casper_fork_slashable` (CasperFork-Witness.v), and `finalization_without_slashing` (CasperRecordReading.v).

Wrong framing to take off the table: "a blockchain is immutable truth." Finality is revocable at a price, the price is named, and a final block can hold a false statement.

And the obvious next question: is the 2017 design what Ethereum actually runs? Partly, and the paper's own title is the clue. It's called *Casper the Friendly Finality Gadget*, and its first sentence says it's a finality system that *overlays* a chain still choosing its blocks by proof of work. A gadget, bolted on top. In 2020 a larger group, Buterin among them, wrote up *Gasper*, which joins Casper's finality to a fork-choice rule called LMD GHOST, and on 15 September 2022, in the switch people called the Merge, Ethereum stopped choosing blocks by proof of work at all and went over to proof of stake. So the checkpoints and the two slashing rules above are one part of a system that holds real money today, and the transcription is of that part's safety proof as the 2017 design states it. Everything between the two, the fork choice and the rest of the running protocol, is the gap between my photocopy and the kitchen that I walk through near the end of this chapter. Does any of that move the price? No. Finalizing is still free for an honest validator; what costs is contradicting yourself.

12.2 Gas: the one that prices the write

Every operation on Ethereum costs *gas*, paid in ether by whoever sent the transaction. The yellow paper, Ethereum's formal specification, gives the reason in one line: every computation is fee-bearing so the network can't be abused and halting questions are sidestepped. The record I care about is a nonzero slot in the shared storage, which every full node in the world (every computer keeping its own complete copy of the chain) has to keep forever.

And here it is: writing a fresh storage slot from zero to nonzero costs twenty thousand gas, by the table in the paper's appendix, plus twenty-one hundred for touching a *cold* slot, one the transaction hasn't touched yet (after the first touch it's *warm*). The stated reason is to discourage storage growth, since storage grows every node's database. That's a price on the write, at the moment of the write, and it's the only one in the five.

Where does a number like twenty-one hundred come from? Somebody picked it, so the obvious next question is who, and why, and the answer has a date on it. In September 2016, while Ethereum's developers were meeting at their second conference, in Shanghai, attackers started filling blocks with an operation that was cheap in gas and expensive in fact: it sent every node off to read data from its disk, and nodes

fell behind the chain. The first repair, a fork that October called Tangerine Whistle, raised the prices of the operations the attack had leaned on. The cold-slot charge came later, in a proposal by Vitalik Buterin and Martin Swende called EIP-2929, switched on in the Berlin upgrade of April 2021, and the motivation it gives is still those 2016 attacks. So twenty-one hundred isn't a law of nature or a measurement of anything deep. It's a scar: a price set after somebody found the place where the old price was wrong. That doesn't change the argument; the write is still priced at the moment of the write. It does change how I read the rest of that table: some of its numbers, at least, are somebody's answer to "what did this cost last time," which is a record of a different kind.

Now the leak, and it's Section 6's leak. Set a slot and then clear it *in the same transaction* and you get a refund. Count it through: first write twenty-two thousand one hundred; clearing it costs a hundred and credits a refund of nineteen thousand nine hundred; net, twenty-three hundred. That's `revoked_write_nearly_free`, and `persistent_write_priced` is the other half: leave it set and you pay at least twenty thousand. The write is priced; the *revoked* write nearly isn't. But I have to add what the fragment leaves out, because twenty-three hundred on its own misleads: the realized refund is capped at one fifth of the gas the whole transaction used (that's the 2021 change from Section 6, the one that ended the refund-hoarding token's advantage). So a real set-then-clear pays a lot more than twenty-three hundred. How much more? The cap is a fifth of everything the transaction used, and that includes the twenty-one thousand every transaction pays just for existing. Take a transaction that does nothing but set and clear the slot: it uses 43,200 gas, so the cap is 8,640, so it pays 34,560, which is 13,560 on top of the base. (Look at the slot alone and forget the base, which the rule doesn't do, and you'd get 17,760.) Either way it's far above twenty-three hundred. The counter-level net is twenty-three hundred. The bill is not.

Here's the arithmetic the way the specification has it, so you can redo it with a pencil.

Definition 12.2 (One slot in one transaction). A slot run records the slot's value at the start of the transaction (original), its current value, whether it has been touched (warm), the gas charged so far, and the refund counter. A store of a new value charges 2100 if the slot is cold, plus 20000 if the slot still holds its original value, the new value differs, and the original is 0 (2900 if the original is nonzero), and otherwise 100. It adjusts the refund counter by the specification's rules; in particular, when the new value equals the original value 0, the refund grows by $20000 - 100 = 19900$. The run starts fresh: original and current 0, cold, gas and refund 0. The *net* charge is gas minus refund.

Theorem 12.3 (The persistent write is priced; the revoked one nearly is not). (1) For every list of stores from a fresh slot, if the slot ends nonzero then the net charge is at least 20000. (2) Storing 1 then 0 leaves the slot at 0 with net charge 2300; storing 1 alone has net

charge 22100.

Proof. (1) By induction on the list, with the invariant: while the original value is 0, the net charge is at least 0 when the current value is 0 and at least 20000 when it is not. Each store keeps it, which is a finite case split on whether the slot is warm and which of the original, current and new values are equal. A fresh slot satisfies it. (2) Direct computation: $2100 + 20000 = 22100$ for the first store, 100 for the second, refund 19900, and $22200 - 19900 = 2300$. $\square$

Checked as `persistent_write_priced` and `revoked_write_nearly_free` (EVMStorageGas.v), transcribed from the execution specification's storage instruction and gas table at a pinned commit. The end-of-transaction refund cap of one fifth is not in the fragment.

The fragment: one slot, one transaction, the gas and refund arithmetic transcribed from the Ethereum execution specification at a named commit. No stack, no call context, no other slots, and *no refund cap*; the file says so.

Then the wrapper, which is a different kind of object, and the walk near the end is about exactly that difference. It's a *gas schedule* in the abstract: any states, any steps, a cost per step, a yes/no commitment flag, and a charge predicate saying which steps get billed. It formalises no EVM (the Ethereum virtual machine) at all. It is Section 7's result with the whole-run ceiling (exact about the amount, not just about which steps pay) renamed, and the file says that too. Its two halves are worth reading separately. Leave one committing step unbilled, and a one-step run commits for nothing (`unde`
`rcharged_opcode_admits_free_commitment`), so the floor fails outright; one missed flip is enough. Bill one step that doesn't commit, and the schedule charges more than the number of commits on that one-step run (`overcharge_breaks_ex`
`actness`), so the ceiling fails. And the class isn't empty: in any system, the bill that charges one exactly on a step that raises the flag and nothing on any other meets both halves (Section 7's step price).

And the wrapper itself, which is the cleanest thing in this section and has the least Ethereum in it.

Definition 12.3 (A gas schedule). A *gas schedule* has states, instructions, a step, a cost $c(s, i) \in \mathbb{N}$ that may depend on the state, a Boolean commitment flag on states, and a Boolean charge predicate $\chi(s, i)$, with two premises: a charged step costs at least 1, and an uncharged step costs 0. A step *commits* when it takes the flag from false to true. For a run t from s , let $c(t)$ be its total cost and $n(t)$ its number of committing steps. The schedule has the *floor* when $c(t) \geq n(t)$ for every run, and *no overcharge* when $c(t) \leq n(t)$ for every run. It has *unit pricing* when $c(s, i) = 1$ if $\chi(s, i)$ and 0 otherwise.

Theorem 12.4 (Gas schedule exactness). *For every gas schedule: it has the floor and no*

overcharge if and only if $\chi(s, i)$ holds exactly when the step from s by i commits, and it has unit pricing. Moreover, (1) if some committing step from an uncommitted state is uncharged, that one-step run costs 0 and the schedule has no certification floor; (2) if some charged step does not commit, the schedule overcharges.

Proof. The equivalence is Section 7's characterization with the whole-run ceiling, under another name. From right to left, each step's cost is 1 on a committing step and 0 otherwise, so a run's cost counts its commitments. From left to right, apply the floor and no overcharge to one-step runs: a committing step must cost at least 1, so it is charged; a charged step costs at least 1 and may not exceed its commitment count, so it commits and costs exactly 1. Parts (1) and (2) are those one-step runs read the other way. $\square$

Checked as `gas_schedule_exactness`, which proves both directions, `undercharged_opcode_admits_free_commitment`, `undercharged_opcode_breaks_certification_floor` and `overcharge_breaks_exactness` (`GasMetering.v`); `gas_schedule_exactness` is `exact_commitment_pricing_characterization` (`CommitmentPredicateAdequacy.v`) carried over unchanged.

Wrong framing: "gas is a fee for truth." Gas prices storage growth and computation. It says nothing about whether what's stored is true.

12.3 Attestation: hash chains, and a quote about the past

A trusted platform module, the small chip on most laptop motherboards, keeps a handful of registers that can only be *extended*: the new value is a hash of the old value together with a digest (a short hash) of whatever software just loaded. You can't set them; you can only fold more in. A *quote* is the chip signing those register values together with a number the asker chose, a *nonce*, so the asker knows the quote is fresh. Intel's SGX does the equivalent for an *enclave*, a walled-off region a program runs in. The record is the extended register.

What's priced? Nothing in the specification. Reversal is *infeasible*, not expensive: you'd have to invert a hash or steal a key. That's the difference between a price and a wall, and it's easy to blur. A price can sit at three levels. It can be economic, a fee or a deposit the protocol takes (Casper and gas are here). It can be cryptographic, a reversal made infeasible by a hardness assumption, which is a wall and not an exact price (attestation and transparency are here). Or it can be physical, energy spent, which none of these five specifications touches. And the record's permanence is only as long as the register lives: a platform's registers are cleared when it resets, so the permanence is per boot.

Why the registers, then, and not something that lasts? Because the quote is what a remote asker reads, and the registers are what it signs. The same chip has longer-lived records, and they're closer to the cup than anything else in this chapter. Version 2.0 of its specification has storage slots of a kind it calls a bit field, changed only by a

command that ORs new bits into the old value: a bit can be set and never cleared, which is $r \vee h$ from Section 10, on a laptop motherboard. It has counters that can only be incremented, and a counter's first increment starts it at the largest value any counter on that chip has ever held, so deleting one and making a new one can't wind it back. And those slots are rationed. Bryan Parno and his coauthors, building a system called Memoir in 2011, noted that the specification of the day only required a counter to be able to tick once every five seconds, and that the chip's storage was expected to survive about a hundred thousand writes in its whole life, so writing once a second would wear it out in under twenty-eight hours. Memoir writes to it twice per boot. Permanent, finite and rationed: the honest record has a budget, the way Theorem 10.10's chain of k -bit words can only climb k times. Phones and small controllers go one floor further down, to fuses like the PROMs in Section 10: a version number kept as a row of fuses, one more blown at each security update, and boot code that refuses anything older. An industry guide on it tells designers to make sure there are enough fuse bits to last the number of updates expected over the device's lifetime, which is part (5) of that same theorem, written as a design rule.

What all of this is for is *rollback*. Memoir's example is a password checker restored to an old saved state to undo a user's password change, and in 2017 a system called ROTE did the same job for Intel's enclaves by having several machines vouch for each other's counters, with the stated result that the only way left to break it is to reset every one of them to its start. In 2016 Raoul Strackx and Frank Piessens's Ariadne got the cost of that job down to flipping a single bit of non-volatile memory per state update, which they call the theoretical lower limit. Notice who the attacker is in every one of these: not a bad program, but whoever gets to put an old state back. A record kept by the step can't see that attacker, which is why every guarantee in Part IV is stated from a clean start, and a restored start is outside all of them.

Replicated services put the same kind of record somewhere else: inside each replica, where a replica that's been taken over still can't make it say two different things. Byung-Gon Chun and coauthors' attested append-only memory (A2M, 2007) is a log a replica can only add to; Dave Levin and coauthors' TrInc (2009) shrinks that to a trusted counter; MinBFT (Giuliana Veronese and coauthors, 2013) uses a trusted counter to reach agreement with $2f + 1$ replicas instead of the usual $3f + 1$ when f of them can lie; and Damysus (J  r  mie Decouchant, David Kozhaya, Vincent Rahli and Jiangshan Yu, 2022) has a small trusted service, the Checker, that keeps a monotonic counter of where the protocol is. Allen Clement, Flavio Junqueira, Aniket Kate and Rodrigo Rodrigues (2012) proved what such a record buys, and the answer is a good warning for this book. On its own, stopping a replica from saying two different things doesn't lower the $3f + 1$ bound in an asynchronous network at all. Add signatures that a third party can check, and it turns any protocol that survives crashes into one that survives liars, with no extra replicas. The way I read it, a record helps the others only as far as they can check what it says, which is Section 7's point from the other

side.

There's a fourth level under those three, and it's the one that always pays. At the logical level, Section 4's room of chairs applies to any of these records: on a finite state space, a step that switches a record on somewhere, and never switches it off anywhere, forgets something. That needs no premise about money, keys or heat. The economic, cryptographic and physical payments each need a premise of their own about their level (a fee rule, a hardness assumption, Landauer's principle), and they hold exactly where that premise holds.

Theorem 12.5 (The logical level always pays). *For any states S , instructions I , step and Boolean reading: if a duplicate-free list names every state, the instruction i never turns the reading from on to off, and i turns it on at some state s , then the step by i is not injective.*

Checked as `permanent_write_has_logical_payment` (PricingPhysicsAudit.v), from `permanent_at_flip_is_not_injective` (PermanentRecordPricing.v); the proof is Theorem 4.1 with permanence required only of the one instruction i .

What the quote can't do, which is the thing people get wrong: it says what was *measured*. It can't say the measured code is correct, and it can't say anything about runtime state that was never folded in. A correctly measured buggy program attests beautifully. The fragment proves that exact limit: one register, a symbolic hash, a quote as a pair, and no function of the quote decides a runtime flag the quote ignores (`quote_cannot_attest_unmeasured_state`), which is Section 7's cube again. And a companion shows that a sign-and-verify interface *alone* gives no authenticity (a scheme where verify always says yes passes the interface), so a cryptographic premise has to be added; the fragment doesn't model signatures, formats, or a freshness check (the nonce is just a field the verifier fixes).

The same limit, turned outward, is the honest edge of the opening chapter's four conditions. Every one of them is about a thing the machine holds: same property, same thing, same edition, honest check. For a claim about the world (the oven is hot, the parcel arrived) there's no edition the machine controls. Whatever reads the world is a window onto it, and "nothing changed between the check and the commit" stops being something a step can enforce and becomes a premise about the sensor. So a record can establish claims about the machine's own state, and claims about the world only through named premises about whatever measured them. A correctly recorded sensor reading certifies the sensor.

Definition 12.4 (Platform and quote). Fix any functions $H : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ (the extend hash) and $\delta : \mathbb{N} \rightarrow \mathbb{N}$ (the digest); nothing is assumed about either. A *platform* is a measurement log (a list of numbers) and a Boolean runtime flag. Its register is the log folded from 0 by H . The *quote* with nonce n is the pair $(\delta(\text{register}), n)$.

Theorem 12.6 (The quote cannot attest unmeasured state). *For every nonce n there is no function D from quotes to Booleans with $D(\text{quote}_n(P)) = \text{flag}(P)$ for every platform P .*

Proof. Take two platforms with the empty log, one flagged true and one false. Their quotes are equal, since the quote never reads the flag, so D would have to return both values on one input. $\square$

Checked as `quote_cannot_attest_unmeasured_state` (TPMQuoteGap.v). In the same file, `quote_decides_measured_claims` says any Boolean claim about the retained digest can be read off the quote. The companion is `tpm_interface_authenticity_refuted` (TPMQuoteAuthenticity.v): a scheme whose verify always says yes satisfies the sign-and-verify interface while accepting a signature no key produced.

12.4 Certificate transparency: visible, not priced

The certificates that make your browser show a padlock are issued by hundreds of authorities, and a bad or compromised one can issue a certificate for a domain it shouldn't. Certificate Transparency makes authorities submit every certificate to public append-only logs built as Merkle trees, so that anyone can check that a certificate was logged and that the log never rewrote its past. It was specified in RFC 6962 and revised as RFC 9162 in 2021; the logs people run still mostly follow 6962, and the fragment I transcribe is 9162's. A Merkle tree is Section 3's hash chain folded into a tree: hash each certificate (those are the *leaves*), hash the leaf hashes in pairs, hash those in pairs, and so on up to one *root* hash. To prove one leaf is in, the log hands you only the few hashes sitting beside the climb from that leaf to the root, the *path*, and you redo the climb yourself. A second check, *consistency*, shows that a bigger tree is a smaller one with leaves added on the end and nothing rewritten. The record is a leaf in the tree.

Priced? Nothing. The RFC says it in its first section: logs don't *prevent* misissuance; they let interested parties *detect* it afterward. And a log that shows different views to different people (a *split view*) can defeat the auditing, so logs are still trusted parties. Visibility, not price.

Which raises the obvious question: if nothing's priced, why does any authority bother logging? Nothing in the RFC makes it. The answer is a browser. Chrome's team set the rule, enforced from Chrome 68 in July 2018, that every certificate issued after 30 April 2018 had to meet its transparency policy, and a publicly trusted certificate that doesn't is simply not trusted by Chrome: the visitor gets a full-page warning instead of the site. So there is a price after all, just not anywhere in the specification. It lives in one vendor's browser, and it's the price of being shown to that browser's users at all. Is that the bottom? Not quite, because it means the strongest lever on this record is a policy one company can change, which is a fact about who holds the stamp and not about the tree. It changes nothing in the table at the end of this chapter, which

asks what each specification prices. But it's worth keeping two sentences apart: "the specification prices nothing" and "nothing is priced."

Do the inclusion proof by hand, because it's the most satisfying ten minutes in this section. Take the RFC's own seven-leaf example tree and verify that leaf `d3` is in it from the path the log hands you: three hashes and a rule about which side to hash on, depending on whether your position is odd. Three hashing steps and you land on the root. The checked work runs the RFC's own examples (`rfc9162_example_inclusion_d3` and its siblings, including the awkward lone-leaf case) through a transcription of the RFC's verifier control flow with a symbolic hash. Verifier logic only: no signatures, no gossip (auditors comparing the views they were shown), no misbehaving log.

Definition 12.5 (Inclusion check, RFC 9162 section 2.1.3.2). Given a leaf index m , a tree size N , a leaf hash, a root hash and a path, the check rejects unless $m < N$. Otherwise it walks the path from the leaf with two counters starting at m and $N - 1$, hashing each path element on the left or the right according to the counters as the RFC specifies and halving the counters, and accepts when the second counter ends at 0 and the result equals the root. A *symbolic hash* makes a leaf hash $\ell(x)$ and a node hash $v(a, b)$ free constructors, so two hashes are equal only when they are built the same way.

Theorem 12.7 (The RFC's own examples pass). *With the symbolic hash, in the RFC's seven-leaf tree with leaves $d_0, \dots, d_6$ (writing d_k also for its leaf hash $\ell(d_k)$), the inclusion check accepts leaf 3 with path $[d_2, v(d_0, d_1), v(v(d_4, d_5), d_6)]$, and likewise leaves 0, 4 and 6 with their paths; the consistency check from size 4 to size 7 accepts. For every input with $m = N$, the inclusion check rejects.*

Proof. The examples are evaluation of the transcribed functions on constructor terms. The boundary case is the first test, $m < N$, failing. $\square$

Checked as `rfc9162_example_inclusion_d3`, its siblings `rfc9162_example_inclusion_d0`, `rfc9162_example_inclusion_d4` and `rfc9162_example_inclusion_d6`, `rfc9162_example_consistency_4_7`, and `rfc9162_inclusion_boundary_safe` (RFC9162Merkle.v). The append-only history is `list append`, and `ct_extension_preserves_entries` says appending keeps every old entry.

12.5 Proof-carrying code: no price, no permanence

Necula's idea from Section 8: ship untrusted code with a proof that it obeys the host's policy; the host computes what would need to be true and checks the proof with a small checker. The record is the checked proof, and it's held only for the moment of loading. Priced? No; checking is cheap by design, that's the whole point, and the producer paid to build the proof off the host. Permanent? No; nothing's written. The fragment is a three-instruction memory-bounds language with the policy "every

address is below the limit”: the checker accepts exactly when the condition holds, and a certificate implies the condition. One example is checked on top of that: a program that reads address two under a limit of two is rejected. Not the paper’s real case study, and no bridge to anything else in this book. And the wrong framing: a program that writes garbage to address zero passes if zero is below the limit. The proof is of the stated policy and nothing else.

And what was the paper’s real case study, the one my fragment isn’t? Network packet filters: the little programs an operating system runs on incoming network traffic to decide which program gets to see which packet. They run inside the kernel, the most trusted part of the system, and they have to be fast, so they’re exactly the code you’d most like to take from a stranger and least like to trust. In the 1996 paper that first set out the idea, Necula and Peter Lee wrote several filters by hand in the assembly language of a DEC Alpha, shipped each with its proof, and reported a one-time cost of 1 to 3 milliseconds to check the proof, after which the filter ran with no checks at all. That’s “checking is cheap by design” with a number on it. It changes nothing about my fragment, which still isn’t that case study, and it doesn’t add a price: the milliseconds are the checker’s work, not a toll on anything being written down.

Definition 12.6 (The bounds fragment). Instructions are `READ  $a$` , `WRITE  $a\ v$` , and `HALT`. For a memory limit L , the policy allows `READ  $a$` and `WRITE  $a\ v$` when $a < L$, and always allows `HALT`. The *verification condition* of a program is that the policy allows every instruction in it. A *certificate* is a list of per-instruction policy facts, one for each instruction in order. The *checker* tests the policy on each instruction and returns the conjunction.

Theorem 12.8 (The checker is the verification condition). *For every limit L and program P : the checker accepts P exactly when the verification condition holds; any certificate for P gives the verification condition; and with $L = 2$ the checker rejects `READ 2` followed by `HALT`.*

Proof. The first two are induction on the program and on the certificate, one instruction at a time. The third is a computation: $2 < 2$ fails. □

Checked as `pcc_checker_accepts_iff_vc`, `pcc_certificate_implies_vc`, and `pcc_unsafe_program_rejected` (NeculaPCC.v, definitions in NeculaPCC-Target.v).

12.6 One example, three levels

Before I lay all five side by side, let me walk one of them up and down the three levels from the start of the chapter, because every claim in here lives at exactly one level and it’s easy to let a claim slide up a floor without noticing. Gas, since it’s the one place all three line up. Think of a restaurant’s real recipe, my photocopy of one page of it, and

a cookbook chapter on what any recipe has to contain. I can check my photocopy line by line. That tells you nothing about what the kitchen cooks tonight.

The real specification. Ethereum’s execution specification says, in a document that runs real money, what a storage write costs, what clearing it refunds, and how the refund is capped at the end of the transaction. None of what I checked is at this level. What the network actually charges is whatever the clients people run actually compute, and the evidence for that is the clients and the chain, not this book.

The transcription. I copied one mechanism out of that document, one slot in one transaction, gas and refund arithmetic only, at a named commit (Definition 12.2, Theorem 12.3). What the result tells you: *if* the storage rule is what I copied, then leaving a fresh slot set costs at least twenty thousand, and set-then-clear nets twenty-three hundred at the counter. What it can’t tell you: anything about the refund cap (not copied, which is why the bill came out so different from the counter earlier), the stack, calls, other slots, or whether any client implements the rule as written. The boundary of the claim is the edge of the copy.

The wrapper. The gas schedule of Definition 12.3 isn’t Ethereum. It’s any system with steps, a cost per step and a yes/no commitment flag, plus one thing *stipulated*: a charge predicate, with the premise that a charged step costs at least one and an uncharged one costs nothing. That premise is the whole of what the wrapper adds, and it’s added by hand, not found in any specification. Given it, the theorem says the bill is exact, never letting a commit go free and never billing a non-commit, exactly when the charge predicate is the commit predicate at unit price. What that tells you is what exact pricing would have to look like, in any system. It doesn’t say Ethereum meets it, and Ethereum doesn’t: real gas bills computation that commits nothing, which the transcription shows in numbers. So the wrapper prices a definition, the transcription prices a copied rule, and neither prices the chain.

The other four sit on the same three levels, and only one more wrapper lives in this chapter. Casper’s transcription is the accountable-safety proof above; its wrapper is a small gadget with the stake at risk handed in. A finalize step that risks nothing turns the record on for free, which breaks Section 3’s toll (the file calls that free forgery), and one that risks at least one unit has a floor (`nothing_at_stake_is_free_for_gery, slashing_finality_floor, PoSFinality.v`). It records exposure to loss, not destroyed collateral, has no Casper checkpoints in it at all, and is priced by stipulation, the same way the gas wrapper is. Attestation, transparency and proof-carrying code have transcriptions that prove what their verifiers do and no price theorem, because their specifications carry no price. Their wrappers are Section 7’s verifier argument told in each system’s vocabulary, with a colliding pair behind each one, and that argument is all they carry here.

How to knock each one down, since the book’s rule is that every claim gets a way to break it. For the gas transcription: a list of stores from a fresh slot that ends nonzero

with net charge under twenty thousand. For the gas wrapper: a schedule that never lets a commit go free and never bills a non-commit, whose charge predicate still differs from the commit predicate at some step. For the finality gadget: a finalizing trace from unfinalized to finalized whose charged stake is zero while the gadget claims a floor. For the attestation fragment: a verifier reading only the quote’s fields that decides a runtime flag the quote ignores.

12.7 The same table, five ways

Lay them out. Casper prices *reversal* (slashing). Gas prices *the write* (and refunds the revocation, capped). Attestation and transparency price *nothing*, and make reversal infeasible or visible instead. Proof-carrying code prices nothing and keeps nothing.

System	What its specification prices	What the transcription checks
Casper finality	reversal, by slashing	a finalization fork means a slashable third
Gas	the write, refunded on revocation	a persistent write nets at least 20000
Attestation	nothing; reversal is infeasible	the quote can’t decide what it never measured
Transparency	nothing; misissuance is visible	the RFC’s own inclusion examples pass
Proof-carrying code	nothing, and nothing is kept	the checker is the verification condition

Each row proves what its own transcription says. Reading a row as a claim about a deployed system takes evidence from that system, and no row shows that certification is the one event every model has to price.

So the thing my toy charges for, the moment the record is written, is charged for in one of five real systems, and in that one the designers had to patch the place where the charge could be clawed back. I find that neither a vindication nor a refutation. It’s a fact about what people built, and what they built was mostly walls (hash chains, signatures) and penalties for contradiction, not tolls on commitment. Whether a toll on commitment is a *better* design than a wall is not something I can prove, and I won’t pretend the five say so. What I read in them, and it’s a reading of five fragments I chose, is that the record axis is real in each of them (each has a bit that goes one way) and that the price axis is mostly empty.

That leaves the question Section 10 raised and Section 13 has to face. Five systems, five different events chosen as “the record”: finalized, committed, attested, logged, checked. Is there anything that picks out one event as *the* one worth recording, or is it a free choice every time? I have a guess. It’s a conjecture, and I’ll show you how to kill it.

13 The pointer conjecture

Here's where I am. A record over any base is a latch on an event (Section 10), and the event is a free choice (Section 10 again, the two cups on one die). Five real systems each picked a different event (Section 12). The toll in this book sits on certification. Why that one? I don't have a proof. I have a conjecture, and the honest shape of a conjecture is: here's the claim, here's where I got the idea, here's what's checked, here's what isn't, and here's how to kill it. In that order.

13.1 The claim

Of all the events a machine could put on its meter, the ones its price list charges for and its counter adds up, certification is the one whose records get *copied*: checked, stored, and relied on across independent places that share no other state. Rival events leave no such trail. If that's true, certification isn't a free choice; it's the one event the world around the machine keeps redundant copies of, and that's a reason to price it and not the others.

There's a second handle on the same question, and it comes from Section 4. The permanent-certificate theorem never uses the word certification for anything special: given Section 4's rule that forgetting is charged, it prices any yes-or-no reading that, once on, stays on, on a machine with finite memory. So it points at a narrower candidate: the events to price are the ones that write a record no later step can take back. Certification is one of them. As an exact answer that candidate is false, because a merging step can also revoke (Section 4's three cards). Whether the records that proliferate are the permanent ones, and whether proliferation picks out anything the merge criterion doesn't, is what the two handles have to settle between them.

Two holes in that, which I can see from here, so you don't have to find them for me. The first is a missing premise. Even if the world copies certification records more than any other kind, "copied" doesn't give you "priced" on its own. Something has to connect them, and the best candidate I have is a premise about incentives: a record that many independent parties copy and lean on is exactly the record worth forging, so it's the one where a price buys the most protection. That's a reasonable premise. It's also a premise, and as I've stated the conjecture, it's missing. The second hole is a circle. If "certification" just means "the record other people rely on," then of course it's the one that gets relied on and copied, and the claim says nothing. So the event has to be picked out without mentioning reliance: here it's a reading going up for good right after a check of the same claim passed, which you can find in a run without asking who reads it. Whether the world copies *that* event more than its rivals is then a real question, and it could come out no.

That's the whole conjecture. Now the rest of what a conjecture owes you.

13.2 Where the shape comes from, and how much of it survived

The shape is from physics, and I want to tell you where exactly, because I borrowed a word and I didn't get to keep the whole thing.

Quantum mechanics has a famous problem: a system can be in a blend of states, and when you measure it you find one definite state, and nobody agrees on what "measuring" did. In 1981 Wojciech Zurek asked a sharper question: of all the ways a measuring device *could* end up, why does it end up recording one particular kind of quantity (the position of a pointer on a dial, say) and not some blend? His answer, developed over twenty years and summarised in a 2009 paper called "Quantum Darwinism," is that the *environment* decides. The air, the light, the table, everything the device touches, is constantly interacting with it, and only certain states of the device survive that interaction; those are the *pointer states*. And more: the environment doesn't just select them, it *copies* them. The light bouncing off a dial carries the dial's reading to every corner of the room, redundantly, so that a thousand observers can each read a different scrap of the environment and all agree on where the pointer is, without touching the dial. Objectivity, on this view, *is* redundancy: a property is objective when many independent fragments of the world each carry it.

And it isn't only Zurek's say-so. Fernando Brandão, Marco Piani and Paweł Horodecki (2015) proved that something like it is generic: under any quantum dynamics at all, observers who learn about a system only through its environment get at most classical information, about one and the same measurement of the system. That's the theorem behind the word "objective," and it's a statement about separate fragments, which is exactly the part I didn't keep. Xiao-Liang Qi and Daniel Ranard (2021) sharpened it into a statement about every possible evolution of system and environment: everywhere in the environment outside a region of fixed size, whatever an observer can learn about the system is classical, read off one fixed measurement. A statement about every evolution and every input is the quantum cousin of what the definitions below ask of an observer, that it record the event in every state, not only the ones that happen to turn up.

That's the shape I borrowed. Certification is to a machine's ecosystem (the machine plus everything around it that reads, stores and relies on its records) what the pointer reading is to the room: the thing that gets copied everywhere. It's a good shape and I think it's the right intuition. Here's what survived into the checked work, and I'm going to be exact because the word "redundant" is doing less than it does for Zurek.

Zurek's redundancy is *graded* (each fragment carries almost all the information, with some small loss) and it requires the fragments to be *disjoint*: different observers reading different parts of the environment. The checked definition is the zero-loss case with *no disjointness at all*. An ecosystem is a state type, some number of observers, and one bit per observer; an observer *records* an event when its bit is true exactly when the event holds; an event *proliferates* when every observer records it. Nothing in that says two

observers draw on separate parts of the state. They could all be reading the same bit. The file says so itself. So I kept the word and not the independence that makes Zurek's version mean "objective." That's a gap between the conjecture's prose and its formal object, and the prose is where the conjecture lives.

The independence I dropped is exactly what the conjecture's closest published neighbour keeps. Jess Riedel, Zurek and Michael Zwolak (2016) took histories of a closed quantum system, the story a whole sequence of events tells, and asked which of them the world keeps records of in many separate pieces of the environment. Redundant records in separate pieces make a history consistent, so ordinary yes-or-no reasoning about it works, and they pick out a small set of histories from the huge number that are merely consistent: in their words, an objective past. That's "what proliferates is what counts" in quantum form, with the disjointness in it, and it came before this book. Halliwell (1999) had already carried the link between records and consistent histories from pure starting states to mixed ones, where the records still exist but are necessarily imperfect.

There's a deeper mismatch in the borrowing, and it runs against the book, so it goes here. In Zurek's picture the whole thing, system and device and environment together, evolves by quantum mechanics' rule for closed systems, and that rule never sends two states to one; every step can be run backwards. So, seen from the whole, the environment's copying is just that, copying: the dial's reading gets written onto fresh light, nothing anywhere is forgotten, and it's Door one of Section 4 with the rest of the universe as the endless strip. That's the operation this book says is free, Bennett writing on a blank page. The forgetting shows up only when you look at the dial alone and ignore what the light carried off. Then a blend of pointer readings decays into a plain either-or, and two different blends can decay into the same either-or: two states, one result, a merge in the coarse picture. So the opening chapter's "every settling forgets" fits the dial's own view and not the view from the whole, and the thing the conjecture leans on, proliferation, is the free half of the story. What that does to the conjecture: if there's a price, it isn't on the copying. It's on whatever makes the copy final from the dial's side, and the claim above, as written, points at the copy.

A published no-go cuts closer still. Philippe Allard Guérin, Veronika Baumann, Flavio Del Santo and Časlav Brukner (2021) showed that, under assumptions that look natural in quantum mechanics, an observer's own perceptions of her outcome at two different times can't share one reality: no joint chance for the two, depending linearly on the starting state, reproduces what unitary quantum theory predicts for each. A record that persists, for the one holding it, is exactly what gets into trouble once the whole is treated as one-to-one. That's not a theorem about my machine, which isn't quantum, but it's the strongest published reason I know to hold the word "permanent" loosely when a physicist is in the room.

Which leaves the question to ask of any borrowed shape: has anybody checked the

original? For a long time quantum Darwinism was a theory without a lab. Then, in 2018 and 2019, three groups working separately, in Italy, China and Germany, went looking for its fingerprint. The German one, at Ulm, with Zurek among the authors, watched the spin of a single defect inside a diamond and the carbon atoms around it, and saw the redundancy the theory predicts: the defect's state turned up copied in its surroundings, and it didn't take much of the surroundings before more of them told you nothing new. The other two did it with photons. Is that the bottom? Not by a long way, and the people involved said so. Zurek himself pointed out that the photon experiments make their copies artificially and that the diamond's environment is small, and others noted that all three are schematic versions of a real environment and don't rule out rival accounts. So the shape I borrowed has passed some first tests in a lab, at the scale of a handful of atoms or photons, with the caveats written on it. None of that touches the conjecture here, which is about machines and never rested on quantum physics. It changes the picture: what I borrowed is a lead somebody is checking, not a settled fact I get to lean on.

13.3 What's checked

Definitions, a toy, a survey, and two refutations of things that would have been stronger.

The definitions above, plus “unique pointer among rivals”: the event proliferates and none of a listed set of rival events does. Uniqueness is *relative to the list*, necessarily, since any event that holds on exactly the same states as a proliferating event proliferates whenever it does: the same fact under another name, so someone has to say which names count as rivals. And the conjecture itself, as a schema: for every ecosystem in some class, its metered event is the unique pointer among its rivals. The class and the rivals are inputs. Nothing about them is proved.

Here's the formal object, small enough to hold in one hand.

Definition 13.1 (Ecosystem, records, proliferates, pointer). An *ecosystem* is a type of states, a number N of observers, and for each observer i a Boolean fragment F_i on states. Observer i *records* an event E (a property of states) when, for every state s , $E(s)$ holds exactly when $F_i(s)$ is true. E *proliferates* when every observer $i < N$ records it. E is the *unique pointer among* a list of rival events when E proliferates and no rival in the list does. Given a class C of ecosystems, and for each member a designated metered event and a list of rivals, the *pointer criterion holds* for C when every member's metered event is the unique pointer among its rivals.

Nothing in the definition makes two fragments read different parts of the state, and “metered” is only a label: the definition has no metering in it.

Conjecture 13.1 (The pointer conjecture). *Let C be the class of ecosystems that faithfully*

model deployed record-keeping disciplines, observed through independent carriers (separate parts of the world, each holding its own copy), with each discipline's certification event as its metered event and the other events a step rule could meter as rivals. Then the pointer criterion holds for C.

The conjecture leans on four notions left without precise mathematical definitions: what it is to “faithfully model” a system, which systems count as “deployed,” what makes carriers “independent,” and which events “a step rule could meter.” That’s why it’s a conjecture and not a theorem with a missing proof.

A toy: three observers on a replicated ledger, where certification proliferates and “work done” doesn’t. The file labels it a sanity check, and it is one, not evidence.

Theorem 13.2 (The toy). *Let the toy have states (a certified bit, a work count), three observers, and every fragment equal to the certified bit. Then “certified” is the unique pointer among the one rival “work at least 1”.*

Proof. Every fragment is the certified bit, so every observer records “certified” by definition. For the rival, take the state with the bit false and work 1: the event holds and observer 0’s fragment is false, so observer 0 does not record it. $\square$

Checked as `toy_cert_unique_pointer` and `toy_work_not_proliferating` (`PointerObservable.v`).

A survey of twelve events, one checked theorem: for each of the five systems in Section 12, the designated event (finalized, committed, attested, logged, checked) proliferates in the observer map I wrote for that system, and a rival (proposer work, a scratch counter, measurement noise, submission latency, prover steps) doesn’t. The five together are one statement (`five_labeled_models_have_selected_pointer`), an instance of the schema above. Plus a symmetric message-authentication code, which does *not* proliferate, and a digital signature, which does. Twelve rows. And a thirteenth that I think is the most important one: swap which bit every observer copies, and the criterion picks the *other* event. The criterion selects whatever the observer map copies. The map is where the argument lives, and I wrote the maps. So the five “confirmations” are five instances of one construction with the fields renamed, and I’d rather say that twice than let it pass once.

Theorem 13.3 (The survey, and the swap). *(1) In each of the five observer models (proof-of-stake, gas, attestation, transparency, proof-carrying), the designated event proliferates and the designated rival does not; in the symmetric-MAC model the authentic-tag event does not proliferate; in the signature model the authentic-signature event does. (2) Let the states be pairs of bits (first, second), with three observers whose fragments all equal the second bit. Then “second is true” is the unique pointer among the one rival “first is true”.*

Proof. Each confirmation holds by definition: each observer's fragment is true exactly when the event holds. Each refutation exhibits one state where the event holds and some observer's bit is down. In (2) that state is (true, false). $\square$

Checked as `twelve_candidate_measurements_checked` and `swapped_event_is_pointer_checked` (`RecordProliferationSurvey.v`, statements in `RecordProliferationSurveyTarget.v`); the five designated models together are `five_labeled_models_have_selected_pointer` (`PointerObservableReductions.v`), an instance of the indexed form of the criterion, `pointer_criterion_holds_indexed`.

Two stronger claims, both refuted, and I'd ask you not to merge them because they fail for different reasons.

The first stronger claim: *every forgery-resistant commitment proliferates*. If that were true, certification would be the pointer because it's the commitment. It's refuted in three observer models (deniable authentication, where only the designated verifier's bit tracks the event; a symmetric MAC, where the key holders share a secret and nobody else can check; an object capability, an unforgeable token that grants access to whoever holds it) and confirmed in two (a public log; a digital signature). Build the first with three observers: give one a copy of the event's bit and the other two a constant false. The one records it; it doesn't proliferate. The checked half is the map computation. Whether my maps are *fair* to the real designs is argued in prose, and you should argue back.

Theorem 13.4 (One blind observer is enough). (1) *In any ecosystem, if some observer $i < N$ has $F_i(s_0)$ false at a state s_0 where E holds, then E does not proliferate.* (2) *Let the deniable model have states (authentic, verifier-accepts), three observers, observer 0's fragment the authentic bit and observers 1 and 2 constantly false. Then observer 0 records "authentic", and "authentic" does not proliferate. The symmetric-MAC and object-capability models also fail to proliferate their events; the public-log and signature models proliferate theirs.*

Proof. (1) Recording would require $E(s_0) \iff F_i(s_0)$, and the left side is true while the right is false. (2) Observer 0's fragment is true exactly when the event holds. Apply (1) with observer 1 and the state (true, true). The other three refutations apply (1) to a blind observer of their own model, and the two confirmations hold by definition. $\square$

Checked as `blind_observer_blocks_proliferation` and `labeled_model_verdicts` (`PointerObservableCounterexamples.v`). The theorem is about the observer maps. That a third party in deniable authentication really cannot be convinced is the modelling claim, and the model has no transcript, key or verifier in it to check that against.

The second stronger claim: *agreement plus accuracy plus locality imply permanence*. That is, if all observers agree, each sees the event exactly, and each updates by the same

local rule with no coordinator, then the recorded event can't be undone. Refuted by the smallest thing imaginable: one bit, two observers who both see it, and the rule "flip." All three conditions hold and true becomes false in one step (`toggle_game_refutes_strong_pointer_necessity`). Do it on paper; it takes a minute and it's humbling. What *does* give permanence is durability, "an observer's true view stays true," added as a premise, and then it follows in a few lines (`durable_consensus_implies_permanence`). So durability is the premise doing the work, and durability is Section 4's permanence wearing a different coat. Agreement doesn't buy it. Do the paper version again with one change: the rule is "stay true once true" instead of "flip," and the bit can't fall, because the rule won't let it, not because the observers agree. (The machine I build later has a reading that's durable in exactly this sense: its flag never comes down, by its step rule (Theorem 14.5).) None of this rules out a narrower theorem with real assumptions about the network, the adversary or durability; it rules out these two strong forms.

Definition 13.2 (Observation game). A *game* has states, a number N of observers, a Boolean event e , a Boolean view V_i for each observer, and a step map. For $i, j < N$ it has *consensus* when $V_i = V_j$ on every state; *authenticity* when $V_i(s) = e(s)$ for every state and every $i < N$; *coordinator-free update* when one function ℓ on Booleans has $V_i(\text{step}(s)) = \ell(V_i(s))$ for every state and every $i < N$; *durable views* when $V_i(s)$ true gives $V_i(\text{step}(s))$ true for every $i < N$; and a *permanent event* when $e(s)$ true gives $e(\text{step}(s))$ true.

Theorem 13.5 (Agreement does not buy permanence; durability does). (1) *It is false that every game with $N > 0$, consensus, authenticity and coordinator-free update has a permanent event.* (2) *Every game with $N > 0$, authenticity and durable views has a permanent event.*

Proof. (1) The toggle game: states are Booleans, two observers, event and both views the identity, step "flip". Consensus and authenticity hold trivially, the local rule is "flip", and the state true steps to false. (2) If $e(s)$ is true, observer 0's view is true by authenticity, stays true by durability, and authenticity turns it back into $e(\text{step}(s))$. $\square$

Checked as `toggle_game_refutes_strong_pointer_necessity` and `durable_consensus_implies_permanence` (`EcosystemGame.v`, definitions in `EcosystemGameTarget.v`).

13.4 What isn't checked, and must stay that way on the page

That certification *in deployed systems* is singled out by redundant proliferation across *independent* carriers. The proposed criterion has no precise definition of "independent" or "deployed," and no cost law connecting proliferation to price. Every correspondence between a Coq observer map and a real system is a sentence I wrote. The conjecture is a conjecture, and anyone who says the formal object proves it is reading more than is there.

13.5 How to kill it

Three ways, in the order I'd try them.

One: find a deployed discipline that prices a *non*-proliferating event. A real system with a toll on something nobody copies. That breaks "the priced event is the copied one."

Two: find a rival event that proliferates *as well as* certification in a real ecosystem. Then certification isn't unique and the conjecture's "the one" is wrong. Or do it in a model: show that something like a multiplication count or a head reversal proliferates by the same standard I used, so the criterion picks out more than one event and decides nothing.

Three: formalise an ecosystem honestly, with independence defined, where certification *fails* the test. The symmetric-MAC row is already most of the way there for a different claim; push it.

The checked work has already refuted two stronger forms of its own claim, which I take as a sign the method works and as a warning about the main claim.

13.6 The end of the logic

That's Part III, and the end of the logic that started in Part II. Here's what it earned. A toll on raising a flag is forced by three premises (Section 4). Learning is free, committing isn't (Section 5). The counter is arithmetic once the schedule is fixed (Section 6). A window with a colliding pair in it can't price the flag exactly (Section 7). Entitlement is an object, a check, and a paid event, and, given lists, a tree and a covering that someone supplies, it costs the index bits it commits to (Section 8). Insight is a committed relation, not a feeling (Section 9). Over any base a record is a latch, with two free choices, the price list and the event (Section 10). No inside procedure whose flip a substrate can represent decides a shortcut property, given a recursion theorem, which the nat family has by construction and L has by proof; a record carries the commitment instead (Section 11). Of five specifications, read through my transcriptions, one prices the write (Section 12). And which event to record is a conjecture with a borrowed shape and a listed set of refutations (Section 13).

The obvious question to ask of all that is whether the toll is the only way. A theorem saying so would be the best thing this book could hold: a short list of requirements that any account of what a step costs has to meet, whatever field it comes from, with the toll and the counter the only accounting left standing. I looked for that theorem, and the answer I can give you is no, and the no is small enough to check on three states. Call them blank, once and twice, with once and twice reading yes. The flip takes blank to once and leaves the other two where they are. The smooth takes both once and twice to twice and leaves blank alone. The smooth forgets something, so the price on merging from Section 4 charges it a mark. It raises nothing, so the toll leaves it free. Now try the requirements that sound neutral: no charge on a move

that changes nothing, none on a move that forgets nothing, the same bill when you rename the states, a total that adds up along a run, a bill that looks only at the step, and one mark on the flip. The toll, read as the bill that charges the flip and nothing else, meets every one of them on this machine. So does the price on merging. They disagree on the smooth, so that list doesn't pick between them. What picks the toll is a requirement that names the record: over any run the total is never more than the number of times the reading went up. With the floor, that leaves exactly one bill, one mark per raise and nothing else (Section 7), and it rules the price on merging out, because the smooth costs that price a mark and raises nothing. So the uniqueness is real and it's small. It holds once you've said which reading you're charging for, and which reading deserves the charge is the pointer conjecture, which is still a conjecture.

The uniqueness theorems that do exist are about pricing information loss, and they're worth a look because they say what kind of bill the price on merging is. John Baez, Tobias Fritz and Tom Leinster (2011) showed that Shannon entropy is the only measure of information loss, for maps that preserve a probability measure, that is functorial, convex-linear and continuous. János Aczél, Bruno Forte and C. T. Ng (1974) showed that the entropies with a short list of natural properties are exactly the linear combinations of Shannon's and Hartley's, where Hartley's is the log of the count, so even for merges the choice between the average and the worst case is a real choice. Both price merges, not raises, so neither picks out the toll, and that's the three-state example in a bigger room. On the other side of the fence, Luciano Pomatto, Philipp Strack and Omer Tamuz (2023) proved a uniqueness result for what it costs to *acquire* information, with constant marginal costs. That's a price on producing a signal, a different bill from the free learning of Section 5.

What none of that is: a machine. Every theorem so far quantifies over "any states, any steps," or is about a small copy of somebody else's system, or points forward at a thing I keep promising. That's the abstract part of the book, and it's done. From the next page on I stop saying "any machine" and build one, mine: the least I can make that still shows the whole model, with every part named and every price written down, so that everything above finally has something to bump into, and then one fixed program on it that runs every machine of the model whose workings a program can compute. If a statement up there doesn't mean what I think it means, that's where it'll show. Part IV is the thing. Let's see if it holds weight.

Part IV

The machine

14 The smallest honest machine

Parts I to III never built anything. They said “any states, any steps,” and whenever a picture was needed, a board or some chairs or a cup stood in. Now I have to build a machine, and the first worry I have about that is a slightly embarrassing one. I built a big one first, the one from the origin story, with dozens of instructions, some picked for reasons and some picked because I had to pick something, and theorems held of it. What does that show? That the model works, or that I’m good at stacking instructions until the proofs go through? A big machine is a wonderful place to hide a mistake. It’s a garage so full of stuff that you could lose a car in it.

So the machine in this book is a small one. The least I can build that still shows the whole model, every piece of it, with nothing waved at. If the idea only stands up with dozens of instructions holding it, it’s their idea and not the model’s. If it stands up on six, the six are the idea, and you can hold all of them in your head at once, which is more than I can say for most things I’ve built.

14.1 What the model actually asks for

Here’s what Parts II and III say the model adds to a machine. A *record*: a flag that reads no or yes, “certified” or not. A *toll*: the step that raises the flag costs something (Section 3). A *ledger*: the running total of those costs, a whole number that only climbs (Section 6). And Section 10 proved the thing this whole chapter stands on: over any base at all, an honest record is a *latch*, a bit that switches on at an event the base decides and never switches off (Theorem 10.1). The record watches the base. It doesn’t compute anything the base couldn’t compute without it.

Push on that for a second, because it cuts the job in half. If the record adds no computing power, then all the computing power comes from the base, and the question “how small can the machine be” splits into two questions that don’t touch each other. For the base: what’s the smallest thing that can compute everything there is to compute? For the record: what’s the smallest set of pieces that makes the flag *mean* something?

One honest asterisk on “don’t touch,” because the machine below makes it matter. What’s blind is the flag and the ledger: the base never reads either of them. But the pieces that make the flag mean something (a table of checked facts, a slot for a commitment, a trap that stops a refused run) have to sit where the step can read them, which is in the base, and they do steer the run: a failed check stops a program dead. The latch theorem holds wherever you draw the line between base and record, which is exactly why it can’t tell you where the meaning lives. On this machine the meaning lives on the base side of the line, in the bookkeeping, and the flag on top reports it.

The record question, the second of those two, is the one Part III left open, and it's open on purpose. Section 9 said that testing a claim and committing to it are two different acts, and that nothing in the five parts of the model makes the second wait for the first. Read Section 3's definition again with that in mind: a machine where you pay one unit and raise the flag over nothing in particular obeys every rule in it. That machine is honest about the price and silent about what was bought. A receipt for nothing. The toll alone can't fix that, because the toll is a rule about *cost*, and "was this earned" is a question about *order*: did a check happen, of this claim, before the commitment, with nothing changing underneath in between?

This machine is my attempt to make the order part of the machine, in the smallest place I could find to do it. I'll show you the pieces, then the proofs, then the places it falls short, which there are, and which I'd rather you hear from me.

14.2 Where the counters come from

The base first, and the base isn't mine. It's Marvin Minsky's. In a 1961 paper in the *Annals of Mathematics*, and then in his 1967 book *Computation: Finite and Infinite Machines*, Minsky studied a machine so bare it's almost a joke. Picture a few bins of pebbles and a numbered list of instructions. There are only two kinds of instruction: put a pebble in bin c and go to the next line; or, if bin c has a pebble, take one out and jump to line j , and if it's empty, just go to the next line. That's it. No memory, no tape, no arithmetic beyond plus one and minus one. When the line you're on has no instruction, the machine stops.

What Minsky showed is that two bins are enough to compute anything a Turing machine can compute, as long as you're willing to pack the input in a strange way. The trick is to keep many numbers inside one number, as the powers of different primes ($2^x 3^y 5^z$ holds three numbers in one), and to do arithmetic on the exponents by multiplying and dividing with the second bin as scratch space. It is as slow as it sounds. Exponentially slow. A two-counter machine doing your taxes would finish sometime after the sun doesn't. But "universal" is a claim about what's *possible*, not about what's pleasant, and on that claim two bins of pebbles are as strong as anything.

Who cares? Minsky did, because the question then was where the edge of computation sits and how little machinery reaches it. You should, because it means the base of this chapter's machine hides nothing: if it can compute everything with plus one, minus one and a jump, then nothing about the record can be smuggling computing power in through some instruction I made fancy. And I care because it's the smallest universal base I know of that is already in a Coq library, so the hard part (proving that its halting question is undecidable) is someone else's checked work and not my say-so.

And the pebbles aren't only my picture. They were somebody's actual machine, the same year. In one issue of the Canadian Mathematical Bulletin, September 1961, Z. A. Melzak described a computer made of distinguishable holes and an unlimited supply of pebbles, with one three-place operation, and a few pages later Joachim Lambek

trimmed it to two moves: put a pebble in hole X ; take one out of hole X if there's one to take. He called it an *infinite abacus*, and early on he writes "Our method is probably not new," which is a hard sentence to beat. He also tells anybody who distrusts infinity not to worry about all those holes, since any one computation only ever uses finitely many of them. Is that two inventions or one? Lambek says his was a modification of Melzak's, so call it one and a half, with Minsky's as a separate find the same year. And then Lambek went somewhere you wouldn't guess: in 1986, with Philip Scott, he wrote a book showing that typed lambda calculi and a certain kind of category are essentially the same thing. Pebbles to category theory in one career, twenty-five years end to end. Does any of it change the argument? No. The library and I use Minsky's version and Minsky's name. But a bare machine that several people reach for in the same year looks less like anybody's invention and more like a shape that was sitting there.

The title of Minsky's 1961 paper is the clue I can't leave alone. It's about "Post's problem of tag." Who's Post, and what's tag? Emil Post, in 1921, on a fellowship at Princeton, was playing with about the simplest rewriting game there is. Take a word of 0s and 1s. Look at its first letter, chop the first three letters off the front, and stick something on the end: 00 if the letter you looked at was 0, 1101 if it was 1. Repeat, and stop when fewer than three letters are left. Try it at the table. Start from 1000: you get 01101, then 0100, then 000, then 00, and it stops. Start from 100: you get 1101, 11101, 011101, 10100, 001101, and then 10100 again, and it goes round forever. Post wanted a rule that tells you, for every starting word, which of those happens. He worked at it for months and later called it hopeless, and as far as I can tell it still is: a hundred years on, nobody knows whether that one little game, with that one rule, can be decided (Stephen Wolfram's 2021 article on it is where I'd start). Minsky's paper showed that tag games in general can compute anything, which is why Post's name is on it. It didn't settle Post's own game. And the chase goes one step further. In 2008 Liesbeth De Mol turned the famous $3n + 1$ problem (halve an even number, triple an odd one and add one, and ask whether you always get down to 1) into a small tag game of the same kind. So is Post's game secretly as hard as everything, or just stubborn? I can't tell you, and as far as the published work goes, nobody can yet.

None of that changes the machine below. It changes the picture. The two-counter machine came out of a hunt for the smallest thing that can't be predicted, and the smallest things are exactly where the open questions are still sitting.

One word clash, before it bites. Minsky's machine is usually called a *counter machine*, and its bins are *counters*. Section 6 also had a counter: the ledger μ , the number that only climbs. Same word, two completely different things, and I'd love to say I set that up on purpose. In this chapter, *counter* means one of Minsky's two bins, called A and B , and the climbing total is always *the ledger* μ .

14.3 The machine

Here's the whole state. I'll say each piece in words first.

- Two counters, *A* and *B*. Each holds a whole number, with no upper limit.
- A *version* for each counter, also a whole number with no limit. It starts at 0 and goes up by one every time its counter is written. Think of it as the counter's edition number: a new value is a new edition.
- A *line number*, which instruction comes next, counting from 1 the way Minsky does (more on that choice just below).
- A *fact table*: a list of facts. A *fact* is three things stuck together: a property, a counter, and a version. "*A* is even, as of edition 3" is a fact. The table has room for sixteen. Why sixteen? It's a pick, with no derivation behind it, and what the pick costs is in the last subsection of this chapter, the one on what the machine doesn't do.
- A *channel*: either empty, or holding one fact. This is where a commitment sits.
- A *trap*: one bit, down at the start. When it goes up, the machine has refused something, and it stays up.

Counting lines from 1 sounds like the least interesting choice in the chapter, and it's one of the oldest arguments in programming, with a famous referee. In 1982 Edsger Dijkstra wrote one of the numbered notes he circulated to colleagues, EWD 831, titled "Why numbering should start at zero." His case goes through ranges first. Of the four ways to write "2 up to 12" with less-than signs, he picks the one that takes the bottom in and leaves the top out, $2 \leq i < 13$: two ranges that meet end to end then share one number written once, and the length is just the top minus the bottom. Start a sequence of N things at zero and it's $0 \leq i < N$, and each thing's number is how many things came before it. He reports that programmers on the language Mesa, which allowed all four conventions, found the other three a steady source of clumsiness and mistakes, and he points out that FORTRAN starts at one. Does that settle it? It settles which convention is tidier, by an argument you can check. It doesn't settle which convention somebody else's work used, and that's the trap: Minsky numbers lines from 1, and so does this machine, because matching the library's two-counter machine line for line is worth more here than tidiness. Is zero the right place to start counting? I can't tell you. Dijkstra's note doesn't claim to either; it shows which way makes fewer mistakes, which is a different and more useful thing.

Everything in that list is the *core*, the part the base moves. Outside the core sit the two pieces from the model: the ledger μ and the flag. The machine never reads either of them to decide what to do next. That's not a promise, it's how the step is written, and it shows up as a theorem below.

The properties a fact can name come from a tiny fixed language with three words: *is*

zero; is even; and is at least n , for any whole number n you like. Each one comes in two forms. There's the *checker*, a little program that computes yes or no: compare with 0, look at the last bit, compare with n . And there's the *meaning*, the plain arithmetic statement the checker is supposed to be checking: the number is 0; the number is two times something; n is less than or equal to the number. Why both? Because a checker is code, and code can be wrong in ways that look right, and "the checker says yes" is only worth anything once somebody has proved it says yes exactly when the meaning is true. That proof is one line per property, and it's the first block below.

Why can a machine check a claim at all? Because a claim with a precise meaning is something you *evaluate*: plug in the values and turn the crank. George Boole worked out in 1847 that you can do arithmetic on claims this way, treating "and" like multiplication and "or" like a kind of addition. Why multiplication? Try it at the table. Write 1 for true and 0 for false, and multiply: one times one is one, and anything times zero is zero. That's "and" to the letter. Then the stranger thing Boole noticed. Say " x and x " and you've said nothing new; it's just x . In numbers that's $x \times x = x$, and he made it a law of his system, and it's a law ordinary numbers mostly break. Which numbers equal their own square? Two times two is four, no. A half times a half is a quarter, no. Zero, yes. One, yes. That's the whole list. So when Boole wrote his bigger book in 1854, *The Laws of Thought*, he asked you to imagine an algebra whose letters take the values 0 and 1 and nothing else. Every checker on this machine answers in that algebra.

And the 1847 book has a backstory worth a detour. Boole was largely self-taught and spent about fifteen years as a schoolmaster, and what pulled him back to logic, early in 1847, was a loud public quarrel between Augustus De Morgan and the Scottish philosopher Sir William Hamilton over who had first thought of a small technical idea in the old logic of syllogisms. Two well-known men fighting over credit, and a schoolmaster watching from the side picks up the subject and writes the little book this paragraph is about. Few people remember what the quarrel was over. Every checker in this chapter runs on what the bystander wrote.

Then push on it, because Boole's "or" isn't quite the one a checker uses. His plus only made sense for things that can't both hold at once, so " x or y " with both of them true had no meaning in his system. In 1864 William Stanley Jevons, in a book called *Pure Logic*, read "or" as "either or both," which gives " x or x is x ," the same trick as before, and he wrote to Boole saying plus ought to work that way. Boole wouldn't have it, since it broke the tie to ordinary algebra his whole system stood on. The "or" every programming language uses is Jevons's. Does that change anything about the checker? No; the crank turns the same way. It changes the picture: the "or" a checker uses is the one the man who invented the algebra turned down.

Hanging a checked claim on a point in a computation is an old move too. Tony Hoare, in 1969, built a whole way of reasoning about programs out of it: attach a claim to

a *point* in a program (“whenever execution reaches here, this is true”), and reason forward from claim to claim. Robert Floyd had done it two years earlier on flowcharts. Their claims were checked on paper by the person writing the proof. This machine’s version is a property of one counter at one edition, checked by a step while it runs.

And is two years earlier the bottom? Not even close, and the next step down is a paper this book already leaned on. In June 1949, at a Cambridge conference held for the new EDSAC computer, Alan Turing gave a talk three pages long, “Checking a large routine,” the same note the counter chapter used for stopping (Section 6). Stopping was the second half of it. The first half is this move, twenty years before Hoare: the programmer should make “definite assertions which can be checked individually,” and the correctness of the whole should follow from them. His example program computes a factorial using nothing but addition, and his picture for why you’d bother is one you can do on the back of an envelope. Add up a long column of four-digit numbers and hand someone only the total, and they have to redo the whole sum at one sitting, because the carries tie every column to the one before it. Hand them the total of each column as well, and the job falls apart into small checks they can do one at a time. That’s a claim hung on a point in the middle of the work, so the person checking never has to hold all of it at once. The paper got its due mostly later: in 1984 F. L. Morris and C. B. Jones republished it, with its misprints corrected, as an early program proof by Alan Turing. Does any of that change the machine? No. It changes the date on the idea, and it tells you something I like: the first person to ask how you’d check a program put the claims in the middle of it, not at the end.

Now the six instructions.

Instruction	What it does, when the trap is down	Cost
INC c	Adds one to counter c , adds one to its version, goes to the next line.	0
DEC $c\ j$	If c is above 0: subtracts one, adds one to its version, jumps to line j . If c is 0: just goes to the next line.	0
HALT	Nothing. A program that reaches it stops.	0
CHECK $p\ c$	Runs the checker for p on c ’s value. If it says yes and the table has fewer than sixteen facts, writes the fact (p, c, c ’s current version) into the table and goes to the next line. Otherwise, trap.	1

Instruction	What it does, when the trap is down	Cost
COMMIT $p\ c$	Looks in the table for exactly the fact $(p, c, c$'s current version), field by field. If it's there, puts it in the channel and goes to the next line. Otherwise, trap.	1
CERTIFY	If the channel holds a fact, goes to the next line, and the flag goes up. Otherwise, trap.	1

The first three are Minsky's machine, plus a HALT so a program can say it's done. The last three are the record's. Read them as a chain: you can only certify with a commitment in the channel, you can only commit to a fact in the table, and the only way into the table is a check that passed. Three links, and each one costs.

One of what? One unit of μ , the same unit as Section 3's toll. Why one and not ten? Because one is the smallest positive whole number, which is honestly the whole defence, and Section 4 is the section where I tried to do better than that for the flag. That argument doesn't reach this machine, though (the end of the chapter says why), so here the one is a choice, full stop.

Where does that one come from, though? It's older than Coq by about a century. In 1888 Richard Dedekind published a little book whose title asks what numbers are and what they're for, and in 1889 Giuseppe Peano published a booklet, written in Latin, which is usually counted as the first statement of the axioms for the whole numbers that people still teach: there's a first number, and every number has a next one, and a few rules about how "next" behaves. The two were worked out independently, and the axioms often go by both names. (Peano's booklet is also where the sign $\in$ that you met in Section 2 is usually said to come from.) Coq builds its whole numbers exactly that way, out of zero and "the next one" and nothing else, so inside Coq the number three really is $S(S(S\ O))$, the next of the next of the next of zero. What that changes here is small and I like it: the one a CHECK costs isn't a number pulled from a hat. It's one application of the smallest step the number system has. You can't charge less than one S and still charge something.

The zeros matter just as much: computing is free on this ledger. Plus one, minus one and jumping around cost nothing at all. Only the three acts that make a claim cost, and each of them costs whether it passes or traps. Failing is not a discount.

The version rule. Here's why versions exist, with numbers. Start with $A = 4$, version 0. CHECK *is even* A passes, and the table gets the fact $(\text{even}, A, 0)$. Now INC A : A becomes 5, and its version becomes 1. Now COMMIT *is even* A . The machine looks for $(\text{even}, A, 1)$, the claim about the current edition. The table only has $(\text{even}, A, 0)$.

No match. Trap. The old fact is still sitting in the table, still true about edition 0, and completely useless for committing to anything about edition 1. A fact is about one edition of one counter, and an edition never comes back, because versions only go up and never wrap. That's the whole trick against stale evidence: you can't check a number, change it, and then commit to the check.

That's the opening chapter's time-of-check-to-time-of-use bug, the file checked and then swapped, and the database answer to it is Kung and Robinson's: at commit time, check that nothing you read was written in the meantime, and back out if something was. The version rule is that check, done by the step, on one counter. It also shows where the guarantee ends. It works because one step does every write and every write bumps a version. A second writer that changed a counter without bumping its version, a second processor or a device writing memory on its own, would walk right around it, and none of the theorems below are about a machine with one of those.

Processor designers put a cousin of this check in silicon. Jensen, Hagensen and Broughton, at Lawrence Livermore in 1987, described a pair of instructions now called load-linked and store-conditional: the load reads a word and the processor starts watching it, and the store goes through only if nothing has written that word since; otherwise it fails and the program tries again. It's tempting to say CHECK is the load and COMMIT is the conditional store, and it's wrong in two places. COMMIT doesn't write the counter it checked, and the thing doing the watching, the fact, sits in a table the step keeps, not with one processor. The hardware version does cover the second writer, because the watching happens on the memory everyone shares, which is exactly the case the version rule leaves out.

The real kinship is older, and it's somewhere else. Picture the table and the counters shared by many processors, with CHECK, COMMIT and CERTIFY each done as one indivisible step. Maurice Herlihy (1991) ranked shared objects by how many processors each one can bring to agree on a single value with nobody ever waiting on anybody, and that object would sit at the top, able to bring any number to agree. The reason isn't that COMMIT is a store-conditional. It's that the object contains Serge Plotkin's sticky bit (1989), a cell the first writer jams for good and everyone after can read: a write that kills every fact about a counter, plus a commitment other processors can see, makes one, and the trap makes another. Take the version rule away, let failures change nothing, and let checked properties only ever turn true, and everything in the object only grows; then it drops to the bottom rung, the level of plain memory, which is the shared-memory face of what Joseph Hellerstein and Peter Alvaro (2020) call CALM, where monotone means nobody has to coordinate. The toll changes neither answer. That split is argued here on paper, not checked, and neither half is mine: the top is Plotkin's and the bottom is the monotone fact that field already knows.

Process-calculus people have a statement of the same split. Johannes Åman Pohjola and Joachim Parrow (2016) show that no model whose parallel composition is mono-

tonic, where a neighbouring process can never stop a step, has a consensus process, and one of their examples of a model that does is test-and-set, an instruction that writes a cell and reports what was there, which is a sticky bit under another name. A store that can refuse permanently, as the version rule does to a stale commit, sits on the non-monotonic side. I haven't written that encoding either, so it's a pointer.

Security people know the pattern by a third name. A check that blesses a value, and a write that takes the blessing away, is close kin to dynamic taint tracking, where a label rides along with the data and a sanitizing step changes it; the labels go back to Dorothy Denning's 1976 lattice model of how information is allowed to flow. How much a monitor like that has to remember is a studied question too: Chamseddine Talhi, Nadia Tawbi and Mourad Debbabi (2008) worked out which rules a monitor can still enforce when its memory is bounded. The version rule is one small, specific member of that family, done by the step.

Notice what *isn't* a write. DEC on a counter that's already 0 doesn't change the counter, so it doesn't change the version. It's a test that happens to move the line number.

The fact table. Facts go in and never come out. Nothing erases them and nothing overwrites them. When the table is full, a seventeenth CHECK doesn't push an old fact out to make room; it traps. That's a choice with a bill attached, and I'll put the bill in the section on what this machine doesn't do.

Traps. Any failure puts the trap up and changes nothing else. Once it's up, every instruction leaves the core alone. A stored program stops at a trap. But I also prove things about *any* list of instructions, not just programs, because an adversary doesn't have to run my programs; they can feed the machine whatever they want. For those lists, an instruction after a trap is still charged its cost, and still changes nothing.

Now the exact version.

Definition 14.1 (Properties, checker, meaning). A *property* is one of: Zero, Even, or $\text{Ge}(n)$ for a whole number n . The *checker* $\text{eval}(p, v)$ computes yes or no: $v = 0$ by equality, evenness by the parity function, $n \leq v$ by comparison. The *meaning* $\text{holds}(p, v)$ is the arithmetic statement: $v = 0$; there is an m with $v = 2m$; $n \leq v$.

Lemma 14.1 (The checker says what it means). *For every property p and every whole number v , $\text{eval}(p, v) = \text{yes}$ if and only if $\text{holds}(p, v)$.*

Proof. Split on the three properties. Equality with zero returns yes exactly at zero; comparison with n returns yes exactly when $n \leq v$. For evenness, the parity checker returns yes at 0, no at 1, and the same answer at $v + 2$ as at v . Induction in steps of two gives yes exactly at $0, 2, 4, \dots$, the numbers $2m$. Those are precisely the three meanings. $\square$

Checked as `eval_iff` (EarnedCore.v).

Definition 14.2 (Core, state, claim, start). A *core* k holds two counters A, B and their versions (whole numbers), a line number, a fact table (a list of facts, a fact being a triple of a property, a counter and a version), a channel (empty or one fact), and a trap bit. Write $\text{val}(k, c)$ and $\text{ver}(k, c)$ for counter c 's value and version. A *state* s is a core, written $\text{core}(s)$, with a ledger $\mu(s)$ and a flag $\text{cert}(s)$. The *claim* p of c in k is the fact $(p, c, \text{ver}(k, c))$. The table's room is 16. The start $\text{start}(a, b)$ has counters a and b , versions 0, line 1, an empty table, an empty channel, the trap down, $\mu = 0$ and the flag down. A state is *clean* when its table and channel are empty and its flag is down.

Definition 14.3 (Instructions, costs, the step). The instructions are `INC  $c$` , `DEC  $c$   $j$` , `HALT`, `CHECK  $p$   $c$` , `COMMIT  $p$   $c$` and `CERTIFY`, with effects on a core with the trap down as in the table above. Passing any check, commitment or certification requires the trap to be down. With that condition, `CHECK` passes when $\text{eval}(p, \text{val}(k, c)) = \text{yes}$ and the table holds fewer than 16 facts, `COMMIT` passes when the claim p of c is in the table, and `CERTIFY` passes when the channel holds a fact; a failing one sets the trap and changes nothing else. On a core with the trap up, every instruction changes nothing. Costs: `CHECK`, `COMMIT` and `CERTIFY` cost 1, the rest 0. One step on a state is

$$\text{exec}(s, i) = (\text{cexec}(\text{core}(s), i), \mu(s) + \text{cost}(i), \text{cert}(s) \vee \text{fires}(\text{core}(s), i)),$$

where cexec is the core's step and $\text{fires}(k, i)$ is yes exactly when i is `CERTIFY` and passes at k . For any list t of instructions (a *trace*), $\text{run}(t, s)$ applies them in order and $\text{cost}(t)$ adds their costs. A *program* is a list of instructions numbered from 1; a program run fetches the instruction at the line number and executes it, and stops when the trap is up, the line holds `HALT`, or the line holds nothing.

Defined as `cexec`, `exec`, `run`, `total_cost`, `run_prog`, `start` and `Minimal.EarnedCore.clean_start` (EarnedCore.v).

Read the step line slowly, because the rest of the chapter leans on its shape. The core moves by its own rule and nothing else. The ledger adds the cost. The flag becomes "it was up, or this is a `CERTIFY` that passed." The core's move doesn't mention μ or the flag anywhere. The base is blind to the record by construction.

A worked run, the shortest one there is. The program: `CHECK is zero A`; `COMMIT is zero A`; `CERTIFY`. Start at $\text{start}(0, b)$, for any b . Line 1: A is 0, the checker says yes, the table is empty so there's room, and the fact $(\text{zero}, A, 0)$ goes in. $\mu = 1$. Line 2: the claim about A 's current edition is $(\text{zero}, A, 0)$, and it's right there. Into the channel. $\mu = 2$. Line 3: the channel holds a fact, the flag goes up. $\mu = 3$. Line 4 has no instruction, so the program stops. Three paid acts, in order, a flag that's up, and a channel that names

the last earned commitment, which here is $(zero, A, 0)$.

Now start the same program at $start(5, b)$. Line 1: A is 5, the checker says no, the trap goes up, $\mu = 1$, and the program stops right there. The flag is down, and you can run it as long as you like: it stays down. That second run is the point of the first. The same three instructions certify from some starts and refuse from others, because the check could have failed, and from most starts it does. A check like that has a no pile, which is the only kind the opening chapter counts as establishing anything.

Could I have picked an even dumber claim? Sure, and the machine would have let me. CHECK *at least 0* A passes on every number there is, so CHECK, COMMIT, CERTIFY on it raises the flag for 3 from every start (stop the presses: a number is at least zero). The step doesn't police that. It checks order, aboutness and that the checker tells the truth; whether the question was worth asking isn't something it looks at. A check that can't fail establishes nothing, and this machine will charge you three and raise the flag for it anyway. Spotting that is the reader's job, with the property in hand, and it's on the list at the end of the chapter.

Others walked into the same line from communication. Oded Goldreich, Brendan Juba and Madhu Sudan (2012) built a theory of communication aimed at goals instead of messages, with a partner you may not understand, and the key notion is *sensing*: being able to tell, from what you see, whether you're making progress. For a goal that ends in one final verdict, the sensing has to be safe, never reporting success that didn't happen. A check that can fail and never lies, standing in front of a verdict you can't take back. That's their theorem in their setting, and it's the same shape.

14.4 It computes everything two counters can

First the base, because if the base can't compute, nothing else here matters.

Definition 14.4 (Two-counter program, compile, window). A *two-counter program* M is a list of instructions, each either MINC c or MDEC $c\ j$. A configuration is a triple $(line, a, b)$. One step of M : if the line holds no instruction, M has *stopped*; MINC adds one to c and goes to the next line; MDEC subtracts one from c and goes to line j if c is above 0, and goes to the next line if c is 0. $mrunch(n, M, x)$ is n steps from x , staying put once stopped. The *compile* of M replaces each MINC c by INC c and each MDEC $c\ j$ by DEC $c\ j$. The *window* of a core k is the triple $(line\ number, val(k, A), val(k, B))$.

Theorem 14.2 (Step for step). *For every two-counter program M and every core k with the trap down: M has stopped at $window(k)$ if and only if the program $compile(M)$ has halted at k ; and if M steps from $window(k)$ to y , then one program step of $compile(M)$ from k lands on a core with window y and the trap still down. So for every n , n program steps of $compile(M)$ from k land on a core whose window is $mrunch(n, M, window(k))$, with the trap down.*

Proof. Look up the instruction at the line. If there is none, both sides stop. If it's MINC c , the compiled INC c adds one to c and goes to the next line, which is exactly the window's move (the version also moves, but the window doesn't show versions). If it's MDEC $c\ j$, split on whether c is 0, and both sides do the same thing in each case. Neither instruction can trap. The run statement is induction on n , one step at a time. $\square$

Checked as `simulation_step` and `simulation_run` (EarnedCore.v).

Theorem 14.3 (Halting matches). *For every two-counter program M and whole numbers a, b : M stops after some number of steps from $(1, a, b)$ if and only if the program `compile`(M) halts after some number of program steps from `start`(a, b).*

Proof. The start has window $(1, a, b)$ and its trap is down. Apply Theorem 14.2 after any number n of steps: the two configurations still have the same window, the trap is still down, and one is stopped exactly when the other is halted. The same n witnesses termination in either direction. $\square$

Checked as `halting_correspondence` (EarnedCore.v).

Now the part that isn't mine. Proving that halting is undecidable for two-counter machines is real work, and it's already done, checked, in a shared Coq project called the Coq Library of Undecidability Proofs (presented by Yannick Forster, Dominique Larchey-Wendling and coauthors at CoqPL 2020). The two-counter result in it is Larchey-Wendling's, reached from his and Forster's work on Minsky machines (CPP 2019) by way of FRACTRAN. Their two-counter machine has the same two instructions as Minsky's, with the same rule for zero. So the only thing I have to prove is that their machine and my INC/DEC fragment are the same machine, step for step, and then borrow their theorem.

One honest wrinkle about what "undecidable" means in that library, because it isn't the textbook sentence. Every function you can write in Coq is a program, so "decidable" there means "some Coq function answers it." Coq by itself can't prove that *no* Coq function answers a question without an extra assumption about what programs are. So the library says the strongest thing it can without one: if anything decided this, then the questions on which a single-tape Turing machine *doesn't* halt could be listed by a program, which is the classic thing that can't be done. That's their convention, and the theorem below inherits it exactly.

Theorem 14.4 (Halting is undecidable for this machine). *The library's two-counter machine agrees with the INC/DEC fragment: its step, its stop, and its termination match those of the compiled program, so a library problem (P, a, b) halts if and only if the compiled program halts from `start`(a, b). Consequently, the question "does program P halt from `start`(a, b)?" is*

undecidable in the library's sense: a decider for it would make the complement of single-tape binary Turing machine halting enumerable.

Proof. The agreement is a case split on the library's instruction at the current line, and then induction along the run, ending in Theorem 14.3. The compile map is then a reduction from the library's two-counter halting problem, which the library proves undecidable, and the library's own lemma carries undecidability across a reduction. $\square$

Checked as `mm2_step_iff, mm2_stop_iff, mm2_terminates_iff, mm2_halt_i`
`ng_iff` and `earned_core_halting_undecidable` (`EarnedCoreLinks.v`), the last
applying the library's `MM2_HALTING_undec`.

What that buys, and what it doesn't. It buys: the base of this machine is a universal machine in Minsky's sense, and nobody (in the library's sense) can predict in general what it'll do. What it doesn't buy is a fast machine, or one where your input arrives conveniently. The counters start holding a and b , given directly; there's no encoding layer in my files. Minsky's prime-power packing is how you'd get from "any Turing machine" to "two counters," and that construction is his and the library's, not something I re-proved.

Is "given directly" just a convenience, or does it bite? It bites, and somebody found exactly where. In 1972 Rich Schroepel, in a short memo from the MIT AI Lab, proved that a two-counter machine started with n in one counter and zero in the other can't compute 2^n . So the packing Minsky needs is precisely the thing two plain counters can't do for themselves. Hand the small machine its inputs straight and there are functions of them it can't compute; hand it the inputs already packed and it computes anything a Turing machine can. That's no contradiction with the undecidability above, which ranges over every program and every starting pair, packed ones included. The base hides nothing. It just needs its input pre-chewed, and the chewing is somebody else's.

14.5 It's a certification system, and the flag is a latch

Now the record. Three facts first, all of them read straight off the step line.

Theorem 14.5 (Ledger, toll, permanence). *For every state s and instruction i , $\mu(\text{exec}(s, i)) = \mu(s) + \text{cost}(i)$, and for every trace t , $\mu(\text{run}(t, s)) = \mu(s) + \text{cost}(t)$. If $\text{cert}(s)$ is down and $\text{cert}(\text{exec}(s, i))$ is up, then i is CERTIFY, it passes at $\text{core}(s)$, and $\text{cost}(i) \geq 1$. If $\text{cert}(s)$ is up, so is $\text{cert}(\text{exec}(s, i))$.*

Proof. Read the ledger component of Definition 14.3: it adds exactly $\text{cost}(i)$. Repeated substitution along a trace gives the sum of its costs, or, equivalently, induction on the length of the trace does. The flag's new value is its old value or $\text{fires}(\text{core}(s), i)$. A false-to-true change therefore requires that event, whose definition requires a passing

CERTIFY. CERTIFY costs 1. An already true flag stays true under the same disjunction. $\square$

Checked as `mu_conservation`, `mu_conservation_trace`, `only_certify_certifies`, `a2` and `cert_permanent` (EarnedCore.v).

That third clause in the middle is Section 3's toll, for this machine. Which means this machine *is* one of the things Section 3 was about, and Section 3's theorem applies to it without my lifting a finger.

Theorem 14.6 (An instance of the certification system). *The states, the instructions, the step exec, the cost and the flag, with the toll of Theorem 14.5, form a certification system (Definition 3.1). So the universal floor (Theorem 3.1) holds here: if $\text{cert}(s_0)$ is down and $\text{cert}(\text{run}(t, s_0))$ is up, then $\text{cost}(t) \geq 1$.*

Proof. Take the state and instruction sets of Definitions 14.2 and 14.3, their step and cost, and the flag as the reading. These provide the five data fields of Definition 3.1. Theorem 14.5 supplies its one required rule. Applying Theorem 3.1 gives the claimed bound. $\square$

Checked as `earned_cs` and `earned_core_floor` (EarnedCoreLinks.v); `earned_core_floor` is `universal_nfi_any_substrate`, the checked name of Section 3's theorem, applied to `earned_cs`. The same floor is proved directly, without the link, as `nfi_floor` (EarnedCore.v).

Then Section 10. Section 10 said: take any base, and any record that answers only to the base and never comes back down, and the record is a latch. This machine has a base: a program together with a core, stepping by the program step. And it has a record, the flag, with the ledger riding along. So the question is whether it's an *honest* extension in Section 10's five-rule sense (Definition 10.2), and if it is, Section 10 does the rest.

Theorem 14.7 (The flag is a latch of the base). *Let the base have states $(\text{program}, \text{core})$, stepping by the program step, starting at $\text{start}(a, b)$'s core for any a, b , and halting when the program has halted. Map each $(\text{program}, \text{state})$ to $(\text{program}, \text{core})$. Then the machine, run as a closed machine on $(\text{program}, \text{state})$, is an honest extension of that base, and so its flag factors as the latch (Theorem 10.1) of the event "the next instruction is a CERTIFY that passes."*

Proof. The rules of Definition 10.2 one at a time. Driven by the computation: the next flag is the old flag or "the next instruction is a passing CERTIFY," which is a function of the base state and the old flag (that's the step line). Ledger: μ only adds. Toll: Theorem 14.5. Permanence: Theorem 14.5. A reachable write: the three-line program

of the worked run raises the flag from a start. Section 10's theorem then gives the factorization. $\square$

Checked as `earned_core_honest` and `earned_core_is_latch` (EarnedCore-Links.v), the latter by `record_axis_is_latch_holds`; the step-level facts are `cert_latch` and `base_blind` (EarnedCore.v).

So far, nothing new. The toll, the floor and the latch are things Part III already promised about any machine of this kind. The next subsection is the part Part III couldn't give.

14.6 Commitments are earned

Here's the shape before the words. Pick any run from a clean start, any list of instructions at all, chosen by somebody who'd love to cheat. Look at the moment a COMMIT would pass. Then, somewhere earlier in that run, there's a CHECK of exactly the same claim that passed and paid, at the same edition of the counter, and between that CHECK and now, nothing touched the counter. Not "probably." Not "on the programs I wrote." On every list of instructions anybody can write.

To say "nothing touched it" exactly, one word.

Definition 14.5 (Untouched). A list of instructions *mid*, run from a state *s*, leaves counter *c* *untouched* when, for every way of splitting *mid* as *t*₁, then *i*, then *t*₂, executing *i* at the core reached by running *t*₁ from *s* leaves both `ver(·, c)` and `val(·, c)` as they were.

Defined as `untouched` (EarnedCore.v).

Theorem 14.8 (A commitment was checked first). *Let s_0 be clean and pre any trace after which COMMIT $p\ c$ would pass. Then pre splits as pre_1 , then CHECK $p\ c$, then mid , where: the CHECK passes at the core reached by pre_1 ; CHECK $p\ c$ and COMMIT $p\ c$ each cost at least 1; c 's version after pre_1 equals its version after pre ; and mid , run from just after the CHECK, leaves c untouched.*

Proof. COMMIT passes, so the claim $(p, c, \text{ver}(c))$ for the current version is in the table. By the no-forging invariant (Theorem 14.12 below), that fact was written by a passing CHECK $p\ c$ at some point pre_1 , with the version c had then; since it's the current version too, the two versions are equal. Versions never go down, and any instruction that changes a counter's value also raises its version, so an instruction in *mid* that touched *c* would have left *c*'s version higher at the end than at the CHECK. It isn't. So nothing in *mid* touched *c*. $\square$

Checked as `earned_commitment_provenance` (EarnedCore.v).

Theorem 14.9 (A raised flag was checked, then committed, then certified). *Let s_0 be clean and t any trace with $\text{cert}(\text{run}(t, s_0))$ up. Then for some p and c , t is*

$$pre_1, \text{ CHECK } p \ c, \ mid_1, \text{ COMMIT } p \ c, \ mid_2, \text{ CERTIFY}, \ post,$$

where the CHECK passes at its state, the COMMIT passes at its state, the CERTIFY passes at its state, c has the same version at the CHECK as at the COMMIT, and mid_1 leaves c untouched.

Proof. The flag starts down, so some first instruction raised it, and by Theorem 14.5 it's a passing CERTIFY. A CERTIFY passes only with a fact in the channel, the channel starts empty, and only a passing COMMIT puts a fact there, so some earlier COMMIT $p \ c$ passed. Theorem 14.8 at that COMMIT gives the CHECK, the equal version, and the untouched stretch. $\square$

Checked as `earned_certification_provenance` (EarnedCore.v).

And the failure side, so you can see the refusals are clean and not half-done:

Theorem 14.10 (Unearned steps trap and change no claim). *If COMMIT $p \ c$ would not pass at a core k , executing it puts the trap up and leaves the channel and the table as they were. If CERTIFY would not pass at a state s , executing it puts the trap up and leaves the flag as it was.*

Proof. If the trap is already up, the core is unchanged and its trap remains up. Otherwise, the failed COMMIT takes the failure branch of the instruction table: set the trap, leave every other core field alone. A failed CERTIFY takes the same branch, and its fires value is false. The flag update is then its old value or false, which leaves it unchanged. The instruction's charge still applies to the ledger. $\square$

Checked as `unearned_commit_traps` and `uncommitted_certify_traps` (EarnedCore.v).

If that rings a bell, it should. It's Milner's move from the opening chapter: in LCF a theorem is a value that only the rules of the logic can build, so nobody can just print one (Michael Gordon, Milner and Christopher Wadsworth wrote the system up in 1979, and Gordon told its history in 2000). Here the thing nobody can print is one bit, and what guards it is one step rule instead of a type checker. Same architecture, much lower down.

A provenance theorem is also an itemised receipt, and the itemised receipt has a history that's exactly this gap. James Ritty ran a saloon in Dayton, Ohio, and believed some of his staff were pocketing money. On a steamboat trip to Europe in 1878 he watched a mechanism that counted the turns of the ship's propeller, and wondered

whether something like it could count the money instead. In 1879 he and his brother John patented a “Cash Register and Indicator,” and they sold a version under the wonderful name Ritty’s Incorruptible Cashier. The first model was this book’s counter (Section 6) with brass on it: press a key, the amount goes onto a dial showing the day’s running total. No drawer, no paper. At closing time you hold the dial up against the cash. And that’s all a total can do. It tells you *that* the money is short, never which sale went missing or who rang it up. The paper came later. The patents passed to other owners and in 1884 to John H. Patterson’s National Cash Register Company, and the story told about that company is that it added a paper roll: one copy kept for the shop’s own books, one handed to the customer. That last part I only have from histories of the company and of the machine, so hold it the way you’d hold any company’s story about itself. The order is right either way: first a total that can tell you something’s wrong, then a list that tells you where. The ledger μ is the dial. Theorem 14.9 is the list: which CHECK, of which claim, about which edition, before which COMMIT.

Read Theorem 14.9 once more for what it *doesn’t* say. Between the CHECK and the COMMIT, the counter is untouched. Between the COMMIT and the CERTIFY, it isn’t promised anything. You can check A is even, commit to it, add one to A , and then certify. The flag goes up, and what it stands on is a commitment about the edition that was current when you committed. The certificate is historical: “this was checked and committed about this edition,” not “this is still true.” I think that’s the right meaning for a certificate (a birth certificate doesn’t expire when the baby grows), but it’s a meaning, and you should know which one you’re getting.

14.7 What a passing check means

Provenance says a CHECK happened. It doesn’t yet say the CHECK was *right*, and that’s a separate question with a separate proof. A machine could have a perfectly ordered chain of check, commit, certify, all standing on a checker that says yes to everything. So: what does a fact in the table actually tell you about the counter?

Theorem 14.11 (Checker soundness). *Let s_0 be clean and t any trace, and let k be the core of $\text{run}(t, s_0)$. For every fact (p, c, v) in k ’s table with $v = \text{ver}(k, c)$, the property holds of the counter’s current value: $\text{holds}(p, \text{val}(k, c))$. In particular, if COMMIT p c would pass at k , then $\text{holds}(p, \text{val}(k, c))$.*

Proof. Carry one sentence through every instruction: every fact in the table has a version no higher than its counter’s, and if its version equals its counter’s, its property holds of its counter’s value. A clean start has an empty table, so the sentence holds there. A passing CHECK adds a fact at the current version, and the checker said yes, so by Lemma 14.1 the property holds. A write raises one counter’s version, so every old fact about that counter falls strictly behind and the second half no longer applies to it. Nothing else changes a value or a version. The sentence holds after every instruction,

so it holds at the end. The part about COMMIT is the case where the claim is in the table at the current version. □

Checked as `checker_soundness` and `committed_claim_holds` (EarnedCore.v), through the invariant `sound` and its one-step form `sound_step`.

So a live fact means exactly this: one of three arithmetic properties holds of one counter’s current value. Not “the program is correct.” Not “the number is interesting.” The property, of the value, now. That’s all “checked” means on this machine, and it’s enough to be true.

So who checked the checker? Nothing on the record did. The record carries the run’s half of the story: a CHECK, then a COMMIT of the same claim about the same edition, then a CERTIFY. The other half, that the checker says yes exactly when the property holds, isn’t in any run. It’s Lemma 14.1, a proof about the little program the step calls, checked once by Coq and priced by nothing on the ledger. So a claim established here stands on two legs: a fact about this run, carried by the record, and a fact about the checker, carried by a proof that whoever reads the record has to accept. That’s the Münchhausen trilemma from the opening chapter taking its third horn on purpose: the chain of reasons stops at a proof, named and on the page, where you can decide whether you’d have stopped there too. And the entitlement it gives is the reader’s. The machine doesn’t know anything; it keeps the order.

The question in that paragraph is older than any checker, and its most famous wording isn’t about what you’d guess. “Who will guard the guardians?” gets quoted about governments and police and, lately, proof assistants, and it often gets pinned on Plato. Push on that. Plato’s *Republic* has the worry, sort of: in Book III, Glaucon says it would be absurd for a guardian to need a guardian, which is close in words and opposite in mood, a man calling the question silly. The Latin, *quis custodiet ipsos custodes*, is from the Roman satirist Juvenal’s sixth satire, and it’s a joke about a husband. His friends tell him to bolt his wife in and set guards on her; the line asks who’ll watch the guards, since she’ll start with them. And it gets better. In 1899 an Oxford undergraduate, E. O. Winstedt, found extra lines of that satire in a medieval manuscript in the Bodleian Library, lines the other copies lack, partly overlapping the familiar ones, and the famous question sits in that tangle. Scholars have argued ever since about whether those lines are Juvenal’s at all. So the most quoted sentence about trusting the checker comes with a doubtful source of its own. Does that change anything here? No. The only answer I know that works is de Bruijn’s (Section 17 has it): you never get rid of the last guard, you make the last guard small enough to watch yourself. Lemma 14.1 is one line per property. Euclid had the instinct: five postulates you can hold in your head, and everything else meant to follow from them.

That split, a fixed checker proved sound from outside and a process that only gets to act when the checker says yes, has a name in the writing on self-improving programs.

Jürgen Schmidhuber’s Gödel machine (2003) rewrites its own code only after its built-in proof searcher has proved the rewrite useful, and Benja Fallenstein and Nate Soares (2015) survey agents that reason about their successors through formal proofs a fixed checker accepts. Same architecture, aimed at a different question: they want to know when a machine may trust a smarter version of itself, and this machine only wants to know whether a flag was earned. Section 11 says why the checker’s proof has to live outside.

Soundness rules out a lying checker, not a pointless question: *at least 0* has a perfectly sound checker and an empty no pile, as the worked run showed.

14.8 Nothing gets forged

The proof of Theorem 14.8 leaned on an invariant I promised to show you. Here it is. The fear it answers: could some clever sequence of instructions get a fact into the table without a CHECK writing it? A DEC with a funny jump target, a COMMIT after a trap, something I didn’t think of? The answer is checked one instruction at a time, which is the only way I trust myself to answer it.

Definition 14.6 (Earned fact; no forgery). After a trace t from s_0 , a fact $f = (p, c, v)$ is *earned* when t is *pre*, then CHECK $p\ c$, then *mid*, where the CHECK passes at the core reached by *pre*, f is exactly the claim p of c at that core, and, if f is live at the end (v equals c ’s version after t), *mid* leaves c untouched. The trace has *no forgery* when every fact in the table after it is earned.

Theorem 14.12 (No forging). *For every core k , instruction i and fact f in the table after executing i at k : either f was already in k ’s table, or i is CHECK of f ’s property and counter, it passes at k , and f is the claim it writes. Consequently, if a trace has no forgery, so does that trace followed by any one instruction; and from every clean start, every trace has no forgery.*

Proof. The first part is a case for each of the six instructions: only a passing CHECK adds anything, and it adds exactly its claim. For the second, a new fact is earned by the CHECK that just wrote it; an old fact stays earned, and if it’s still live after the new instruction, its counter’s version didn’t move, so the new instruction didn’t touch the counter and the untouched stretch grows by one. A clean start has an empty table, which has no forgery, and the step carries it along any trace. $\square$

Checked as `facts_step`, `no_forging_step` and `no_forging` (EarnedCore.v).

14.9 A full table traps

Sixteen facts of room, and no eviction. Two halves to prove: nothing ever leaves, and nothing ever overflows.

Theorem 14.13 (Bounded table, no eviction). *Every instruction keeps every fact already*

in the table. If the table holds 16 or more facts, CHECK $p\ c$ puts the trap up and leaves the table as it was. If the table holds at most 16 facts, it still holds at most 16 after any instruction.

Proof. With a raised trap no instruction changes the table. With the trap down, only CHECK can change it, and a successful CHECK adds one fact without removing an entry. Success requires a previous length strictly below 16, so the new length is at most 16. At length 16 or greater that branch is unavailable: CHECK raises the trap and preserves the table. Every other instruction preserves it as well. $\square$

Checked as `facts_keep`, `full_table_traps` and `facts_bounded_step` (EarnedCore.v).

Why trap instead of making room? Because “making room” means deciding which fact dies, and a fact that can die is a fact whose absence you have to explain later. A table that only grows is a history you can read back, and the provenance proofs above lean on exactly that. The bill for it is real, though, and it’s in the last subsection of this chapter, the one on what the machine doesn’t do.

14.10 The window can’t see the record

Section 7 was about windows: a view of a state that shows some of it, and the theorem that a window which can’t tell two states apart can’t price the step between them exactly (Theorem 7.2). Here’s the smallest pair I could build where the window is Minsky’s view of the machine (line number and the two counters) and the record is everything else.

Run two traces from `start(0,0)`. The *earned* run is CHECK *is zero* A; COMMIT *is zero* A; CERTIFY. The *idle* run is DEC A 0, three times. In the idle run, A is 0 every time, so each DEC just goes to the next line and writes nothing. Both runs end on line 4, with $A = 0$ and $B = 0$, both versions 0. Through the window, identical. One of them has paid 3, holds a fact, could commit again, and has its flag up. The other has paid nothing, holds nothing, can’t commit, and its flag is down. Two photographs, same picture, different people.

Theorem 14.14 (Same window, different record). *The earned run and the idle run from `start(0,0)` end at cores with the same window and the same versions of A and B, and with different ledgers, different flags, different tables, and different answers to “would COMMIT is zero A pass.”*

Proof. In the earned run, CHECK adds (Zero, A, 0), COMMIT puts that fact in the channel, and CERTIFY raises the flag. The ledger is $1 + 1 + 1 = 3$. No counter is written. In the idle run each DEC sees zero, so it advances the line without writing a counter or raising a version. Its three costs are zero and it adds no fact or commitment. Thus both final windows are (4, 0, 0) with versions (0, 0). The earned table contains the

current zero claim and the idle table is empty, giving the stated difference in whether another COMMIT would pass, as well as in the ledger and flag. $\square$

Checked as `receipt_separation` (EarnedCore.v).

Theorem 14.15 (No window oracle). *There is no function g from windows to whole numbers with $g(\text{window}) = \mu$ at the end of every trace from $\text{start}(0,0)$. Likewise there is no function from windows to yes or no that gives the flag at the end of every such trace, and none that gives whether COMMIT is zero. A would pass.*

Proof. Such a g would have to give both 3 and 0 on the one window the two runs of Theorem 14.14 share. The same for the flag and the commit right. $\square$

Checked as `no_mu_oracle`, `no_cert_oracle` and `no_commit_oracle` (EarnedCore.v).

This is Section 7's shape, made of pebbles: the record lives off the window. Anybody who only watches the counters and the line number, which is everything Minsky's machine *is*, can't tell you what was paid, what was committed, or whether the flag is up. Not because they looked carelessly. Because it isn't there to see.

Two things the same in everything you look at and still two things: philosophy has an old fight about exactly that, and it's worth a detour, because it tells you which side of the fight these theorems stand on. Leibniz said it can't happen, not for real things. There are no two individuals indiscernible from each other, he wrote to Samuel Clarke in 1716, and he told a story to go with it. A clever gentleman of his acquaintance, talking with him in front of the Electress Sophia in the garden at Herrenhausen, thought he could find two leaves perfectly alike. She dared him to. He "ran all over the Garden a long time" looking, and it was "to no purpose." You can run his search tonight with a handful of anything, leaves or coins or grains of rice: look hard enough and every pair differs somewhere. Then in 1952 Max Black pushed back with a universe you can't build at the table. Nothing in it but two iron spheres, the same size, the same colour, the same everything, some distance apart. Name a difference between them. There isn't one, and there are still two of them. People have argued about Black's spheres ever since, and I'm not going to settle it.

Here's why I don't have to. Leibniz's claim is about *every* property a thing has. The theorems above are about three: the line and the two counters. The two runs aren't indiscernible in his sense at all: one has a fact in its table and its flag up, and the other doesn't, and Coq tells them apart in a line. They're indiscernible to a window, which is a much cheaper thing, and it's the thing that matters for a machine, because an observer of a machine only ever gets a window. Leibniz's gentleman would always find his difference if he could look at everything. It doesn't help him if all he's allowed to look at is three things about each leaf.

There's a horse in this, too, if you squint. In Germany in the early 1900s a mathematics teacher named Wilhelm von Osten showed off a horse called Clever Hans, who answered arithmetic questions by tapping a hoof, and got them right, in front of crowds. Same answers out as a horse that could count: that's the idle run's matching window. Then the psychologist Oskar Pfungst stopped looking at the answers and started changing the conditions, and published what he found in 1907. Put blinders on Hans, or have him asked by someone who didn't know the answer, and the arithmetic fell apart. Hans was reading the questioner: tiny, unintended movements of the head and body as the hoof reached the right number. That's the probe. Pfungst didn't argue with the answers. He asked a question the answers couldn't fake, one that needed the real thing to be there, the way a COMMIT needs a checked fact to be in the table: run COMMIT *is zero A* after each of the two runs, and one passes and the other traps (Theorem 14.14). And the part of the story I like best: nobody was lying. Von Osten was never persuaded by Pfungst and kept showing his horse. A matching window doesn't need a cheat behind it, which is why you want a probe and not an accusation.

14.11 The exact price is three

Section 3 said a raised flag costs at least one. On this machine that's true and too modest. A raised flag needs a CERTIFY, a CERTIFY needs a COMMIT, a COMMIT needs a CHECK, and each of those is its own instruction costing one. So at least three. One of what: three units of μ , one per paid act, each one the "one" I picked back in the table. And three isn't a guess that happens to hold; it's the number of links in the chain.

Theorem 14.16 (The minimum price of a certificate). *If s_0 is clean and $\text{cert}(\text{run}(t, s_0))$ is up, then $\text{cost}(t) \geq 3$ and $\mu(\text{run}(t, s_0)) \geq \mu(s_0) + 3$. For programs: if the flag is up after n program steps from $\text{start}(a, b)$, then $\mu \geq 3$ there.*

Proof. Theorem 14.9 finds a CHECK, a COMMIT and a CERTIFY at three different places in t , each costing 1, and every other instruction costs at least 0. The ledger statement is Theorem 14.5. A program run is the run of the instructions it actually executes, so the program form follows. $\square$

Checked as `certified_run_min_cost` and `program_certified_min_cost` (EarnedCore.v).

Theorem 14.17 (Three is tight). *For all whole numbers a and b , run the program CHECK is zero A; COMMIT is zero A; CERTIFY for 4 program steps from $\text{start}(a, b)$. If $a = 0$, it ends with the flag up and $\mu = 3$ exactly. If $a \neq 0$, it ends with the flag down and the trap up, and for every n , n program steps from $\text{start}(a, b)$ leave the flag down.*

Proof. If $a = 0$, the checker for Zero says yes at 0, so the first instruction adds (Zero, A, 0) to the empty table and charges 1. No counter changes, so the second

instruction finds that same fact, puts it in the channel and charges 1. The third finds the channel occupied, raises the flag and charges 1. The line is now 4, beyond the three-instruction program, and the fourth program step adds no charge. The final ledger is exactly 3. If $a \neq 0$, the checker says no, so the first instruction sets the trap and leaves the flag down. A program run takes no step once the trap is up, so every later program step leaves the state, flag included, exactly where it is. $\square$

Checked as `earned_run_check_can_fail` and `earned_run_refused_forever` (EarnedCore.v), with the program named `earned_run`, and the lemma `run_prog_trapped` for a trapped run standing still. The version whose check can't fail, `CHECK at least 0 A; COMMIT at least 0 A; CERTIFY`, certifies from every start for exactly 3 (`min_cost_tight`, with that program named `witness`). So the floor of three is reachable from any start at all, and that's the machine not policing trivial checks, not the machine establishing something.

That's the worked run from the start of the chapter, now as a theorem. The floor is 3, and a run whose check could have failed pays exactly 3, so 3 is the price, not a bound with slack in it.

Don't mistake that 3 for a measure of how hard a claim is to certify. It's the same 3 for every claim, easy or hard, because only the three acts that make a claim are priced and the counters work out whatever the check needs for free. A cost of certifying that grows with the claim would have to price the checking itself, and that's a different question with its own literature. Guy Blanc, Caleb Koch, Jane Lange and Li-Yang Tan (2022) studied one version of it under the name the query complexity of certification: how many questions an algorithm has to ask a black-box function to hand back a small certificate for its answer. Their count moves with the function. Mine doesn't move at all.

And that's every piece of the model, standing on six instructions: a base that computes everything two counters can, a ledger that's exact, a toll the universal floor applies to, a flag that's a latch of the base, a window that can't see the record, and a commitment that can't happen unless a check of the same claim, on the same edition, came first and was right. Every result above is closed: Coq reports no assumptions for any of them. That isn't my say-so. The file ends by asking Coq to print the assumptions of each headline theorem, and the book's test suite compiles it and fails unless every one comes back closed. (The project-wide audit behind the crosswalk doesn't list these files, which is why its rows mark them differently.) The base file uses nothing but Coq's standard library, so you can check it with nothing else installed; the link file adds only the book's own abstract records and the library of undecidability proofs.

14.12 What it takes to be Thiele-complete

The front matter promised a definition of Thiele-complete with teeth, so here it is. First, why it needs teeth. The easy definition goes: a Thiele machine is Thiele-complete

when its base can compute anything a Turing machine can. Sounds fine, and it's nearly empty. Take Minsky's two-counter machine, charge one mark for every move, and pick any yes-or-no question about its configuration as the record. Every move costs one, so every move that raises the record costs one, so it obeys the toll. That's a Thiele machine with a universal base: a clock bolted to a computer. A machine whose record never goes up passes too, since a toll on a raise that never happens is never broken. Neither one earns anything, and a name both of them get for free isn't saying much.

So the definition below asks for what the small machine actually does, written so that any machine at all can be held up against it. You hand over a reading of the machine, a list of things, and then the reading has to pass four clauses.

Definition 14.7 (Thiele-complete). A *machine* has states, moves, a step, a cost for each move and a yes/no reading of each state, the *record*. An *interface* for it hands over: a *universal base*, meaning a two-counter configuration shown by each state (its *window*), a class of *live* states, a move for each two-counter instruction, and a loaded state for each start $(1, a, b)$; a type of *claims*, each with a *meaning*, a statement about a state; a *checker* that says yes or no to a claim at a state; for each claim, a relation "the thing this claim is about is unchanged" between two states; a set of *clean* states; a *ledger*, a whole number for each state; and a *kind* for each move: a base move, CHECK of a claim, COMMIT of a claim, or CERTIFY. The machine is *Thiele-complete* when some interface meets four clauses. (a) *Universal base*. Each two-counter instruction's move is a base move; from every live state it acts on the window exactly as the instruction acts on a configuration, and the state stays live; each loaded state shows $(1, a, b)$, is live and is clean; no base move changes the record; and a record that's up stays up under every move. (b) *Earned record*. Clean states have the record down. Every run from a clean state that ends with the record up contains a CHECK of some claim c that passes, then a COMMIT of c with the thing c is about unchanged at every state in between, then a CERTIFY, with the record down just before that CERTIFY and up just after it. A passing check means its claim's meaning holds of the state it checked, and a meaning that holds keeps holding while the thing it's about is unchanged. (c) *Exact toll*. Base moves cost 0, each CHECK, COMMIT and CERTIFY costs exactly 1, and each move adds its cost to the ledger. (d) *Non-vacuity*. Some claim c has a CHECK, a COMMIT and a CERTIFY such that those three moves alone, from a loaded state, raise the record exactly when c 's meaning holds there; and c holds at some loaded state and fails at another. A Thiele machine whose base is universal, with no more asked, is *weakly Thiele-complete*.

Defined as `thiele_interface`, `thiele_complete_with` and `thiele_complete` (ThieleComplete.v), with the four clauses as `universal_base_clause`, `earned_record_clause`, `exact_toll_clause` and `non_vacuity_clause`, and the easy notion as `weakly_thiele_complete`.

Read the four clauses as four plain sentences. (a) It computes anything a Turing machine can, and the computing can't reach the record: the base moves are the two-counter machine, and they leave the record exactly where it was. (b) The record goes up only on top of the small machine's chain: a check that passed, a commitment to the same claim with nothing changed underneath, then a certificate. (c) Computing is free and each of the three acts costs one, so the ledger counts acts, not steps. (d) Somewhere there's a claim whose check could have said no. A machine whose every claim is "at least 0" doesn't qualify, and neither does one with nothing to certify.

And look where the checker's honesty sits. In (b), "a passing check means its claim holds" is a field of the interface, proved once, outside every run, exactly where it sat for the small machine (Lemma 14.1). The definition doesn't ask the machine to vouch for its own checker, and Section 11 says why nothing of arithmetic strength could. Nor does it ask the checker to be something a machine could run. Clause (b) wants a yes or no for every claim at every state, sound for the claim's meaning, and nothing in it says that answer has to be computable. Every instance in this book has a checker that's an actual program, so anything those machines certify could be listed by a program that runs them. But hand the definition a checker that's an oracle, one that knows which programs run forever, and it's still met, and the machine certifies claims no program could ever confirm. So "a Thiele-complete machine certifies exactly the claims a program could list" is true only with "and its checker is computable" added, and the definition doesn't require that.

One word needs pinning before the theorems, because this book uses "exact" for two different things and they don't agree. In clause (c), an *exact toll* means each of the three acts, CHECK and COMMIT included, costs exactly one, so the ledger counts the acts that make a claim. Back in Section 6, the *exact schedule* (`a2_and_aara_iff_exact`) was something else: one mark on the step that raises the flag and nothing on any other step. Measured against the flag alone, the small machine isn't exact in that second sense, since CHECK and COMMIT raise no flag and cost one each; measured the way (c) asks, it is. Neither is wrong. They answer different questions: does the ledger count the acts, or does it charge the flag and only the flag? When I say exact toll I mean the first, and exact schedule means the second.

Theorem 14.18 (What every Thiele-complete machine has). *Let a machine be Thiele-complete through an interface. (1) It obeys the toll and has a universal base, so it's weakly Thiele-complete; its base runs every two-counter program step for step, with halting matched. (2) Along any run, the ledger rises by exactly the number of CHECK, COMMIT and CERTIFY moves in it. (3) A run from a clean state that ends with the record up contains at least three of those moves and raises the ledger by at least 3. (4) Some CHECK fails at some loaded state, on a claim that is false there, and some run from some loaded state raises the record. (5) On a run from a clean state, the move that raises the record is a CERTIFY. (6) Every move costs 0 or 1, and some move costs 0.*

Proof. (1) A raise of the record isn't made by a base move, by (a), so it's a record move, which costs one by (c). Running a program on the base is induction on its steps, each matched by (a). (2) is (c), added up along the run. (3) Clause (b) finds a CHECK, a COMMIT and a CERTIFY at three different places, and (2) counts them. (4) is (d): the three moves certify exactly where the claim holds, the claim fails at some loaded state, and the check can't pass there because (b) says a passing check means the claim holds; at the loaded state where it holds, the three moves raise the record. (5) Clause (b) puts the record down right before a CERTIFY and up right after it, and the record never comes down, so that CERTIFY is the first and only move where it rises. (6) Every move is a base move or a record move, and (a) supplies base moves. $\square$

Checked as `thiele_complete_is_weak`, `base_runs_every_program`, `base_halting_correspondence`, `ledger_counts_record_moves`, `certificate_costs_three`, `check_can_fail`, `some_run_certifies`, `only_certify_raises`, `complete_costs_at_most_one` and `complete_has_free_move` (ThieleComplete.v).

The last line of (6) is the one that kills the clock.

Theorem 14.19 (A clock is a Thiele machine and isn't Thiele-complete). *Take the two-counter machine, let every move cost 1, and take as the record any yes/no question about its configuration whatsoever; or take a flag that switches on, and stays on, the first time any chosen test of the configuration says yes. Either way the machine obeys the toll and has a universal base, so it's weakly Thiele-complete, and it is not Thiele-complete. More generally, no machine whose every move costs at least 1 is Thiele-complete.*

Proof. Every move costs one, so any raise costs at least one: the toll holds for any record. The machine's own moves are the two-counter instructions, which gives the universal base. But a Thiele-complete machine has a free move, by Theorem 14.18(6), and this one has none. $\square$

Checked as `clock_weakly_thiele_complete`, `clock_not_thiele_complete`, `latch_clock_weakly_thiele_complete` and `latch_clock_not_thiele_complete` (ThieleComplete.v).

That's short, and it's short because clause (c) does exactly the job it was written for: it says computing is free, and a clock charges for computing. The other clauses earn their keep too, and that's worth checking, because a definition with a clause nothing needs is a definition hiding something.

Theorem 14.20 (Each clause does work). *(1) Take a free two-counter base with a ledger beside it, and one extra move TICK that costs 1 and raises a flag. It obeys the toll, and read with TICK as CERTIFY it meets the universal base clause and the exact toll clause; it is not*

Thiele-complete, and more generally no machine with a move that raises the record from every state is. (2) Take a free two-counter base and a record that never rises. It obeys the toll and meets clauses (a), (b) and (c); it is not Thiele-complete, and more generally no machine whose record never rises is.

Proof. (1) From a clean state, TICK alone raises the record. Clause (b) would need a CHECK, a COMMIT and a CERTIFY inside that one-move run, and one move can't be three. (2) Clause (d) needs a run from some loaded state that raises the record, and none does. $\square$

Checked as `paid_latch_weakly_thiele_complete`, `paid_latch_meets_base_and_toll`, `paid_latch_not_thiele_complete`, `one_move_record_excluded`, `silent_weakly_thiele_complete`, `silent_meets_base_record_toll`, `silent_not_thiele_complete` and `never_certifies_excluded` (ThieleComplete.v).

The paid latch is the receipt for nothing from the opening chapter, wearing a universal base: honest about the price, silent about what was bought. The definition throws it out.

Now the other direction. A definition nobody meets is as useless as one everybody meets.

Theorem 14.21 (The small machine, the generic one and the sorted one are Thiele-complete). *(1) This chapter's machine is Thiele-complete, read with INC and DEC as base moves, CHECK $p\ c$ and COMMIT $p\ c$ as record moves on the claim " p holds of c ," CERTIFY as CERTIFY, the window as (line, A , B), "unchanged" as "same value and same version of c ," clean states as clean starts and the ledger as μ . (2) So is the same machine over any property language with an exact equality test and a checker proved to mean what it says, as long as some property is true of one number and false of another. (3) In particular the machine whose property is "this counter, read as a list, is sorted" is Thiele-complete. (4) Every Thiele-complete machine is a certification system (Definition 3.1) and obeys the universal floor; for this chapter's machine, that system agrees step for step and cost for cost with the one of Theorem 14.6, and for the generic machine with the generic machine's own.*

Proof. (1) Clause (a) is Theorem 14.2 together with the step line: INC and DEC never touch the flag, and the flag never comes down. Clause (b) is Theorem 14.9 with checker soundness (Lemma 14.1), and a meaning about c keeps holding while c 's value doesn't move. Clause (c) is the price table. Clause (d) is Theorem 14.17 with the claim " A is zero," which holds at $(0, b)$ and fails at $(5, b)$. (2) and (3) are the same argument with the property language left open, and the sorted claim holds of the list numbered 18, which is $[1, 2]$, and fails of 20, which is $[2, 1]$. (4) is (1) of Theorem 14.18 and Section 3's floor. $\square$

Checked as `earned_core_thiele_complete`, `earned_generic_thiele_complete` and `sorted_machine_thiele_complete` (ThieleComplete.v), and `thiele_complete_floor`, `earned_complete_agrees` and `generic_complete_agrees` (EarnedGenericLinks.v).

And here's the thing every machine that meets the definition can't help doing. It's Section 7's shape again, only this time it isn't about one machine. It's about all of them.

Theorem 14.22 (Every Thiele-complete machine hides its record and its ledger from its window). *Let a machine be Thiele-complete through an interface, and let its window be the two-counter configuration its universal base shows. (1) For every state s there's a loaded state and one free base move from it that lands on a state with s 's window, with the record down and the ledger unchanged. (2) There's a loaded state and two runs from it that end with the same window, one with the record up and one with it down; and likewise two runs from one loaded state that end with the same window and ledgers at least 3 apart. (3) So no function of the window gives the record at the end of every run from every loaded state, and no function of the window gives the ledger. (4) The same holds of the certification system built from the machine.*

Proof. Theorem 14.18(4) gives a loaded state and a run from it that raises the record, and (3) says that run paid at least 3. Say it ends with window (line ℓ , x , y). Two-counter instructions can steer any configuration to any other: empty A , fill it to $x + 1$, empty B , fill it to y , then one DEC of A that jumps to line ℓ . Those are base moves, so from the same loaded state they reach the same window, cost nothing, and leave the record down, by (a) and (c). That's two runs, one window, two records and two ledgers, which is (2). A function of the window would have to give both answers on that one window, which is (3). For (1), load $(x + 1, y)$ and take just the last DEC. (4) is (3) read through the certification system, whose runs are the machine's. $\square$

Checked as `complete_every_window_printed`, `complete_hides_record`, `complete_hides_ledger`, `complete_no_record_oracle`, `complete_no_ledger_oracle` and `thiele_complete_hides` (ThieleCompleteWindow.v), and `complete_cs_window_blind` (EarnedGenericLinks.v).

Notice how little that took, and say it out loud so nobody mistakes it for more. It's the fibre fact from Section 7 plus one observation: a universal base can print any window for free. Which is the point. The window here is everything a Turing machine would call the machine, the line and the counters. On a Thiele-complete machine, that's never enough to tell you what was established or what it cost, and it can't be, because computing is free and can't touch the record. A richer window, one that shows the flag, is a different window, and nothing above says it fails.

14.13 One machine that runs them all

The small machine raises its flag only by checking, committing and certifying, and it pays for that. The next question is whether one machine can stand in for all of them,

the way Turing's universal machine stands in for every Turing machine. Here's what I mean by that. A *universal Thiele machine* is one fixed machine running one fixed program that runs any machine of a stated class as a guest, with the guest's record kept and the guest's toll collected by the host's own steps, not by a convention some program running on it has to keep.

And I want it the hard way. The easy way is to give the host a move that means "take one step of the guest," and then "it runs any guest" is true because I wrote it that way. There's a host like that later in this section, and it's useful as a picture, but it isn't Turing's kind of universal. Turing's universal machine has no instruction for "do what the machine on the tape would do." It has the same kind of instructions every other Turing machine has, and one fixed table, and the table does the reading and the deciding and the carrying out. That's the bar here: one fixed program, written in the small machine's own six kinds of instruction, with no step of the guest built into anything.

Turing's own explanation is worth having in your head, because the program below copies its shape. In the 1936 paper the universal machine's tape starts with the machine to imitate, written out as a string of letters, its standard description, and after it the imitated machine's complete configurations, one after another: where its head is, what state it's in, what's on its tape. To take one step, the universal machine finds the last configuration, looks through the description line by line, comparing, until it finds the line for that state and that symbol, and then writes out the next configuration the way that line says. Then it goes back and does it again. It leaves marks on squares so it doesn't lose its place. Every bit of that is ordinary instructions on an ordinary tape, and nothing in its table knows any particular machine.

Turing's is the first universal program, and the first one people actually used came in through the side door. John McCarthy, writing up his new language Lisp around 1960, wanted to show it was neater than Turing machines, so he wrote a universal function for it, *eval*, the language describing how to evaluate itself, and showed it was shorter and easier to follow than a universal Turing machine. In his own history of Lisp, that was for the paper, a thing to read; the notation for writing programs as data was made up for the paper with no thought of using it in practice. Then S. R. Russell, Steve Russell, noticed that eval could serve as an interpreter and coded it by hand, and there was a running language with an interpreter that nobody had planned. McCarthy adds that the surprise froze some decisions he'd made "rather lightheartedly" for the paper. So the trick has a habit: written to be read, it ends up being run. The program below goes the other way round, on purpose. It's written to run, and the proofs check that what runs is what the page says.

The host. The program runs on a machine with the small machine's instructions and the small machine's prices: INC, DEC, HALT for free, and CHECK, COMMIT, CERTIFY for one each, with the same table of sixteen facts, the same channel, trap,

ledger and flag. One change: instead of two counters it has a counter for every whole number, each with its own version. No new kind of instruction, no new price.

Definition 14.8 (The host machine). The *host machine* is the machine of Definition 14.3 with one change: it has a counter, with its version, for every whole number r (a *register*), and $\text{INC } r$, $\text{DEC } r$, $\text{CHECK } p \ r$ and $\text{COMMIT } p \ r$ name one register each. Its property language has one property, *Slot*. Write $\langle m, n \rangle$ for the number $(2n + 1) \cdot 2^m$; every positive number is $\langle m, n \rangle$ for exactly one pair, and 0 is no pair. Number the small machine's properties $\text{Zero} \mapsto 0$, $\text{Even} \mapsto 1$, $\text{Ge}(n) \mapsto n + 2$. The checker of *Slot* says yes at a value x exactly when $x = \langle m, v \rangle$ and the small machine's checker says yes to the property numbered m at v ; its meaning is the same with *holds* in place of the checker.

Defined as the machine of `EarnedMulti.v`, with `pair`, `unpair`, `pcode`, `heval` and `holds` (`UniversalCodes.v`).

Be clear about the one place the guest's machine shows up in the host. The host's single property knows how to read the small machine's three-word property language: "this number is the pair (property, value) and the property holds of the value." It doesn't know the small machine's step. That's the counter version of Turing's universal machine knowing the alphabet its guests write in.

Two facts that force the design. Why more counters? Because of two facts, and the program looks the way it does because of them.

The first is about reading. On this machine the only way to find out what's in a counter is to take pebbles out (*DEC*) and put them back (*INC*), and both of those are writes, so the counter's version goes up. A fact is about one version. So a counter the program reads can never hold a live fact, and the guest's counters are exactly the ones the program has to read, to decide which way a *DEC* jumps and to check a property. The host can't keep the guest's facts on its copies of the guest's counters.

The second says that even if reading were free, a fixed program couldn't. It's the pigeonhole.

Theorem 14.23 (No host fact on an exact copy of the guest's counter). *Take any host property language with an exact equality test and any checker, a finite list Q of host properties, and any translation of the small machine's properties into members of Q whose host check passes whenever the guest's check passes on the same number. Then there are $n \neq m$, with $\text{Ge}(n)$ and $\text{Ge}(m)$ translated to the same host property, such that the guest program $\text{CHECK } \text{Ge}(n) \ A$; $\text{COMMIT } \text{Ge}(m) \ A$ traps within two steps from every start, while on every host state with the trap down and room in its table, and some register holding a number at least n , the translated *CHECK* and *COMMIT* on that register both pass and put the committed claim in the channel.*

Proof. A fixed program mentions finitely many properties, so Q is finite, and the translation sends the infinitely many $Ge(0), Ge(1), \dots$ into it: two of them land on the same member. The guest's COMMIT asks for $Ge(m)$, and its table can only hold $Ge(n)$, so it traps (or its CHECK already trapped). The host sees one property twice: its CHECK passes, since the guest's would, and its COMMIT finds exactly what that CHECK wrote. $\square$

Checked as `pigeonhole_collision` and `no_exact_copy_host` (UniversalNoCopy.v), with the same statement on the host's certification system as `multi_cs_no_exact_copy` (UniversalInterpreterLinks.v).

So the host keeps the guest's facts somewhere else: in *slots*, extra registers used for nothing but holding facts, one per checked fact, each holding the pair (property number, value) that the guest checked. The pigeonhole doesn't bite a slot, because the property lives in the slot's number, not in the instruction: CHECK Slot on a slot asks about whatever pair is sitting there, and there are infinitely many pairs. A slot is written only when the program is moving a fact into it. And when the guest writes the counter a fact was about, the program gives every slot about that counter a nudge, one INC and one DEC, which leaves the number alone and pushes the version up, so the host's facts go stale at exactly the moment the guest's do.

The program. Here's U the way Turing would describe it, one register at a time and then one job at a time. Registers 0 and 1 hold the guest's A and B . Register 2 holds the guest's whole program as one number: each instruction gets a number (an opcode paired with its operands, INC as 0, DEC as 1, HALT as 2, CHECK as 3, COMMIT as 4, CERTIFY as 5), and the list of them is packed with $\langle \cdot, \cdot \rangle$, the empty list as 0. Register 3 is the guest's line. Registers 4 to 13 are scratch. Registers 14 and 15 count how many CHECKs on A and on B have passed. Registers 16 to 47 are *mirrors*, sixteen for each guest counter, and registers 48 to 79 are the slots, sixteen for each guest counter, because the guest's table holds sixteen. Register 80 is a dead end that's never written.

Each guest step goes like this. At line 1 the program copies the program number and the guest's line into scratch, putting the originals back as it goes, because copying a counter on this machine means emptying it into two others and refilling it. If the guest's line is 0, or past the end of the program, it stops. Otherwise it peels instructions off the front of the copy, one per line, until it holds the one at the guest's line, and splits that into opcode and operands. Then a chain of DECs on the opcode sends it to one of six handlers.

- INC c : INC the guest's counter c , nudge every slot about c and zero its mirror, add one to the guest's line, back to line 1.
- DEC $c\ j$: DEC the guest's counter c . If it was 0, add one to the guest's line. If it wasn't, nudge every slot about c , zero its mirrors, and set the guest's line to j . Back to line 1.

- HALT: stop.
- CHECK $p\ c$: let k be the count of passed CHECKs on c . If k is below 16, build $\langle \text{number of } p, \text{value of } c \rangle$ in slot k of c , run CHECK Slot on that slot, set its mirror to the number of p plus one, add one to the count and to the guest's line, and go back to line 1. If k is 16, run CHECK Slot on the dead end, which holds 0, which is no pair, so it traps, just as the guest's seventeenth CHECK does.
- COMMIT $p\ c$: look through the sixteen mirrors of c for one equal to the number of p plus one. At the first match, run COMMIT Slot on that slot. With no match, run COMMIT Slot on the dead end, which traps.
- CERTIFY: run CERTIFY, add one to the guest's line, back to line 1.

Why the mirrors? COMMIT has to find the slot holding the claim, and it can't read the slot to find out, because reading would make the fact stale. So beside every slot sits a mirror the program is allowed to read: the property's number plus one while the slot's fact is live, 0 once the guest has written the counter. If a guest CHECK fails, the host's CHECK on the slot fails too, since the slot holds exactly the pair the guest checked, and both machines trap.

Count what costs anything. Fetching, unpacking, copying, nudging, branching: all INC and DEC, all free, the way computing is free on the small machine. The only paid instructions are the CHECK, COMMIT and CERTIFY sites, and each guest CHECK, COMMIT or CERTIFY makes the program run exactly one of them. So the host's ledger lands on the guest's, mark for mark. And when the guest's flag goes up, the host's goes up by the host's own CERTIFY, standing on the host's own COMMIT and CHECK on a slot. It isn't copying a bit. It runs the same chain on its own registers.

Definition 14.9 (The program U , and loading a guest). U is the fixed list of 3728 host instructions described above, starting at line 1 and mentioning registers 0 to 80; exactly 69 of them cost anything. For a guest program P and a start (x, y) , the *loaded host* $\text{hload}(P, x, y)$ has register 0 at x , register 1 at y , register 2 at the number of P , register 3 at 1 and every other register at 0, every version 0, line 1, an empty table and channel, the trap down, ledger 0 and flag down. Write $\text{grun}(m)$ for m program steps of P from $\text{start}(x, y)$, and $\text{hrun}(n)$ for n program steps of U from $\text{hload}(P, x, y)$.

Defined as $U, \text{hload}, \text{prog_code}, \text{grun}$ and hrun (UniversalLayout.v, UniversalSim.v, UniversalRun.v), with the 69 paid sites listed exactly by U_paid_sites (UniversalLayout.v).

Theorem 14.24 (U keeps pace with its guest). *For every guest program P , start (x, y) and number of guest steps m , there's a number of host steps n such that one of two things holds. Either the host is back at line 1 with $m \leq n$ and related to the guest: registers 0 and 1 hold*

the guest's A and B, register 3 holds the guest's line, the guest's trap is down, the two ledgers are equal, the two flags are equal, and the host's table and channel hold, fact for fact, slot facts matching the guest's facts and channel. Or both have stopped, with the same counters, the same trap, the same ledger, the same flag, and matching tables and channels. And from any related pair where the guest hasn't stopped, one guest step is matched by at least one host step that lands related again, or with both stopped.

Proof. The one-step part is a case for each thing the guest can do at its line: stop; INC; DEC, taken or not; CHECK, passing, failing, or with a full table; COMMIT, found or not; CERTIFY, passing or not. For each, the matching stretch of U is proved to land where the case says. The relation carries the bookkeeping that makes the next case go through: each count register equals the number of slots in use for its counter, each slot in use holds (property number, checked value), a slot's fact is current exactly when the matching guest fact is, a live slot's mirror is the property's number plus one and a stale one's is 0, and unused slots, unused mirrors and the dead end are 0. A freshly loaded host is related to the guest's start. The run statement is induction on m . $\square$

Checked as `hload_rel` and `U_step` (UniversalSim.v), and `U_simulation` (UniversalRun.v).

Theorem 14.25 (Halting and answers match). *For every P , x and y , the guest's run of P from $\text{start}(x, y)$ halts if and only if U 's run from $\text{hload}(P, x, y)$ halts. And whenever the guest has halted at step m and U at step n , the host's registers 0 and 1 hold the guest's A and B, and the two traps, the two ledgers and the two flags are equal.*

Proof. Theorem 14.24: a halted guest is matched by a stopped host, and a host that has stopped can't be back at line 1 with the guest still running, because line 1 isn't a stop. A halted machine stays halted, so the two halting points carry the same values. $\square$

Checked as `universal_halting` and `universal_output` (UniversalRun.v).

Theorem 14.26 (The host's flag is the guest's, and its ledger is exact). *(1) The guest's flag is up at some step of its run if and only if the host's flag is up at some step of U 's run. (2) For every m there's an n at which the two are related or both stopped, with $\mu(\text{hrun}(n)) = \mu(\text{grun}(m))$. And for every n there's an m with $\mu(\text{grun}(m)) \leq \mu(\text{hrun}(n)) \leq \mu(\text{grun}(m + 1))$: in between matching points, the host's ledger never gets ahead of the guest's next step.*

Proof. (1) Flags are equal at matching points and never come down. (2) Ledgers are equal at matching points; inside the stretch of U that runs one guest instruction, at most the one paid site of that instruction runs, so the host's ledger sits between the guest's before and after it. $\square$

Checked as `universal_flag_iff` and `universal_ledger_exact` (Universal-Run.v).

That's a third sense of "exact," and it's the plainest of the three: two ledgers, one number.

Theorem 14.27 (The host earns its own flag). *If the host's flag is up after n steps of U from $\text{hload}(P, x, y)$, then the instructions U executed contain, in order, a passing CHECK Slot on a slot, then a passing COMMIT Slot on the same slot, with the slot untouched and its version unchanged in between, then the passing CERTIFY that raised the flag; the slot held $\langle \text{number of } p, v \rangle$ at the CHECK, with $\text{holds}(p, v)$. And the guest's own run from $\text{start}(x, y)$ contains a passing CHECK $p\ c$, then, with c untouched and its version unchanged, a passing COMMIT $p\ c$, then a passing CERTIFY, with c holding v at its CHECK.*

Proof. By Theorem 14.26 the guest's flag rises too; find the guest step where it rises, a passing CERTIFY, and the related host point just before it. The guest's channel there names a fact, and the relation says the host's channel names the matching slot fact. The host's own flag rises at its own first passing CERTIFY, and nothing paid runs between the related point and that CERTIFY, so the host's channel hasn't moved: it names that slot fact. The host machine's version of Theorem 14.9 gives the host's CHECK and COMMIT on that slot. The record kept with the slot says what pair it held and when, and the checker's meaning gives $\text{holds}(p, v)$. The same theorem on the guest's run, up to the step where its flag rose, gives the guest's chain on the same p and c , and the record says c held v at its CHECK. $\square$

Checked as `universal_earned` (UniversalRun.v).

Theorem 14.28 (The host is Thiele-complete, and U 's halting is undecidable). *(1) The host machine, with Slot as its property and one move per instruction, is Thiele-complete (Definition 14.7), and U 's run from $\text{hload}(P, x, y)$ is a run of it. (2) Read as a certification system it obeys the universal floor; a run of U that raises the host's flag costs at least 3 there, and at least 1 on the system built from its Thiele-completeness, which agrees with it step for step and cost for cost. (3) U halts from $\text{hload}(P, x, y)$ exactly when P halts from $\text{start}(x, y)$, so the question "does U halt from this loaded host?" is undecidable in the library's sense of Theorem 14.4.*

Proof. (1) The base is registers 0 and 1, run by INC and DEC; the claims are "Slot holds of register r "; the rest is the host machine's own versions of the small machine's theorems, and for non-vacuity, register 0 loaded with 1, which is $\langle 0, 0 \rangle$, the pair of Zero and 0, against register 0 loaded with 0, which is no pair. (2) is Theorem 14.21(4) and Theorem 14.27. (3) is Theorem 14.25, and the identity map reduces the small machine's halting question, undecidable by Theorem 14.4, to this one. $\square$

Checked as `universal_thiele_complete` and `U_run_on_host_machine` (`UniversalRun.v`), and `multi_cs_floor`, `interp_U_certified_floor`, `interp_U_complete_floor`, `interp_complete_agrees`, `interp_halting_iff` and `interp_halting_undecidable` (`UniversalInterpreterLinks.v`).

Put it together, plainly. One fixed program, 3728 instructions of the small machine's six kinds, with no guest step built into anything, runs every program of the small machine from every start. It halts exactly when its guest halts, with the guest's answer in its first two registers. Its flag goes up exactly when the guest's would, by its own check, its own commit and its own certificate. Its ledger is the guest's, mark for mark, wherever the two line up. And the machine it runs on is Thiele-complete. Since the small machine runs every two-counter program, *U* runs every computation. That's the universal Thiele machine of this book, for the small machine's programs; the next subsection carries it to every computably presented Thiele machine.

The fine print, before you take that anywhere. The guests start clean, from `start(x, y)`; the whole-run theorems are about loaded hosts, and only the one-step theorem is about any related pair. The host's property reads the small machine's three-word language, so a guest with a different property language needs a different host property; the host of the next subsection has one, a single property that reads a universal claim language. Sixteen slots per counter is the guest's table size, not a choice of the host's. And it's slow: every guest step unpacks the program number from the front, on a machine where copying a number means moving it a pebble at a time.

Enforcement in the step is the reference monitor's move from the opening chapter, Anderson's 1972 move: one mechanism everything has to pass through. What's put together here is a priced, earned record under that move, carried to every guest by one fixed program. The move itself is Anderson's. Kevin Hamlen, Greg Morrisett and Fred Schneider (2006) sorted out which policies mechanisms like that can enforce at all, and their paper is the standard place to see why enforcing a property of every run escapes the Rice-style walls that deciding one runs into.

Why make such a fuss about proofs that walk every case? Because the first universal machine had bugs. It's the machine in Turing's 1936 paper, and nobody gets to call Turing careless. In 1947 Emil Post, in an appendix to a paper on a problem of Thue, pointed out that the construction leaned on a convention it hadn't earned, and he patched a small defect in how the universal machine writes out each complete configuration. Around the same time Donald Davies, who joined the National Physical Laboratory in 1947, where Turing was designing a computer, worked through the universal machine's tables and found more mistakes. (It's the same Davies who later named the *packet* in packet switching.) His corrections were finally printed in Jack Copeland's collection *The Essential Turing* in 2004. They may be some of the first bugs ever written, in a program for a computer that didn't exist yet. That's a lovely claim and a hard one to check, so take it as a lead and not a fact. What it changes here: a

universal machine is exactly the kind of construction where a careful person leaves a hole, U has 3728 lines for a hole to hide in, and its theorems are checked by Coq one case at a time, which is the kind of check that catches Davies’s kind of slip. It doesn’t catch a definition that says the wrong thing. That’s why the definitions are on the page, for you.

14.14 Every computably presented Thiele machine, on one host

U runs the small machine’s programs, and through them every computation. But a Thiele machine from somewhere else has its own states, its own moves, its own prices and its own reading, and running it through the two-counter encoding keeps its computation and nothing else. So the question Turing’s machine raises for this book is the bigger one. Take any Thiele machine whose workings can be computed. Is there one fixed program that runs every such machine as a guest, keeping its halting, raising its own flag exactly when the guest’s reading would rise, on an earned chain of its own, and keeping a ledger equal to the guest’s prices and not to the host’s flat ones? There is, within limits I’ll list, and here it is.

First the class, said plainly. A *computably presented* Thiele machine is a certification system in Section 3’s sense (states, moves, a step, a price for each move, a yes/no reading) together with a *driver* that picks the next move from the current state or says stop, and a way of writing states and moves as whole numbers. “Computably” means four little programs, written in the plainest language of computable functions there is, the one built from zero, plus one, picking an argument, composing, recursion and search: one computes the driver, one the step, one the price, and one the reading, and the reading’s program answers yes or no on every state there is. Run from a start s_0 , the driver’s moves give one run. Its *ledger* after n moves is the sum of their prices, and its *latch* after n moves says whether the reading has been yes at any state so far.

The host needs one thing the small machine doesn’t have: a paid move that certifies nothing. Without it the host can only charge one per CHECK, COMMIT or CERTIFY, and a guest whose moves cost seven can’t be billed seven. So the host machine gets a seventh instruction, PAY: it costs one, goes to the next line, and changes nothing else; it never raises the flag. Read through the definition of Thiele-complete, PAY is a check of a claim that never holds, so the host stays Thiele-complete with it (`pu_universal_thiele_complete`, `UniversalPRun.v`), and its floor of three for a raised flag still holds with PAY present (`pu_multi_certified_run_min_cost`, `EarnedMultiPriced.v`).

The program is U with one more handler: a seventh branch on the guest’s opcode, for PAY, that runs the host’s own PAY once. Call it U_P ; it’s 3735 instructions, with 70 paid sites, the 69 of U plus the PAY. The guest it runs is a program of the small machine with PAY added, and that guest is where the presented machine goes. Its four programs are compiled into two counters, by the packing Minsky used, with the vendored library’s compiler from computable functions to counter machines doing the hard part; the guest keeps the presented machine’s state coded in its counter B , takes

one of its moves by running the compiled step, pays the move's price in PAYs, and, when the compiled reading says yes, checks, commits and certifies. The host's one property is fixed too: a claim is either one of the small machine's three words about a number, or "run reading program r on the coded state v ," and the host's checker evaluates that with one fixed routine for every presented machine there is. So nothing in U_P knows any particular presented machine. The machine arrives as data, the way Turing's guests arrived on the tape.

Theorem 14.29 (One host runs every computably presented Thiele machine). *Let M be a computably presented Thiele machine and s_0 a start state. Load U_P with the guest compiled from M 's presentation and s_0 coded in. Then: (1) U_P halts exactly when M 's run stops. (2) For every n such that M hasn't stopped before move n , there's a host time t at which the host's register for the guest's counter B decodes to M 's state after n moves, the host's ledger is M 's ledger after n moves plus a surcharge, and the host's flag is M 's latch after n moves; and if M stops at n , such a t has U_P halted as well. (3) The host's flag is up at some time exactly when M 's reading is yes at some state of its run. (4) If the host's flag is up at time t , the host's own instructions up to t contain a passing CHECK on a slot, then a passing COMMIT on the same slot with the slot untouched in between, then the CERTIFY that raised the flag; and the slot held the claim "reading program r says yes of v " for M 's own reading program r , that claim holds, and v decodes to a state of M 's run where the reading is yes.*

Proof. Two layers, each proved on its own. The bottom layer is the compiler: for every presented machine, its guest program's run passes through points that code M 's states in order, with the guest's ledger equal to M 's ledger plus the surcharge and the guest's flag equal to M 's latch, and the guest's own flag stands on an earned chain whose claim is M 's reading program applied to the coded state where the reading first said yes. The top layer is U_P , proved the way U was in the previous theorems, with the PAY case added: it keeps pace with any guest of this kind, halting and flag matched, ledger equal at matching points, and its flag earned on a slot. Compose the two. The earned part takes one more step: the host's chain gives a guest CHECK that passed, and the guest's earned chain is unique enough to pin that CHECK to the compiler's, so its claim is M 's reading at the first state where it said yes. $\square$

Checked as `presented_universal` (PresentedUniversal.v), whose five parts are also stated separately as `presented_universal_halting`, `presented_universal_points`, `presented_universal_halt_point`, `presented_universal_flag_iff` and `presented_universal_earned`; the host layer is `pu_universal_halting`, `pu_universal_flag_iff`, `pu_universal_ledger_exact` and `pu_universal_earned` (UniversalPRun.v).

Now the surcharge, because that's where the ledger stops being exact, and it can't be helped. The host raises its flag the only way the host machine allows, by a CHECK, a COMMIT and a CERTIFY, and those cost three. If M 's first move that turns its reading

to yes costs at least three, the host pays the difference in PAYs and nothing more is owed. If it costs less, the host can't match it.

Theorem 14.30 (Exact, within two, and no better). *Let M, s_0 and the loaded host be as in Theorem 14.29, and let M 's reading be no at s_0 . (1) For every n before M stops there's a host time at which the host's ledger lies between M 's ledger after n moves and that plus two, with the state decoded and the flag matched as before. (2) If moreover every move that first turns M 's reading to yes costs at least three, the surcharge is zero and the host's ledger equals M 's exactly at those times. (3) If M 's latch is up after n moves while its ledger is below three, then no run of the host machine from a clean start, under any program at all, ends with its flag equal to that latch and its ledger raised by exactly M 's ledger.*

Proof. (1) and (2): the guest pays M 's prices in PAYs, a unit at a time, except at the move where M 's reading first says yes. There it has to run its CHECK, COMMIT and CERTIFY, three units, and it makes up the rest of the move's price in PAYs, so it pays the larger of the price c and three. The price is at least one, because the move raised a reading (the toll), so the excess is at most two, and none when $c \geq 3$. (3): a raised flag on the host machine costs at least three from a clean start, whatever the program, and M 's ledger is less. $\square$

Checked as `presented_universal_surcharge_le_two`, `presented_universal_exact` and `presented_universal_no_exact_below_three`, with the corollary for U_p itself, `presented_universal_U_no_exact_below_three` (`PresentedUniversal.v`). If the reading is already yes at the start, the host has to raise its flag at once, and that costs three on top of a ledger of nothing.

And one machine run all the way, so the class isn't empty. Its states are whole numbers, its one move adds one and costs three, its driver always picks that move, and its reading is "at least three." The four programs are written out from the library's building blocks and proved to compute those four things. Loaded into U_p from 0, the host never halts, its flag goes up, at the host time matching move n its ledger is exactly $3n$ and its flag is up exactly when $n \geq 3$, and the raised flag stands on a host chain whose slot claims the machine's own reading of a state at least three (`pu_demo_computably_presented`, `pu_demo_never_halts`, `pu_demo_flag_rises`, `pu_demo_exact` and `pu_demo_earned`, `PresentedDemo.v`).

Now the limits, each of which is in the statement, said plainly.

- **Readings that are decided.** The reading's program has to answer yes or no on every state. A reading a program could only confirm, waiting forever on the no cases, isn't covered, and neither is a reading no program computes at all.
- **One run per start.** The driver picks the move, so the guest is deterministic. A machine whose next move is somebody's free choice gets in only with that

choice written into its driver.

- **The record is the latch.** The host's flag is up once M 's reading has been yes at any state of the run, which is exactly the latch of Section 10. A reading that goes yes and then back to no is carried as "it was yes once."
- **The ledger is exact up to the surcharge.** At most two, zero when the first raising move costs three or more, and no program on this host can do better for a raise that costs less than three.
- **Prices in whole units, paid one at a time.** A move that costs a million is a million PAYs. Nothing here is fast, and the compiler's packing is Minsky's, so it's slower still.
- **Proved, not run.** The demonstration machine's numbers come from theorems, not from running U_P on it; the compiled programs are far too big to step through by hand, and some of the library's pieces don't compute inside the checker at all.

14.15 A simpler host, with the guest step built in

Before U there's a much simpler host, and I keep it in the book because it's easy to read and it shows the copy rule in one line. Its universality is by construction: it has a move that runs one guest step by calling the small machine's own step. So it isn't the universal Thiele machine of this book; U is. Read it as a picture of what U does the long way, and as a host that starts from any guest state at all, not just a clean start.

It carries a guest inside its own state: the guest's program as data, the guest's counters, its table, its ledger, its flag. The host has one extra move, run one guest step. In that one move it looks at the guest's flag, takes the step, looks again, and if the flag went from down to up, the host's own copy of that flag goes up in the same move, and the host pays for it.

Definition 14.10 (The host). A *guest* is a program P of this chapter's machine together with a state g of it (Definition 14.3). A *host state* h has six parts: its own core $\text{core}(h)$, its ledger $\mu(h)$, its own flag $\text{own}(h)$, the guest program $P(h)$ held as a list of instructions, the guest state $g(h)$, and the *mirror* $m(h)$, a yes/no copy of the guest's flag. The host has two kinds of move. OWN i runs the instruction i on the host's own core, costs what this chapter's machine charges for i , and raises own exactly when that machine's flag would rise. GSTEP b costs b . If the host's core is trapped it does nothing else. Otherwise it moves the host to its next line and, if the guest's next instruction costs at most b , runs that instruction on $g(h)$; in the same move the mirror becomes $m(h) \vee (\neg \text{cert}(g(h)) \wedge \text{cert}(g'))$, where g' is the new guest state. The host's reading is $\text{own}(h) \vee m(h)$. Its stored program is $U_0 = [\text{GSTEP } 1; \text{OWN } (\text{INC } A); \text{OWN } (\text{DEC } A \ 1)]$: one guest step, then two moves that bring the host back to its first line. (The Coq file calls it U ; I write U_0 here so

nobody mixes it up with the universal program U .)

Every instruction of the small machine costs at most one, so GSTEP 1 runs every guest instruction there is. Why charge the budget b , and not the cost of whatever instruction got fetched? Because charging the budget keeps the price a function of the move alone, which is what the certification system of Part II asks for: you read the price of a host move off the move, the way you read the small machine's prices off its table, without peeking inside the guest. The bill for that is easy to see. A host that steps a free guest instruction, an INC say, still pays one for the move.

So the host's ledger is closer to a clock than to a receipt for its guests: one unit per guest step, whatever the step was. It never falls behind the guest's own bill (part (3) of Theorem 14.33 below), and it matches that bill only when every guest step happens to cost one. Could a cleverer host charge exactly? Not with a price read off the move alone. GSTEP 1 is the same move whether the guest's next instruction is a free INC or a paid CHECK, so one price can't match both. A host that charges exactly has to be built the other way, with no guest step to put a price on: that's U , which pays only on the record moves it runs itself, so its ledger equals its guest's (Theorem 14.26).

And one thing about the definition before the theorems, so you know which of them carry weight. GSTEP runs the guest's instruction by calling the small machine's own step on the guest's state: the guest's step is a primitive of the host. So "the host runs any guest" (Theorem 14.31) is true by construction. The host was built to do exactly that, and the proof is mostly unfolding. The theorems with something in them come after it. The mirror always equals the guest's flag (Theorem 14.32). Nothing raises the mirror unless the guest itself passes a CERTIFY inside a guest step the host paid for (Theorem 14.34). No OWN move, in any host program at all, touches the guest's state or the mirror (the second sentence of that same theorem), and no host move of either kind rewrites the guest's program (`hexec_gprog`, `UniversalThiele.v`). And the mirror is tied to the guest's earned chain: if it's up, the guest's own run contains a passing CHECK, then a COMMIT of the same claim, then a CERTIFY (`host_mirror_earned`, `UniversalThiele.v`). So this host runs any guest by construction and keeps the guest's record by theorem, with a copy of the guest's flag rather than a flag it earns itself. U does both the long way: it runs the guest with its own instructions, and it earns its own flag.

Theorem 14.31 (The host runs any guest). *(1) If the host's core is not trapped, then after n moves GSTEP 1 the guest state is the guest program's own run of n steps from $g(h)$, the guest program is unchanged, and the host is still not trapped. (2) If the host's core is at line 1 and not trapped, then $3n$ steps of the program U_0 leave the guest exactly where n steps of its own program put it. (3) For every two-counter program M and inputs a, b : M halts from $(1, (a, b))$ exactly when the host program made of OWN moves of the compiled M halts, from a host whose core starts at (a, b) , whatever guest that host carries.*

Proof. (1) Induction on n : a host that isn't trapped stays untrapped through a guest step, and with budget 1 the guest step is the guest's own step. (2) One pass of U_0 is a guest step, then INC A and DEC A back to line 1, so it is (1) once per pass. (3) A host program made only of OWN moves is the small machine, core for core, and Theorem 14.3 does the rest. $\square$

Checked as `universal_simulation`, `universal_program_simulation` and `host_thiele_complete` (UniversalThiele.v).

Theorem 14.32 (The copy is the record). *Say a host state h agrees when $m(h) = \text{cert}(g(h))$. Every host move takes an agreeing state to an agreeing state, so every host trace and every stored host program does too, and a freshly loaded host agrees. In particular, if h agrees and is not trapped, the mirror after n moves GSTEP 1 is the guest's flag after n steps of its own program.*

Proof. One move at a time. An OWN move touches neither the mirror nor the guest. A trapped GSTEP touches neither. An untrapped one sets the mirror to the old mirror or the crossing, and the guest's flag is a latch (Theorem 14.5), so if the old mirror equalled the old flag, the new mirror equals the new flag. The last sentence is this together with Theorem 14.31. $\square$

Checked as `record_agreement_step`, `record_agreement`, `record_agreement_program`, `hload_agrees` and `simulated_record` (UniversalThiele.v).

Theorem 14.33 (The host's step collects the guest's toll). *(1) If h agrees, is not trapped, and the guest step that budget b allows takes the guest's flag from down to up, then the mirror is down before GSTEP b and up after it, the host's ledger rose by exactly b , and $b \geq 1$. (2) In the run of (1) of Theorem 14.31, if the guest's flag goes up between its steps n and $n + 1$, the mirror goes up between the host's moves n and $n + 1$, and that move costs 1. (3) For every host trace t , $\mu(g(\text{run}(t, h))) + \mu(h) \leq \mu(\text{run}(t, h)) + \mu(g(h))$: the host's ledger grows at least as fast as the guest's.*

Proof. (1) The guest's flag rises only on a passing CERTIFY (Theorem 14.5), which costs 1, and the budget let it run, so $b \geq 1$; the mirror was the guest's flag, so it was down, and the move sets it to the crossing. (2) is (1) at the matching move. (3) Induction on t : a guest step that runs costs the guest at most the budget the host paid, and every other move leaves the guest's ledger alone. $\square$

Checked as `toll_enforced_by_host_step`, `every_guest_crossing_is_a_host_crossing`, `host_pays_for_guest`, `host_cost_covers_guest` and `simulated_cost` (UniversalThiele.v).

Theorem 14.34 (No free host certification). *If the mirror is down at h and up after a host move i , then i is GSTEP b with $b \geq 1$, the host was not trapped, the guest's next instruction was CERTIFY, and the guest's flag went from down to up in that move. A trace made only of OWN moves leaves the mirror and the guest state exactly where they were. And the host's own flag rises only by an OWN CERTIFY that passes on the host's core.*

Proof. An OWN move copies the mirror unchanged, and a trapped GSTEP does too, so the rise is an untrapped GSTEP b whose guest step crossed; a crossing is a passing CERTIFY, which costs 1 and needed the budget. The own flag moves by this chapter's step line. $\square$

Checked as `no_free_host_certification_step`, `no_free_host_certification`, `no_free_host_certification_program`, `guestless_runs_leave_mirror` and `own_record_only_by_certify` (UniversalThiele.v).

Theorem 14.35 (The host is a Thiele machine). *For every host state and move, if the host's reading goes from no to yes, the move costs at least 1; the same holds for the mirror alone and for the own flag alone. So the host read through $\text{own} \vee m$, the host read through m alone, and the guest class are certification systems (Definition 3.1), and any host trace that raises the host's reading costs at least 1.*

Proof. The mirror's half is Theorem 14.34; the own flag's half is the small machine's toll on the host's core. The trace form is Section 3's universal floor. $\square$

Checked as `host_toll`, `host_own_toll` and `host_mirror_toll` (UniversalThiele.v), and `host_cs`, `host_mirror_cs`, `guest_cs` and `host_nfi` (UniversalThieleLinks.v).

Remark 14.1 (Two guests, run). Load the guest CHECK is zero A; COMMIT is zero A; CERTIFY; HALT with both counters at 0. After two moves GSTEP 1 the mirror is down; after three it is up, the guest's flag is up, the host's own flag is still down, and the host has paid 3. Under the program U_0 the mirror is down at host step 6 and up at step 7, with 3 paid. The guest CERTIFY; CERTIFY traps on its first step: after five moves GSTEP 1 the mirror is down, the guest is trapped, and the host has paid 5 for nothing.

Checked as `demo_certifies`, `demo_program_certifies` and `demo_forgery_fails` (UniversalThiele.v).

That copy can't be faked. The proof walks every host move there is, and the copy only rises in a guest step where the guest itself ran CERTIFY and passed. A host program that never touches the guest leaves the copy exactly where it was. The copy always equals the guest's flag, the host's ledger always grows at least as fast as the

guest's, and a three-instruction loop runs any small-machine program one guest step per pass. Here the mirror, the crossing test and the charge sit in the host's step rule, so Theorems 14.33 and 14.34 hold for every sequence of host moves, including ones that never run a guest. That's the simple version. Everything it gets from its built-in move, U gets from ordinary instructions, plus an exact ledger and a flag of its own.

14.16 The machine versus the program it runs

All three hosts in this chapter come down to one distinction, and it's worth saying on its own because people run the two halves together: *the machine doing the stepping* (what the machine is) and *the program it runs* (what runs on it). The toll of Section 3 is a property of the machine doing the stepping. A rule a program keeps holds as long as the program keeps it.

Here's the picture I keep for it. Play Monopoly on the counter of a real bank. The game has a bank of its own, with its own rules, and one of them is printed in the rules sheet: the Bank never goes broke, and if it runs out of money the Banker may issue more by writing on any ordinary paper. Inside the game that's fine; the players agreed to it. The real bank's teller, standing right there, has no opinion about any of it. Push a thousand in Monopoly money across her counter and her ledger doesn't move, because her rules are about her money, not the game's. The real bank is the machine doing the stepping. The game is a program it happens to be hosting. A rule that lives in the game holds as long as the players keep to it. A rule that lives in the bank holds whatever the players do.

(And since the game is on the table: its box told an origin story for decades, that an out-of-work salesman named Charles Darrow invented it during the Depression. The board is older. Elizabeth Magie patented The Landlord's Game in 1904, with a square track, railroads and a square that sends you to jail, and described it as a practical demonstration of the system of land-grabbing. Darrow sold his version to Parker Brothers in 1935, and the company bought Magie's patent for a few hundred dollars. A story told by the company selling the product is a lead to check, every time. It changes nothing about banks. It's a good reminder of who writes the story on the box.)

A fixed machine can enforce an invariant in its own transition table, and that includes a Turing machine. Give its tape a ledger track and a certification cell, and write a table that never sets the cell without bumping the ledger: that machine obeys the toll in its step relation. It's an instance of the abstract certification system, not a counterexample to it. What it doesn't do, built the ordinary way, is enforce the toll on the programs it runs. A universal machine whose table was written only to interpret carries the simulated certification bit as plain data on its tape. If the host is programmable, so that other code shares that tape with the guest, any of that code can flip the bit without paying, because the host's step has no opinion about it. So an ordinary programmable computer, as built, doesn't enforce the toll for its programs; it isn't that it can't. The

simpler host above enforces it by having its own step mirror the guest's bit and charge for its crossing (Theorems 14.32 and 14.34), and the universal programs U and U_P are fixed tables that go one better: they earn their own flag, by their own check, commit and certificate, exactly when the guest's would rise, and pay on their own ledger what the guest pays (Theorems 14.27 and 14.29). That's a statement about which reading a step rule prices. It isn't a claim that Turing machines can't price one.

Computing ran into this exact gap in hardware, and how it got out is worth the detour. In 1974 Gerald Popek and Robert Goldberg asked when a machine can run guests, a whole operating system inside another, efficiently and safely. Their answer is one sentence you could check against any instruction set. Call an instruction *sensitive* if it reads or changes the machine's own settings, the things the host has to keep for itself, and *privileged* if running it outside the host's own mode makes the machine stop and hand control to the host, which is called a trap. For the kind of machine they modelled: if every sensitive instruction is privileged, a host can be built. The guest runs at full speed, and whenever it touches something that matters, the machine itself stops it and lets the host decide. That's a host whose step has an opinion.

Then the obvious next question: did the machines people actually bought pass the test? The most common one didn't. In 2000 John Scott Robin and Cynthia Irvine went through every instruction of Intel's Pentium and found seventeen that are sensitive and not privileged; POPE, which loads the processor's flags, is the one usually pointed at. A guest could run them and the host would never hear about it. The fixes came two ways: VMware rewrote the guest's code on the fly so the troublesome instructions never ran as written, and in the middle of the 2000s Intel and AMD added hardware support so they'd trap. Does that change anything here? Not the theorems. Popek and Goldberg were after speed and isolation, not a record and a price, so their result isn't mine and mine isn't theirs. It changes the picture, though. The universal programs sit where they'd want a host to sit: the host's own steps collect each guest's toll and earn each guest's record again. And an industry learned the same lesson the expensive way: a rule binds the guest only if it lives in the step underneath.

The smallest checked version of "a ledger beside an ordinary machine" is a Turing machine configuration with a counter next to it, where each step adds one, plus one more if it writes a nonblank symbol. The counter doesn't disturb the machine.

Theorem 14.36 (A counter beside a Turing machine changes nothing). *Let a Turing machine with transition table δ step a configuration c to $\text{tm}_n(\delta, c)$ after at most n steps (stopping when δ has no entry). Pair each configuration with a counter, and step the pair by stepping the configuration and adding the step's charge. Then for every n , δ and pair (c, m) , the configuration part after at most n steps of the pair is $\text{tm}_n(\delta, c)$. In particular this holds from $(c, 0)$.*

Proof. By induction on n . For $n = 0$ both sides are c . For $n + 1$, if δ has no entry at

c , both runs stop and both sides are c . Otherwise the pair steps to (c', m') where c' is exactly the Turing machine's next configuration, whatever m' is, and the induction hypothesis at (c', m') finishes it. $\square$

Checked as `thiele_simulates_turing_gen` and `thiele_simulates_turing` (ProperSubsumption.v). The “Thiele” in those names is a local record, a Turing configuration and a number; it is not a Thiele machine in this book's sense, and the file says not to cite it as one simulating Turing machines. A counter beside a machine is the scorekeeper's tally from Section 3: honest about the machine, and kept by whoever holds the pencil.

One limit on all of this, so it isn't read as more. Certification is one event. Other kinds of observable a machine might carry would each need their own event, their own evidence and their own cost law, and nothing here supplies those.

14.17 Part III's results, run on this machine

Part III proved its counts and its collisions for any system with the named pieces, and it left a promise: that the machine of this part would be put through them. Here it is. The eight-state machine of Section 4 gets run by the small machine, price for price. Structural entitlement, the count of what narrowing a list of rivals costs, is proved for every Thiele-complete machine, with the small machine as a worked member, a larger search on the host that has a counter for every number, and a run that shows the count's one awkward hypothesis, the covering, can't be dropped. And the time a check saves gets an exact exchange rate, for the part of the machine that has no record. (The no-bare-verifier result is in Section 7, next to the collision it comes from.)

The eight-state machine, run by the small machine. Look at the small machine through a window that shows two things: its line number read modulo four, which I'll call its *slot*, and its flag. Call a state *live* when its trap latch is down, the fact “ $A \geq 0$ ” about A 's current version is in its table, and its channel holds a commitment, which is what CERTIFY needs in order to run.

Definition 14.11 (The eight-state machine, compiled). Stamp is the single instruction CERTIFY. Next-slot is the single instruction INC B . Jump to slot a , for a one of 0, 1, 2, 3, is the four instructions COMMIT ($A \geq 0$) A ; COMMIT ($A \geq 0$) A ; INC B ; DEC B a . The *setup* is the two instructions CHECK ($A \geq 0$) A ; COMMIT ($A \geq 0$) A .

The two COMMITs in a jump change nothing but the line and the channel, and each costs one, which is where the jump's price of two comes from. The INC makes B positive, so the DEC takes one off and jumps to line a for nothing, and B is back where it was.

Theorem 14.37 (The small machine runs the eight-state machine). (1) For every *live*

state s and every move m of the eight-state machine, running m 's instructions from s ends in a live state whose window is the eight-state machine's step of s 's window by m , and the ledger has risen by exactly m 's charge: one for stamp, two for jump, zero for next-slot. (2) The same holds for any list of moves: the window after the compiled list is the eight-state machine's run of the list, the state is live, and the ledger has risen by the sum of the charges. (3) So a run of compiled moves from a live state that takes the flag from down to up has raised the ledger by at least one. (4) From any start (a, b) , the setup ends in a live state with window (slot 3, flag down) and the ledger at 2; after it, any compiled list runs as on the eight-state machine from (slot 3, flag down), with the ledger at 2 plus the sum of the charges; and if the flag is up at the end, the whole run, setup included, contains the earned chain of Theorem 14.9.

Proof. (1) By cases on the move. On a live state CERTIFY runs, because a committed claim is on the channel; it raises the flag, moves the line on by one and charges one. INC B moves the line on by one and charges nothing, and it touches only B and B 's version, so the fact about A and the commitment stand and the state stays live. In a jump, each COMMIT finds its fact in the table and succeeds, charging one, and the INC and the DEC are free and end at line a , whose slot is a ; the flag isn't touched. (2) is (1) over the list, by induction. (3) The eight-state machine's own toll (Section 4) says a run of its moves that takes the flag from down to up is charged at least one, and (2) carries that charge across. (4) CHECK ($A \geq 0$) passes at every value of A and the COMMIT after it passes, which takes line 1 to line 3, slot 3, with two marks paid; then (2) applies. A raised flag was raised by a CERTIFY, and Theorem 14.9 gives the chain. $\square$

Checked as `frag_runs_finite_machine`, `frag_pays_finite_price`, `frag_runs_finite_trace`, `frag_certification_paid`, `frag_setup_live`, `frag_program_runs` and `frag_program_earned` (FragmentSmall.v), with the room argument for the eight-state machine itself as `frag_fin_toll` and `frag_cert_system` in the same file. Two reservations, said now. The setup's claim is true of every value, so its CHECK can't fail; it's there because the small machine won't CERTIFY without a checked, committed claim, not as a test. And the jump's price of two is two paid COMMITs, a choice a program makes that the machine allows and doesn't force. The window is a projection, too: the small machine's full state keeps far more than eight states' worth.

And the whole machine is not the piece.

Theorem 14.38 (The small machine's merges). (1) A move of the small machine that takes the flag from down to up is CERTIFY; CERTIFY merges two different full states into one (a state with the flag down and the same state with the flag up land on the same state), and it costs at least one. (2) DEC A 2 also merges two different full states, and it costs nothing. So it is false of the small machine that every move that merges is priced.

Proof. (1) Only CERTIFY changes the flag, and it costs one. The two states with the flag down and up, and a committed claim, are the same state after CERTIFY. (2) Take a core at line 1 with A at 1 at version 0, and the same core with A at 0 at version 1. DEC A 2 takes the first to $A = 0$ at version 1, line 2, a write; it takes the second to line 2 with A still 0 at version 1, because DEC on zero is no write. Same state, price zero. $\square$

Checked as `frag_small_certify_merges`, `frag_small_certifying_step_priced_merge`, `frag_small_dec_free_merge` and `frag_small_not_merge_priced` (FragmentSmall.v). It's the same fact as the last item of Section 14's list of what the machine doesn't do, now with names: the chairs force a price on every merge, this machine's schedule puts the floor on the flag only, and that's why the eight-state piece is a piece.

Structural entitlement, for every Thiele-complete machine. Theorem 8.3 was about any certification system and handed the run's payments in as a bill. For a Thiele-complete machine (Definition 14.7) the bill is known. Call the CHECK, COMMIT and CERTIFY moves of a run its *record moves*; each costs exactly one, so the ledger rises by the number of record moves, and clause (b) says what a raised record must contain.

Theorem 14.39 (Structural entitlement represented). *Let a machine be Thiele-complete through an interface. Take a clean start s_0 and a run t from it; a type of rivals and a type of observations with an exact equality test; an observation obs and a representative observation r of each rival; a decoder d from runs to observations; lists Ω , Ω' of rivals and a decision tree T . Suppose (M1) Ω' strictly narrows Ω , as in Theorem 8.3, and (M2) some dropped rival shows an observation different from every survivor's; (M3) the run ends with the record up and the decoded run $d(t)$ is shown by some survivor; (M4) $\text{depth}(T)$ is at most the number of record moves in t ; (M5) $|\Omega'| > 0$; and (M6) the covering is by fibres: there is a map F from rivals to lists of rivals such that every rival x of Ω lies in $F(u)$ for some survivor u with $r(x) = r(u)$, the lengths of the lists $F(u)$ over the survivors add up to at least $|\Omega|$, and no $F(u)$ is longer than $L(T)$. Then (1) the membership test of Ω' is strictly stronger than that of Ω ; (2) t contains the earned chain of clause (b), with the meaning of the claim true at the CHECK and at the COMMIT; (3) $\lceil \log_2 |\Omega| \rceil - \lceil \log_2 |\Omega'| \rceil$ is at most the number of record moves of t ; (4) the number of record moves of t is exactly the ledger's rise; and (5) the ledger rose by at least 3.*

Proof. (1) is Theorem 8.3(1). (2) Clause (b) applied to a run from a clean state that ends with the record up gives the chain, and a passing check means the claim's meaning holds, which keeps holding while the thing it's about is unchanged. (3) The fibres cover: summing their lengths gives $|\Omega| \leq L(T) \cdot |\Omega'|$, and Theorem 8.3's count finishes it, with the record moves, which are the paid steps, in place of the charged steps. (4) is clause (c) added up. (5) is Theorem 14.18(3). $\square$

Checked as `ent_representation` (EntitlementSmall.v), whose Coq statement is

printed in Section E. It's a checklist, as I said above, and the small print is the same kind: (M3)'s second half feeds no conclusion, only the record's being up is used; the gap in (M1) is unused, only the inclusion; and the first clause of (M6), the representative observation, isn't used by the count either. What the machine adds to Theorem 8.3 is conclusions (2), (4) and (5): the payment is the machine's own three acts, each at exactly one, and a raised record carries the earned chain, which is more than the older phrase "a structure was added." The packaged form, a record of the fifteen things a shortcut has to supply, with a theorem that every inhabitant lands in the bound, is `ent_shortcut` and `ent_every_shortcut_lands_here`. The count with the tree thrown away, in the form I use below, is `ent_complete_tree_bound`.

Two further statements, about the run's own questions and about the record clause.

Theorem 14.40 (A floor with no supplied tree). *Take a machine and an interface that meets the exact-toll clause (c) of Definition 14.7, so each record act costs exactly one. Call the questions of a run the claims of its CHECK moves, and say two rival states answer alike when every one of those claims gets the same verdict at both (taken at the rivals themselves, not at the states the run passed through). If, for a list Ω of rival states and a number $m > 0$, the rivals of Ω that answer alike with any one rival, itself included, are never more than m , then $\lceil \log_2 |\Omega| \rceil - \lceil \log_2 m \rceil$ is at most the number of CHECK moves of the run, which is at most its number of record moves, which is the ledger's rise.*

Proof. A run's questions give each rival an answer list, and answer lists are lists of Booleans of length equal to the number of CHECK moves, so there are at most 2^c answer classes among the rivals, c the number of CHECK moves. The classes have at most m members each, so $|\Omega| \leq 2^c \cdot m$, and the rounded logarithms follow. The rest is clause (c). $\square$

Checked as `ent_questions_floor` (EntitlementSmall.v). The tree has gone: it is the run's own questions. There are limits, and they're the ones in the statement. The verdicts are taken at the rivals as they start, not at the states the run had reached when it asked, and a free branch, DEC on zero, can still narrow a list for nothing. So this prices a narrowing that the paid checks explain, and no more.

Theorem 14.41 (What the record adds). *Let a machine be Thiele-complete through an interface. (1) From a clean start, the record is up at the end of a run exactly when the run contains the earned chain, and a run with the record up raised the ledger by at least 3. (2) Conclusions (1), (3) and (4) of Theorem 14.39 hold without a clean start and without the record being up, for a machine that only meets the exact-toll clause (c): the hypotheses (M1), (M2), (M4), (M5) and (M6), and the decoded run being shown by some survivor, are all it needs.*

Checked as `ent2_record_adds`, `ent2_upgrade_iff` and `ent2_observed_re`

presentation (EntitlementMore2.v). So the record doesn't make narrowing cost more. It adds the earned chain and a floor of 3, and that's all it adds: a run that narrows without ever raising the record pays its index bits and no more.

Weights, and the partition form. Two more pieces of the same count, both checked. First, suppose the rivals aren't equally many: give each a whole-number weight, a multiplicity, and call the sum of the weights of a list its *mass*. The count survives with masses in place of lengths. If the prior's mass is at most $L(T)$ times the posterior's, the rounded index bits of the two masses differ by at most $\text{depth}(T)$; so for any certification system a run that paid at least that many charged steps has a bill at least the drop, and for a Thiele-complete machine the representation theorem goes through with masses and the same five conclusions. On such a machine the ledger rises by the same number from every start, the number of record moves, so the expected rise over a weighted prior is its mass times the record moves, and the drop in rounded bits times the prior's mass is at most that expected rise, with no averaging argument. The weights are whole numbers cross-multiplied, not probabilities. Checked as `ent2_weighted_bits`, `ent2_wreduction_covers`, `ent2_weighted_cs`, `ent2_weighted_representation`, `ent2_rise_independent`, `ent2_expected_rise_exact` and `ent2_weighted_expected_bound` (EntitlementMore2.v). Second, the covering by fibres has a plainer form that doesn't need the clause that the fibres add up to the prior. Say the rivals are *partitioned* by the representative observation when every rival looks like some survivor under it and no observation class, counted inside the prior, has more than $L(T)$ members. The fibre lengths then add up to at least the prior, because every rival is in the class of some survivor, so a partition gives the covering, and the count holds under it. The converse is false: there are coverings that aren't partitions. Checked as `ent2_cover_sum`, `ent2_partition_old_iff`, `ent2_partition_to_reduction`, `ent2_partition_bits` and `ent2_reduction_not_partition` (EntitlementMore2.v).

A worked member on the small machine. Eight rivals: the starts (a, b) with $a \in \{0, 1, 2, 3\}$ and $b \in \{0, 1\}$, counter A at a and counter B at b . The run is CHECK (Even) A ; CHECK $(A \geq 2) A$; COMMIT $(A \geq 2) A$; CERTIFY, from $(2, 0)$. The program raises the flag from a rival exactly when a is even and at least 2, so exactly from $(2, 0)$ and $(2, 1)$, and the posterior is those two. The decoder reads one "passed" for each CHECK in the run's receipts; the representative observation is " B is 0," which isn't constant; the fibres are the four rivals that share B ; the tree is the complete tree of depth 2.

Proposition 14.42 (The eight-cup member). *For every a and b , the run raises the flag from (a, b) exactly when a is even and at least 2. All hypotheses of Theorem 14.39 hold, with eight rivals narrowed to two: the test of the posterior is strictly stronger; the index bits are $3 - 1 = 2$; the run has 4 record moves and the ledger rises by exactly 4; and the earned chain is in the run. The run's own questions (Theorem 14.40) are its 2 CHECKs, so the floor is met with equality. With the two CHECKs alone, the record stays down, the ledger rises by 2, and*

the bound $2 \leq 2$ still holds.

Checked as `ent_small_certifies_iff`, `ent_small_posterior_is_certified`, `ent_small_instance`, `ent_small_questions` (`EntitlementSmall.v`) and `ent2_observed_small_instance` (`EntitlementMore2.v`), with `ent_earned_complete` with saying the small machine meets the definition through its own interface. It's a real narrowing, not the cups of Section 8 in a costume, and the posterior is exactly the set of rivals the run certifies, so here the lists are tied to the run by a proof and not supplied.

A search through n bits. The same shape at a size you can't enumerate, on the host that has a counter for every number (Definition 14.8). That host is Thiele-complete too, with the property language of Zero, Even and At-least- n , read at registers (`ent2_machine_complete`, `MultiThiele2.v`). Hide $n = k + m$ bits, one in each of registers 0 to $n - 1$. The program asks k questions, one per bit of the first k : CHECK "at least 1" of register i for a 1, CHECK "is 0" for a 0. Then it COMMITs the first claim and CERTIFYs. The prior is every n -bit value; the posterior is the 2^m values that begin with the answers; the tree is the complete tree of depth k .

Theorem 14.43 (A search through n bits). *For $1 \leq k \leq 16$ and any m : from a world holding a value v of at least k bits, the program certifies exactly when v begins with its answers, so the posterior is exactly the set of rivals it certifies from; the test of the posterior is strictly stronger than the prior's; the earned chain is in the run; the prior has 2^{k+m} members and the posterior 2^m , so the index bits are exactly k ; the run has exactly $k + 2$ record moves and the ledger rises by exactly $k + 2$; and the run's own questions are k CHECKs, so Theorem 14.40 is met with equality. With the COMMIT and the CERTIFY left off, the questions alone narrow 2^{k+m} values to 2^m , the record stays down, and the ledger rises by exactly k . And no run that raises the record contains more than 16 CHECKs before its CERTIFY, so with 17 questions no start certifies.*

Proof. Each question is a CHECK of whether one register holds a given bit, which passes exactly when it does; a failing CHECK traps. So the program certifies exactly when every answer is right, which is when v begins with the answers. The counting is the count of bit strings, and the representation theorem applies with every hypothesis a theorem. The cap is the fact table: it holds sixteen facts, never evicts, and a CHECK on a full table traps (Section 14), so a seventeenth CHECK before the CERTIFY ends the run in a trap. $\square$

Checked as `ent2_search_certifies_iff`, `ent2_search_posterior_is_certified`, `ent2_bit_search`, `ent2_search_lands`, `ent2_search_questions_floor`, `ent2_search_observed`, `ent2_questions_cap` and `ent2_no_search_past_16` (`BitSearch2.v`, `BitSearchMember2.v`, `BitSearchObserved2.v`). The sixteen is the machine's, not the program's: it's the same ceiling as the item "Sixteen checks

per run, ever” in the list of what the machine doesn’t do (Section 14), and it’s why this search has $k \leq 16$ however many bits are hidden. Unasked bits can be any number. The lists, the tree and the fibres are supplied here, as in the representation theorem, and tied to the run only for this member. The covering still prices the size of the drop, not its reason.

The covering can’t be dropped. It’s a hypothesis, and one might hope it’s a nuisance one could remove. It isn’t, and the small machine shows it. Fix $n \geq 4$, take the 2^n starts with A below 2^n as the rivals, and run the three-move chain on the claim “ $A \geq 2^n - 1$ ”: CHECK, COMMIT, CERTIFY. It certifies exactly from $A = 2^n - 1$, so the posterior is one start, n index bits are gone, and the ledger rose by 3. So for $n \geq 4$ the claim “the index-bit drop is at most the ledger’s rise” is false for this run, and no tree of depth at most 3 covers the narrowing, whatever the representative observation. One property with a big constant in it carries n bits in a single yes-or-no question; what the bound prices is a narrowing whose classes are balanced, a tree with enough leaves, not any narrowing. Checked as `ent2_uncovered_posterior`, `ent2_uncovered_claim_false` and `ent2_no_cheap_covering` (CoveringNeeded2.v).

The time a check saves. The question is whether counter A is at least m . There are two ways to find out. The *free* way uses base moves only, INC and DEC, which cost nothing. But a base move reads A one unit at a time: a DEC takes a unit off A or finds it empty. After n moves, a run that started with A at least n has never seen A empty, so it can’t have behaved differently for different such A except by a constant shift in A itself. The *paid* way is CHECK ($A \geq m$), COMMIT it, CERTIFY.

Theorem 14.44 (The time tax). (1) No program of two-counter instructions that runs fewer than m moves has its final line number and its final B determine whether $A \geq m$ at every start; in particular none tells $A = m - 1$ from $A = m$. (2) A program of m DEC’s and INC’s (a ladder: DEC A and on success go on to the next pair, otherwise INC B) decides it in exactly m moves, from the line number and B , and costs 0. (3) The paid chain of three instructions raises the flag exactly when $A \geq m$ and has a ledger of exactly 3, whatever m is. (4) So if a ledger unit is worth λ moves, the paid route costs $3 + 3\lambda$ and the free route costs at least m , and the paid route is cheaper exactly when $3 + 3\lambda < m$.

Proof. (1) Take the starts $(m - 1, b)$ and (m, b) . When $n < m$ both have $A \geq n$, so in the first n moves no DEC finds A empty in either run, and the two runs differ only by the shift of one unit in A . They end at the same line with the same B , though $A \geq m$ holds of one and not the other. (2) Run the ladder: each pair either takes a unit off A and moves on, or finds A empty and falls through to the INC. The line after m moves is $2m + 1$ exactly when all m units were there. It’s all DEC and INC, so it costs 0. (3) is the price table: CHECK ($A \geq m$) passes exactly when $A \geq m$, a failing CHECK traps and the flag stays down, and the three moves cost three. (4) is arithmetic on (1) to (3). $\square$

Checked as `ent2_free_run_shift`, `ent2_free_cannot_decide_fast`, `ent2_ladder_decides`, `ent2_machine_free_cannot_decide`, `ent2_machine_ladder_decides`, `ent2_ladder_free`, `ent2_chain_decides` and `ent2_time_tax` (`TimeTax2.v`), the first three for two-counter configurations and the next three for the same programs compiled and run on the machine. The exchange is sharper than a comparison of two particular programs: the free side is a lower bound over every program of INC and DEC, not a choice of program, and the paid side is constant. The limits: the bound is for programs that make no record move, INC and DEC with falling off the end as stopping, and not for every Thiele-complete machine, because the definition doesn't forbid a base move that changes a counter by more than one unit, and such a machine is a different machine, one the bound doesn't speak for. And the two routes put their answers in different places, the free one in the line number and the paid one in the flag; this compares two ways to answer one question, not two certifications.

14.18 The diagonal, on this machine's programs

Section 11 left the diagonal standing on any substrate, on a little family built to fit it, and on L, and said a machine is one more substrate of that kind once it supplies two things: programs that can be handed to programs as numbers, and a recursion theorem for the transformations it can carry out. This machine's universal program reads a whole guest program out of one register, with a numbering of programs that goes there and back (Definition 14.9), and halting is undecidable here in the library's sense (Theorem 14.4). What's left are the two classical results that turn a universal machine into a place where the diagonal bites: Rice's theorem and Kleene's recursion theorem. They come with a wrinkle I should put first, because it decides which machine they're about.

The wrinkle is the one from Section 14. The small machine takes its inputs straight into two counters, and Schroeppe showed two plain counters can't even compute 2^n from n . Rice's reduction has to keep a program's input safe while a copy of a halting instance runs. Kleene's construction has to run an evaluator while keeping its input. Two counters have nowhere to keep it. So the two theorems land where they're true and where I could prove them. The host machine of Definition 14.8, the one with a counter for every number that U runs on, is a fine programming system: its programs are numbers, an input sits in register 1, and there's room to spare. On it both theorems hold in full. On the two-counter machine itself, Rice's theorem holds for what a program does from the clean start, where there's no input to keep. That's what's proved; what isn't is at the end of the subsection.

The host machine's programs are numbered the way U numbers a guest's. Write $\langle m, n \rangle$ for $(2n + 1) \cdot 2^m$. INC r is $\langle 0, r \rangle$, DEC r is $\langle 1, \langle r, j \rangle \rangle$, HALT is $\langle 2, 0 \rangle$, CHECK r is $\langle 3, r \rangle$, COMMIT r is $\langle 4, r \rangle$, CERTIFY is $\langle 5, 0 \rangle$, and a program is the list code: the empty program is 0, and an instruction x followed by a program t is $\langle x, \text{code}(t) \rangle$. Every number decodes to some program, and decoding a program's code gives the program

back. Say a host program *outputs* y on input x when, started with x in register 1 and every other register 0, with a clean table, it halts with y in register 0.

Rice first, because it's the shorter road. Henry Rice proved in 1953 that no program decides any property of what programs *do*, as opposed to how they're written, unless every program has it or none does. The road from halting is a single detour. Take a property of what programs do, a program y that has it, and a program n that doesn't. Given an instance of two-counter halting, a Minsky program H with starting counters a and b , build a host program that first loads a and b into two fresh registers, then runs a copy of H moved onto those registers, then empties both, then runs y . If H halts on (a, b) the built program behaves like y and has the property. If H runs forever the built program never stops on any input, and it behaves like the program that loops, which lacks the property (if the looping program has it instead, swap the roles of y and n). Everything the built program does before y is INC and DEC, which pay nothing and touch no fact, channel, trap latch or flag. So a decider for the property would be a decider for halting, and the library already says what that would cost: the complement of single-tape Turing-machine halting would become something a program could list, the same synthetic sense of "undecidable" as Theorem 14.4. Two words carry the weight, and they're the ones to check in the statement: what "behaves the same" sees, and where the program starts.

Definition 14.12 (Behaving the same, on the host). A host program *ends* in a state when it halts there, from the start with x in register 1 and the rest clean; a trap counts as stopping. Two host programs *behave the same* when for every input x , any state one ends in has a state the other ends in that agrees with it on every register's value, the fact table, the channel, the trap latch, the ledger and the flag, and the same the other way round. The line number and the versions aren't compared. A property of host programs is *extensional* when behaving the same carries it from one program to the other.

Theorem 14.45 (Rice's theorem on the host machine). *Let Π be an extensional property of host programs that some program has and some program lacks. Then Π is undecidable in the library's sense.*

Proof. The reduction above. The prefix before y leaves every register but the two fresh ones as it found them, and empties those, so when H halts the built program ends in whatever state y ends in from the same input, with the same ledger, table, channel, trap and flag. When H doesn't halt, the built program stops on no input. The reduction is from the complement of halting, so no excluded middle is needed; the proof splits on the decider's own answer for the looping program. $\square$

Checked as `sm_host_rice`, with the reduction's two halves as `sm_rice_prog_halts` and `sm_rice_prog_diverges` (`SmHostRice.v`). A property that depends only

on the partial function a program computes, whether it stops and what it leaves in register 0, is extensional, so it's covered; the files also derive that case again from the recursion theorem below, as `sm_host_rice_fun` (`SmKleene.v`).

On the small machine itself the same road works from the clean start, and only from there. Say two two-counter programs *behave the same from the clean start* when, whatever state one halts in from counters $(0,0)$, the other halts in a state with the same counters, trap latch, ledger and flag, and the same the other way round. The fact table isn't compared, because the prefix moves the versions of A and B by fixed amounts and the facts carry versions. There is no input to keep, since the program starts at $(0,0)$, and H 's instance is loaded straight into A and B .

Theorem 14.46 (Rice's theorem for the small machine, from the clean start). *Let Π be a property of two-counter programs that's carried from a program to any program that behaves the same from the clean start, held by some program and lacked by some other. Then Π is undecidable in the library's sense. In particular "halts from the clean start" is undecidable.*

Checked as `sm_guest_rice` and `sm_guest_halting_clean_undecidable` (`SmGuestRice.v`). What this does not say is that no program decides a property of what two-counter programs do on *every* input. The standard reduction can't reach that, for Schroepel's reason, and it isn't proved.

Kleene's recursion theorem is the deeper one, and it's a construction, which is Section 11's point about Kleene's little gem (Moschovakis's phrase): the content is the building, and the contradiction in Section 11 was just the last line. It says: for every transformation F of programs that some program can carry out on numbers, there's a program p that computes the same thing as $F(p)$. The shape of the building is the old one. From the program that carries out F , build an *evaluator* that reads a number, treats it as a program's code, specialises that program to its own code, runs F 's program on the result, and runs whatever program comes back. The specialising step is what logicians call s-m-n: fix one input of a program and get a new program that needs one input fewer. Then the witness is the evaluator specialised to its own code. Run on any input, it builds its own code, hands it to F 's program, gets the code of $F(p)$, and runs that.

Here's each piece on the host machine, with its exact premise.

Theorem 14.47 (s-m-n, and a universal program for host programs). (1) *For a number c and a host program V , let $\text{spec}(c, V)$ be c copies of INC 2 followed by V with its jumps moved to follow them. Then for all x, y , $\text{spec}(c, V)$ outputs y on input x if and only if V , started with x in register 1 and c in register 2, halts with y in register 0.* (2) *There is a host program W made only of INC, DEC and HALT such that, for all c, x, y , W started with x in register 1 and c in register 2 halts with y in register 0 exactly when the host program numbered c outputs y*

on input x .

Proof. (1) The prefix writes register 2 c times and nothing else, so it moves that register's version from 0 to c . A fact records a version, so after the prefix every later version of register 2, and the version in every fact the run records about register 2, is c more than in a clean run. A CHECK or a COMMIT reads versions only to ask whether two versions of the same register are equal, and adding c to both keeps the answer, so the two runs take the same steps. The run is V 's run from the stated start, moved down by c lines. (2) is the evaluator. A finite algorithm decodes a number into a program, runs it step by step on given registers, and, searching over the number of steps, returns the output of the first halting run. I wrote that algorithm out in Coq, turned it into a term of the lambda calculus L , and ran the library's compiler from L to Minsky programs, which gives a program of two-counter instructions that a host program can be read off from, instruction for instruction. The tools supply the program that computes the evaluator. The fixed-point argument below is not theirs. $\square$

Checked as `sm2_smn_all` (`SmSmnAll.v`), an if-and-only-if in both directions that carries no premise on register 2, and `sm_host_universal` (`SmKleene.v`); the narrower form of (1), with the premise, is still there as `sm_smn`. The numbering is `sm_hcode`, `sm_hdecode` and `sm_hdecode_hcode` (`SmCodes.v`). Be exact about one thing: U is not that universal program. U runs two-counter guests, and two-counter guests can't run the evaluator. W runs host programs, and it can, because it's a host program on a machine with registers.

Theorem 14.48 (Kleene's recursion theorem on the host machine). *(1) For every host program T there is a host program e such that for all x, y : e outputs y on input x if and only if, for some d , T outputs d on the number of e , and the program numbered d outputs y on x . (2) So for every transformation F of host programs that some host program carries out on numbers (on the number of p it outputs the number of $F(p)$, for every p), there is a host program e such that for all x, y , e outputs y on x if and only if $F(e)$ does. (3) The same on numbers: for every function f on numbers that some host program computes, some number e has the program numbered e and the program numbered $f(e)$ output the same on every input.*

Proof. (1) Let E be the following program of INC and DEC, built the same way as W from one finite evaluator, with T 's number t written in. On inputs x and c it computes the number of `spec( $c, P_c$ )`, where P_c is the program numbered c , runs T on that number to get d , and then runs the program numbered d on x . Let c_0 be the number of E , and let $e = \text{spec}(c_0, E)$. By (1) of Theorem 14.47, e on input x does what E does on x and c_0 . And E on c_0 builds the number of `spec( $c_0, P_{c_0}$ )`, which is the number of e , since P_{c_0} is E . So e hands its own number to T , gets d , and runs d on x . (2) and (3) are (1) with T carrying out F or f . $\square$

Checked as `sm_second_recursion`, `sm_kleene` and `sm_kleene_codes` (`SmKleene.v`). That fixed point is of the *function*: e and $F(e)$ stop on the same inputs and leave the same number in register 0. Their ledgers, flags and fact tables can differ, because the marks e pays are the evaluator's. I'd like a fixed point that agrees on the record too, and the next two theorems say how much of that can be had: the ledger and the flag, yes, and every register, no.

Theorem 14.49 (The recursion theorem, up to the record). *Let F be a transformation of host programs that some host program T carries out on numbers. Then there is a host program e such that, for every input x , any state e ends in has a state $F(e)$ ends in that agrees with it on register 0, the trap latch, the ledger, the flag, the number of facts in the table, and whether the channel is empty, and the same the other way round. So e stops on x exactly when $F(e)$ does, and the ledgers are equal exactly, with no surcharge. T may itself check, commit and certify.*

Proof. The construction is Theorem 14.48's, with a different evaluator. The first step is to see that a record is small. From a clean start, a host program that stops has a record I can read off five numbers: what it left in register 0; how many facts the table holds; 1 if the trap latch is up and 0 if not; 1 if the flag is up and 0 if not; and what is left of the ledger once the facts, the trap and the flag are paid for, which I'll call the commits. The record obeys a short rulebook. The table holds at most sixteen facts. The latch is a single bit. A commit needs a fact behind it, a certify needs a commit, and the channel is empty exactly when no commit was paid. The ledger is the facts, plus the commits, plus the flag, plus the trap, and that's all of it. Those are the only properties of a record the replay will use, and I proved them by following the plain machine alongside the real one, one step at a time.

The second step is five blocks, one for each number. Each block reads the code of T , runs T on the number of the specialised program, runs the program that comes out on x , and returns its number. All of that happens on plain data: the blocks are programs of INC and DEC, built the way W was, from an evaluator written out in Coq, turned into a term of L and compiled to Minsky programs. So if T has marks of its own, they are simulated and never paid.

The third step is the payment. The evaluator puts 1 in a fresh register, which satisfies the host's one property, and CHECKs it once for every fact. It COMMITs to that claim once for every commit. It CERTIFYs once if the flag is up. It copies the first number into register 0. And if the trap latch is up, it CHECKs a register that holds 0, which fails, and the host traps. Everything before the payments is INC and DEC and costs nothing, and the payments cost what the record costs, so the ledger is exact. Specialise the evaluator to its own number, as before. $\square$

Checked as `sm2_kleene_obs` (`SmFixedPoint.v`), with the five numbers as `sm2_final_numbers` and the evaluator's specification as `sm2_ev_spec` (`SmTally.v`) and

the replay program as `sm2_VG_forward` and `sm2_VG_halt` (`SmFixed.v`), where the trap latch is up exactly when the fifth number is 1 and the flag exactly when the third is positive; `sm2_kleene_fun` gives back the version that compares only the function. Two limits, and the second one is a theorem. First, e pays after it has computed everything. It is not the way U runs a guest, mirroring every move: e pays the counts, CHECKs, then COMMITs, then the CERTIFY, whatever order $F(e)$ used, so the order of the record's events isn't preserved, and for a program $F(e)$ that never stops, e never stops and pays nothing, though $F(e)$ may pay without bound. The relation says nothing about states in the middle of a run. Second, no fixed point agrees on every register, or on what the facts name.

Theorem 14.50 (No fixed point of full behaviour). *There is a transformation F of host programs, carried out by a host program, such that no host program e behaves the same as $F(e)$ in the sense of Definition 14.12.*

Proof. Let $F(p)$ be the one-line program `INC  $r$` , with r one above the number of p . A host program carries this out on numbers; I got it from the same route through L, not by hand. A program's number is at least every register it names: the number of an instruction is above its register, and the code of a list is above every instruction in it. So e can't name r , the register $F(e)$ writes. On input 0 the run of $F(e)$ ends with 1 in register r , and if e stops at all, it ends with 0 there. If e doesn't stop, then it already differs from $F(e)$, which does. Either way they don't behave the same. $\square$

Checked as `sm2_no_exact` and `sm2_reg_bound` (`SmNoExact.v`). A variant of F that also CHECKs register r would give $F(e)$ a fact about r that e can't have, which is why I say the facts can't be compared by what they name either. I haven't checked that variant in Coq; the theorem above is what is checked. What survives is what Theorem 14.49 states, and that is as much as can be had.

And with the recursion theorem proved, the wall of Section 11 goes up on this machine without a premise left over. Call a property of host programs *functional* when it depends only on the partial function a program computes.

Theorem 14.51 (The diagonal on the host machine). *Let Π be a functional property of host programs that some program has and some program lacks. (1) Read “ f is representable” as “some host program carries f out on numbers.” Then no function d from programs to Booleans whose flip is representable is right about Π on every program: this is Theorem 11.1, with its recursion premise a theorem here. (2) No host program decides Π outright: there is no program D that, on the number of every program, halts with a number in register 0 that's 1 exactly when the program has Π .*

Proof. (1) The flip, send programs that d calls good to the program lacking Π and the others to the one having it, is carried out by a host program by hypothesis. Theo-

rem 14.48(2) gives a program e that computes what its image computes; Π holds of e exactly when it holds of the image, and the image lacks it exactly when d calls e good, so d is wrong at e . (2) From a deciding program D build a host program that computes D 's flip on numbers, by the same route through L and the Minsky compiler, reading the Boolean off D 's runs with the standard library's unbounded search; then (1) applies. This one is an outright negation, not the synthetic notion. $\square$

Checked as `sm_no_inside_decider` and `sm_no_host_decider` (`SmKleene.v`, `SmDecider.v`).

Rice and Kleene are the ingredients the diagonal of Section 11 asks a substrate for, and the wall they put up doesn't care about the record: the base does the computing and the flag watches. A program's ledger and flag are among what "behaves the same" sees on the host, so no program decides a nontrivial property of those either, and the fixed point is up to the record, which is as much as can be had. What's still open is Kleene's recursion theorem for programs of the two-counter machine on plain inputs, and Rice's theorem for what a two-counter program does on every input; I haven't proved either, and I haven't tried to refute them.

14.19 What the universal machine covers, and what it doesn't

Here's what that covers, said plainly, a sentence at a time. The guests of U are this chapter's programs, every one of them, from every start. That's the stated class. It's the same kind of class Turing's universal machine has: a fixed instruction set, and a finite program held as data, here packed into one number the way Turing packed his into a description number. Each guest keeps its computation, its halting, its answer, its record and its prices, exactly: the host halts when it halts, its flag goes up when the guest's would, on a chain of its own, and its ledger is the guest's.

The guests of U_P are every Thiele machine whose driver, step, prices and reading are computed by programs, with the reading decided on every state. Each keeps its computation and its halting; its reading becomes the host's flag as a latch, raised on an earned chain whose claim is the guest's own reading; and its prices become the host's ledger, exactly, except for a surcharge of at most two at the move where the reading first says yes, which no program on that host can avoid when that move costs less than three (Theorems 14.29 and 14.30). Picture a clerk who'll stamp anybody's receipt. He copies it onto his own forms, bills at your rates to within two, and the stamp at the bottom is one he earned by checking your own reading, not one he copied.

What's outside the class is outside the theorems. A machine whose reading no program decides (an oracle checker, say, which the definition of Thiele-complete allows) doesn't come in. Nor does a machine whose next move is nobody's computation. Nor does a guest that starts with a table of facts nobody checked: the guests start where their presentation says, and the host from a clean start.

And here's why insisting on no built-in step is the point. A plain interpreter on an

ordinary computer runs guests too, but then the guest's record is data the interpreter's program updates by convention, which is exactly the position of the ordinary programmable computer from the front matter. That's a fact about how such interpreters are built, not about what they can do: U and U_P are fixed interpreters, and their record isn't kept by convention. Nothing in them raises the host's flag except the host's own CERTIFY, and the host's step won't let that pass without a COMMIT, or the COMMIT without a CHECK on a slot nobody has touched. The guest's record isn't data the host updates. It's a record the host earns again, under the same rule, on its own registers.

14.20 What it doesn't do

Here's where it falls short, in plain words, because a small machine is only honest if its edges are as visible as its middle.

- **The check price is flat.** CHECK costs 1 whether it's asking "is this 0" or "is this at least a billion." The property language is fixed and each property is one evaluation, so "pays for its work" here means that flat price, not a price that grows with how hard the question was. A machine that charged by the difficulty of the check would be a different machine, and I haven't built it.
- **DEC on zero isn't a write.** It doesn't change the counter, so it doesn't change the version. That's the right call for versions (nothing changed), but it means DEC on zero is a free way to move the line number without leaving any mark. The idle run in the window section uses exactly that.
- **After a trap, a trace still pays.** On programs, a trap stops the run. On arbitrary traces, instructions after a trap are each charged their cost and none of them changes the core. I chose to keep charging so that the ledger stays the plain sum of what was issued (Theorem 14.5 has no exceptions); the price of that choice is that a trapped machine can still be billed for nothing.
- **"Checked" means what the property language can say.** On this machine that's is zero, is even and is at least n , and a live fact means one of those holds of one counter's current value. It doesn't mean the program is correct, or that anything follows from the property, or that the property was worth checking. So can a proved machine of this kind check "sorted," the claim the opening chapter's table runs on? Yes, through an encoding. A copy of this machine, instruction for instruction, with its property language left open, has provenance, checker soundness, no forging and the floor of three proved once for every property language that comes with an exact equality test and a checker proved equal to its meaning. Give it "the list this counter encodes is sorted" as a property and it certifies a sorted list for three and refuses an unsorted one forever (`sorted_run_certifies`, `sorted_run_certifies_iff` and `sorted_run_refused_forever`, `EarnedGeneric.v`). The copy is written out again in its own file, not derived from this one by a theorem; putting the two side by side is how you check they're the same machine. What makes that a check of the list, and not of

a number, is the proof that the checker means sortedness. Without that proof, “sorted” would be a number some unconstrained code wrote into a counter, and the record would be honest about the counter and silent about the list.

- **The certificate is historical.** Between the COMMIT and the CERTIFY, the counter can change, and the flag still goes up. The flag says a claim was checked and committed about an edition, not that the claim is still true.
- **One latch, and a channel that moves on.** The flag records that an earned chain happened, once. After that it stays up, and later CERTIFYs change nothing. The channel is the part that names a fact: with the flag up from a clean start, it names a fact that some earlier passing COMMIT committed (`certified_channel_earned` and `channel_names_last_commit`, `EarnedCore.v`). But a later commitment replaces it, so at the end it names the last earned commitment, which needn’t be the one the flag rose on. Several claims need a latch each.
- **Sixteen checks per run, ever.** Facts never leave and the table holds sixteen, so a run gets at most sixteen passing CHECKs in its whole life, and the seventeenth traps. A certificate needs only one, so this costs nothing for the theorems, but it’s a hard wall for any program that wants to keep checking forever, and this machine’s small version of the opening chapter’s “you can’t keep all of it.”
- **The clean start matters.** Provenance, soundness, no forging and the price of three are all stated from a clean start. Hand the machine a table pre-loaded with facts nobody checked, and of course none of that follows. The theorems are about what the machine produces, not about what you smuggle in at the door. Spell the door out once: the theorems cover any list of instructions anybody can issue, from a clean start, on a step that runs as written, with nothing else writing to the machine. They say nothing about whoever sets or restores the start, about a reset, about a device that misbehaves, or about whoever carries the flag’s value to you.
- **The universal programs have stated guests.** U runs every one of this machine’s programs from every start, with halting, answers, flag and ledger matched exactly. U_P runs every computably presented Thiele machine, with its reading decided by a program, its record carried as a latch, and its ledger matched to within a surcharge of two that can’t be avoided in general. A machine outside that class, with a reading no program decides, isn’t covered by either. The simpler host with the built-in guest step pays one per guest step whatever the step was, so its ledger is at least the guest’s and usually more; U ’s is exact, and U_P ’s is exact up to the surcharge.
- **Slow, and fed directly.** Universality is Minsky’s, with the inputs placed straight in the counters. Getting from an arbitrary Turing machine to two counters goes through his exponential packing, which my files don’t redo; the compiler behind U_P uses the library’s version of it, and nothing about it is fast.

- **The step doesn't police trivial checks.** CHECK *at least 0* passes on every value, so CHECK, COMMIT, CERTIFY on it raises the flag for three from any start (the note after Theorem 14.17). A check that can't fail establishes nothing, and the machine raises the flag for it anyway. The step checks order, aboutness and an honest checker, and it leaves "was that worth asking" to whoever reads the record.
- **The toll is chosen here, not forced.** Section 4's argument forces a toll on a finite machine that pays for every merge. This machine is neither. Its counters have no ceiling, so it's infinite. And its free instructions merge states: take a core at line ℓ with c at 1 and version w , and a core at line $j - 1$ with c at 0 and version $w + 1$, everything else alike, and DEC $c\ j$ sends both to line j with c at 0 and version $w + 1$, for nothing. So for this machine the price of one is a design choice the chairs don't force.

14.21 What this chapter adds up to

So Part IV has one machine, and it isn't the model. It's a Thiele machine in the front matter's sense, an instance of the certification system with the toll proved: `earned_cs` (Theorem 14.6). It's Thiele-complete, in the sense of Definition 14.7, the sense no clock meets (Theorems 14.19 and 14.21): it computes whatever a Turing machine can, its computing is free and can't touch the record, and its record is earned. And the universal Thiele machine runs on a host made of its own kinds of instruction: one fixed program, U , with no guest step built in, that runs every one of its programs with halting, flag and ledger matched exactly and earns its own flag (Section 14.13); and the same program with one paid move added, U_p , that runs every computably presented Thiele machine, carrying its reading as a latch, its prices to within two, and a flag earned on the guest's own reading (Section 14.14).

If you ever find yourself wondering whether a property of this machine is the model's or just the machine's, hold it up against Section 3's certification system, the latch of Section 10 and the four clauses of Thiele-complete. What follows from those holds of every machine that meets them; the rest (sixteen slots, three properties, a flat price of one) is this machine's choice. Six instructions, and a seventh for a host that has to bill somebody else's prices. If the model needs more than that, this is where it'll show.

What's left of Part IV is a check worth earning, and then what any of this costs against physics.

15 The CHSH check

The small machine checks whether a number is zero, even, or at least something. That's a fine way to show the chain, and a dull thing to certify. Here's a check worth earning, built from zero, and then the one thing the small machine does with it, which is smaller and more exact than "it checks for quantum behaviour," a sentence that's tempting and wrong. The short version, so you know where this is going: the check

never computes the famous number and never writes down the famous bound. It tests seven facts about integers in eight counts. If they pass, a certain five-by-five table has a property called positive semidefinite, and that property forces the famous number under the famous bound. The wall is a consequence, not an ingredient. Everything else is explaining those sentences.

But first, why is a quantum game in a book about records at all? Not for the physics. Nothing in this section tests nature, and the check can't tell honest play from a made-up tally (the guarantee paragraph below says exactly how). It's here for three reasons, all about the record. It's a check whose content is striking enough to tempt anyone into thinking that certifying it should cost more than certifying something dull, and on the small machine it doesn't: a raised flag costs three, whatever the claim (Section 14). It sets free observation against a priced check: a round of the game is written down for nothing, and judging the pile is one of the three paid acts. And it's an integer test small enough to be a property in the small machine's sense, a checker that's a little program and a meaning that's a statement, with a proof that the two agree, which is exactly what Section 14's generic machine asks of any property before it lets a flag stand on it.

15.1 The game

Two players, Alice and Bob, in separate rooms. A referee flips two coins and hands Alice a question x , zero or one, and Bob a question y , zero or one. Alice answers a , zero or one; Bob answers b . No talking during the game. They may agree on a plan beforehand, as detailed as they like. They win a round when their answers *differ exactly when both questions were one*. In words: if either question is zero, match; if both are one, differ.

Try to win every time. A plan is a list: what Alice says to x equals zero, what she says to x equals one, and the same two for Bob. Four bits. Now write the four requirements the rounds impose: a_0 equals b_0 (both asked zero); a_0 equals b_1 ; a_1 equals b_0 ; a_1 differs from b_1 . Chain the first three: a_0 equals b_0 equals a_1 , and a_0 equals b_1 , so b_0 equals b_1 . But the fourth says a_1 differs from b_1 , and a_1 equals b_0 equals b_1 . Contradiction. One of the four must break. So any plan loses at least one of the four question pairs, and if the referee asks all four equally often, the best a plan can do is three out of four. Seventy-five percent. Do it the long way too: there are sixteen plans (four bits), and if you write them all out and count wins you'll find eight win three rounds and eight win one, and none wins four (the broken requirements always come in odd numbers, which you can see by adding up the four equations: the left sides cancel to zero and the right sides to one). Sharing a coin before the game to pick a plan at random doesn't help; a mixture of numbers no larger than three-quarters is no larger than three-quarters. Arthur Fine proved in 1982 that the whole "pre-agreed answers, no influence between rooms" story is *exactly* the mixtures of those sixteen lists, one scoreboard for all four questions at once; the sixteen are the whole classical story.

The number. Recode answers zero and one as plus one and minus one. For each question pair, let E be the average, over many rounds, of Alice's answer times Bob's. The CHSH quantity is S equals E_{00} plus E_{01} plus E_{10} minus E_{11} . For one fixed plan each win adds one to S and each loss subtracts one, so S is wins minus losses: a three-win plan has S equal to two, a one-win plan has minus two. With equal question frequencies, the win rate is one half plus S over eight. So "seventy-five percent" and " S at most two" are the same fact, and the second is the one that got the name.

The ceiling is easy to reach and costs nothing to write down. Tally one round per question pair with both players answering zero every time: after the recoding every answer is plus one, so all four correlators are one, the all-ones plan, which matches on every pair and so wins the three pairs that ask for a match, $S = 1 + 1 + 1 - 1 = 2$. On the small machine a tally like that is a few INCs on counters, and INC is free. Nothing in it says the rounds were played; anybody can tally anything for nothing. The other side, that no finite mixture of plans beats two, is Theorem F.2 in Appendix F, one direction of Fine's result.

15.2 Where it came from, and why anyone cared

So who started this, and why would anybody care whether two rooms can win a coin game? The trail starts in 1935, with Einstein, Podolsky and Rosen asking whether quantum mechanics describes everything that's real about a pair of particles. Their argument: if measuring one particle lets you predict the other's result with certainty, the result must already have been fixed, and the theory leaves that out. The wall they had in mind was a pre-agreed list of answers, the plan above. And the first thing worth checking is whether the paper even says what Einstein meant. By his own account, not quite. Podolsky did the writing, and that June Einstein wrote to Schrödinger that the essential thing had been "smothered by formalism." The person who lays that letter out is Arthur Fine, the same Fine whose theorem closed the sixteen plans above, in his encyclopedia entry on the argument. So even the founding paper is a lead to check, not a verdict.

For thirty years that was philosophy. In 1964 John Bell turned it into arithmetic: if each particle carries pre-agreed answers and nothing travels between the rooms, certain averages can't exceed a fixed number, and quantum theory predicts they do. And where does a paper like that get published? In a brand-new journal called *Physique Physique Fizika*, which charged authors no page fees and in fact paid them by the word, and which folded in 1968 after four volumes. The story told in Andrew Whitaker's biography of Bell is that the journal gave out no free copies, so Bell spent his payment buying copies of his own paper to mail to other physicists. What his inequality rules out is the *pair* of assumptions together, called local realism (answers exist before you ask, and no influence across the gap), not either one alone. In 1969 Clauser, Horne, Shimony and Holt rewrote it as the four-average form above, S at most two, which could be tried in a lab with imperfect equipment.

Then the obvious next question: did anybody actually try it? Freedman and Clauser tried it in 1972; Aspect's group did it in 1982 with the measurement settings switched while the photons were in flight. Before that, in 1975, Alain Aspect went to see Bell at CERN to tell him what he planned, and by Aspect's own account Bell heard him out in silence and then asked, first thing, whether he had a permanent position. Aspect did. Bell's reason, as Aspect tells it, was that most physicists would take the experiment for a waste of time, or take him for a crackpot. And in 2015 three groups (Hensen's in Delft with electrons a kilometre apart; Giustina's; Shalm's) closed the two main escape routes at once. Clauser, Aspect and Zeilinger got the 2022 Nobel for it. So in under fifty years the subject went from a question you needed a permanent job to risk to a Nobel. The lab number sits above two. A remaining escape, that the referee's coins were secretly correlated with the particles, is a philosophical one and I'm not going to pretend it's closed.

What quantum play does: if the players share an entangled pair and choose what to measure from their question, the best they can do is S equals two times the square root of two, about 2.83, a win rate of about eighty-five percent. Boris Tsirelson showed in 1980 that that's a *ceiling*: quantum correlations violate Bell, but only up to there.

And the wall above that, which is the one I find most instructive. In 1994 Popescu and Rohrlich asked: what if you only demand that nothing can be *signalled*, that Alice's answers on their own have the same odds whatever Bob is asked, and the other way round? Then you can write down a "box" (the PR box, after their initials) that wins *every* round (its answers differ exactly when both questions are one, by fiat, each answer alone a fair coin), S equals four, and nothing about it lets either room send a message. So relativity alone doesn't explain the quantum ceiling; the PR box obeys relativity and beats 2.83. The order is: two for plans, 2.83 for quantum, four for no-signalling.

And the box has the same kind of history as everything else in this book: somebody got there first. Popescu and Rohrlich's initials stuck, but the big review of the field by Nicolas Brunner and four coauthors credits boxes like it to Peter Rastall and, separately, to Leonid Khalfin and Tsirelson himself, both in 1985, nine years before. Then the question that really matters: if relativity isn't what stops things at 2.83, what is? People have tried candidates. In 2005 Wim van Dam showed that with enough PR boxes, Bob could work out any yes-or-no function of both their inputs after Alice sends him a single bit, which would make the whole question of how much two parties need to talk collapse, and that's a strong hint the boxes are too good to be real. In 2009 Marcin Pawłowski and five coauthors proposed a principle they called information causality: Bob can't get more information about Alice's data than the bits she actually sends him. Ordinary and quantum physics both obey it, and any theory that beats 2.83 breaks it. It recovers part of the quantum boundary, not all of it. Which principle fixes 2.83 is still an open question, and this book won't answer it. The check below is a fence at the middle step.

15.3 Wrong turns to take off the table

Every one of these gets said with a straight face, and every one is easy to say by accident, so here they are, each with what's actually true beside it.

"Faster than light." Nothing travels. Each room alone sees fair coins; the correlation shows up only when you lay the two columns side by side afterward. "Quantum means random." The average S is fixed by the state; what's random is each single answer; the plans are the less random side. "A Bell violation proves nonlocality." It rules out local realism, both assumptions together, and every real test has loopholes with names. "A tally above two proves nonlocality." No: S is a *population* average, and one finite tally can land above two by luck. Four rounds, one per question pair, with coin-flip answers, reach S equals four one time in sixteen. That gap between a population bound and a finite sample is the problem Zhang, Glancy and Knill work through for real Bell tests (Physical Review A, 2011 and 2013), and the book doesn't solve it. The way they solve it is worth knowing, because it has the shape of this book's record. Turn each round into a small bet that no local plan can expect to win, even a plan that remembers every earlier round, multiply the bets as the rounds come in, and Jean Ville's inequality from 1939 says the chance the product ever climbs past one over δ is at most δ . So a flag can go up for good the moment the product crosses, with its error stated up front. It needs one thing the small machine doesn't have: settings drawn underneath the program, not chosen by it. And how much can one round tell you? Wim van Dam, Richard Gill and Peter Grünwald (2005) measured exactly that, and of the two-party proofs they compare, CHSH is the strongest by far, at a few hundredths of a bit per round, so a convincing record takes hundreds of rounds, not four. The checked work proves only the narrow thing, that a tally above two is not explained by *one fixed plan replayed every round* (`chsh_violation_rules_out_locally_factorizable_coupling`), and that narrow thing is what the sentence has to stay. And "the PSD test proves the data is quantum." (PSD is short for positive semidefinite, the property of the five-by-five table from the short version; the next part builds both.) It doesn't, and the direction matters. Passing implies the five-by-five table, pinned in a way the next part explains, is PSD, and that implies S squared is at most eight; both are proved. Whether the correlators of a PSD table like that are the correlators of some quantum strategy is the cited Tsirelson, Landau and Masanes result, and I haven't proved it here. Failing says nothing: the check is sound and not complete, and quantum behaviour can fail it. Every one of the sixteen plans fails it, and I'll show you why. What it guarantees is in the last part of this section.

15.4 The five-by-five, and what positive semidefinite means

Here's the mathematics the check rests on, and it's one idea: a table of numbers *could be a table of dot products*. Take a two-by-two symmetric table with a and c on the diagonal and b off it. It's positive semidefinite (PSD) when a and c are at least zero and a times c minus b squared is at least zero. Check two by hand: ones on the diagonal and one half off, the determinant is three quarters, passes; ones on the diagonal and 1.2 off,

the determinant is negative, fails. The reading: PSD means the table could be the dot products of some arrows, and 1.2 is too big a dot product for two arrows of length one. That's the whole idea, generalised to bigger tables.

Now the table. Rows and columns labelled I, A0, A1, B0, B1 (the identity, Alice's two questions, Bob's two). Diagonal all ones. The I row and column all zero (that's "zero marginals": the table takes each answer alone to average to zero, a fair coin; the counts record no marginals and the check doesn't test this). A0 with A1 zero, B0 with B1 zero (no round asks Alice both questions, so no tally holds that average, and the table sets it to zero). Those zeros are choices, they pin the table to one slice, and I'll come back to what they cost. And the four Alice-Bob cells hold the four correlators E. That's a five-by-five with four free numbers in it. It splits into the one in the corner and a four-by-four block made of an identity and the two-by-two grid E of correlators, and PSD of that block is the same as saying *E shrinks every vector* (E is a contraction), which is three inequalities: one minus E_{00}^2 minus E_{10}^2 at least zero; one minus E_{01}^2 minus E_{11}^2 at least zero; and their product at least the square of $(E_{00}E_{01} + E_{10}E_{11})$.

Run it. All four correlators 0.6 except E11 at minus 0.6, S equals 2.4: the two diagonal terms are 0.28 and 0.28, the cross term zero, passes; above two, below 2.83. All four at one over root two (E11 negative): diagonals zero, cross term zero, passes with no slack, S equals two root two exactly, the 2.83; nudge any entry's size up and a diagonal goes negative. The all-ones plan (one, one, one, one): first diagonal is one minus one minus one, negative, *fails*, though S equals two and it's a perfectly legal classical play. The PR box (one, one, one, minus one): fails the same way. A fence at the middle step, and a fence that keeps out every one of the sixteen plans: each has all four correlators at plus or minus one, so the first diagonal is always one minus one minus one. Only diluted mixtures of plans get through.

That point, one over root two in size in every entry, is four real numbers, and no tally of rounds ever lands on it. A tally's correlators are ratios of whole numbers, and the point needs each of them to be one over root two, which would make root two a fraction, and you killed that fraction with a pencil in Part I. That point and its mirror image, every sign flipped, are the only places the bound is reached (Remark F.2 in the appendix). So a passing tally always has S squared strictly below eight. It can get as close as you like: the (17, 3) tally in the five examples below has S squared equal to 7.84, and tallies of the same shape with a fraction nearer one over root two get nearer still. Two root two is the least bound the check can promise, and no tally the check accepts ever reaches it.

In 2007 Navascués, Pironio and Acín showed this is the first rung of a ladder: bigger tables of expected products, each PSD, each a necessary condition for quantum behaviour when its entries are left free, each rung tighter, and (their 2008 paper) converging to the quantum set (strictly, the set from commuting operators, which can

differ from the tensor-product set in general scenarios). The check's table is the first rung with the marginals and the two within-party entries pinned to zero. Pinning is a stronger demand than the rung makes, so passing puts you inside the first rung, and a quantum point can still fail. Landau in 1988 and Masanes in 2003 found the exact boundary for this two-questions-each case, and it's not a straight line at 2.83 but a curved surface (the arc-sine form) touching it at one point. So the obvious next question: if every rung is tighter than the last, does some rung finally land exactly on the quantum set and stop? A 2026 paper by Chaturvedi says no: no *finite* rung of the ladder captures the whole quantum set even in this simplest case, which is the caution this book keeps about the ladder, confirmed from outside.

15.5 The check, in integers

Eight counts: for each of the four question pairs, one counting rounds where the answers matched and one where they differed. A round is one bump to one count, after checking all four inputs are bits; nothing is tested and no marginals are recorded. Then the check reads the eight and tests seven integer facts. For each pair, let N be same plus different and D be same minus different (so E is D over N). The seven: the four N are strictly positive (every pair was sampled; a pair with no rounds is rejected, not read as a correlator of zero); A , which is N_{00} squared N_{10} squared minus D_{00} squared N_{10} squared minus D_{10} squared N_{00} squared, is at least zero; B , the same with the other column, is at least zero; and with C equal to $D_{00} D_{01} N_{10} N_{11}$ plus $D_{10} D_{11} N_{00} N_{01}$, C squared is at most A times B . Those are the three contraction inequalities with the denominators cleared. No division, no reals, no square root, no S . Try it: counts (same, different) of (4, 1), (4, 1), (4, 1), (1, 4) give E of 0.6, 0.6, 0.6, minus 0.6; N is five and D is three (minus three for the last); A is 175, B is 175, C is zero; passes. That's the first example above, the S equals 2.4 one, in integers.

The theorem chain, in words: a passed check gives PSD of the five-by-five built from the counts; contraction and PSD are equivalent, both directions; and PSD forces S squared at most eight. The names are in the statements below. The underlying algebra is one identity: four times the sum of the four correlators squared, minus S squared, is a sum of six squares (*chsh_gap_is_sum_of_squares*), so if the row sums of squares are at most one, S squared is at most eight (*tsirelson_squared*), and the point one over root two reaches it (*tsirelson_tight*). That's Tsirelson's bound as *algebra*, with no Hilbert space (the mathematical home of quantum states) in it.

Here is the chain as three statements. The derivations are in Appendix F; this is what they prove.

Definition 15.1 (Correlators from the counts; the pinned matrix). For eight counts and a question pair xy , let $E_{xy} = (\text{same}_{xy} - \text{diff}_{xy}) / (\text{same}_{xy} + \text{diff}_{xy})$, read as 0 when both are 0. For reals E_{xy} , $M(E)$ is the symmetric five-by-five matrix on I, A_0, A_1, B_0, B_1 with ones on the diagonal, E_{xy} in the cells (A_x, B_y) and (B_y, A_x) , and zero everywhere

else (Definition F.5). M is PSD when $v^T M v \geq 0$ for every real vector v , and $S(E) = E_{00} + E_{01} + E_{10} - E_{11}$.

Theorem 15.1 (The integer check implies the pinned matrix is PSD). *If eight counts pass the seven integer facts, then $M(E)$ for their correlators is symmetric and PSD.*

Proof. The four positivity facts make every denominator positive, so each E_{xy} is D_{xy}/N_{xy} , and dividing the other three facts by positive squares turns them into the three contraction inequalities (Lemma F.15). Contraction is equivalent to PSD of $M(E)$ by a Schur complement on the four-by-four block (Corollary F.13). $\square$

Checked as `column_contractive_check_witness_sound` with `npa_psd_iff_column_contractive` (Theorem F.16 and Corollary F.13 in the appendix).

Theorem 15.2 (PSD forces the Tsirelson bound). *For all real $E_{00}, E_{01}, E_{10}, E_{11}$: if $M(E)$ is symmetric and PSD, then $S(E)^2 \leq 8$.*

Proof. A PSD matrix with unit diagonal has every three-by-three principal minor at least zero. On rows A_0, B_0, B_1 that minor is $1 - E_{00}^2 - E_{01}^2$, and on rows A_1, B_0, B_1 it is $1 - E_{10}^2 - E_{11}^2$, so the four squares sum to at most 2. The identity $4 \sum_{xy} E_{xy}^2 - S^2 = \text{a sum of six squares}$ then gives $S^2 \leq 4 \cdot 2 = 8$. $\square$

Checked as `npa_psd_implies_tsirelson_bound`, with the row bounds `npa_psd_zero_marginal_implies_row_bounds` and the identity `chsh_gap_is_sum_of_squares` (TsirelsonFromAlgebra.v); this is Theorem F.14 in the appendix.

Corollary 15.3 (A passed check means the bound). *If eight counts pass the seven integer facts, their correlators satisfy $S^2 \leq 8$.*

Proof. Theorem 15.1 supplies positive semidefiniteness of the pinned moment matrix for these four correlators. Theorem F.14 in the derivation appendix applies to every such matrix and gives $S^2 \leq 8$. Its quadratic form calculation is printed there. $\square$

Remark 15.1 (Sound, not complete). The pinned matrix is stricter than the quantum question it is usually read against. PSD of $M(E)$ gives the correlators a PSD completion, since the zeros are one choice of the two free entries (`zero_marginal_implies_elliptope`, `ElliptopeCompletion.v`). The converse fails. The all-ones plan has a PSD completion (`deterministic_strategy_elliptope`), and every deterministic plan is a quantum strategy, a standard fact cited and not proved here; yet its minor on rows A_0, B_0, B_1 is $1 - 1 - 1 = -1$, so $M(E)$ is not PSD there, and a tally of one matching round per pair gives $A = 1 - 1 - 1 = -1$ in the integer check, so the check fails. The second failure is now a named theorem on the machine: the tally of one

matching round per pair is refused forever (Proposition 15.6). So a pass proves the bound and a failure proves nothing about the tally. The free tally from the start of this section is exactly such a tally: it reaches $S = 2$ and the check throws it out.

None of that algebra is new, and I don't want it to read as if it were. Tsirelson's bound has been formalised before, in Isabelle/HOL by Echenim and Mhalla and in Lean's Mathlib, and Navascués, Pironio and Acín already analyse this very matrix by Schur complement and discuss the level above it. What I think is mine here is smaller: an integer decision procedure on the counts, and a checked link from "the check passed" to PSD to the bound.

15.6 The check, earned on the small machine

Now put it on the machine. A check on the small machine is a property: a checker, which is a little program that says yes or no, a meaning, which is a statement, and a proof that the two agree. The version of the small machine with its property language left open (Section 14, Theorem 14.21) takes any property that comes with those three things, and for every one of them a raised flag stands on a passing CHECK of one claim, a COMMIT of the same claim with nothing touched in between, and a CERTIFY, for three. So here's the plan for the game. Pack the eight counts into one counter, as a list, the way the sorted-list machine packs a list into one number (Section 14). Let the checker be the seven integer facts read off the unpacked counts. And let the meaning be what the facts are for: every pair was sampled, and the pinned five-by-five is positive semidefinite. Then the proof the generic machine asks for, that the checker says yes exactly when the meaning holds, is the chain above with the denominators cleared in both directions, and a raised flag on that machine means: these counts, at this edition of their counter, passed the check, somebody committed to that with nothing changed underneath, and the correlators they give satisfy $S^2 \leq 8$. Here is that, stated exactly, and it's stronger than the chain above in one way I want to say once and plainly. The earlier theorems said that a passed check makes the pinned matrix PSD. The check is also complete for it: it passes *if and only if* every pair was sampled and the pinned matrix is PSD, because clearing the denominators works in both directions. The statement is over the host machine of Definition 14.8, the small machine with a counter for every number, with its property language filled in by one property. A CHECK reads one counter, which is why the eight counts have to travel in one.

Definition 15.2 (The packed tally, and the property PCHSH). Read a whole number as a list in the list code (0 is the empty list, and $2^x(2y + 1)$ is x followed by the list y), and take as its *tally* the first eight entries, in the order $\text{same}_{00}, \text{diff}_{00}, \text{same}_{01}, \text{diff}_{01}, \text{same}_{10}, \text{diff}_{10}, \text{same}_{11}, \text{diff}_{11}$, a missing entry reading 0. Every number is some tally. The property PCHSH has the checker "the tally passes the seven integer facts" and the meaning "every question pair was sampled, and the

pinned matrix $M(E)$ of the tally's correlators is symmetric and PSD."

Theorem 15.4 (The check is exact). *(1) The seven integer facts hold of a tally if and only if every pair was sampled and $M(E)$ is symmetric and PSD. (2) For every whole number v , the checker of PCHSH says yes at v if and only if the meaning holds of v 's tally. (3) Reading the number that packs a tally gives that tally back. (4) If the facts hold, the score of the tally satisfies $S^2 \leq 8$ and $|S| \leq 2\sqrt{2}$.*

Proof. (1) The forward half is Theorem 15.1. For the backward half, PSD gives the three contraction inequalities (Corollary F.13); with every N positive, multiplying them through by the squares of the denominators gives $A \geq 0$, $B \geq 0$ and $C^2 \leq AB$, which is Lemma F.15 read the other way. (2) is (1) with the packing, since every number is a tally. (3) is the code and its decoder being inverse. (4) is (1) forward, then Theorem 15.2. $\square$

Checked as `small_chsh_check_iff`, `small_chsh_clear_denominators`, `small_chsh_psd_iff_contractive`, `small_chsh_psd_tsirelson`, `small_chsh_check_tsirelson`, `small_chsh_eval_iff` and `small_chsh_tally_of_code` (`SmallChshCheck.v`, `SmallChshMachine.v`). Part (4) is one checked theorem now, not a composition of two. The old remark that no single theorem states the two together is out of date.

Theorem 15.5 (A raised flag stands on the check). *On the machine above, let t be a trace from a clean start s_0 that ends with the flag up. Then, for some counter c ,*

$$t = \text{pre} \cdot \text{CHECK PCHSH } c \cdot \text{mid}_1 \cdot \text{COMMIT PCHSH } c \cdot \text{mid}_2 \cdot \text{CERTIFY} \cdot \text{post},$$

and with v the value of c after pre: the tally of v passes the seven integer facts; every pair was sampled and the pinned matrix is symmetric and PSD; its score satisfies $S^2 \leq 8$ and $|S| \leq 2\sqrt{2}$; and c still held v when the COMMIT ran. The same holds for any stored program: if its run from a start raises the flag, the trace it executed has that shape. Conversely, for a counter c holding the packing of a tally t , the three instructions CHECK PCHSH c , COMMIT PCHSH c , CERTIFY raise the flag if and only if t passes the check, and then the ledger reads exactly 3; if t fails, the trap goes up at the CHECK and no number of further steps raises the flag.

Proof. A raised flag from a clean start has the earned chain (Theorem 14.9, which holds for this machine as for the small one): a passing CHECK, then a COMMIT of the same claim with the counter's edition unchanged, then a CERTIFY. A passing CHECK of PCHSH at c means the checker says yes at c 's value, which is the meaning by Theorem 15.4(2), and the bound is (4) of the same theorem. An unchanged edition means an unchanged value. For the converse, the CHECK passes exactly when the tally passes; if it does, COMMIT and CERTIFY find their facts and the three acts cost

three; if it fails, the trap latch goes up, and from a trapped core no instruction changes anything (Theorem 14.10). $\square$

Checked as `small_chsh_flag_implies_tsirelson`, `small_chsh_program_flag_implies_tsirelson`, `small_chsh_chain_certifies_iff`, `small_chsh_tally_certifies_iff`, `small_chsh_tally_certifies` and `small_chsh_tally_refused_forever` (`SmallChshMachine.v`); the first is printed in Section E. That's the one thing the machine does with the check, as the section's opening said: a raised flag can stand on it. Five tallies, each a theorem with its counts written out, show the check isn't vacuous and where its edges are.

Proposition 15.6 (Five tallies). *Counts are given as (same, diff) for the pairs 00, 01, 10, 11. (4,1), (4,1), (4,1), (1,4) has $S = 12/5$: the check passes, the flag rises, the ledger reads 3. (17,3), (17,3), (17,3), (3,17) has $S = 14/5 = 2.8$, just under $2\sqrt{2}$, with $S^2 = 7.84$: it passes, the ledger reads 3. (9,1), (9,1), (9,1), (1,9) has $S = 16/5$ with $S^2 > 8$: refused forever, whatever follows. (1,0), (1,0), (1,0), (0,1), the PR box, has $S = 4$: refused forever. (1,0), (1,0), (1,0), (1,0), the all-ones classical plan, has $S = 2$: refused forever.*

Checked as `small_chsh_demo_12_5_certifies`, `small_chsh_demo_14_5_certifies`, `small_chsh_demo_16_5_refused_forever`, `small_chsh_demo_pr_box_refused_forever` and `small_chsh_demo_all_ones_refused_forever` (`SmallChshMachine.v`). The last one is the check being stricter than the bound, as a named theorem: a perfectly legal classical play with $S = 2$ is turned away.

What the theorem doesn't say stays what it was. The tally is *placed*, not played: the theorem starts from a counter that already holds the packing, and no instruction plays a round, so a program can put any code in a counter for free. There are no marginals. The pinned matrix is stricter than the quantum set, failing says nothing, and there is no margin for sampling noise at Tsirelson's point.

A tally stays free. Each count is a counter, a round is an INC on one of them, and INC costs nothing, so a pile of trials costs nothing to write down; the packing into one counter is a program's business, and in the theorem the packed counter is simply there at the start. What costs is the claim about the pile. Statistics are free; the claim about them is what costs.

So here, in one paragraph, is what the check guarantees. If it passes, then every question pair was sampled, the four empirical ratios satisfy three polynomial inequalities, hence the fixed five-by-five is PSD, hence S squared is at most eight. It does *not* guarantee that the tally came from honest play (a program chooses its own trial inputs and can tally whatever it likes for nothing, including four rounds that put S at four). It does not guarantee any marginal is zero (no marginals are recorded and none are tested). It does not guarantee any quantum state exists, or that the correlators lie in

the quantum set rather than a larger fence. And it fails the tally of every single plan, honest classical play included. It has no margin for sampling noise either. Honest play at Tsirelson’s point aims at correlators of plus or minus one over root two, and there every pair of squared correlators the check adds up comes to exactly one, the edge of what it accepts. So an honest tally aimed at that point sits right on the line, and ordinary luck over a finite number of rounds pushes it over a good share of the time, and the check fails. That’s my arithmetic about where the point sits, not a theorem in the files. The check certifies a *shape* of the tally. What the shape means is a question for whoever supplied the rounds.

And notice which way the fence faces. The most practical use anyone has found for a Bell test, certified randomness, needs the opposite bound. Stefano Pironio and colleagues (2010) showed that a Bell violation, with settings the devices can’t choose, certifies that the outputs are random, and Rotem Arnon-Friedman, Frédéric Dupuis, Omar Fawzi, Renato Renner and Thomas Vidick (2018) made the accounting tight with what they call entropy accumulation. A string printed in advance passes no such test, so it’s the cleanest case I know, outside this book, of a check that can fail separating output that was established from output that was printed. That needs S bounded from below, with an error probability. This check bounds S from above and fails every single plan, so it’s a fence that keeps out the super-quantum and the fixed plan, not a nonlocality test. The deployed version exists, too: Gautam Kavuri and colleagues (2025) run a public randomness beacon whose outputs each trace back through a Bell test to a public record, the nearest working system I know to a certified claim written down for good. And records cut both ways there. Jonathan Barrett, Roger Colbeck and Adrian Kent (2013) showed that a device which keeps its own memory of earlier rounds can leak it through outputs it makes public later, so reusing a device can undo a guarantee the user thought was settled. A permanent record in the wrong box is an attack.

15.7 A rational stand-in for an irrational number

One number you’ll see in the files: 5657 over 2000, which is 2.8285, and two root two is 2.82842... It’s the four-decimal round-up, chosen so that rational code can test “above the Tsirelson bound” without square roots, and it’s safe because 5657 squared is 32,001,649 and eight times 2000 squared is 32,000,000, so the fraction sits above root eight by about seven parts in a hundred thousand (`rational_tsirelson_bound`). It’s used in a few rational theorems. It is *not* used in the integer check, and the real-number theorems use root eight.

15.8 Two more tests, as mathematics

The second rung: a nine-by-nine table on I , the four questions, and the four Alice-Bob *products*, with the correlators and five extra free numbers for three- and four-body entries. Its PSD is again equivalent to a contraction-like condition (`q1ab_psd_iff_column_contractive`), with a cascade of two Schur steps down to a four-by-four

that Sylvester’s criterion closes. Sylvester’s name on a test about matrices fits in a way I enjoy, because the word *matrix* is his. In 1850 James Joseph Sylvester used it for an “oblong arrangement of terms” out of which you cut square blocks and take their determinants, and *matrix* was already an old word for the place something grows from. That’s exactly what the test does: it cuts the top-left square blocks out of the big grid, one row deep, two, three, four (the leading minors), and asks each one’s determinant to be positive. Was he the first to state the test? The name doesn’t settle that; names stick to people for all kinds of reasons, and I’d treat this one as a lead. And a fact that’s easy to list without noticing what it means: with all five extra numbers at zero, the condition certifies a strict *subset of the classical set* (the ball where the four correlators’ squares sum to at most one), forcing S at most two (`qlab_check_at_gzero_implies_classical_bound`); it can’t certify anything above the classical line. To get a point above two you have to supply the extras (`qlab_certifies_a_superclassical_chsh_point`, S equals 2.4 with one extra at minus one half). Appendix F has the algebra.

The two sizes are one test, not two. A single positive-semidefinite predicate takes the size as a parameter, and the five-by-five and the nine-by-nine are it at two sizes.

Definition 15.3 (The size-generic PSD test). A *claim* is a top index n and a matrix M indexed by naturals. It is *realizable* when $M_{ij} = M_{ji}$ for all $i, j \leq n$ and $\sum_{i \leq n} \sum_{j \leq n} v_i M_{ij} v_j \geq 0$ for every real vector v . The CHSH claim of E is top index 4 with the matrix $M(E)$ of Definition 15.1; the level-1+AB claim uses top index 8 and the nine-by-nine matrix.

Corollary 15.7 (The level-1 test is the size-five case). *For all real $E_{00}, E_{01}, E_{10}, E_{11}$: the three contraction inequalities hold if and only if the CHSH claim of E is realizable. The same holds at size nine for the level-1+AB contraction condition and its claim.*

Proof. Realizability at top index 4 unfolds to symmetry and PSD of the five-by-five, by computing the finite sums, and that is equivalent to contraction by Corollary F.13. At size nine the same unfolding meets Theorem F.19. Nothing new is proved about the matrices; the point is that one predicate covers both. $\square$

Checked as `column_contractive_iff_general_realizable` (`GenRealizability.v`) and `column_contractive_qlab_iff_general_realizable`.

The third thing: the *elliptope*, the set of correlators for which *some* choice of the two missing entries (A_0 with A_1 , B_0 with B_1) makes the five-by-five PSD. Every one of the sixteen plans is inside it (`deterministic_strategy_elliptope`), mixtures stay inside (`elliptope_convex`), Tsirelson’s point is inside (`elliptope_tsirelson`), the PR box is out (`pr_box_not_elliptope`), and it’s strictly bigger than the classical set (`beyond_classical_elliptope`). There’s a decider for it with a soundness

proof, in two branches: a strict Sylvester check with the completion supplied, and a rational factorisation certificate of the completed matrix, checked in integers. What’s proved is that a pass puts the correlators inside the set and that no choice of inputs gets the PR box through (`elliptope_check_full_sound`, `elliptope_full_gate_never_accepts_pr_box`), and, by computation, that some named points pass, the all-ones plan and a point that the cited arc-sine boundary puts exactly on the quantum edge among them (`gate_accepts_all_ones`, `gate_accepts_pythagorean_boundary`). The file’s header also argues that every point with a rational PSD completion has such a certificate; that is a comment, not a theorem. It’s a decider with a soundness proof, as mathematics; none of the small machine’s properties is the elliptope check.

15.9 What isn’t proved

The arc-sine curve. That any of these sets equals what quantum mechanics can reach. That any state or measurement achieves 2.83. Those are the cited external results, and the files say so in their scope notes. Also: the finite-mixture theorem (`fine_theorem` in the source) proves only one direction, that a finite local model satisfies the bound; Fine’s actual theorem is an equivalence, so “Fine’s theorem is proved” would be wrong and I haven’t said it. And one more thing about tallies, which the guarantee paragraph above already used: a free tally is not thereby classical (the four rounds with S equal to four). The correlators (one, one, one, minus one) are the PR box, not a classical play; the classical vertex that fails the check is (one, one, one, one).

That’s the game, the number, the two walls, and the check. Last of the machine sections: what all this costs against physics.

16 The counter against physics

Section 4 did the physics argument once: on a finite machine a permanent flag merges states, Landauer’s principle (a named premise) prices merges, and the toll follows. I’m not going to do it again. This section is about what a machine’s counter, the small machine’s μ included, has to do with any of that, and the answer is: less than the word “Landauer” in a file name suggests, and I’d rather you hear the exact amount from me.

16.1 The calibration is a unit conversion

There’s a function in the source that multiplies the counter by a chosen energy per mark, and a theorem that a state with one mark, at the scale $kT \ln 2$ per mark, carries $kT \ln 2$ of energy (`calibrated_mu_landauer_energy`, proving the predicate `mu_landauer_calibration` for every k and T). Read that for exactly what it is: a unit conversion, true by definition, with the scale as a free input. Nothing in it picks the scale. The theorem would be just as true with the scale set to a dollar a mark, and I’d rather say that than let the name “Landauer” in it do any work. What *would* connect marks to joules is the premise in Section 4, `landauer_heat`, and that one is an assumption with a name that nothing proves. And even as a scale, $k_B T \ln 2$ a mark

is the floor for one particular squeeze, a halving. Corollary 4.9's floor for a flip that brings k states into a room already holding m is $k_B T \ln \frac{m+k}{m}$, and for one newcomer into a big room that's far below a halving, roughly $k_B T / m$. Price every mark at $k_B T \ln 2$ and you can overstate the floor of a single narrow flip by a factor that grows with m .

Remember the constant from Boltzmann's grave in Part II, the exchange rate between a count and heat? It has a newer chapter, and it makes the point sharper than I can. The k_B in $k_B T \ln 2$ turns a temperature into an energy, and for most of its history it was a measured number, with error bars, pinned down by careful experiments. Since 20 May 2019 it isn't. That day the international system of units was redefined so that Boltzmann's constant is exactly 1.380649×10^{-23} joules per kelvin, by definition, and the kelvin is whatever makes that true. So the conversion from temperature to energy has been a convention since then, in the strictest sense: nobody can measure it wrong any more. What's still physics is everything around it, whether wiping a bit at a temperature really sheds that much heat, which is what the Lyon bead was measuring. The calibration theorem is the same kind of move as that redefinition. It fixes a conversion, and the physics, if there is any, is somewhere else.

Definition 16.1 (Energy at a scale). For a real number c (joules per mark) and a state s , the energy of s at scale c is $\mu(s) \cdot c$. The *Landauer calibration* at k_B and T is the statement that every state with $\mu(s) = 1$ has energy $k_B T \ln 2$ at scale $k_B T \ln 2$.

Theorem 16.1 (The calibration is a unit conversion). (1) For all real k_B and T , the Landauer calibration at k_B and T holds. (2) There are two energy readings of the state, at scales 1 and 2, that disagree on every state with one mark.

Proof. (1) Substitute $\mu(s) = 1$: $1 \cdot k_B T \ln 2 = k_B T \ln 2$, by ring arithmetic, for any k_B and T , positive or not. (2) At $\mu(s) = 1$ the two readings are 1 and 2. $\square$

Checked as `calibrated_mu_landauer_energy` (`PricingPhysicsAudit.v`), which proves `mu_landauer_calibration` (`PricingPhysicsTarget.v`) for all k_B and T , and `mu_has_no_intrinsic_joule_value` (`PricingPhysicsAudit.v`). The physics is the premise of Section 4, restated here once in the form the next theorems test. In words first: a *dissipation* hands each instruction a whole number of heat units, each unit worth $kT \ln 2$. Landauer's principle, in that currency, says an instruction that takes at least one state where some yes-or-no property is false to a state where it's true, and never sends a true one back to false, dissipates at least one unit. The calibration says no instruction dissipates more units than the counter charges it. One word about that word: "dissipation" here means heat handed to the surroundings, not waste, in the sense Section 4 took from Sagawa: a bill for where the forgotten bit goes, not a tax on doing it badly. Two premises, nothing proved about either, and the only question is whether they can sit in the same room:

Premise 16.2 (Landauer, in units of the counter). A *dissipation* is a function δ from instructions to naturals, counted in units of $k_B T \ln 2$. Landauer’s principle, in this vocabulary (Premise 4.5 of Section 4): every instruction that collapses a yes-or-no property has $\delta(i) \geq 1$, where i *collapses* P when some state with P false is sent to a state with P true and every state with P true is sent to one with P true. The *calibration*: $\delta(i) \leq \text{cost}(i)$ for every instruction. Neither is proved anywhere; the theorems below ask whether they can hold together.

16.2 The premises have to be able to hold together

Here’s the one piece of this section that I think is good physics discipline, and it’s a consistency test. Suppose you want to claim the counter tracks dissipation. You need two premises: that a positively charged step dissipates something positive, and an upper calibration that bounds dissipation by the charge. Can those two hold at once? There’s a theorem saying exactly when (`calibrated_model_exists_iff_positive_cost`): a model satisfying both exists precisely when every operation you’ve chosen to count (the statement below calls those *eligible*) costs at least one. I should say how cheap that is. The model the proof builds takes the dissipation to be the cost itself, so the theorem shows the two premises don’t contradict each other, not that the counter tracks any dissipation. Still, it does bite on a schedule: if you count a zero-cost step, no such model can exist. The one place the tree shows the premises holding is a one-bit reset, where resetting costs one and idling costs nothing (`bit_model_has_satisfiable_calibration`).

Now point the test at the small machine of Section 14, counting every step that collapses some yes-or-no property, and it fails. Its free instructions merge states (the last item of the list at the end of that chapter has the pair: two cores that one DEC sends to the same core, for nothing), so a step that collapses the classes of “the line number is j ” costs zero there, and (1) below says no dissipation can charge it one and stay under its cost. That’s my reading of the instruction table, not a separate theorem. The machine prices the certifying chain and leaves its other merges free, so the premise pair holds for the steps the toll is about and fails for the whole instruction set. That’s the honest shape.

Theorem 16.2 (When the premises can hold together). (1) *For any set of operations, any predicate “eligible” on them, and any cost, a dissipation with $\delta(i) \geq 1$ on every eligible i and $\delta(i) \leq \text{cost}(i)$ on every i exists if and only if every eligible operation costs at least one.* (2) *On a one-bit machine with a reset of cost 1 and an idle of cost 0, taking “eligible” to mean “sends two different states to one”, such a dissipation exists.*

Proof. (1) If such a δ exists, $1 \leq \delta(i) \leq \text{cost}(i)$. Conversely, take $\delta = \text{cost}$. (2) The reset is the only operation that sends two states to one, and it costs 1; apply (1). $\square$

Checked as `calibrated_model_exists_iff_positive_cost`, with the definition `calibrated_positive_model`, and `bit_model_has_satisfiable_calibration` (`ObservationPolicy.v`).

But now the history escape from Section 4, pointed at a machine. Keep the previous state in a history list (push it onto the front each step; below, $s :: h$ is the list h with s put on the front) and every fixed-instruction step becomes injective, while the step you can see stays the same; the line number can collapse to one place while the bigger state keeps its past. The free merge, lifted to a machine with history, isn't a merge. So "this step merged, therefore it dissipated" is not a sentence about the visible step alone; it's a sentence about the whole state including whatever history is retained, and a physical model has to say what's retained. The permanent-certificate theorem in Section 4 supplies that for the certifying step on finite memory. For the rest of a machine's steps it's unsaid.

Theorem 16.3 (Keeping history makes every step injective). *Let the history step send a pair (s, h) , a state and a list of past states, to $(\text{apply}(s, i), s :: h)$. For any step function, the history step with a fixed instruction is injective, and its first component is the original step.*

Proof. Undoing the history step, by popping the list, returns the old pair, so two pairs with the same image are equal. $\square$

Checked as `retained_history_step_injective` (`ObservationPolicy.v`).

16.3 The calorimeter protocol: a specification, not a measurement

Here is how one *would* measure this, and it's here because it makes the boundary exact. Take a two-state register with energies zero and a gap, a population p in the excited state, and the simplest possible rate equation for how p changes in one time step (a rate for climbing into the excited state, a rate for falling out of it, and the length of the step). For the canonical reset (population one half to zero, with the rates chosen so the equation closes), the heat delivered to the bath is exactly half the gap (`canonical_reset_heat_exact`). Choose the gap to be twice $kT \ln 2$ and that heat is exactly $kT \ln 2$ (`selected_gap_gives_landauer_heat`).

Read that twice, because the temptation is to read it as a derivation of Landauer's number and it's the opposite. The gap was *chosen* to make the heat come out as Landauer's number. The same populations and rates with half that gap give half the Landauer value, and there's a theorem exhibiting identical population dynamics with different heat (`master_equation_does_not_fix_heat_scale`). The rate equation fixes the *shape*; the energy scale is a selected Hamiltonian (the physicist's name for the rule that says what energy each state has), and selecting it is not deriving it.

Definition 16.3 (The two-state protocol). For a gap Δ , the register's mean energy at excited population p is $\Delta \cdot p$. One step of the rate equation is

$$p' = p + \delta t (k_{01}(1 - p) - k_{10} p),$$

and the heat to the bath over a step from p to p' is defined as $\Delta p - \Delta p'$. The canonical reset takes $p = \frac{1}{2}$, $\delta t = 1$, $k_{01} = 0$, $k_{10} = 1$, ending at $p' = 0$, and counts one logical event.

Theorem 16.4 (The heat is fixed by the chosen gap). (1) The canonical numbers satisfy the rate equation. (2) For every Δ , the canonical reset's heat is $\Delta/2$. (3) For all k_B , T , the gap $\Delta = 2k_B T \ln 2$ gives heat $k_B T \ln 2$. (4) If $k_B > 0$ and $T > 0$, the gap $k_B T \ln 2$ gives heat strictly below $k_B T \ln 2$. (5) There are two different gaps for which the same canonical populations satisfy the rate equation and the heats differ.

Proof. All five are field arithmetic on the definitions: $\frac{1}{2} + 1 \cdot (0 - \frac{1}{2}) = 0$; $\Delta \cdot \frac{1}{2} - \Delta \cdot 0 = \Delta/2$; and $\ln 2 > 0$ for (4). For (5) take gaps 1 and 2. $\square$

Checked as `canonical_reset_satisfies_master_equation`, `canonical_reset_heat_exact`, `selected_gap_gives_landauer_heat`, `smaller_gap_refutes_unconditional_landauer_floor` (given $k_B > 0$ and $T > 0$) and `master_equation_does_not_fix_heat_scale` (`CalorimeterProtocol.v`), with the definitions in `CalorimeterProtocolTarget.v`.

What ties it to a machine: the protocol counts one logical event (the file defines that count as the constant one), and one is what the small machine's CERTIFY charges, the last of the three paid acts behind a raised flag (Section 14). That's a match of two numbers, both one by choice, and nothing more. There is no theorem mapping the protocol's two register states to any machine state. The file's "heat to the bath" is the register's mean energy before minus after, so it also assumes the bath receives what the register loses, and no Coq says that. So a device test would have to measure the gap, the populations, the rates, and the bath heat independently; separately establish thermal admissibility and detailed balance (that the up and down rates stand in the ratio the temperature dictates); and separately establish that the device's physical reset is the machine's certify event. The protocol is an executable experimental specification. It is not a measured device, and it is not an unconditional conversion of marks to joules, and anyone who tells you otherwise is reading a file name.

16.4 What permanence adds to Landauer: nothing new, said plainly

Here's a fair question: is there a prediction the permanent-certificate theorems make that Landauer's principle alone doesn't? No, and the reason is short. The smallest case, one certified state and one flipping in, is "set a bit to one whatever it held," which is the reset Landauer described and Bérut measured; Landauer's own name for it is "restore to one." The general bound, $kT \ln((m \text{ plus } k) \text{ over } m)$ for squeezing m

plus k equally likely states into m , is Landauer’s formula for that squeeze. What the permanence theorems add is the *derivation*: which premises about a machine (finite, permanent) make the certifying step a reset, so that Landauer applies to it. That’s a result about the machine, not about physics, and it’s the right size.

16.5 What this section is

A calibration that’s a premise; a consistency check the premises pass on the steps the toll is about and fail on the whole instruction set; a history that makes any step injective; and a measurement protocol that’s a specification with a selected scale. The physics in this book is in Section 4, as a named premise with its critics named. Here there’s accounting, and I’ve tried to call every line by its right name. That’s the end of the machine. What’s left is what it took to check any of this, and where the checking stops.

Part V

Checking it, and its edge

17 What Coq is, and why

Coq is a proof assistant, which is a phrase I had to go and look up. Underneath the phrase it’s a programming language, but one whose type system is strict enough to do a thing ordinary languages can’t: you write a proof down as a program, and the compiler refuses it unless the program genuinely matches the statement you claimed. You can’t talk it into a maybe.

The name is a joke three ways at once, and it’s worth knowing before you say it out loud in English. *Coq* is French for rooster. It’s also a nod to Thierry Coquand, who worked out the logic it started from with Gérard Huet, and to that logic’s initials, CoC, for the Calculus of Constructions; and there’s a French habit of naming research tools after animals. The English slang reading did what you’d guess. By the development team’s own account, some people, women students especially, ended up never learning the tool because of its name. So in March 2025 it got a new one, the Rocq Prover, after Rocquencourt, the Inria site where it was first built, and after the Roc, the giant bird of legend, so the bird stayed. Everything in this book was checked with Coq 8.18, from before the rename, so that’s the name I use. The rooster changes nothing about what got checked.

17.1 Types as claims

Here’s the idea, and it’s one idea. In a normal language a type says what kind of value a variable holds: in Python, “ x is an int” means x is a whole number. In Coq, types are *claims*. The type “ n is greater than zero” is the type of *proofs* that n is posi-

tive. A function from “ n is greater than zero” to “ n plus one is greater than zero” is a proof that if n is positive, n plus one is too, and the compiler checks that the function really has that type, the way it would check that a function from integers to strings really returns strings. This correspondence between proofs and programs has a name, Curry-Howard, after two people who noticed it separately decades apart, and Coq’s foundation, the Calculus of Inductive Constructions, from Coquand, Huet and Paulin-Mohring in the late 1980s, is a type system built to exploit it. The “inductive” part means you define your own recursive types (numbers, lists, trees, machine states) inside the system instead of having them handed down as primitives, so the foundation is explicit; nothing is smuggled.

“Decades apart” undersells it, and the gap is worth a side trip. Haskell Curry noticed in 1934 that the type of a function from A to B reads like the claim “ A implies B ,” and that the types his functions could have lined up with the claims you can prove; he and Robert Feys pushed it further in 1958. William Howard wrote the sharper version in 1969, and then it went around as a photocopied manuscript until 1980, when it was finally printed in a collection of essays for Curry. So the idea under the tool this book is checked with spent its first eleven years as a stack of photocopies. Is that the bottom? No. Reading a proof of “if A then B ” as a method that turns any proof of A into a proof of B is older still: the intuitionists had it in the 1920s and 1930s, Brouwer, Heyting and Kolmogorov, which is why people also call it the BHK interpretation. Where it really starts I can’t tell you. It changes nothing in what the checker checks. It changes how I’d tell the story: nobody sat down one day and invented “proofs are programs.” It kept getting noticed, by people coming from different directions, until somebody wrote it down plainly.

The compiler checks whether the types line up. If your proof has a hole and you write “trust me,” it rejects you. But here’s the part people skip, and it’s the part that matters: the compiler checks that the proof matches the *statement*. It does not check that the statement says what you *meant*. Hand it a statement that means nothing and it’ll prove it for you, three characters, no complaint. So a proof being accepted is not the same as the thing being true. It means this proof matches this statement, and whether that’s the right statement is on me, not the machine. Section 3 said a short version of this and I’m saying it again because every section since has leaned on it: the checked names in this book are a promise that the *proof* is there, and the sentences around them are my claim about what the statement *says*, and the second can be wrong while the first is fine, which is why so many sections of this book stop to say exactly what a statement’s hypotheses are.

17.2 Why I used it

Not to make anything true. I used it because I don’t trust my own eye to catch a gap in an argument I want to believe. I spent months convinced by my own reasoning, pushed the proof through, and watched a key step come apart in my hands. The machine caught where I’d fooled myself. That’s what it’s for. Gap-catcher. Not the

proof. The proof is the argument, and an argument has to close in a way a person can follow, or it hasn't closed at all, which is why this book exists in prose and not as a list of theorem names.

Two precedents show that a proof assistant is a serious tool and not a toy. CompCert, Xavier Leroy's C compiler, written and proved correct in Coq: the assembly it emits behaves as the source does, for programs with defined behaviour. And Georges Gonthier's 2005 proof of the four-colour theorem, a century-old problem whose first proof (in 1976) needed a computer to check thousands of cases and was distrusted for exactly that reason, redone in Coq so that the *checker* was the small, auditable thing and the cases were the program. That's de Bruijn's criterion applied to mathematics itself: the thing you have to trust is the small kernel, not the big proof. The name is Henk Barendregt's, after Nicolaas de Bruijn, whose Automath project, begun in 1967, kept the part of a proof checker you have to trust small: a system meets the criterion when its proofs can be checked by a small independent program that a person could read.

The 1976 proof deserves its own side trip. Kenneth Appel and Wolfgang Haken did it at the University of Illinois, and the mathematics department changed the slogan on its postage meter to "Four colors suffice." The slogan it replaced had been on the department's mail since the middle 1960s, and it said that $2^{11213} - 1$ is prime, a prime Don Gillies, in the same university's computer science department, had found. Think about that for a second. For years on end a mathematics department stamped its mail with results nobody could have checked by hand, first one and then the other, and both were right. So was the distrust of the 1976 proof wrong? I don't think so. The worry was never that the computer lied. It was that nobody could read what had been checked, and a check you can't look at is hard to lean on. Gonthier's answer was de Bruijn's: make the part you have to read small enough to read.

17.3 What "zero Admitted" means

"Zero Admitted" is literal. Nowhere in the proof tree did I write "trust me" and walk on. Every step the checker could see, it saw, end to end, with no "and the rest is obvious." A hygiene fact about the gap-catcher, not a verdict on the argument: it says I didn't let myself skip a step where the machine was watching. It does not say the statements are the right statements.

What it doesn't mean, listed. It doesn't mean the theorems prove physical reality; they prove things about the machine as formalised, and whether that describes the world is answered by experiments (Section 4, Section 16). It doesn't mean Coq's own foundation is proved consistent; it's believed consistent, and that can't be proved from inside, by Gödel (Section 11), so the foundation is trusted, and "trusted" has substance: it's the same kernel CompCert and the four-colour proof stand on, and a bug in it would take all of them down together, which is a kind of safety in numbers. And it doesn't mean no axioms: the proofs use a handful of standard-library assumptions,

and the assumption audit names them: functional extensionality on dependent functions (two functions that agree everywhere are equal); two decidability principles that come with the standard library's real numbers, used by the matrix proofs behind the game check; and, in some theorems, the law of excluded middle (every proposition is true or false). The count, from the full audit: the full-corpus probe covers 5,369 named theorems across 308 files: 4,882 close under the global Coq context outright, and 487 use only Coq standard-library assumptions. Zero project-local axioms appear in any of the 5,369 dependency trees. The small machine's files sit outside that probe, under `minimal/`: the small machine, its version for any property language, the simpler host, the definition of Thiele-complete and its window theorem, the host machine with a register for every whole number, the numbers that write a guest program down, and the pigeonhole (Section 14). Each ends by printing the assumptions of every headline theorem in it, and the test suite compiles every one of them and fails unless every one of those lines says the theorem is closed under Coq's own rules, with no axiom at all. A lemma a closed theorem uses is closed with it. So for those files "nothing assumed" means exactly that, and the evidence is the files' own printout, not the big audit. The universal program's own files, which lean on the vendored library's counter-machine blocks, sit in the kernel tree with the other link files, and they also end by printing their headline theorems' assumptions; so do the files of the priced host, the compiler from computable functions and the theorem for every computably presented Thiele machine. In plain words: every theorem in the audit either needs nothing at all or leans only on those standard-library assumptions, none of mine. Those numbers belong to the audit run that produced them; the shape doesn't move between runs, the counts do.

17.4 What you're trusting, in one place

If you accept every checked theorem in this book, here is what you're still taking on trust, all of it, so you don't have to collect it from five chapters.

1. The statements. The checker confirms a proof proves its statement. It doesn't confirm the statement says what my sentence about it says. That's why the book prints the important ones in full, premises and all, and why Appendix E prints the Coq text of the load-bearing ones beside a line on whether my sentence says exactly that, more, or less. A machine check supplements that reading.
2. Coq's kernel and its foundation, and the standard assumptions above.
3. The tools under Coq. Coq's own program, the compiler it was built with, the operating system it ran on and the processor under all of them are trusted. This is the list Ken Thompson's lecture from Section 11 was about. I didn't write Coq, or OCaml, or any of the rest, so by his moral I can't fully trust any of them, and neither can you. What I can do is keep the list short and print it, so you know exactly which tools you're being asked to believe have no spare key cut into them.

4. The tested parts. The test suite is evidence that the scripts around the proofs do what's claimed on the runs tried, not proof that they always do. Edsger Dijkstra had a sentence for this, and the version in his 1976 book starts by admitting he'd already said it many times: program testing "can be quite effective for showing the presence of bugs, but is hopelessly inadequate for showing their absence." (I'm quoting it as John Harrison quotes it, from that book.) That's why this item says *tested* and not *proved*.
5. The physics. Anywhere a theorem has Landauer's principle or a calorimeter in it, the physical premise is named and assumed (Section 4, Section 16), never derived.
6. Whoever sets the start and whoever reads the record. Every guarantee about the record is about programs run from a clean start on a step that does what its definition says. The operator who chooses the start, a reset, a device running the machine, and anyone reading a report they didn't watch being made are outside it (Section 20 lists the whole fence).

17.5 Thompson's knife, again

Every link on that list is a place where Ken Thompson's compiler from Section 11 could sit unseen: a translator that plants something and re-plants it when it translates itself. Coq's own binary, the compiler that built it, the operating system under both. I don't believe any of them does. I can't prove it, and the discipline of this section is to say so rather than to let "formally verified" cover a chain whose bottom is trust.

And a planted knife isn't the only way a trusted link goes wrong. Sometimes nobody plants anything and the thing at the bottom is just wrong. In 1994 Thomas Nicely, a mathematics professor at Lynchburg College who was computing with twin primes and their reciprocals, found that his new Intel Pentium got some divisions wrong. The example everybody passes around is 4,195,835 divided by 3,145,727, which the bad chips got wrong in the fourth decimal place. Intel's explanation was missing entries in a table inside the divider: five of them, it said, dropped by a script. Intel set aside about \$475 million to cover it.

Push on the five, because that's the part that got retold. In 2024 Ken Shirriff went to die photographs of an affected chip, read the table's layout off the silicon, and counted sixteen missing entries, left out by a mistake in the table's mathematical definition, of which only five ever mattered. So the story that got repeated for thirty years had the effect right and the cause off, and it took somebody reading the actual silicon to say so. Floating-point arithmetic went on to become one of the places where formal verification earned its keep in industry; John Harrison, who did that work at Intel, brings up this very bug in a 2007 survey of it. Does any of that touch this book? Only at the bottom of the list above. Every proof here was checked by a program running on some processor, and nobody here has read that processor's tables off the silicon either. That's why the processor is on the list as trusted, and not as something kinder.

17.6 The auditor on top

Beyond the type checker, the project has a custom proof auditor I called the Inquisitor, because a thing that reads every proof looking for ways you’ve cheated deserves a name you won’t forget. It runs on every commit, and its source names about a hundred distinct rules. The main families: no holes (no Admitted, no admit, no give-up, no unproved axiom or parameter in project code); vacuity heuristics (no theorem whose conclusion is literally True or a tautology; no theorem of the shape “P implies P”; existence claims backed by a constant witness get flagged, and a flag is a reason to look, not a verdict, since some constructions like a fixed interpreter program are *supposed* to be fixed and get judged on what they compute); physics-definition heuristics (no physical quantity defined as a placeholder constant, no physics claim without an invariance condition); scope enforcement (the proof tree is split into tiers, the kernel tier may not import from research-side files, and every active file must reach the cost or semantics foundation through its imports, or it doesn’t get to claim the kernel’s pedigree); and comment hygiene (no unfinished or historical markers in maintained source). Its rules are heuristics. They point at places to look. They don’t prove a theorem means what it should, and they can’t promise every empty premise got caught.

So none of this is foolproof, the checker and the auditor included, and I don’t treat it as if it were. That’s the reason some of the theorems are runs. The worked examples in Part IV, the earned run that pays exactly three and the one that traps for good, the two guests run on the simpler host, the demonstration machine run on the universal host, are computed by Coq from the definitions, not described, and the tests are written to fail when a headline theorem quietly stops coming back closed. A proof can be hollow in a way nobody spots. A definition that has to produce the numbers on the page, run by the checker itself, is harder to fool, and when it disagrees, it tells you where to look.

17.7 Checking the companion formalization

The mathematical claims are the statements printed in this book, with their stated hypotheses and the arguments given here. Running software is an additional way to check their formal counterparts. It isn’t a prerequisite for reading the argument.

A reproducible check needs a fixed copy of the proof sources, dependencies and probes. Changing that copy changes the object being checked; a later software result cannot silently revise a theorem in this edition. The procedure below describes the companion proof build. With Python 3, GNU make, Coq 8.18 and its standard library and CSDP installed, run this from the root of a checkout of the sources at a fixed revision:

```
python3 scripts/reproduce_coq.py
```

That copies the sources into a fresh directory and works only from the copy: it builds the part of the vendored Coq Library of Undecidability Proofs the proofs use, regen-

erates the build file, rebuilds every proof from source, runs the contract probes that re-check the named theorems and their assumptions, and runs Coq’s independent checker over the result. It reuses no compiled proof from the checkout, and it writes its pass file only after every stage passes. A failed run resumes from the snapshot it captured, and a change to the sources needs a fresh run:

```
python3 scripts/reproduce_coq.py --resume \
  --output artifacts/reproduction/RUN_DIRECTORY
```

The run has to pass. Don’t stop at the pass: read the theorem types and their global dependencies, because a closed report doesn’t make a theorem’s own premises go away.

That’s the checker. What’s left is to tell you how to break what it checked.

18 How to break this

Every main claim in this book comes with the exact shape of the thing that would kill it. I went to the trouble on purpose. A claim you can’t break is a claim that wasn’t saying anything, and I’d rather hand you a loaded gun pointed at my own work than a sentence too vague to aim at. So these are written as instructions: here is the trace to find, the object to build, the theorem to exhibit. Build one and the proof tree falls, and I mean that as an invitation.

The invitation is an old one, and people have put prices on it. Donald Knuth gives the first person to find each error in his books one hexadecimal dollar, 0x\$1.00, which is \$2.56, deposited these days at the Bank of San Serriffe, a bank that doesn’t exist, since real checks turned out to be an open door for fraud. Why pay strangers to find your mistakes? His own page says it plainly enough: a careful reader makes the book better for everyone who reads it next. This chapter doesn’t come with a bank, real or made up. What it comes with is, for every claim, a description of the exact object that would make it fall, so you know before you start what counts as winning.

Before the list, the honest part about it. Most targets below are theorems Coq has checked, and knocking one of those over head-on would take a bug in Coq’s kernel. That can happen, but it isn’t where I’d put my money. The places a reader can actually win are aimed at me, not at Coq, and there are three. A sentence of mine that says more than the Coq statement it cites (Appendix E puts the load-bearing ones side by side to make that easy). A Coq definition that doesn’t capture the English word it’s named for: on the small machine, “is at least 0” is a check that can’t fail, and the step certifies it anyway, so “certified” there doesn’t mean “found something out” (Section 14 says so, and that’s the kind of gap to hunt). And a list of premises that only its own example satisfies.

18.1 The abstract claims (Parts II and III)

The universal floor (Section 3). Build a certification system that satisfies the one-step toll and give a run from uncertified to certified with total cost zero. `universal_nfi_any_substrate` falls. If you instead drop the toll, you’ve built a free-certification system, not a counterexample.

The toll from premises (Section 4). Exhibit a finite, duplicate-free list of every state, a certificate no step revokes, and an *injective* step that switches it on. `permanent_flip_is_not_injective` falls, and the derivation of the toll with it. An infinite state space, a revocable certificate, or an unpriced merge is not a counterexample; the file already builds one of each. For the squeeze bound, a finite permanent machine with a cost meeting the halving premise where one step switches k states on beside m certified ones and two to the cost times m is less than m plus k ; `permanent_flips_compression_bound` falls. For merge-or-revoke, a finite machine and an injective instruction that switches a reading on without switching it off anywhere; `flip_merges_or_revokes` falls. A move that merges and also revokes isn’t a counterexample to anything; it’s the three cards. For the eight-state machine, a move of it that merges two states and costs nothing; `fin_merging_priced` falls.

The bill in bits (Section 4). Exhibit a finite machine with a permanent certificate, a move that switches k states on beside m certified ones (k at least one), and a spread of probability over those m plus k states whose image under the move has entropy above log base two of m . `permanent_step_entropy_ceiling` falls. For the general form, a step and a spread under which two different states, each with some probability, land on the same state while the entropy doesn’t drop; `entropy_drop_pos_of_support_merge` falls. A heat measurement on a real device below what the bound asks doesn’t touch either theorem. It refutes the premise `landauer_heat` for that device, which is Landauer’s principle, and that’s a fight with physics, not with the counting. And a certifying step from a state already known that gives off no heat isn’t a counterexample: `known_state_flip_forces_no_heat` says that case owes nothing.

Learning is free (Section 5). Show that the four-state demon’s measure step merges two states, or that its observer’s candidate list doesn’t shrink. `observer_narrowing_can_be_free` falls. A different price schedule that charges the measure step is allowed and isn’t a counterexample; the claim is “not forced,” not “always free.”

The counter (Section 6; on the small machine, Section 14). Find a state of the small machine and an instruction such that the ledger after is anything other than the ledger before plus that instruction’s cost; the loud version is a ledger that goes down. `mu_conservation` falls. The uniqueness half, that only one bookkeeping starts at zero and follows the schedule, is argued in Section 6 and not checked; a reachable state where two such bookkeepings disagree breaks the argument there.

Windows (Section 7). Exhibit a window and a price computed from the observed before

and after that meets the floor and never overcharges, on a machine where two steps share their before and after pictures and only one certifies. `shadow_cannot_price_exactly` falls. A window that shows certification is not a counterexample; the theorem says exactly that such a window prices exactly. For the three escapes, a verifier of one of the three kinds that's wrong about the pair, under its own relation; the argument after Definition 7.1 falls.

Entitlement (Section 8). Produce a strict narrowing of a candidate list, with a distinguishing observation, that does not yield a strictly stronger membership test; the card argument falls. Or exhibit a run with a tree no deeper than its paid steps, covering the narrowing, whose counter rose by less than the index bits lost; Theorem 8.2 falls. On every Thiele-complete machine: a run from a clean start that ends with the record up, meets the hypotheses of Theorem 14.39, and has fewer record moves than the index bits it narrowed, or has no earned chain in it; `ent_representation` falls, and the weighted form (`ent2_weighted_representation`) falls the same way. A narrowing with a big drop and a small bill is not a counterexample unless a tree no deeper than the record moves covers it: the covering is a hypothesis, and `ent2_uncovered_claim_false` says why it has to stay one. On the host machine, a run that raises the record after more than sixteen CHECKS; `ent2_questions_cap` falls. On the small machine, a program of INC and DEC that runs fewer than m moves and tells $A = m - 1$ from $A = m$ by its line number and B ; `ent2_free_cannot_decide_fast` falls.

The latch (Section 10). Exhibit an extension of a base whose record is driven by the computation and never switches off, and which is not a latch on any event. `record_axis_is_latch_holds` falls. A record that switches off, or one driven by a hidden clock, is the file's own pair of counterexamples, not yours.

The diagonal (Section 11). Two routes. A decider for some substrate with a shortcut predicate whose flip the substrate can represent and which answers correctly on every program; the generic proof fails at a named step. For the nat family, a d that correctly decides the predicate of the substrate built from d . L has its recursion theorem by proof, so it gets a target of its own: a map on L 's terms, computable by an L term, with no term that behaves like its own image; `L_recursion_theorem` falls. On the host machine: a host program that decides an extensional property of host programs that some program has and some lacks, or a map of host programs that a host program carries out, with no host program that computes the same function as its image; `sm_host_rice` or `sm_kleene` falls. On the small machine: a decider for a property of what two-counter programs do from the clean start that some program has and some lacks; `sm_guest_rice` falls. A decider for such a property over every input of two-counter programs is not a counterexample to anything claimed, since that isn't proved either way. A map of host programs that a host program carries out, with no host program that stops when its image stops and agrees with it on register 0, the trap latch, the ledger, the flag, the number of facts and whether the channel is empty, is a

counterexample to `sm2_kleene_obs`. A map whose fixed point disagrees with its image on some other register is not one: no fixed point agrees on every register (`sm2_no_exact`).

The pointer conjecture (Section 13). It's a conjecture, so the three routes are its own: a deployed discipline pricing a non-proliferating event; a rival event proliferating as well as certification in a real ecosystem; an honest formalisation with independence defined where certification fails the test.

One concession goes with it, and I make it in full. The window results, the separation of the two runs, and the counter's uniqueness don't single out certification. Their proofs use only that the step carries something the classical picture doesn't determine, so the same shape of result holds for any such field. Nothing in Parts II to IV proves that certification is the one event a model is *forced* to price; Section 4 forces a price on every merge, certifying or not. That's the conjecture, and Section 13 is the case I make for it.

18.2 The machine claims (Part IV)

The small machine (Section 14). Four targets. From a clean start, a run that raises the flag without an earlier passing CHECK, COMMIT and CERTIFY of the same claim, with the counter's version unchanged between the CHECK and the COMMIT; `earned_certification_provenance` falls, and `earned_commitment_provenance` for a COMMIT alone. From a clean start, a trace that leaves a fact in the table nobody checked; `no_forging` falls. From a clean start, a run that ends certified for a total charge under three; `certified_run_min_cost` falls, and a run at exactly three is not a counterexample: CHECK "A is 0", COMMIT, CERTIFY from a start where A is 0 is one, and the same program from any other start is refused forever (`earned_run_check_can_fail`, `earned_run_refused_forever`). A check that can't fail certifying for three isn't a counterexample either; the step doesn't police trivial checks, and the book says so. A two-counter machine and a program whose run on the small machine disagrees with it step for step, or halts when it doesn't; `simulation_step` or `halting_correspondence` falls.

Thiele-complete (Section 14). A machine that charges at least one for every move and still meets the four clauses; `clock_not_thiele_complete` falls. A Thiele-complete machine and a function of its window that gives its record at the end of every run from every loaded state; `complete_no_record_oracle` falls. A clause none of the excluded machines needs: drop it and show the paid latch or the silent machine still fails, and `paid_latch_not_thiele_complete` or `silent_not_thiele_complete` has been proved by the wrong clause, which is worth a letter.

The universal program (Section 14.13). A program of the small machine and a start on which *U* halts and the guest doesn't, or the other way round; `universal_halting` falls. One on which the flags disagree, or the ledgers differ at a matching point; `universal_flag_iff` or `universal_ledger_exact` falls. A run of *U* whose raised

flag doesn't stand on a passing CHECK, then COMMIT, of one slot, then CERTIFY; universal_earned falls.

Every computably presented machine (Section 14.14). A computably presented Thiele machine and a start on which U_P 's flag goes up while the machine's reading never says yes, or the other way round; presented_universal_flag_iff falls. With the reading starting at no, a matching point where U_P 's ledger is more than two above the machine's; presented_universal_surcharge_le_two falls. A program on the host machine, from a clean start, that matches a raise costing less than three exactly; presented_universal_no_exact_below_three falls. A machine whose reading no program decides isn't a counterexample. It's outside the class, and the book says so.

The classical CHSH bound (Section 15). Exhibit four fixed bits and a tally consistent with that one plan, every pair sampled, scoring above two. chsh_violation_rules_out_locally_factorizable_coupling falls, and so does known mathematics. A randomised plan landing past two by luck doesn't count; the theorem is about one fixed plan, and Section 15's coin-flip example shows the luck.

The integer check (Section 15). The game check has two walls, and here's how to knock each one over. A correlator that passes the three contraction inequalities whose five-by-five is not PSD (forward), or a PSD five-by-five of that shape whose correlators aren't contractive (reverse); npa_psd_iff_column_contractive falls. A correlator whose pinned five-by-five is PSD (contractive, by the equivalence) with S squared above eight; npa_psd_implies_tsirelson_bound falls. The PR box and the classical mixture (1, 0, 1, 0) are *not* counterexamples to any of these. Both fail contraction by plain arithmetic (the table in Appendix F.4), which is what the theorems need them to do, not what would break them. On the machine: a raised flag from a clean start with no passing CHECK of the packed-counts property before it, or whose counts give $S^2 > 8$; small_chsh_flag_implies_tsirelson falls. A tally that passes the seven facts while its pinned matrix isn't PSD, or one with every pair sampled and a PSD pinned matrix that fails them; small_chsh_check_iff falls. The all-ones tally and the PR box being refused are not counterexamples; those refusals are the point.

The nine-by-nine (Section 15). A point and five extra numbers where the contraction-like condition holds and the nine-by-nine isn't PSD, or the other way round; qlab_psd_iff_column_contractive falls. A point where the condition holds with all five extras at zero and S is above two; qlab_check_at_gzero_implies_classical_bound falls.

The elliptope completion (Section 15). This one's a shape, so you're trying to push a point in or out of it. Three targets, one per theorem, plus the decider beside it. Signs a_0, a_1, b_0, b_1 , each squaring to one, and a vector on which the completed matrix of that plan goes negative; deterministic_strategy_elliptope falls. A correlator with some positive semidefinite completion and S squared above eight; elliptope

`pe_tsirelson` falls, and with it the claim that one completion can't stretch both Cauchy-Schwarz budgets at once. Real numbers x and y that make the PR box's completed matrix positive semidefinite; `pr_box_not_elliptope` falls. And for the decider that sits beside it: inputs on which it says yes while the correlators have no PSD completion at all (S squared above eight is one way to show that); `elliptope_check_full_sound` falls. The all-ones plan and $(1, 0, 1, 0)$ aren't counterexamples here either, from the other side: the completion admits them (`turing_point_elliptope` for the second), which is what the theorems require.

The five models (Section 12). Each fragment has a theorem with a named falsifier: a finality fork with nobody slashable; a gas schedule meeting floor and no-overcharge whose charging predicate differs from the commit predicate on some step; one of the RFC's own example proofs that the transcribed inclusion check rejects, or an input with leaf index equal to tree size that it accepts; a program the bounds checker accepts whose verification condition fails. Each kills one fragment. None of them, and I want this clear, says anything about the real system beyond the fragment.

The audit (Section 17). Run the assumption probe on any theorem the book cites and find a project-local axiom or parameter in its dependencies, and the claim of zero of them falls. For the small machine, its hosts and the universal programs, compile any of their files and find a printed assumption line that isn't closed. Run the Inquisitor and read what it says; it fails on high and medium findings. A new finding means somebody looks. It doesn't automatically mean a theorem is false.

18.3 What's mine and what isn't

This is not a claim to have been first. Appendix C does the sorting item by item, with sources: what was known, what's added, and what kind of addition it is. Here's the short form, and I'd welcome correction on any boundary.

Borrowed, with names. As the opening chapter warned, I usually had the hunch first and found these names after, and that order doesn't make any of it mine. Everything in Part I (every source is cited where it's used), the category theory (Eilenberg, Mac Lane) included. Minsky's two-counter machines and the Coq library that proves their halting undecidable and compiles computable functions into them (Forster, Larchey-Wendling and coauthors). Turing, Church, Kleene, Rice, Cantor, Gödel, the diagonal pattern, Lawvere's unification, and the mechanized versions of them in HOL4, Lean, and Coq (Norrish, Carneiro, Forster, Kunze, Smolka). Bell, CHSH, Tsirelson, Wehner, NPA, Landau, Masanes, Fine, the elliptope, and the machine-checked proofs of Tsirelson's bound that came before mine (Echenim and Mhalla in Isabelle, and Lean's `mathlib`). Landauer, Bennett, Szilard, Maxwell, and their critics. The potential method (Tarjan) and its mechanizations (Nipkow, Charguéraud and Pottier), Lamport's clocks, Floyd and Hoare. Gettier, Nacula, Cook, Hartley, Shannon, Rissanen. Zurek. The guarantee about every program: the reference monitor (Anderson), execution monitors (Schneider), and monitors checked at every instruction and verified

in Coq (Azevedo de Amorim and coauthors). The counting under the permanence results: Poincaré’s recurrence, as Zermelo used it. The seL4 and CompCert discipline. The facts under the permanence results are Landauer’s own example and Shannon’s bounds, and “set a bit to one whatever it held” is my way of saying his RESTORE TO ONE.

Formal reconstructions. The Coq development defines the state types, transition systems, real-matrix predicates, encodings, and proof terms this book uses. When I say a theorem is checked, I mean the stated proposition has a kernel-checked proof under its printed premises. That doesn’t make the underlying idea mine, and it doesn’t make it new.

What this project proved, inside its stated interfaces. The abstract floor, which is the potential method with one particular potential. The derivation of the toll from finiteness, permanence, and a named price, in counted states and in bits, with each premise’s counterexample built. The eight-state machine on which every one of those premises is a theorem. The window that can’t price the flag exactly, and the small machine’s own colliding pair. The latch form over any base, with its two counterexamples and converse. The small machine’s earned record (checked, then committed, then certified, about the same edition, nothing forged, three at least), its exact ledger, its universality through Minsky, the definition of Thiele-complete that every clock fails and the window blindness of every machine meeting it, the universal program U with its guest’s halting, flag and ledger matched exactly and its own flag earned, the priced program U_P that runs every computably presented Thiele machine with its reading latched, its flag earned on the guest’s own reading and its ledger exact up to an unavoidable surcharge of two, and the simpler host’s mirror, its isolation and its refusal to certify for free. The substrate diagonal and the L results, with Rice’s and Kleene’s theorems on the machine’s host and Rice’s from the clean start on the two-counter machine. The integer check, exact for the pinned matrix being PSD (and so a sufficient test for the bound, not a test for quantum behaviour), the equivalence with contraction, Tsirelson’s bound from PSD, which is known mathematics checked again here, and the check earned on the host, a raised flag standing on it. Structural entitlement for every Thiele-complete machine, with its worked members, its floor that takes no supplied tree and its proof that the covering can’t be dropped; the eight-state machine run by the small machine; the time a check saves; and the collision of every bare-transcript verifier on every Thiele-complete machine. All of it with premises printed. Of that list, the toll derivation as a rule on a ledger (Ouldrige and Wolpert priced the same merge in entropy first), the pricing form of the window fact, and the latch form are the only pieces I’d call new statements, and only in the sense that I didn’t find them stated where I looked. The fourth thing I’d name is built and checked rather than new in idea: a record that’s priced, and on the small machine earned, under the reference monitor’s every-program move, which is old and isn’t mine. Appendix C names the same four the same way, with the prior work beside each.

Interpretation, which is mine and not proved. That structure is what a computation establishes and the toll is the right thing to price. That certification is the pointer event. That the record axis “belongs in the step.” Those are proposals, with the arguments I’ve given, and the proposals are where I’d aim first if I were you.

19 The claim ledger

Here’s everything this book claims, one line each: what kind of claim it is, and where it lives. “Proved” means proved under the premises printed in that section, and nothing more. If a line here ever says more than its section does, the section is right and the line is wrong. “By unfolding” after it means the proof does little more than read the answer off the definitions, a case split whose every branch is a line: true, checked, and carrying almost nothing the definitions didn’t already say. If you’re counting results, count those differently; I’d rather mark them than let them pad the list.

Ledger is a good word for it, and it’s older than bookkeeping. Around 1400 a *ledger* was a book that lay permanently in one place, a big church breviary, say, from the old verb *leggen*, to lay; the account-book sense turns up in the 1500s. A book that stays put. That’s the job here too. The home sections are long and they wander, and this is the one place where every claim lies still, so you can come back and find it where you left it.

19.1 Claims about the model

First the model. Every line in this table is about any system that has the pieces the line names: a state and a step, plus whatever the result asks for, a reading and the toll, a counter and a price list, a window, programs. None of them is about my machine. The table runs over several pages, in reading order.

Here’s one way to use it, and it takes five minutes. Pick the sentence in this book you trust least. Find its line. Read the Kind column before anything else. Then go to the home section and check that the line says no more than the section does. If it says more, you’ve found my mistake, and the end of this section says what I’d like done about it.

Claim	Kind	Home
The early engine derived physics	not claimed; it ran a simulation	Section 1
Structure: relations, kept under chosen changes, carried by any stuff, relative to a frame	choice (my overlay of the sources, not something they proved together)	Section 2
Turing’s four parts have no slot for “these belong together”	not claimed; I looked and didn’t find one, which is weaker	Section 2

Claim	Kind	Home
Equivalence relation, partition	definition	Section 2, Definition 2.1
Sameness and sorting are one object	argued, not checked in Coq (textbook, Halmos)	Section 2, Theorem 2.1
A partition is the start, smallest by the count of names	choice	Section 2
Nobody prices vouching for a sorting	not claimed; only that I found no one who does	Section 2

Claim	Kind	Home
Certification system: states, moves, step, cost, reading	definition	Section 3, Definition 3.1
The toll	choice	Section 3, Definition 3.1
The universal floor: no to yes costs at least one	proved (premise: the toll)	Section 3, Theorem 3.1
The floor speaks about the world, or computing power	not claimed	Section 3
A computation is in scope just by being one	not claimed; someone has to supply the five parts and prove the toll	Section 3
On a finite machine, a permanent flip merges	definition; proved (no price premise)	Section 4, Definition 4.1, Theorem 4.1
Merge pricing	assumed (premise)	Section 4, Premise 4.2
Finite, permanent and merge pricing give the toll	proved (premise: merge pricing)	Section 4, Theorem 4.2
Each premise has a counterexample	proved (the strip, the coin, the stamp); “free” is read off the pictures	Section 4
A flip merges or revokes	proved	Section 4, Theorem 4.3
Forced to be priced exactly when it merges	definition; proved (premise: move names decidable)	Section 4, Definition 4.3, Theorem 4.4
Permanent records are forced; the converse fails	proved (the three cards)	Section 4, Corollary 4.5
Halving price	assumed (premise)	Section 4, Premise 4.4
The squeeze bound	proved (premise: halving price)	Section 4, Theorem 4.6

Claim	Kind	Home
An eight-state machine meets every premise, so the toll holds on it outright	proved	Section 4
The small machine runs the eight-state machine through a window, move for move, paying exactly its charges	proved (from live states; the setup's claim can't fail)	Section 14.17, Theorem 14.37
The small machine as a whole is not merge-priced	proved	Section 14.17, Theorem 14.38
The halving price holds exactly when no pile-up exceeds 2^{cost} ; the eight-state price list is the least that meets it; the small machine is not halving-priced	proved (finite machines; Landauer's price premise stays named)	Section 4

Claim	Kind	Home
Forgetting nothing removes no bits	proved	Section 4, Theorem 4.7
A permanent flip removes at least $\log_2 \frac{m+k}{m}$ bits	proved	Section 4, Theorem 4.8
Landauer's principle	assumed (premise)	Section 4, Premise 4.5
Heat floor of a permanent flip	proved (premises: Landauer, $k_B T \geq 0$)	Section 4, Corollary 4.9
Positive heat for any full spread	proved (premises: Landauer, $k_B T > 0$)	Section 4, Corollary 4.10
Entropy pricing gives the toll	definition; proved (premise: entropy pricing)	Section 4, Definition 4.6, Theorem 4.11
A known state owes nothing	proved	Section 4, Theorem 4.12
The bits removed, one term per state	proved	Section 4, Theorem 4.13
Landauer's principle is true of the world	not claimed; a named premise, with its critics named	Section 4
Any machine worth building is of this kind	not claimed	Section 4
Permanence is the job to pick	choice	Section 4

Claim	Kind	Home
The machine's own narrowing is priced	proved (premise: halving price)	Section 5, Theorem 5.1
An observer can learn for free	definition; proved (one schedule that allows it)	Section 5, Definition 5.1, Theorem 5.2
Learning is free on every schedule	not claimed	Section 5

Claim	Kind	Home
Three states are the minimum for free teaching	proved	Section 5, Theorem 5.3
"No Free Insight" names the committed kind	choice (the name kept, the scope stated)	Section 5
Watching narrows like a quantum measurement	not claimed	Section 5
A counter kept by a price list	definition	Section 6, Definition 6.1
Conservation	argued, not checked in Coq	Section 6, Theorem 6.1
Only one honest count	argued, not checked in Coq	Section 6, Theorem 6.2
The fold on lists is unique; it reaches states only under a condition	argued, not checked in Coq	Section 6
Initiality picks the prices, or makes marks physical	not claimed	Section 6
The toll is the potential method	definition; proved	Section 6, Definition 6.2, Theorem 6.3
The potential method is new here	not claimed (it's Tarjan's)	Section 6
The counter is the cost-semantics frameworks' writer	proved for the writer (<code>run_writer_is_run_and_cost</code>): a run in the writer over whole numbers returns the final state and the total cost	Section 6
The counter embeds the cost-semantics frameworks' type systems and their before-the-run checking of grades	not claimed; by design the grade is written by the step during the run	Section 6

Claim	Kind	Home
A window that forgets the counter can't recover it	argued, not checked in Coq	Section 6
A decoder needs constant fibres	proved (any types)	Section 7, Theorem 7.1
No exact window price across a collision	proved (premise: a collision)	Section 7, Theorem 7.2

Claim	Kind	Home
On the toy, no verifier on bare transcripts is sound and complete, and none that reads only the bare part of a richer transcript is either, for any pair like it	proved, over arbitrary types	Section 7
Full-state, commitment and response verifiers are sound and complete for their explicit explanation relations; commitment requires binding and honesty	proved; verification cost is stipulated, and no proof reads it	Section 7, Definition 7.1, Theorem 7.3
On every Thiele-complete machine, no verifier on bare transcripts is sound and complete for the record, or for a ledger rise of 3	proved	Section 7, Theorem 7.4
The one-round “ask” models a cheating prover	not claimed	Section 7
The window fact is mysticism, or Rice	not claimed	Section 7
Entitlement: object, checked relation, paid event	definition (the book's word, not in Coq)	Section 8, Definition 8.1
A decision tree has at most two to its depth leaves	definition; proved	Section 8, Definition 8.2, Lemma 8.1

Claim	Kind	Home
Paid narrowing, counted	argued with the counter; proved with the bill of the charges in its place (premises: a covering tree, paid steps)	Section 8, Theorem 8.2
Structural entitlement over any certification system: a strictly stronger test, a paid crossing, a bill that covers the index bits	proved (premises: covering tree, paid steps)	Section 8, Theorem 8.3
Structural entitlement for every Thiele-complete machine: stronger test, earned chain, index bits at most the record moves, ledger rise at least 3	proved (premises M1 to M6)	Section 14.17, Theorem 14.39
A floor with no supplied tree, from the run's own CHECKs	proved (answers taken at the rivals as they start)	Section 14.17, Theorem 14.40
What the record adds to the price bound: the earned chain and a floor of 3	proved	Section 14.17, Theorem 14.41
The weighted count and the partition form	proved; weights are whole numbers, not probabilities	Section 14.17
The eight-cup member, and a search through n bits with up to 16 questions	proved; the 16 is the machine's fact table	Section 14.17, Proposition 14.42, Theorem 14.43
The covering hypothesis can't be dropped	proved, by a counterexample on the small machine	Section 14.17
The time tax: no free program of fewer than m moves decides $A \geq m$, and three paid moves do	proved, for programs of INC and DEC	Section 14.17, Theorem 14.44
The lists, tree and fibres are read off the run	not claimed, except in the worked members	Section 14.17
Entitlement is truth, or permission	not claimed	Section 8

Claim	Kind	Home
The count prices why candidates were dropped	not claimed; it prices the size of the drop	Section 8
Certified insight event, raw observation	definition	Section 9, Definition 9.1
The toll prices only the crossing, flat in depth	fact read off the definitions	Section 9, Remark 9.1
Sorting moves by whether the reading moves	choice	Section 9
Nothing makes the crossing wait for a check	fact read off the definitions	Section 9, Remark 9.2
A paid flag proves its claim true	not claimed	Section 9

Claim	Kind	Home
Base, cover, honest extension, latch	definition	Section 10, Definition 10.1, Definition 10.2
Over any base, an honest record is a latch	proved	Section 10, Theorem 10.1
Every reachable latch is an honest record	proved	Section 10, Theorem 10.2
A finite reversible base can't write a record	proved; an endless history can	Section 10, Theorem 10.3
Two records are two latches	proved	Section 10, Lemma 10.4
The toggle: driven, not permanent, no latch	proved	Section 10, Lemma 10.5
The clock: permanent, not driven	proved	Section 10, Lemma 10.6
Honest records agree up to the schedule	argued, not checked in Coq	Section 10
The event is a free choice	argued, not checked in Coq (the converse applied twice)	Section 10, Corollary 10.7
The schedule doesn't fix the probabilities	proved, with the branching latch refuted	Section 10, Theorem 10.8
One state space up to renaming	not claimed	Section 10

Claim	Kind	Home
The latch on a Turing machine, a register machine, L	definition; proved (and the relation is an equivalence)	Section 10, Definition 10.4, Theorem 10.9
Weak base equivalence connects those bases	not claimed	Section 10
A growing record is a family of latches	definition; proved, and one latch isn't enough	Section 10, Definition 10.5, Theorem 10.10

Claim	Kind	Home
Substrate; shortcut predicate	definition	Section 11, Definition 11.1, Definition 11.2
No inside decider with a representable flip	proved (premise: the substrate's recursion theorem)	Section 11, Theorem 11.1
The nat family: every decider misses program two	proved	Section 11, Theorem 11.2
An outside decider gets the nat family right	argued, not checked in Coq (the pencil argument)	Section 11
Recursion, Rice and halting for L	definition; proved	Section 11, Definition 11.3, Theorem 11.3
Undecidable means unknowable; the wall separates Turing degrees	not claimed	Section 11
Casper setting; one storage slot; a gas schedule	definition (transcribed; the gas schedule's charge premise stipulated)	Section 12, Definition 12.1, Definition 12.2, Definition 12.3
Accountable safety	proved (premises: quorum intersection, one parent)	Section 12, Theorem 12.1
The quorum premise can hold; one vote finalizes free	proved	Section 12, Theorem 12.2
A persistent write nets at least 20000	proved, on the transcription	Section 12, Theorem 12.3
Gas is exact exactly when commits pay one	proved (premise: the stipulated charge predicate)	Section 12, Theorem 12.4
A permanent write on a finite machine merges	proved	Section 12, Theorem 12.5

Claim	Kind	Home
Platform and quote; inclusion check; bounds fragment	definition	Section 12, Definition 12.4, Definition 12.5, Definition 12.6
The quote can't attest what it never measured	proved	Section 12, Theorem 12.6
Claim	Kind	Home
The RFC's inclusion examples pass	proved (symbolic hash)	Section 12, Theorem 12.7
The proof-carrying checker is the verification condition	proved	Section 12, Theorem 12.8
A transcription is the real system	not claimed	Section 12
A toll is a better design than a wall	not claimed	Section 12
Ecosystem, proliferation, pointer; observation game	definition	Section 13, Definition 13.1, Definition 13.2
Certification is the event whose records proliferate	conjecture	Section 13, Conjecture 13.1
In the toy, "certified" is the unique pointer	proved	Section 13, Theorem 13.2
The survey; the swap shows the verdict follows the observer maps	proved	Section 13, Theorem 13.3
One blind observer stops proliferation	proved	Section 13, Theorem 13.4
Agreement doesn't buy permanence; durable views do	proved	Section 13, Theorem 13.5
The formal objects prove the conjecture	not claimed; "independent" and "deployed" are defined nowhere	Section 13
The floor raised to a threshold	proved (premises: the witness grows by at most the cost and meets the threshold)	Appendix B, Theorem B.1

19.2 Claims about the machine

Part IV built one machine, and the ledger keeps it apart from the model. It's the small one of Section 14, two counters and six instructions, where commitments are earned and universality is proved, with the definition of Thiele-complete that every

machine can be held against, the universal program U on a host made of the same instruction kinds, and the priced program U_p that runs every computably presented Thiele machine. Every line in the next tables is about those machines, or about every machine meeting that definition, or about the game check and the physics section that close Part IV, and nothing else, under the premises printed in its section.

Claim	Kind	Home
Checker and arithmetic meaning agree	proved on the minimal machine, under the stated premises	Section 14 , 14.1
Two-counter simulation and halting correspondence	proved on the minimal machine, under the stated premises	Section 14 , 14.2
Halting is undecidable by the vendored two-counter reduction	proved on the minimal machine, under the stated premises	Section 14 , 14.4
Exact ledger, permanent certification and toll	proved on the minimal machine, under the stated premises, by unfolding	Section 14 , 14.5
Instance of the abstract certification system	proved on the minimal machine, under the stated premises	Section 14 , 14.6
Certification factors as a latch over the core base	proved on the minimal machine, under the stated premises	Section 14 , 14.7
A passing commitment has a prior matching check at the same version	proved on the minimal machine, under the stated premises	Section 14 , 14.8
Certification has a check, commitment and certification in that order	proved on the minimal machine, under the stated premises	Section 14 , 14.9
Unearned commitment and uncommitted certification trap	proved on the minimal machine, under the stated premises	Section 14 , 14.10
Current-version checked claims are true	proved on the minimal machine, under the stated premises	Section 14 , 14.11
No stored fact is forged on a clean-start trace	proved on the minimal machine, under the stated premises	Section 14 , 14.12
Bounded table keeps facts and traps when full	proved on the minimal machine, under the stated premises	Section 14 , 14.13

Claim	Kind	Home
Equal computational windows can hide different records	proved on the minimal machine, under the stated premises	Section 14, 14.14
The window cannot recover the ledger, flag or commitment permission	proved on the minimal machine, under the stated premises	Section 14, 14.15
Clean-start certification costs at least three	proved on the minimal machine, under the stated premises	Section 14, 14.16
A certificate whose check could have failed costs exactly three, and when that check fails the program is refused forever	proved on the minimal machine, under the stated premises	Section 14, 14.17
The step refuses trivial checks	not claimed; “is at least 0” passes every value and certifies for three too	Section 14
Provenance, checker soundness, no forging and the floor of three hold for any property language whose checker is proved to mean what it says, a sorted list read out of a counter included	proved on the minimal machine (<code>sorted_run_certifies_iff</code> and <code>sorted_run_refused_forever</code> , <code>EarnedGeneric.v</code>)	Section 14
The channel names the last earned commitment, and a later commit can replace it	proved on the minimal machine (<code>certified_channel_earned</code> and <code>channel_names_latest_commit</code> , <code>EarnedCore.v</code>)	Section 14
The guarantees hold against whoever sets the start, a reset, or a second writer that doesn’t bump versions	not claimed; they are about programs from a clean start, one step at a time	Section 14, Section 20

Claim	Kind	Home
Thiele-complete means a universal base that is free and can't touch the record, an earned record with a checker proved to mean its claims, an exact toll of one mark per check, commit and certificate, and a claim whose check can fail	definition	Section 14, 14.7
Every Thiele-complete machine obeys the toll, runs every two-counter program, counts its record moves on its ledger, pays at least three per certificate, has a check that fails and a run that certifies, and raises its record only by a CERTIFY	proved, for every machine and every interface meeting the definition	Section 14, 14.18
A clock (every move charged, any reading or any latch as the record) obeys the toll with a universal base and is not Thiele-complete; nor is any machine whose every move costs something	proved	Section 14, 14.19
Each clause does work: a paid latch meets the base and toll clauses and fails, and a silent machine meets three clauses and fails	proved	Section 14, 14.20
The small machine, the same machine over any property with a proved checker, and the sorted-list machine are Thiele-complete	proved on the minimal machine	Section 14, 14.21

Claim	Kind	Home
Every Thiele-complete machine hides its record and its ledger from its base window	proved, for every machine and every interface meeting the definition	Section 14 , 14.22
No fixed host program can keep the guest's facts on an exact copy of a guest counter	proved, for any host property language and any translation into a finite list of host properties	Section 14.13 , 14.23
One fixed program U , in the small machine's own instruction kinds and with no guest step built in, runs every small-machine program from every start, step for step, with halting and answers matched	proved on the host machine	Section 14.13 , 14.24 , 14.25
U 's flag rises exactly when the guest's would, and its ledger equals the guest's at every matching point	proved on the host machine	Section 14.13 , 14.26
U 's raised flag stands on U 's own check, commit and certificate on a slot, and on the guest's own earned chain for the same claim	proved on the host machine	Section 14.13 , 14.27
U 's host is Thiele-complete, and whether U halts on a loaded guest is undecidable	proved on the host machine; undecidable in the library's sense	Section 14.13 , 14.28
The simpler host runs any small-machine program as a guest, and halts with any two-counter program	proved on the minimal machine, by construction: its guest step runs the guest's own step as a primitive; the guest program is held as a list	Section 14.13 , 14.31
The simpler host's mirror always equals the guest's flag	proved on the minimal machine	Section 14.13 , 14.32

Claim	Kind	Home
The simpler host's step collects the guest's toll, and its ledger covers the guest's	proved on the minimal machine	Section 14.13 , 14.33
The simpler host's mirror rises only in a guest step that passed CERTIFY	proved on the minimal machine, for every sequence of host moves	Section 14.13 , 14.34
The simpler host's own instructions never touch the guest	proved on the minimal machine	Section 14.13 , 14.34
The simpler host's raised mirror means the guest's own run checked, committed and certified the same claim	proved on the minimal machine (<code>host_mirror_earned</code> , <code>UniversalThiele.v</code>)	Section 14.13
The simpler host is a certification system	proved on the minimal machine	Section 14.13 , 14.35
The simpler host's ledger equals the guest's	not claimed; a guest step with budget b costs b whatever the guest instruction cost, so its ledger only covers the guest's; U 's is exact	Section 14.13 , 14.33

Claim	Kind	Home
One fixed program U_P , U with one paid move added, runs every computably presented Thiele machine: halting matched, the state decoded at every matching point, the host's flag up exactly when the guest's reading has been yes, and the host's ledger the guest's plus a surcharge	proved on the priced host	Section 14.14 , 14.29

Claim	Kind	Home
U_P 's raised flag stands on its own check, commit and certificate on a slot whose claim is the guest's own reading of a state where it said yes	proved on the priced host	Section 14.14 , 14.29
The surcharge is at most two when the reading starts at no, and zero when the first raising move costs at least three	proved	Section 14.14 , 14.30
No program on the host machine matches a raise that cost less than three exactly	proved, for every program from a clean start	Section 14.14 , 14.30
The priced host is Thiele-complete, with the paid move read as a check that never passes	proved	Section 14.14
A demonstration machine (one move of price three, reading "at least three") runs on U_P with its ledger exactly $3n$	proved, not run	Section 14.14
A machine whose reading no program decides, or whose next move no program computes, runs on U_P	not claimed; outside the class	Section 14.14
A guest's reading that goes back to no is carried as no	not claimed; the host's flag is the latch of the reading	Section 14.14
A counter beside a Turing machine changes nothing	proved	Section 14.16 , 14.36
A Turing machine can't price a reading	not claimed	Section 14.16

Claim	Kind	Home
Rice's theorem on the host machine, for properties that depend on everything a program ends with (registers, table, channel, trap, ledger, flag)	proved; undecidable in the library's sense	Section 14.18 , Theorem 14.45
Rice's theorem for two-counter programs, from the clean start	proved; the fact table isn't compared	Section 14.18 , Theorem 14.46
s-m-n, and a universal host program for host programs	proved, with no premise on register 2	Section 14.18 , Theorem 14.47
Kleene's recursion theorem on the host machine	proved; the fixed point is of the function	Section 14.18 , Theorem 14.48
Kleene's recursion theorem up to the record on the host machine: stopping, register 0, trap latch, ledger exactly, flag, number of facts, whether the channel is empty	proved; the host program pays after it has computed, not step for step, and simulates T	Section 14.18 , Theorem 14.49
A fixed point of full behaviour (every register, what the facts name) on the host machine	refuted for registers; for what the facts name, only a variant is argued, not checked	Section 14.18 , Theorem 14.50
No inside decider with a representable flip, and no host decider, for functional properties of host programs	proved (the recursion premise is a theorem there)	Section 14.18 , Theorem 14.51
Rice and Kleene for two-counter programs on plain inputs	not claimed; not proved	Section 14.18

Claim	Kind	Home
The integer check makes the pinned matrix PSD	definition; proved	Section 15, Definition 15.1, Theorem 15.1
PSD forces $S^2 \leq 8$	proved (algebra only)	Section 15, Theorem 15.2
A passed check means $S^2 \leq 8$	proved, as the composition of the two	Section 15, Corollary 15.3
A PSD pinned matrix has a PSD completion	proved	Section 15, Remark 15.1
Every deterministic plan fails the check	argued, not checked in Coq	Section 15, Remark 15.1
The level-1 test is the size-five case	definition; proved	Section 15, Definition 15.3, Corollary 15.7
A tally above two isn't one plan replayed	proved	Appendix F, Theorem F.3
A tally above two proves nonlocality; a pass proves the data quantum	not claimed	Section 15
A finite local model meets the bound	proved; the converse, and so Fine's full theorem, isn't claimed	Appendix F, Theorem F.2
The elliptope gate is sound; the PR box fails it	proved	Appendix F, Theorem F.29
The arc-sine boundary; the quantum set; reaching $2\sqrt{2}$	not claimed (cited, not proved)	Appendix F
The integer route to the bound is new mathematics	not claimed; the route is NPA's	Section 15
The integer check passes exactly when every pair was sampled and the pinned matrix is PSD	proved	Section 15, Theorem 15.4
A raised flag on the host machine stands on a passing CHECK of the packed counts, a COMMIT and a CERTIFY; the chain certifies for 3 exactly when the check passes	proved	Section 15, Theorem 15.5

Claim	Kind	Home
Five tallies: 12/5 and 14/5 certify; 16/5, the PR box and the all-ones plan are refused forever	proved	Section 15, Proposition 15.6
The tally was played, or was honest, or puts a bound on nature	not claimed; it is placed in a counter for free	Section 15
The calibration is a unit conversion	definition; proved	Section 16, Definition 16.1, Theorem 16.1
Landauer in counter units; dissipation at most cost	assumed (premise)	Section 16, Premise 16.2
The premises fit when every counted step costs at least one, and fit a one-bit reset	proved	Section 16, Theorem 16.2
The small machine's whole instruction set fails the premise pair	argued, not checked in Coq (its free steps merge)	Section 16
Keeping history makes every step injective	proved	Section 16, Theorem 16.3
The heat is fixed by the chosen gap	definition; proved	Section 16, Definition 16.3, Theorem 16.4
The counter is physical dissipation, in joules	not claimed	Section 16
Permanence predicts more than Landauer	not claimed	Section 16
Zero Admitted; a handful of standard assumptions	fact from the assumption audit, true of the run that printed it	Section 17
The small machine's files, the Thiele-complete files and the universal programs' files assume nothing	fact from each file's own printed assumptions, which the test suite requires to be closed for the files under minimal/	Section 17
The theorems prove physical reality, or Coq's consistency	not claimed	Section 17

19.3 What I propose and haven't proved

And then there's what I'm proposing. These are mine. They reach further than any theorem in here, and nothing in the tables puts an unavoidable physical price on every observable change.

- That the axis belongs in the step (Section 20, Section 21).
- That the usual account of a computation is a shadow of the step (Section 21).
- That certification is the event worth recording, which is the pointer conjecture (Section 13).
- That this is a structural account of computation and not just a cost model (Section 20).
- That the toll is the right thing to price, against a wall or a penalty for contradiction (Section 12, Section 21).

And three things left open: whether Landauer's critics are right, which would leave every theorem standing and one premise unowned by the world (Section 4); whether certification really is the pointer event (Section 13); and whether there's a smaller start than a partition, by a ruler someone can state (Section 2).

That's the whole ledger. If you find a claim in this book that isn't on it, or a line that says more than its home does, that's my mistake, and I'd want to hear about it.

20 Scope

This is the line around what I've proved. Inside it: an accounting law for one event, a derivation of that law from premises each with its own counterexample, a set of windows that provably lose it, a record axis that's a latch over any base, a diagonal that holds by proof on L, a six-instruction machine on which commitments are earned and which computes what two counters can, one fixed program on it that runs every computably presented Thiele machine, and an integer check that implies a bound by algebra. Outside it: everything that follows. This is the fence. I built it myself, so lean on it before you trust it.

G. K. Chesterton has a famous fence, in a 1929 book called *The Thing*. A reformer finds a fence across a road, can't see what it's for, and wants it cleared away; the wiser one tells him to go off and find out why it was put there, and only then come back and argue for tearing it down. This section runs the other way round. Every post below comes labelled with what it's there for, the overclaim it stops, so you don't have to go off and find out first. Pull up any post you like. Just look at what walks through the gap when you do.

Physical interpretation of the counter. The ledger is unique once the start and schedule are chosen. That doesn't make it entropy, and it doesn't make it energy. A physical

bridge needs a model of which states carry information, premises that can all hold at once, and a real argument tying the formal charge to dissipation. The permanent-certificate theorems supply the first for an abstract finite machine with a permanent reading. The two premises are shown able to hold together whenever every counted step costs at least one, and for a one-bit reset. The small machine's free steps merge states and are charged nothing, which breaks the bridge for its whole instruction set (Section 16), and the calorimeter protocol shows that population dynamics don't fix the energy scale. No joule value attaches to one mark without thermal and device premises (Section 4, Section 16). In short: once you pick the start and the price list, the number is forced by arithmetic, and it isn't energy. The rest of that paragraph is the list of what I'd need before it could be.

Class separation. The witnesses show specific windows losing specific fields. They don't stop a conventional simulator from keeping those fields in its encoding. No separation of standard complexity classes follows from the collisions (Section 7, Section 14). I'll be blunt about why I'm careful here. Whether P equals NP is one of the seven problems the Clay Mathematics Institute put a million dollars on, at a meeting in Paris in May 2000. One of the seven has been solved since, the Poincaré conjecture, by Grigori Perelman. P versus NP hasn't. If a two-state collision on my machine settled it, somebody would be owed a million dollars, and it wouldn't be me. A window that loses a field is a fact about that window. A complexity class is a fact about what every possible program can do within a budget, and nothing in these collisions reaches that far.

Trace initiality, and what it doesn't settle. The universal property of the trace fold is about evaluating lists of instructions: pick where a target starts, and there's exactly one way to fold a list into it (Section 6, argued there). The counter's uniqueness assumes a schedule. Exact pricing assumes an event, and a reference count that says what charging too much means (Section 7). Those results make the construction precise. None of them shows that the schedule I chose, or the event I chose, is physically necessary.

Other observables. Certification is one worked example. A family of other observations might turn up other structural laws, but each would need its own events, its own evidence, and its own cost rules, justified on their own terms. The window and oracle results are about what specific maps keep. They don't show infinitely many independent dimensions, zero mutual information, or a cost for changing coordinates, and piling predicates on top of each other doesn't give a topology, a metric, or independence (Section 14.16).

Program predicates versus state observations. The state-level collision (Section 7's wall) and the program-level diagonal (Section 11's wall) live in different domains. Linking them would take a theorem of its own. The nat family and L (Section 11) keep their explicit quantifiers and representability premises.

Entitlement bounds. The count is handed its tree and its covering, along with the payment condition. It doesn't dig the tree out of an arbitrary run, and it isn't a Shannon bound on every trace (Section 8). On every Thiele-complete machine the payment is the record moves, each exactly one mark, and a floor that takes no tree exists when the questions are the run's own CHECKs; but its answers are taken at the rivals as they start, a free branch can still narrow for nothing, the lists, tree and fibres are supplied except in the worked members, and the covering can't be dropped. The search through n bits is capped at 16 questions by the machine's fact table, and the time tax is a lower bound for programs of INC and DEC, not for every Thiele-complete machine (Section 14.17).

Quantum behaviour. The fixed-matrix equivalence is about one completion with zero marginals and zero within-party moments. The ellipsope lets the within-party moments move, at the level of correlators. Full behaviours with marginals, and operator realisations, need identifications this book doesn't make. A sound check isn't complete for those bigger sets, and the ellipsope decider is mathematics, not one of the small machine's properties (Section 15).

The record axis. The proved structural result is: a permanent record driven by a base computation is a latch on an event; on a finite machine its write merges states; and the named windows don't recover it. Certification is one instance, not a theorem selecting the unique event every system must record (Section 10, Section 13). A finite monotone multi-valued record decomposes into threshold latches, not into one Boolean latch. Revocation, including revocation priced from the other side as in Casper, lies outside the permanent-record axis; a price on revocation doesn't transfer to a price on the write (Section 12). Probabilistic records don't inherit the uniqueness result either. Two honest random records can have the same branches and the same charges and still give those branches different weights, one choosing evenly and one two to one, so the schedule doesn't fix the probabilities (`schedule_determines_probabilities_refuted`). And no single deterministic latch explains a record that can branch both ways from the same observation (`deterministic_latch_handles_branching_refuted`). Weak base equivalence is defined and is an equivalence; no two of the four bases are related by it in the checked work, and the latch form holds on each separately, so it doesn't tell them apart (Section 10). In short: a record that stays up is a latch, and on a finite machine its write merges states; which event deserves the latch isn't proved; records that can come back down lie outside all of this (they're limit computability's, and Section 10 points at Ershov's ladder), and random ones lose the uniqueness.

Who it's safe against. Every guarantee about the record is about programs: any list of instructions, run from a clean start on a step that does what its definition says, one writer stepping at a time. That covers the buggy program and the hostile one. It doesn't cover whoever sets the start (a snapshot put back, a table loaded with facts nobody checked), a reset or a power cycle, whoever picks the program, the wire that

carries the report, someone who flips the flag with a probe, the tools that built the step, or a second writer that changes a counter without bumping its edition. Those are the operator, the device and the tool chain, and they're outside this fence. A record read by someone who didn't watch it being made is only as good as their reason to believe the box that shows it. The ways across that I know of, a quote over measured state and a proof that travels on its own, are in Section 7 and Section 12, and this book builds neither.

The diagonal on the machine. What's proved for the small machine's programs is that halting is undecidable in the library's synthetic sense, for them and for the universal program (Section 14). L has its recursion theorem, Rice and halting by proof (Section 11). On the host machine, the one with a counter for every number, Rice's theorem holds for properties that depend on everything a program ends with (every register, the table, the channel, the trap, the ledger and the flag, not the line number or the versions), for programs run on an input in register 1; s-m-n holds with no premise on register 2; and Kleene's recursion theorem holds for the transformations a host program carries out on numbers, with a fixed point that agrees with its image on stopping, register 0, the trap latch, the ledger, the flag, the number of facts and whether the channel is empty, and cannot agree on every register. On the two-counter machine itself only Rice's theorem is proved, for what a program does from the clean start, with the fact table not compared. Both theorems for two-counter programs on plain inputs are not proved (Section 14.18).

The five models. Each is a fragment transcribed from a specification, with a named theorem and a named limit: the Casper transcription has no deposits or fork choice; the gas fragment omits the refund cap; the attestation fragment has no signatures; the transparency verifier covers control flow, not collision resistance; the proof-carrying checker is a three-instruction role model. None is a claim about the real system beyond its fragment (Section 12).

The machine. Part IV has one machine, and here's its fence. The small machine of Section 14 (two counters, six instructions) is where commitments are earned: a raised flag went through a passing check, a commitment and a certification of the same claim, nothing is forged, and the cheapest certificate costs exactly three, a price reached by a check that could have failed (the step doesn't police trivial checks: a check that can't fail certifies for three too). It's also where universality is proved, by matching Minsky's two-counter machines step for step and borrowing their undecidable halting from the Coq library. It's Thiele-complete under a definition every clock-style machine fails, and every machine meeting that definition hides its record and ledger from its own window. The universal Thiele machine is one fixed program, U , on a host with the small machine's instruction kinds and more counters, with no guest step built in: it runs every program of the small machine from every start, halts when the guest halts, raises its own flag by its own check, commit and certificate exactly when the guest's flag would rise, and keeps a ledger equal to the guest's at every matching

point. With one paid move added, one fixed program, U_P , runs every computably presented Thiele machine: its step, prices and reading computed by programs, the reading decided on every state, the run fixed by a driver. The host halts when the guest stops, its flag is the latch of the guest's reading, raised on its own earned chain whose claim is that reading, and its ledger is the guest's plus a surcharge of at most two when the reading starts at no, a surcharge no program on that host can avoid for a raise that costs less than three. A machine whose reading no program decides is outside. A simpler host with a guest step built in runs guests by construction and keeps a mirror of the guest's flag; its ledger only covers the guest's. Its check price is flat, its fact table holds sixteen, and its certificate is historical (Section 14 lists the rest).

The comparison machines. The small machines I built to compare against are fragments too. The random-access memory result covers an addressed list memory with load, increment and decrement on cells (`concrete_ram_write_reads_back` and its neighbours), and it's about that machine. The register machine Section 10 puts the latch on (`record_axis_is_latch_on_ram_holds`) is a different small machine, the one with indirect addressing, and that theorem is about it. The reversible-machine result covers arithmetic update cores: add and subtract, each undone by the other, with or without wrapping at a modulus (`janus_like_unbounded_inverse`, `janus_like_bounded_inverse`). Neither is a theorem about real memories or full reversible languages. The cost-semantics results are local correspondences, not full cost-framework embeddings: the trace's cost is a writer cost and a grade, and the toll is the potential method with one potential (Section 6). Nobody's whole type system gets embedded either.

Outward consequences. I looked for results about real systems that would need the record axis. Five narrow ones are proved, each as a two-state countermodel: a certificate-transparency client's local tree head can't decide whether the world's log is consistent (`ct_local_view_insufficient`); a quote check that matches only the composite digest accepts a mismatched selection (`tpm_selection_binding_is_necessary`); a local suffix can't decide a trusted anchor (`weak_subjective_suffix_insufficient`); an acknowledgement without a durable commit record can be lost (`wal_ack_requires_durability`); and a current local log can't decide what happened before it (`audit_local_snapshot_insufficient`). Each reduces to a known indistinguishability or durability argument. Five stronger candidates would need protocol or hardware semantics I haven't formalised, and I've proved nothing about them. So none of this yields a new outside result that both needs the record axis and is ready to use.

Where each result lives. Every system that follows the toll, whatever its states: the floor, at most as many certifications as cost, and the potential-method reading. Every finite machine whose price covers merges: the toll for permanent records, and which instructions every such price must charge. Every finite machine paying per halving: the machine's own narrowing bound. Every machine extending a base with a driven,

permanent record: the latch. Every window through which a certifying and a non-certifying step look alike: no exact price from the window. Every machine meeting the definition of Thiele-complete: a free move, at least three per certificate, and a window blind to the record and the ledger. Any eight counts: the integer check and its PSD bridge, which are mathematics. This machine only: its six instructions and their prices, its sixteen slots, its own colliding pair, and the encodings its universal programs use.

What I'm proposing beyond that. That the axis belongs in the step; that certification is the event worth recording; that this is a structural account of computation and not just a cost model. Those reach further than any theorem here, and no theorem here puts an unavoidable physical price on every observable change. They're the argument, and the argument is what the rest of the book was. What kind of claim is that? A proposal about which distinctions an account of computation ought to draw, and no proof settles an "ought". The proofs show the proposal hangs together, follows from premises I've named, and costs little (six instructions on the small machine); whether anyone should adopt it is argued, not proved. Here's what would weaken it: if every use people have for the record turned out to be recoverable from a log the program writes plus a correctness proof about the program, or if the real systems keep choosing walls over tolls for good reasons, as the five in Section 12 mostly did.

21 Conclusion

I started with one word and a machine I couldn't explain. Here's where the word ended up, and where the machine did.

Structure turned out to be relations among things, kept under chosen changes, carried by whatever stuff, always relative to a frame. The start I chose, the smallest one by the count of names, was a sorting of a pile, and the thing nobody I read had priced was the moment you stand behind the sorting. That was Part I, and it ended with a question.

The question got an exact form in Part II: five parts, one rule, one theorem you proved at the table. Then the rule stopped being only my preference. On a finite machine where the flag stays up, raising it forgets something, and if forgetting is charged for, the way Landauer's principle charges for it in its worst case, the toll follows. That's a derivation under three premises you can refuse one at a time, each with its counterexample built, not a proof that every computation has to adopt the rule; and the charge on forgetting is a named premise with its critics named. Learning can be free; committing to a record that stays put can't be, on those premises. That's as far as physics goes in this book, and I've tried to keep it exactly that far.

Part III was the logic, and here's what it earned, in the order it was earned. The counter is arithmetic once the schedule is chosen. A window in which a raising step and a quiet one look the same can't price the raise exactly, by a fact about functions far older than the machine. Entitlement is defined as an object, a checked relation, and

a paid event; given lists, a tree and a covering somebody supplies, a run that paid a charged step for every level of the tree has paid for the index bits it commits to (no run is shown to have to pay them); on every Thiele-complete machine the same count is a theorem about its record moves, with the earned chain inside the run, and the count can't do without the covering; and it is a license to lean on a claim, not a proof that the claim is true. Insight, in the sense the toll prices, is a committed relation, not a feeling, and its floor is flat: at least one, however deep. The claim behind the flag is something a check could have refused, and in the bare model nothing makes the commit wait for the check. Over any base, an honest record is a latch, with two free choices, the event and the price list; "only one substrate" is false and "one structure, many price lists" is true once the event is fixed. No procedure the substrate itself can represent decides the shortcut predicate, given a recursion theorem, by the same diagonal that broke Frege's floor, and the machine's answer is to record the commitment rather than automate the judgement. Of five record disciplines I modelled in fragments, one prices the write. And which event to record is a conjecture, with its refutations listed.

Part IV built one machine, two counters and six instructions, and on it the model stands whole: a raised flag was checked, then committed, then certified, on the same edition of the same counter, nothing gets forged, the cheapest certificate whose check could have failed costs exactly three (the step doesn't police trivial checks), and it computes everything Minsky's two-counter machines can, so its halting is undecidable. Earned commitments and universality are proved there. So is what Thiele-complete has to mean: a universal base that's free and can't touch the record, an earned record, one mark for each check, commit and certificate, and a check that can fail. Every clock, every machine that charges for computing, fails it; the small machine meets it; and every machine that meets it hides its record and its ledger from its own window. And so is a universal Thiele machine, with its scope said plainly: one fixed program, U , written in the small machine's own six kinds of instruction on a host with more counters, with no guest step built in, that runs every small-machine program from every start, halts exactly when its guest halts, raises its own flag by its own check, commit and certificate exactly when the guest's would, and keeps a ledger equal to the guest's. And then the bigger version of the same thing: with one paid move added, one fixed program, U_P , runs every computably presented Thiele machine, a machine from anywhere whose step, prices and reading are computed by programs. It halts when the guest stops, raises its own flag on its own earned chain exactly when the guest's reading has said yes, with the guest's own reading as the claim it checked, and keeps the guest's prices to within a surcharge of at most two, which no program on that host can avoid for a raise that costs less than three. An ordinary programmable computer, as built, doesn't do that for the programs it runs; U and U_P show a fixed interpreter can. All of it is proved in Coq with nothing assumed: each file prints the assumptions of every headline theorem, and every line comes back closed. Then a check worth earning, the two-player game with an integer test whose pass forces Tsirelson's bound, and what the counter has to do with physics, which is less than

the word Landauer in a file name suggests. The places where the machine came up short of the idea stayed on the page: its free steps merge states, so the toll on it is a choice the chairs don't force, and a check that can't fail certifies for three like any other. Part V was how to check it and how to break it. Every result carries the shape of the thing that would kill it, gathered in one place (Section 18): a run from uncertified to certified with the counter never rising; a contractive correlator whose pinned matrix isn't positive semidefinite; a certification system that follows the toll and certifies at total cost zero; a function on numbers that correctly decides the shortcut predicate of the substrate built from it. Any one of them brings down the theorem it matches. I hunted. I came up empty. For a theorem Coq has checked, that's no surprise: finding one would take a bug in the checker. The places a reader can really win are where my sentence says more than the statement does, or a definition doesn't catch the word it's named for, and Section 18 starts with those.

If you asked what's new here, I wouldn't give you one thing, because there isn't one. There are four, and each comes with its fence and its neighbours. First, the toll isn't only my preference: on a finite machine a reading nothing turns off can only be turned on by a step that merges states, so if merges are priced, raising it costs (Theorem 4.1, Theorem 4.2). The counting under it is the finite cousin of the recurrence argument Zermelo threw at Boltzmann in 1896, and the price on a merge is Landauer's, as a premise. Ouldrige and Wolpert (2023) had already put a physical price on that merge for finite automata, so what's here is the toll as a rule on a ledger. Second, a window that shows a raising step and a quiet one the same can't price the raise exactly (Theorem 7.2); that a price computed from what a window shows can't tell apart two steps the window shows the same is elementary, and what's here is the pricing form. Third, a record driven by the computation that never switches off is a latch on one event, over any base (Theorem 10.1); latches and history variables are old, and the theorem covers only records that never come down. Fourth, a record that's priced, and on the small machine earned (Theorem 14.9), under the move of guaranteeing something about every program. That move isn't mine at all. It's the reference monitor's (Anderson, 1972), made precise for monitors that watch a run (Schneider, 2000), and verified in Coq for monitors checked at every instruction (Azevedo de Amorim and coauthors, 2015). What's put together under it here, built and checked rather than new in idea, is the check bound to the edition it looked at, the toll paid on top, and universal programs that carry both to every program of the small machine and to every computably presented Thiele machine, earning their own flag as they go. Appendix C sorts all four with sources.

And two things I'd point at if you asked how far the checking reaches. Neither is a fifth new thing.

In the game: an integer check on eight counts forces a five-by-five matrix to be positive semidefinite, and from there the Tsirelson bound follows, and the chain never once reaches for physics. No Hilbert space, no operators, no observables. The route through

the matrix isn't mine; it's the first rung of a ladder Navascués, Pironio and Acín built in 2007, and the semidefinite road to Tsirelson's bound is older than that. What's here is the check that runs in whole numbers, and the checked link from "the check passed" to the bound. It's algebra you can follow line by line, and the assumption audit on that link finds exactly three standard axioms from Coq's own library (two that come with its real numbers, and functional extensionality) and nothing of mine, which is how anyone can see that no physics was smuggled in. And on the small machine's host a raised flag can stand on that check: from a clean start a raised flag contains a passing CHECK of the packed counts, a COMMIT and a CERTIFY, and the check passes exactly when every pair was sampled and the five-by-five is positive semidefinite (Section 15).

In the model: the diagonal rules out a correct decider whenever its flip can be expressed inside the substrate and the substrate supplies the fixed point; in L the fixed point is proved from the reduction rules and the diagonal is Rice's theorem. On the small machine's host, Kleene's recursion theorem and Rice's theorem are proved for its own programs, so the wall stands there by proof, with its premise a theorem: no host program decides a property of what a program computes (Section 14.18).

What's open, by my count. Whether the toll is the *right* thing to price, as against a wall or a penalty for contradiction, which is what the five real systems mostly built. Whether certification is the pointer event. Whether Landauer's critics are right, which would leave every theorem standing and one premise unowned by the world. And the question I'd most like someone to take from me: is there a smaller start than a partition, by a ruler they can state? I said in Section 2 it would change this book. It would.

And the two computers from the first pages, the sorter and the copier, holding the same bits. Walk the warehouse again. On an ordinary machine there's still no lamp for what the sorter did, and nothing in this book changes that. On the small machine there is one: a flag and a counter you can read off the state, and a theorem that, from a clean start, the flag can't be up unless a check of that very claim, about that very edition, passed first, where a claim can be any property whose checker is proved to mean what it says, "this counter, read as a list, is sorted" among them (`sorted_run_certifies_iff` and `sorted_run_refused_forever`, `EarnedGeneric.v`). That's the one lamp I set out to light. It's a lamp for whoever can read this machine's state. A "tests passed" sheet fits it only when the test is a checker proved to mean what it says; the sticker and the very sure AI from the first pages ask something else, whether to believe a report you didn't watch being made, and nothing in this book answers that (Section 20). Whether every machine ought to have that lamp is the argument, not the proof.

So that's the Thiele Machine. An abstract model with five parts and one rule, an argument made with it that the usual account of a computation is a shadow and the

axis it leaves out belongs in the step, and one small machine to check the argument against, with one fixed program on it that runs every machine of the model whose workings a program can present. The theorems are what they are, with their premises printed. The interpretation is mine, and I've tried to say at every turn which is which, because the first time through I was the only one checking, and I've met me, and that's not enough people. Now there's you.

A Things I had to learn first

I came in with none of the formal fields this work turned out to need. By the time the proofs compiled I'd needed enough of half a dozen of them that I had to put down somewhere what those fields had words for. This appendix is that place. None of it is the argument. It's what I needed to be able to make it. Everything below is what I understood after I went and read about it; I'm not claiming to teach you these subjects, and if you already know a part, skip it.

A.1 Bits, bytes, and what a CPU sees

A bit is one digit that's either zero or one, the smallest fact a machine can hold. Everything else is bits stacked up. Eight of them make a byte, which lands you any number from 0 to 255, and the first thing that genuinely surprised me is that the same byte is also a letter: a convention called ASCII simply agrees to call 65 the letter A, 48 the digit 0, 32 a space. Nothing in the byte knows which it is. The meaning is a convention laid on top.

You can see the convention's seams with a pencil. Write 65 in binary: 1000001. Knock off the first two digits and what's left, 00001, is 1, and A is the first letter. B is 66, which ends in 00010, two. Now lower case: a is 97, which is 1100001, the same pattern with one more bit switched on, the one worth 32. Every capital and its small letter differ in exactly that bit. That's on purpose. The first version of the code, in 1963, had no lower case at all; when lower case arrived, in the 1967 revision, it was placed so the two cases differ by one bit, which made matching them, and building keyboards and printers, simpler. Nothing in the byte knows it's a letter, but somebody made the letters easy for a circuit to tell apart.

Everything in this subsection is about real CPUs. None of it belongs to the abstract model, which has a kind of state and a step rule and nothing about widths or slots, and the small machine of Part IV needs very little of it: its two counters have no width at all, and hold any whole number.

A word is a chunk of bits the CPU treats as one unit, these days usually 32 bits (0 to about 4 billion) or 64 (0 to about 18 quintillion). The abstract model doesn't care which width.

A register is a named storage slot inside the CPU that holds one word, the fastest place to read or write. Memory is a longer list of word-sized slots accessed by index, the

address. The program counter is one specific register that tracks where in the program the machine is; after every instruction it moves forward by one unless the instruction says jump. On the small machine it's called the line number. A stack pointer is a register pointing at the top of a region used for nested calls. A port is a named channel for talking to the outside world.

Modular arithmetic means counting wraps at a maximum: with a maximum of 16, 17 wraps to 1. Bitwise operations work on the bits of a value one position at a time: AND is 1 only when both bits are, OR when either is, XOR (exclusive or) exactly when they differ. A shift moves every bit left or right, which multiplies or divides by a power of two. Popcount is the count of one-bits in a value. Hexadecimal is base 16, digits 0 to 9 then A to F, written with a 0x prefix: 0xFF is 255.

A trap, on a CPU, means execution goes to a fixed handler instead of advancing, the chip's "this went wrong." On the small machine a trap is one bit that goes up when an instruction refuses, and then the core stops moving. An ISA, instruction set architecture, is the list of every instruction a CPU recognises, each identified by an opcode, a numeric tag the CPU decodes. The small machine's has six. The abstract model has none; it has a step rule.

A.2 The notation the formal parts use

I don't think in these symbols, and I had to learn to read them. The natural numbers are 0, 1, 2, and so on, written with a double-struck $\mathbb{N}$; the integers add the negatives; the reals are the whole number line. A function from a set A to a set B is written with an arrow. For a finite set, vertical bars around it mean its size; for a number, its absolute value. A big sigma is a sum over every member of a set. A delta before a quantity means the change in it. An upside-down A means "for every"; a backwards E means "there exists." A double arrow is implication; a double-headed one is "if and only if." A small circle between two functions is composition, apply the right one first. Angle brackets around a random quantity mean its long-run average, which I picked up from the Bell-test papers. In this book a circled plus is bitwise XOR, and the quantum sense of a circled times never appears, because the game section does its work in plain integer arithmetic. A superscript T on a matrix is its transpose, rows and columns swapped, and I with a subscript n is the identity matrix.

A.3 Coq, in just enough detail

Coq is a proof assistant: a programming language whose type system is expressive enough that proofs are programs and the compiler checks them (Part V has the long version). A Theorem is a named statement followed by a proof; a Lemma is the same thing used for an intermediate result; a Definition introduces a named value; an Inductive type is one with a finite list of constructors, the only ways to build a value of it (Coq's natural numbers have two: zero, and successor of another natural). A tactic is a step inside a proof: induction (base case, then step), destruct (split into cases), reflexivity (prove a thing equals itself), simpl (compute), lia and lra and nra

(arithmetic solvers for integers, linear reals, and polynomial reals), `psatz` (a certificate search for polynomial inequalities). `Qed` ends a proof and asks `Coq` to check the whole chain. `Admitted` is the keyword for “I claim this without proof,” and it appears nowhere in the active tree. A Hypothesis inside a Section is a named premise that becomes an argument of every theorem that uses it when the Section closes; “closed under the global context” doesn’t make it go away. `Print Assumptions` lists the global axioms a theorem depends on; on the kernel tier it shows only standard-library ones. A total function always returns a value, and `Coq` requires every function to be total; the small machine’s step function is, traps included. “Parametric in K ” means the proof works for any K ; the small machine’s table size, sixteen, is a literal constant, and the proofs are about that literal.

A.4 Category theory beyond Part I

Section 2 sketches objects, arrows, composition, identities, associativity, and the identity laws. Beyond that: a functor is a structure-preserving map from one category to another, sending objects to objects and arrows to arrows while respecting composition and identities; think of it as a translation rule between two worlds of structure. An embedding is a functor that throws nothing away, and faithful is the precise version for arrows: two distinct arrows never land on one. Conservative is the version for inverses: if the image of an arrow has an inverse, the arrow already had one, so the functor can’t manufacture an isomorphism (an arrow with an inverse) that wasn’t there. The two properties are different, and neither gives the other. Read the natural numbers under addition as a category with one object and an arrow for each number, and the same for the integers. The inclusion of the first into the second is faithful, since distinct naturals stay distinct integers, but not conservative: 1 has no inverse among the naturals, and its image has one, minus 1. The other way round, the functor from the integers onto the trivial one-arrow group is conservative, since every arrow of the integers already has an inverse, but not faithful: it sends every integer to the same arrow. An embedding needs faithfulness specifically. A monoidal product combines two objects or two arrows into one, with a unit and laws that make the combining behave. An initial object has exactly one arrow into every object; the counter’s uniqueness in this book is a uniqueness-on-reachable-states fact, not a category-theoretic initial object. Yoneda’s lemma is where the categorical engine first clicked for me, the night I stopped borrowing other people’s runtimes, and nothing in the formal development uses it.

A.5 Bell-test vocabulary

A correlation between two outcomes is the average of their product; with outcomes plus or minus one it lies between minus one and one, plus one meaning always match, minus one always differ. A correlator is one such average in a labelled setting, `E00` for Alice’s setting 0 and Bob’s 0. The marginal of a joint distribution is the distribution of one side alone with the other averaged out; zero-marginal means each side’s outcome is plus or minus one equally often on average. A local hidden-variable model is one

where both outcomes are fixed by a shared random variable agreed before the game plus each side's own setting; Bell's theorem says no such model produces certain correlations quantum mechanics does. No-signalling means neither side's marginal depends on the other's setting. Quantum-realizable means compatible with some quantum state and measurements; the NPA hierarchy gives a ladder of necessary tests, each a positive-semidefiniteness condition on a moment matrix, and no single rung is sufficient in general. The PR box is the no-signalling correlation that wins the game every time, stronger than quantum mechanics allows; nature doesn't provide it. Cauchy-Schwarz is the inequality that a dot product is bounded by the product of the lengths. The Tsirelson bound is two root two, the largest game score quantum mechanics allows; the classical bound is two and the PR bound is four. (If you go looking for Boris Tsirelson's 1980 paper, look under Cirel'son, the spelling the journal printed, or you may not find it.) The elliptope is the set of correlators that can be filled out into a positive-semidefinite matrix.

A.6 Other terms I use without rebuilding

Provenance is the recorded chain of where an object came from; on the small machine, a commitment's provenance is the passing check that came before it (Section 14). Simulation, in computer science, is one system imitating another's input-output behaviour; a Turing machine simulates any system of the model, my machine included, by writing the whole state out on its tape, and the difference this book is about is what a window on that tape shows. That's real, and it's exactly what people reach for when they want to wave the whole question away: if a tape can stand in for the state, what was the difference? The tape carries a picture of the state, not the state, the way a photograph of a fire keeps you exactly as warm as any other photograph. The counter and the toll live in the step rule, not in the bytes, and an encoding copies the bytes. A Turing machine built with the toll in its own transition table is a different case: it's an instance of the model, not a simulation of it. What no machine can do is make a simulated rule binding just by writing it down on tape.

Projection is a function that drops information on purpose. The window on the small machine keeps three things, the line number and the two counters, which are what Minsky's machine already has, and that's why projecting lands you back in ordinary computation. It drops everything else, including the parts doing the new work: the versions, the table of checked facts, the channel, the ledger and the flag. Nothing is wrong with it. It's a perfectly good function. It just costs you what made the state more than a pair of counters. A quotient is the operation of declaring two things the same when they agree on what you chose to look at and working with the classes instead. A projection does exactly that: it doesn't just drop fields one at a time, it glues together states that the dropped fields kept apart. The fibre of a function over a value is everything the function sends to that value. An injection sends no two inputs to the same output; a surjection covers every output. Decidable means an algorithm always halts with a yes-or-no answer; keep the claim and the algorithm

apart, since the algorithm decides the claim only once there's a proof that its answer matches. Polynomial time means running time bounded by a power of the input size; P is the problems solvable that way, NP the problems where a candidate solution can be *checked* that way, and whether the two are the same is open.

"Kernel" wears three hats in this book, and I'd rather point that out than let it ambush you. First, mathematics: the kernel of a map is what it sends to the identity, and for a projection the useful version is the relation "these two states project to the same thing." Second, the proofs, where the hat comes in two sizes: the kernel tier is the part of the Coq development the main theorems live in (Section 17), holding the model's theorems and the link files that tie the small machine to them, and the files are where the model is written down, not the model itself; and Coq's own kernel is the small checker that accepts or rejects every proof. Third, the operating-system kernel, which I never use and mention only so you know it isn't meant. The sentence around the word always says which one is.

A.7 Matrices

A matrix is positive semidefinite (PSD) when $v^T M v \geq 0$ for every vector v ; the game check is all about one matrix being PSD. The determinant of a square matrix is a single number that is zero exactly when the matrix squashes some direction flat. For a two-by-two matrix with rows (a, b) and (c, d) it's $ad - bc$. For a bigger one, expand along a row: for each entry, delete its row and column, take the determinant of what's left, multiply by the entry, alternate the signs, and add up; the recursion bottoms out at two by two. A principal submatrix is what you get by choosing some of the indices and keeping only those rows and the same columns; its determinant is a principal minor. A symmetric matrix is PSD exactly when every principal submatrix is, every one of them, and the game check uses the three-by-three ones (Appendix F.4). The Schur complement turns a question about a block matrix into a question about a smaller one. For a symmetric block matrix with corner block P , off-diagonal block Q and other corner R , it's $R - Q^T P^{-1} Q$, and when P is positive definite the whole matrix is PSD exactly when that complement is (Lemma F.8). Positive definite matters; invertible isn't enough. Take $P = -1$ and every other entry zero: the complement is zero, which is PSD, and the whole matrix isn't.

A.8 Borrowed ideas the book pushes on

These get their full story in the chapter that uses them; here's the one-line version for when you've forgotten which one Landauer was.

The pigeonhole principle: put more pigeons than holes and some hole holds two. Section 4's room with too few chairs is that, and so is the proof that a permanent flag can only be raised by a step that forgets something. Landauer's principle (Rolf Landauer, 1961): a step that wipes information, lowering the entropy of what a machine holds by some number of bits, gives off at least Boltzmann's constant times the temperature times the natural log of two in heat per bit. In this book it's always a premise with

its name on it, never derived (Section 4, Section 16). Entropy, in Claude Shannon's sense: for a spread of chances over states, roughly the average number of yes-or-no questions you'd need to pin the state down, so an even spread over n states has $\log_2 n$ bits. A spread, in this book, is just a list of chances that add up to one. Maxwell's demon (Section 5): a little creature that watches molecules and sorts fast from slow, seemingly for free; Charles Bennett's answer was that the measuring can be free and the wiping of the demon's notes is what costs. The potential method (Robert Tarjan, 1985): pay for a run with a quantity stored in the state, so a step's bill plus the change in the stored quantity is what you count; Section 6 reads the toll as that method with one particular potential.

The halting problem (Alan Turing, 1936): no program decides, for every program and input, whether it stops. Kleene's recursion theorem (Stephen Kleene, 1938): for any transformation of programs that can itself be computed, some program behaves exactly like its own transform. Rice's theorem (Henry Rice, 1953): any property that depends only on what a program does, and isn't true of all programs or of none, can't be decided by a program. The diagonal is the move under all three: build the thing that does the opposite of what the table says about it (Section 11). L is a small lambda calculus, a language made of nothing but functions applied to functions, which Yannick Forster and Gert Smolka built in Coq and which Section 11 uses as a tool. A hash is a short fingerprint of a longer piece of data, made so that finding two inputs with the same fingerprint is supposed to be hard; a hash chain folds each new record into the fingerprint of the last, and a Merkle tree folds them pairwise into one root (Section 3, Section 12).

A.9 Words this book builds

These are mine, or mine in the sense that I pinned them down for this book. Each is built properly in the chapter named; this is a reminder, not the build.

A few words first, because everything else hangs off them and each has exactly one meaning. The *model* (or the abstract model) is a kind of state and a rule for stepping, plus exactly the extra pieces a given theorem asks for: a certification reading and the toll, or a counter and a price list, or a window, or programs written as states and a recursion theorem. The name Thiele Machine belongs to the model. Parts I to III live there, and every claim in them is about any system with the pieces it names. A *substrate* is the model's shape with whatever extension a theorem names, and nothing else; the word never means my machine, and never the side of a simulation that does the stepping (*Let me be straight with you*). *My machine* (also the machine I build, the build, the small machine, the smallest honest machine, Section 14) is the concrete machine of Part IV that fills in those pieces at once: two counters with their versions, a table of checked facts, a channel, a trap, a ledger and a flag, six instructions and their prices, with its hosts and universal programs. The *scaffolding* is that machine and everything built on it, and Part IV lives there. A *Thiele machine*, lower case, is any system proved to be an instance of the certification system (Definition 3.1), five parts

and the toll; the paper toys are, and so is the small machine. *Weakly Thiele-complete* means a Thiele machine whose base is Turing-universal; that bar is nearly empty, since a clock that charges one for every move pays the toll for any record. *Thiele-complete* means more (Definition 14.7): a universal base whose moves are free and can't touch the record, a record raised only after a passing check, a commitment to the same claim and a certificate, one mark for each of those three acts, and a claim whose check could fail. Every clock-style machine fails it; the small machine meets it; and every machine that meets it hides its record and ledger from its own window. A *universal* Thiele machine is one fixed program that runs any machine of a stated class as a guest, with the guest's record earned again and its toll paid by the host's own steps. The ones in this book are U (Section 14.13), a fixed program in the small machine's instruction kinds, with no guest step built in, that runs every small-machine program with halting, flag and ledger exact, and U_p (Section 14.14), the same program with one paid move added, that runs every *computably presented* Thiele machine, one whose step, prices and reading are computed by programs with the reading decided on every state, keeping its reading as a latch and its prices to within an unavoidable two. The simpler host with a built-in guest step runs guests by construction and keeps a mirror of the guest's flag. A computer running the small machine isn't a Thiele machine; it implements one, inside bounds, the way a laptop implements a Turing machine. When a word below belongs to only one of the two layers, the entry says which.

A *partition* is a sorting of a pile into groups, nothing in two groups and nothing left over (Section 2). A *certification system* is a state type, moves, a step, a cost per move, and a yes-or-no reading, plus one rule, the *toll*: a move that turns the reading from no to yes costs at least one (Section 3). The files call that rule A2 in one of their two numbered lists (Appendix B). A move *merges* when two different starting states land on the same end; one that never does is *injective*, it forgets nothing. A reading is *permanent* when no move turns it from yes back to no (Section 4).

The *counter*, which I also call the *ledger*: in the model, a number carried in the state that starts at zero and that every step raises by exactly that step's charge (Section 6). My machine has exactly one, the ledger μ (Section 14). Two other things get called counters, and each chapter says which: Minsky's two bins on the small machine, and the eight counts that tally the game's rounds (Section 15). Neither is the ledger. A *price list* says what each move costs; the model takes it as given, and my machine's table of what each instruction costs is in Section 14. A *receipt* is a total that lives in the state, only climbs, equals the sum of the charges on every path including the failures, and is the only such total for its price list (Section 6 gives the argument for any system with a counter and a price list; on my machine the sum is a checked theorem, Section 14). In the model there is just a *certification reading*, yes or no (Section 3); on the small machine it's the *flag*, set by a CERTIFY that passes and never cleared.

These belong to the small machine only (Section 14). A *two-counter machine* is Minsky's: two counters holding whole numbers with no upper limit, and two instructions: add

one, and subtract one and jump if the counter is above zero, or else go on to the next line; already enough that nobody can decide in general whether one halts. The small machine is a two-counter machine plus HALT and three instructions for the record, six in all: INC, DEC, HALT, CHECK, COMMIT and CERTIFY. A counter's *version* is its edition number: it starts at 0 and goes up by one every time the counter is written, so an edition never comes back. A *fact* is a property, a counter and a version together, such as "*A is even, as of edition 3*"; the properties come from a fixed language of three, *is zero*, *is even* and *is at least n*. The same theorems also hold for any property language whose checker is proved to mean what it says, and one such language reads a counter as a list and asks whether it's sorted (`sorted_run_certifies_iff` and `sorted_run_refused_forever`, `EarnedGeneric.v`). The *fact table* holds up to sixteen facts, and the only way in is a CHECK that passed; nothing ever leaves it. The small machine's *channel* holds one committed fact or nothing, and COMMIT puts a fact there only if the table has it about the counter's current edition. It names the last commitment earned that way, and a later COMMIT replaces it (`certified_channel_earned` and `channel_names_last_commit`, `EarnedCore.v`). The *trap* is a bit that goes up when an instruction refuses, and then nothing in the core moves again. A commitment is *earned* when a passing check put its fact in the table first, about the same edition, with no write to that counter in between; on the small machine every flag raised from a clean start is earned, and that's a theorem. CHECK, COMMIT and CERTIFY cost one each and everything else costs nothing, so the cheapest certificate costs exactly three, and a check that could have failed reaches it: CHECK *is zero* on *A*, COMMIT, CERTIFY, from a start where *A* is 0. The step doesn't police trivial checks; *is at least 0* passes every value and certifies for three too. The *host machine* is the small machine with a counter, called a *register*, for every whole number, and one property, Slot, true of a number that packs a property of the small machine together with a value it holds of. A *slot* is a register of the host used for nothing but holding one checked fact about the guest, and its *mirror* is a register beside it, safe to read, that says which claim the slot holds while its fact is live. *U* is the fixed program of 3728 host instructions that runs every small-machine program; it reads guest instructions out of one register, keeps the guest's counters in two others, and puts every checked fact in a slot, because on this machine reading a counter writes it. The universal program's *guest* is a program of the small machine, held as one number in a register of the host and run one instruction at a time by *U*'s fetch and decode. *U_p* is *U* with one more handler, for PAY, an instruction that costs one and does nothing else; its guests are programs compiled from computably presented Thiele machines (Section 14.14).

A *window* is any function that shows you part of a state (Section 7); a *shadow*, said loosely, is what a window leaves you with. A *collision* is two steps that look the same through a window, before and after, where one raises the flag and the other doesn't (Section 7). *Entitlement* is what a run has when it carries an object, a checked relation between that object and a claim, and a paid event that put the object in place: a license to lean on the claim, not a proof that it's true (Section 8). A *latch* is a record

that switches on the first time some event holds and never switches off (Section 10). The diagonal's *substrate* (Section 11) is the model's shape extended with programs, a run that may give nothing back, a counter that never falls, a way to write programs as states, the transformations the substrate counts as its own, and its own recursion theorem for those; the proof of the diagonal uses neither the counter nor the toll.

B Every assumption in one place

This book leans on a small number of rules that nothing proves, plus a handful of record types that carry rules as fields. I've named each one where it's used. Here they are side by side, because a reader who wants to break this book should be able to see every place where it says "suppose," in one list, with what each "suppose" buys.

One thing to undo first, and it's my fault. The checked work numbers the rules its records carry with bare labels, A2, A3 and so on, as if they were axioms. The book uses "the toll" for the rule that matters and avoids the bare labels; this appendix maps the labels for anyone reading the files.

B.1 The records that carry a rule

Definition B.1 (Certification system). The record `CertificationSystem` (`UniversalCertificationCost.v`) has five data fields, a state type, a move type, a step function, a cost per move in $\mathbb{N}$, and a yes/no reading, and one proof field, `cs_cert_costs`: a move that turns the reading from no to yes costs at least one. This proof field is the toll of Section 3, and in this record's numbering it is called A2. Every result stated "for any certification system" assumes it, because the record cannot be built without it.

Definition B.2 (Quantitative certification system). The record `QuantitativeCertificationSystem` (`QuantitativeNoFI.v`) extends a certification system with a *witness*, a function W from states to $\mathbb{N}$, and a threshold $N \in \mathbb{N}$, and adds two proof fields: (A3, `qcs_cost_bounds_witness`) $W(s) + \text{cost}(i) \geq W(\text{step}(s, i))$, the witness grows by at most what the move cost; and (A5, `qcs_cert_threshold_witness`) a state that reads yes has $W(s) \geq N$. An optional extension, `QuantitativeCertificationSystem_full`, adds (A4, `qcsf_witness_nondecreasing`) $W(s) \leq W(\text{step}(s, i))$.

Theorem B.1 (The floor raised to a threshold). *In a quantitative certification system, every run from a state with witness 0 (this starting condition is called A6) to a state that reads yes has total cost at least N .*

Proof. Adding A3 along the run telescopes: the witness at the end is at most the witness at the start plus the total cost, so at most the total cost. A5 says the end witness is at least N . Neither A4 nor the toll is used. $\square$

Checked as `universal_nfi_quantitative` (`QuantitativeNoFI.v`). A3 and A4 bound opposite directions, and neither implies the other: witness 1, cost 0, witness

0 satisfies A3 and breaks A4, and witness 0, cost 0, witness 1 does the reverse. The record is a general shape; the small machine’s own floor of three is proved directly (Theorem 14.16), not through this record.

Here are the labels side by side, so none of them is left without a meaning.

Label	In the certification records
A2	the toll: field <code>cs_cert_costs</code> of <code>CertificationSystem</code>
A3	the witness grows by at most the cost: field <code>qcs_cost_bounds_witness</code>
A4	the witness never drops: field <code>qcsf_witness_nondecreasing</code> of the full record
A5	yes needs a witness at the threshold: field <code>qcs_cert_threshold_witness</code>
A6	the run starts with witness zero: a hypothesis of <code>universal_nfi_quantitative</code>

B.2 The premises the derivations take

These are definitions used as hypotheses. Nothing proves them in general; each theorem that uses one says so in its statement.

Premise	What it says	Where it’s used	Checked name
Finite	a list with no repeats contains every state	the room argument, Section 4	<code>finite_states</code>
Permanent	no move turns the reading from yes to no	the room argument	<code>permanent</code> (and <code>permanent_at</code> for one move)
Merge pricing	every move that merges costs at least one	the toll derived, Section 4	<code>merging_steps_priced</code>
Halving price	each unit of cost pays for at most one halving of distinct states	the squeeze bound; narrowing along a run, Section 5	<code>compression_priced</code>
Entropy pricing	each move costs at least the bits it removes from any spread	the third road to the toll	<code>entropy_priced</code>
Landauer’s principle	removing ΔH bits gives off heat $\geq k_B T \ln 2 \cdot \Delta H$	the heat floor; with $k_B T \geq 0$ (or > 0) as a separate hypothesis on each theorem	<code>landauer_heat</code>

Premise	What it says	Where it's used	Checked name
Recursion (generic)	every transformation the substrate can represent has a fixed point	the diagonal, Section 11	field <code>recursion_theorem</code> of the class <code>Substrate</code>
Commitment contract	a commitment bit is set only when the claim holds, and whenever it holds	the commitment escape, Section 7	stated in Definition 7.1
Computably presented	four programs compute the driver, the step, the price and the reading, and the reading's program answers on every state	the universal host for presented machines, Section 14.14	<code>cg_presentation</code>
Clean start	an empty fact table, an empty channel and the flag down	every provenance theorem about the small machine, Section 14	<code>clean_start</code>

Three remarks on the table. First, the recursion premise holds by construction for the `nat` family and by proof for `L` (Section 11) and for the programs of the small machine's host (Section 14.18). Second, the class `Substrate` (`Substrate.v`) carries a different rule from the toll, `mu_monotone`: the counter never goes down on a step. Neither implies the other, and the diagonal uses neither. Third, merge pricing is Landauer's principle taken in its worst case over what the machine might be holding: it charges every merging move, while the principle in its general form charges only for uncertainty actually removed, when the step can be driven by a protocol timed from outside it (Theorem 4.12); under time-symmetric control even a known state pays for a move that doesn't undo itself (Riechers, Boyd, Wimsatt and Crutchfield, 2020).

B.3 The axioms of the checker itself

Beyond the premises above, the checked proofs rest on Coq's own foundation and on a handful of standard-library axioms, which the assumption audit lists theorem by theorem: functional extensionality on dependent functions; two decidability principles for the real numbers (used wherever real analysis is, for example in the game check of Section 15); and the law of excluded middle, in some theorems. No axiom or parameter of the project's own appears in the audited theorems. Section 17 says what that does and doesn't mean.

C What's mine, and what isn't

Every chapter of this book leans on somebody, and I've named them where I lean. This appendix does a different job, and it's the one where I'd most like to be caught out early rather than late. For each thing the book could be accused of claiming as its own, it says what was already known, what this work actually defines, builds, proves, or connects, and what kind of contribution that is. Six kinds, and they aren't the same claim: a *new result* (a statement nobody had); a *new proof* of something known; a *mechanization* (a known result checked by a machine, here Coq); an *implementation* (something built that runs); a *synthesis* (known pieces put together under one question); and a *conjecture*. Every kind can be worth doing. None of them is the others. And if you read down the list and notice that "new result" hardly shows up, good. You're reading it right.

Two warnings about my own evidence before the list. First, "I didn't find it" is a fact about my searches, not about the literature. I searched with every tool I had, and some tools will cheerfully invent a citation if you let them, so where it matters I say where I looked, and you should assume it may well be sitting in a paper I'd need a few more lifetimes of homework to read. Second, nothing below is inferred from how many theorems there are, from the audit's clean bill, or from a good story about who came before. Those tell you the work was checked. They don't tell you it's new. Thousands of checked theorems could, in principle, be thousands of careful ways of agreeing with Tarjan.

C.1 1. The universal floor and the potential method

Known: the potential method of amortized analysis, a function on states whose drop bounds the cost of a run, is Tarjan's (1985), and it has been mechanized more than once (Nipkow in Isabelle/HOL, 2015 and 2019; Charguéraud and Pottier's time credits in Coq, 2019), built into type systems for automatic resource analysis (Hoffmann and coauthors, 2011 onward), and recast in cost-aware type theories (Niu, Sterling, Grodin and Harper, 2022). Mine: choosing the potential that is one when a designated reading says no and zero when it says yes, so that "the crossing costs" (the toll) and "every run that crosses paid" (the floor) are the potential method's two halves (Section 6). Kind: a mechanization of a known technique with a particular choice of potential. The choice is the point: it makes "was certified" something the account must carry. The theorem is not new mathematics. If it looks clever, that's Tarjan's cleverness; my part was picking which number to plug in.

Go back to Tarjan's paper and there's a sentence that sits right on the line this book draws. He opens with the dictionary (to amortize is to put money aside at intervals, as in a sinking fund) and then gives two views of the method. In the banker's view the computer is coin-operated, and each operation is handed credits to spend or save; in the physicist's view, which he credits to Daniel Sleator, every state has a potential and you count its changes. Then he stops to say that the credits are only an accounting

device: the programs that actually run contain no mention of credits or debits. That's exactly right for what he wanted, a bound proved about a program from outside. It's also exactly the gap. In his method the coins live in the proof. The one move I chose was to put them in the step, where no program takes a step without it being counted. Same arithmetic, different place.

C.2 2. The toll from finiteness, permanence, and a price on merges

Known: that steps which merge states (no single-valued inverse) are what Landauer's principle charges for is Landauer's (1961), sharpened by Bennett (1973, 1982, 2003; the words "the erasure of a bit or the merging of two computation paths" are from the 2003 abstract); generalizations and the dependence on the input distribution are Maroney's (2009), Sagawa's (2014), Wolpert's; the critique is Earman and Norton's and Norton's. Write-once memory is an old topic in coding theory (Rivest and Shamir, 1982), about reuse rather than price, and Chris Heegard (1985) worked out the capacity of permanent memory with noise, again a question of how much fits and not of what it costs. Statisticians priced permanence on finite memory in their own currency: in Hellman and Cover's analysis of hypothesis testing with finite memory (1970), a machine that settles its answer by falling into states it never leaves does worse than one that doesn't, so a decision held for good costs memory states, not heat. And the counting under the theorem is older than any of that, twice over. Theorem 4.1 is the finite, discrete cousin of the recurrence argument Zermelo, using Poincaré's theorem of 1890, turned on Boltzmann in 1896, aimed at a reading instead of at entropy, and Bocchieri and Loinger gave the quantum recurrence in 1957. In automata form it is classical: a finite machine whose moves are all one-to-one reshuffles recognises only group languages, and the inputs that raise a permanent reading form a language that is never one of those unless it is empty (Pin's 1992 paper on reversible automata is the reference I'd check; automata theorists treat the step as folklore), and going quantum doesn't change it (Brodsky and Pippenger, 2002: quantum finite automata that measure once, at the end, accept with bounded error exactly the group languages). The closest of all is recent. Ouldrige and Wolpert (2023) proved that if the smallest automaton for a language has a step that merges states, every automaton for it does, and recognising it on a device that repeats one local step produces entropy, while a language whose smallest automaton only reshuffles can be recognised for free; one of their worked examples is the strings with no b , the complement of the latch's language. That's the core with a physical price already on it. And the logarithm in the heat floor matches the absolute-irreversibility bound of Murashita, Funo and Ueda (2014), which Manzano, Kardeş, Roldán and Wolpert (2024) carried over to computation, but as a mirror and not a special case: the squeeze is ordinary Landauer heat, whose logarithm equals their bound for the free expansion that undoes it, and absolute irreversibility reaches the commit itself only on a fixed-rate device starting from a known state. Constructor theory states a cousin of the same core for devices that run in a cycle: Marletto, Vedral and coauthors (2022) and Violaris and Marletto (2022) built an irreversible task out of the quantum homogeniser and argued that

erasure may cost more than Landauer’s bound. Section 4 tells all of it. Mine: the price as a named rule on a ledger, a premise that merges cost at least a mark, which turns “the flip merges” into “the flip costs” on any machine whether or not anybody’s physics is right, and the toll derived from it (Theorem 4.2); what comes with that form, namely the characterisation of the moves every merge-pricing schedule must charge (the merging moves, so the core again in pricing words), the merge-or-revoke dichotomy for a single move, the squeeze and entropy forms for a permanent flip, and a built counterexample for each premise; and the mechanization of all of it (Section 4). Kind: the core is classical, not mine: Zermelo’s in physics, folklore in automata theory, with Pin’s paper as the place to check it. The pricing form isn’t wholly mine either: priced in entropy, for automata, it’s Ouldrige and Wolpert’s, derived from the physics instead of assumed. What’s mine is the price as a named rule on a ledger, the toll derived from it, its single-move and counted forms, the counterexamples, and the mechanized proofs, and the physics is a named premise and is Landauer’s. Searches: the core turned up in the places just named, and its physical price in Ouldrige and Wolpert; I didn’t find the ledger form or the derived toll stated anywhere I looked, and nobody’s search covers a whole field, mine least of all; I found no mechanization of Landauer’s principle in Coq, Isabelle, or Lean.

C.3 3. Learning free, committing priced

Known: that measurement can be done at no fundamental cost and the cost sits in erasing the record is Bennett’s resolution of Maxwell’s demon (1982), with the quantitative trade-off by Sagawa and Ueda (2009). The closest physics I found to the line between learning and committing is Ouldrige and ten Wolde’s fundamental costs of persistent copies (2017; with Govern, 2017), copies that stay correlated with their originals after the two come apart; theirs is heat for a correlation in chemistry, and Section 5 says how it differs. The sharp chemical form, that a persistent copy grown at equilibrium carries no information about its template, is Poulton, ten Wolde and Ouldrige’s (2019), and that the bill can be split between measuring and erasing in more than one way is Fahn’s (1996). Mine: the four-state demon and the three-state ring as checked models, and the two-state impossibility (Section 5). Kind: mechanization and illustration of a known physical point inside my ledger. Not a new physical claim. Bennett did the hard part in 1982; I built a toy with four dots on it and got Coq to agree with him.

C.4 4. Windows that can’t price what they hide

Known: a function of an observation is constant on the observation’s fibres; that’s elementary, and the systems version of it is Moore’s distinguishing experiments (1956), Kalman’s observability (1960), and the completeness question of abstract interpretation (Cousot and Cousot, 1977; Giacobazzi, Ranzato and Scozzari, 2000); in control theory it is diagnosability and opacity of discrete-event systems (Sampath and coauthors, 1995; Saboori and Hadjicostis, 2007), where a collision is a hidden event no observer can diagnose without delay; in economics it is the hard-evidence view of

mechanism design (Green and Laffont, 1986; Bull and Watson, 2007), where what a mechanism can tell apart is limited by the evidence on offer; and its sharp form for fingerprints is Naor and Rothblum’s lower bound for checking a stored file (2005); in distributed systems it is Haeberlen and Kuznetsov’s (2009), that only omission and commission faults can be detected from outside, and in philosophy it is Easwaran’s transferability (2009). Mine: stating the pricing version (no window price is exact across a collision), exhibiting a colliding pair for the small machine’s window, and the theorem that every Thiele-complete machine has such a pair for the window its base shows (Section 7, Section 14). Kind: mechanization of an elementary lemma, plus specific facts about the machine and the definition.

C.5 5. Structural entitlement and its bound

Known: decision-tree counting bounds (a tree that tells N cases apart has depth at least $\log_2 N$); certifying algorithms that hand over checkable certificates (McConnell, Mehlhorn, Näher and Schweitzer, 2011); proof-carrying code (Necula, 1997); the gap between justified true belief and knowledge (Gettier, 1963). Mine: the definition of entitlement as an object, a checked relation, and a paid event on the record, and the count that, given lists, a tree and a covering, a run that paid for the tree’s depth pays at least the index bits it commits to (Section 8). Kind: synthesis; the counting idea is standard, the assembly and its vocabulary are this work’s, the pricing policy is assumed, not derived, and the count is mechanized: in its bill form over any certification system, and for the record moves of every Thiele-complete machine, with the earned chain in the run, a floor that takes the run’s own checks in place of a tree, and a counterexample showing the covering can’t be dropped (Section 14.17).

C.6 6. The record axis is a latch

Known: monotone, write-once state is old: the latch (Eccles and Jordan, 1918), sticky bits (Plotkin, 1989), grow-only state in replicated data types (Shapiro and coauthors), history variables and stuttering in refinement (Abadi and Lamport, 1991), append-only authenticated sets for transparency logs, which fit revocation as a once-only event (Tomescu and coauthors, 2019), and, in program logic, monotonic state, where a property witnessed once on state that only moves forward can be recalled at any later point, with a soundness proof and a working home in F^* (Ahman, Fournet, Hrițcu, Maillard, Rastogi and Swamy, 2018): the type-theory name for a record that stays up. Past the finite case, inductive and quasi-inductive definitions (Richter and Aczel, 1974; Burgess, 1986) and the flag trick of infinite time Turing machines (Hamkins and Lewis, 2000) are the transfinite versions of a record that only grows and a toggle, and Danos and Krivine’s (2005) reversible calculus needs an action declared irreversible in order to commit. Mine: the exact statement that a record driven by the computation that never switches off is, over any base, a latch on one event, with its converse and a counterexample for each of the two conditions, and the argument that honest records agree up to the price list while the stronger uniqueness fails (Section 10). Kind: a small structural lemma, I think not stated before in this form (I found no prior statement

in the searches I ran), built from old notions; the latch form is mechanized and the uniqueness is argued.

C.7 7. The diagonal and L

Known: Kleene’s recursion theorem and Rice’s theorem; Lawvere’s fixed-point form (1969) and Yanofsky’s retelling (2003); mechanized recursion theorems and Rice (Norris in HOL4, 2011; Carneiro in Lean; Forster, Kunze, Smolka and coauthors in Coq, including L itself, 2017, and the Coq Library of Undecidability). Mine: the diagonal stated once over any substrate that supplies a recursion theorem; and Rice and halting proved again for L from its reduction rules (Section 11); and the same two theorems built for the small machine’s host, with Rice’s also from the clean start on the two-counter machine (Section 14.18). Kind: checked instances, not new theorems of computability. They hold for any model with universality and s-m-n, and the work is in building those for this machine.

C.8 8. The game check

Known: that positive semidefiniteness of a correlation matrix gives the Tsirelson bound is Tsirelson’s (1980) and Wehner’s semidefinite proof (2006); the moment-matrix hierarchy, the Schur-complement analysis of this very level-one matrix, the arcsine characterization, and the level $1 + AB$ are Navascués, Pironio and Acín’s (2007, 2008), with Landau (1988) and Masanes (2003); and Tsirelson’s bound has already been checked by machine, in Isabelle/HOL (Echenim and Mhalla, 2023) and in Lean’s mathlib. Rational sum-of-squares certificates are Peyrl and Parrilo’s (2008). Mine: a decision procedure in integer arithmetic on outcome counts; a checked chain from “the counts pass” to “the pinned level-one matrix is positive semidefinite” to “ $S^2 \leq 8$ ”; proofs of the contraction equivalence by explicit test vectors; and the level- $1 + AB$ cascade and an ellipsope gate in whole numbers, with a soundness proof (Section 15 and its derivation appendix). Kind: mechanization, with a new proof of a known equivalence in one slice. I found no prior formalization taking counts as input, and that’s a report on where I looked, not on everything that exists. The same check earned on the small machine’s host, a raised flag standing on it, is built and checked, not new in idea (Section 15).

C.9 9. Five record disciplines, one question

Known: each system and its guarantee belongs to its designers: Casper (Buterin and Griffith, 2017), Ethereum’s storage gas and refunds (EIP-2200, EIP-3529), extend-only TPM registers, Certificate Transparency (RFC 9162), proof-carrying code. Mine: asking all five whether the write is priced, with a small checked fragment of each (Section 12). Kind: synthesis and an organizing reading; the fragments are not verifications of those systems.

C.10 10. The pointer conjecture

Known: Zurek’s quantum Darwinism (2009), with Ollivier, Poulin and Zurek (2004) and Blume-Kohout and Zurek (2006): preferred states are the ones whose records proliferate redundantly in the environment. Mine: borrowing that shape for the question of which event deserves a record, with definitions, a toy, and refutations of two stronger forms (Section 13). Kind: conjecture. No claim about quantum physics.

C.11 11. The counter as a receipt

Known: one-way counters that give order and termination are old: Turing’s 1949 note on checking a large routine, Floyd’s ranking functions (1967), Lamport’s clocks (1978). Carrying a run’s cost beside its result is the writer cost of Danner and Licata and the graded effect semantics of Katsumata (2014), with a working-language version in Orchard, Liepelt and Eades (2019). The receipt side of time credits is Mével, Jourdan and Pottier’s (2019): in their Coq-checked extension of the Iris logic every step uses up a credit and produces a receipt, receipts prove lower bounds, and enough of them prove that a counter can’t overflow in any feasible time, which is the closest published neighbour of a total that only climbs. Mine: putting three properties side by side (the total only climbs, it equals the sum of the charges on every path including the failures, and it is the only such total for a given price list), the argument for any system with a counter and a price list, and the checked sum on the small machine (Section 6, Section 14). Kind: synthesis; the abstract statements are argued in the text and are not checked in Coq, and on the small machine the sum is. Against the graded frameworks the answer is split: the counter is their writer exactly (`run_writer_is_run_and_cost`), and their type systems, bounding relations and before-the-run checking of grades are not embedded, because the grade here is written by the step during the run (Section 6). On loop-free programs the counter agrees with a fixed grade exactly, and where a run’s cost depends on its data it holds the exact number that a fixed grade can only bound; that’s argued in Section 6, not checked, and cost-aware type theories such as calf get exactness back with grades that depend on the inputs.

C.12 12. The small machine

Known: two-counter machines are Minsky’s (1961, 1967), and the undecidability of their halting is checked in the Coq Library of Undecidability (Forster, Larchey-Wendling and coauthors, 2019, 2020). The guarantee about every program is older than any of this, and it isn’t mine either. It’s the reference monitor (Anderson, 1972): a check that sits in the path of every operation, so no program gets around it. Schneider (2000) worked out what a monitor that watches a run can enforce: only safety properties, the kind a bad run breaks at some finite moment, and “no CERTIFY without an earlier earned COMMIT” is one of those. Azevedo de Amorim and coauthors (2015) verified, in Coq, monitors that check tags at every instruction. The shape of “earned,” a certificate nothing can make except through the rules, is Milner’s LCF, where a theorem is a value only the logic’s rules can build (Gordon, Milner and Wadsworth, 1979; Gordon, 2000); the small machine is that architecture at the level of single in-

structions. “No certify without an earlier commit” is also a correspondence assertion in the sense of Woo and Lam (1993), which Gordon and Jeffrey (2003) check with a type system, and a store of commitments kept by the rules and not the speaker is Hamblin’s (1970); the difference from the first is again that here the step enforces it. A check that blesses a value and a write that takes the blessing away is close kin to dynamic taint tracking, whose labels go back to Denning’s lattice model of information flow (1976), with the blessing itself, controlled endorsement, given a type system and a security theorem by Cecchetti, Myers and Arden (2017), and what a monitor can enforce with bounded memory is Talhi, Tawbi and Debbabi’s question (2008). A small monotone record inside each replica of a replicated service is A2M (Chun and coauthors, 2007), TrInc (Levin and coauthors, 2009), MinBFT (Veronese and coauthors, 2013) and Damysus (Decouchant and coauthors, 2022), and Clement, Junqueira, Kate and Rodrigues (2012) proved what such a record buys and what it doesn’t. The signature route to the same goal, which needs no trusted step, is IA-CCF (Shamis and coauthors, 2021): replicas sign and log their protocol messages in the ledger, so blame lands on individual members however many of them misbehave. And “check once, issue a capability the system trusts, rely on it later” is built and deployed in Garg and Pfenning’s proof-carrying file system (2010), whose logic lets a policy’s consequences change with time and state, and in Sirer and coauthors’ logical attestation for the Nexus operating system (2011), unforgeable statements about programs used as credentials; those are the closest systems I’ve found to check, commit, certify. Mine: a six-instruction machine on two counters, with a version for each counter, a bounded fact table and a channel; its theorems that a committed fact was checked first about the same edition, that nothing is forged, that nothing is evicted, that the channel names the last earned commitment (`certified_channel_earned` and `channel_names_last_commit`, `EarnedCore.v`), and that the cheapest certificate whose check could have failed costs exactly three and no less (the step doesn’t police trivial checks, so one that can’t fail costs three too); the same theorems for any property language whose checker is proved to mean what it says, a sorted list read out of a counter included (`sorted_run_certifies_iff` and `sorted_run_refused_forever`, `EarnedGeneric.v`); and the step-for-step match with the library’s machine, so halting is undecidable for it (Section 14). Kind: an instance of monitor enforcement, implemented and mechanized, with two things on top that I didn’t find put together where I looked: the check is bound to the edition of the thing it checked, and a toll is paid for the commitment. The universality and the undecidability are borrowed, and the other properties are short consequences of the step rule. Whether the design is new I can’t say, and I make no claim beyond the proofs.

C.13 13. The universal host

Known: Turing’s universal machine (1936), which runs any machine given as data, and the universal register machines that run two-counter programs. Keeping a guest and the code running beside it out of each other’s state is what a virtual machine monitor promises (Popek and Goldberg, 1974). Compiling computable functions into

counter machines is checked in the Coq Library of Undecidability. Mine: one fixed program, U , written in the small machine's own six kinds of instruction on a host with a register for every whole number, with no guest step built in, that runs every program of the small machine from every start, halts exactly when its guest does, raises its own flag by its own check, commit and certificate exactly when the guest's would rise, and keeps a ledger equal to the guest's at every matching point; its host machine is Thiele-complete, and whether U halts on a loaded guest is undecidable (Section 14.13). Two small facts force its shape, and they're stated there: reading a counter on this machine writes it, and a finite program can't keep the guest's facts on a copy of the guest's counter, by the pigeonhole. Beside it, a simpler host with a built-in guest step runs guests by construction and keeps a mirror of the guest's flag, with the mirror, isolation and earned-chain theorems. An ordinary programmable computer, as built, doesn't do this for the programs it runs, because the code beside a program shares its memory and can flip anything; U is a fixed interpreter that does. Kind: implementation and mechanization of a design whose shape is Turing's universal machine, with the record and the toll carried through. The universality is borrowed from the small machine and from Minsky. And the same program with one paid move added, U_P , runs every computably presented Thiele machine, with its halting matched, its reading carried as a latch on the host's own earned flag whose claim is that reading, and its prices to within a surcharge of at most two that no program on the host can avoid in general (Section 14.14); the compiler that carries the guest onto two counters is the library's. Kind for that part: implementation and mechanization again. That a universal machine runs any computable machine is Turing's; what's carried through here is the record, earned again, and the prices, exact up to the surcharge, and I didn't find that put together where I looked.

C.14 14. Thiele-complete, defined so a clock fails

Known: Turing completeness, and the observation every theorist makes in four seconds, that a toll alone is cheap: a machine that charges for every step pays any toll. Enforcement mechanisms have been compared and sorted before, and they aren't one notion seen several ways: James Morris (1973) separated access limitation from authentication in programming languages, Wai Wong (1995) recorded and checked HOL proofs so a theorem could carry its derivation, and Kevin Hamlen, Greg Morrisett and Fred Schneider (2006) classified which policies execution monitors, program rewriters and static analysis can each enforce, and they differ. Mine: a definition of Thiele-complete with four clauses (a free universal base that can't touch the record, an earned record, an exact toll on the three record acts, and a check that can fail), the proofs that every clock-style machine and a paid latch and a silent machine fail it while the small machine, the same machine over any property with a proved checker, and the universal program's host meet it, and the theorem that every machine meeting it hides its record and ledger from its own window (Section 14). Kind: a definition with short consequences. The window theorem is the fibre fact of item 4 plus one observation, that a universal base prints any window for free. I'm not claiming the

definition is the only reasonable one, or that LCF kernels, reference monitors and proof-carrying code are all Thiele-complete machines in disguise; Hamlen, Morrisett and Schneider’s sorting says such mechanisms genuinely differ, and the book doesn’t have a robustness theorem to set against that.

C.15 What that adds up to

Read straight down the list and the honest summary is this. The mathematics underneath this book is mostly known, and some of it is very old. What’s mine is the question (should an account of computation carry what was certified and what certifying cost, as part of how it steps), and the answer isn’t one result. It’s four, and I name them the same way wherever they come up. The toll derived from finiteness, permanence and a price on merges (item 2; the counting is classical, the finite cousin of Zermelo’s recurrence objection and folklore about permutation automata, and Ouldrige and Wolpert already priced it in entropy for automata, so what’s mine is the price as a rule on a ledger and the toll it gives, and the price is Landauer’s, as a premise). The pricing form of the window fact (item 4; the fact itself is elementary). The latch form of a record that never comes down (item 6; latches and history variables are old). And a record that’s priced, and on the small machine earned, under the reference monitor’s every-program move (items 12, 13 and 14; the move is Anderson’s, Schneider’s and the micro-policy authors’, and what’s added is the check bound to its edition, the toll on top, a definition of Thiele-complete that a clock fails, and universal programs that carry both to every small-machine program and to every computably presented Thiele machine, earning their own flag). The first is a ledger form of a core that’s classical and already priced in physics; the second and third are small exact statements I didn’t find stated where I looked; the fourth is built and checked rather than new in idea. Under all four sits a large body of mechanized proof that ties those statements to one small machine and the programs that run on it, and my best attempt at saying which is which, which you should check as hard as anything else in here. Item 2’s core was Zermelo’s and Pin’s before it was mine, and its price in physics was Ouldrige and Wolpert’s. If someone shows me its ledger form, the pricing form in item 4, or item 6 stated earlier, I’ll move it up the list and be glad to, and a little embarrassed, in that order.

D Every result and where it is checked

Every numbered result in this book is in the table below, in the order it appears, with the Coq names that carry it, the file each lives in, and what the last full assumption audit printed for it. “none” means the theorem is closed: its proof uses nothing beyond Coq’s own rules. The other entries name the standard-library assumptions it uses: FE is functional extensionality on dependent functions, RD is the pair of decidability principles that come with the real numbers, and EM is excluded middle. “def” marks a definition, which has nothing to assume. “closed (file)” marks a theorem of the small machine’s files under minimal/ (EarnedCore.v, EarnedGeneric.v, UniversalThiele.v, ThieleComplete.v, ThieleCompleteWindow.v, EarnedMulti.v, UniversalCodes.v and

UniversalNoCopy.v, with the files that carry the entitlement, fragment, verifier, search, time-tax and recursion results of Part IV: EntitlementSmall.v, FragmentSmall.v, VerifierSmall.v, MultiThiele2.v, BitSearch2.v, BitSearchMember2.v, BitSearchObserved2.v, EntitlementMore2.v, CompressionSmall2.v, TimeTax2.v, CoveringNeeded2.v and the recursion files SmHostBlocks.v, SmHostRice.v, SmCodes.v, SmInterp.v, SmEvalL.v, SmFuel.v, SmMMAHost.v, SmKleene.v, SmDecider.v and SmGuestRice.v), which the project-wide audit doesn't list: each file ends by printing the assumptions of every headline theorem, the test suite compiles it and fails unless every one of those lines says closed, and a lemma that a printed theorem uses is closed with it. The universal program's files under coq/kernel/foundation (UniversalLayout.v, UniversalSim.v, UniversalRun.v and UniversalInterpreterLinks.v) are in the kernel tree, so the project-wide audit covers them, and they also print their own headline theorems' assumptions; so do the priced host's files and the presented-machine files (UniversalPRun.v, PresentedUniversal.v, PresentedDemo.v and their neighbours). "+N" after a file means *N* other files in the tree define a name spelled the same way; the row gives the file this result uses. A premise or a conjecture has no Coq name on purpose: it's assumed or open, and the section that states it says so.

This table is an index of companion formalizations. The statements, hypotheses and arguments are in the numbered sections it points to. An audit status reports the recorded check; running a later version does not change a claim in this book.

Result	Sec.	Coq name and file	Axioms
Definition 2.1: Equivalence relation, partition	§2	stated in the text	
Theorem 2.1: Sameness and sorting are one object	§2	argued in the text, not checked in Coq	
Definition 3.1: Certification system	§3	CertificationSystem (UniversalCert ificationCost.v)	def
Theorem 3.1: The universal floor	§3	universal_nfi_any_substrate (Univ ersalCertificationCost.v) cs_state (UniversalCertificationCos t.v) cs_instr (UniversalCertificationCos t.v) cs_step (UniversalCertificationCost. v) and 3 more	none def def def
Definition 4.1: Finite, permanent, injective	§4	permanent (PermanentCertification.v) step_injective (PermanentCertificat ion.v)	def def
Theorem 4.1: A permanent flip merges	§4	permanent_flip_is_not_injective (PermanentCertification.v)	none

Result	Sec.	Coq name and file	Axioms
Premise 4.2: Merge pricing	§4	merging_steps_priced (PermanentCertification.v)	def
Theorem 4.2: The toll from finiteness, permanence, and merge pricing	§4	a2_from_merging_price_and_permanence (PermanentCertification.v) permanent_certification_trace_flip_or (PermanentCertification.v) certification_system_from_merging_price (PermanentCertification.v)	none none def
Theorem 4.3: Merge or revoke	§4	flip_merges_or_revokes (PermanentRecordPricing.v) permanent_at_flip_is_not_injective (PermanentRecordPricing.v)	none none
Definition 4.3: Forced to be priced	§4	forced_priced (PermanentRecordPricing.v)	def
Theorem 4.4: Forced price is merging	§4	forced_priced_iff_merges (PermanentRecordPricing.v)	none
Corollary 4.5: Permanent records are forced	§4	forced_priced_iff_merges (PermanentRecordPricing.v) permanent_record_write_is_forced_priced (PermanentRecordPricing.v) forced_price_without_permanent_record (PermanentRecordPricing.v)	none none none
Premise 4.4: Halving price	§4	compression_priced (PermanentRecordPricing.v)	def
Theorem 4.6: The squeeze bound	§4	a2_from_compression_price_and_permanence (PermanentRecordPricing.v) permanent_flips_compression_bound (PermanentRecordPricing.v) permanent_flips_log_bound (PermanentRecordPricing.v) permanent_flips_collapse_at_least (PermanentCertification.v)	none none none none
In the text: an eight-state machine meets every premise	§4	fin_finite, fin_permanent, fin_merging_priced, fin_a2_from_merging_price, fiber_bound_compression (FiniteCertMachine.v)	none
In the text: each move's price is the halvings in its largest squeeze; the halving price holds exactly when no pile-up exceeds 2^{cost}	§4	frag_price_is_squeeze (FragmentSmall1.v) ent2_compression_priced_iff_fibres (CompressionSmall2.v) ent2_frag_compression_priced (CompressionSmall2.v) ent2_frag_toll_by_compression (CompressionSmall2.v) ent2_frag_cost_minimal (CompressionSmall2.v) and 1 more	closed (file) closed (file) closed (file) closed (file) closed (file)

Result	Sec.	Coq name and file	Axioms
Definition 14.11: The eight-state machine, compiled	§14	stated in the text	
Theorem 14.37: The small machine runs the eight-state machine	§14	frag_runs_finite_machine (Fragmen tSmall.v) frag_pays_finite_price (FragmentSm all.v) frag_runs_finite_trace (FragmentSm all.v) frag_certification_paid (FragmentS mall.v) and 5 more	closed (file) closed (file) closed (file) closed (file)
Theorem 14.38: The small machine's merges	§14	frag_small_certify_merges (Fragme ntSmall.v) frag_small_certifying_step_price d_merge (FragmentSmall.v) frag_small_dec_free_merge (Fragme ntSmall.v) frag_small_not_merge_priced (Frag mentSmall.v) and 1 more	closed (file) closed (file) closed (file) closed (file)
Theorem 4.7: A move that forgets nothing removes no bits	§4	step_entropy_invariant_if_inject ive (PermanentCertificationEntropy. v)	FE, RD, EM
Theorem 4.8: Entropy after a permanent flip	§4	permanent_step_entropy_ceiling (P ermanentCertificationEntropy.v) permanent_flip_uniform_entropy_d rop (PermanentCertificationEntropy. v)	FE, RD, EM FE, RD, EM
Premise 4.5: Landauer's principle, as a premise	§4	landauer_heat (PermanentCertificati onEntropy.v)	def
Corollary 4.9: Heat floor of a permanent flip	§4	permanent_step_entropy_ceiling (P ermanentCertificationEntropy.v) permanent_flip_uniform_entropy_d rop (PermanentCertificationEntropy. v) entropy_le_log_support (PermanentC ertificationEntropy.v) permanent_flip_heat_floor (Perman entCertificationEntropy.v) and 1 more	FE, RD, EM FE, RD, EM FE, RD, EM FE, RD, EM
Corollary 4.10: Positive heat for any full spread	§4	permanent_flip_full_support_entr opy_drop_positive (PermanentCertif icationEntropy.v) permanent_flip_full_support_heat _positive (PermanentCertificationEn tropy.v)	FE, RD, EM FE, RD, EM

Result	Sec.	Coq name and file	Axioms
Definition 4.6: Entropy-priced	§4	stated in the text	
Theorem 4.11: The toll from entropy pricing	§4	a2_from_entropy_price_and_permanence (PermanentCertificationEntropy.v) entropy_priced_trace_floor (PermanentCertificationEntropy.v) certification_system_from_entropy_price (PermanentCertificationEntropy.v)	FE, RD, EM FE, RD, EM def
Theorem 4.12: A known state owes nothing	§4	known_state_step_removes_no_entropy (PermanentCertificationEntropy.v) known_state_flip_forces_no_heat (if and only if) (PermanentCertificationEntropy.v)	FE, RD, EM FE, RD, EM
Theorem 4.13: The bits removed, one term per state	§4	entropy_drop_as_point_sum (PermanentCertificationEntropy.v) entropy_drop_nonneg (PermanentCertificationEntropy.v) entropy_drop_pos_of_support_merge (PermanentCertificationEntropy.v)	FE, RD, EM FE, RD, EM FE, RD, EM
Theorem 5.1: The machine's own narrowing is priced	§5	run_narrowing_priced_log (KnowledgeNarrowing.v)	none
Definition 5.1: What an observer knows	§5	knowledge (KnowledgeNarrowing.v)	def
Theorem 5.2: An observer can learn for free	§5	observer_narrowing_can_be_free (KnowledgeNarrowing.v) demon_observer_learns (KnowledgeNarrowing.v) demon_machine_spread_kept (KnowledgeNarrowing.v) measure_forgets_nothing (KnowledgeNarrowing.v) and 2 more	none none none none
Theorem 5.3: Three states are the minimum for free teaching	§5	free_incremental_narrowing_with_three (KnowledgeNarrowingMinimal.v) no_free_incremental_narrowing_below_three (KnowledgeNarrowingMinimal.v) demon_refutes_incremental (KnowledgeNarrowingMinimal.v)	none none none
Definition 6.1: A counter kept by a price list	§6	stated in the text	

Result	Sec.	Coq name and file	Axioms
Theorem 6.1: Conservation	§6	argued in the text, not checked in Coq	
Theorem 6.2: Only one honest count	§6	argued in the text, not checked in Coq	
Definition 6.2: Potential, amortized cost	§6	amortized_nonneg (CostSemanticsCom parison.v) cert_potential (CostSemanticsCompar ison.v)	def def
Theorem 6.3: The toll is the potential method	§6	a2_iff_nonnegative_amortized_cos t (CostSemanticsComparison.v) potential_telescoping (CostSemanti csComparison.v) nfi_by_potential (CostSemanticsCom parison.v) certification_system_is_potential _method (CostSemanticsComparison.v) and 2 more	none none none none
Theorem 7.1: Decoders need constant fibres	§7	decoding_requires_fiber_constanc y (ObservationPolicy.v) selected_representatives_give_de coder (ObservationPolicy.v)	none none
Theorem 7.2: No exact window price across a collision	§7	shadow_floor_overcharges (ShadowPr icing.v) shadow_cannot_price_exactly (Shad owPricing.v) window_showing_reading_prices_ex actly (ShadowPricing.v) window_showing_reading_has_no_co llision (ShadowPricing.v)	none none none none
Definition 7.1: Three richer transcripts and their explanations	§7	stated in the text	
Theorem 7.3: Three conditional escapes	§7	ver_full_state_escape (VerifierSma ll.v) ver_commitment_escape (VerifierSma ll.v) ver_honest_meets_contract (Verifi erSmall.v) ver_unchecked_breaks_contract (Ve rifierSmall.v) and 1 more	closed (file) closed (file) closed (file) closed (file)
In the text: the same collision survives into any richer transcript that keeps it	§7	ver_collision_blocks (VerifierSma ll.v) ver_no_factor (VerifierSmall.v)	closed (file) closed (file)

Result	Sec.	Coq name and file	Axioms
Theorem 7.4: No bare verifier on a Thiele-complete machine	§7	ver_complete_no_record_verifier (VerifierSmall.v) ver_complete_no_ledger_verifier (VerifierSmall.v) ver_small_no_ledger_verifier (VerifierSmall.v) ver_corollary (VerifierSmall.v) ver_small_corollary (VerifierSmall.v)	closed (file) closed (file) closed (file) closed (file) closed (file)
Definition 8.1: Entitlement	§8	stated in the text	
Definition 8.2: Decision tree	§8	stated in the text	
Lemma 8.1: A tree is no wider than two to its depth	§8	decision_tree_leaves_le_pow2_depth (DecisionTreeBound.v) decision_tree_log2_up_leaf_bound (DecisionTreeBound.v)	none none
Theorem 8.2: Paid narrowing, counted	§8	argued in the text with the counter; the bill form is ent_cs_count (EntitlementSmall.v)	closed (file)
Theorem 8.3: Structural entitlement, over any certification system	§8	ent_narrowing_strengthens (EntitlementSmall.v) ent_cs_count (EntitlementSmall.v) ent_cs_entitlement (EntitlementSmall.v) ent_leaves_le_pow2 (EntitlementSmall.v) and 1 more	closed (file) closed (file) closed (file) closed (file) closed (file)
Theorem 14.39: Structural entitlement represented	§14	ent_representation (EntitlementSmall.v) ent_every_shortcut_lands_here (EntitlementSmall.v) ent_complete_tree_bound (EntitlementSmall.v)	closed (file) closed (file) closed (file)
Theorem 14.40: A floor with no supplied tree	§14	ent_questions_floor (EntitlementSmall.v)	closed (file)
Theorem 14.41: What the record adds	§14	ent2_record_adds (EntitlementMore2.v) ent2_upgrade_iff (EntitlementMore2.v) ent2_observed_representation (EntitlementMore2.v)	closed (file) closed (file) closed (file)

Result	Sec.	Coq name and file	Axioms
In the text: weights and the partition form	§14	ent2_weighted_bits (EntitlementMore2.v) ent2_weighted_cs (EntitlementMore2.v) ent2_weighted_representation (EntitlementMore2.v) ent2_weighted_expected_bound (EntitlementMore2.v) ent2_partition_to_reduction (EntitlementMore2.v) and 7 more	closed (file) closed (file) closed (file) closed (file) closed (file) closed (file)
Proposition 14.42: The eight-cup member	§14	ent_small_certifies_iff (EntitlementSmall.v) ent_small_posterior_is_certified (EntitlementSmall.v) ent_small_instance (EntitlementSmall.v) ent_small_questions (EntitlementSmall.v) and 2 more	closed (file) closed (file) closed (file) closed (file)
Theorem 14.43: A search through n bits	§14	ent2_mmachine_complete (MultiThiele2.v) ent2_bit_search (BitSearchMember2.v) ent2_search_certifies_iff (BitSearchMember2.v) ent2_search_questions_floor (BitSearchMember2.v) ent2_questions_cap (BitSearch2.v) and 4 more	closed (file) closed (file) closed (file) closed (file) closed (file) closed (file)
In the text: the covering can't be dropped	§14	ent2_uncovered_posterior (CoveringNeeded2.v) ent2_uncovered_claim_false (CoveringNeeded2.v) ent2_no_cheap_covering (CoveringNeeded2.v)	closed (file) closed (file) closed (file)
Theorem 14.44: The time tax	§14	ent2_free_run_shift (TimeTax2.v) ent2_free_cannot_decide_fast (TimeTax2.v) ent2_ladder_decides (TimeTax2.v) ent2_chain_decides (TimeTax2.v) ent2_time_tax (TimeTax2.v) and 3 more	closed (file) closed (file) closed (file) closed (file) closed (file) closed (file)

Result	Sec.	Coq name and file	Axioms
Definition 9.1: Certified insight event, raw observation	§9	stated in the text	
Remark 9.1: What the toll prices, and what it doesn't	§9	argued in the text	
Remark 9.2: Nothing makes the crossing wait	§9	argued in the text	
Definition 10.1: Base, record-carrying machine, cover	§10	stated in the text	
Definition 10.2: Honest extension, latch	§10	stated in the text	
Theorem 10.1: The record is a latch	§10	record_axis_is_latch_holds (StructuralRecordAxis.v) BaseMachine (StructuralCoreAnyBase.v) BaseCover (StructuralCoreAnyBase.v) computation_driven (StructuralCoreAnyBase.v) and 6 more	none def def def
Theorem 10.2: Every reachable latch is an honest record	§10	latch_core_honest (RecordAxisDiscrimination.v)	none
Theorem 10.3: Records on reversible machines	§10	history_latch_injective (RecordAxisDiscrimination.v) history_latch_honest (RecordAxisDiscrimination.v) finite_reversible_cannot_write (RecordAxisDiscrimination.v) permanent_flip_is_not_injective (PermanentCertification.v)	none none none none
Lemma 10.4: Two records are two latches	§10	record_pair_is_two_latches_holds (StructuralRecordAxis.v)	none
Lemma 10.5: The toggle	§10	toggle_not_permanent (StructuralRecordAxis.v) toggle_not_latch (StructuralRecordAxis.v)	none none
Lemma 10.6: The clock	§10	toggle_computation_driven (StructuralRecordAxis.v) toggle_not_permanent (StructuralRecordAxis.v) toggle_not_latch (StructuralRecordAxis.v) clock_record_permanent (StructuralRecordAxis.v) and 1 more	none none none none

Result	Sec.	Coq name and file	Axioms
Corollary 10.7: The event is a free choice	§10	argued in the text, not checked in Coq	
Theorem 10.8: The schedule doesn't fix the probabilities	§10	<code>schedule_determines_probabilities_refuted (ProbabilisticRecord.v)</code> <code>deterministic_latch_handles_branching_refuted (ProbabilisticRecord.v)</code>	none none
Definition 10.3: Three concrete bases	§10	stated in the text	
Definition 10.4: Weak base equivalence	§10	stated in the text	
Theorem 10.9: Outside bases, and the relation	§10	<code>weak_base_equiv_refl_holds (CrossBaseGranularity.v)</code> <code>weak_base_equiv_sym_holds (CrossBaseGranularity.v)</code> <code>weak_base_equiv_trans_holds (CrossBaseGranularity.v)</code> <code>weak_match_left_runs (CrossBaseGranularity.v)</code> and 5 more	none none none none
Definition 10.5: A growing record	§10	stated in the text	
Theorem 10.10: Growing records are families of latches	§10	<code>growing_record_decomposes_holds (GrowingRecord.v)</code> <code>thresholds_determine_record_holds (GrowingRecord.v)</code> <code>record_price_iff_threshold_price_holds (GrowingRecord.v)</code> <code>one_latch_refuted (GrowingRecord.v)</code> and 1 more	none none none none
Definition 11.1: Substrate	§11	stated in the text	
Definition 11.2: Shortcut predicate	§11	stated in the text	
Theorem 11.1: No inside decider	§11	<code>structural_shortcut_undecidable (if and only if) (StructuralUndecidability.v)</code> <code>Substrate (Substrate.v)</code> <code>WithShortcutPredicate (StructuralUndecidability.v)</code>	none def def
Theorem 11.2: The nat family	§11	<code>nat_self_undecidable (NatSubstrateInstance.v)</code> <code>nat_recursion_theorem (NatSubstrateInstance.v)</code> <code>nat_structural_shortcut_undecidable (if and only if) (NatSubstrateInstance.v)</code>	none none none

Result	Sec.	Coq name and file	Axioms
Definition 11.3: L, as built here	§11	stated in the text	
Theorem 11.3: Recursion, Rice and halting for L	§11	second_recursion (LRecursion.v) L_recursion_theorem (LRecursion.v) L_rice (LRecursion.v) L_halting_undecidable (LRecursion.v) and 2 more	none none none none
Definition 12.1: Casper setting and votes	§12	stated in the text	
Theorem 12.1: Accountable safety	§12	accountable_safety (CasperFFG.v) conflicting_records_are_priced (CasperRecordReading.v)	none none
Theorem 12.2: The premise can hold, and the write is free	§12	casper_fork_exists (CasperForkWitness.v) casper_fork_slashable (CasperForkWitness.v) finalization_without_slashing (CasperRecordReading.v)	none none none
Definition 12.2: One slot in one transaction	§12	stated in the text	
Theorem 12.3: The persistent write is priced; the revoked one nearly is not	§12	persistent_write_priced (EVMStorageGas.v) revoked_write_nearly_free (EVMStorageGas.v)	none none
Definition 12.3: A gas schedule	§12	stated in the text	
Theorem 12.4: Gas schedule exactness	§12	gas_schedule_exactness (if and only if) (GasMetering.v) undercharged_opcode_admits_free_commitment (GasMetering.v) undercharged_opcode_breaks_certification_floor (GasMetering.v) overcharge_breaks_exactness (GasMetering.v) and 1 more	none none none none
Theorem 12.5: The logical level always pays	§12	permanent_write_has_logical_payment (PricingPhysicsAudit.v) permanent_at_flip_is_not_injective (PermanentRecordPricing.v)	none none
Definition 12.4: Platform and quote	§12	stated in the text	

Result	Sec.	Coq name and file	Axioms
Theorem 12.6: The quote cannot attest unmeasured state	§12	quote_cannot_attest_unmeasured_state (TPMQuoteGap.v) quote_decides_measured_claims (TPMQuoteGap.v) tpm_interface_authenticity_refuted (TPMQuoteAuthenticity.v)	none none none
Definition 12.5: Inclusion check, RFC 9162 section 2.1.3.2	§12	stated in the text	
Theorem 12.7: The RFC's own examples pass	§12	rfc9162_example_inclusion_d3 (RFC9162Merkle.v) rfc9162_example_inclusion_d0 (RFC9162Merkle.v) rfc9162_example_inclusion_d4 (RFC9162Merkle.v) rfc9162_example_inclusion_d6 (RFC9162Merkle.v) and 3 more	none none none none
Definition 12.6: The bounds fragment	§12	stated in the text	
Theorem 12.8: The checker is the verification condition	§12	pcc_checker_accepts_iff_vc (NeculaPCC.v) pcc_certificate_implies_vc (NeculaPCC.v) pcc_unsafe_program_rejected (NeculaPCC.v)	none none none
Definition 13.1: Ecosystem, records, proliferates, pointer	§13	stated in the text	
Conjecture 13.1: The pointer conjecture	§13	open	
Theorem 13.2: The toy	§13	toy_cert_unique_pointer (PointerObservable.v) toy_work_not_proliferating (PointerObservable.v)	none none
Theorem 13.3: The survey, and the swap	§13	twelve_candidate_measurements_checked (RecordProliferationSurvey.v) swapped_event_is_pointer_checked (RecordProliferationSurvey.v) five_labeled_models_have_selected_pointer (PointerObservableReductions.v) pointer_criterion_holds_indexed (PointerObservableReductions.v)	none none none def

Result	Sec.	Coq name and file	Axioms
Theorem 13.4: One blind observer is enough	§13	blind_observer_blocks_proliferation (PointerObservableCounterexamples.v) labeled_model_verdicts (PointerObservableCounterexamples.v)	none none
Definition 13.2: Observation game	§13	stated in the text	
Theorem 13.5: Agreement does not buy permanence; durability does	§13	toggle_game_refutes_strong_pointer_necessity (EcosystemGame.v) durable_consensus_implies_permanence (EcosystemGame.v)	none none
Definition 14.1: Properties, checker, meaning	§14	stated in the text	
Lemma 14.1: The checker says what it means	§14	eval_iff (EarnedCore.v)	closed (file)
Definition 14.2: Core, state, claim, start	§14	stated in the text	
Definition 14.3: Instructions, costs, the step	§14	cexec (EarnedCore.v) exec (EarnedCore.v) run (EarnedCore.v, +10) total_cost (EarnedCore.v, +1) and 2 more	def def def def
Definition 14.4: Two-counter program, compile, window	§14	stated in the text	
Theorem 14.2: Step for step	§14	simulation_step (EarnedCore.v) simulation_run (EarnedCore.v)	closed (file) closed (file)
Theorem 14.3: Halting matches	§14	halting_correspondence (EarnedCore.v)	closed (file)
Theorem 14.4: Halting is undecidable for this machine	§14	mm2_step_iff (EarnedCoreLinks.v) mm2_stop_iff (EarnedCoreLinks.v) mm2_terminates_iff (EarnedCoreLinks.v) mm2_halting_iff (if and only if) (EarnedCoreLinks.v) and 1 more	none none none none
Theorem 14.5: Ledger, toll, permanence	§14	mu_conservation (EarnedCore.v, +2) mu_conservation_trace (EarnedCore.v, +1) only_certify_certifies (EarnedCore.v, +1) a2 (EarnedCore.v, +3) and 1 more	none closed (file) closed (file) closed (file) and 1 more

Result	Sec.	Coq name and file	Axioms
Theorem 14.6: An instance of the certification system	§14	earned_core_floor (EarnedCoreLinks.v) universal_nfi_any_substrate (UniversalCertificationCost.v) nfi_floor (EarnedCore.v, +1) earned_cs (EarnedCoreLinks.v)	none none closed (file) def
Theorem 14.7: The flag is a latch of the base	§14	earned_core_honest (EarnedCoreLinks.v) earned_core_is_latch (EarnedCoreLinks.v) record_axis_is_latch_holds (StructuralRecordAxis.v) cert_latch (EarnedCore.v) and 1 more	none none none closed (file)
Definition 14.5: Untouched	§14	untouched (EarnedCore.v)	def
Theorem 14.8: A commitment was checked first	§14	earned_commitment_provenance (EarnedCore.v)	closed (file)
Theorem 14.9: A raised flag was checked, then committed, then certified	§14	earned_certification_provenance (EarnedCore.v)	closed (file)
Theorem 14.10: Unearned steps trap and change no claim	§14	unearned_commit_traps (EarnedCore.v) uncommitted_certify_traps (EarnedCore.v)	closed (file) closed (file)
Theorem 14.11: Checker soundness	§14	checker_soundness (EarnedCore.v) committed_claim_holds (EarnedCore.v) sound_step (EarnedCore.v) sound (EarnedCore.v)	closed (file) closed (file) closed (file) def
Definition 14.6: Earned fact; no forgery	§14	stated in the text	
Theorem 14.12: No forging	§14	facts_step (EarnedCore.v) no_forging_step (EarnedCore.v) no_forging (EarnedCore.v)	closed (file) closed (file) closed (file)

Result	Sec.	Coq name and file	Axioms
Theorem 14.13: Bounded table, no eviction	§14	facts_keep (EarnedCore.v) full_table_traps (EarnedCore.v) facts_bounded_step (EarnedCore.v)	closed (file) closed (file) closed (file)
Theorem 14.14: Same window, different record	§14	receipt_separation (EarnedCore.v, +1)	closed (file)
Theorem 14.15: No window oracle	§14	no_mu_oracle (EarnedCore.v, +2) no_cert_oracle (EarnedCore.v, +2) no_commit_oracle (EarnedCore.v)	closed (file) closed (file) closed (file)
Theorem 14.16: The minimum price of a certificate	§14	certified_run_min_cost (EarnedCore .v) program_certified_min_cost (Earne dCore.v)	closed (file) closed (file)
Theorem 14.17: Three is tight	§14	earned_run_check_can_fail (Earned Core.v) earned_run_refused_forever (Earne dCore.v) earned_run (EarnedCore.v) min_cost_tight (EarnedCore.v)	closed (file) closed (file) def closed (file)
Definition 14.7: Thiele-complete	§14	thiele_interface (ThieleComplete.v) thiele_complete_with (ThieleComple te.v) thiele_complete (ThieleComplete.v) weakly_thiele_complete (ThieleComp lete.v)	def def def def
Theorem 14.18: What every Thiele-complete machine has	§14	thiele_complete_is_weak (ThieleCom plete.v) base_halting_correspondence (Thie leComplete.v) certificate_costs_three (ThieleCom plete.v) complete_has_free_move (ThieleComp lete.v) and 6 more	closed (file) closed (file) closed (file) closed (file)

Result	Sec.	Coq name and file	Axioms
Theorem 14.19: A clock is a Thiele machine and isn't Thiele-complete	§14	clock_weakly_thiele_complete (ThieleComplete.v)	closed (file)
		clock_not_thiele_complete (ThieleComplete.v)	closed (file)
		latch_clock_weakly_thiele_complete (ThieleComplete.v)	closed (file)
		latch_clock_not_thiele_complete (ThieleComplete.v)	closed (file)
Theorem 14.20: Each clause does work	§14	paid_latch_meets_base_and_toll (ThieleComplete.v)	closed (file)
		paid_latch_not_thiele_complete (ThieleComplete.v)	closed (file)
		silent_meets_base_record_toll (ThieleComplete.v)	closed (file)
		silent_not_thiele_complete (ThieleComplete.v)	closed (file)
		and 4 more	
Theorem 14.21: The small machine, the generic one and the sorted one are Thiele-complete	§14	earned_core_thiele_complete (ThieleComplete.v)	closed (file)
		earned_generic_thiele_complete (ThieleComplete.v)	closed (file)
		sorted_machine_thiele_complete (ThieleComplete.v)	closed (file)
		thiele_complete_floor (EarnedGenericLinks.v)	none
		and 2 more	
Theorem 14.22: Every Thiele-complete machine hides its record and its ledger from its window	§14	complete_every_window_printed (ThieleCompleteWindow.v)	closed (file)
		complete_no_record_oracle (ThieleCompleteWindow.v)	closed (file)
		complete_no_ledger_oracle (ThieleCompleteWindow.v)	closed (file)
		complete_cs_window_blind (EarnedGenericLinks.v)	none
		and 3 more	
Definition 14.8: The host machine	§14	pair (UniversalCodes.v)	def
		pcode (UniversalCodes.v)	def
		heval (UniversalCodes.v)	def
		hholds (UniversalCodes.v)	def
Theorem 14.23: No host fact on an exact copy of the guest's counter	§14	pigeonhole_collision (UniversalNoCopy.v)	closed (file)
		no_exact_copy_host (UniversalNoCopy.v)	closed (file)
		multi_cs_no_exact_copy (UniversalInterpreterLinks.v)	none

Result	Sec.	Coq name and file	Axioms
Definition 14.9: The program U , and loading a guest	§14	<code>U (UniversalLayout.v, +2)</code> <code>hload (UniversalSim.v, +1)</code> <code>prog_code (UniversalCodes.v)</code> <code>U_paid_sites (UniversalLayout.v)</code>	<code>def</code> <code>def</code> <code>def</code> <code>none</code>
Theorem 14.24: U keeps pace with its guest	§14	<code>hload_rel (UniversalSim.v)</code> <code>U_step (UniversalSim.v)</code> <code>U_simulation (UniversalRun.v)</code>	<code>none</code> <code>none</code> <code>none</code>
Theorem 14.25: Halting and answers match	§14	<code>universal_halting (UniversalRun.v)</code> <code>universal_output (UniversalRun.v)</code>	<code>none</code> <code>none</code>
Theorem 14.26: The host's flag is the guest's, and its ledger is exact	§14	<code>universal_flag_iff (UniversalRun.v)</code> <code>universal_ledger_exact (UniversalRun.v)</code>	<code>none</code> <code>none</code>
Theorem 14.27: The host earns its own flag	§14	<code>universal_earned (UniversalRun.v)</code>	<code>none</code>
Theorem 14.28: The host is Thiele-complete, and U 's halting is undecidable	§14	<code>universal_thiele_complete (UniversalRun.v)</code> <code>U_run_on_host_machine (UniversalRun.v)</code> <code>interp_halting_undecidable (UniversalInterpreterLinks.v)</code> <code>interp_U_certified_floor (UniversalInterpreterLinks.v)</code> and 4 more	<code>none</code> <code>none</code> <code>none</code> <code>none</code>
Definition 14.10: The host	§14	<code>hexec (UniversalThiele.v)</code> <code>U (UniversalThiele.v)</code>	<code>def</code> <code>def</code>
Theorem 14.31: The host runs any guest	§14	<code>universal_simulation (UniversalThiele.v)</code> <code>universal_program_simulation (UniversalThiele.v)</code> <code>host_thiele_complete (UniversalThiele.v)</code>	<code>closed (file)</code> <code>closed (file)</code> <code>closed (file)</code>
Theorem 14.32: The copy is the record	§14	<code>record_agreement_step (UniversalThiele.v)</code> <code>record_agreement (UniversalThiele.v)</code> <code>simulated_record (UniversalThiele.v)</code>	<code>closed (file)</code> <code>closed (file)</code> <code>closed (file)</code>
Theorem 14.33: The host's step collects the guest's toll	§14	<code>toll_enforced_by_host_step (UniversalThiele.v)</code> <code>every_guest_crossing_is_a_host_crossing (UniversalThiele.v)</code> <code>host_cost_covers_guest (UniversalThiele.v)</code>	<code>closed (file)</code> <code>closed (file)</code> <code>closed (file)</code>

Result	Sec.	Coq name and file	Axioms
Theorem 14.34: No free host certification	§14	no_free_host_certification_step (UniversalThiele.v) guestless_runs_leave_mirror (UniversalThiele.v) own_record_only_by_certify (UniversalThiele.v)	closed (file) closed (file) closed (file)
Theorem 14.35: The host is a Thiele machine	§14	host_toll (UniversalThiele.v) host_nfci (UniversalThieleLinks.v) host_cs (UniversalThieleLinks.v)	closed (file) none def
Remark 14.1: Two guests, run	§14	demo_certifies (UniversalThiele.v) demo_program_certifies (UniversalThiele.v) demo_forgery_fails (UniversalThiele.v)	closed (file) closed (file) closed (file)
In the text: the same chain for any property with a proved checker, “sorted” included	§14	generic_earned_certification_provenance (EarnedGeneric.v) generic_checker_soundness (EarnedGeneric.v) generic_no_forging (EarnedGeneric.v) generic_certified_run_min_cost (EarnedGeneric.v) sorted_run_certifies_iff (EarnedGeneric.v) sorted_run_refused_forever (EarnedGeneric.v)	closed (file)
In the text: the channel names the last earned commitment	§14	certified_channel_earned (EarnedCore.v) channel_names_last_commit (EarnedCore.v)	closed (file)
In the text: a raised mirror means the guest’s own run earned it	§14	host_mirror_earned (UniversalThiele.v)	closed (file)
Theorem 14.29: One host runs every computably presented Thiele machine	§14	presented_universal (PresentedUniversal.v) presented_universal_halting, presented_universal_points, presented_universal_halt_point, presented_universal_flag_iff, presented_universal_earned (PresentedUniversal.v) pu_universal_ledger_exact, pu_universal_earned, pu_universal_thiele_complete (UniversalPRun.v)	none none none

Result	Sec.	Coq name and file	Axioms
Theorem 14.30: Exact, within two, and no better	§14	presented_universal_surcharge_le_two, presented_universal_exact, presented_universal_no_exact_below_three, presented_universal_U_no_exact_below_three (PresentedUniversal.v)	none
In the text: a demonstration machine on the priced host	§14	pu_demo_computably_presented, pu_demo_never_halts, pu_demo_flag_rises, pu_demo_exact, pu_demo_earned (PresentedDemo.v)	none
Theorem 14.36: A counter beside a Turing machine changes nothing	§14	thiele_simulates_turing_gen (ProperSubsumption.v)	none
Definition 14.12: Behaving the same, on the host	§14.18	sm_hequiv (SmHostBlocks.v)	def
		sm_hext (SmHostRice.v)	def
Theorem 14.45: Rice's theorem on the host machine	§14.18	sm_host_rice (SmHostRice.v)	closed
		sm_rice_prog_halts (SmHostRice.v)	(file)
		sm_rice_prog_diverges (SmHostRice.v)	closed
			(file)
		sm_host_rice_fun (SmKleene.v)	closed
			(file)
			closed
			(file)
Theorem 14.46: Rice's theorem for the small machine, from the clean start	§14.18	sm_guest_rice (SmGuestRice.v)	closed
		sm_guest_halting_clean_undecidable (SmGuestRice.v)	(file)
			closed
			(file)
Theorem 14.47: s-m-n, and a universal program for host programs (each an if and only if)	§14.18	sm2_smn_all (SmSmnAll.v)	closed
		sm_smn (SmKleene.v)	(file)
		sm_host_universal (SmKleene.v)	closed
		sm_hdecode_hcode (SmCodes.v)	(file)
			closed
			(file)
			closed
			(file)
Theorem 14.48: Kleene's recursion theorem on the host machine (equivalence of what the two compute)	§14.18	sm_second_recursion (SmKleene.v)	closed
		sm_kleene (SmKleene.v)	(file)
		sm_kleene_codes (SmKleene.v)	closed
			(file)
			closed
			(file)

Result	Sec.	Coq name and file	Axioms
Theorem 14.49: The recursion theorem, up to the record (latch and flag up exactly when the numbers say)	§14.18	sm2_kleene_obs (SmFixedPoint.v)	closed
		sm2_kleene_fun (SmFixedPoint.v)	(file)
		sm2_final_numbers (SmTally.v)	closed
		sm2_ev_spec (SmTally.v)	(file)
		sm2_VG_forward (SmFixed.v)	closed
		sm2_VG_halt (SmFixed.v)	(file)
		sm2_while (SmLoops.v)	closed
		sm2_mma_ctx (SmMMAOff.v)	(file)
			closed
			(file)
			closed
			(file)
			closed
			(file)
Theorem 14.50: No fixed point of full behaviour	§14.18	sm2_no_exact (SmNoExact.v)	closed
		sm2_reg_bound (SmNoExact.v)	(file)
			closed
Theorem 14.51: The diagonal on the host machine (no decider right exactly when the property holds)	§14.18	sm_no_inside_decider (SmKleene.v)	closed
		sm_no_host_decider (SmDecider.v)	(file)
			closed
Definition 15.1: Correlators from the counters; the pinned matrix	§15	stated in the text	
Theorem 15.1: The integer check implies the pinned matrix is PSD	§15	column_contractive_check_witness	FE, RD
		_sound (CHSHColumnCheck.v)	FE, RD
		column_contractive_check_witness	FE, RD
		_npa_psd (CHSHColumnCheck.v)	
Theorem 15.2: PSD forces the Tsirelson bound	§15	npa_psd_iff_column_contractive (if and only if) (CHSHColumnCheck.v)	
		npa_psd_implies_tsirelson_bound (CHSHColumnCheck.v)	FE, RD
		npa_psd_zero_marginal_implies_row_bounds (CHSHColumnCheck.v)	FE, RD
		chsh_gap_is_sum_of_squares (TsirelsonFromAlgebra.v)	
Corollary 15.3: A passed check means the bound	§15	the composition of Theorem 15.1 and Theorem 15.2	

Result	Sec.	Coq name and file	Axioms
Definition 15.2: The packed tally, and the property PCHSH	§15	small_chsh_tally (SmallChshCheck.v) small_chsh_tally_of (SmallChshMachine.v) small_chsh_meaning (SmallChshCheck.v)	def def def
Theorem 15.4: The check is exact	§15	small_chsh_check_iff (SmallChshCheck.v) small_chsh_clear_denominators (SmallChshCheck.v) small_chsh_psd_iff_contractive (SmallChshCheck.v) small_chsh_check_tsirelson (SmallChshCheck.v) small_chsh_eval_iff (SmallChshMachine.v) and 2 more	FE, RD FE, RD FE, RD FE, RD FE, RD
Theorem 15.5: A raised flag stands on the check	§15	small_chsh_flag_implies_tsirelson (SmallChshMachine.v) small_chsh_program_flag_implies_tsirelson (SmallChshMachine.v) small_chsh_chain_certifies_iff (SmallChshMachine.v) small_chsh_tally_certifies (SmallChshMachine.v) small_chsh_tally_refused_forever (SmallChshMachine.v) and 1 more	FE, RD FE, RD FE, RD FE, RD FE, RD
Proposition 15.6: Five tallies	§15	small_chsh_demo_12_5_certifies (SmallChshMachine.v) small_chsh_demo_14_5_certifies (SmallChshMachine.v) small_chsh_demo_16_5_refused_forever (SmallChshMachine.v) small_chsh_demo_pr_box_refused_forever (SmallChshMachine.v) small_chsh_demo_all_ones_refused_forever (SmallChshMachine.v)	FE, RD FE, RD FE, RD FE, RD
Remark 15.1: Sound, not complete	§15	zero_marginal_implies_elliptope (ElliptopeCompletion.v) deterministic_strategy_elliptope (ElliptopeCompletion.v)	RD FE, RD
Definition 15.3: The size-generic PSD test	§15	stated in the text	
Corollary 15.7: The level-1 test is the size-five case	§15	column_contractive_iff_generalizable (GenRealizability.v) column_contractive_qlab_iff_generalizable (GenRealizability.v)	FE, RD FE, RD

Result	Sec.	Coq name and file	Axioms
Definition 16.1: Energy at a scale	§16	stated in the text	
Theorem 16.1: The calibration is a unit conversion	§16	calibrated_mu_landauer_energy (PricingPhysicsAudit.v) mu_has_no_intrinsic_joule_value (PricingPhysicsAudit.v) mu_landauer_calibration (PricingPhysicsTarget.v)	FE, RD, EM FE, RD def
Premise 16.2: Landauer, in units of the counter	§16	assumed, not proved	
Theorem 16.2: When the premises can hold together	§16	calibrated_model_exists_iff_positive_cost (ObservationPolicy.v) bit_model_has_satisfiable_calibration (ObservationPolicy.v) calibrated_positive_model (ObservationPolicy.v)	none none def
Theorem 16.3: Keeping history makes every step injective	§16	retained_history_step_injective (ObservationPolicy.v)	none
Definition 16.3: The two-state protocol	§16	stated in the text	
Theorem 16.4: The heat is fixed by the chosen gap	§16	canonical_reset_satisfies_master_equation (CalorimeterProtocol.v) canonical_reset_heat_exact (CalorimeterProtocol.v) selected_gap_gives_landauer_heat (CalorimeterProtocol.v) smaller_gap_refutes_unconditional_landauer_floor (given $0 < k_B$ and $0 < T$) (CalorimeterProtocol.v) and 1 more	FE, RD FE, RD FE, RD, EM FE, RD, EM
Definition B.1: Certification system	§B	CertificationSystem (UniversalCertificationCost.v) cs_cert_costs (UniversalCertificationCost.v)	def def
Definition B.2: Quantitative certification system	§B	QuantitativeCertificationSystem (QuantitativeNoFI.v) qcs_cost_bounds_witness (QuantitativeNoFI.v) qcs_cert_threshold_witness (QuantitativeNoFI.v) QuantitativeCertificationSystem_full (QuantitativeNoFI.v)	def def def def
Theorem B.1: The floor raised to a threshold	§B	universal_nfi_quantitative (QuantitativeNoFI.v)	none
Definition F.1: Settings, outcomes, correlators	§F	stated in the text	

Result	Sec.	Coq name and file	Axioms
Definition F.2: Counters	§F	WitnessCounts (CHSHColumnCheck.v)	def
Lemma F.1: Deterministic strategies	§F	local_S_2_deterministic (BoxCHSH.v) deterministic_strategy_chsh_bounded (MinorConstraints.v)	none FE, RD
Definition F.3: Finite factorizable model, is_factorizable	§F	is_factorizable (MinorConstraints.v)	def
Theorem F.2: fine_theorem, one direction only	§F	fine_theorem (MinorConstraints.v) factorizable_CHSH_classical_bounded (MinorConstraints.v)	FE, RD FE, RD
Remark F.1: What the name does and does not cover	§F	fine_theorem (MinorConstraints.v) local_box_CHSH_bound (MinorConstraints.v)	FE, RD FE, RD
Definition F.4: WCLocallyConsistent	§F	WCLocallyConsistent (CHSHStatisticalBridge.v)	def
Theorem F.3: A tally above 2 is not one replayed strategy	§F	chsh_violation_rules_out_locally_factorizable_coupling (CHSHCouplingBridge.v) locally_consistent_classical_bound (CHSHCouplingBridge.v) chsh_stat_from_wc (CHSHStatisticalBridge.v)	none none def
Lemma F.4: The six-square identity, chsh_gap_is_sum_of_squares	§F	chsh_gap_is_sum_of_squares (TsirelsonFromAlgebra.v)	FE, RD
Theorem F.5: Row-bound form of Tsirelson's bound	§F	tsirelson_squared (TsirelsonFromAlgebra.v) tsirelson_from_row_bounds (TsirelsonGeneral.v) tsirelson_bound_abs (TsirelsonGeneral.v)	FE, RD FE, RD FE, RD
Theorem F.6: Tightness	§F	tsirelson_tight (TsirelsonFromAlgebra.v) tsirelson_achievable (TsirelsonGeneral.v)	FE, RD FE, RD
Remark F.2: Where equality can happen	§F	argued in the text	
Lemma F.7: A rational constant above the bound, rational_tsirelson_bound	§F	rational_tsirelson_bound (TsirelsonFromAlgebra.v)	FE, RD
Definition F.5: The pinned level-1 matrix	§F	zero_marginal_npa (NPAMomentMatrix.v) NPAMomentMatrix (NPAMomentMatrix.v) npa_to_matrix (NPAMomentMatrix.v)	def def def

Result	Sec.	Coq name and file	Axioms
Lemma F.8: Schur complement, the standard form	§F	argued in the text, not checked in Coq	
Definition F.6: Column contractive, zero_marginal_column_contractive	§F	zero_marginal_column_contractive (CHSHColumnCheck.v)	def
Lemma F.9: The 2x2 test, psd2_quadratic_form_nonneg	§F	psd2_quadratic_form_nonneg (CHSHColumnCheck.v)	FE, RD
Theorem F.10: Contractive implies PSD	§F	zero_marginal_npa_column_contractive_implies_psd (CHSHColumnCheck.v) PSD5 (ConstructivePSD.v)	FE, RD def
Lemma F.11: Discriminant, quadratic_nonneg_discriminant	§F	quadratic_nonneg_discriminant (ConstructivePSD.v)	FE, RD
Theorem F.12: PSD implies contractive	§F	npa_psd_implies_column_contractive (CHSHColumnCheck.v) npa_quad5_test_col0 (CHSHColumnCheck.v) npa_quad5_test_col1 (CHSHColumnCheck.v) npa_quad5_test_schur (CHSHColumnCheck.v)	FE, RD FE, RD FE, RD FE, RD
Corollary F.13: The equivalence	§F	npa_psd_iff_column_contractive (if and only if) (CHSHColumnCheck.v) column_contractive_iff_npa_psd (CHSHColumnCheck.v) PSD5 (ConstructivePSD.v) npa_psd (NPAMomentMatrix.v)	FE, RD FE, RD def def
Theorem F.14: PSD implies Tsirelson	§F	npa_psd_implies_tsirelson_bound (CHSHColumnCheck.v) npa_psd_implies_tsirelson_bound_abs (CHSHColumnCheck.v) psd_3x3_determinant_nonneg (ConstructivePSD.v) npa_psd_zero_marginal_implies_row_bounds (CHSHColumnCheck.v)	FE, RD FE, RD FE, RD FE, RD
Remark F.3: What PSD of the pinned matrix means	§F	zero_marginal_implies_elliptope (ElliptopeCompletion.v)	RD
Definition F.7: column_contractive_check_witness	§F	column_contractive_check_witness (CHSHColumnCheck.v)	def

Result	Sec.	Coq name and file	Axioms
Lemma F.15: The seven facts are the three conditions with denominators cleared	§F	argued in the text, not checked in Coq	
Theorem F.16: Soundness of the integer check	§F	column_contractive_check_witness_sound (CHSHColumnCheck.v) column_contractive_check_witness_npa_psd (CHSHColumnCheck.v)	FE, RD FE, RD
Definition F.8: qlab_moment_matrix	§F	qlab_moment_matrix (QuantumPartitionPSD_1AB.v)	def
Definition F.9: The residual, qlab_residual	§F	qlab_residual (QuantumPartitionPSD_1AB.v) column_contractive_qlab (QuantumPartitionPSD_1AB.v)	def def
Lemma F.18: Three squares plus the residual, quad9_qlab_sos_decomposition	§F	quad9_qlab_sos_decomposition (QuantumPartitionPSD_1AB.v)	FE, RD
Theorem F.19: PSD at level 1+AB, qlab_psd_iff_column_contractive	§F	qlab_psd_iff_column_contractive (if and only if) (QuantumPartitionPSD_1AB.v) column_contractive_qlab_implies_psd9 (QuantumPartitionPSD_1AB.v) psd9_implies_column_contractive_qlab (QuantumPartitionPSD_1AB.v) PSD9 (QuantumPartitionPSD_1AB.v) and 1 more	FE, RD FE, RD FE, RD def
Lemma F.20: Fraction-free Schur step	§F	sym6_Schur_identity (QuantumPartitionPSD_1AB.v) sym5_Schur_identity (QuantumPartitionPSD_1AB.v) sym6_qf_nonneg_from_pd (QuantumPartitionPSD_1AB.v) sym5_qf_nonneg_from_pd (QuantumPartitionPSD_1AB.v)	FE, RD FE, RD FE, RD FE, RD
Lemma F.21: Sylvester for 4x4, sym4_qf_nonneg_from_pd	§F	sym4_qf_nonneg_from_pd (QuantumPartitionPSD_1AB.v) sym4_LDLT_identity (QuantumPartitionPSD_1AB.v) sym4_d1 (QuantumPartitionPSD_1AB.v) sym4_d4 (QuantumPartitionPSD_1AB.v) and 1 more	FE, RD FE, RD def def

Result	Sec.	Coq name and file	Axioms
Definition F.10: What the integer check computes	§F	g12345_COMMON_Z (CHSHColumnCheck.v) cleared_g12345_H11_Z (CHSHColumnCheck.v) cleared_g12345_H56_Z (CHSHColumnCheck.v) schur_step_Z (CHSHColumnCheck.v) qlab_g12345_check_z_kernel (CHSHColumnCheck.v)	def def def def
Theorem F.22: The full 1+AB check implies PSD	§F	qlab_g12345_caller_witness_z_abs_sound (QuantumPartitionPSD_1AB.v) qlab_g12345_minors_witness_implies_psd9 (QuantumPartitionPSD_1AB.v)	FE, RD FE, RD
Remark F.4: Interior only	§F	argued in the text	
Theorem F.23: The zero-moment check sits inside the classical bound	§F	qlab_check_at_gzero_forces_unit_ball (QuantumPartitionPSD_1AB.v) qlab_check_at_gzero_implies_classical_bound (QuantumPartitionPSD_1AB.v) qlab_residual_g_zero_decomp (QuantumPartitionPSD_1AB.v) column_contractive_check_qlab_sound_at_g_zero (QuantumPartitionPSD_1AB.v) and 2 more	FE, RD FE, RD FE, RD FE, RD
Definition F.11: Completion and membership	§F	completed_npa (ElliptopeCompletion.v) elliptope_realizable (ElliptopeCompletion.v)	def def
Theorem F.24: Deterministic plans are inside, deterministic_strategy_elliptope	§F	deterministic_strategy_elliptope (ElliptopeCompletion.v)	FE, RD
Theorem F.25: Convexity and finite mixtures	§F	elliptope_convex (ElliptopeCompletion.v) elliptope_finite_mixture (ElliptopeCompletion.v) lhv_mixture_elliptope (ElliptopeCompletion.v) elliptope_zero (ElliptopeCompletion.v)	FE, RD FE, RD FE, RD FE, RD
Theorem F.26: Tsirelson inside the completion set, elliptope_tsirelson	§F	elliptope_tsirelson (ElliptopeCompletion.v) psd_cauchy_schwarz (ElliptopeCompletion.v)	FE, RD FE, RD

Result	Sec.	Coq name and file	Axioms
Corollary F.27: The PR box is outside, <code>pr_box_not_elliptope</code>	§F	<code>pr_box_not_elliptope</code> (ElliptopeCompletion.v)	FE, RD
Theorem F.28: Strictly beyond the classical bound, <code>beyond_classical_elliptope</code>	§F	<code>beyond_classical_elliptope</code> (ElliptopeCompletion.v)	FE, RD
Definition F.12: The two branches	§F	<code>elliptope_pd_check</code> (ElliptopeGate.v) <code>elliptope_ldl_check</code> (ElliptopeGate.v) <code>ldl_equations</code> (ElliptopeGate.v) <code>elliptope_check_full</code> (ElliptopeGate.v)	def def def def
Theorem F.29: Soundness of the gate, <code>elliptope_check_full_sound</code>	§F	<code>elliptope_check_full_sound</code> (ElliptopeGate.v) <code>elliptope_full_gate_never_accepts_pr_box</code> (ElliptopeGate.v) <code>elliptope_pd_check_sound</code> (ElliptopeGate.v) <code>elliptope_ldl_check_sound</code> (ElliptopeGate.v) and 3 more	FE, RD FE, RD FE, RD FE, RD

E The exact words of fifteen theorems

This book states its theorems in its own words, and Section 17 says why that’s a risk: the checker confirms a proof matches its statement, not that my sentence matches the statement. So here are the statements themselves for seventeen of the results the argument leans on hardest. Each is copied from its file with the comments taken out and nothing else changed, and each comes with its file, a plain reading, and a line on whether the book’s printed version says exactly that, more, or less. You don’t need to read Coq to use this. Put the printed theorem next to the block and check that every premise in the block turns up in the sentence. That’s the whole test. These are the statements this book was checked against; if a file differs later, the book means the one printed here.

E.1 The universal floor

`universal_nfi_any_substrate`, `coq/kernel/nfi/UniversalCertificationCost.v`, with the record it quantifies over.

```
Record CertificationSystem := mk_cert_system {
  cs_state : Type;
  cs_instr : Type;
  cs_step  : cs_state -> cs_instr -> cs_state;
```

```

cs_cost  : cs_instr -> nat;
cs_cert  : cs_state -> bool;
cs_cert_costs :
  forall (s : cs_state) (i : cs_instr),
    cs_cert s = false ->
    cs_cert (cs_step s i) = true ->
    cs_cost i >= 1;
}.

Theorem universal_nfi_any_substrate :
  forall (CS : CertificationSystem)
    (trace : list (cs_instr CS))
    (s0 : cs_state CS),
  cs_cert CS s0 = false ->
  cs_cert CS (cs_run CS trace s0) = true ->
  cs_total_cost CS trace >= 1.

```

In plain words: a certification system is a record of a state type, a move type, a step, a cost and a yes-or-no reading, plus one more field, `cs_cert_costs`, which is the toll: a move that takes the reading from false to true costs at least one. Since the toll is a field, every system the theorem talks about has it. The theorem: start at a state reading false, run any list of moves (`cs_run` applies them in order, `cs_total_cost` adds their costs), end at a state reading true, and the costs add up to at least one. Theorem 3.1 says exactly this; the rule Definition 3.1 states as the toll is the record's last field here.

E.2 A permanent flip merges

`permanent_flip_is_not_injective`, `coq/kernel/nfi/PermanentCertification.v`, inside a section that fixes the states, the moves, the step and the reading.

```

Variables (S I : Type).
Variable step : S -> I -> S.
Variable cert : S -> bool.

Definition finite_states (all : list S) : Prop :=
  NoDup all /\ forall s, In s all.

Definition permanent : Prop :=
  forall s i, cert s = true -> cert (step s i) = true.

Definition step_injective (i : I) : Prop :=
  forall a b, step a i = step b i -> a = b.

Theorem permanent_flip_is_not_injective :
  forall all s i,
    finite_states all ->
    permanent ->
    cert s = false ->
    cert (step s i) = true ->
    ~ step_injective i.

```

In plain words: `finite_states all` says the list *all* has no repeats and every state is on it, so there are finitely many. `permanent` says once the reading is true, every move keeps it true. `step_injective i` says move *i* never sends two different states to the same one. The theorem: on a finite machine with a permanent reading, a move that takes some state from false to true isn't injective, so it merges two states. Theorem 4.1 says exactly this, with "merges" meaning "isn't injective".

E.3 The toll from finiteness, permanence and merge pricing

`a2_from_merging_price_and_permanence`, the same file and section, with a cost on moves added.

```
Variable cost : I -> nat.
```

```
Definition merging_steps_priced : Prop :=
  forall i, ~ step_injective i -> cost i >= 1.
```

```
Definition a2_holds : Prop :=
  forall s i, cert s = false -> cert (step s i) = true -> cost i >= 1.
```

```
Theorem a2_from_merging_price_and_permanence :
  forall all,
    finite_states all ->
    permanent ->
    merging_steps_priced ->
    a2_holds.
```

In plain words: `merging_steps_priced` says every move that isn't injective costs at least one, and `a2_holds` is the toll. The theorem: finite, permanent and merges priced give the toll. Theorem 4.2 says this and one thing more, that the machine with that cost is then a certification system the universal floor applies to. That second half isn't in this statement. It's the definition right after it in the file (`certification_system_from_merging_price`), which fills the record's toll field with this theorem, so the printed form is true, and it leans on two Coq items, not one.

E.4 No exact window price across a collision

`shadow_cannot_price_exactly`, `coq/kernel/nfi/ShadowPricing.v`, inside a section that fixes the states, moves, step, reading and a window `obs`.

```
Variables (S I O : Type).
```

```
Variable step : S -> I -> S.
```

```
Variable cert : S -> bool.
```

```
Variable obs : S -> O.
```

```
Definition flips (s : S) (i : I) : bool :=
  negb (cert s) && cert (step s i).
```

```
Definition shadow_cost (price : O -> O -> nat) (s : S) (i : I) : nat :=
  price (obs s) (obs (step s i)).
```

```

Definition meets_floor (cost : S -> I -> nat) : Prop :=
  forall s i, flips s i = true -> cost s i >= 1.

```

```

Definition never_overcharges (cost : S -> I -> nat) : Prop :=
  forall s i, flips s i = false -> cost s i = 0.

```

```

Definition shadow_collision : Prop :=
  exists s1 i1 s2 i2,
    obs s1 = obs s2 /\
    obs (step s1 i1) = obs (step s2 i2) /\
    flips s1 i1 = true /\
    flips s2 i2 = false.

```

```

Theorem shadow_cannot_price_exactly :
  shadow_collision ->
  forall price,
    ~ (meets_floor (shadow_cost price) /\
      never_overcharges (shadow_cost price)).

```

In plain words: `flips s i` says move i takes s from false to true. A shadow price is any function of what the window shows before and after a step. `meets_floor` says every flip costs at least one, `never_overcharges` says every step that isn't a flip costs nothing, and a `shadow_collision` is two steps that look the same through the window, before and after, one a flip and one not. The theorem: given a collision, no shadow price does both. Theorem 7.2 bundles this with two neighbours from the same file, that a price meeting the floor charges some non-flip at least one (`shadow_floor_overcharges`) and that a window showing the reading prices exactly (`window_showing_reading_prices_exactly`). This statement is the middle clause, and the book says exactly what it says.

E.5 The record is a latch

`record_axis_is_latch_holds`, `coq/kernel/foundation/StructuralRecordAxis.v`;
the definitions come from `StructuralCoreAnyBase.v` and `StructuralCoreCover.v`.

```

Definition record_permanent (M : RCM) : Prop :=
  forall s, rc_cert M s = true -> rc_cert M (rc_next M s) = true.

```

```

Definition computation_driven (M : RCM) (B : BaseMachine) (C : BaseCover M B)
  : Prop :=
  exists f : b_state B -> bool -> bool,
    forall m, rc_cert M (rc_next M m) = f (base_state M B C m) (rc_cert M m).

```

```

Definition HonestBaseExtension (M : RCM) (B : BaseMachine) (C : BaseCover M B)
  : Prop :=
  computation_driven M B C /\ ledger_carried M /\ rc_a2 M /\
  record_permanent M /\ reachable_record_write M.

```

```

Definition latch_next (B : BaseMachine) (h : b_state B -> bool)
  (x : b_state B * bool) : b_state B * bool :=
  (b_next B (fst x), orb (snd x) (h (fst x))).

```

```

Definition latch_factorization (M : RCM) (B : BaseMachine) (C : BaseCover M B)
  (h : b_state B -> bool) : Prop :=

```



```
forall m,
  (base_state M B C (rc_next M m), rc_cert M (rc_next M m)) =
  latch_next B h (base_state M B C m, rc_cert M m).
```

```
Definition record_axis_is_latch : Prop :=
  forall M B C, HonestBaseExtension M B C ->
    exists h, latch_factorization M B C h.
```

```
Theorem record_axis_is_latch_holds : record_axis_is_latch.
```

In plain words: here a machine M steps by itself (rc_next gives the one next state), carries a reading rc_cert and a ledger, and sits over a base B through a cover C , a map down to the base's states that commutes with the step. $\text{computation_driven}$ says the reading's next value is some function f of the base state and the reading's current value. *Honest* means driven, a ledger that never falls, the toll, a permanent reading, and some reachable step that writes it. latch_next is the latch of an event h : the base steps, and the record becomes "already on, or h holds now." The theorem: every honest extension evolves exactly as the latch of some event. Theorem 10.1 says a little more than this statement: it names the event, $h(b) = f(b, \text{off})$, where the statement only says one exists. The proof builds exactly that event (its first move is $\text{exists } (\text{fun } b \Rightarrow f \ b \ \text{false})$), so the printed form is what the proof shows, not only what the statement says. And the proof uses two of the five honesty conditions, *driven* and *permanent*; the other three are carried along and never used.

E.6 A raised flag was checked, then committed, then certified

`earned_certification_provenance, minimal/EarnedCore.v.`

```
Theorem earned_certification_provenance : forall s0 tr,
  clean_start s0 -> cert (run tr s0) = true ->
  exists pre1 p c mid1 mid2 post,
    tr = pre1 ++ CHECK p c :: mid1 ++ COMMIT p c :: mid2 ++ CERTIFY :: post /\
    check_ok (core_of (run pre1 s0)) p c = true /\
    commit_ok (core_of (run (pre1 ++ CHECK p c :: mid1) s0)) p c = true /\
    certify_ok (core_of (run (pre1 ++ CHECK p c :: mid1 ++ COMMIT p c :: mid2) s0))
      = true /\
    ver (core_of (run pre1 s0)) c = ver (core_of (run (pre1 ++ CHECK p c :: mid1) s0)) c /\
    untouched (run (pre1 ++ [CHECK p c]) s0) mid1 c.
```

In plain words: the clean-start premise on s_0 says the fact table and the channel are empty and the flag is down, and $\text{run } \text{tr } s_0$ runs a list of instructions. The theorem: from a clean start, if the flag is up at the end, the list splits as some start, CHECK $p \ c$, a middle, COMMIT of the same $p \ c$, another middle, CERTIFY, and the rest; the CHECK, the COMMIT and the CERTIFY each pass where they run; the counter's version at the CHECK equals its version at the COMMIT; and nothing in the first middle touched that counter. Theorem 14.9 says exactly this.

E.7 Checker soundness

`checker_soundness, minimal/EarnedCore.v.`

```
Theorem checker_soundness : forall s0 tr f,
  clean_start s0 ->
  let k := core_of (run tr s0) in
  In f (facts k) -> f_ver f = ver k (f_ctr f) -> holds (f_prop f) (val k (f_ctr f)).
```

In plain words: from a clean start, after any list of instructions, every fact in the table that's about its counter's current version is true of that counter's current value, where holds is the property's meaning written without the checker. Theorem 14.11 says exactly this in its first sentence; its "in particular" is the next corollary in the file, committed_claim_holds.

E.8 No forging

no_forging, minimal/EarnedCore.v, with the two definitions it uses.

```
Definition earned (s0 : state) (tr : list instr) (f : fact) : Prop :=
  exists pre mid,
    tr = pre ++ CHECK (f_prop f) (f_ctr f) :: mid /\
    check_ok (core_of (run pre s0)) (f_prop f) (f_ctr f) = true /\
    f = claim (core_of (run pre s0)) (f_prop f) (f_ctr f) /\
    (f_ver f = ver (core_of (run tr s0)) (f_ctr f) ->
      untouched (run (pre ++ [CHECK (f_prop f) (f_ctr f)]) s0) mid (f_ctr f)).
```

```
Definition no_forger (s0 : state) (tr : list instr) : Prop :=
  forall f, In f (facts (core_of (run tr s0))) -> earned s0 tr f.
```

```
Theorem no_forging : forall s0 tr, clean_start s0 -> no_forger s0 tr.
```

In plain words: a fact is earned when a passing CHECK of exactly that property and counter wrote it somewhere in the list, it is exactly the claim that CHECK made, and, if it's still about the counter's current version, nothing touched the counter since. no_forger says every fact in the table at the end is earned. The theorem: from every clean start, every list of instructions leaves no forger. That's the last clause of Theorem 14.12, with Definition 14.6 saying the same as earned; the one-step clause before it is facts_step.

E.9 The minimum price of a certificate

certified_run_min_cost, minimal/EarnedCore.v.

```
Theorem certified_run_min_cost : forall s0 tr,
  clean_start s0 -> cert (run tr s0) = true ->
  total_cost tr >= 3 /\ mu (run tr s0) >= mu s0 + 3.
```

In plain words: from a clean start, a list of instructions that ends with the flag up cost at least three in total, and the ledger rose by at least three. Theorem 14.16 says exactly this in its first sentence; its form for stored programs is program_certified_min_cost.

E.10 Three, reached by a check that could have failed

earned_run_check_can_fail and earned_run_refused_forever, minimal/EarnedCore.v, with the program they run.

```
Definition earned_run : list instr := [CHECK PZero CA; COMMIT PZero CA; CERTIFY].
```

```
Theorem earned_run_check_can_fail : forall a b,
  (a = 0 ->
    cert (run_prog 4 earned_run (start a b)) = true /\
```

```

mu (run_prog 4 earned_run (start a b)) = 3) /\
(a <> 0 ->
  cert (run_prog 4 earned_run (start a b)) = false /\
  err (core_of (run_prog 4 earned_run (start a b))) = true).

```

```

Theorem earned_run_refused_forever : forall n a b,
  a <> 0 -> cert (run_prog n earned_run (start a b)) = false.

```

In plain words: `earned_run` is CHECK “ A is 0”, COMMIT the same, CERTIFY, and `start a b` puts a in A and b in B with an empty table. The first theorem: run it for four program steps; if a is 0 the flag is up and the ledger is exactly 3, and if a isn’t 0 the flag is down and the trap is up. The second: if a isn’t 0, the flag is down after any number of steps. So three is reached by a check that could have failed, and when it fails nothing ever certifies. Theorem 14.17 says this. The older witness in the same file, `min_cost_tight`, reaches three with “ A is at least 0”, a check that can’t fail; the step doesn’t police that, and Section 14 says so.

E.11 Thiele-complete, the four clauses

`thiele_complete`, `minimal/ThieleComplete.v`, with the clauses it is made of.

`load I a b` is the interface’s loaded state for the start $(1, a, b)$; `run M` runs a list of moves from a state.

```

Definition universal_base_clause : Prop :=
  (forall i, ti_kind I (ub_compile (ti_base I) i) = KBase) /\
  (forall a b, ti_clean I (load a b)) /\
  (forall s m, ti_kind I m = KBase -> m_record M (m_step M s m) = m_record M s) /\
  (forall s m, m_record M s = true -> m_record M (m_step M s m) = true).

```

```

Definition earned_chain (s0 : m_state M) (tr : list (m_move M)) : Prop :=
  exists pre c chk mid1 cmt mid2 crt post,
    tr = pre ++ chk :: mid1 ++ cmt :: mid2 ++ crt :: post /\
    ti_kind I chk = KCheck c /\ ti_kind I cmt = KCommit c /\ ti_kind I crt = KCertify /\
    ti_check I (run M pre s0) c = true /\
    (forall t1 t2, mid1 = t1 ++ t2 ->
      ti_same I c (run M pre s0) (run M (pre ++ chk :: t1) s0)) /\
    m_record M (run M (pre ++ chk :: mid1 ++ cmt :: mid2) s0) = false /\
    m_record M (run M (pre ++ chk :: mid1 ++ cmt :: mid2 ++ [crt]) s0) = true.

```

```

Definition earned_record_clause : Prop :=
  (forall s, ti_clean I s -> m_record M s = false) /\
  (forall s0 tr, ti_clean I s0 -> m_record M (run M tr s0) = true -> earned_chain s0 tr) /\
  (forall s c, ti_check I s c = true -> ti_meaning I c s) /\
  (forall c s s', ti_same I c s s' -> ti_meaning I c s -> ti_meaning I c s').

```

```

Definition record_move (m : m_move M) : nat :=
  match ti_kind I m with KBase => 0 | _ => 1 end.

```

```

Definition exact_toll_clause : Prop :=
  (forall m, m_cost M m = record_move m) /\
  (forall s m, ti_ledger I (m_step M s m) = ti_ledger I s + m_cost M m).

```

```

Definition non_vacuity_clause : Prop :=
  exists c chk cmt crt,
    ti_kind I chk = KCheck c /\ ti_kind I cmt = KCommit c /\ ti_kind I crt = KCertify /\
    (forall a b, m_record M (run M [chk; cmt; crt] (load a b)) = true <->
      ti_meaning I c (load a b)) /\

```

```

    (exists a b, ti_meaning I c (load a b)) /\
    (exists a b, ~ ti_meaning I c (load a b)).

Definition thiele_complete_with : Prop :=
  universal_base_clause /\ earned_record_clause /\ exact_toll_clause /\ non_vacuity_clause.

Definition thiele_complete (M : machine) : Prop :=
  exists I : thiele_interface M, thiele_complete_with I.

Theorem clock_not_thiele_complete : forall rd, ~ thiele_complete (clock rd).

```

In plain words: M is a machine (states, moves, step, cost, record) and I an interface for it (a universal base, claims with meanings, a checker, “unchanged,” clean states, a ledger, and a kind for each move). The first clause says the compiled two-counter moves are base moves, loaded states are clean, base moves leave the record alone, and the record never comes down; the universal base’s own fields, not printed here, say the compiled moves act on the window exactly as the two-counter instructions act. `earned_chain` is a passing check of a claim, then a commit of the same claim with the thing it’s about unchanged at every point between, then the certify that raises the record. The earned-record clause says clean states have the record down, every run from a clean state that raises it contains that chain, a passing check means the claim holds, and a claim that holds keeps holding while its thing is unchanged. The exact toll: base moves cost 0, record moves 1, and the ledger adds each move’s cost. Non-vacuity: some claim’s bare chain raises the record from a loaded state exactly when the claim holds there, and the claim holds at one loaded state and fails at another. `thiele_complete` asks for one interface meeting all four. `clock` is the two-counter machine with every move costing 1 and any reading of its configuration as the record. Definition 14.7 and Theorem 14.19 say exactly this.

E.12 The universal program’s flag and ledger

`universal_halting`, `universal_flag_iff`, `universal_ledger_exact` and `universal_thiele_complete`, `coq/kernel/foundation/UniversalRun.v`.

```

Theorem universal_halting : forall P x y,
  (exists m, E.halted P (E.core_of (grun P x y m))) <->
  (exists n, M.halted U (M.core_of (hrun P x y n))).

Theorem universal_flag_iff : forall P x y,
  (exists m, E.cert (grun P x y m) = true) <-> (exists n, M.cert (hrun P x y n) = true).

Theorem universal_ledger_exact : forall P x y,
  (forall m, exists n,
    ((rel P (grun P x y m) (hrun P x y n) /\ m <= n) /\
     rel_halt P (grun P x y m) (hrun P x y n)) /\
    M.mu (hrun P x y n) = E.mu (grun P x y m)) /\
  (forall n, exists m,
    E.mu (grun P x y m) <= M.mu (hrun P x y n) <= E.mu (grun P x y (S m))).

Theorem universal_thiele_complete : TC.thiele_complete host_machine.

```

In plain words: `grun P x y m` is m program steps of the small machine’s program P from start x y ; `hrun P x y n` is n program steps of the one fixed host program

U from the host loaded with P , x and y . The first theorem: the guest halts at some step exactly when U does. The second: the guest's flag is up at some step exactly when the host's is. The third: for every guest step count there's a host step count where the two are related, or both stopped, with equal ledgers; and every host point's ledger lies between the guest's at some step and at the step after. `rel` and `rel_halt` are the relations Theorem 14.24 spells out. The fourth: the host machine, the small machine with a register for every whole number and the one property `PSlot`, read one move per instruction, is Thiele-complete in the sense of the previous subsection. Theorems 14.25, 14.26 and 14.28 say exactly this.

E.13 The copy is the record, on the simpler host

`record_agreement`, `minimal/UniversalThiele.v`.

```
Definition agrees (h : hstate) : Prop := mrec h = E.cert (gst h).
```

```
Theorem record_agreement : forall tr h, agrees h -> agrees (hrun tr h).
```

In plain words: the host keeps a mirror bit `mrec` beside the guest's state `gst`, and `agrees h` says the mirror equals the guest's own flag. The theorem: a host state that `agrees` still `agrees` after any list of host moves, whatever they are. Theorem 14.32 says this for lists of moves and adds stored programs and loading, which are `record_agreement_program` and `hload_agrees`.

E.14 No free host certification, on the simpler host

`no_free_host_certification`, `minimal/UniversalThiele.v`, with the host's moves and their costs.

```
Inductive hinstr : Type :=
| OWN (i : E.instr)
| GSTEP (b : nat).
```

```
Definition hcost (i : hinstr) : nat :=
  match i with OWN j => E.cost j | GSTEP b => b end.
```

```
Definition crossing (g g' : E.state) : bool := negb (E.cert g) && E.cert g'.
```

```
Theorem no_free_host_certification : forall tr h,
  mrec h = false -> mrec (hrun tr h) = true ->
  exists pre b post, tr = pre ++ GSTEP b :: post /\ b >= 1 /\
    crossing (gst (hrun pre h)) (gst (hexec (hrun pre h) (GSTEP b))) = true.
```

In plain words: a host move is either `OWN i`, one of the small machine's own instructions on the host's core, or `GSTEP b`, one guest step with budget b , which costs b whatever the guest instruction cost. `crossing g g'` says the guest's flag went from down to up. The theorem: if the mirror is down before a list of host moves and up after it, then somewhere in the list there's a `GSTEP b` with b at least one at which the guest's flag crossed. Theorem 14.34 prints the one-move form, `no_free_host_certification_step`, which also says the host wasn't trapped and the guest's next instruction was `CERTIFY`; this is the form for whole lists that it lifts to. Neither says

what the guest did before its CERTIFY. That a raised mirror means the guest's own run checked and committed first is the next theorem.

E.15 A raised mirror stands on the guest's own earned chain, on the simpler host

`host_mirror_earned`, `minimal/UniversalThiele.v`.

```
Theorem host_mirror_earned : forall tr a b P x y,
  mrec (hrun tr (hload a b P x y)) = true ->
  exists n gtr prel p c mid1 mid2 post,
    gtr = E.trace_of n P (E.start x y) /\
    gst (hrun tr (hload a b P x y)) = E.run gtr (E.start x y) /\
    gtr = prel ++ E.CHECK p c :: mid1 ++ E.COMMIT p c :: mid2 ++ E.CERTIFY :: post /\
    E.check_ok (E.core_of (E.run prel (E.start x y))) p c = true /\
    E.commit_ok (E.core_of (E.run (prel ++ E.CHECK p c :: mid1) (E.start x y))) p c
      = true /\
    E.certify_ok (E.core_of (E.run (prel ++ E.CHECK p c :: mid1 ++ E.COMMIT p c :: mid2)
      (E.start x y))) = true /\
    E.ver (E.core_of (E.run prel (E.start x y))) c
      = E.ver (E.core_of (E.run (prel ++ E.CHECK p c :: mid1) (E.start x y))) c /\
    E.untouched (E.run (prel ++ [E.CHECK p c]) (E.start x y)) mid1 c.
```

In plain words: load the host with a guest program P and a guest start (x, y) (`hload` gives the guest a clean start), run any list of host moves, and suppose the mirror is up at the end. Then for some number of steps n , the guest's state is its own program's run of n steps, and the instructions that run executed split exactly the way Theorem 14.9 splits a trace: a passing CHECK, a COMMIT of the same claim at the same version with the counter untouched between, then a passing CERTIFY. So the host's mirror can't be up unless the guest's own program earned it. The book's sentences citing `host_mirror_earned` say this in words; the Coq says a little more than they need, since the chain sits inside the instructions the guest program itself executed and not just in some list.

E.16 Structural entitlement, represented

`ent_representation`, `minimal/EntitlementSmall.v`, with the definitions it leans on.

```
Definition ent_strict_sublist (post prior : list X) : Prop :=
  incl post prior /\ exists x, In x prior /\ ~ In x post.

Definition ent_distinguishes (obs : X -> O) (w : X) (post : list X) : Prop :=
  forall t, In t post -> obs w <> obs t.

Definition ent_member (eqb : O -> O -> bool) (obs : X -> O) (L : list X) (o : O) : bool :=
  existsb (fun t => eqb (obs t) o) L.

Definition ent_stronger (P Q : O -> bool) : Prop :=
  forall o, P o = true -> Q o = true.

Definition ent_strictly_stronger (P Q : O -> bool) : Prop :=
  ent_stronger P Q /\ exists o, P o = false /\ Q o = true.

Definition ent_reduction (r : X -> O) (T : ent_tree) (prior post : list X) : Prop :=
  exists F : X -> list X,
    (forall x, In x prior -> exists t, In t post /\ In x (F t) /\ r x = r t) /\
    length prior <= fold_right Nat.add 0 (map (fun t => length (F t)) post) /\
    (forall t, In t post -> length (F t) <= ent_leaves T).

Definition ent_certified {M : machine} (I : thiele_interface M) {O : Type}
  (s0 : m_state M) (tr : list (m_move M))
  (d : list (m_move M) -> O) (P : O -> bool) : Prop :=
  m_record M (run M tr s0) = true /\ P (d tr) = true.

Theorem ent_representation :
  forall (M : machine) (I : thiele_interface M) (X O : Type)
    (s0 : m_state M) (tr : list (m_move M))
    (d : list (m_move M) -> O) (obs r : X -> O) (eqb : O -> O -> bool)
```

```

    (prior post : list X) (T : ent_tree),
  thiele_complete_with I ->
    (forall o1 o2, eqb o1 o2 = true <-> o1 = o2) ->
  ent_strict_sublist post prior ->
  (exists w, In w prior /\ ~ In w post /\ ent_distinguishes obs w post) ->
  ti_clean I s0 ->
  ent_certified I s0 tr d (ent_member eqb obs post) ->
  ent_depth T <= record_moves I tr ->
  0 < length post ->
  ent_reduction r T prior post ->
  ent_strictly_stronger (ent_member eqb obs post) (ent_member eqb obs prior) /\
  earned_chain I s0 tr /\
  (exists pre c chk mid1 cmt rest,
    tr = pre ++ chk :: mid1 ++ cmt :: rest /\
    ti_kind I chk = KCheck c /\ ti_kind I cmt = KCommit c /\
    ti_meaning I c (run M pre s0) /\ ti_meaning I c (run M (pre ++ chk :: mid1) s0)) /\
  Nat.log2_up (length prior) - Nat.log2_up (length post) <= record_moves I tr /\
  record_moves I tr = ti_ledger I (run M tr s0) - ti_ledger I s0 /\
  ti_ledger I s0 + 3 <= ti_ledger I (run M tr s0).

```

In plain words: take a machine and an interface through which it is Thiele-complete, any type of rivals X and of observations O with an exact equality test, a run t from a clean state s_0 , and a decoder d that reads the run's receipts, its list of moves. Hypotheses in order: the posterior list is inside the prior and drops at least one rival; some dropped rival shows an observation no survivor shows; the start is clean; the run ends with the record up and the posterior's membership test accepts the decoded run; the tree is no deeper than the number of record moves in the run; the posterior is non-empty; and the covering is by fibres, as `ent_reduction` spells out. Conclusions in order: the posterior's test is strictly stronger; the run contains the earned chain; that chain's claim holds at its CHECK and at its COMMIT; the index bits are at most the record moves; the record moves are exactly the ledger's rise; and the rise is at least 3. Theorem 14.39 says exactly this, with the hypotheses numbered (M1) to (M6) and the conclusions numbered (1) to (5), and it folds the chain and its meanings into one conclusion where the Coq lists them as two.

E.17 A raised flag stands on the CHSH check

`small_chsh_flag_implies_tsirelson`, `SmallChshMachine.v`, over the host machine with the one property `small_chsh_PCHSH`.

```

Theorem small_chsh_flag_implies_tsirelson :
  forall (s0 : @state small_chsh_prop) (tr : list (@instr small_chsh_prop)),
  clean_start s0 ->
  cert (run small_chsh_prop_eqb small_chsh_eval tr s0) = true ->
  exists pre1 c mid1 mid2 post,
    tr = pre1 ++ CHECK small_chsh_PCHSH c :: mid1
      ++ COMMIT small_chsh_PCHSH c :: mid2 ++ CERTIFY :: post /\
    let v := vals (core_of (run small_chsh_prop_eqb small_chsh_eval pre1 s0)) c in
    let t := small_chsh_tally_of v in
    small_chsh_check t = true /\
    small_chsh_meaning t /\
    (small_chsh_score t * small_chsh_score t <= 8)%R /\
    (Rabs (small_chsh_score t) <= 2 * sqrt 2)%R /\
    vals (core_of (run small_chsh_prop_eqb small_chsh_eval
      (pre1 ++ CHECK small_chsh_PCHSH c :: mid1) s0)) c = v.

```

In plain words: start from a clean start and run any list of instructions of the host machine whose only property is the CHSH one, and suppose the flag is up at the end.

Then the list splits as some instructions, a CHECK of the property at some counter c , some more, a COMMIT of the same property at the same counter, some more, a CERTIFY, and the rest. Take the value v that counter c held just before the CHECK, and read it as a tally of eight counts. That tally passes the seven integer facts (`small_chsh_check`); it means what the check is for: every question pair sampled and the pinned five-by-five symmetric and positive semidefinite (`small_chsh_meaning`); its score S has $S^2 \leq 8$ and $|S| \leq 2\sqrt{2}$; and counter c still held v when the COMMIT ran. Theorem 15.5 says exactly this in its first part. It says more in two places that have their own Coq theorems, the stored-program form and the converse that the three instructions certify exactly when the check passes, and the real numbers in the statement are why it carries the three standard-library assumptions printed in Section D.

E.18 The recursion theorem up to the record, and no better

`sm2_kleene_obs` and `sm2_no_exact`, `SmFixedPoint.v` and `SmNoExact.v`, with the relation they use. Here M is the host machine's module, `hendsPxs` says program P , started with x in register 1 and everything else clean, stops (a trap counts as stopping) in state s , and `sm_computes_mapTF` says that on the number of every program p , T stops with the number of Fp .

```
Definition sm2_obs (s t : hstate) : Prop :=
  M.vals (M.core_of s) 0 = M.vals (M.core_of t) 0 /\
  M.err (M.core_of s) = M.err (M.core_of t) /\
  M.mu s = M.mu t /\ M.cert s = M.cert t /\
  length (M.facts (M.core_of s)) = length (M.facts (M.core_of t)) /\
  (M.chan (M.core_of s) = None <-> M.chan (M.core_of t) = None).
```

```
Definition sm2_obs_equiv (P Q : list hinstr) : Prop :=
  forall x,
    (forall s, hends P x s -> exists t, hends Q x t /\ sm2_obs s t) /\
    (forall t, hends Q x t -> exists s, hends P x s /\ sm2_obs s t).
```

```
Theorem sm2_kleene_obs : forall (F : list hinstr -> list hinstr) (T : list hinstr),
  sm_computes_map T F -> exists e, sm2_obs_equiv e (F e).
```

```
Theorem sm2_no_exact :
  exists (F : list hinstr -> list hinstr) (T : list hinstr),
    sm_computes_map T F /\ forall e, ~ hequiv e (F e).
```

In plain words: `sm2_obsst` says two final states agree on register 0, the trap latch, the ledger, the flag, the number of facts and whether the channel is empty, and nothing else. `sm2_obs_equivPQ` says that on every input, any state P ends in has a state Q ends in that agrees in that sense, and the same the other way round, which includes that P stops on an input exactly when Q does. The first theorem: for every transformation F of host programs that some host program T carries out on numbers, there is a program e that agrees with $F e$ in that sense. T is any host program, and may itself check, commit and certify. The second: there is a transformation F , carried out by a host program, such that no program e behaves the same as $F e$ in the sense of Definition 14.12, which compares every register, the fact table, the channel, the trap latch, the ledger and the

flag. Theorems 14.49 and 14.50 say these two. The book’s printed version says exactly this, with the line number and versions left out of both comparisons, as the definition does.

F The CHSH derivation, in full

This appendix supports Section 15. That section tells the story; this one does the algebra. Each item gives the exact statement with all its hypotheses, the steps a reader needs to check it, and the Coq name with its file. Nothing here identifies any matrix with a quantum state. Where the reading “quantum” appears, it is a cited result from outside the kernel.

F.1 A. Setting and notation

Definition F.1 (Settings, outcomes, correlators). Alice receives a setting $x \in \{0, 1\}$ and Bob a setting $y \in \{0, 1\}$. Each answers a bit, recoded as an outcome $a, b \in \{+1, -1\}$ (bit 0 becomes $+1$, bit 1 becomes -1). The correlator E_{xy} is the average of the product ab over the rounds with settings (x, y) . The CHSH quantity is

$$S = E_{00} + E_{01} + E_{10} - E_{11}.$$

I collect the four correlators into the 2×2 matrix

$$E = \begin{pmatrix} E_{00} & E_{01} \\ E_{10} & E_{11} \end{pmatrix},$$

whose rows are indexed by Alice’s setting and whose columns by Bob’s. The rows of E are (E_{00}, E_{01}) and (E_{10}, E_{11}) ; the columns are (E_{00}, E_{10}) and (E_{01}, E_{11}) .

The same polynomial S appears under three names in the source: `CHSH` (kernel/quantum/TsirelsonGeneral.v), `CHSH_value` (kernel/quantum/TsirelsonFromAlgebra.v) and `chsh_S` (kernel/quantum/ElliptopeCompletion.v).

Definition F.2 (Counters). Eight counts sit in the record `WitnessCounts` (kernel/quantum/CHSHColumnCheck.v): for each setting pair, `samexy` counts rounds whose two answer bits matched and `diffxy` counts rounds whose bits differed. A round adds one to exactly one of the eight. Write

$$N_{xy} = \text{same}_{xy} + \text{diff}_{xy}, \quad D_{xy} = \text{same}_{xy} - \text{diff}_{xy}, \quad E_{xy} = \frac{D_{xy}}{N_{xy}}.$$

A matched round contributes $ab = +1$ and a differing round $ab = -1$, so D_{xy} is the sum of the products and D_{xy}/N_{xy} is their average. Since $|D_{xy}| \leq N_{xy}$, every correlator from counters lies in $[-1, 1]$. In Coq the integers are `chsh_n_z` and `chsh_d_z` (ker-

nel/quantum/CHSHColumnCheck.v). When a bucket is empty, the real-valued correlator `state_bucket_correlation` (kernel/quantum/CHSHColumnCheck.v) and the rational one `chsh_correlator_q` (kernel/quantum/CHSHStatisticalBridge.v) both return 0 by convention. The CHSH score of a tally, `chsh_stat_from_wc` (CHSHStatisticalBridge.v), adds the first three correlators and obtains $-E_{11}$ by swapping the two counters of the last bucket.

F.2 B. The classical bound

Lemma F.1 (Deterministic strategies). *Let $a_0, a_1, b_0, b_1 \in \{+1, -1\}$. Then $-2 \leq a_0b_0 + a_0b_1 + a_1b_0 - a_1b_1 \leq 2$.*

Proof. Factor: $a_0b_0 + a_0b_1 + a_1b_0 - a_1b_1 = a_0(b_0 + b_1) + a_1(b_0 - b_1)$. If $b_0 = b_1$ then $b_0 + b_1 = \pm 2$ and $b_0 - b_1 = 0$; if $b_0 \neq b_1$ the other way round. In both cases exactly one of the two brackets is ± 2 and the other is 0, so the expression is ± 2 times a sign, which is $+2$ or -2 . There are sixteen sign choices, and every one of them gives exactly ± 2 , never 0 or ± 4 . $\square$

The Coq proofs do not factor; they split into the sixteen sign cases and compute. The rational statement is `local_S_2_deterministic` (kernel/quantum/BoxCHSH.v): for $f_A, f_B : \mathbb{N} \rightarrow \mathbb{Q}$ with every value ± 1 , the absolute value of $f_A(0)f_B(0) + f_A(0)f_B(1) + f_A(1)f_B(0) - f_A(1)f_B(1)$ is at most 2. The real statement is `deterministic_strategy_chsh_bounded` (kernel/quantum/MinorConstraints.v), with conclusion $-2 \leq \dots \leq 2$.

Definition F.3 (Finite factorizable model, `is_factorizable`). A real-valued correlation function $E(a, b, x, y)$ is factorizable (kernel/quantum/MinorConstraints.v) when there exist response functions $A(x, \lambda)$, $B(y, \lambda)$, weights $p(\lambda)$ and a natural number $\lambda_{\max}$ such that: $0 \leq p(\lambda) \leq 1$ for $\lambda \leq \lambda_{\max}$; $\sum_{\lambda=0}^{\lambda_{\max}} p(\lambda) = 1$; every $A(x, \lambda)$ and every $B(y, \lambda)$ is -1 or $+1$; and for all a, b, x, y , $E(a, b, x, y) = \sum_{\lambda=0}^{\lambda_{\max}} p(\lambda)A(x, \lambda)B(y, \lambda)$. The outcome arguments a, b are dummies; only correlations are modelled.

Theorem F.2 (`fine_theorem`, one direction only). *If E is factorizable, then $-2 \leq E(0,0,0,0) + E(0,0,0,1) + E(0,0,1,0) - E(0,0,1,1) \leq 2$ (kernel/quantum/MinorConstraints.v; restated as `factorizable_CHSH_classical_bound`).*

Proof. Substitute the factorization: $S = \sum_{\lambda} p(\lambda)S_{\lambda}$, where S_{λ} is the deterministic expression of Lemma F.1 at the signs $A(\cdot, \lambda), B(\cdot, \lambda)$. Each $S_{\lambda} \in [-2, 2]$, the weights are nonnegative and sum to 1, so $-2 = \sum_{\lambda} p(\lambda)(-2) \leq S \leq \sum_{\lambda} p(\lambda) \cdot 2 = 2$. $\square$

Remark F.1 (What the name does and does not cover). Fine's 1982 theorem is an equivalence between several descriptions of local models and the Bell inequalities. $\mathfrak{f}$

`ine_theorem` proves one direction: a finite factorizable model obeys the bound. It proves no converse, and it covers only finitely many hidden states ($\lambda \leq \lambda_{\max}$), not an arbitrary probability space. The transfer to rational boxes is `local_box_CHSH_bound` (`kernel/quantum/MinorConstraints.v`): if `box_correlations B` is factorizable then the real image of $|S(B)|$ is at most 2.

The finite-tally statement concerns one fixed strategy replayed on every recorded round.

Definition F.4 (`WCLocallyConsistent`). For bits $a_0, a_1, b_0, b_1 \in \{0, 1\}$ and a counter record w (`kernel/quantum/CHSHStatisticalBridge.v`), w is locally consistent with (a_0, a_1, b_0, b_1) when, for each setting pair (x, y) : if $a_x = b_y$ then $\text{diff}_{xy} = 0$, otherwise $\text{same}_{xy} = 0$; and every bucket is sampled, $N_{00}, N_{01}, N_{10}, N_{11} > 0$.

Theorem F.3 (A tally above 2 is not one replayed strategy). *Let w be a counter record whose rational score `chsh_stat_from_wc w` is greater than 2. Then for all bits a_0, a_1, b_0, b_1 , w is not locally consistent with (a_0, a_1, b_0, b_1) . This is `chsh_violation_rules_out_locally_factorizable_coupling` (`kernel/quantum/CHSHCouplingBridge.v`). Its contrapositive form, `locally_consistent_classical_bound`, states that a locally consistent record has score of absolute value at most 2.*

Proof. Under local consistency each sampled bucket is pure: if $a_x = b_y$ every round in the bucket matched, so $E_{xy} = N_{xy}/N_{xy} = +1$; otherwise every round differed and $E_{xy} = -1$. In signs, $E_{xy} = \alpha_x \beta_y$ with $\alpha_x = (-1)^{a_x}$ and $\beta_y = (-1)^{b_y}$. The score is then the deterministic expression of Lemma F.1, which is at most 2 in absolute value. A score above 2 contradicts this. $\square$

The theorem rules out one explanation, a single four-bit plan used on every round. It says nothing about a local model that redraws its hidden state every round; Section 15 gives a four-round coin-flip tally with score 4.

F.3 C. Tsirelson's bound from algebra

Lemma F.4 (The six-square identity, `chsh_gap_is_sum_of_squares`). *For all real a, b, c, d ,*

$$\begin{aligned} 4(a^2 + b^2 + c^2 + d^2) - (a + b + c - d)^2 \\ = (a - b)^2 + (a - c)^2 + (a + d)^2 + (b - c)^2 + (b + d)^2 + (c + d)^2. \end{aligned}$$

(`kernel/quantum/TsirelsonFromAlgebra.v`, closed by `ring`.)

Proof. Expand the square on the left: $(a + b + c - d)^2 = a^2 + b^2 + c^2 + d^2 + 2ab + 2ac -$

$2ad + 2bc - 2bd - 2cd$. So the left side is

$$3(a^2 + b^2 + c^2 + d^2) - 2ab - 2ac + 2ad - 2bc + 2bd + 2cd.$$

On the right, each variable appears in exactly three of the six brackets (a in $a - b$, $a - c$, $a + d$; b in $a - b$, $b - c$, $b + d$; c in $a - c$, $b - c$, $c + d$; d in $a + d$, $b + d$, $c + d$), so the squared terms add to $3(a^2 + b^2 + c^2 + d^2)$. The cross terms are $-2ab$ from $(a - b)^2$, $-2ac$ from $(a - c)^2$, $+2ad$ from $(a + d)^2$, $-2bc$ from $(b - c)^2$, $+2bd$ from $(b + d)^2$ and $+2cd$ from $(c + d)^2$. Both sides agree term by term. $\square$

Since the right side is a sum of squares, $S^2 \leq 4 \sum_{xy} E_{xy}^2$ for any four reals; this is `cauchy_schwarz_chsh` (`TsirelsonFromAlgebra.v`).

Theorem F.5 (Row-bound form of Tsirelson's bound). *For all real $E_{00}, E_{01}, E_{10}, E_{11}$ with $E_{00}^2 + E_{01}^2 \leq 1$ and $E_{10}^2 + E_{11}^2 \leq 1$, $S^2 \leq 8$. This is `tsirelson_squared` (`kernel/quantum/TsirelsonFromAlgebra.v`) and `tsirelson_from_row_bounds` (`kernel/quantum/TsirelsonGeneral.v`). The absolute form $|S| \leq \sqrt{8} = 2\sqrt{2}$ is `tsirelson_bound_abs` (`TsirelsonGeneral.v`).*

Proof. Adding the two row bounds gives $\sum_{xy} E_{xy}^2 \leq 2$. By Lemma F.4, $S^2 \leq 4 \sum_{xy} E_{xy}^2 \leq 8$. $\square$

Theorem F.6 (Tightness). *The point $E_{00} = E_{01} = E_{10} = 1/\sqrt{2}$, $E_{11} = -1/\sqrt{2}$ satisfies both row bounds and has $S = \sqrt{8}$. This is `tsirelson_tight` (`TsirelsonFromAlgebra.v`) and `tsirelson_achievable` (`TsirelsonGeneral.v`).*

Proof. Each row has squared length $1/2 + 1/2 = 1$, and $S = 4/\sqrt{2} = 2\sqrt{2} = \sqrt{8}$. $\square$

Remark F.2 (Where equality can happen). This step is argued here, not checked in Coq. Equality $S^2 = 8$ under the row bounds forces equality in both steps of the proof above: the six squares must all vanish, which gives $a = b = c = -d$, and the rows must be tight, which gives $2a^2 = 1$. So the point of Theorem F.6 and its negative are the only places the bound is reached. A tally's correlators are ratios of whole numbers and $2a^2 = 1$ has no such solution, so no tally reaches the bound; the integer check's passing tallies stay strictly below it and come as near as you like.

Lemma F.7 (A rational constant above the bound, `rational_tsirelson_bound`). $\sqrt{8} < 5657/2000$ (`kernel/quantum/TsirelsonFromAlgebra.v`).

Proof. Both numbers are positive, so it suffices to compare squares. $5657^2 = 32,001,649$ and $8 \cdot 2000^2 = 32,000,000$, so $(5657/2000)^2 = 32,001,649/4,000,000 > 8 = (\sqrt{8})^2$. If

$\sqrt{8} \geq 5657/2000$ held, squaring both nonnegative sides would give $8 \geq (5657/2000)^2$, a contradiction. $\square$

The constant is the four-decimal round-up of $2\sqrt{2} = 2.8284271\dots$; the gap is about 7.3×10^{-5} . It lets rational code test “above the Tsirelson bound” without a square root. It plays no part in the integer check of item E.

F.4 D. The zero-marginal 5x5 moment matrix

Definition F.5 (The pinned level-1 matrix). `zero_marginal_npa` (`kernel/quantum/NPAMomentMatrix.v`) fills the general record `NPAMomentMatrix` with all four one-party entries and both within-party entries set to zero, and `npa_to_matrix` lays it out on rows and columns I, A_0, A_1, B_0, B_1 :

$$M(E) = \begin{array}{c} \\ I \\ A_0 \\ A_1 \\ B_0 \\ B_1 \end{array} \begin{array}{ccccc} & I & A_0 & A_1 & B_0 & B_1 \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & E_{00} & E_{01} \\ 0 & 0 & 1 & E_{10} & E_{11} \\ 0 & E_{00} & E_{10} & 1 & 0 \\ 0 & E_{01} & E_{11} & 0 & 1 \end{pmatrix} \end{array}$$

Positive semidefiniteness in the source is defined by quadratic forms, not eigenvalues: `PSD5` (`kernel/quantum/ConstructivePSD.v`) says $v^\top M v \geq 0$ for every $v \in \mathbb{R}^5$, and `npa_psd` (`NPAMomentMatrix.v`) is “symmetric and `PSD5`”. In block form,

$$M(E) = \begin{pmatrix} 1 & 0 \\ 0 & G \end{pmatrix}, \quad G = \begin{pmatrix} I_2 & E \\ E^\top & I_2 \end{pmatrix}.$$

The corner 1 is decoupled, so $M(E)$ is PSD exactly when G is.

Lemma F.8 (Schur complement, the standard form). *Let P be symmetric positive definite.*

The symmetric block matrix $\begin{pmatrix} P & Q \\ Q^\top & R \end{pmatrix}$ is PSD if and only if $R - Q^\top P^{-1} Q$ is PSD.

Proof. For u, w of the right sizes,

$$\begin{pmatrix} u \\ w \end{pmatrix}^\top \begin{pmatrix} P & Q \\ Q^\top & R \end{pmatrix} \begin{pmatrix} u \\ w \end{pmatrix} = (u + P^{-1} Q w)^\top P (u + P^{-1} Q w) + w^\top (R - Q^\top P^{-1} Q) w.$$

The first term is never negative because P is positive definite, and it is zero at $u = -P^{-1} Q w$. So the form is nonnegative for all (u, w) exactly when the second term is nonnegative for all w . $\square$

The Coq proofs do not invoke this lemma abstractly. They carry out its one instance with $P = I_2$, $Q = E$, $R = I_2$, where the complement is

$$I_2 - E^\top E = \begin{pmatrix} 1 - E_{00}^2 - E_{10}^2 & -(E_{00}E_{01} + E_{10}E_{11}) \\ -(E_{00}E_{01} + E_{10}E_{11}) & 1 - E_{01}^2 - E_{11}^2 \end{pmatrix}.$$

The diagonal of $E^\top E$ holds the squared lengths of the columns of E , which is why the conditions below are column conditions.

Definition F.6 (Column contractive, `zero_marginal_column_contractive`). (`kernel/quantum/CHSHColumnCheck.v`) Write $p = 1 - E_{00}^2 - E_{10}^2$, $q = 1 - E_{01}^2 - E_{11}^2$ and $s = E_{00}E_{01} + E_{10}E_{11}$. The correlators are column contractive when

$$p \geq 0, \quad q \geq 0, \quad pq - s^2 \geq 0.$$

Lemma F.9 (The 2x2 test, `psd2_quadratic_form_nonneg`). (`kernel/quantum/CHSHColumnCheck.v`) If $\alpha \geq 0$, $\delta \geq 0$ and $\alpha\delta - \beta^2 \geq 0$, then $\alpha u^2 + 2\beta uw + \delta w^2 \geq 0$ for all real u, w .

Proof. If $w = 0$ the form is $\alpha u^2 \geq 0$. Otherwise put $t = u/w$, so the form is $w^2(\alpha t^2 + 2\beta t + \delta)$. If $\alpha = 0$, then $-\beta^2 \geq 0$ forces $\beta = 0$ and the bracket is $\delta \geq 0$. If $\alpha > 0$, $\alpha t^2 + 2\beta t + \delta = \alpha(t + \beta/\alpha)^2 + (\alpha\delta - \beta^2)/\alpha$, a sum of two nonnegative terms. $\square$

Theorem F.10 (Contractive implies PSD). If $p \geq 0$, $q \geq 0$ and $pq - s^2 \geq 0$, then `PSD5` holds of $M(E)$. This is `zero_marginal_npa_column_contractive_implies_psd` (`kernel/quantum/CHSHColumnCheck.v`).

Proof. Write $v = (v_0, x_0, x_1, y_0, y_1)$. Reading off the entries of $M(E)$,

$$v^\top M(E) v = v_0^2 + x_0^2 + x_1^2 + y_0^2 + y_1^2 + 2x_0(E_{00}y_0 + E_{01}y_1) + 2x_1(E_{10}y_0 + E_{11}y_1).$$

Complete the squares in x_0 and x_1 :

$$\begin{aligned} v^\top M(E) v &= v_0^2 + (x_0 + E_{00}y_0 + E_{01}y_1)^2 + (x_1 + E_{10}y_0 + E_{11}y_1)^2 \\ &\quad + p y_0^2 - 2s y_0 y_1 + q y_1^2. \end{aligned}$$

To check the last three terms, expand the two squares: they contribute $(E_{00}y_0 + E_{01}y_1)^2 + (E_{10}y_0 + E_{11}y_1)^2$ beyond the cross terms, which is $(E_{00}^2 + E_{10}^2)y_0^2 + 2s y_0 y_1 + (E_{01}^2 + E_{11}^2)y_1^2$; subtracting it from $y_0^2 + y_1^2$ leaves $p y_0^2 - 2s y_0 y_1 + q y_1^2$. The first three terms are squares. The last is the 2×2 form with $\alpha = p$, $\beta = -s$, $\delta = q$, nonnegative by Lemma F.9. $\square$

Lemma F.11 (Discriminant, `quadratic_nonneg_discriminant`). (`ker-`

nel/quantum/ConstructivePSD.v) If $a + 2bt + ct^2 \geq 0$ for every real t , then $b^2 \leq ac$.

Proof. If $c < 0$ the quadratic tends to $-\infty$, so $c \geq 0$. If $c = 0$, the line $a + 2bt$ is nonnegative for every t only if $b = 0$, and then $b^2 = 0 = ac$. If $c > 0$, evaluate at $t = -b/c$: $a - b^2/c \geq 0$, so $b^2 \leq ac$. (The Coq proof is the same case analysis, with explicit witnesses t for the contradictions.) $\square$

Theorem F.12 (PSD implies contractive). *If PSD5 holds of $M(E)$, then $p \geq 0$, $q \geq 0$ and $pq - s^2 \geq 0$. This is `npa_psd_implies_column_contractive` (*kernel/quantum/CHSHColumnCheck.v*).*

Proof. Feed the quadratic form three test vectors, written in the order (I, A_0, A_1, B_0, B_1) ; their values are the lemmas `npa_quad5_test_col0`, `npa_quad5_test_col1` and `npa_quad5_test_schur` (*CHSHColumnCheck.v*). Each is the choice $x = -Ey$ that zeroes both completed squares in the proof of Theorem F.10, so only $y^\top (I_2 - E^\top E)y$ is left.

First, $v = (0, -E_{00}, -E_{10}, 1, 0)$ gives $v^\top Mv = p$, so $p \geq 0$.

Second, $v = (0, -E_{01}, -E_{11}, 0, 1)$ gives $v^\top Mv = q$, so $q \geq 0$.

Third, for each real t , $v(t) = (0, -(E_{00}t + E_{01}), -(E_{10}t + E_{11}), t, 1)$ gives $v(t)^\top Mv(t) = pt^2 - 2st + q \geq 0$. Apply Lemma F.11 with $a = q$, $b = -s$, $c = p$: $s^2 \leq pq$. $\square$

Corollary F.13 (The equivalence). *For all real $E_{00}, E_{01}, E_{10}, E_{11}$, PSD5 of $M(E)$ holds if and only if the correlators are column contractive: `npa_psd_iff_column_contractive` (*kernel/quantum/CHSHColumnCheck.v*). With symmetry packaged in, `column_contractive_iff_npa_psd` states contractivity if and only if `npa_psd` holds of `zero_marginal_npa` at the four correlators.*

The bound itself goes through rows rather than columns.

Theorem F.14 (PSD implies Tsirelson). *If `npa_psd` holds of `zero_marginal_npa` at $E_{00}, E_{01}, E_{10}, E_{11}$, then $S^2 \leq 8$, and $|S| \leq \sqrt{8}$. These are `npa_psd_implies_tsirelson_bound` and `npa_psd_implies_tsirelson_bound_abs`.*

Proof. `psd_3x3_determinant_nonneg` (*kernel/quantum/ConstructivePSD.v*) says a symmetric PSD matrix with unit diagonal has every 3×3 principal minor $1 - x^2 - y^2 - z^2 + 2xyz \geq 0$. On rows A_0, B_0, B_1 the entries are $x = E_{00}, y = E_{01}, z = 0$, so $E_{00}^2 + E_{01}^2 \leq 1$. On rows A_1, B_0, B_1 the same gives $E_{10}^2 + E_{11}^2 \leq 1$. This is `npa_psd_zero_marginal_implies_row_bounds`. Then Theorem F.5 applies, through `tsirelson_from_minors` (*kernel/quantum/TsirelsonGeneral.v*). $\square$

Here are the three conditions and the score at a few points, all by direct arithmetic. In this table, $a = 3/5$, $b = 4/5$ and $t = 1/\sqrt{2}$.

point ($E_{00}, E_{01}, E_{10}, E_{11}$)	S	p	q	$pq - s^2$	contractive
(0, 0, 0, 0)	0	1	1	1	yes
(0.6, 0.6, 0.6, -0.6)	2.4	0.28	0.28	0.0784	yes
($a, b, b, -a$)	2.8	0	0	0	yes, with no slack
($t, t, t, -t$)	$2\sqrt{2}$	0	0	0	yes, with no slack
(1, 1, 1, 1), all-ones plan	2	-1	-1		no
(1, 0, 1, 0), a classical mixture	2	-1	1		no
(1, 1, 1, -1), PR box	4	-1	-1		no

Every deterministic plan has all four correlators equal to ± 1 , so $p = 1 - 1 - 1 = -1$ and every one of the sixteen fails.

Remark F.3 (What PSD of the pinned matrix means). $M(E)$ is the level-1 moment matrix with the two within-party entries $\langle A_0 A_1 \rangle$ and $\langle B_0 B_1 \rangle$ pinned to zero. Its PSD is sufficient for the correlators to admit some PSD completion (item G, `zero_marginal_implies_elliptope`), and it is not necessary: the all-ones plan has a completion but fails here. So a check built on $M(E)$ is sound for the completion set and not complete for it, and it rejects some honest classical tallies. That correlators with a PSD completion are exactly those reachable by quantum measurements is Tsirelson's theorem, cited and not proved here (item H).

F.5 E. The integer check

Definition F.7 (`column_contractive_check_witness`). (`kernel/quantum/CHSHColumnCheck.v`) From the counters of Definition F.2, with all arithmetic in $\mathbb{Z}$, let

$$\begin{aligned}
 A &= N_{00}^2 N_{10}^2 - D_{00}^2 N_{10}^2 - D_{10}^2 N_{00}^2, \\
 B &= N_{01}^2 N_{11}^2 - D_{01}^2 N_{11}^2 - D_{11}^2 N_{01}^2, \\
 C &= D_{00} D_{01} N_{10} N_{11} + D_{10} D_{11} N_{00} N_{01}.
 \end{aligned}$$

The check is the conjunction of seven facts, in this order: $0 < N_{00}$, $0 < N_{01}$, $0 < N_{10}$, $0 < N_{11}$, $0 \leq A$, $0 \leq B$, and $C \cdot C \leq A \cdot B$.

There is no division, no real number, no square root and no S in it.

Lemma F.15 (The seven facts are the three conditions with denominators cleared). As-

sume $N_{00}, N_{01}, N_{10}, N_{11} > 0$ and put $E_{xy} = D_{xy}/N_{xy}$. Then, with p, q, s as in Definition F.6,

$$p = \frac{A}{N_{00}^2 N_{10}^2}, \quad q = \frac{B}{N_{01}^2 N_{11}^2}, \quad s = \frac{C}{N_{00} N_{01} N_{10} N_{11}},$$

$$pq - s^2 = \frac{AB - C^2}{(N_{00} N_{01} N_{10} N_{11})^2}.$$

Because every denominator is positive, $0 \leq A \iff p \geq 0$, $0 \leq B \iff q \geq 0$ and $C^2 \leq AB \iff s^2 \leq pq$.

Proof. For p : $1 - D_{00}^2/N_{00}^2 - D_{10}^2/N_{10}^2$; multiply by $N_{00}^2 N_{10}^2$ to get $N_{00}^2 N_{10}^2 - D_{00}^2 N_{10}^2 - D_{10}^2 N_{00}^2 = A$. For q the same with columns 01 and 11, multiplied by $N_{01}^2 N_{11}^2$. For s :

$$\frac{D_{00}}{N_{00}} \frac{D_{01}}{N_{01}} + \frac{D_{10}}{N_{10}} \frac{D_{11}}{N_{11}} = \frac{D_{00} D_{01} N_{10} N_{11} + D_{10} D_{11} N_{00} N_{01}}{N_{00} N_{01} N_{10} N_{11}} = \frac{C}{N_{00} N_{01} N_{10} N_{11}}.$$

Then $pq = AB/(N_{00}^2 N_{10}^2 N_{01}^2 N_{11}^2) = AB/(N_{00} N_{01} N_{10} N_{11})^2$ and $s^2 = C^2/(N_{00} N_{01} N_{10} N_{11})^2$; subtract. Multiplying an inequality by a positive number preserves it in both directions. $\square$

The Coq source states the direction it needs:

Theorem F.16 (Soundness of the integer check). *If `column_contractive_check_witness w = true`, then the four correlators `state_bucket_correlation samexy diffxy` of w are column contractive. This is `column_contractive_check_witness_sound (kernel/quantum/CHSHColumnCheck.v)`, and `column_contractive_check_witness_npa_psd` (same file) carries it on to PSD of the pinned matrix.*

Proof. The first four conjuncts make every N_{xy} positive, so the empty-bucket convention never applies and each correlator is D_{xy}/N_{xy} . The integer facts lift to the reals by the injection $\mathbb{Z} \rightarrow \mathbb{R}$, which preserves order. Lemma F.15 turns them into the three conditions. $\square$

Corollary F.17 (A passed check implies the Tsirelson bound). *If `column_contractive_check_witness w = true`, the correlators of w satisfy $S^2 \leq 8$. It is the composition of Theorem F.16, Corollary F.13 and Theorem F.14.*

On the small machine the same conclusion is one checked theorem, `small_chsh_check_tsirelson` (`SmallChshCheck.v`), and with the earned form `small_chsh_flag_implies_tsirelson` (`SmallChshMachine.v`); see Section 15.

A worked tally. Take $(\text{same}, \text{diff}) = (4, 1), (4, 1), (4, 1), (1, 4)$ for the pairs 00, 01, 10, 11. Then $N_{xy} = 5$ for all four, $D_{00} = D_{01} = D_{10} = 3$, $D_{11} = -3$, and the correlators are

0.6, 0.6, 0.6, -0.6 with $S = 2.4$. Compute:

$$A = 25 \cdot 25 - 9 \cdot 25 - 9 \cdot 25 = 625 - 225 - 225 = 175,$$

$$B = 25 \cdot 25 - 9 \cdot 25 - 9 \cdot 25 = 175,$$

$$C = 3 \cdot 3 \cdot 5 \cdot 5 + 3 \cdot (-3) \cdot 5 \cdot 5 = 225 - 225 = 0.$$

All seven facts hold: four positive N , $0 \leq 175$ twice, and $0 \leq 175 \cdot 175$. Dividing back, $p = 175/625 = 0.28$ and $s = 0$, the second row of the table in item D.

Two more, for contrast. One round per pair from the all-ones plan, $(1, 0)$ four times, gives $N = D = 1$ everywhere and $A = 1 - 1 - 1 = -1 < 0$: the check fails. The tally $(4, 1), (9, 1), (9, 1), (1, 4)$ gives correlators $3/5, 4/5, 4/5, -3/5$, with $N = 5, 10, 10, 5$ and $D = 3, 8, 8, -3$: $A = 25 \cdot 100 - 9 \cdot 100 - 64 \cdot 25 = 0$, $B = 100 \cdot 25 - 64 \cdot 25 - 9 \cdot 100 = 0$ and $C = 3 \cdot 8 \cdot 10 \cdot 5 + 8 \cdot (-3) \cdot 5 \cdot 10 = 0$. It passes with no slack, at $S = 2.8$.

F.6 F. Level 1+AB

The source indexes Alice's observables A_1, A_2 and Bob's B_1, B_2 ; I keep the book's A_0, A_1, B_0, B_1 , so the Coq e_{11} is E_{00} here, and so on. The five extra parameters are named in the record `Q1ABMoments` (kernel/quantum/QuantumPartitionPSD_1AB.v): γ_1 for $\langle A_0 A_1 B_0 \rangle$, γ_2 for $\langle A_0 A_1 B_1 \rangle$, γ_3 for $\langle A_0 B_0 B_1 \rangle$, γ_4 for $\langle A_1 B_0 B_1 \rangle$, γ_5 for $\langle A_0 A_1 B_0 B_1 \rangle$. These are labels only; the file builds no operators.

Definition F.8 (`qlab_moment_matrix`). (kernel/quantum/QuantumPartitionPSD_1AB.v) The 9×9 real matrix on rows and columns $I, A_0, A_1, B_0, B_1, A_0 B_0, A_0 B_1, A_1 B_0, A_1 B_1$, with the entries exactly as the Coq definition places them:

	I	A_0	A_1	B_0	B_1	$A_0 B_0$	$A_0 B_1$	$A_1 B_0$	$A_1 B_1$
I	1	0	0	0	0	E_{00}	E_{01}	E_{10}	E_{11}
A_0	0	1	0	E_{00}	E_{01}	0	0	γ_1	γ_2
A_1	0	0	1	E_{10}	E_{11}	γ_1	γ_2	0	0
B_0	0	E_{00}	E_{10}	1	0	0	γ_3	0	γ_4
B_1	0	E_{01}	E_{11}	0	1	γ_3	0	γ_4	0
$A_0 B_0$	E_{00}	0	γ_1	0	γ_3	1	0	0	γ_5
$A_0 B_1$	E_{01}	0	γ_2	γ_3	0	0	1	$-\gamma_5$	0
$A_1 B_0$	E_{10}	γ_1	0	0	γ_4	0	$-\gamma_5$	1	0
$A_1 B_1$	E_{11}	γ_2	0	γ_4	0	γ_5	0	0	1

Two conventions are fixed by this layout. First, the four-body cells carry opposite signs: $+\gamma_5$ at $(A_0 B_0, A_1 B_1)$ and $-\gamma_5$ at $(A_0 B_1, A_1 B_0)$. The file states this is a convention of the real matrix, not derived from operators, and guards it with `qlab_four_body_cells_are_conjugate` (QuantumPartitionPSD_1AB.v). Second, the matrix keeps the

level-1 pinning: every cell whose label reduces to a marginal or to $\langle A_0 A_1 \rangle$ or $\langle B_0 B_1 \rangle$ is zero, for instance $(A_0, A_0 B_0)$, $(A_0 B_0, A_0 B_1)$ and $(A_0 B_0, A_1 B_0)$. Symmetry is `qlab_moment_matrix_symmetric`.

Write a vector as $(v_I, v_{A_0}, v_{A_1}, v_{B_0}, v_{B_1}, v_{00}, v_{01}, v_{10}, v_{11})$, where v_{xy} is the coordinate of $A_x B_y$. Put

$$\begin{aligned} L_1 &= E_{00}v_{00} + E_{01}v_{01} + E_{10}v_{10} + E_{11}v_{11}, \\ L_2 &= E_{00}v_{B_0} + E_{01}v_{B_1} + \gamma_1 v_{10} + \gamma_2 v_{11}, \\ L_3 &= E_{10}v_{B_0} + E_{11}v_{B_1} + \gamma_1 v_{00} + \gamma_2 v_{01}. \end{aligned}$$

Definition F.9 (The residual, `qlab_residual`). (`QuantumPartitionPSD_1AB.v`)

$$\begin{aligned} R &= v_{B_0}^2 + v_{B_1}^2 + v_{00}^2 + v_{01}^2 + v_{10}^2 + v_{11}^2 \\ &\quad + 2v_{B_0}(\gamma_3 v_{01} + \gamma_4 v_{11}) + 2v_{B_1}(\gamma_3 v_{00} + \gamma_4 v_{10}) \\ &\quad + 2\gamma_5(v_{00}v_{11} - v_{01}v_{10}) - L_1^2 - L_2^2 - L_3^2. \end{aligned}$$

The predicate `column_contractive_qlab` says $R \geq 0$ for all six real inputs.

Lemma F.18 (Three squares plus the residual, `quad9_qlab_sos_decomposition`). (`QuantumPartitionPSD_1AB.v`, closed by `ring`.) For every vector,

$$v^T M_9 v = (v_I + L_1)^2 + (v_{A_0} + L_2)^2 + (v_{A_1} + L_3)^2 + R.$$

Proof. The coordinates v_I, v_{A_0}, v_{A_1} appear in the form only through their diagonal squares and the cross terms $2v_I L_1, 2v_{A_0} L_2, 2v_{A_1} L_3$, read off rows I, A_0, A_1 of the matrix. Completing each square adds $L_1^2 + L_2^2 + L_3^2$, which the residual subtracts. What is left in R is the diagonal of the remaining six coordinates and the cross terms among them, read off the lower right 6×6 block. $\square$

Theorem F.19 (PSD at level 1+AB, `qlab_psd_iff_column_contractive`). (`QuantumPartitionPSD_1AB.v`) For all real E_{xy} and $\gamma_1, \dots, \gamma_5$, PSD9 of the matrix of Definition F.8 holds if and only if `column_contractive_qlab` holds. The directions are `column_contractive_qlab_implies_psd9` and `psd9_implies_column_contractive_qlab`.

Proof. If $R \geq 0$ everywhere, Lemma F.18 writes the form as three squares plus R . Conversely, given the six residual inputs, choose $v_I = -L_1, v_{A_0} = -L_2, v_{A_1} = -L_3$; the three squares vanish and PSD gives $R \geq 0$. This is the level-1 argument of item D one size up. $\square$

As a quadratic form in $(v_{B_0}, v_{B_1}, v_{00}, v_{01}, v_{10}, v_{11})$, numbered 1 to 6, the residual has the symmetric coefficient matrix H with these upper entries. They are the `q12345_H11` to `q12345_H66` definitions (`QuantumPartitionPSD_1AB.v` onward), and `q12345_s`

`ym6_qf_equals_residual` proves the form equals R . Each entry can be read back off R by collecting the coefficient of $v_i v_j$.

$$\begin{aligned}
H_{11} &= 1 - E_{00}^2 - E_{10}^2, & H_{22} &= 1 - E_{01}^2 - E_{11}^2, \\
H_{12} &= -(E_{00}E_{01} + E_{10}E_{11}), & H_{13} &= -E_{10}\gamma_1, \\
H_{14} &= \gamma_3 - E_{10}\gamma_2, & H_{15} &= -E_{00}\gamma_1, \\
H_{16} &= \gamma_4 - E_{00}\gamma_2, & H_{23} &= \gamma_3 - E_{11}\gamma_1, \\
H_{24} &= -E_{11}\gamma_2, & H_{25} &= \gamma_4 - E_{01}\gamma_1, \\
H_{26} &= -E_{01}\gamma_2, & H_{33} &= 1 - E_{00}^2 - \gamma_1^2, \\
H_{34} &= -(E_{00}E_{01} + \gamma_1\gamma_2), & H_{35} &= -E_{00}E_{10}, \\
H_{36} &= \gamma_5 - E_{00}E_{11}, & H_{44} &= 1 - E_{01}^2 - \gamma_2^2, \\
H_{45} &= -\gamma_5 - E_{01}E_{10}, & H_{46} &= -E_{01}E_{11}, \\
H_{55} &= 1 - E_{10}^2 - \gamma_1^2, & H_{56} &= -(E_{10}E_{11} + \gamma_1\gamma_2), \\
H_{66} &= 1 - E_{11}^2 - \gamma_2^2.
\end{aligned}$$

The top left 2×2 block is $I_2 - E^\top E$ from item D.

The full check does not decide $R \geq 0$ directly. It proves the stronger fact that H is positive definite, by two Schur steps and a Sylvester test, all fraction-free.

Lemma F.20 (Fraction-free Schur step). *For a symmetric $n \times n$ matrix h and any vector v , with $S_{ij} = h_{11}h_{ij} - h_{1i}h_{1j}$ for $i, j \geq 2$ (here S is the Schur matrix, not the score),*

$$h_{11} v^\top h v = \left(\sum_j h_{1j} v_j \right)^2 + \sum_{i,j \geq 2} S_{ij} v_i v_j.$$

So if $h_{11} > 0$ and the $(n-1) \times (n-1)$ matrix S gives a nonnegative form, then h does too. The instances used are `sym6_Schur_identity` (`QuantumPartitionPSD_1AB.v`) and `sym5_Schur_identity`, with the conclusions `sym6_qf_nonneg_from_pd` and `sym5_qf_nonneg_from_pd`.

Proof. Expand the square: $h_{11}^2 v_1^2 + 2h_{11}v_1 \sum_{j \geq 2} h_{1j}v_j + (\sum_{j \geq 2} h_{1j}v_j)^2$. The left side is $h_{11}^2 v_1^2 + 2h_{11}v_1 \sum_{j \geq 2} h_{1j}v_j + h_{11} \sum_{i,j \geq 2} h_{ij}v_i v_j$. The difference is $\sum_{i,j \geq 2} (h_{11}h_{ij} - h_{1i}h_{1j})v_i v_j$. Dividing by $h_{11} > 0$ keeps the sign. $\square$

Lemma F.21 (Sylvester for 4×4 , `sym4_qf_nonneg_from_pd`). (`QuantumPartitionPSD_1AB.v`) Let h be symmetric 4×4 with leading principal minors d_1, d_2, d_3, d_4 (`sym4_d1` to `sym4_d4`). If all four are strictly positive, then $v^\top h v \geq 0$ for all v .

Proof. `sym4_LDLT_identity` (closed by `ring`) is the fraction-free factorization

$$d_1^2 d_2 d_3 v^\top h v = d_1 d_2 d_3 P_1^2 + d_1 d_3 Q_1^2 + Q_2^2 + d_1^2 d_2 d_4 v_4^2,$$

where $P_1 = \sum_j h_{1j} v_j$, $Q_1 = (h_{11} h_{22} - h_{12}^2) v_2 + (h_{11} h_{23} - h_{12} h_{13}) v_3 + (h_{11} h_{24} - h_{12} h_{14}) v_4$, and Q_2 is the third linear form (`sym4_Q2`). With every $d_k > 0$ the right side is nonnegative and the factor on the left is positive. $\square$

The cascade for the full check runs on H . Step one: $H_{11} > 0$ and the 5×5 matrix $S_{ij}^{(6)} = H_{11} H_{ij} - H_{1i} H_{1j}$, $2 \leq i, j \leq 6$, gives a nonnegative form. Step two: $S_{22}^{(6)} > 0$ and the 4×4 matrix $S_{ij}^{(5)} = S_{22}^{(6)} S_{ij}^{(6)} - S_{2i}^{(6)} S_{2j}^{(6)}$, $3 \leq i, j \leq 6$, gives a nonnegative form. Step three: the four leading minors of $S^{(5)}$ are positive. The real predicates are `sym5_pd_interior`, `sym6_pd_interior` and the witness `qlab_g12345_minors_witness`, which implies `column_contractive_qlab` and then `PSD9 (qlab_g12345_minors_witness_implies_psd9)`.

Definition F.10 (What the integer check computes). The supplied moments arrive as bucket pairs: $\gamma_k = N_{g_k} / D_{g_k}$ with $N_{g_k} = \text{same}_{g_k} - \text{diff}_{g_k}$ and $D_{g_k} = \text{same}_{g_k} + \text{diff}_{g_k}$. Watch the letters: for these five buckets the source swaps them, so here N is the difference and D the count, the reverse of item A. I keep the source's letters so the names below match the files. Let $K = (N_{00} N_{01} N_{10} N_{11} D_{g_1} D_{g_2} D_{g_3} D_{g_4} D_{g_5})^2$ (`g12345_COMMON_Z`, `kernel/quantum/CHSHColumnCheck.v`). The check computes the 21 integers KH_{ij} (`cleared_g12345_H11_Z` to `cleared_g12345_H56_Z`), the 15 integers of $S^{(6)}$ at scale K^2 and the 10 integers of $S^{(5)}$ at scale K^4 , each by `schur_step_Z`, $h_{11} h_{ij} - h_{1i} h_{1j}$ on integers. The decider `qlab_g12345_check_z_kernel` is the conjunction of: $N_{xy} > 0$ for the four pairs; $D_{g_k} > 0$ for $k = 1..5$; $-D_{g_k} < N_{g_k} < D_{g_k}$ for $k = 1..5$; and six strict positivities, namely $KH_{11} > 0$, $K^2 S_{22}^{(6)} > 0$, and the four leading minors of the integer $S^{(5)}$. The full check `qlab_g12345_full_integer_check_kernel` is `column_contractive_check_witness` of item E and this decider.

The bridge lemmas, for example `cleared_g12345_H11_Z_bridge` (`QuantumPartitionPSD_1AB.v`), prove that each cleared integer, read as a real, equals the corresponding real quantity times the stated positive power of K . A $k \times k$ leading minor of a matrix scaled by $c > 0$ scales by c^k . So every integer positivity holds exactly when the real one does.

Theorem F.22 (The full 1+AB check implies PSD). *Let the counts make every N_{xy} and every D_{g_k} positive, and let `qlab_g12345_check_z_kernel` pass on them. Then the matrix of Definition F.8 at the four correlators and $\gamma_k = N_{g_k} / D_{g_k}$ is symmetric and PSD9. This is `qlab_g12345_caller_witness_z_abs_sound` followed by `qlab_g12345_minors_witness_implies_psd9` (`kernel/quantum/QuantumPartitionPSD_1AB.v`).*

Proof. The bucket conjuncts make every denominator positive, the bridges carry the six integer positivities to the real cascade, and Lemma F.20 twice plus Lemma F.21 give $R \geq 0$, hence PSD9 by Theorem F.19. $\square$

Remark F.4 (Interior only). Every test in the cascade is strict, so a passing check certifies that H is positive definite, not merely semidefinite. A point where H is PSD but singular cannot pass, and neither can a moment with $|\gamma_k| = 1$. For example, the tally of item E with correlators $(3/5, 4/5, 4/5, -3/5)$ has $H_{11} = 1 - 9/25 - 16/25 = 0$, so the full 1+AB check fails on it for every choice of γ , though the level-1 check of item E accepts it. The source proves soundness only; no converse is stated.

A worked example of the cascade, in exact rational arithmetic (argued here, not a Coq Example). Take the correlators $(0.6, 0.6, 0.6, -0.6)$ and $\gamma = (0, 0, 0, 0, -1/2)$; as buckets, γ_1 to γ_4 are $(1, 1)$, giving $0/2$, and γ_5 is $(1, 3)$, giving $-2/4$, both inside the strict bounds. Then $H_{11} = H_{22} = 7/25$, $H_{12} = 0$, and every entry H_{1j} , H_{2j} with $j \geq 3$ is zero. So $S_{22}^{(6)} = (7/25)^2 = 49/625$, and $S^{(5)}$ is a positive multiple of the lower right block of H :

$$S^{(5)} = \frac{343}{781250} \begin{pmatrix} 32 & -18 & -18 & -7 \\ -18 & 32 & 7 & 18 \\ -18 & 7 & 32 & 18 \\ -7 & 18 & 18 & 32 \end{pmatrix},$$

where $50H_{33} = 32$, $50H_{34} = 50H_{35} = -18$, $50H_{36} = 50(-1/2 + 0.36) = -7$, $50H_{45} = 50(1/2 - 0.36) = 7$, and $50H_{46} = 50H_{56} = 18$. The leading minors of the integer matrix are 32, 700, 15000 and 140625, all positive. With $H_{11} = 7/25 > 0$ and $S_{22}^{(6)} > 0$, all six tests pass, and with item E's $A = B = 175$, $C = 0$ the full check passes. At $\gamma_5 = 0$ instead, the same block becomes $I_4 - \mathbf{e}\mathbf{e}^T$ with $\mathbf{e} = (0.6, 0.6, 0.6, -0.6)$, whose third leading minor is negative ($\frac{1}{25^3}$ times -1250), and the check fails.

That last fact is general. The zero-moment check `column_contractive_check_q1ab_kernel` (kernel/quantum/CHSHColumnCheck.v) takes no γ buckets; it is the level-1 check together with `sum_E_sq_check_witness`, the integer form of $E_{00}^2 + E_{01}^2 + E_{10}^2 + E_{11}^2 \leq 1$ with denominators cleared by $(N_{00}N_{01}N_{10}N_{11})^2$.

Theorem F.23 (The zero-moment check sits inside the classical bound). *If `column_contractive_check_q1ab w = true` (`QuantumPartitionPSD_1AB.v`, the same function), then the four correlators of w satisfy $E_{00}^2 + E_{01}^2 + E_{10}^2 + E_{11}^2 \leq 1$ (`q1ab_check_at_gzero_forces_unit_ball`) and $|S| \leq 2$ (`q1ab_check_at_gzero_implies_classical_bound`).*

Proof. At $\gamma = 0$ the residual splits into the level-1 form in (v_{B_0}, v_{B_1}) and $|\mathbf{v}|^2 - \langle \mathbf{e}, \mathbf{v} \rangle^2$ in $\mathbf{v} = (v_{00}, v_{01}, v_{10}, v_{11})$, where $\mathbf{e} = (E_{00}, E_{01}, E_{10}, E_{11})$ (`q1ab_residual_g_zero_decomp`). The check gives column contractivity at $\gamma = 0$ (`column_contractive_ch`

eck_q1ab_sound_at_g_zero). Test at $v_{B_0} = v_{B_1} = 0$ and $\mathbf{v} = \mathbf{e}$: with $r = |\mathbf{e}|^2 \geq 0$, the residual is $r - r^2 \geq 0$, so $r \leq 1$. Then Lemma F.4 gives $S^2 \leq 4r \leq 4$. $\square$

So the zero-moment check certifies a strict part of the classical region: no point above $|S| = 2$, and none of the sixteen deterministic points either, since each has $|\mathbf{e}|^2 = 4$. A point above 2 needs supplied moments. `q1ab_certifies_a_superclassical_chsh_point` (`QuantumPartitionPSD_1AB.v`) proves PSD9 at correlators $(3/5, 3/5, 3/5, -3/5)$, $\gamma = (0, 0, 0, 0, -1/2)$, with $S = 12/5$, through the γ_5 slice: its real condition `q1ab_g5_caller_witness` asks $-1 < \gamma_5 < 1$ and

$$(1 - \gamma_5)[(E_{00} + E_{11})^2 + (E_{01} - E_{10})^2] + (1 + \gamma_5)[(E_{00} - E_{11})^2 + (E_{01} + E_{10})^2] \leq 2(1 - \gamma_5^2).$$

At this point the left side is $\frac{3}{2} \cdot 0 + \frac{1}{2} \cdot (\frac{36}{25} + \frac{36}{25}) = \frac{36}{25} = 1.44$ and the right side is $2 \cdot \frac{3}{4} = 1.5$. The integer version, `q1ab_g5_kernel_check_accepts_superclassical`, runs the γ_5 check on the counters $(4, 1), (4, 1), (4, 1), (1, 4)$ and the bucket $(1, 3)$ and returns `true` by computation.

Each of the four 1+AB checks (`CHSHColumnCheck.v`) contains `column_contractive_check_witness` as a conjunct, so a passing 1+AB check also passes the level-1 check and inherits $S^2 \leq 8$ from item E. That is a composition I did by reading the definitions; the source has no single theorem stating it.

F.7 G. The elliptope completion

Definition F.11 (Completion and membership). `completed_npa` (`kernel/quantum/ElliptopeCompletion.v`) is the record of item D with the within-party entries set to free reals: x in the (A_0, A_1) cells and y in the (B_0, B_1) cells, marginals still zero. The correlators are elliptope realizable (`elliptope_realizable`) when some x, y make the completed matrix symmetric and PSD.

$$M(E; x, y) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & x & E_{00} & E_{01} \\ 0 & x & 1 & E_{10} & E_{11} \\ 0 & E_{00} & E_{10} & 1 & y \\ 0 & E_{01} & E_{11} & y & 1 \end{pmatrix}$$

Its quadratic form at $v = (v_0, \dots, v_4)$ is $\sum_k v_k^2 + 2xv_1v_2 + 2yv_3v_4 + 2E_{00}v_1v_3 + 2E_{01}v_1v_4 + 2E_{10}v_2v_3 + 2E_{11}v_2v_4$ (`completed_quad_expand`). The pinned matrix is the case $x = y = 0$, so PSD of $M(E)$ implies membership: `zero_marginal_implies_elliptope`.

Theorem F.24 (Deterministic plans are inside, `deterministic_strategy_ellip`)

`tope`). (*ElliptopeCompletion.v*) If $a_0^2 = a_1^2 = b_0^2 = b_1^2 = 1$, then $(a_0b_0, a_0b_1, a_1b_0, a_1b_1)$ is elliptope realizable, with $x = a_0a_1$ and $y = b_0b_1$.

Proof. With these entries the form equals $v_0^2 + (a_0v_1 + a_1v_2 + b_0v_3 + b_1v_4)^2$. Expanding the second square gives $a_0^2v_1^2 + \dots$ on the diagonal, which is $v_1^2 + \dots$ by the unit squares, and the cross terms $2a_0a_1v_1v_2 = 2xv_1v_2$, $2a_0b_0v_1v_3 = 2E_{00}v_1v_3$, and so on for all six pairs. A sum of two squares is never negative. $\square$

The all-ones plan is the instance `classical_tightness_witness_elliptope`, and the classical mixture $(1, 0, 1, 0)$ is `turing_point_elliptope`, at $x = 1, y = 0$, with form $v_0^2 + (v_1 + v_2 + v_3)^2 + v_4^2$.

Theorem F.25 (Convexity and finite mixtures). For $0 \leq \lambda \leq 1$, if E and F are elliptope realizable, so is $\lambda E + (1 - \lambda)F$ (`elliptope_convex`, *ElliptopeCompletion.v*). For finitely many members with nonnegative weights summing to at most 1, the weighted sum is realizable (`elliptope_finite_mixture`). Every finite mixture of deterministic plans is realizable (`lhv_mixture_elliptope`).

Proof. The form is affine in (E, x, y) for each fixed v . Take the completion $(\lambda x_1 + (1 - \lambda)x_2, \lambda y_1 + (1 - \lambda)y_2)$; the form at the mixture is the same mixture of two nonnegative forms. The finite version is induction on the number of members, with the shortfall in weight absorbed by the zero point (`elliptope_zero`). The last statement combines this with Theorem F.24. $\square$

Theorem F.26 (Tsirelson inside the completion set, `elliptope_tsirelson`). (*ElliptopeCompletion.v*) If E is elliptope realizable then $S^2 \leq 8$.

Proof. For a symmetric PSD matrix the bilinear form obeys Cauchy-Schwarz, $\mathcal{B}(u, w)^2 \leq \mathcal{Q}(u) \mathcal{Q}(w)$ (`psd_cauchy_schwarz`), because $\mathcal{Q}(u + tw) = \mathcal{Q}(u) + 2t\mathcal{B}(u, w) + t^2\mathcal{Q}(w) \geq 0$ for all t and Lemma F.11 applies. Take $u_1 = e_{A_0}$ and $w_+ = e_{B_0} + e_{B_1}$: $\mathcal{B}(u_1, w_+) = E_{00} + E_{01}$, $\mathcal{Q}(u_1) = 1$, $\mathcal{Q}(w_+) = 2 + 2y$. Take $u_2 = e_{A_1}$ and $w_- = e_{B_0} - e_{B_1}$: $\mathcal{B}(u_2, w_-) = E_{10} - E_{11}$, $\mathcal{Q}(w_-) = 2 - 2y$. So $c_1^2 \leq 2 + 2y$ and $c_2^2 \leq 2 - 2y$ with $c_1 = E_{00} + E_{01}$, $c_2 = E_{10} - E_{11}$, and both budgets are nonnegative. Then $c_1^2 c_2^2 \leq (2 + 2y)(2 - 2y) = 4 - 4y^2 \leq 4$, so $c_1 c_2 \leq 2$, and $S^2 = (c_1 + c_2)^2 = c_1^2 + 2c_1 c_2 + c_2^2 \leq (2 + 2y) + 4 + (2 - 2y) = 8$. $\square$

The completion can make a row of E longer than 1 (the all-ones plan has rows of squared length 2), which is why this proof does not use the row bounds of item D.

Corollary F.27 (The PR box is outside, `pr_box_not_elliptope`). (*ElliptopeCompletion.v*) $(1, 1, 1, -1)$ is not elliptope realizable, since its $S^2 = 16 > 8$.

Theorem F.28 (Strictly beyond the classical bound, `beyond_classical_elliptope`). (*ElliptopeCompletion.v*) $(3/5, 3/5, 3/5, -3/5)$ is elliptope realizable and has $S = 12/5 > 2$.

Proof. At $x = y = 0$ the form equals $v_0^2 + (v_1 + \frac{3}{5}v_3 + \frac{3}{5}v_4)^2 + (v_2 + \frac{3}{5}v_3 - \frac{3}{5}v_4)^2 + \frac{7}{25}v_3^2 + \frac{7}{25}v_4^2$. Check: the two middle squares give v_3^2 coefficient $\frac{9}{25} + \frac{9}{25} = \frac{18}{25}$, plus $\frac{7}{25}$ is 1; the v_3v_4 terms are $+\frac{18}{25}$ and $-\frac{18}{25}$ and cancel; the cross terms are $2 \cdot \frac{3}{5}v_1v_3$, $2 \cdot \frac{3}{5}v_1v_4$, $2 \cdot \frac{3}{5}v_2v_3$ and $-2 \cdot \frac{3}{5}v_2v_4$, as the entries require. $\square$

By Theorem F.2 no finite factorizable model has $S > 2$, so this point is not a finite mixture of plans, while Theorem F.25 puts every such mixture inside. The completion set therefore strictly contains the finite classical set, and the PR box lies outside it. The source does not state that the completion set lies inside the no-signalling set.

The gate (`kernel/quantum/ElliptopeGate.v`) is an integer decider with a soundness proof. Its completion entries arrive as bucket pairs, and an empty bucket is read as 0/1 (`bden`), so unlike item E it does not require every pair to be sampled.

Definition F.12 (The two branches). `elliptope_pd_check` (*ElliptopeGate.v*): with the completion supplied as buckets and all entries scaled to integers by a positive common factor K , the four leading minors of the 4×4 block on A_0, A_1, B_0, B_1 are strictly positive. `elliptope_ldl_check`: with the completion and a rational certificate supplied (six subdiagonal entries over a common positive denominator, four pivots $d_k \geq 0$ over another), the ten entry equations of $M = LDL^T$ hold in cross-multiplied integer arithmetic (`ldl_equations`). `elliptope_check_full` is their disjunction.

Theorem F.29 (Soundness of the gate, `elliptope_check_full_sound`). (*ElliptopeGate.v*) If `elliptope_check_full` returns true on a counter record, any completion buckets and any certificate, then the four correlators of the record are elliptope realizable. And `elliptope_full_gate_never_accepts_pr_box`: on the PR box tally the gate returns false for every choice of the remaining arguments.

Proof. The first branch reduces to Lemma F.21 through the same fraction-free identity (`elliptope_pd_check_sound`). The second is a weighted sum of squares: given the ten equations, the form is $\sum_k d_k p_k^2$ with $d_k \geq 0$ (`elliptope_ldl_check_sound`). The PR statement is soundness followed by Corollary F.27. $\square$

Computed examples, each closed by reflexivity: `gate_accepts_all_ones`, the all-ones plan, through the second branch at $x = y = 1$; `gate_accepts_turing_point`; `gate_accepts_beyond_classical`, the 12/5 point, through the first branch at $x = y = 0$; `gate_accepts_pythagorean_boundary`, the point $(3/5, 4/5, 4/5, -3/5)$, through the second branch with pivots $(1, 1, 0, 0)$.

The gate is a decider with a soundness proof, and that's all it is: no property of the small machine is built from it.

F.8 H. What is not proved

The following are cited results or open, and none of them is a Coq theorem in this development.

The arcsine boundary. For two settings and two outcomes per side, the correlators reachable by quantum measurements are characterized by $|\arcsin E_{00} + \arcsin E_{01} + \arcsin E_{10} - \arcsin E_{11}| \leq \pi$ and its sign variants (Tsirelson, Landau 1988, Masanes 2003). No arcsine appears in any Coq statement; the only mentions are comments in `ElliptopeGate.v`. The claim that $(3/5, 4/5, 4/5, -3/5)$ lies on that boundary is such a comment: $\arcsin \frac{3}{5} + \arcsin \frac{4}{5} = \frac{\pi}{2}$ by the 3-4-5 triangle, so the sum is π . It is correct arithmetic about a cited characterization.

Equality with the quantum set. That the completion set of item G is exactly the set of quantum correlators is Tsirelson's representation theorem, cited in the header of `ElliptopeCompletion.v` and not proved. That PSD of any of these matrices is necessary for quantum behaviour is the NPA reading, also cited.

Achievability. No state and no pair of measurements is constructed. Theorem F.6 is a point of $\mathbb{R}^4$, not an experiment.

Completeness of the checks. Every check here is proved sound, never complete. The 1+AB cascade is strict and misses boundary points. The comment in `ElliptopeGate.v` that the second branch accepts every point with a rational PSD completion is a mathematical remark about LDL^T factorizations, not a theorem in the file; nor is there a theorem that the four 1+AB checks accept nested sets.

Finite levels. For full behaviours, marginals included, the NPA levels are relaxations. Chaturvedi (arXiv 2607.14569, 2026) shows that no finite NPA level characterizes the complete quantum set even in this simplest Bell scenario.

Statistics and physics. Nothing here says how many rounds make a tally informative, and nothing says the counters came from honest play; a program chooses its own trial inputs.

F.9 I. Prior work

Tsirelson proved the bound $2\sqrt{2}$ for quantum correlations in 1980 and showed that quantum two-outcome correlators are inner products of real unit vectors. Landau (1988) and Masanes (2003, arXiv quant-ph/0309137) gave the exact characterization for two settings and two outcomes. Wehner (2006, arXiv quant-ph/0510076) proved Tsirelson bounds, CHSH among them, by a semidefinite program on the 4×4 Gram matrix of those vectors. Navascués, Pironio and Acín (2007, arXiv quant-ph/0607119; 2008, arXiv 0803.4290) built the hierarchy of moment-matrix tests; their 2008 paper applies a Schur complement to this level-1 matrix, recovers the arcsine condition, and

discusses the 1+AB level. Formal proofs of Tsirelson’s bound already exist, by Echenim and Mhalla in Isabelle/HOL (arXiv 2306.12535) and in Lean’s mathlib (tsirelson_inequality in Mathlib.Algebra.Star.CHS), and exact rational sum-of-squares certificates go back at least to Peyrl and Parrilo (2008). So the moment-matrix route to $2\sqrt{2}$ is standard. What is specific here is narrower: an integer certificate computed from outcome counts, and a mechanized chain from “the counts pass” to PSD of the pinned matrix and on to $S^2 \leq 8$. I did not find that chain elsewhere, and I make no claim beyond it. On the host machine of the small machine, the same chain is earned: a raised flag from a clean start stands on a passing check of the packed counts, and the check passes exactly when the pinned matrix is positive semidefinite and every pair was sampled (Section 15, Theorems 15.4 and 15.5). The converse direction of the integer check, integers from PSD, is checked there too, and I make no claim beyond that.

G Constructing the quoting operation

The proof in Section 11 uses a program that takes the code of a term and returns the code of that code. Here is its construction. Nothing in the operation lets a program inspect itself by magic: its input is an ordinary tree encoded as a closed lambda term.

G.1 Numerals and syntax as data

For readability, the formulas below give names to bound variables. Renaming a bound variable changes nothing; the underlying syntax uses the de Bruijn indices of Definition 11.3. Applications associate to the left. Write $X \Downarrow Y$ here for a finite sequence of the weak call-by-value steps ending in the value Y .

Scott numerals are the following closed values:

$$\bar{0} = \lambda z. \lambda s. z, \quad \overline{n+1} = \lambda z. \lambda s. s \bar{n}.$$

Thus a numeral selects either the supplied zero case or the supplied successor case, passing its predecessor to the latter. The cases below are values, so choosing one does not prematurely run the other.

The three syntax constructors, operating on already encoded children, are

$$\begin{aligned} V &= \lambda n. \lambda v. \lambda a. \lambda l. v \ n, \\ A &= \lambda p. \lambda q. \lambda v. \lambda a. \lambda l. a \ p \ q, \\ L &= \lambda p. \lambda v. \lambda a. \lambda l. l \ p. \end{aligned}$$

For a variable of index n , an application and an abstraction, respectively, define their

codes by the following resulting values:

$$\begin{aligned}\lceil \#n \rceil &= \lambda v. \lambda a. \lambda l. v \bar{n}, \\ \lceil p q \rceil &= \lambda v. \lambda a. \lambda l. a \lceil p \rceil \lceil q \rceil, \\ \lceil \lambda p \rceil &= \lambda v. \lambda a. \lambda l. l \lceil p \rceil.\end{aligned}$$

In particular, $V \bar{n}$, $A \lceil p \rceil \lceil q \rceil$ and $L \lceil p \rceil$ reduce to these three codes. A code is closed even when the term it describes has a free variable: the index of that variable is stored as a numeral. It is data inside the code, not a free variable of the code.

G.2 Recursion without running under a lambda

The usual self-application trick has to delay its recursive call in a call-by-value language. Use

$$W = \lambda x. \lambda f. \lambda v. f (\lambda a. x x f a) v, \quad \text{rec}(F) = \lambda v. W W F v.$$

When F and u are closed values, beta reduction gives

$$\text{rec}(F) u \longrightarrow W W F u \longrightarrow^* F (\lambda a. W W F a) u = F (\text{rec}(F)) u.$$

The recursive argument is a lambda, hence already a value. It starts work only when F applies it to an argument. That is the whole reason this construction survives the call-by-value rule.

G.3 Quoting numerals

Let $E_j = \lceil \#j \rceil$ for $j = 0, 1, 2$, and let $Z = \lceil \bar{0} \rceil$, obtained directly from the finite syntax of $\bar{0} = \lambda \lambda. 1$. Write $L^k(X)$ for k nested applications of the syntax builder L to X ; it is an abbreviation for constructing k lambda nodes, not a new operation of the language.

Define

$$F_N = \lambda r. \lambda n. n Z (\lambda k. L^2(A E_0(r k))), \quad Q_N = \text{rec}(F_N).$$

Then $Q_N \bar{n} \Downarrow \lceil \bar{n} \rceil$, by induction on n . At zero the Scott numeral selects Z . At a successor it passes $\bar{n}$ to the second case; the induction hypothesis supplies the code of $\bar{n}$, and the builders produce the code of $\lambda \lambda. (\#0 \bar{n})$, which is exactly the syntax of $\overline{n+1}$.

G.4 Quoting terms

For a proposed recursive call r , define three handlers:

$$\begin{aligned}H_V &= \lambda n. L^3(A E_2(Q_N n)), \\ H_A(r) &= \lambda p. \lambda q. L^3(A (A E_1(r p)) (r q)), \\ H_L(r) &= \lambda p. L^3(A E_0(r p)).\end{aligned}$$

Now set

$$F_Q = \lambda r. \lambda t. t H_V (H_A(r)) (H_L(r)), \quad Q = \text{rec}(F_Q).$$

The handlers are lambdas and so are values. The Scott code of the input chooses one of them and hands it the encoded children. Induction on the represented term proves $Q \ulcorner t \urcorner \Downarrow \ulcorner \ulcorner t \urcorner \urcorner$:

- For $t = \#n$, the first handler constructs three lambda nodes over an application of variable 2 to the quoted numeral. That is the syntax of $\lambda\lambda\lambda.(\#2\bar{n})$, the code of $\#n$.
- For $t = pq$, the induction hypotheses quote both child codes. The second handler constructs three lambdas over variable 1 applied to those two codes. That is the syntax of the application code.
- For $t = \lambda p$, the last handler constructs three lambdas over variable 0 applied to the quoted child code. That is the syntax of the abstraction code.

Each recursive call works on a proper child of the finite input tree, or on a smaller numeral. Each builder takes finitely many beta steps. The induction therefore establishes termination as well as the value returned.

The `mkapp` used in Section 11 is the builder A here (not the term called A in that proof). With Q and that builder written out, its self-referential term has every piece on the page. The companion names are `mk_app_enc`, `rec_spec`, `Qn_spec` and `Q_spec` (`LRecursion.v`); the construction above is the argument those names identify.

H Sources and mathematical dependencies

These works identify the principal mathematical dependencies and the intellectual background named in the book. The definitions, statements and worked examples remain on its pages. A published result about thermodynamic cost, quantum correlations or universal computation is not an additional theorem proved by this manuscript.

The universal programs and the compiler that puts every computably presented Thiele machine on the universal host use effective computability and counter-machine simulation; their construction is given in Section 14.13 and Section 14.14, and the counter and lambda references identify the established machinery under them. The thermodynamic references do not remove the explicit physical premise of the heat bound. The environmental-record papers motivate Section 13; they do not prove this book's pointer conjecture.

H.1 Structure, information and knowledge

Euclid. Elements, Books I and X. Definitions, postulates, common notions and commensurability.

Plato. Timaeus; Theaetetus. The receptacle and the discussion of knowledge as true judgement with an account. Philosophical antecedents, not premises of the formal

results.

Aristotle. Physics II.1; Metaphysics VII.3. Matter, form and substance.

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